Moduli, Deformation Theory, and Stacks

— ***EYNTKA*** **Series** —

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Contents

Introduction

A moduli problem is asking whether a class of geometric objects (or a class of isomorphism classes of geometric objects) that is paramaterized by some invariant (fixed genus, closed subscheme of $\mathbb{P}^n$ with a fixed Hilbert polynomial, vector bundles of a fixed rank, etc.) can be paramaterized by a geometric object (ex. a variety, a scheme, or other new objects defined in this textbook when schemes are insufficient). The parameterisation by the geometric object records how the objects vary: a family of objects over a base S should be precisely a morphism from S to the space, so that a curve in the space is a one-parameter family and the local structure at a point measures the ways the corresponding object can deform. When such a space exists, it converts questions about an entire class of objects into geometry on a single object.

The obstacle that shapes the whole subject is the presence of automorphisms. An object with a nontrivial automorphism cannot be separated from its symmetries by a space of points alone, and the "naïve" moduli space of such objects either fails to exist or discards the information the automorphisms carry. The book is organized around three responses to this difficulty, one for each of its parts. The first part formulates a moduli problem as a functor of points and asks when that functor is representable by a scheme; it builds the construction toolkit: the Hilbert and Quot schemes and geometric invariant theory, that produces moduli spaces in the representable case. The second part linearizes the local question through deformation theory: the tangent space and the obstruction space of a moduli problem at an object are cohomology groups of that object, and they determine the dimension of the moduli space and whether it is smooth. The third part enlarges the notion of space to the algebraic stack, the geometric object that remembers automorphisms and represents the moduli problems that schemes cannot; Artin's criteria, fed by the deformation theory of the second part, certify when a moduli problem is such a stack. The final chapter applies these to the moduli of curves, where a single object carries a deformation theory, a coarse space built by invariant theory, and a stack structure coalescing automorphism information.

Prerequisites

The reader is assumed to be at home with schemes, quasi-coherent sheaves, and sheaf cohomology at the level of a first graduate course in algebraic geometry, together with a working knowledge of homological algebra. These prerequisites, and a few more specialized inputs, are cited as machinery where they are used rather than re-derived, with references to the companion volumes of this series: [12] for scheme theory and sheaf cohomology, [10] for derived functors and Ext, [4] for group schemes and their actions, and [9] for the Galois cohomology behind the arithmetic threads. Étale-cohomological facts about the Brauer group, needed only in the chapter on gerbes, are stated and attributed to the co-requisite [8]. A small number of theorems whose proofs lie beyond a first course, Artin's approximation theorem, the Keel-Mori theorem, and the stable reduction theorem, are stated and used in full with their proofs cited; everything else is proved.

The four movements of the book are largely cumulative and are best read in order, though a reader interested primarily in stacks could begin with the third part after absorbing the language of the first. What the reader should carry away is a single habit of thought: a moduli problem is a functor; when automorphisms obstruct its representability the remedy is to remember them, in a stack; and the local geometry of the resulting space is deformation theory, a computation in the cohomology of one object.

Part I

Moduli Problems and Their Construction

Whether Moduli spaces exist is the theme of this Part. The idea, due to Grothendieck, is to record the moduli problem as a functor: to each scheme S one assigns the set of families over S, with pullback making the assignment contravariant, and a *moduli space* is then a geometric object (ex. for us this chapter almost always a scheme) that *represents* this functor ([12, the Functor of Points]). The first chapter develops this language, separates fine from coarse moduli spaces, and meets the obstruction, objects with automorphisms, that prevents many natural moduli functors from being representable at all.

The remaining chapters build the machinery that produces moduli spaces when they exist. The next constructs the Hilbert and Quot schemes, Grothendieck's representable functors, from which most moduli spaces are cut out as locally closed subschemes; the last develops geometric invariant theory, which forms quotients by group actions and so produces moduli spaces of objects taken up to isomorphism. Together they are the construction toolkit that the deformation theory of the next Part refines and the stack-theoretic language of the final Part generalizes when schemes are not enough.

1

Moduli Problems and Representability via Schemes

A moduli problem asks for a space that parametrizes a class of geometric objects, but the phrase "the space of all such objects" conceals a real question: in what sense is that collection a scheme? This chapter answers it in the language Grothendieck introduced for the purpose. A moduli problem is encoded as a functor that assigns to each scheme S the set of families of objects over S; a moduli space, when one exists, is a scheme that *represents* this functor, and the universal family such a scheme carries makes the correspondence between its points and the objects being classified exact. section 1.1 sets up this dictionary and the notion of a fine moduli space, and section 1.2 carries it out in full for the Grassmannian, the fundamental example in which everything works as one would wish.

Representability is the *exception* rather than the rule: the obstruction is visible in the most classical moduli problem of all, that of elliptic curves. An object with nontrivial automorphisms admits families that are fiberwise constant yet globally "twisted", and no scheme can distinguish them from the trivial family. The best one can then hope for in terms of schemes is a *coarse moduli space*, which records the points but forgets how objects vary. Section 1.3 isolates this automorphism obstruction and works the j-line in detail, section 1.4 shows how adding rigidifying structure restores representability, and section 1.5 assembles the criteria that decide representability in general and open onto the deformation theory of the next Part.

1.1 Moduli Problems as Functors of Points

A first answer to constructing a moduli space is to take the *set* of isomorphism classes and look for a scheme structure on it. The difficulty is in determining how such a family varies. Sometimes, this can be eminently visible as in the case of elliptic curves paramaterized by the j-line, however over more complicated invariant (ex. Hilbert Polynomials of a closed-scheme) the moduli space is not so evident.

The remedy is to record, for every base scheme S, the set of families over S. This assignment is a functor, and the language for turning such a functor into geometry is the *functor of points* ([12, the Functor of Points]): a scheme is determined by the sets of maps into it from all other schemes, so a moduli problem is representable by scheme exactly when its functor of families agrees with the maps-into functor of some scheme. This section makes that precise. It defines the moduli functor, defines a fine moduli space as a scheme representing it, and extracts from the Yoneda lemma the object that makes representability useful in practice: a single universal family through which every other family is pulled back in exactly one way.

The starting datum of any moduli problem is a rule that says what a *family* over a base S is and how families pull back along a morphism of bases:

Definition 1.1.1: Moduli Problem (Functor)

A *moduli problem* is a contravariant functor $\mathcal{F}$

$$\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

that to each scheme S (of finite type over the base field k) one assigns a set

$$\mathcal{F}(S) = \{\,\text{families of the objects over } S\,\}/\cong$$

where "family of objects" is the datum of the problem taken up to isomorphism of families[a]; a morphism $f : T \to S$ is assigned the *pullback* map

$$f^*\colon \mathcal{F}(S) \longrightarrow \mathcal{F}(T)$$

carrying a family over S to its restriction over T.

[a] in definition ref:HERE this is formalized as a fibered-category of groupoids

The functor $\mathcal{F}$ carries more information than the set of isomorphism classes, which is recovered as $\mathcal{F}(\operatorname{Spec} k)$: the extra data is the collection $\mathcal{F}(S)$ for positive dimensional S together with the pullback maps, and this is precisely the record of how objects vary. A family over S is a single object that deforms as one moves through S; pullback along $f\colon T \to S$ reindexes that variation by T. Contravariance is the reflection of a basic geometric fact that a family over S restricts to a family over any T mapping into S, so a map of bases $T \to S$ induces a map of families in the opposite direction. The whole subject is the study of when this functor is geometric, and how to build the geometry when it is not.

The definition is only as concrete as the phrase "family of the objects" is made to be, so it is worth fixing that phrase in the two cases that run through this chapter.

Example 1.1.2: Families of Subschemes and of Line Bundles

1. Fix a scheme X over k. The *moduli of closed subschemes of* X assigns to S the set

$$\mathcal{F}(S) = \{\, Z \subseteq X \times_k S \text{ closed, flat over } S\,\}$$

of closed subschemes of $X \times_k S$ that are flat over S. A morphism $f\colon T \to S$ pulls a family Z

back to $f^*Z = Z \times_S T \subseteq X \times_k T$, the base change of Z along f; flatness is preserved under base change, so f^*Z is again a family over T. Over a point $\operatorname{Spec} k$ a family is a single closed subscheme of X, so $\mathcal{F}(\operatorname{Spec} k)$ is the set of closed subschemes of X. This is the functor whose representing object, when it exists, is the Hilbert scheme ([12, Hilbert Schemes]).

2. The *Picard problem* assigns to S the set

$$\mathcal{F}(S) = \{\,\text{line bundles on } X \times_k S\,\}/\cong$$

of line bundles up to isomorphism, with $f\colon T \to S$ acting by pullback $(\mathrm{id}_X \times f)^*$ of line bundles. Over a point this is $\operatorname{Pic}(X)$, the group of line bundles on X; over a general S it records how a line bundle on X can be deformed as S varies.

The rest of the section asks the single question a moduli functor poses, namely whether it is the functor of points of a scheme. The best possible answer is that $\mathcal{F}$ is, on the nose, the functor $h_M = \operatorname{Hom}(-, M)$ of maps into a scheme M. When this happens, points of M are objects, maps into M are families, and the geometry of M is the geometry of the moduli problem. This is the notion of a *fine moduli space*.

Definition 1.1.3: Fine Moduli Space (Set-Valued)

A *fine moduli space* for the moduli functor $\mathcal{F}$ is a scheme M together with an isomorphism of functors

$$\eta\colon \mathcal{F} \xrightarrow{\sim} h_M = \operatorname{Hom}(-, M)$$

where $h_M\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ sends S to the set $\operatorname{Hom}(S, M)$ of morphisms $S \to M$. When such an M exists, $\mathcal{F}$ is said to be *representable* ([12, Representable Functor]), and M *represents* $\mathcal{F}$.

The isomorphism η is the content of the definition, not the scheme M alone: it consists of a bijection $\eta_S\colon \mathcal{F}(S) \to \operatorname{Hom}(S, M)$ for every S, compatible with pullback, so that a family over S is literally the same datum as a morphism $S \to M$. Taking $S = \operatorname{Spec} k$ recovers the underlying set-theoretic promise, that isomorphism classes of objects biject with k-points of M; but the definition asks for this in families, for all S at once, which is why it captures variation and the naive point-set does not. A fine moduli space, when it exists, is unique up to unique isomorphism, since a representing object of a functor is: if M and M' both represent $\mathcal{F}$, then $h_M \cong \mathcal{F} \cong h_{M'}$, and the Yoneda lemma turns this isomorphism of functors into a unique isomorphism $M \cong M'$ of schemes.

The word "isomorphism of functors" is doing real work and is easy to underestimate, so before extracting its main consequence it helps to see one case where the representing scheme and the isomorphism are both completely explicit.

Example 1.1.4: Projective Space as a Fine Moduli Space

Fix $n \geq 0$ and let $\mathcal{F}$ be the functor of *rank-1 locally free quotients of the trivial rank-$(n+1)$ bundle*: to a scheme S assign

$$\mathcal{F}(S) = \left\{ \mathcal{O}_S^{n+1} \twoheadrightarrow \mathcal{L} \;:\; \mathcal{L} \text{ a line bundle on } S \right\} \Big/ \cong$$

where two surjections $q\colon \mathcal{O}_S^{n+1} \twoheadrightarrow \mathcal{L}$ and $q'\colon \mathcal{O}_S^{n+1} \twoheadrightarrow \mathcal{L}'$ are identified if there is an isomorphism $\mathcal{L} \cong \mathcal{L}'$ carrying q to q'. A morphism $f\colon T \to S$ acts by pullback: f^*q is the surjection $\mathcal{O}_T^{n+1} = f^*\mathcal{O}_S^{n+1} \twoheadrightarrow f^*\mathcal{L}$, again a line-bundle quotient because pullback is right exact and preserves local freeness. Then $\mathbb{P}^n = \mathbb{P}^n_k$ represents $\mathcal{F}$, with universal quotient the tautological surjection

$$\mathcal{O}_{\mathbb{P}^n}^{n+1} \twoheadrightarrow \mathcal{O}_{\mathbb{P}^n}(1)$$

onto the twisting line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$.

To verify the universal property directly, unwind what a T-point of $\mathbb{P}^n$ is. A morphism $g\colon T \to \mathbb{P}^n$ is the same as a line bundle $\mathcal{L} = g^*\mathcal{O}(1)$ on T together with $n+1$ global sections $s_0, \ldots, s_n \in \Gamma(T, \mathcal{L})$ generating $\mathcal{L}$ at every point, modulo the choice of trivialization of $\mathcal{L}$; the sections are the pullbacks $s_i = g^*(x_i)$ of the homogeneous coordinates. Packaging the sections into a map is exactly the surjection

$$q_g\colon \mathcal{O}_T^{n+1} \longrightarrow \mathcal{L}, \qquad (a_0, \ldots, a_n) \longmapsto \textstyle\sum_i a_i s_i$$

which is surjective precisely because the s_i generate $\mathcal{L}$ everywhere. This assignment $g \mapsto q_g$ is the pullback of the tautological quotient along g, since $\mathcal{O}(1)$ is generated by $x_0, \ldots, x_n$ and $q_g = g^*\big(\mathcal{O}_{\mathbb{P}^n}^{n+1} \twoheadrightarrow \mathcal{O}_{\mathbb{P}^n}(1)\big)$. Conversely, a line-bundle quotient $q\colon \mathcal{O}_T^{n+1} \twoheadrightarrow \mathcal{L}$ furnishes generating sections $s_i = q(e_i)$ and hence a unique morphism $g\colon T \to \mathbb{P}^n$ with $q_g = q$, by the universal property of projective space as Proj. The two constructions are mutually inverse and compatible with further pullback, so

$$\mathcal{F}(T) \xrightarrow{\sim} \operatorname{Hom}(T, \mathbb{P}^n)$$

is a natural isomorphism, and $\mathbb{P}^n$ is a fine moduli space for rank-1 quotients of $\mathcal{O}^{n+1}$.

The example is the template for the whole chapter: a moduli functor is pinned to a scheme by exhibiting one distinguished family over that scheme and showing every other family comes from it by pullback in a unique way. Here the distinguished family is $\mathcal{O}_{\mathbb{P}^n}^{n+1} \twoheadrightarrow \mathcal{O}(1)$, and the phrase "comes from it by pullback in a unique way" is the statement $\mathcal{F} \cong h_{\mathbb{P}^n}$. That this pattern is not special to $\mathbb{P}^n$ but is forced by the definition of representability is the content of the next result.

Representability was defined as an isomorphism $\mathcal{F} \cong h_M$ of functors, which is a statement about all schemes S at once and so seems to require checking infinitely many bijections. The Yoneda lemma collapses this to a single piece of data: one family over M itself. The mechanism is that a natural transformation out of h_M is completely determined by where it sends the identity morphism id_M, and running this in reverse shows that an isomorphism $\mathcal{F} \cong h_M$ is the same as a family over M with a universal property. That family is what one constructs in practice, and what one produces when asked to "exhibit the moduli space".

Proposition 1.1.5: Universal Family

Let M be a scheme and $\mathcal{F}$ a moduli functor. Giving an isomorphism of functors $\eta\colon \mathcal{F} \xrightarrow{\sim} h_M$ is equivalent to giving a family $\mathcal{U} \in \mathcal{F}(M)$, the *universal family*, with the following universal property: for every scheme S and every family $\xi \in \mathcal{F}(S)$ there is a unique morphism $g\colon S \to M$ with $g^*\mathcal{U} = \xi$. In particular, $\mathcal{F}$ is representable by M if and only if such a universal family exists, and then $\mathcal{U}$ corresponds under η to the identity $\mathrm{id}_M \in h_M(M)$.

***Proof*:**
The argument is the Yoneda lemma ([12, Yoneda Lemma]) applied to the functor $\mathcal{F}$ and the representable functor h_M; the proof records what each direction of the equivalence produces so that the universal family and the isomorphism are explicit.

Step 1: From an isomorphism to a universal family. Suppose given a natural isomorphism

$\eta\colon \mathcal{F} \xrightarrow{\sim} h_M$. In particular $\eta_M\colon \mathcal{F}(M) \to h_M(M) = \mathrm{Hom}(M, M)$ is a bijection, so there is a unique family

$$\mathcal{U} := \eta_M^{-1}(\mathrm{id}_M) \in \mathcal{F}(M)$$

the preimage of the identity morphism. Fix a family $\xi \in \mathcal{F}(S)$ and set $g := \eta_S(\xi) \in \mathrm{Hom}(S, M)$. Naturality of η with respect to the morphism $g\colon S \to M$ is the commuting square

$$\begin{array}{ccc} \mathcal{F}(M) & \xrightarrow{\eta_M} & \mathrm{Hom}(M, M) \\ \big\downarrow g^* & & \big\downarrow g^* \\ \mathcal{F}(S) & \xrightarrow{\eta_S} & \mathrm{Hom}(S, M) \end{array}$$

where the right-hand g^* is precomposition with g. Chase the family $\mathcal{U} \in \mathcal{F}(M)$ in the top-left corner around the square. Across the top it maps to $\eta_M(\mathcal{U}) = \mathrm{id}_M$, and then down to $g^*(\mathrm{id}_M) = \mathrm{id}_M \circ g = g$; down the left it maps to $g^*\mathcal{U}$, and then across to $\eta_S(g^*\mathcal{U})$. Equality of the two routes yields $\eta_S(g^*\mathcal{U}) = g = \eta_S(\xi)$, and η_S is injective, so

$$g^*\mathcal{U} = \xi$$

This exhibits ξ as a pullback of $\mathcal{U}$. For uniqueness, suppose $g'\colon S \to M$ also satisfies $(g')^*\mathcal{U} = \xi$. Running the same naturality square for g' gives $\eta_S((g')^*\mathcal{U}) = g'$, hence $g' = \eta_S(\xi) = g$. Thus g is the unique morphism with $g^*\mathcal{U} = \xi$, and $\mathcal{U}$ has the claimed universal property.

Step 2: From a universal family to an isomorphism. Conversely, suppose given a family $\mathcal{U} \in \mathcal{F}(M)$ such that for every S and every $\xi \in \mathcal{F}(S)$ there is a unique $g\colon S \to M$ with $g^*\mathcal{U} = \xi$. Define, for each S, a map

$$\theta_S\colon \mathrm{Hom}(S, M) \longrightarrow \mathcal{F}(S), \qquad g \longmapsto g^*\mathcal{U}$$

which is a well-defined map of sets because pullback of families is defined. Naturality of $\theta = (\theta_S)_S$ is functoriality of pullback: for $f\colon T \to S$ and $g\colon S \to M$,

$$\theta_T(g \circ f) = (g \circ f)^*\mathcal{U} = f^*(g^*\mathcal{U}) = f^*\theta_S(g)$$

so θ is a natural transformation $h_M \to \mathcal{F}$. The universal property says exactly that each θ_S is a bijection: given $\xi \in \mathcal{F}(S)$, existence of a g with $g^*\mathcal{U} = \xi$ is surjectivity of θ_S, and uniqueness of that g is injectivity. A natural transformation that is bijective on every object is an isomorphism of functors, so $\theta\colon h_M \xrightarrow{\sim} \mathcal{F}$ is an isomorphism; its inverse $\eta = \theta^{-1}\colon \mathcal{F} \xrightarrow{\sim} h_M$ is the sought isomorphism.

Step 3: The two constructions are mutually inverse. It remains to check that starting from η, producing $\mathcal{U} = \eta_M^{-1}(\mathrm{id}_M)$, and then producing θ returns η^{-1}, and symmetrically. From Step 1, for $g \in \mathrm{Hom}(S, M)$ set $\xi = g^*\mathcal{U}$; the uniqueness there gives $\eta_S(g^*\mathcal{U}) = g$, that is $\eta_S \circ \theta_S = \mathrm{id}_{\mathrm{Hom}(S,M)}$, so $\theta = \eta^{-1}$. In particular $\mathcal{U} = \theta_M(\mathrm{id}_M) = \eta_M^{-1}(\mathrm{id}_M)$ corresponds to id_M under η, as claimed. The two passages are therefore inverse to one another, giving the asserted equivalence between isomorphisms $\mathcal{F} \cong h_M$ and universal families, completing the proof.

The proposition is the reason the functorial viewpoint is workable, not only correct. To prove a scheme M is a fine moduli space one need not construct a bijection $\mathcal{F}(S) \cong \mathrm{Hom}(S, M)$ for every S by hand; it suffices to write down one family $\mathcal{U}$ over M and verify that every family is a unique pullback of it. In the projective-space example (example 1.1.4) the universal family is the tautological quotient $\mathcal{O}^{n+1} \twoheadrightarrow \mathcal{O}(1)$, and the verification that every quotient over T is a unique pullback is precisely the universal property of $\mathbb{P}^n$ that was checked there; the proposition explains why that single verification is enough. The universal family also carries a concrete geometric meaning: its fiber over a point $m \in M$ is the object that m classifies, so $\mathcal{U}$ is a single geometric family whose fiber over each point of the moduli space is the corresponding object. This is the sense in which a fine moduli space "carries the objects on its back".

Representability is a strong demand, and the structural constraint it imposes is exactly where the interesting moduli problems fail it. A functor of the form h_M is not an arbitrary functor: it satisfies a gluing condition inherited from morphisms of schemes, and a moduli functor that violates this condition cannot be representable no matter how the objects are arranged.

The motivating paragraph for the warning is the following observation about morphisms. A morphism $S \to M$ is a local datum: to give it is to give morphisms $S_i \to M$ on the opens of a cover that agree on overlaps, because M is glued from its structure sheaf. Hence h_M is a *Zariski sheaf*: a compatible family of T-points on a cover glues to a unique T-point. Any representable moduli functor must inherit this, and the obstruction to it is the presence of automorphisms.

1.1.6 Warning: Automorphisms Obstruct Representability A representable functor h_M is necessarily a *Zariski sheaf*: for any scheme S with a Zariski open cover $\{S_i\}$, a collection of morphisms $g_i\colon S_i \to M$ with $g_i|_{S_i\cap S_j} = g_j|_{S_i\cap S_j}$ glues to a unique $g\colon S \to M$ restricting to each g_i. Consequently, if a moduli functor $\mathcal{F}$ fails the sheaf condition, it cannot be representable. Objects with nontrivial automorphisms manufacture exactly this failure. Suppose an object X_0 over k has a nontrivial automorphism. One can then build a family over a base S whose restriction to each member S_i of an open cover is the constant family $X_0 \times S_i$, so that the family is trivial locally on the cover, yet whose gluing over the overlaps is twisted by the automorphism so that the family is not globally the constant family $X_0 \times S$. Such a family has isomorphic fibers over every point of S but is not itself constant. Under any candidate isomorphism $\mathcal{F} \cong h_M$ its class in $\mathcal{F}(S)$ would have to glue, from the locally-constant data, to the class of the constant family, and the two are distinct elements of $\mathcal{F}(S)$: the sheaf condition is violated, and no fine moduli space exists.
This is the central obstruction of the chapter. It is the reason the moduli of elliptic curves has no fine moduli space and must be replaced by the coarse j-line (section 1.3), and the sheaf-theoretic diagnosis given here is sharpened into a working representability criterion in section 1.5. The class of examples alluded to, locally trivial but globally twisted families, is made fully explicit in example 1.3.4.

The warning identifies the fault line that organizes the entire book. When $\mathcal{F}$ is a sheaf and is locally representable, it is representable, and the Hilbert and Quot schemes of the next chapter provide the fundamental such functors. When $\mathcal{F}$ fails to be a sheaf because its objects have automorphisms, one has three responses, and the book pursues all of them in turn: settle for a coarse moduli space that co-represents $\mathcal{F}$ (section 1.3), rigidify the problem by adding structure to kill the automorphisms (section 1.4), or enlarge the category of geometric objects to stacks, in which automorphisms are remembered rather than fatal. The functor of points is the frame in which all three are stated.

Exercise 1.1.1

1. Work out the functor of points of $\mathbb{P}^1$ directly from example 1.1.4. Describe $\mathcal{F}(S)$ for the functor of rank-1 locally free quotients of $\mathcal{O}_S^2$, show that a T-point of $\mathbb{P}^1$ is the same as a pair of sections (s_0, s_1) of a line bundle $\mathcal{L}$ on T generating $\mathcal{L}$ at every point, taken modulo isomorphism of $\mathcal{L}$, and check that on $T = \operatorname{Spec} k$ this recovers the usual k-points $[a_0 : a_1]$ of

$\mathbb{P}^1$.

2. Let X_0 be an object over k with a nontrivial automorphism $\sigma\colon X_0 \to X_0$, and let $S = S_1 \cup S_2$ be a scheme with two opens whose intersection $S_{12} = S_1 \cap S_2$ is nonempty. Using σ, describe how to glue the constant families $X_0 \times S_1$ and $X_0 \times S_2$ over S_{12} into a family ξ over S that has X_0 as every fiber but need not be isomorphic to the constant family $X_0 \times S$. Explain, referring to box 1.1.6, why the existence of such a ξ shows that the set of isomorphism classes $\mathcal{F}(\operatorname{Spec} k)$ alone cannot determine a scheme structure representing $\mathcal{F}$.

3. Suppose $\mathcal{F}$ is represented by a scheme M with universal family $\mathcal{U}$, and let $\xi \in \mathcal{F}(S)$ be a family that happens to be the pullback $h^*\mathcal{U}$ of $\mathcal{U}$ along some morphism $h\colon S \to M$. Prove, using only proposition 1.1.5, that h is the *unique* morphism $S \to M$ classifying ξ, and deduce that the classifying map of the universal family $\mathcal{U}$ itself is id_M.

4. A scheme M is given as a fine moduli space for a functor $\mathcal{F}$ by an isomorphism $\eta\colon \mathcal{F} \xrightarrow{\sim} h_M$. Explain how to read off the universal family $\mathcal{U} \in \mathcal{F}(M)$ from η, and carry this out for $M = \mathbb{P}^n$ and the quotient functor of example 1.1.4: identify which element of $\mathcal{F}(\mathbb{P}^n)$ is $\eta_{\mathbb{P}^n}^{-1}(\mathrm{id}_{\mathbb{P}^n})$.

5. Define a functor $\mathcal{G}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ by $\mathcal{G}(S) = \{\text{invertible global functions on } S\} = \Gamma(S, \mathcal{O}_S)^\times$, with pullback the obvious restriction. Show that $\mathcal{G}$ is a Zariski sheaf, and prove that $\mathcal{G}$ is representable by the scheme $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ by exhibiting the universal family and verifying the universal property of proposition 1.1.5. Contrast this with the functor $\mathcal{G}'(S) = \Gamma(S, \mathcal{O}_S)^\times/(\text{constants } k^\times)$ and argue that $\mathcal{G}'$ fails the sheaf condition, hence is not representable.

1.2 Fine Moduli Spaces: the Grassmannian

The previous section made the abstract case: a moduli problem is a functor $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ (definition 1.1.1), a fine moduli space is a scheme M with $\mathcal{F} \cong h_M$ (definition 1.1.3), and representability is equivalent to the existence of a universal family (proposition 1.1.5). What that section could not yet supply is a moduli problem interesting enough to matter and simple enough to solve by hand from beginning to end. This section supplies it. The Grassmannian $\mathrm{Gr}(d, n)$, whose points are the d-dimensional quotient spaces of a fixed n-dimensional vector space, is the model fine moduli problem: the functor it represents is written down in one line, the representing scheme is built explicitly from affine charts, the universal family is exhibited concretely, and the whole space is shown to be projective through the Plücker embedding. Every later construction in this book, most immediately the Hilbert and Quot schemes of the following chapter, is cut out of a Grassmannian, so the effort spent here is repaid many times over.

The design choice that makes the construction work is to phrase a point of the Grassmannian not as a subspace but as a *quotient*. A d-dimensional subspace $W \subseteq k^n$ and the quotient $k^n \twoheadrightarrow k^n/W$ carry the same information, but the quotient behaves far better in families: a surjection of a trivial bundle onto a locally free sheaf is a functorial datum requiring no side condition, whereas "a locally free subsheaf" must carry the extra condition that the quotient be locally free too (so that the sub is a *subbundle*, not just a subsheaf). Phrasing points as quotients builds that condition in from the start. This is the same bookkeeping that made $\mathbb{P}^n$ the moduli space of rank-1 quotients of $\mathcal{O}^{n+1}$ in example 1.1.4, and the Grassmannian is the direct generalization from rank 1 to rank d.

The functor of points is stated for an arbitrary base scheme S: a family over S is a rank-d locally free quotient of the trivial rank-n sheaf $\mathcal{O}_S^n$, and the whole point of working with quotients is that pulling back a surjection of sheaves along a morphism $T \to S$ again gives a surjection, so the assignment is contravariant with no further hypotheses.

Definition 1.2.1: The Grassmannian Functor

Fix integers $0 \le d \le n$. The *Grassmannian functor* $\underline{\mathrm{Gr}}(d,n)\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ assigns to a scheme S the set

$$\underline{\mathrm{Gr}}(d,n)(S) = \left\{ \mathcal{O}_S^n \overset{q}{\twoheadrightarrow} \mathcal{Q} \;\middle|\; \mathcal{Q} \text{ locally free of rank } d \right\} \Big/ \cong$$

of surjections of $\mathcal{O}_S$-modules from the trivial sheaf $\mathcal{O}_S^n$ onto a locally free sheaf $\mathcal{Q}$ of rank d, where two surjections $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ and $q'\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}'$ are identified when there is an isomorphism $\varphi\colon \mathcal{Q} \xrightarrow{\sim} \mathcal{Q}'$ with $\varphi \circ q = q'$. For a morphism $f\colon T \to S$ the pullback map sends the class of q to the class of $f^*q\colon \mathcal{O}_T^n \twoheadrightarrow f^*\mathcal{Q}$.

Because f^* is right exact, f^*q is again surjective, and $f^*\mathcal{Q}$ is again locally free of rank d, so the pullback map is well defined and $\underline{\mathrm{Gr}}(d,n)$ is a functor. Two remarks fix the meaning of the equivalence relation. First, the identification by isomorphisms of $\mathcal{Q}$ is forced by the moduli philosophy of definition 1.1.1: a family is an equivalence class, and here the class of q remembers only the kernel $\ker q \subseteq \mathcal{O}_S^n$ together with the fact that the quotient is a bundle. Second, over a field, $\underline{\mathrm{Gr}}(d,n)(k)$ is exactly the set of d-dimensional quotient spaces of k^n, equivalently the set of $(n-d)$-dimensional subspaces $\ker q \subseteq k^n$, which is the classical Grassmannian of $(n-d)$-planes. The functor is the family-version of that set.

The subbundle picture is the same datum read backwards, and it is worth recording because half the literature (and half of this book) prefers it. Given a quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ locally free of rank d, the kernel $\mathcal{S} = \ker q$ is a locally free subsheaf of rank $n-d$ whose quotient is locally free, hence a *subbundle*; conversely a rank-$(n-d)$ subbundle $\mathcal{S} \hookrightarrow \mathcal{O}_S^n$ has locally free quotient of rank d. The two descriptions are interchanged by the short exact sequence

$$0 \to \mathcal{S} \to \mathcal{O}_S^n \xrightarrow{q} \mathcal{Q} \to 0 \tag{1.1}$$

and neither is more basic than the other. The convention here takes the quotient $\mathcal{Q}$ as primary and calls $\mathcal{S}$ the associated sub.

Definition 1.2.2: The Grassmannian

The *Grassmannian* $\mathrm{Gr}(d,n)$ is the scheme constructed in theorem 1.2.4 below, characterized by the property that it represents the functor $\underline{\mathrm{Gr}}(d,n)$ of definition 1.2.1.

Before proving representability, it helps to see the charts concretely in the smallest new case, so that the abstract cover in the proof reads as bookkeeping for a picture already understood.

Example 1.2.3: Charts on the Grassmannian of Lines in 3-Space

Take $d=2$, $n=3$, so $\mathrm{Gr}(2,3)$ parametrizes rank-2 quotients of $\mathcal{O}_S^3$, equivalently rank-1 subbundles (lines) of $\mathcal{O}_S^3$. Over a field a point is a line $\ell \subseteq k^3$ through the origin, spanned by a nonzero vector (a_0,a_1,a_2) up to scale, so $\mathrm{Gr}(2,3)(k)$ is exactly $\mathbb{P}^2$, the set of such lines. The three charts are the loci where $a_0 \neq 0$, $a_1 \neq 0$, $a_2 \neq 0$; on the first, scaling to $a_0 = 1$ leaves the two free coordinates $(a_1,a_2) \in \mathbb{A}^2$, and these are the affine coordinates on that chart. So $\mathrm{Gr}(2,3) \cong \mathbb{P}^2$, its dual realization as lines rather than planes. This is the case $n = 3$ of the identification $\mathrm{Gr}(n-1, n) = \mathbb{P}^{n-1}$, which is the duality of exercise 1.2.1 (6) applied to $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$; the three charts here are indexed by the nonzero coordinate of the spanning line vector, which under that duality corresponds to the U_I charts of theorem 1.2.4 indexed by the two-subsets $I \subseteq \{1, 2, 3\}$.

The representability theorem is the technical heart of the section. Its proof is the archetype for every later representability argument: cover the functor by open subfunctors, each of which is manifestly representable by an affine space, then check that these charts glue to a scheme representing the whole. The charts are indexed by d-element subsets of the coordinates, and the chart U_I is the locus where a given family, restricted to those d coordinates, is already an isomorphism onto the quotient.

The one input this argument needs, beyond the definitions, is the criterion that a functor which is a Zariski sheaf and admits an open cover by representable subfunctors is itself representable. That criterion is developed in full generality later (see theorem 1.5.6); here the covering subfunctors are so explicit that the gluing can be exhibited by hand, and the proof below does exactly that rather than invoking the general statement. This keeps the argument self-contained and shows the mechanism the general theorem abstracts.

Theorem 1.2.4: Representability of the Grassmannian

For $0 \le d \le n$ there is a scheme $\mathrm{Gr}(d, n)$, smooth and projective over $\mathbb{Z}$ (hence over any base) of relative dimension $d(n-d)$, representing the functor $\underline{\mathrm{Gr}}(d, n)$ of definition 1.2.1. It is covered by $\binom{n}{d}$ affine charts, each isomorphic to $\mathbb{A}^{d(n-d)}$.

***Proof*:**
The construction proceeds by building the covering subfunctors, identifying each with an affine space, and gluing. Throughout, $[n] = \{1, \ldots, n\}$, and for a d-element subset $I \subseteq [n]$ the inclusion of the coordinates indexed by I is written $\iota_I : \mathcal{O}_S^d \to \mathcal{O}_S^n$.

Step 1: The covering subfunctors. For each d-element subset $I \subseteq [n]$ define a subfunctor $\underline{U}_I \subseteq \underline{\mathrm{Gr}}(d, n)$ by declaring, for a family $q : \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ over S, that $[q] \in \underline{U}_I(S)$ when the composite

$$\mathcal{O}_S^d \xrightarrow{\iota_I} \mathcal{O}_S^n \xrightarrow{q} \mathcal{Q}$$

is an isomorphism. This condition is invariant under the equivalence relation of definition 1.2.1: replacing q by $\varphi \circ q$ replaces the composite by $\varphi \circ (q \circ \iota_I)$, still an isomorphism. It is also stable under pullback, since f^* preserves isomorphisms, so $\underline{U}_I$ is a genuine subfunctor.

The condition defining $\underline{U}_I$ is open: the composite $q \circ \iota_I$ is a map between locally free sheaves of the same rank d, and the locus in S where such a map is an isomorphism is the non-vanishing locus of its determinant, an open subscheme. A family q over S therefore lies in $\underline{U}_I(T)$ after pullback along $f : T \to S$ if and only if f factors through this open subscheme, which is the statement that $\underline{U}_I \hookrightarrow \underline{\mathrm{Gr}}(d, n)$ is an open subfunctor.

Step 2: The charts cover. A family $q : \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ lies in $\underline{U}_I(S)$ for suitable I at least locally on S. Fix a point $s \in S$; the surjection q of a rank-n trivial bundle onto a rank-d bundle has, at the residue field $\kappa(s)$, a surjection $\kappa(s)^n \twoheadrightarrow \mathcal{Q} \otimes \kappa(s)$ of a rank-n space onto a rank-d space. Some d of the standard basis vectors of $\kappa(s)^n$ map to a basis of the d-dimensional target, that is, $q \circ \iota_I$ is an isomorphism at s for some I. By openness of the isomorphism locus (Step 1), $q \circ \iota_I$ is an isomorphism on a neighborhood of s. Thus the pullback of q to that neighborhood lies in $\underline{U}_I$, and the $\underline{U}_I$ cover $\underline{\mathrm{Gr}}(d, n)$ in the sense that every family is, locally on its base, in some chart.

Step 3: Each chart is an affine space. Fix I. Order the $n-d$ complementary coordinates as $j_1, \ldots, j_{n-d}$, the elements of $[n] \setminus I$. The claim is that $\underline{U}_I$ is represented by the affine space $\mathbb{A}^{d(n-d)}$ with coordinate ring $k[x_{ab}]$, where a ranges over I and b ranges over $[n] \setminus I$, that is, by the space of $d \times (n-d)$ matrices.

Given a family $[q] \in \underline{U}_I(S)$, the composite $\theta = q \circ \iota_I \colon \mathcal{O}_S^d \xrightarrow{\sim} \mathcal{Q}$ is an isomorphism, and $\theta^{-1} \circ q \colon \mathcal{O}_S^n \to \mathcal{O}_S^d$ is a surjection restricting to the identity on the coordinates I. Its remaining entries assemble into a $d \times (n-d)$ matrix $M(q)$ over $\mathcal{O}_S(S)$, whose (a, b) entry records the image in the a-th standard basis vector of $\mathcal{O}_S^d$ of the b-th complementary coordinate. Replacing q by $\varphi \circ q$ replaces θ by $\varphi \circ \theta$ and leaves $\theta^{-1} \circ q$ unchanged, so $M(q)$ depends only on the class $[q]$ and gives a well-defined map

$$\underline{U}_I(S) \longrightarrow \mathrm{Hom}(S, \mathbb{A}^{d(n-d)}) = \{d \times (n-d) \text{ matrices over } \mathcal{O}_S(S)\}$$

natural in S. Conversely, a matrix M over $\mathcal{O}_S(S)$ determines the surjection $q_M \colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{O}_S^d$ that is the identity on the coordinates I and has entry M_{ab} on the complementary coordinate b; this q_M visibly lies in $\underline{U}_I(S)$, with $\theta = \mathrm{id}$, and $M(q_M) = M$. The two constructions are mutually inverse and natural, so $\underline{U}_I \cong h_{\mathbb{A}^{d(n-d)}}$. By the Yoneda direction of proposition 1.1.5, $\underline{U}_I$ is representable by $U_I := \mathbb{A}^{d(n-d)}$.

Step 4: Gluing. For two subsets I, J the overlap subfunctor $\underline{U}_I \cap \underline{U}_J$ is the locus in $U_I \cong \mathbb{A}^{d(n-d)}$ where, in addition, $q \circ \iota_J$ is an isomorphism. In the coordinates of U_I a family is the surjection q_M; the composite $q_M \circ \iota_J$ is the $d \times d$ submatrix of the full $d \times n$ matrix $(\,\mathrm{Id}_I \,|\, M\,)$ on the columns indexed by J, so $\underline{U}_I \cap \underline{U}_J$ is the principal open subset of U_I where the determinant of that submatrix is invertible, an open subscheme $U_{IJ} \subseteq U_I$. The transition isomorphism $U_{IJ} \xrightarrow{\sim} U_{JI}$ is read off by recomputing the normalized matrix relative to the coordinates J: it sends the matrix normalized on I to the matrix normalized on J, obtained by left-multiplying $(\,\mathrm{Id}_I \,|\, M\,)$ by the inverse of its J-submatrix and deleting the resulting identity block. These transition maps are morphisms of schemes (entries are rational functions with the relevant determinant inverted) and satisfy the cocycle condition on triple overlaps, because on $U_{IJ} \cap U_{IK}$ all three normalizations describe the *same* surjection q read in three coordinate frames, so passing $I \to J \to K$ agrees with $I \to K$.

By the gluing construction for schemes, the U_I glue along the U_{IJ} to a scheme $\mathrm{Gr}(d, n)$. The subfunctors $\underline{U}_I$ glue correspondingly to a functor which, by Steps 1 and 2, is $\underline{\mathrm{Gr}}(d, n)$, and each $\underline{U}_I$ is represented by the chart U_I; a family over an arbitrary S is one that restricts to a chart family on each piece of a cover of S, and the transition maps guarantee these glue to a unique morphism $S \to \mathrm{Gr}(d, n)$. Hence $\mathrm{Gr}(d, n)$ represents $\underline{\mathrm{Gr}}(d, n)$.

Step 5: Dimension, smoothness, projectivity. Each chart is $\mathbb{A}^{d(n-d)}$, so $\mathrm{Gr}(d, n)$ is smooth of dimension $d(n-d)$ and covered by $\binom{n}{d}$ such charts. The construction used only $\mathbb{Z}$-coefficients (the transition maps are integral rational functions), so $\mathrm{Gr}(d, n)$ is defined and smooth over $\mathbb{Z}$, hence over any base by base change. Projectivity is the content of proposition 1.2.7, which realizes $\mathrm{Gr}(d, n)$ as a closed subscheme of a projective space. This completes the proof.

The theorem is the first fine moduli space of the book realized from scratch. What the reader should take from it is less the statement than the shape of the argument, because it recurs: a moduli functor is representable when it can be covered by open subfunctors each of which is an affine space of parameters, and the parameters are the freedom left after normalizing a family into a standard form. Here the standard form is "the identity on a chosen set of d coordinates," the remaining freedom is a $d \times (n-d)$ matrix, and the transition between two normalizations is a determinant computation.

The Hilbert and Quot schemes of the next chapter are built by exhibiting them as closed subfunctors of Grassmannian functors, so representability there is inherited from the theorem just proved.

The case $d = 1$ recovers the running example of the previous section. A rank-1 locally free quotient of $\mathcal{O}_S^n$ is exactly the datum classified by $\mathbb{P}^{n-1}$, and unwinding the definitions gives $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$, matching $\mathbb{P}^{n-1}$ as the moduli space of rank-1 quotients of $\mathcal{O}^n$ from example 1.1.4 (there stated for $\mathcal{O}^{n+1}$, giving $\mathbb{P}^n$; the index shift is the passage from $n+1$ to n). The charts of the theorem specialize to the standard affine charts of projective space: the subset $I = \{i\}$ is the locus where the i-th coordinate of the quotient is nonzero, and the $1 \times (n-1)$ matrix of remaining entries is exactly the tuple of affine coordinates on that chart. So the Grassmannian is the common generalization of projective space to higher-rank quotients, and $\mathbb{P}^{n-1}$ is its rank-1 shadow.

Representability produces, through proposition 1.1.5, a universal family on $\mathrm{Gr}(d, n)$: the family corresponding to the identity morphism $\mathrm{Gr}(d, n) \to \mathrm{Gr}(d, n)$. Naming it and its associated sub is worthwhile, because these two bundles are the source of nearly every natural construction on the Grassmannian, from its tangent bundle to the tautological classes computed against it in intersection theory.

Definition 1.2.5: Universal Quotient and Universal Sub

On $G = \mathrm{Gr}(d, n)$ the *universal quotient bundle* $\mathcal{Q}$ is the locally free sheaf of rank d appearing in the family $q_{\mathrm{univ}} \colon \mathcal{O}_G^n \twoheadrightarrow \mathcal{Q}$ that corresponds, under the isomorphism $\underline{\mathrm{Gr}}(d, n) \cong h_G$ of theorem 1.2.4, to the identity morphism $\mathrm{id}_G \in h_G(G)$. Its kernel $\mathcal{S} = \ker q_{\mathrm{univ}}$, a locally free sheaf of rank $n - d$, is the *universal subbundle* (or *tautological subbundle*). They sit in the universal short exact sequence

$$0 \to \mathcal{S} \to \mathcal{O}_G^n \xrightarrow{q_{\mathrm{univ}}} \mathcal{Q} \to 0$$

on $\mathrm{Gr}(d, n)$.

The universal property is the statement of proposition 1.1.5 specialized to this family: for every scheme S and every rank-d locally free quotient $q \colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}'$, there is a unique morphism $\varphi_q \colon S \to \mathrm{Gr}(d, n)$ with $\varphi_q^* q_{\mathrm{univ}} = q$ (up to the identification of definition 1.2.1), that is, $\varphi_q^* \mathcal{Q} \cong \mathcal{Q}'$ compatibly with the surjection from $\mathcal{O}_S^n$. Every family of d-dimensional quotients is a pullback of the one universal family, in exactly one way. This is what it means for $\mathrm{Gr}(d, n)$ to be a *fine* moduli space in the sense of definition 1.1.3: the universal bundle is a single geometric object over the moduli space from which all others are cut by pullback.

The tautological name comes from the field-point picture. Over the classical Grassmannian, the fiber of $\mathcal{S}$ at a point $[\ell]$ (a subspace $\ell \subseteq k^n$) is the subspace ℓ itself, and the fiber of $\mathcal{Q}$ is the quotient k^n/ℓ; the subbundle is "tautologically" the space its point names. The next example makes the universal bundle explicit in charts, staying within the promised distance of the definition.

Example 1.2.6: The Universal Sequence in a Chart

On the chart $U_I \cong \mathbb{A}^{d(n-d)}$ of theorem 1.2.4 the universal quotient is the free sheaf $\mathcal{O}_{U_I}^d$, with universal surjection $q_{\mathrm{univ}}|_{U_I} = (\,\mathrm{Id}_I \big| M\,) \colon \mathcal{O}_{U_I}^n \twoheadrightarrow \mathcal{O}_{U_I}^d$, where $M = (x_{ab})$ is the matrix of chart coordinates. The universal sub $\mathcal{S}|_{U_I}$ is the kernel, freely generated by the $n - d$ columns

$$s_b = e_b - \sum_{a \in I} x_{ab}\, e_a \qquad (b \in [n] \setminus I)$$

where $e_1, \ldots, e_n$ is the standard basis of $\mathcal{O}_{U_I}^n$: each s_b is annihilated by $(\,\mathrm{Id}_I \big| M\,)$ since its I-coordinates $-x_{ab}$ exactly cancel the contribution of its complementary coordinate e_b. Over the overlap of two charts these local frames differ by the transition matrices of Step 4 of the proof, which is the concrete content of $\mathcal{Q}$ and $\mathcal{S}$ being globally defined bundles rather than trivial ones.

The Grassmannian was built as a smooth scheme covered by affine charts, but its charts alone do not reveal that it is projective; the affine cover exhibits it as separated and of finite type, not as a closed subscheme of any $\mathbb{P}^N$. Projectivity comes from a single global construction: send a d-dimensional subspace to its top exterior power, a line in $\wedge^d k^n$, which is a point of the projective space $\mathbb{P}(\wedge^d k^n)$. In families this is the map classified by the line bundle $\wedge^{n-d}\mathcal{S}^\vee$ (or equivalently $\wedge^d \mathcal{Q}$), and the resulting morphism turns out to be a closed embedding whose image is cut out by explicit quadratic equations. This is the mechanism that makes the Grassmannian, and through it every Hilbert and Quot scheme, a projective variety.

The functorial version uses the quotient. A rank-d quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ induces on top exterior powers a surjection $\wedge^d q\colon \wedge^d \mathcal{O}_S^n = \mathcal{O}_S^{\binom{n}{d}} \twoheadrightarrow \wedge^d \mathcal{Q}$ onto a line bundle, which is precisely the datum classifying a morphism $S \to \mathbb{P}(\wedge^d k^n) = \mathbb{P}^{\binom{n}{d}-1}$ (a rank-1 quotient of a trivial bundle, in the sense of example 1.1.4). This assignment is natural in S, hence a morphism of functors, hence a morphism of the representing schemes.

Proposition 1.2.7: The Plücker Embedding

The morphism

$$\mathrm{p}\colon \mathrm{Gr}(d,n) \longrightarrow \mathbb{P}\Big(\bigwedge^d k^n\Big) = \mathbb{P}^{\binom{n}{d}-1}$$

induced by $q \mapsto \wedge^d q$ is a closed embedding. Its image is the zero locus of the quadratic *Plücker relations*: writing the homogeneous coordinates as $p_{i_1\cdots i_d}$ indexed by d-element subsets $i_1 < \cdots < i_d$ of $[n]$ (the coordinates on the decomposable vectors $e_{i_1} \wedge \cdots \wedge e_{i_d}$), the image is cut out by

$$\sum_{t=0}^{d} (-1)^t \, p_{i_1\cdots i_{d-1} j_t} \, p_{j_0 \cdots \widehat{j_t} \cdots j_d} = 0 \tag{1.2}$$

for all $i_1 < \cdots < i_{d-1}$ and $j_0 < \cdots < j_d$ in $[n]$, where $\widehat{j_t}$ denotes omission. Consequently $\mathrm{Gr}(d,n)$ is projective over $\mathbb{Z}$, hence over any base.

***Proof*:**
The argument has three parts: the map is a morphism (already established above), it is a locally closed embedding checked on charts, and its image is exactly the Plücker locus, which is closed. Work with the sub $\mathcal{S}$ of rank $n-d$; equivalently one may phrase everything through $\wedge^d \mathcal{Q}$, and the two give the same map since $\wedge^d \mathcal{Q} \cong \wedge^n \mathcal{O}_S^n \otimes (\wedge^{n-d}\mathcal{S})^\vee$ is the same line bundle up to the trivial twist $\wedge^n \mathcal{O}_S^n \cong \mathcal{O}_S$.

Step 1: Injectivity on points and the decomposable coordinates. On $\bar{k}$-points, p sends a subspace $W = \ker q \subseteq \bar{k}^n$ of dimension $n-d$ (equivalently the quotient of dimension d) to the line $\wedge^d(\bar{k}^n/W)$, whose Plücker coordinates $p_{i_1\cdots i_d}$ are the $d \times d$ minors of a $d \times n$ matrix representing q. A subspace is recovered from these minors: the coordinate $p_{i_1\cdots i_d}$ is nonzero exactly when $q \circ \iota_I$ is an isomorphism for $I = \{i_1 < \cdots < i_d\}$, that is, exactly on the chart U_I, and on that chart the ratios of minors recover the normalized matrix M of theorem 1.2.4. Hence distinct points of $\mathrm{Gr}(d,n)$ have distinct images, so p is injective on points; the same computation applies with S-valued families, giving injectivity of the map of functors.

Step 2: A closed embedding on each chart. Fix I and work over the standard affine chart $D_+(p_I) \subseteq \mathbb{P}^{\binom{n}{d}-1}$ where the coordinate p_I is nonzero. By Step 1, $\mathrm{p}^{-1}(D_+(p_I)) = U_I$. On U_I the family is $(\,\mathrm{Id}_I | M\,)$, and its $d \times d$ minor on columns J is a polynomial in the entries x_{ab} of M; normalizing so that $p_I = 1$, the minor on the columns I with one index a replaced by a complementary index b equals $\pm x_{ab}$ exactly. Thus the coordinate functions p_J/p_I on $D_+(p_I)$ pull back to polynomials in the x_{ab} that include, among them, all $d(n-d)$ coordinates $\pm x_{ab}$ themselves. The comorphism

$$k\big[\, p_J/p_I \big| J \,\big] \longrightarrow k[x_{ab}] = \mathcal{O}(U_I)$$

is therefore surjective (the generators x_{ab} are in the image), so $\mathrm{p}\,|_{U_I} : U_I \to D_+(p_I)$ is a closed immersion. A morphism that is a closed immersion over each member of an open cover of the target, with the preimages covering the source, is a locally closed immersion; here the images $\mathrm{p}(U_I) = \mathrm{p}(\mathrm{Gr}(d,n)) \cap D_+(p_I)$ cover $\mathrm{p}(\mathrm{Gr}(d,n))$, so p is an immersion onto a locally closed subscheme.

Step 3: The image is the closed Plücker locus. It remains to identify the image with the closed subscheme $Y \subseteq \mathbb{P}^{\binom{n}{d}-1}$ defined by (1.2), which will show the immersion is closed. The relations (1.2) are exactly the condition that a vector $\omega \in \wedge^d k^n$ be *decomposable*, $\omega = v_1 \wedge \cdots \wedge v_d$ for some $v_i \in k^n$: this is the classical decomposability criterion of multilinear algebra: a d-vector is decomposable if and only if its coordinates satisfy the quadratic Plücker relations. It enters here only to describe the underlying point set, since the scheme-theoretic identification of the image is carried out independently at the chart level below. Every point of $\mathrm{p}(\mathrm{Gr}(d,n))$ is decomposable by construction, so $\mathrm{p}(\mathrm{Gr}(d,n)) \subseteq Y$. Conversely, a decomposable line $[v_1 \wedge \cdots \wedge v_d]$ with the v_i linearly independent spans a d-dimensional subspace, hence a point of $\mathrm{Gr}(d,n)$ mapping to it, so $Y \subseteq \mathrm{p}(\mathrm{Gr}(d,n))$ on points. Scheme-theoretically, $Y \cap D_+(p_I)$ is defined by the relations (1.2) normalized at $p_I = 1$, and these solve to express every p_J/p_I as a polynomial in the $\pm x_{ab}$ minors, giving $Y \cap D_+(p_I) \cong \mathbb{A}^{d(n-d)} = U_I$; so the immersion $U_I \hookrightarrow D_+(p_I)$ has image exactly $Y \cap D_+(p_I)$. Therefore p is a closed immersion with image Y.

Since $\mathrm{Gr}(d,n)$ is thereby a closed subscheme of $\mathbb{P}^{\binom{n}{d}-1}$, it is projective over the base, completing the proof.

The proposition closes the account of $\mathrm{Gr}(d,n)$ as a fine moduli space: it is representable (theorem 1.2.4), it carries a universal family (definition 1.2.5), and it is projective (through the Plücker embedding just constructed). The three facts are not independent decorations. The Plücker embedding is itself the classifying map of the line bundle $\wedge^d \mathcal{Q}$ built from the universal quotient, so projectivity is read off the universal family. This is the pattern the book will exploit repeatedly: a moduli space is projective because a natural line bundle on it, assembled from the universal family, is very ample.

The Plücker relations (1.2) deserve a concrete instance, since for $d = 1$ or $d = n-1$ they are vacuous (the Grassmannian is a projective space and there is nothing to cut out) and only become substantive at $d = 2$, $n = 4$.

Example 1.2.8: The Plücker Quadric $\mathrm{Gr}(2,4)$

Take $d=2$, $n=4$: $\mathrm{Gr}(2,4)$ parametrizes rank-2 quotients of $\mathcal{O}_S^4$, equivalently rank-2 subbundles, equivalently (over a field) 2-planes through the origin in k^4, or lines in $\mathbb{P}^3$. Its dimension is $d(n-d)=2{\cdot}2=4$, and the Plücker space is $\mathbb{P}(\wedge^2 k^4)=\mathbb{P}^5$ with coordinates $p_{12},p_{13},p_{14},p_{23},p_{24},p_{34}$.

The charts. There are $\binom{4}{2}=6$ charts U_I, one for each pair $I\subseteq\{1,2,3,4\}$. On U_{12} a family is normalized to the 2×4 matrix

$$\begin{pmatrix} 1 & 0 & x_{13} & x_{14} \\ 0 & 1 & x_{23} & x_{24} \end{pmatrix}$$

whose four entries $x_{13},x_{14},x_{23},x_{24}$ are the affine coordinates on $U_{12}\cong\mathbb{A}^4$; here the first two columns (indices $I=\{1,2\}$) are the identity block. The Plücker coordinates are the 2×2 minors: $p_{12}=1$, $p_{13}=x_{23}$, $p_{14}=x_{24}$, $p_{23}=-x_{13}$, $p_{24}=-x_{14}$, and $p_{34}=x_{13}x_{24}-x_{14}x_{23}$. The chart U_{34} normalizes on the last two columns instead, and its coordinates are related to those of U_{12} by inverting the relevant minor, as in Step 4 of theorem 1.2.4.

The single relation. For $d=2$, $n=4$ the relations (1.2) reduce to one quadric. Here $d-1=1$, so there is a single i-index, and $d+1=3$, so there are three j-indices; the only choice up to the labeling is $i_1=1$ and $(j_0,j_1,j_2)=(2,3,4)$, and the sum over $t=0,1,2$ gives $p_{12}p_{34}-p_{13}p_{24}+p_{14}p_{23}$:

$$p_{12}\,p_{34}-p_{13}\,p_{24}+p_{14}\,p_{23}=0 \tag{1.3}$$

Substituting the chart values verifies it: $1\cdot(x_{13}x_{24}-x_{14}x_{23})-x_{23}\cdot(-x_{14})+x_{24}\cdot(-x_{13})=x_{13}x_{24}-x_{14}x_{23}+x_{14}x_{23}-x_{13}x_{24}=0$. Thus $\mathrm{Gr}(2,4)$ is the smooth quadric fourfold in $\mathbb{P}^5$ cut out by (1.3), the smallest Grassmannian that is not itself a projective space.

Exercise 1.2.1

1. Show directly from definition 1.2.1 that the assignment sending a rank-d quotient $q\colon\mathcal{O}_S^n\twoheadrightarrow\mathcal{Q}$ to its kernel subbundle $\mathcal{S}=\ker q$ defines an isomorphism of functors between $\underline{\mathrm{Gr}}(d,n)$ and the functor $S\mapsto\{\text{rank-}(n-d)\text{ subbundles of }\mathcal{O}_S^n\}$. (A rank-$(n-d)$ subbundle means a subsheaf $\mathcal{S}\hookrightarrow\mathcal{O}_S^n$, locally a direct summand, with locally free quotient.)

2. Using the chart $U_I\cong\mathbb{A}^{d(n-d)}$ of theorem 1.2.4, compute the transition map $U_{12}\cap U_{13}\to U_{13}$ explicitly for $\mathrm{Gr}(2,3)$: write the coordinates on U_{13} as rational functions of the coordinates on U_{12}, and identify the open locus of U_{12} on which the transition is defined.

3. Verify that $\mathrm{Gr}(1,n)=\mathbb{P}^{n-1}$ by comparing functors of points: show that a rank-1 locally free quotient $\mathcal{O}_S^n\twoheadrightarrow\mathcal{L}$ is exactly the datum that classifies a morphism $S\to\mathbb{P}^{n-1}$, and match the affine charts of theorem 1.2.4 with the standard charts of $\mathbb{P}^{n-1}$. Relate this to example 1.1.4.

4. On $\mathrm{Gr}(2,4)$, write down the two-plane $W\subseteq k^4$ corresponding to the Plücker point $[p_{12}:p_{13}:p_{14}:p_{23}:p_{24}:p_{34}]=[1:0:0:0:0:0]$, and verify it satisfies (1.3). Which chart U_I does this point lie in, and what are its affine coordinates there? Then check whether the point $[1:0:0:0:0:1]$ satisfies (1.3), and interpret the answer geometrically.

5. Show that the universal quotient bundle $\mathcal{Q}$ on $\mathrm{Gr}(1,n)=\mathbb{P}^{n-1}$ is the line bundle $\mathcal{O}(1)$, and hence that the Plücker embedding of proposition 1.2.7 is the identity in this case. (Use example 1.1.4 to identify the universal rank-1 quotient.)

6. Prove that $\mathrm{Gr}(d,n)$ and $\mathrm{Gr}(n-d,n)$ are isomorphic as schemes, and describe the isomorphism

at the level of functors of points. (Consider the dual: a rank-d quotient of $\mathcal{O}_S^n$ has a rank-$(n-d)$ kernel; dualize the short exact sequence (1.1).)

1.3 Coarse Moduli Spaces and the Automorphism Obstruction

The Grassmannian of section 1.2 is the fortunate case: its moduli functor is representable, and the representing scheme carries a universal family from which every other family is pulled back. Most moduli problems are not so obliging. The functor one writes down at the start almost never has a fine moduli space, and the reason is structural rather than accidental. It is already visible in the oldest moduli problem of all, the classification of elliptic curves up to isomorphism, where the culprit can be named precisely: the objects being classified have nontrivial automorphisms.

This section isolates that obstruction and extracts from the wreckage the best geometric object that survives. When no scheme represents $\mathcal{F}$, one may still ask for a scheme M whose $\bar{k}$-points match the isomorphism classes of objects and which receives a map from $\mathcal{F}$ that is as close to an isomorphism as the automorphisms permit. Such an M is a *coarse* moduli space. It remembers which objects there are but forgets how families of them twist, and this forgetting is exactly what a scheme is forced to do in the presence of automorphisms. The elliptic-curve moduli problem is worked in full: its coarse space is the affine j-line $\mathbb{A}^1_j$, and the same automorphisms that produce the coarse space also produce explicit families with pairwise isomorphic fibers that are not isomorphic as families, which is a direct proof that no fine moduli space can exist.

The warning issued at the end of section 1.1 (box 1.1.6) was that a representable functor is a sheaf, so a moduli functor that fails the sheaf condition cannot be representable. The failure mechanism is worth stating in the abstract before meeting it in an example. Suppose $\mathcal{F}$ is a moduli functor in the sense of definition 1.1.1, and suppose an object x of the moduli problem has an automorphism $\alpha \neq \mathrm{id}$. Take the constant family $x \times S$ over a base S. Both id and $\alpha \times \mathrm{id}_S$ are automorphisms of this family, and they are distinct. If $\mathcal{F}$ were represented by a scheme M (definition 1.1.3), a family over S would be the same data as a morphism $S \to M$, and the automorphisms of the family would be the automorphisms of that morphism, of which there is exactly one, the identity. A family that admits two distinct self-isomorphisms therefore cannot correspond to a morphism into any scheme. The automorphism is the obstruction, and it does not go away by choosing a cleverer scheme.

There is a weaker demand a scheme can meet. Rather than asking M to represent $\mathcal{F}$, ask only that M receive a natural map from $\mathcal{F}$ that is initial among all such maps to schemes, and that on geometric points this map be a bijection onto the isomorphism classes. This is the notion of a coarse moduli space.

Definition 1.3.1: Coarse Moduli Space

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a moduli functor over k (definition 1.1.1). A *coarse moduli space* for $\mathcal{F}$ is a scheme M over k together with a natural transformation

$$\eta\colon \mathcal{F} \to h_M$$

satisfying the following two conditions.

1. *Universality.* For every scheme N over k and every natural transformation $\theta\colon \mathcal{F} \to h_N$, there is a unique morphism of schemes $g\colon M \to N$ with $\theta = h_g \circ \eta$, where $h_g\colon h_M \to h_N$ is postcomposition with g.
2. *Bijection on geometric points.* The map

$$\eta(\operatorname{Spec} \bar{k})\colon \mathcal{F}(\operatorname{Spec} \bar{k}) \to h_M(\operatorname{Spec} \bar{k}) = M(\bar{k})$$

is a bijection.

A moduli functor is said to *admit a coarse moduli space* when such a pair (M, η) exists.

Condition (1) is a universal property, so it pins down the pair (M, η) up to unique isomorphism when it exists: the coarse space is the closest scheme-valued functor h_M underneath $\mathcal{F}$, or in categorical language, M corepresents $\mathcal{F}$ (h_M is the universal object under $\mathcal{F}$ in the category of representable functors). Condition (2) supplies the geometric content that a bare universal property would not: without it, the constant map from $\mathcal{F}$ to a point would satisfy (1) vacuously in trivial cases, and one wants M to parametrize the isomorphism classes. Together the two conditions say that M is a scheme whose $\bar{k}$-points *are* the isomorphism classes of objects, compatibly with all families, and that among schemes carrying such a map M is the finest. What M does not do is carry a universal family: there is a map $\mathcal{F} \to h_M$ but in general no inverse, so a morphism $S \to M$ need not come from any single family over S. A coarse moduli space records the objects but not the moduli of families.

Proposition 1.3.2: A Fine Moduli Space is Coarse

Let $\mathcal{F}$ be a moduli functor over k, and suppose $\mathcal{F}$ is represented by a scheme M in the sense of definition 1.1.3, with a natural isomorphism $\Phi\colon \mathcal{F} \xrightarrow{\sim} h_M$. Then (M, Φ) is a coarse moduli space for $\mathcal{F}$.

***Proof*:**
Both defining conditions of definition 1.3.1 must be checked for the pair (M, Φ).

Step 1: Universality. Let N be any scheme over k and $\theta\colon \mathcal{F} \to h_N$ a natural transformation. Since Φ is a natural isomorphism it has an inverse $\Phi^{-1}\colon h_M \to \mathcal{F}$, and $\theta \circ \Phi^{-1}\colon h_M \to h_N$ is a natural transformation between representable functors. By the Yoneda lemma ([12, Yoneda Lemma]), natural transformations $h_M \to h_N$ correspond bijectively to morphisms $M \to N$, so there is a unique morphism $g\colon M \to N$ with $h_g = \theta \circ \Phi^{-1}$. Composing on the right with Φ gives $h_g \circ \Phi = \theta$, which is the required factorization. Uniqueness of g is immediate: any g' with $h_{g'} \circ \Phi = \theta$ satisfies $h_{g'} = \theta \circ \Phi^{-1} = h_g$, and Yoneda forces $g' = g$.

Step 2: Bijection on geometric points. The component of Φ at $\operatorname{Spec} \bar{k}$ is a bijection $\mathcal{F}(\operatorname{Spec} \bar{k}) \to$

$h_M(\operatorname{Spec} \bar{k})$ because Φ is a natural isomorphism, so every component is an isomorphism of sets. This is precisely condition (2).

Both conditions hold, so (M, Φ) is a coarse moduli space for $\mathcal{F}$, completing the proof.

The proposition says that fineness is strictly stronger than coarseness: a fine moduli space is automatically a coarse one, with the coarsening map η taken to be the representing isomorphism Φ itself. The converse fails, and the gap between the two notions is exactly the phenomenon this section exists to explain. A coarse space exists in many cases where a fine one does not, and when a fine space does exist the two coincide. This is why the search for a moduli space proceeds in two stages: first ask for the ideal object (a fine space), and when the automorphism obstruction blocks it, retreat to the coarse space, which measures how much was lost.

The loss is not abstract. It manifests as families that a fine moduli space would have to distinguish but a coarse one cannot, because the families have identical fibers everywhere yet are not isomorphic. Such families are the fingerprint of a nontrivial automorphism, and they deserve a name.

Definition 1.3.3: Isotrivial Family

Fix a class of objects with a notion of family, and let x_0 be a fixed object over k. A family $\pi\colon X \to S$ over a k-scheme S is *isotrivial* (with fiber type x_0) if

1. every geometric fiber $X_{\bar{s}}$, for $\bar{s}$ a geometric point of S, is isomorphic to x_0; yet
2. $X \to S$ is not isomorphic, as a family over S, to the constant family $x_0 \times S \to S$.

A family satisfying (1) alone is said to have *constant fibers*; the point of the definition is that (1) does not force triviality.

An isotrivial nontrivial family is the local-to-global gap made concrete. Pointwise it looks exactly like the constant family $x_0 \times S$, since every fiber is a copy of x_0, but globally the copies are stitched together with a twist that no fiberwise isomorphism can undo. A moduli functor sends both this family and the constant family to the *same* data over each geometric point (both assign x_0 to every point), so any natural map $\mathcal{F} \to h_M$ to a scheme sends the two families to the same morphism $S \to M$. If $\mathcal{F}$ were representable, families would be determined by their classifying morphism, and the two families would have to be isomorphic, contradicting nontriviality. The existence of a single isotrivial nontrivial family thus rules out a fine moduli space. The mechanism producing such families is always the same: an automorphism of x_0 used as gluing data.

Example 1.3.4: An Isotrivial Nontrivial Family of Elliptic Curves

Work over an algebraically closed field k with $\operatorname{char} k \neq 2$, and let E be an elliptic curve over k with j-invariant $j(E) \notin \{0, 1728\}$, so that $\operatorname{Aut}(E) = \{\pm 1\}$, generated by the inversion automorphism $\iota = [-1]\colon E \to E$, $P \mapsto -P$ (the automorphism group and the special values $0, 1728$ are recorded in [7]). The involution ι is nontrivial because E has points with $2P \neq 0$, on which $\iota(P) = -P \neq P$; indeed the fixed points of ι are exactly the 2-torsion $\{P \mid 2P = 0\}$, a proper closed subscheme, so $\iota \neq \mathrm{id}$ on the rest of E.

Let $S = \operatorname{Spec} R$ where $R = k[t, t^{-1}]$, so $S = \mathbb{G}_m$ is the punctured affine line, and consider its connected double cover

$$\varphi\colon \tilde{S} = \operatorname{Spec} k[u, u^{-1}] \to S, \qquad \varphi^{\#}(t) = u^2$$

which is an étale $\mathbb{Z}/2\mathbb{Z}$-torsor for char$k \neq 2$; the nontrivial deck transformation is $\sigma\colon u \mapsto -u$. Form the constant family $E \times \tilde{S} \to \tilde{S}$ and glue it to itself over the two sheets of φ using the automorphism ι: concretely, take the quotient

$$X = \big(E \times \tilde{S}\big)/(\iota \times \sigma)$$

by the free $\mathbb{Z}/2\mathbb{Z}$-action that acts as ι on the fiber and as σ on the base. Since σ is free on $\tilde{S}$, the action is free, and the projection $E \times \tilde{S} \to \tilde{S}$ descends to a morphism $\pi\colon X \to \tilde{S}/\sigma = S$. This is the *quadratic twist* of E along the cover φ.

Every geometric fiber of π is isomorphic to E. Fix a geometric point $\bar{s}$ of S; its two preimages $\bar{u}, \sigma(\bar{u})$ in $\tilde{S}$ each carry the fiber E, and the gluing identifies $E \times \{\bar{u}\}$ with $E \times \{\sigma\bar{u}\}$ via ι. The fiber $X_{\bar{s}}$ is therefore a single copy of E, so condition (1) of definition 1.3.3 holds with fiber type $x_0 = E$. In particular every fiber has the same j-invariant $j(E)$.

The family $X \to S$ is not the constant family $E \times S$. Suppose it were, say via an S-isomorphism $\psi\colon E \times S \xrightarrow{\sim} X$. Pulling back along φ trivializes both sides over $\tilde{S}$, and ψ becomes an automorphism of $E \times \tilde{S}$ over $\tilde{S}$ commuting with the two descent data. The descent datum for the constant family is the identity gluing $\mathrm{id} \times \sigma$, while that for X is $\iota \times \sigma$; an isomorphism of the two families is a $\tilde{S}$-automorphism β of $E \times \tilde{S}$ with $\beta \circ (\mathrm{id} \times \sigma) = (\iota \times \sigma) \circ \beta$, that is, a family of automorphisms $\beta_u \in \mathrm{Aut}(E)$ with $\beta_{-u} = \iota \circ \beta_u$. Since $\mathrm{Aut}(E) = \{\pm 1\}$ is a constant (hence étale) group scheme and $\tilde{S} = \mathbb{G}_m$ is connected, the classifying morphism $u \mapsto \beta_u$ from $\tilde{S}$ to $\mathrm{Aut}(E)$ is constant, so β_u is a single automorphism independent of u, forcing $\beta = \iota \circ \beta$, hence $\iota = \mathrm{id}$. This contradicts $\iota \neq \mathrm{id}$, which holds for every elliptic curve. (The hypothesis $j(E) \neq 0, 1728$ is not needed for $\iota \neq \mathrm{id}$; it enters only to pin $\mathrm{Aut}(E)$ to exactly $\{\pm 1\}$, so that the classifying morphism lands in a constant group and the constancy argument applies cleanly.) So no such ψ exists, and condition (2) holds.

The family $X \to S$ is therefore isotrivial and nontrivial. Its very existence proves that the elliptic-curve moduli functor $\mathcal{M}_{1,1}$ (a moduli functor in the sense of definition 1.1.1) is not representable: a fine moduli space would send $X \to S$ and $E \times S \to S$, which have identical fibers everywhere, to the same classifying morphism $S \to M$, and would then be forced to declare the two families isomorphic. They are not.

The construction pinpoints where representability breaks. Every ingredient is standard, an elliptic curve, its inversion automorphism, and an étale double cover, and the twist is assembled from nothing more than the automorphism ι acting as gluing data over a nonsimply-connected base. The same recipe works whenever the objects of a moduli problem have a nontrivial automorphism and the base has a nontrivial torsor for the relevant automorphism group. This is the general shape of the automorphism obstruction: automorphisms of objects become gluing data for locally trivial but globally twisted families, and the moduli functor, which sees only fibers, cannot separate them.

Having seen that $\mathcal{M}_{1,1}$ is not representable, the task is to identify the coarse space that survives. Over $\bar{k}$ the isomorphism classes of elliptic curves are classified by a single number, the j-invariant, and the scheme whose points are those numbers is the affine line. This affine line, the *j-line*, is the coarse moduli space.

1.3.5 Note: the moduli functor $\mathcal{M}_{1,1}$ Throughout this section $\mathcal{M}_{1,1} \colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ denotes the moduli functor of elliptic curves in the sense of definition 1.1.1: for a k-scheme S, $\mathcal{M}_{1,1}(S)$ is the set of isomorphism classes of families $(E \to S, \sigma)$ with $E \to S$ smooth and proper of relative dimension 1, geometric fibers of genus 1, and σ a section. Pullback of families makes it contravariant. The j-invariant of a single elliptic curve over a field is developed in [7].

Theorem 1.3.6: The j-Line is a Coarse Moduli Space

Let k be an algebraically closed field. The affine line $\mathbb{A}^1_j = \operatorname{Spec} k[j]$, the *$j$-line*, is a coarse moduli space for the moduli functor $\mathcal{M}_{1,1}$ of elliptic curves, with the natural transformation

$$\eta \colon \mathcal{M}_{1,1} \to h_{\mathbb{A}^1_j}$$

sending a family $(E \to S) \in \mathcal{M}_{1,1}(S)$ to the morphism $S \to \mathbb{A}^1_j$ classifying its fiberwise j-invariant. The pair $(\mathbb{A}^1_j, \eta)$ satisfies both conditions of definition 1.3.1. However, $\mathcal{M}_{1,1}$ is not representable, so $\mathbb{A}^1_j$ is not a fine moduli space.

***Proof*:**

The j-invariant of a single elliptic curve over k takes every value of k and determines the isomorphism class over $\bar{k} = k$: two elliptic curves over an algebraically closed field are isomorphic if and only if they have the same j-invariant, and every element of k arises as a j-invariant ([7]). This standard input is used freely below.

Step 1: The natural transformation η is well defined. For a family $E \to S$ the assignment $s \mapsto j(E_s)$ is a morphism of schemes $j_E \colon S \to \mathbb{A}^1_j$. Concretely, working locally where $E \to S$ is given by a Weierstrass equation with coefficients in $\mathcal{O}_S(U)$ and discriminant a unit, the j-invariant is the universal rational expression in those coefficients, which is a regular function on U; these glue to a global morphism $j_E \colon S \to \mathbb{A}^1_j$ independent of the chosen Weierstrass presentation, since j is an isomorphism invariant. Setting $\eta_S(E \to S) = j_E$ gives a map $\mathcal{M}_{1,1}(S) \to h_{\mathbb{A}^1_j}(S)$, and naturality in S holds because forming a Weierstrass presentation and computing j commute with base change: for $f \colon T \to S$ one has $j_{f^*E} = j_E \circ f$, which is the equation $h_{\mathbb{A}^1_j}(f) \circ \eta_S = \eta_T \circ \mathcal{M}_{1,1}(f)$.

Step 2: Bijection on geometric points. A $\bar{k}$-point of $\mathbb{A}^1_j$ is an element of $k = \bar{k}$, and $\eta(\operatorname{Spec} \bar{k})$ sends an elliptic curve over $\bar{k}$ to its j-invariant. By the classification input, this map $\mathcal{M}_{1,1}(\operatorname{Spec} \bar{k}) \to \mathbb{A}^1_j(\bar{k}) = k$ is a bijection: it is surjective because every value of k is a j-invariant, and injective because equal j-invariants force an isomorphism over $\bar{k}$. This is condition (2) of definition 1.3.1.

Step 3: Universality. Let N be a scheme over k and $\theta \colon \mathcal{M}_{1,1} \to h_N$ a natural transformation; a morphism $g \colon \mathbb{A}^1_j \to N$ with $h_g \circ \eta = \theta$ must be produced, and shown unique. No hypothesis is placed on N: it need not be separated, so the argument is arranged so that every step rests on the naturality of θ with respect to *scheme* morphisms (restriction along open immersions and base change of families), never on the determination of a morphism into N by its values on closed points. The strategy is to realize g over an open cover of $\mathbb{A}^1_j$ by applying θ to tautological families, then to glue and to verify the factorization by functoriality.

Cover $\mathbb{A}^1_j$ by finitely many affine opens U_i each carrying a Weierstrass family $\mathcal{E}_{U_i} \to U_i$ whose fiberwise j-invariant is the restriction of the coordinate j. Such a cover exists: away from $j = 0$ and $j = 1728$ the family $y^2 = x^3 + a(j)x + b(j)$ with $a(j), b(j)$ regular and chosen so the fiberwise j-invariant equals the coordinate provides one open, and small opens around 0 and 1728 carrying suitable Weierstrass equations (with the two special j-values omitted from each of the other charts) complete the cover; on each U_i the class $\mathcal{E}_{U_i} \in \mathcal{M}_{1,1}(U_i)$ has $\eta_{U_i}(\mathcal{E}_{U_i}) = \iota_i$, the open immersion $\iota_i \colon U_i \hookrightarrow \mathbb{A}^1_j$, by Step 1. Applying θ gives a morphism $g_i := \theta_{U_i}(\mathcal{E}_{U_i}) \in h_N(U_i) = \mathrm{Hom}(U_i, N)$.

The g_i agree on overlaps as *morphisms of schemes*. Write $U_{ij} = U_i \cap U_j$ with open immersions $r_i \colon U_{ij} \hookrightarrow U_i$ and $r_j : U_{ij} \hookrightarrow U_j$. Both restricted families $r_i^* \mathcal{E}_{U_i}$ and $r_j^* \mathcal{E}_{U_j}$ over U_{ij} have fiberwise j-invariant the coordinate $j|_{U_{ij}}$, but they need not be isomorphic as families over U_{ij}: two Weierstrass families with equal j-invariant function are forms of one another, and can differ by a nontrivial quadratic twist recorded by a class in $H^1_{\text{ét}}(U_{ij}, \underline{\mathrm{Aut}}(\mathcal{E}))$ for the automorphism group scheme $\underline{\mathrm{Aut}}(\mathcal{E})$ of the fiber, which is exactly the twisting phenomenon that example 1.3.4 exhibits over $\mathbb{G}_m$ (reducedness of U_{ij} does not kill it, since $\mathbb{G}_m$ is reduced yet carries the nontrivial twist). The two families do become isomorphic after an étale cover, however: any two elliptic curves over a base with the same j-invariant are locally isomorphic in the étale topology, since a twist is split by the finite étale cover trivializing its class. Choose a surjective étale morphism $p \colon U'_{ij} \to U_{ij}$ over which $p^* r_i^* \mathcal{E}_{U_i}$ and $p^* r_j^* \mathcal{E}_{U_j}$ are isomorphic as families over U'_{ij}, hence equal in $\mathcal{M}_{1,1}(U'_{ij})$. Naturality of θ along $r_i \circ p$ and $r_j \circ p$ then gives, in $\mathrm{Hom}(U'_{ij}, N)$,

$$g_i \circ r_i \circ p = \theta_{U'_{ij}}(p^* r_i^* \mathcal{E}_{U_i}) = \theta_{U'_{ij}}(p^* r_j^* \mathcal{E}_{U_j}) = g_j \circ r_j \circ p$$

so the two morphisms $g_i \circ r_i$ and $g_j \circ r_j$ from U_{ij} to N agree after the base change p. Since p is faithfully flat of finite presentation, it is an epimorphism of schemes, and two morphisms into any scheme N that agree after such a base change are equal; hence $g_i \circ r_i = g_j \circ r_j$ in $\mathrm{Hom}(U_{ij}, N)$, an identity valid whether or not N is separated (the descent uses only that p is an fppf epimorphism, not any separatedness of N). The morphisms g_i therefore glue to a single morphism $g \colon \mathbb{A}^1_j \to N$ with $g \circ \iota_i = g_i$.

This g satisfies $h_g \circ \eta = \theta$. Both are natural transformations $\mathcal{M}_{1,1} \to h_N$, so it suffices to check that $(h_g \circ \eta)_S(E \to S) = \theta_S(E \to S)$ for every family $E \to S$; equivalently, that $g \circ j_E = \theta_S(E \to S)$ as morphisms $S \to N$, where $j_E = \eta_S(E \to S)$. The families $\{j_E^{-1}(U_i)\}$ cover S by open subschemes, so it suffices to verify equality after restriction to each $V_i := j_E^{-1}(U_i)$. On V_i the morphism j_E factors as $j_E|_{V_i} = \iota_i \circ f_i$ through a morphism $f_i \colon V_i \to U_i$, and $f_i^* \mathcal{E}_{U_i}$ is a Weierstrass family over V_i whose fiberwise j-invariant equals $j_E|_{V_i}$, the same as that of $E|_{V_i} \to V_i$. As on the overlaps, the two families $E|_{V_i}$ and $f_i^* \mathcal{E}_{U_i}$ need not be isomorphic over V_i (they may differ by a twist), but they become isomorphic after a surjective étale cover $q \colon V'_i \to V_i$, so they agree in $\mathcal{M}_{1,1}(V'_i)$. Naturality of θ along q and $f_i \circ q$ gives

$$\theta_{V'_i}(q^* E|_{V_i}) = \theta_{V'_i}(q^* f_i^* \mathcal{E}_{U_i}) = g_i \circ f_i \circ q = g \circ \iota_i \circ f_i \circ q = g \circ j_E|_{V_i} \circ q$$

so $\theta_{V_i}(E|_{V_i}) \circ q = (g \circ j_E|_{V_i}) \circ q$ as morphisms $V'_i \to N$; since q is a faithfully flat epimorphism these agree already over V_i, giving $\theta_{V_i}(E|_{V_i}) = g \circ j_E|_{V_i}$. Since these agree on the open cover $\{V_i\}$ of S, $\theta_S(E \to S) = g \circ j_E$ as required, so $h_g \circ \eta = \theta$.

Uniqueness of g is forced by the same functoriality, with no appeal to separatedness. If g' also satisfies $h_{g'} \circ \eta = \theta$, then applying both sides to the tautological family $\mathcal{E}_{U_i}$ over U_i gives $g' \circ \iota_i = g' \circ \eta_{U_i}(\mathcal{E}_{U_i}) = \theta_{U_i}(\mathcal{E}_{U_i}) = g_i = g \circ \iota_i$ as morphisms $U_i \to N$, that is, $g'|_{U_i} = g|_{U_i}$. Since $\{U_i\}$ is an open cover of $\mathbb{A}^1_j$ and morphisms of schemes into N form a Zariski sheaf (two morphisms agreeing on each member of an open cover of the source are equal, for any target N), $g' = g$. This establishes condition (1).

Step 4: $\mathcal{M}_{1,1}$ is not representable, so $\mathbb{A}^1_j$ is not fine. If $\mathcal{M}_{1,1}$ were representable by a scheme M, then by proposition 1.3.2 the pair (M, Φ) would be a coarse moduli space, and by the uniqueness of coarse spaces (the universal property of definition 1.3.1 determines M up to unique isomorphism) one would have $M \cong \mathbb{A}^1_j$ with Φ an isomorphism $\mathcal{M}_{1,1} \cong h_{\mathbb{A}^1_j}$. But example 1.3.4 exhibits, for $\operatorname{char} k \neq 2$, two families over $S = \mathbb{G}_m$, the constant family $E \times S$ and its quadratic twist $X \to S$, with the same fiberwise j-invariant, hence $\eta_S(E \times S) = \eta_S(X) = j_E$ the same morphism $S \to \mathbb{A}^1_j$. Under an isomorphism $\Phi \cong \eta$ these two families would be equal in $\mathcal{M}_{1,1}(S)$, that is, isomorphic as families over S, contradicting the nontriviality proved in example 1.3.4. Therefore $\mathcal{M}_{1,1}$ is not representable. When $\operatorname{char} k = 2$ the quadratic-twist construction of example 1.3.4 does not apply verbatim, since the double cover $u \mapsto u^2$ is inseparable; non-representability still holds, now witnessed by the extra automorphisms of the supersingular curve (whose j-invariant is $0 = 1728$ in characteristic 2), which produce a nontrivial twist by the same recipe over a base carrying an étale torsor for its larger automorphism group ([7]). In every characteristic, then, the coarse space $\mathbb{A}^1_j$ is not a fine moduli space, as we sought to show.

The theorem is the prototype every later chapter measures itself against. It exhibits both halves of the moduli story at once: a genuine geometric object, the j-line, that correctly parametrizes isomorphism classes, and a precise accounting of what that object cannot see, namely the twisting of families recorded in example 1.3.4. The coarse space is not a failure but a compromise forced by the automorphisms: it is the best scheme available, and proposition 1.3.2 guarantees it would upgrade to a fine space the instant the automorphisms were removed. The diagnosis also points at the cure. The obstruction is that $\mathcal{M}_{1,1}$ fails the sheaf condition (box 1.1.6) because étale-locally trivial families need not be globally trivial, and the twisting is governed by the automorphism group $\operatorname{Aut}(E)$. Two later strategies repair this: rigidify the objects until their automorphism groups become trivial, which recovers a fine moduli space for the rigidified problem, and enlarge the category of geometric objects to stacks, which remember automorphisms rather than quotienting them away. The first is taken up in section 1.4; the second is the subject of the descent and algebraic-stacks chapters later in the book.

Concretely, the coarse-versus-fine gap can be read off the automorphism groups directly. Over $\bar{k}$ with $\operatorname{char} k \notin \{2, 3\}$, a generic elliptic curve has $\operatorname{Aut}(E) = \{\pm 1\}$ of order 2, while the two special curves have larger groups: $j = 1728$ gives $\operatorname{Aut}(E) = \mathbb{Z}/4\mathbb{Z}$ and $j = 0$ gives $\operatorname{Aut}(E) = \mathbb{Z}/6\mathbb{Z}$ ([7]). The coarse space $\mathbb{A}^1_j$ is a smooth affine line, but the two special points $j = 0$ and $j = 1728$ are where the automorphism group jumps, and it is precisely at those points that a stack-theoretic refinement would record extra structure. The coarse space smooths over this jump; the fine-moduli-theoretic content lives in the automorphism groups the coarse space discards.

Exercise 1.3.1

1. Let (M, η) be a coarse moduli space for a moduli functor $\mathcal{F}$ in the sense of definition 1.3.1. Prove that (M, η) is unique up to unique isomorphism: if (M', η') is another coarse moduli space for $\mathcal{F}$, there is a unique isomorphism $g \colon M \to M'$ of schemes with $h_g \circ \eta = \eta'$.

2. Consider the terminal functor $\mathcal{F}(S) = \{*\}$ (a one-point set for every S). Show that the universality condition (1) of definition 1.3.1 already forces the coarse space to be the reduced point $M \cong \operatorname{Spec} k$ (so here condition (1) determines the geometry and condition (2) holds automatically). Then show that condition (2) alone does *not* pin down M: exhibit a scheme M' over k with a natural transformation $\eta' \colon \mathcal{F} \to h_{M'}$ for which condition (2) holds but condition (1) fails, and explain in one sentence what universality contributes beyond a bijection on geometric points.

3. In the setup of example 1.3.4, suppose instead that E is an elliptic curve with $j(E) = 0$ over $\bar{k}$ with char$k \notin \{2, 3\}$, so that $\mathrm{Aut}(E) = \mathbb{Z}/6\mathbb{Z}$. Using an order-3 automorphism ζ of E and a connected étale $\mathbb{Z}/3\mathbb{Z}$-cover $\tilde{S} \to S = \mathbb{G}_m$, construct an isotrivial nontrivial family whose fibers are all isomorphic to E. Verify conditions (1) and (2) of definition 1.3.3, and explain why the argument requires char$k \neq 3$.

4. Let $\pi\colon X \to S$ be the quadratic twist of example 1.3.4, with $S = \mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$. Compute the classifying morphism $\eta_S(X)\colon S \to \mathbb{A}^1_j$ explicitly, and confirm that it equals $\eta_S(E \times S)$. Deduce that $\eta_S(X)$ and $\eta_S(E \times S)$ take the same value at each closed point of S, a single point of $\mathbb{A}^1_j$, and explain why this equality of classifying morphisms is exactly the statement that $\mathbb{A}^1_j$ cannot distinguish the two families.

5. Prove the converse-type statement implicit in definition 1.3.3: if a moduli functor $\mathcal{F}$ admits *no* isotrivial nontrivial family (that is, every family with constant fibers of a fixed type is isomorphic to the constant family), does it follow that $\mathcal{F}$ is representable? Either prove it or give a counterexample, identifying which of the two conditions for a fine moduli space can still fail.

6. Let $\mathcal{F}$ be a moduli functor with a coarse moduli space (M, η), and suppose $\mathcal{F}$ happens to have *no* nontrivial automorphisms (every object x has $\mathrm{Aut}(x) = \{\mathrm{id}\}$) but is nonetheless not representable by a scheme. Explain, using box 1.1.6, why the existence of a coarse moduli space (M, η) (constructed as in the j-line argument, by hand from an invariant) does not immediately produce a fine moduli space, and name (in prose) the kind of geometric object that can represent such an $\mathcal{F}$.

1.4 Rigidification and Level Structure

The previous section diagnosed a precise failure. The moduli functor of elliptic curves is not representable, and the obstruction is not a technicality of the construction but a feature of the objects themselves: every elliptic curve carries the automorphism $[-1]$, and this automorphism produces families that are locally trivial yet globally nontrivial, exactly the families a representable functor cannot support (theorem 1.3.6). The j-line $\mathbb{A}^1_j$ survives only as a coarse moduli space (definition 1.3.1), recording isomorphism classes on geometric points but carrying no universal family.

There are two responses to this obstruction, and this book pursues both. The deep response, developed in Part III, keeps the objects as they are and enlarges the category of geometric spaces until the functor becomes representable by a stack. The present section pursues the elementary response: change the moduli problem by decorating each object with extra data, chosen so that no nontrivial automorphism fixes the data. An object rigidified in this way has no automorphisms as a decorated object, the mechanism of theorem 1.3.6 shuts off, and the decorated moduli functor becomes fine. The archetype of such decorating data is a *level structure* on an elliptic curve, a choice of basis for the N-torsion. Throughout, N is a positive integer invertible in k, so that the N-torsion is well behaved in every family under consideration.

The guiding principle is that automorphisms are the enemy of representability, so the cure is to add structure on which the automorphisms act freely enough to leave no fixed points but the trivial one. What "structure" means is best fixed by an explicit example before the general notion, and the N-torsion of an elliptic curve supplies it. Recall that for an elliptic curve E over a field in which N is invertible, the group of N-torsion points $E[N] = \{P \in E \mid NP = O\}$ is free of rank two over $\mathbb{Z}/N$, so $E[N] \cong (\mathbb{Z}/N)^2$ ([7]). A basis of $E[N]$ is the same as an isomorphism $(\mathbb{Z}/N)^2 \to E[N]$, and this is the datum to be added.

Definition 1.4.1: Level Structure

Let N be a positive integer invertible in k, and let E be an elliptic curve over a k-scheme S (a smooth proper morphism $E \to S$ of relative dimension one with geometrically connected genus one fibers, equipped with a section $O: S \to E$). The N-torsion $E[N] = \ker([N]: E \to E)$ is then a finite étale group scheme over S, locally on S isomorphic to the constant group $(\mathbb{Z}/N)^2$. A *level-N structure* on E is an isomorphism of S-group schemes

$$\alpha: (\mathbb{Z}/N)^2_S \xrightarrow{\sim} E[N]$$

where $(\mathbb{Z}/N)^2_S$ denotes the constant group scheme on S. Equivalently, over a field it is an ordered basis (P, Q) of $E[N]$ as a $\mathbb{Z}/N$-module, namely α sends the standard generators e_1, e_2 to P, Q.

This construction instantiates a general pattern. A class of geometric objects is *rigidified* by a choice of *rigidifying data*: extra structure attached to each object, functorial in the base, whose only automorphism compatible with a fixed object is the identity of that object. A level-N structure is the instance of this pattern for elliptic curves: it is a coordinate system on the N-torsion, and just as a choice of basis rigidifies a vector space by eliminating its linear automorphisms as ambient symmetries, a level structure is meant to eliminate the automorphisms of the elliptic curve. The role of the hypothesis N invertible in k is to keep $E[N]$ étale of the expected rank; in characteristic p dividing N the p-part of the torsion collapses (an ordinary curve has $E[p] \cong \mathbb{Z}/p$ rather than $(\mathbb{Z}/p)^2$), and the definition would attach the wrong amount of data. The second paragraph abstracts the pattern: the useful feature of a level structure is not that it involves torsion points but that a nontrivial automorphism of E must move it, and any decorating datum with that property will serve the same rigidifying purpose.

The level-N structures assemble into a moduli functor exactly as bare elliptic curves did, now with the decorating datum carried along.

Definition 1.4.2: The Level-N Moduli Functor

Fix N invertible in k. The *level-N moduli functor* $\mathcal{F}_N: \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ assigns to a k-scheme S the set

$$\mathcal{F}_N(S) = \left\{ (E, \alpha) \;\middle|\; \begin{array}{l} E \to S \text{ an elliptic curve,} \\ \alpha: (\mathbb{Z}/N)^2_S \xrightarrow{\sim} E[N] \text{ a level-}N\text{ structure} \end{array} \right\} \Big/ \cong$$

of isomorphism classes of pairs, where an isomorphism $(E, \alpha) \to (E', \alpha')$ is an isomorphism $\varphi: E \to E'$ of elliptic curves over S carrying α to α', meaning $\varphi[N] \circ \alpha = \alpha'$ on N-torsion. Pullback along $f: T \to S$ sends (E, α) to $(f^*E, f^*\alpha)$, making $\mathcal{F}_N$ contravariant.

The condition $\varphi[N] \circ \alpha = \alpha'$ is what distinguishes the level-N problem from the bare one. Two elliptic curves with level structure are identified only by an isomorphism that respects the chosen bases of their N-torsion, so the set $\mathcal{F}_N(S)$ remembers strictly more than the underlying moduli functor $\mathcal{F}$ of elliptic curves: forgetting α defines a natural transformation $\mathcal{F}_N \to \mathcal{F}$ that is generally many to one, since a single curve E admits many level structures. What has been bought by this extra bookkeeping is that the isomorphisms in $\mathcal{F}_N$ are far more rigid, and the next result measures precisely how rigid.

The obstruction to representing $\mathcal{F}$ was that the automorphism group of an object was nontrivial, which forced a family to be gluable from its restrictions in more than one way (box 1.1.6). The decisive property of a level structure is that it collapses the automorphism group of a decorated object to the trivial group, provided N is at least 3. The following lemma isolates this fact; it is the engine of the representability theorem.

Lemma 1.4.3: Level Structures Kill Automorphisms

Let $N \geq 3$ be invertible in k, let E be an elliptic curve over an algebraically closed field, and let α be a level-N structure on E. If $\varphi \in \mathrm{Aut}(E)$ is an automorphism of E as an elliptic curve fixing α, that is $\varphi[N] \circ \alpha = \alpha$, then $\varphi = \mathrm{id}_E$.

***Proof*:**
Write $\varphi[N]$ for the induced automorphism of $E[N]$. The condition $\varphi[N] \circ \alpha = \alpha$ says that $\varphi[N]$ is the identity on $E[N]$, because α is an isomorphism and so right-cancellable. The proof reduces to showing that an automorphism of E acting trivially on $E[N]$ is the identity, for $N \geq 3$.

Step 1: The automorphism group and its faithful action. An automorphism of E as an elliptic curve fixes the origin O, hence is a group automorphism of E ([7]). Every such φ restricts to a $\mathbb{Z}/N$-linear automorphism of the N-torsion, giving a group homomorphism

$$\rho_N \colon \mathrm{Aut}(E) \to \mathrm{Aut}_{\mathbb{Z}/N}(E[N]) \cong \mathrm{GL}_2(\mathbb{Z}/N)$$

the last isomorphism induced by any basis of $E[N]$. The claim is that ρ_N is injective for $N \geq 3$.

Step 2: Injectivity via the degree bound. Let $\varphi \in \ker \rho_N$, so φ acts as the identity on $E[N]$. The endomorphism $\psi = \varphi - \mathrm{id}$ of E then kills $E[N]$, so $E[N] \subseteq \ker \psi$. If $\psi \neq 0$, then ψ is an isogeny (a nonzero endomorphism of an elliptic curve is surjective with finite kernel). Since N is invertible in k, the group $E[N] \cong (\mathbb{Z}/N)^2$ consists of N^2 distinct étale (geometric) points, all lying in $\ker \psi$; the number of geometric points of the kernel is the separable degree $\deg_s \psi$, so $\deg_s \psi \geq N^2$. As $\deg \psi \geq \deg_s \psi$ always, this forces $\deg \psi \geq N^2$.

Now bound $\deg \psi$ from above. The endomorphism ring $\mathrm{End}(E)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field or, in characteristic p, an order in a quaternion algebra; in every case the degree is the norm form, a positive definite quadratic form, and φ is an automorphism, hence a unit, so $\deg \varphi = 1$. The triangle inequality for the norm on $\mathrm{End}(E)$ gives

$$\sqrt{\deg \psi} = \|\varphi - \mathrm{id}\| \leq \|\varphi\| + \|\mathrm{id}\| = \sqrt{\deg \varphi} + \sqrt{\deg(\mathrm{id})} = 1 + 1 = 2$$

so $\deg \psi \leq 4$. Combining $\deg \psi \geq N^2$ with $\deg \psi \leq 4$ yields $N^2 \leq 4$, that is $N \leq 2$, contrary to $N \geq 3$. Therefore $\psi = 0$, that is $\varphi = \mathrm{id}_E$, which shows $\ker \rho_N$ is trivial and ρ_N is injective.

Step 3: Conclusion. The automorphism φ of the hypothesis lies in $\ker \rho_N$ by the reduction in the opening paragraph, and $\ker \rho_N$ is trivial by Step 2. Hence $\varphi = \mathrm{id}_E$, completing the proof.

The numerical heart of the argument is the failure of $[-1]$ to fix a level-N structure once $N \geq 3$. The automorphism $[-1]$, present on every elliptic curve, acts on $E[N]$ as multiplication by -1; this is the identity on $E[N]$ precisely when $2P = O$ for all $P \in E[N]$, that is precisely when $N \leq 2$. So for $N = 1$ or $N = 2$ the automorphism $[-1]$ fixes every level structure and the rigidification fails to remove it, whereas for $N \geq 3$ it moves each level structure, and with the last universal automorphism broken the object becomes rigid. The endomorphism norm inequality in Step 2 is what upgrades this observation from $[-1]$ to *all* automorphisms simultaneously, including the extra automorphisms at $j = 0$ and $j = 1728$ ([7]). This is the exact sense in which a level-N structure "rigidifies" an elliptic curve.

The representability theorem that the lemma powers can now be stated.

Theorem 1.4.4: Representability of Level-N Moduli

Let $N \geq 3$ be invertible in k. The level-N moduli functor $\mathcal{F}_N$ of definition 1.4.2 is representable by a smooth affine k-scheme $Y(N)$: there is an isomorphism of functors $\mathcal{F}_N \cong h_{Y(N)}$, so $Y(N)$ is a fine moduli space (definition 1.1.3) for elliptic curves with level-N structure. In particular $Y(N)$ carries a universal elliptic curve with level-N structure $(\mathcal{E}, \alpha^{\mathrm{univ}})$ through which every family is a unique pullback.

***Proof*:**

The proof has two parts: the removal of the automorphism obstruction, which is the conceptual content and where lemma 1.4.3 enters, and the actual construction of $Y(N)$, which proceeds by cutting the representing scheme out of a space that is already known to be fine.

Step 1: Rigidity of objects. Fix a pair (E, α) over an algebraically closed field. An automorphism of (E, α) in the sense of definition 1.4.2 is an automorphism φ of E with $\varphi[N] \circ \alpha = \alpha$, and lemma 1.4.3 shows the only such φ is id_E. Thus every object of the level-N problem has trivial automorphism group. This is exactly the hypothesis whose failure obstructed representability of the bare functor $\mathcal{F}$ (theorem 1.3.6): with no nontrivial automorphisms there is no room to build a locally trivial but globally nontrivial family, so the obstruction of box 1.1.6 is removed.

Step 2: A rigid family is determined by its restrictions. Make Step 1 quantitative. Let (E, α) and (E', α') be two families over S that become isomorphic over each member of a Zariski (indeed étale) cover $\{S_i \to S\}$. On each overlap S_{ij} the two given isomorphisms φ_i, φ_j differ by an automorphism of the pair $(E|_{S_{ij}}, \alpha|_{S_{ij}})$. The scheme of automorphisms of a rigidified elliptic curve is unramified over the base (an automorphism is rigid, being determined by finitely many torsion conditions), and by Step 1 it has trivial geometric fibers; an unramified group scheme with trivial fibers is the trivial group, so the automorphism in question is the identity and $\varphi_i = \varphi_j$ on S_{ij}. The local isomorphisms therefore agree on overlaps and glue to a global isomorphism $(E, \alpha) \cong (E', \alpha')$. So a family with level structure is determined up to unique isomorphism by its restrictions to a cover, which is the sheaf property that a representable functor must have and that $\mathcal{F}$ lacked.

Step 3: Construction of the representing scheme. It remains to exhibit a scheme representing $\mathcal{F}_N$. Every elliptic curve is a plane cubic in Weierstrass form after a choice of origin and invariant differential, and the coefficients of a Weierstrass equation define a scheme mapping to the affine space of coefficients; the substitutions relating two Weierstrass equations of the same curve form the action of a solvable group G, and the representing scheme is the quotient by this action, which exists as a scheme precisely because the added data makes the action free (no residual automorphisms remain to fix a point). Concretely, fix a Weierstrass family

$$\mathcal{W}: \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

over the affine space $\mathbb{A}^5 = \operatorname{Spec} k[a_1, a_2, a_3, a_4, a_6]$, and let $U \subseteq \mathbb{A}^5$ be the open locus where the discriminant Δ is invertible, so $\mathcal{W}|_U \to U$ is an elliptic curve. Adjoining a level-N structure means adjoining an ordered basis of the N-torsion. The N-torsion $\mathcal{W}[N] \to U$ is finite étale, and the functor on U-schemes assigning to $T \to U$ the set of bases of $(\mathcal{W}[N])_T$ is representable by a scheme $\widetilde{U} \to U$ finite étale (it is a torsor under the constant group $\mathrm{GL}_2(\mathbb{Z}/N)$ over the locus where N is invertible, cut out inside a product of copies of $\mathcal{W}[N]$ by the nondegeneracy conditions). The pair $(\mathcal{W}_{\widetilde{U}}, \alpha^{\mathrm{univ}})$ is then a family with level structure over $\widetilde{U}$, where α^{univ} is the tautological basis.

The redundancy in this presentation is the freedom to change Weierstrass coordinates. Two triples over the same base give the same elliptic curve exactly when they differ by a substitution

$$(x, y) \longmapsto (u^2x + r,\ u^3y + su^2x + t), \qquad u \in \mathbb{G}_m,\ r, s, t \in \mathbb{G}_a$$

and these substitutions form the solvable group $G = \mathbb{G}_m \ltimes \mathbb{G}_a^3$ of the opening sentence, of dimension 4, acting on the coefficient space $\mathbb{A}^5$ and lifting to an action on $\widetilde{U}$ that transports the tautological torsion basis along the coordinate change. Two points of $\widetilde{U}$ give isomorphic pairs (E, α) if and only if they lie in one G-orbit: a coordinate change matching both the curves and their level structures is exactly an element of G carrying one point to the other. By Step 1 such an element is unique when it exists (no automorphisms to spoil uniqueness), which is precisely the statement that G acts *freely* on $\widetilde{U}$ (a nonidentity element fixing a point would be a nontrivial automorphism of the corresponding rigidified pair). A free action of the smooth affine solvable group G on the smooth affine scheme $\widetilde{U}$ admits a geometric quotient

$$Y(N) = \widetilde{U}/G$$

which is again a smooth affine scheme; the quotient collapses the 4 redundant coordinate-change directions, so $Y(N)$ has dimension $\dim \widetilde{U} - \dim G = 5 - 4 = 1$, a curve, as the geometry demands. Its functor of points is by construction

$$h_{Y(N)}(T) = \{\, T \to \widetilde{U} \,\}/\!\sim\, = \{\text{elliptic curves over } T \text{ with level-}N \text{ structure}\}/\!\cong\, = \mathcal{F}_N(T)$$

naturally in T. The chain of equalities is exactly the assertion $\mathcal{F}_N \cong h_{Y(N)}$: the first equality is the definition of the quotient, and the middle equality holds because every pair (E, α) over T is Zariski-locally Weierstrass with a chosen torsion basis, hence Zariski-locally a T-point of $\widetilde{U}$, and these local charts glue uniquely by Step 2. Smoothness of $Y(N)$ follows from smoothness of $\widetilde{U}$ over the smooth base U and the freeness of the G-action. Therefore $Y(N)$ is a fine moduli space for $\mathcal{F}_N$, with universal family the descent of $(\mathcal{W}_{\widetilde{U}}, \alpha^{\mathrm{univ}})$, as we sought to show.

The theorem is the payoff of the entire chapter in miniature: a moduli problem that failed to be representable has been repaired, not by changing the category of geometric spaces but by refining the objects until their automorphism groups vanish. Steps 1 and 2 are the conceptual core and are where lemma 1.4.3 is spent; they say that rigidity of objects implies rigidity of families, which is the sheaf-theoretic condition a representable functor requires. Step 3 then constructs $Y(N)$ concretely, and the construction is instructive: the representing scheme is a free quotient of an étale cover of the Weierstrass parameter space by the solvable group of coordinate changes, a pattern that recurs when GIT builds moduli spaces as quotients in chapter 3. The hypothesis $N \geq 3$ is not a convenience but the exact threshold from lemma 1.4.3: for $N \leq 2$ the universal automorphism $[-1]$ survives, the quotient in Step 3 is not free, and $\mathcal{F}_N$ is again only coarsely representable.

The fine moduli space produced by the theorem has a name and a classical life of its own, and it is worth seeing it as a concrete curve rather than an abstract representing object.

Example 1.4.5: The Modular Curve $Y(N)$

The scheme $Y(N)$ of theorem 1.4.4 is the (affine) *modular curve of full level* N. Over $\mathbb{C}$ it has a transcendental description that makes its geometry visible, though the description has one feature that is easy to miss: an elliptic curve over $\mathbb{C}$ is a complex torus $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ with τ in the upper half-plane $\mathfrak{H}$, and its N-torsion is $\frac{1}{N}(\mathbb{Z}+\mathbb{Z}\tau)/(\mathbb{Z}+\mathbb{Z}\tau)$, with the standard basis $\{\frac{1}{N}, \frac{\tau}{N}\}$. A level-$N$ structure is a choice of ordered basis of this torsion, an *arbitrary* isomorphism $\alpha\colon (\mathbb{Z}/N)^2 \to E[N]$ as in definition 1.4.1, and here a discrete invariant enters. The Weil pairing e_N attaches to α the primitive N-th root of unity $\zeta_\alpha = e_N(\alpha(e_1), \alpha(e_2)) \in \mu_N$, and any isomorphism of pairs preserves ζ_α (the Weil pairing is functorial in the curve), so ζ_α is constant on each connected component of the moduli. There are $\varphi(N) = |(\mathbb{Z}/N)^\times|$ primitive N-th roots of unity, and the change of variables $g \in \mathrm{SL}_2(\mathbb{Z}/N)$ (determinant 1) fixes ζ_α while a general $g \in \mathrm{GL}_2(\mathbb{Z}/N)$ raises it to the $\det g$ power ($\zeta \mapsto \zeta^{\det g}$); the fixed-determinant (*symplectic*) level structures with a given ζ_α are exactly those parametrized by the principal congruence subgroup

$$\Gamma(N) = \ker\big(\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)\big)$$

acting on $\mathfrak{H}$ by fractional linear transformations ([7]). Thus over $\mathbb{C}$ the modular curve is a disjoint union of one copy of $\Gamma(N)\backslash\mathfrak{H}$ for each of the $\varphi(N)$ values of the Weil-pairing invariant,

$$Y(N)(\mathbb{C}) = \coprod_{\zeta \in \mu_N^{\mathrm{prim}}} \Gamma(N)\backslash\mathfrak{H}$$

a smooth affine Riemann surface with $\varphi(N)$ connected components, whose points are exactly the isomorphism classes of elliptic curves with full level-N structure. (Fixing the value ζ_α once and for all, that is passing to symplectic level structures, is the standard way to obtain the connected modular curve $\Gamma(N)\backslash\mathfrak{H}$; the full GL_2-functor of definition 1.4.2 sees all $\varphi(N)$ components at once.) For $N \geq 3$ the group $\Gamma(N)$ acts freely on $\mathfrak{H}$ (it contains no elliptic elements and, crucially, not $-\mathrm{id}$, which is where $N \geq 3$ reappears), so each component is a manifold with no orbifold points, matching the smoothness asserted by the theorem.

The universal family over $Y(N)$ is the descended family whose fiber over the class of τ is the torus $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ with its tautological torsion basis; concretely it is the Weierstrass family $y^2 = 4x^3 - g_2(\tau)x - g_3(\tau)$ built from the Eisenstein series, carried down along $\mathfrak{H} \to \Gamma(N)\backslash\mathfrak{H}$ together with the marked basis of N-torsion. This is a genuine universal object: any family of elliptic curves with level-N structure over a base S is the pullback of $(\mathcal{E}, \alpha^{\mathrm{univ}})$ along the unique classifying map $S \to Y(N)$ furnished by theorem 1.4.4. For $N = 3$ one has $\varphi(3) = 2$, so $Y(3)$ has two connected components, each an affine curve of genus zero, a projective line with a few points removed; the removed points are the cusps, where the elliptic curve degenerates, and they will reappear as the boundary when $\mathcal{M}_{1,1}$ is compactified in the capstone.

The modular curve makes the abstract representability concrete: $Y(N)$ is an ordinary quasi-projective variety, its complex points are $\varphi(N)$ copies of a familiar quotient of the upper half-plane, and it carries a family that specializes to any elliptic-curve-with-level-structure one wishes to name. The appearance of $\Gamma(N)$ ties the algebraic construction to the classical theory of modular forms, where functions on $\Gamma(N)\backslash\mathfrak{H}$ are precisely modular forms of level N ([7]); the moduli-theoretic reading is that a modular form is a section of a line bundle on the fine moduli space $Y(N)$.

Rigidification produced a fine moduli space by enlarging the problem, but the original problem, the classification of elliptic curves without extra data, has not gone away. The relation between the two is exact and useful: the fine space $Y(N)$ maps down to the coarse space of the un-rigidified problem by forgetting the level structure, and the map is the quotient by the group that permutes level structures. This exhibits the j-line as a group quotient of a smooth curve, which is often the most tractable way to study it.

Proposition 1.4.6: Coarse Space as a Quotient of the Rigidified Space

Let $N \geq 3$ be invertible in k, with $k = \bar{k}$. The finite group $\mathrm{GL}_2(\mathbb{Z}/N)$ acts on the fine moduli space $Y(N)$ by changing the level structure: an element $g \in \mathrm{GL}_2(\mathbb{Z}/N)$ sends the class of (E, α) to the class of $(E, \alpha \circ g)$. Forgetting the level structure induces a morphism $Y(N) \to \mathbb{A}^1_j$ to the j-line that is invariant under this action and identifies the j-line with the quotient:

$$\mathbb{A}^1_j \;\cong\; Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$$

and this presents the coarse moduli space $\mathbb{A}^1_j$ of elliptic curves (theorem 1.3.6, definition 1.3.1) as the quotient of the fine rigidified space by the finite group acting on level structures.

***Proof*:**
Step 1: The action and its orbits. For $g \in \mathrm{GL}_2(\mathbb{Z}/N)$ and a level structure $\alpha\colon (\mathbb{Z}/N)^2 \to E[N]$, the composite $\alpha \circ g$ is again an isomorphism $(\mathbb{Z}/N)^2 \to E[N]$, hence another level structure on the same curve E; this defines a right action of $\mathrm{GL}_2(\mathbb{Z}/N)$ on the pairs (E, α) fixing E and permuting the decorating data. Since $\mathrm{GL}_2(\mathbb{Z}/N)$ acts simply transitively on the set of bases of $E[N] \cong (\mathbb{Z}/N)^2$, for a fixed E the orbit of (E, α) under $\mathrm{GL}_2(\mathbb{Z}/N)$ is the set of *all* pairs (E, α') with the same underlying curve. Passing to isomorphism classes, the $\mathrm{GL}_2(\mathbb{Z}/N)$-orbit of the point of $Y(N)$ representing (E, α) consists of exactly the classes of pairs whose curve is isomorphic to E.

Step 2: The forgetful map factors through the quotient. The assignment $(E, \alpha) \mapsto E$ is a natural transformation $\mathcal{F}_N \to \mathcal{F}$ of moduli functors, and post-composing with the coarse map $\mathcal{F} \to h_{\mathbb{A}^1_j}$ of theorem 1.3.6 gives a natural transformation $\mathcal{F}_N \to h_{\mathbb{A}^1_j}$, hence by Yoneda and representability of $\mathcal{F}_N$ a morphism of schemes $\pi\colon Y(N) \to \mathbb{A}^1_j$. Concretely π sends the class of (E, α) to $j(E)$. Since $j(E)$ depends only on E and not on α, the map π is constant on $\mathrm{GL}_2(\mathbb{Z}/N)$-orbits by Step 1, so it is $\mathrm{GL}_2(\mathbb{Z}/N)$-invariant and factors through the quotient scheme $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$, which exists because $\mathrm{GL}_2(\mathbb{Z}/N)$ is finite acting on a quasi-projective scheme, inducing $\bar{\pi}\colon Y(N)/\mathrm{GL}_2(\mathbb{Z}/N) \to \mathbb{A}^1_j$.

Step 3: $\bar{\pi}$ is a bijection on points, hence an isomorphism. On $\bar{k}$-points, Step 1 shows the fibers of π are exactly the $\mathrm{GL}_2(\mathbb{Z}/N)$-orbits: two pairs (E, α), (E', α') map to the same j if and only if $E \cong E'$, if and only if they lie in one orbit. Therefore $\bar{\pi}$ is injective on $\bar{k}$-points. It is surjective because every $j_0 \in \mathbb{A}^1_j(\bar{k})$ is $j(E)$ for some elliptic curve E over $\bar{k}$ (theorem 1.3.6), and E admits a level-N structure over $\bar{k}$ (its N-torsion is $(\mathbb{Z}/N)^2$, so a basis exists), giving a point of $Y(N)$ over j_0. Thus $\bar{\pi}$ is a bijection on $\bar{k}$-points between normal varieties. Upgrading this to an isomorphism requires two inputs beyond the set-theoretic bijection, and both are verified here rather than asserted.

First, $\bar{\pi}$ is birational. The morphism $\pi\colon Y(N) \to \mathbb{A}^1_j$ is finite. To see this, compactify: adding the cusps to each component completes $Y(N)$ to a smooth projective curve $X(N)$, and completing the j-line adds the single point $j = \infty$ to $\mathbb{A}^1_j = \mathbb{P}^1 \setminus \{\infty\}$, giving $X(1) = \mathbb{P}^1$. The forgetful map extends to a nonconstant morphism $\bar{\pi}_c\colon X(N) \to X(1) = \mathbb{P}^1$ of smooth projective curves, and a nonconstant morphism of smooth projective curves is finite. All cusps of $X(N)$ lie over $j = \infty$: the cusps are by construction the points where the elliptic curve degenerates, which is exactly the fiber of $X(N) \to \mathbb{P}^1$ over ∞ ([7]). Restricting the finite morphism $\bar{\pi}_c$ over the cusp-free locus $\mathbb{A}^1_j = \mathbb{P}^1 \setminus \{\infty\}$ recovers $\pi\colon Y(N) \to \mathbb{A}^1_j$, which is therefore finite, being the base change of a finite morphism along the open immersion $\mathbb{A}^1_j \hookrightarrow \mathbb{P}^1$. Over a $\bar{k}$-point j_0 with $j_0 \neq 0, 1728$ the curve E has $\mathrm{Aut}(E) = \{\pm 1\}$ ([7]). By Step 1 the fiber $\pi^{-1}(j_0)$ is the set of isomorphism classes of level structures on E, that is the $|\mathrm{GL}_2(\mathbb{Z}/N)|$ bases of $E[N]$ modulo the action of $\mathrm{Aut}(E) = \{\pm 1\}$; since $\{\pm 1\}$ acts on bases through $\pm\mathrm{id} \in \mathrm{GL}_2(\mathbb{Z}/N)$ without fixed points (as $-\mathrm{id} \neq \mathrm{id}$ for $N \geq 3$), the fiber has exactly $|\mathrm{GL}_2(\mathbb{Z}/N)|/2$ points, each a reduced point because N is invertible in k and the level cover is étale there. This common value is both the degree of π over the ordinary locus and the size of the corresponding $\mathrm{GL}_2(\mathbb{Z}/N)$-orbit, by the orbit-stabilizer count $|\mathrm{GL}_2(\mathbb{Z}/N)|/|\{\pm 1\}|$ with stabilizer exactly the $\{\pm 1\}$ acting through $\mathrm{Aut}(E)$. Hence the generic fiber of $\bar{\pi} = \pi/\mathrm{GL}_2(\mathbb{Z}/N)$ is a single reduced point, so $\bar{\pi}$ is a finite morphism of degree one, that is birational.

Second, $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$ is normal, being the quotient of the smooth (hence normal) $Y(N)$ by a finite group, and $\mathbb{A}^1_j$ is smooth. A finite birational morphism to a normal variety is an isomorphism (this is the form of Zariski's main theorem that does not require separability hypotheses beyond birationality, which was checked above and holds in every characteristic because N is invertible). Therefore $\bar{\pi}$ is an isomorphism $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N) \cong \mathbb{A}^1_j$.

That $\mathbb{A}^1_j$ is the coarse moduli space of elliptic curves is theorem 1.3.6, so the displayed isomorphism presents the coarse space as the finite-group quotient of the fine rigidified space, completing the proof.

The proposition is the reconciliation the section was aiming for. The un-rigidified moduli problem has no fine solution, but it has a coarse one, $\mathbb{A}^1_j$, and the coarse solution is recovered from the fine solution of the rigidified problem by dividing out the choices that the rigidification introduced. The group $\mathrm{GL}_2(\mathbb{Z}/N)$ measures exactly how much extra data a level structure carries, and quotienting by it erases that data. This is a template for the whole subject: a moduli problem with automorphisms is studied by rigidifying to a fine space and then quotienting by the finite group that acts on the rigidifying data, and the quotient is the coarse space. The template has a limitation the reader should carry forward. The quotient $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$ remembers only orbits, so it forgets the stabilizers, the automorphism groups of the curves, exactly the information $\mathbb{A}^1_j$ was already blind to at $j = 0$ and $j = 1728$. The framework that keeps track of the stabilizers while still forming the quotient is the quotient *stack* $[Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)]$, and building it is the task of Part III.

Exercise 1.4.1

1. Show that for $N = 2$ the automorphism $[-1]$ fixes every level-2 structure on every elliptic curve, and conclude that lemma 1.4.3 fails for $N = 2$. Deduce that the level-2 moduli functor $\mathcal{F}_2$ is not representable by a fine moduli space over $\bar{k}$ (with $\operatorname{char} k \neq 2$), even though it carries more data than $\mathcal{F}$.

2. Let $N \geq 3$. Compute the order of $\mathrm{GL}_2(\mathbb{Z}/N)$ for N prime, and verify directly that $\mathrm{GL}_2(\mathbb{Z}/3)$ acts simply transitively on the set of ordered bases of $(\mathbb{Z}/3)^2$. Using proposition 1.4.6, interpret the degree of the forgetful map $Y(3) \to \mathbb{A}^1_j$ over a generic j (where $\mathrm{Aut}(E) = \{\pm 1\}$) and explain the discrepancy between $|\mathrm{GL}_2(\mathbb{Z}/3)|$ and this degree in terms of the residual action of $[-1]$.

3. Prove that the endomorphism-norm inequality used in Step 2 of lemma 1.4.3 is sharp for $N = 2$ by exhibiting, on an arbitrary elliptic curve, an endomorphism $\psi = \varphi - \mathrm{id}$ of degree 4 with φ an automorphism and $E[2] \subseteq \ker \psi$. (Suggestion: take $\varphi = [-1]$.)

4. A level structure is one instance of rigidifying data. For a smooth projective curve C of genus $g \geq 2$, an *n-canonical framing* is a choice of ordered basis of $H^0(C, \omega_C^{\otimes n})$ for $n \geq 3$. Explain, by analogy with definition 1.4.1 and lemma 1.4.3, why such a framing should rigidify the moduli problem of genus-g curves: identify the group that acts on the framings (playing the role of $\mathrm{GL}_2(\mathbb{Z}/N)$) and state what property of $\mathrm{Aut}(C)$ makes the framing kill automorphisms.

5. Show that over $\mathbb{C}$ each connected component $\Gamma(3)\backslash\mathfrak{H}$ of the modular curve $Y(3)$ of example 1.4.5 has genus zero, by computing the index $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(3)]$ and the number of cusps, then applying the Riemann-Hurwitz formula to the covering $\Gamma(3)\backslash\mathfrak{H} \to Y(1) = \mathbb{A}^1_j$. (You may use that $Y(1)(\mathbb{C}) = \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$ has genus zero with one cusp and elliptic points of orders 2 and 3.) Then check that the full forgetful map $Y(3) \to Y(1)$, taken over both components, has degree 24, reconciling with part (b).

1.5 The Functor-of-Points Toolkit

The preceding sections have twice confronted the same fault line. A moduli problem is a functor $\mathcal{F} \colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ (definition 1.1.1), and the best possible outcome is a fine moduli space, a scheme M with $\mathcal{F} \cong h_M$ (definition 1.1.3). Yet functors that arise from geometry routinely fail to be representable, and the failure is not random. The warning issued in box 1.1.6 named the culprit: a functor of points is always a sheaf for the Zariski topology, so any moduli functor that is not such a sheaf cannot be representable. This section turns that observation into working tools. Two questions organize it. First, given a functor, what are the criteria that decide whether it is representable? Second, what first-order information does a functor carry even before that question is settled, and how is that information extracted?

The answer to the first question is a local-to-global principle: a functor representable in the Zariski topology and locally representable on an open cover is representable, exactly as a scheme is glued from affine charts. The answer to the second is the tangent space of a functor, computed by feeding the functor the dual numbers $k[\varepsilon]$. This tangent space is the first invariant of deformation theory, the thread the book picks up after this chapter: where this chapter asks whether a moduli space exists, the deformation theory that follows asks what it looks like near a point, and the tangent space of the functor is where those two investigations meet. The two tools of this section are complementary. The sheaf-and-cover criterion is a global test for existence; the tangent space is a pointwise linearization that survives even when no representing scheme does.

The first tool is the tangent space, and it comes from the dual numbers. A scheme X over k has, at each k-point x, a Zariski tangent space $T_xX = (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$, and the classical characterization of tangent vectors identifies them with morphisms $\operatorname{Spec} k[\varepsilon] \to X$ sending the closed point to x. The dual numbers $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ are the algebraic model of an infinitesimally short arc: a k-point together with a tangent direction. A functor $\mathcal{F}$ of points has no underlying topological space and no local ring, so its tangent space cannot be defined through a cotangent module. What it does have is its value on $\operatorname{Spec} k[\varepsilon]$, and the same tangent vectors reappear there. The device is to compare $\mathcal{F}(k[\varepsilon])$ with $\mathcal{F}(k)$ along the map that sets ε to zero.

Throughout, for a k-algebra A write $\mathcal{F}(A)$ for $\mathcal{F}(\operatorname{Spec} A)$; this is the standard abuse that lets a contravariant functor on schemes be evaluated on rings covariantly. The reduction homomorphism $k[\varepsilon] \to k$, $\varepsilon \mapsto 0$, induces $\mathcal{F}(k[\varepsilon]) \to \mathcal{F}(k)$, and a choice of base point $x \in \mathcal{F}(k)$ singles out a fiber.

Definition 1.5.1: Tangent Space of a Functor

Let $\mathcal{F} \colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a functor and $x \in \mathcal{F}(k)$ a k-point. Let

$$\pi \colon \mathcal{F}(k[\varepsilon]) \longrightarrow \mathcal{F}(k)$$

be the map induced by the k-algebra homomorphism $k[\varepsilon] \to k$ sending $\varepsilon \mapsto 0$. The *tangent space of $\mathcal{F}$ at x* is the fiber

$$T_x\mathcal{F} \;=\; \pi^{-1}(x) \;=\; \{\xi \in \mathcal{F}(k[\varepsilon]) \mid \pi(\xi) = x\}$$

the set of lifts of x to a $k[\varepsilon]$-point of $\mathcal{F}$.

As stated, $T_x\mathcal{F}$ is only a pointed set: it always contains the *constant* lift, the image of x under the section $\mathcal{F}(k) \to \mathcal{F}(k[\varepsilon])$ induced by the structure map $k \hookrightarrow k[\varepsilon]$. For a functor of geometric origin it carries far more structure. When $\mathcal{F} = h_X$ is representable, the fiber $T_x\mathcal{F}$ is precisely the classical Zariski tangent space T_xX, a k-vector space; the identification is exactly the arc description of tangent vectors recalled above, and the full account of why the functorial definition recovers the geometric one is developed in [12, Tangent Space of a Functor]. The vector-space structure does not require representability. It is enough that $\mathcal{F}$ be *reasonable* in the following precise sense.

Proposition 1.5.2: Vector Space Structure on the Tangent Space

Suppose $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ takes the fiber product of k-algebras

$$k[\varepsilon] \times_k k[\varepsilon] \;\cong\; k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$$

to the corresponding fiber product of sets, that is, the natural map

$$\mathcal{F}\big(k[\varepsilon] \times_k k[\varepsilon]\big) \longrightarrow \mathcal{F}(k[\varepsilon]) \times_{\mathcal{F}(k)} \mathcal{F}(k[\varepsilon])$$

is a bijection. Then for every $x \in \mathcal{F}(k)$ the set $T_x\mathcal{F}$ carries a natural k-vector space structure whose zero is the constant lift.

***Proof*:**
Fix $x \in \mathcal{F}(k)$. The argument realizes addition and scalar multiplication as functorial consequences of ring maps out of $k[\varepsilon]$.

Step 1: Addition. The ring $B = k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$ is the fiber product $k[\varepsilon] \times_k k[\varepsilon]$, with the two projections $p_i\colon B \to k[\varepsilon]$ given by $\varepsilon_i \mapsto \varepsilon$ and $\varepsilon_j \mapsto 0$ for $j \neq i$. By hypothesis

$$\mathcal{F}(B) \xrightarrow{\;\sim\;} T_x\mathcal{F} \times T_x\mathcal{F}$$

once one restricts to the fiber over x: an element of the right side is a pair (ξ_1, ξ_2) of lifts of x, and the isomorphism produces a unique $\zeta \in \mathcal{F}(B)$ restricting to ξ_i under $\mathcal{F}(p_i)$. The addition homomorphism

$$\sigma\colon B \to k[\varepsilon], \qquad \varepsilon_1, \varepsilon_2 \mapsto \varepsilon$$

is well defined because it carries the defining relations $(\varepsilon_1, \varepsilon_2)^2 = 0$ to $\varepsilon^2 = 0$, which holds in $k[\varepsilon]$, and it is a k-algebra map reducing to the identity modulo ε. Set $\xi_1 + \xi_2 := \mathcal{F}(\sigma)(\zeta)$. This lands in $T_x\mathcal{F}$ since σ covers $\varepsilon \mapsto 0$, and it is independent of choices because ζ was unique.

Step 2: Scalar multiplication. For $\lambda \in k$ the map $m_\lambda\colon k[\varepsilon] \to k[\varepsilon]$, $\varepsilon \mapsto \lambda\varepsilon$, is a k-algebra homomorphism reducing to the identity modulo ε. Define $\lambda \cdot \xi := \mathcal{F}(m_\lambda)(\xi)$ for $\xi \in T_x\mathcal{F}$. Since m_λ covers $\varepsilon \mapsto 0$, the result again lies in $T_x\mathcal{F}$.

Step 3: The axioms. The vector-space axioms are inherited from identities among the structure maps. Commutativity and associativity of $+$ follow from the symmetry of B under $\varepsilon_1 \leftrightarrow \varepsilon_2$ and from the associativity of the triple analogue $k[\varepsilon_1, \varepsilon_2, \varepsilon_3]/(\varepsilon_1, \varepsilon_2, \varepsilon_3)^2$, which is the iterated fiber product $(k[\varepsilon] \times_k k[\varepsilon]) \times_k k[\varepsilon]$ and so is respected by $\mathcal{F}$ through two applications of the fiber-product hypothesis. The zero element is the constant lift 0_x, the image of x under $k \hookrightarrow k[\varepsilon]$: composing σ with the inclusion of one factor recovers the other summand, so $\xi + 0_x = \xi$. The identity $m_\lambda \circ m_\mu = m_{\lambda\mu}$ gives $\lambda(\mu\xi) = (\lambda\mu)\xi$, and $m_1 = \mathrm{id}$ gives $1 \cdot \xi = \xi$. Distributivity $\lambda(\xi_1 + \xi_2) = \lambda\xi_1 + \lambda\xi_2$ holds because m_λ commutes with σ up to the symmetric substitution $\varepsilon_i \mapsto \lambda\varepsilon_i$, and $(\lambda+\mu)\xi = \lambda\xi + \mu\xi$ follows by applying $\mathcal{F}$ to the map $B \to k[\varepsilon]$, $\varepsilon_1 \mapsto \lambda\varepsilon$, $\varepsilon_2 \mapsto \mu\varepsilon$, and comparing with σ. All verifications are functorial images of ring-level identities, so no further choices intervene, completing the proof.

The hypothesis of proposition 1.5.2 is the functorial shadow of a single geometric fact: two tangent vectors at a point can be added because a pair of tangent directions is the same datum as one map from the "doubled" dual numbers B. Every representable functor satisfies the hypothesis, since h_X turns the fiber-product presentation of B into a fiber product of Hom-sets; the deformation functors studied later satisfy it too, which is why their tangent spaces are vector spaces even when the functors are not representable. This is the sense in which the tangent space survives the failure of representability, and it is the reason first-order deformation theory can proceed by linear algebra.

Example 1.5.3: The Tangent Space to Projective Space at a Point

Take $\mathcal{F} = h_{\mathbb{P}^n}$, the functor represented by projective n-space over k, described in example 1.1.4 as the moduli of rank-one locally free quotients of $\mathcal{O}^{n+1}$. Fix a k-point $x = [a_0 : \cdots : a_n] \in \mathbb{P}^n(k)$, and after reordering assume $a_0 \neq 0$, so x lies in the standard chart $U_0 \cong \mathbb{A}^n$ with affine coordinates $t_i = x_i/x_0$ for $i = 1, \ldots, n$; in these coordinates x is the point $(a_1/a_0, \ldots, a_n/a_0)$, which after rescaling may be written $(c_1, \ldots, c_n)$. A $k[\varepsilon]$-point of $\mathbb{P}^n$ lifting x is a quotient $\mathcal{O}_{k[\varepsilon]}^{n+1} \twoheadrightarrow L$ with L locally free of rank one, reducing modulo ε to the given quotient at x. Because x lies in U_0, such a lift is the same as an A-point of $\mathbb{A}^n$ over $A = k[\varepsilon]$ reducing to x, that is, an n-tuple

$$\left(c_i + b_i\varepsilon\right)_{i=1}^{n}, \qquad b_i \in k$$

The reduction π discards the $b_i\varepsilon$ terms, sending this tuple to $(c_1, \ldots, c_n) = x$, so the fiber $T_x\mathbb{P}^n$ is the set of tuples $(b_1, \ldots, b_n) \in k^n$. Under the addition of proposition 1.5.2 the doubled dual numbers send $\varepsilon_1, \varepsilon_2 \mapsto \varepsilon$, which adds the two tuples coordinatewise, and m_λ scales them; thus $T_x\mathbb{P}^n \cong k^n$ as a k-vector space. This matches the classical Zariski tangent space $T_x\mathbb{P}^n$ of dimension n, computed here without ever writing down $\mathfrak{m}_x/\mathfrak{m}_x^2$.

The tangent space is a pointwise probe. To decide representability one needs a global constraint, and the sharpest one available at this stage is the sheaf condition. Box 1.1.6 flagged, informally, that a representable functor is a Zariski sheaf and that objects with automorphisms break this property. That warning is now upgraded to a proposition with a proof, and strengthened: representability forces the sheaf condition not only for the Zariski topology but for the far finer fppf topology, in which covers may be flat surjective rather than open.

Recall the sheaf condition in functorial form. A functor $\mathcal{F} \colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ is a *sheaf* for a class of covers if, for every cover $\{U_i \to S\}$ in the class, the sequence

$$\mathcal{F}(S) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \times_S U_j)$$

is an equalizer: an element of $\prod_i \mathcal{F}(U_i)$ that agrees on all overlaps $U_i \times_S U_j$ comes from a unique element of $\mathcal{F}(S)$. For the Zariski topology the $U_i \to S$ are open immersions covering S; for the fppf topology they are morphisms, flat and of finite presentation, jointly surjective.

Proposition 1.5.4: Representable Functors are fppf Sheaves

For every scheme X, the functor of points $h_X = \mathrm{Hom}(-, X)$ is a sheaf for the fppf topology on **Sch**, and in particular for the Zariski topology. Consequently, any functor $\mathcal{F}$ that is not a Zariski sheaf is not representable.

***Proof*:**

The Zariski case is elementary and comes first; the fppf case is the content of faithfully flat descent for morphisms, which is imported from the descent chapter.

Step 1: The Zariski sheaf condition. Let $\{U_i \to S\}$ be a Zariski open cover, so the $U_i \subseteq S$ are open subschemes with $\bigcup_i U_i = S$ and $U_i \times_S U_j = U_i \cap U_j$. Given morphisms $f_i \colon U_i \to X$ agreeing on overlaps, $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$, a morphism of schemes is a continuous map of underlying spaces together with a map of structure sheaves, and both data are local on the source. The set-map $|S| \to |X|$ is defined by $s \mapsto f_i(s)$ for any i with $s \in U_i$; agreement on overlaps makes this independent of i, hence well defined and continuous since it is continuous on each open U_i. The sheaf map $\mathcal{O}_X \to f_*\mathcal{O}_S$ is assembled from the $\mathcal{O}_X \to (f_i)_*\mathcal{O}_{U_i}$ by the sheaf axiom on X, again using overlap agreement. The resulting $f\colon S \to X$ restricts to f_i on each U_i, and it is the unique such morphism because a morphism is determined by its restrictions to an open cover. This is exactly the equalizer condition.

Step 2: The fppf sheaf condition. Let $\{U_i \to S\}$ be an fppf cover. Refining, it suffices to treat a single faithfully flat morphism of finite presentation $u\colon U \to S$, since a disjoint union $\coprod_i U_i \to S$ of the covering maps is again faithfully flat and of finite presentation, and the equalizer over the family coincides with the equalizer for $\coprod_i U_i \to S$. The claim is then that

$$\mathrm{Hom}(S, X) \longrightarrow \mathrm{Hom}(U, X) \rightrightarrows \mathrm{Hom}(U \times_S U, X)$$

is an equalizer. A morphism $g\colon U \to X$ equalizing the two maps $U \times_S U \rightrightarrows U$ is a morphism satisfying $g \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$ on $U \times_S U$, that is, a descent datum for the morphism g along the fppf cover u. Faithfully flat descent for morphisms to a scheme asserts that such a g descends uniquely to a morphism $S \to X$: the map on topological spaces descends because u is submersive (a quotient map, being faithfully flat and quasi-compact), and the map on structure sheaves descends because faithfully flat descent is effective for quasi-coherent modules, applied to $\mathcal{O}_X$-algebra structures. Faithfully flat descent for morphisms is the subject of a later chapter, where it is proved in full; invoking it here rather than reproving it avoids circularity, since the descent chapter presupposes the functor-of-points language established in this one. Granting that theorem, the displayed sequence is an equalizer.

Step 3: The contrapositive. If $\mathcal{F} \cong h_X$ for some scheme X, then $\mathcal{F}$ inherits the Zariski sheaf property of h_X by transport of the natural isomorphism. Hence a functor failing the Zariski sheaf condition cannot be isomorphic to any h_X, so it is not representable, as we sought to show.

The proposition is the precise form of the obstruction that recurs throughout the chapter. Box 1.1.6 phrased it as a slogan; proposition 1.5.4 makes it a decision procedure. To show a functor is not representable, exhibit a Zariski cover $\{U_i \to S\}$ and local data $\{f_i\}$ that agree on overlaps but glue to *no* element of $\mathcal{F}(S)$, or glue to more than one. Objects with automorphisms produce exactly this failure: an automorphism lets one twist local families along overlaps so that they agree as isomorphism classes yet cannot be reconciled into a single global family. The j-line seen in theorem 1.3.6 is the prototype, and the worked example below isolates the mechanism in its barest form. The fppf strengthening matters for a reason that becomes central once stacks appear: moduli functors typically *are* sheaves for the fppf or étale topology even when they fail to be representable by a scheme, so the sheaf condition alone does not detect the failure, a point that returns when algebraic stacks are introduced later in the book. What separates a representable functor from a mere fppf sheaf is the second half of the toolkit, the local representability that the next result formalizes.

A scheme is not built all at once; it is glued from affine pieces along open subschemes. The functor of points should be constructible the same way, and it is: a functor that looks representable on an open cover and satisfies the Zariski sheaf condition is globally representable. This is the local-to-global criterion promised at the start, and it is the mechanism by which nearly every moduli space in the book is eventually constructed, the Grassmannian of theorem 1.2.4 among them. The statement requires the notion of an open subfunctor, the functorial counterpart of an open subscheme.

Definition 1.5.5: Open Subfunctor

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a functor. A subfunctor $\mathcal{G} \subseteq \mathcal{F}$ (a functor with $\mathcal{G}(S) \subseteq \mathcal{F}(S)$ for all S, compatibly with restriction) is an *open subfunctor* if for every scheme T and every morphism $T \to \mathcal{F}$ (equivalently, by Yoneda, every $t \in \mathcal{F}(T)$), the fiber product $\mathcal{G} \times_{\mathcal{F}} T$ is representable by an open subscheme of T. A family of open subfunctors $\{\mathcal{G}_i \subseteq \mathcal{F}\}$ is an *open cover* if for every field K over k and every $x \in \mathcal{F}(K)$, there is some i with $x \in \mathcal{G}_i(K)$; equivalently, for every $t \in \mathcal{F}(T)$ the open subschemes $\mathcal{G}_i \times_{\mathcal{F}} T \subseteq T$ cover T.

The definition says that pulling an open subfunctor back along any T-point of $\mathcal{F}$ produces an open subscheme of T, and that these opens cover T as t ranges over the point. When $\mathcal{F} = h_X$ is representable, its open subfunctors are exactly the h_U for $U \subseteq X$ open, and an open cover of $\mathcal{F}$ is a Zariski open cover of X; the definition is the transcription of that situation into functorial language, made so that it can be applied before X is known to exist. With it, the criterion reads as follows.

Theorem 1.5.6: Zariski-Local Criterion for Representability

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a functor satisfying:

1. $\mathcal{F}$ is a sheaf for the Zariski topology; and
2. $\mathcal{F}$ admits an open cover $\{\mathcal{G}_i \subseteq \mathcal{F}\}_{i \in I}$ by open subfunctors, each of which is representable, say $\mathcal{G}_i \cong h_{X_i}$ for a scheme X_i.

Then $\mathcal{F}$ is representable: there is a scheme X with $\mathcal{F} \cong h_X$, obtained by gluing the X_i along the open subschemes cut out by their pairwise intersections $\mathcal{G}_i \cap \mathcal{G}_j$.

***Proof*:**
The proof reconstructs a scheme from the charts X_i and identifies its functor of points with $\mathcal{F}$. The two hypotheses do exactly complementary work: the open cover supplies the charts and their overlaps, and the sheaf condition supplies the unique global sections that make the gluing represent $\mathcal{F}$.

Step 1: The charts and their overlaps. For each i, fix an isomorphism $\mathcal{G}_i \cong h_{X_i}$; by Yoneda the identity of X_i is a distinguished element $u_i \in \mathcal{G}_i(X_i) \subseteq \mathcal{F}(X_i)$, the universal element of the chart. For a pair (i, j) consider the open subfunctor $\mathcal{G}_j \cap \mathcal{G}_i = \mathcal{G}_j \times_{\mathcal{F}} \mathcal{G}_i$. Pulling $\mathcal{G}_j$ back along the X_i-point $u_i \in \mathcal{F}(X_i)$ gives, by the open-subfunctor hypothesis applied to $\mathcal{G}_j$ and $T = X_i$, an open subscheme

$$X_{ij} \;:=\; \mathcal{G}_j \times_{\mathcal{F}} X_i \;\subseteq\; X_i$$

Concretely X_{ij} is the largest open of X_i over which the universal element u_i lands in $\mathcal{G}_j$. Symmetrically $X_{ji} \subseteq X_j$ is defined.

Step 2: The transition isomorphisms. Both $X_{ij} \subseteq X_i$ and $X_{ji} \subseteq X_j$ represent the same functor, namely $\mathcal{G}_i \cap \mathcal{G}_j$: indeed $h_{X_{ij}} = \mathcal{G}_j \times_{\mathcal{F}} \mathcal{G}_i = \mathcal{G}_i \cap \mathcal{G}_j$ by construction, and likewise $h_{X_{ji}} = \mathcal{G}_i \cap \mathcal{G}_j$. Yoneda turns the resulting isomorphism of functors into a unique isomorphism of schemes

$$\varphi_{ji}\colon X_{ij} \xrightarrow{\;\sim\;} X_{ji}$$

Because it is induced by the identity on the common functor $\mathcal{G}_i \cap \mathcal{G}_j$, one has $\varphi_{ii} = \mathrm{id}$ and $\varphi_{ij} = \varphi_{ji}^{-1}$.

Step 3: The cocycle condition. Gluing schemes requires the transition maps to satisfy the cocycle identity on triple overlaps. Restricting to the open subscheme $X_{ijk} \subseteq X_i$ representing $\mathcal{G}_i \cap \mathcal{G}_j \cap \mathcal{G}_k$, the isomorphisms φ_{ji}, φ_{kj}, φ_{ki} are all induced by the identity on the triple-intersection functor $\mathcal{G}_i \cap \mathcal{G}_j \cap \mathcal{G}_k$. Hence

$$\varphi_{ki} = \varphi_{kj} \circ \varphi_{ji} \quad \text{on} \quad X_{ijk}$$

since both sides are the unique isomorphism induced by that identity. The cocycle condition holds, so the gluing datum $\big(X_i, X_{ij}, \varphi_{ji}\big)$ is effective: there is a scheme X with open subschemes $\iota_i \colon X_i \hookrightarrow X$ covering X, such that $\iota_i(X_{ij}) = \iota_j(X_{ji})$ and $\iota_j \circ \varphi_{ji} = \iota_i$ on X_{ij}.

Step 4: A natural transformation $\mathcal{F} \to h_X$. The universal elements $u_i \in \mathcal{F}(X_i)$ are compatible on overlaps: on X_{ij} the element u_i restricts into $\mathcal{G}_j$ (that is what X_{ij} was), and under φ_{ji} it is carried to the restriction of u_j, because both are the universal element of $\mathcal{G}_i \cap \mathcal{G}_j$. Transporting along the ι_i, the sections $u_i \in \mathcal{F}(X_i)$ agree on the overlaps $\iota_i(X_{ij}) = \iota_j(X_{ji})$ inside X. Since $\mathcal{F}$ is a Zariski sheaf and $\{X_i \to X\}$ is a Zariski open cover of X, there is a *unique* $u \in \mathcal{F}(X)$ restricting to u_i on each X_i. By Yoneda, u is a natural transformation $h_X \to \mathcal{F}$, equivalently a morphism $\Phi \colon h_X \to \mathcal{F}$ of functors.

Step 5: Φ is an isomorphism. It suffices to check that $\Phi_T \colon h_X(T) \to \mathcal{F}(T)$ is a bijection for every scheme T; both sides are Zariski sheaves in T (the source by proposition 1.5.4, the target by hypothesis (1)), so bijectivity may be checked locally on T. Given $t \in \mathcal{F}(T)$, hypothesis (2) gives an open cover $\{T_i\}$ of T with $t|_{T_i} \in \mathcal{G}_i(T_i) \cong h_{X_i}(T_i)$; thus $t|_{T_i}$ corresponds to a morphism $g_i \colon T_i \to X_i \xrightarrow{\iota_i} X$, and by construction $\Phi(g_i) = t|_{T_i}$. On the overlap $T_i \cap T_j$ the two morphisms g_i, g_j land in $\iota_i(X_{ij}) = \iota_j(X_{ji})$ and are identified there by φ_{ji}, so $g_i|_{T_i \cap T_j} = g_j|_{T_i \cap T_j}$ as morphisms to X. Because h_X is a Zariski sheaf, the g_i glue to a unique $g \colon T \to X$ with $\Phi(g) = t$; this is surjectivity, with uniqueness of g giving injectivity. Hence Φ_T is a bijection for all T, so $\mathcal{F} \cong h_X$, completing the proof.

The theorem is the workhorse of the entire representability enterprise. It reduces the global question "is $\mathcal{F}$ representable?" to two local ones: does $\mathcal{F}$ glue in the Zariski topology, and can one cover it by pieces already known to be schemes? Both halves are indispensable. Dropping the sheaf condition, one could cover a functor by representable opens whose local charts refuse to glue, exactly the pathology that makes non-separated or non-representable objects; dropping local representability, a Zariski sheaf carries no charts to glue. The Grassmannian is representable precisely because its standard affine charts (example 1.2.8) are representable open subfunctors and the quotient-bundle functor is a Zariski sheaf, and theorem 1.2.4 is in essence this criterion applied to those charts. The same template constructs the Hilbert and Quot schemes in the next chapter, once suitable representable charts are produced. The criterion also clarifies where representability can fail: not in the gluing of charts, which is automatic once they exist, but in the sheaf condition, which is where automorphisms strike.

Proposition 1.5.4 converts non-representability into a concrete search: find a cover and local data that violate the sheaf condition. The following example carries out that search in the simplest nontrivial setting, the naive functor of line bundles up to isomorphism. Its failure is the same one that stops the j-line from being a fine moduli space in theorem 1.3.6, stripped of the geometry so that the mechanism is visible in a single cohomology group.

Example 1.5.7: The Naive Picard Functor is not a Zariski Sheaf

Work over k and consider the naive Picard functor

$$\mathcal{F} \colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}, \qquad \mathcal{F}(S) = \mathrm{Pic}\,(S) = \big\{\text{line bundles } \mathcal{L} \text{ on } S\big\}/\cong$$

the set of isomorphism classes of invertible sheaves on S, with pullback along $f\colon T \to S$ given by f^*. This is a moduli functor in the sense of definition 1.1.1: its objects are line bundles and its morphisms are pullbacks. The claim is that $\mathcal{F}$ is *not* representable, and the obstruction is the passage to isomorphism classes, which forgets the automorphisms $\mathcal{L} \xrightarrow{c} \mathcal{L}$ $(c \in k^\times)$ that every line bundle carries.

To see the sheaf condition fail, take $S = \mathbb{P}^1$ and the standard Zariski cover $S = U_0 \cup U_1$ by the two affine charts $U_0 = \operatorname{Spec} k[t]$, $U_1 = \operatorname{Spec} k[u]$ with $u = t^{-1}$, overlap $U_0 \cap U_1 = \operatorname{Spec} k[t, t^{-1}]$. On each chart every line bundle is trivial up to isomorphism, since $\operatorname{Pic}(\mathbb{A}^1) = 0$; and $\operatorname{Pic}(U_0 \cap U_1) = \operatorname{Pic}(\mathbb{G}_m) = 0$ as well. Hence

$$\mathcal{F}(U_0) = \mathcal{F}(U_1) = \mathcal{F}(U_0 \cap U_1) = \{\,*\,\}$$

are all singletons, so the fiber product $\mathcal{F}(U_0) \times_{\mathcal{F}(U_0 \cap U_1)} \mathcal{F}(U_1)$ is a single point: the trivial bundle on each chart is the unique local datum, and it is automatically compatible on the overlap.

The global set is not a point. Every line bundle on $\mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(n)$ for a unique $n \in \mathbb{Z}$, so $\mathcal{F}(\mathbb{P}^1) = \operatorname{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$, and each of these restricts to the trivial bundle on both charts. The restriction map

$$\mathcal{F}(\mathbb{P}^1) \longrightarrow \mathcal{F}(U_0) \times_{\mathcal{F}(U_0 \cap U_1)} \mathcal{F}(U_1)$$

is therefore $\mathbb{Z} \to \{*\}$, which is not injective. The equalizer condition fails in its *uniqueness* clause: the unique compatible local datum lifts to infinitely many global classes, one for each n, rather than to a single one. Concretely, gluing the two trivial charts by the transition unit $t^n \in k[t, t^{-1}]^\times$ produces $\mathcal{O}_{\mathbb{P}^1}(n)$, and all these gluings have identical restriction to the charts yet are pairwise non-isomorphic globally. Hence $\mathcal{F}$ is not a Zariski sheaf, and by proposition 1.5.4 it is not representable.

The mechanism is the one anticipated by box 1.1.6. Passing to isomorphism classes erases the transition data t^n, which is precisely the automorphism-valued gluing information $\mathcal{O}(U_0 \cap U_1)^\times$ that distinguishes the global bundles; the sheaf condition then cannot recover n from local data that all look trivial. This is the same failure that prevented the j-line from being a fine moduli space in theorem 1.3.6, where isotrivial families with nonconstant transition automorphisms restrict to identical local objects yet are globally distinct. The repair, developed in the stacks Part of the book, is to remember the automorphisms rather than quotient them away, replacing the set-valued functor $\mathcal{F}$ by a groupoid-valued one, that is, a stack. The correct Picard functor, defined as a sheafification and made representable by adding rigidifying data, is itself constructed later; the naive version here fails at the first step, the sheaf condition.

The example isolates the difference between a coarse and a fine moduli space in a single computation. The set $\mathcal{F}(S)$ discards the automorphisms of its objects by passing to isomorphism classes, and that quotient is exactly what destroys the sheaf condition: gluing data differing by transition automorphisms become indistinguishable when restricted to the charts, yet build different global objects. No scheme's functor of points has this defect, by proposition 1.5.4, which is why $\mathcal{F}$ cannot be representable no matter how the geometry is arranged. The tangent-space tool and the gluing criterion together frame the two directions the book now takes. The deformation theory to come linearizes the moduli problem at a point, computing the tangent space of definition 1.5.1 as a cohomology group and asking when deformations are unobstructed; the theory of algebraic stacks that follows it enlarges the target category so that functors like the one above become representable objects, algebraic stacks, whose functor of points is a groupoid keeping track of the automorphisms this example was unable to forget.

Exercise 1.5.1

1. Let $\mathcal{F} = h_{\mathbb{A}^1}$ be the functor represented by the affine line over k, so $\mathcal{F}(A) = A$ for a k-algebra A. Compute $\mathcal{F}(k[\varepsilon])$, the reduction map $\pi \colon \mathcal{F}(k[\varepsilon]) \to \mathcal{F}(k)$, and the tangent space $T_x\mathcal{F}$ at a point $x \in \mathbb{A}^1(k) = k$. Verify directly, using the doubled dual numbers $k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$, that the addition of proposition 1.5.2 recovers ordinary addition on k, so $T_x\mathcal{F} \cong k$.

2. Let $\mathcal{F} = h_G$ for an affine algebraic group G over k, and take $x = e$ the identity. Show that $T_e\mathcal{F}$ with the vector-space structure of proposition 1.5.2 is the Lie algebra $\mathrm{Lie}(G)$ as a k-vector space. (Hint: a $k[\varepsilon]$-point reducing to e is an element of $G(k[\varepsilon])$ mapping to e under $\varepsilon \mapsto 0$; identify these with left-invariant derivations of $\mathcal{O}_{G,e}$.)

3. Prove that a filtered colimit of representable functors need not be representable, but that every open subfunctor of a representable functor h_X is of the form h_U for a unique open subscheme $U \subseteq X$. Deduce that in theorem 1.5.6 the charts X_i may be taken to be affine without loss of generality.

4. Fix a scheme X over k and define the *relative Picard functor* $\mathrm{Pic}_{X/k} \colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ by $\mathrm{Pic}_{X/k}(A) = \ker\big(\mathrm{Pic}\,(X \times_k \mathrm{Spec}\,A) \to \mathrm{Pic}\,(X)\big)$, the line bundles on $X_A := X \times_k \mathrm{Spec}\,A$ trivial along the reduction $X \hookrightarrow X_A$, up to isomorphism (this is a k-pointed functor: $\mathrm{Pic}_{X/k}(k) = \{[\mathcal{O}_X]\}$ is a single point, the class of the trivial bundle). Compute its tangent space $T_{[\mathcal{O}_X]}\mathrm{Pic}_{X/k}$ at that point using definition 1.5.1. Show that a $k[\varepsilon]$-point of $\mathrm{Pic}_{X/k}$ is a line bundle on $X[\varepsilon] = X \times_k \mathrm{Spec}\,k[\varepsilon]$ restricting to $\mathcal{O}_X$, and identify the resulting set with $H^1(X, \mathcal{O}_X)$. (Hint: line bundles are classified by H^1 of the units sheaf, and $1 + \varepsilon\mathcal{O}_X$ is a subgroup of $\mathcal{O}_{X[\varepsilon]}^\times$ isomorphic to the additive group $\mathcal{O}_X$.) Conclude that even though the naive functor $\mathcal{F} = \mathrm{Pic}$ of example 1.5.7 is not representable, the associated relative functor has a well-defined tangent space, a k-vector space, illustrating proposition 1.5.2.

5. Let $\mathcal{F}$ be a Zariski sheaf admitting a cover by two representable open subfunctors $\mathcal{G}_0 \cong h_{X_0}$, $\mathcal{G}_1 \cong h_{X_1}$ with $\mathcal{G}_0 \cap \mathcal{G}_1 \cong h_{X_{01}}$. Write out the scheme X produced by theorem 1.5.6 explicitly when $X_0 = X_1 = \mathbb{A}^1$ and $X_{01} = \mathbb{A}^1 \setminus \{0\} = \mathrm{Spec}\,k[t, t^{-1}]$, for the two gluings $t \mapsto t$ and $t \mapsto t^{-1}$. Identify the two resulting schemes and explain, via the sheaf condition, why one of them fails to be separated.

6. Let $n \geq 2$ be invertible in k and consider the presheaf $\mathcal{F}(S) = \mathcal{O}(S)^\times / (\mathcal{O}(S)^\times)^n$, global units modulo n-th powers. Show that $\mathcal{F}$ fails the fppf sheaf condition already over the single base $S = \mathrm{Spec}\,k$: for a suitable k, exhibit a nontrivial fppf cover $S' \to S$ on which a class becomes trivial though it is nonzero over S. Explain why the Zariski sheaf condition over $\mathrm{Spec}\,k$ imposes no such constraint (a point has no nontrivial Zariski cover), so $\mathcal{F}$ passes the Zariski test at this base while failing the fppf test. Conclude that the fppf half of proposition 1.5.4 is stronger than its Zariski half and rests on descent rather than on open gluing.

2

The Hilbert and Quot Schemes

The previous chapter reduced a moduli problem to a functor and asked when that functor is representable; the Grassmannian was the model case in which the answer is yes. This chapter produces the two representable functors from which most moduli spaces in algebraic geometry are cut out: the Quot scheme, parametrizing quotients of a fixed coherent sheaf, and its special case the Hilbert scheme, parametrizing closed subschemes of a fixed projective scheme with a fixed Hilbert polynomial. Grothendieck's existence theorem, that these functors are representable by projective schemes, is the foundation on which Hilbert schemes, moduli of sheaves, and the quotient constructions of the next chapter all rest.

The engine is flatness, the algebraic form of a family varying continuously, and the Hilbert polynomial is the discrete invariant it holds constant. section 2.1 sets up the Quot and Hilbert functors on this basis; section 2.2 controls which sheaves can occur, through Castelnuovo–Mumford regularity, boundedness, and the flattening stratification; and section 2.3 assembles these into the representability theorem, realizing Quot and Hilb as closed subschemes of a Grassmannian. section 2.4 reads off the local structure, identifying the tangent space to the Hilbert scheme with the global sections of the normal sheaf; this is the first computation of deformation theory and the bridge to the next Part. section 2.5 closes with worked schemes: the Hilbert scheme of points, hypersurfaces, and the Grassmannian recovered as a Quot scheme.

2.1 Flat Families and Hilbert Polynomials

Before any Hilbert or Quot scheme can be built, the moduli problem it represents has to be pinned down, and that requires answering a question the first chapter deferred: what, precisely, is a family of subschemes, or of quotient sheaves? A family should be a collection of objects varying *continuously* over a base S, but “continuously” has to be given algebraic content, since over a scheme there is no metric to appeal to. The content is flatness. A family that is flat over S is one whose members do not jump in a way that changes their discrete invariants as the parameter moves, and the discrete invariant that flatness holds fixed is the Hilbert polynomial.

This section fixes the two notions the whole chapter runs on. It recalls the Hilbert polynomial of a coherent sheaf on projective space, proves that in a flat family the fiberwise Hilbert polynomial is locally constant on the base, and then uses that fact to define the Quot and Hilbert functors, the moduli functors whose representability is the theorem of section 2.3. Flatness is the reason those functors are sheaves in the first place, and the fixed Hilbert polynomial P is the label that carves the boundless functor of all quotients into the bounded, representable pieces $\underline{\mathrm{Quot}}^P$ that individual moduli spaces attach to. The chapter's guiding image, a flat family versus a non-flat one in which the Hilbert polynomial jumps, appears at the end of the section and makes the flatness condition tangible.

The first of the two notions is the Hilbert polynomial, a single element of $\mathbb{Q}[t]$ that packages the numerical invariants of a coherent sheaf on projective space. Fix once and for all a projective scheme $Y \subseteq \mathbb{P}^n_k$ over the base field k, with $\mathcal{O}_Y(1)$ the restriction of the twisting sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$. For a coherent sheaf $\mathcal{F}$ on Y and an integer m, write $\mathcal{F}(m) = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(m)$ for the m-th Serre twist. The one piece of upstream machinery needed here is that the cohomology of a coherent sheaf on a projective scheme is finite dimensional over k and vanishes in high degree after twisting; this is Serre's finiteness and vanishing theorem from [12], which the reader may take as given.

The Euler characteristic packages the cohomology of $\mathcal{F}$ into a single integer.

Definition 2.1.1: Hilbert Polynomial of a Coherent Sheaf

Let $Y \subseteq \mathbb{P}^n_k$ be projective and $\mathcal{F}$ a coherent sheaf on Y. Recall the *Euler characteristic*

$$\chi(\mathcal{F}) = \sum_{i \geq 0} (-1)^i \dim_k H^i(Y, \mathcal{F})$$

a finite sum, since $H^i(Y, \mathcal{F}) = 0$ for $i > \dim Y$ and each term is finite dimensional ([12, Euler Characteristics and Hilbert Polynomials]). The *Hilbert polynomial* of $\mathcal{F}$ is the function of the twisting parameter m given by

$$P_{\mathcal{F}}(m) = \chi\big(\mathcal{F}(m)\big)$$

which agrees with a polynomial in m of degree $\dim \operatorname{Supp} \mathcal{F}$ ([12, Hilbert Polynomial]); for $\mathcal{F} = \mathcal{O}_Y$ it recovers the classical Hilbert polynomial of the homogeneous coordinate ring of Y ([12, Hilbert Polynomial and Hilbert Function]).

The name "polynomial" is justified by a theorem of Snapper and Grothendieck: $P_{\mathcal{F}}$ agrees, for all integers m, with a polynomial in m with rational coefficients, of degree equal to $\dim \operatorname{Supp} \mathcal{F}$. This is proved in full in [12]; the argument is an induction on the dimension of the support using a hyperplane section, and it is the reason the Hilbert polynomial is an object of $\mathbb{Q}[t]$ rather than only a numerical function. Two features make $P_{\mathcal{F}}$ the right invariant for moduli. First, it is computed from cohomology, so it is a deformation invariant rather than a set-theoretic count. Second, for $m \gg 0$ the higher cohomology $H^i(Y, \mathcal{F}(m))$ vanishes for $i > 0$, so $P_{\mathcal{F}}(m) = \dim_k H^0(Y, \mathcal{F}(m))$ counts the twisted global sections, and this ties the polynomial to the classical Hilbert function of a graded module. The leading coefficients encode the geometry: for a subscheme the degree of P is the dimension, and the top coefficient is $(\deg Z)/(\dim Z)!$, so it records the degree of the subscheme in $\mathbb{P}^n$ up to the factorial normalization (the two coincide outright when $\dim Z = 1$, as in the plane-curve example below).

Example 2.1.2: The Hilbert Polynomial of a Plane Curve

Let $C \subseteq \mathbb{P}^2_k$ be a curve of degree d, the zero scheme of a single homogeneous polynomial of degree d, and let $\mathcal{O}_C$ be its structure sheaf. The Hilbert polynomial of $\mathcal{O}_C$ is computed from the ideal-sheaf sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

the ideal sheaf $\mathcal{I}_C = \mathcal{O}_{\mathbb{P}^2}(-d)$ being generated by the degree-d equation. Twisting this exact sequence by $\mathcal{O}(m)$ gives $0 \to \mathcal{O}_{\mathbb{P}^2}(m-d) \to \mathcal{O}_{\mathbb{P}^2}(m) \to \mathcal{O}_C(m) \to 0$, and Euler characteristic is additive on short exact sequences, so

$$\chi(\mathcal{O}_C(m)) = \chi(\mathcal{O}_{\mathbb{P}^2}(m)) - \chi(\mathcal{O}_{\mathbb{P}^2}(m-d))$$

Using $\chi(\mathcal{O}_{\mathbb{P}^2}(m)) = \binom{m+2}{2}$, this gives

$$P_{\mathcal{O}_C}(m) = \binom{m+2}{2} - \binom{m-d+2}{2} = d\,m + \frac{-d^2+3d}{2} = d\,m + 1 - \frac{(d-1)(d-2)}{2}$$

The polynomial has degree 1, recording that C is a curve; its leading coefficient d is the degree of C; and its constant term $1-g$ with $g = \binom{d-1}{2}$ is 1 minus the arithmetic genus, the genus-degree formula read off the Hilbert polynomial. For a conic $d=2$ this reads $P(m) = 2m+1$, and for a cubic $d=3$ it reads $P(m) = 3m$, so that $\chi(\mathcal{O}_C) = P(0)$ equals 1 and 0 respectively, matching $g=0$ and $g=1$.

The example already shows the polynomial doing its job as a bookkeeper: a single element of $\mathbb{Q}[t]$ simultaneously records dimension, degree, and genus. In a moduli problem these are exactly the discrete data one wants held fixed while the geometry deforms, and the next result is what guarantees they *are* held fixed.

With the Hilbert polynomial in hand, the second notion is flatness, and the result binding the two together is that flatness holds the Hilbert polynomial constant along a family. A family of subschemes or sheaves parametrized by S is a single sheaf on $Y \times_k S$, whose restriction to the fiber $Y_s = Y \times_k \kappa(s)$ over a point $s \in S$ is the member of the family sitting over s. For the family to deserve the name, the members should vary continuously, and the algebraic form of continuity is that the sheaf on the total space $Y \times S$ be *flat* over S. Flatness is developed in full in [12]; the following refresher records the one statement this chapter leans on.

Definition 2.1.3: Flat Family over a Base

Let Y be a scheme over k, S a k-scheme, and $p\colon Y \times_k S \to S$ the projection. A coherent sheaf $\mathcal{G}$ on $Y \times_k S$ is a *flat family over* S if $\mathcal{G}$ is flat over S, meaning that for every point of $Y \times_k S$ the stalk $\mathcal{G}_x$ is a flat module over the local ring $\mathcal{O}_{S,p(x)}$. For $s \in S$ the *fiber* of the family is the coherent sheaf

$$\mathcal{G}_s = \mathcal{G} \otimes_{\mathcal{O}_S} \kappa(s)$$

on $Y_s = Y \times_k \kappa(s)$, the restriction of $\mathcal{G}$ to the fiber over s.

Flatness is not visible on any single fiber; it is a condition on how the fibers fit together as s moves. The archetype to keep in mind is that a family of divisors on $\mathbb{P}^1$ is flat exactly when the number of points, counted with multiplicity, is constant, whereas allowing a point to appear or disappear as the parameter passes through a special value destroys flatness. This is the discrete-invariant discipline flatness enforces, and the following proposition is its precise cohomological expression: the entire Hilbert polynomial, not just one of its coefficients, is locally constant along a flat family.

The proof rests on a single import, the theorem on cohomology and base change, which controls how the cohomology of a family interacts with restriction to a fiber. In the form needed here it says: for a flat family $\mathcal{G}$ over S that is proper over S, and for each degree i, there is a coherent sheaf on S computing H^i of the fibers, and after a Serre twist the higher cohomology of the fibers vanishes uniformly, so that the zeroth cohomology becomes locally free of the constant rank one expects. The statement and proof are in [12]; it is the workhorse behind every constancy result in this chapter.

Proposition 2.1.4: Local Constancy of the Hilbert Polynomial

Let $Y \subseteq \mathbb{P}^n_k$ be projective, S a Noetherian k-scheme, and $\mathcal{G}$ a flat family over S of coherent sheaves on Y, that is, a coherent sheaf on $Y \times_k S$ flat over S with proper support over S. For $s \in S$ let $P_s \in \mathbb{Q}[t]$ be the Hilbert polynomial of the fiber $\mathcal{G}_s$ on Y_s, computed with respect to $\mathcal{O}_{Y_s}(1)$. Then the map

$$s \longmapsto P_s$$

is locally constant on S: every point of S has an open neighborhood on which P_s is a fixed polynomial. In particular, if S is connected, the fiberwise Hilbert polynomial is the same for every $s \in S$.

***Proof*:**
The claim is local on S, so it suffices to prove that the function $s \mapsto P_s(m)$ is locally constant on S for each fixed integer m, since a polynomial in $\mathbb{Q}[t]$ of bounded degree is determined by its values at enough integers and the local constancy of each value will then force local constancy of the polynomial. Fix m.

Step 1: Reduce to the vanishing of higher cohomology. Let $p\colon Y \times_k S \to S$ be the projection and $q\colon Y \times_k S \to Y$ the other one, so that $\mathcal{G}(m) := \mathcal{G} \otimes q^*\mathcal{O}_Y(m)$ is the m-twisted family, again flat over S with proper support, and its fiber over s is $\mathcal{G}_s(m)$. By definition

$$P_s(m) = \chi(\mathcal{G}_s(m)) = \sum_{i \geq 0} (-1)^i \dim_{\kappa(s)} H^i(Y_s, \mathcal{G}_s(m))$$

The strategy is to show each individual term $s \mapsto \dim_{\kappa(s)} H^i(Y_s, \mathcal{G}_s(m))$ is locally constant. This is false term by term for arbitrary m, since cohomology can jump on special fibers, but the alternating sum is stable, and the cleanest route to the alternating sum is to twist so hard that all but the zeroth term vanish and then descend.

Step 2: The Euler characteristic is locally constant. By the theorem on cohomology and base change ([12]) applied to the flat proper family $\mathcal{G}(m)$ over S, there is, locally on S, a bounded complex

$$K^\bullet\colon \quad 0 \to K^0 \to K^1 \to \cdots \to K^N \to 0$$

of finite locally free $\mathcal{O}_S$-modules whose cohomology computes the cohomology of the family universally, in the sense that for every base change $T \to S$ the cohomology of $K^\bullet \otimes_{\mathcal{O}_S} \mathcal{O}_T$ computes $H^\bullet$ of the pulled-back family. Specializing $T = \operatorname{Spec} \kappa(s)$, the complex $K^\bullet \otimes \kappa(s)$ computes $H^\bullet(Y_s, \mathcal{G}_s(m))$. Now the Euler characteristic of a bounded complex of finite-dimensional vector spaces equals the Euler characteristic of its cohomology, so

$$\sum_i (-1)^i \dim_{\kappa(s)} H^i(Y_s, \mathcal{G}_s(m)) = \sum_i (-1)^i \dim_{\kappa(s)} \left(K^i \otimes \kappa(s)\right)$$

Each K^i is locally free of some rank r_i over $\mathcal{O}_S$, so $\dim_{\kappa(s)}(K^i \otimes \kappa(s)) = r_i$ is constant on the open where $K^\bullet$ is defined and K^i has that rank. Hence

$$P_s(m) = \sum_i (-1)^i r_i$$

is constant on that open neighborhood of the chosen point. This is where flatness enters: without it the base-change complex $K^\bullet$ of finite locally free modules need not exist, and the Euler characteristic could jump.

Step 3: From one twist to the polynomial. Step 2 gives local constancy of $s \mapsto P_s(m)$ for the fixed m. Running the argument for every m shows that each coefficient of the polynomial, being a fixed $\mathbb{Q}$-linear combination of finitely many values $P_s(m)$, is locally constant, so the polynomial P_s itself is locally constant on S. On a connected S a locally constant function is constant, giving the final assertion, as we sought to show.

The proposition is the technical fact that makes the Hilbert polynomial a usable moduli label. It says that flatness and the Hilbert polynomial are two sides of one coin: a flat family cannot change its members' Hilbert polynomials, and conversely the Hilbert polynomial is exactly the invariant whose constancy flatness certifies. This is why the moduli functors below are indexed by a fixed polynomial P. Over a connected base every member of a flat family carries the same P, so demanding a fixed P is the same as asking for the connected component of the space of all flat families, and it is on such a component that a moduli space has any hope of being of finite type. The converse direction, that a family with locally constant Hilbert polynomial over a reduced base is automatically flat, is a standard criterion proved in [12]; it will be used silently when verifying flatness of explicit families.

With flatness and the Hilbert polynomial in hand, the moduli problems this chapter solves can be written down. The primary one classifies quotient sheaves of a fixed coherent sheaf. Fix a coherent sheaf $\mathcal{F}$ on the projective scheme Y; think of $\mathcal{F}$ as an ambient sheaf whose quotients are to be parametrized. A family of quotients over a base S is a quotient of the pullback of $\mathcal{F}$ to $Y \times S$, and the family is admissible when it is flat over S, so that its Hilbert polynomial is constant, and when that polynomial is a prescribed P.

For a k-scheme S write $Y_S = Y \times_k S$, with projection $p_S \colon Y_S \to Y$, and $\mathcal{F}_S = p_S^*\mathcal{F}$ for the pullback of $\mathcal{F}$ to Y_S. A morphism $g \colon T \to S$ induces $\mathrm{id}_Y \times g \colon Y_T \to Y_S$, and pulling back along it carries data over Y_S to data over Y_T; this is the mechanism of contravariance.

Definition 2.1.5: The Quot Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective, $\mathcal{F}$ a coherent sheaf on Y, and $P \in \mathbb{Q}[t]$ a polynomial. The *Quot functor*

$$\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \colon \mathbf{Sch}_k^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

assigns to a k-scheme S the set of isomorphism classes of surjections of coherent sheaves on $Y_S = Y \times_k S$

$$\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$$

such that $\mathcal{Q}$ is flat over S and each fiber $\mathcal{Q}_s$ has Hilbert polynomial P on Y_s. Two quotients $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ and $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}'$ are identified when there is an isomorphism $\mathcal{Q} \cong \mathcal{Q}'$ commuting with the surjections from $\mathcal{F}_S$, that is, having the same kernel subsheaf of $\mathcal{F}_S$. To a morphism $g \colon T \to S$ the functor assigns the pullback

$$g^* \colon \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(S) \longrightarrow \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(T), \qquad \big(\mathcal{F}_S \twoheadrightarrow \mathcal{Q}\big) \longmapsto \big(\mathcal{F}_T \twoheadrightarrow (\mathrm{id}_Y \times g)^* \mathcal{Q}\big)$$

the surjection obtained by pulling $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ back along $\mathrm{id}_Y \times g$.

Two points make this a well-posed moduli functor rather than a formal gesture. First, pullback preserves the defining conditions: pulling back a surjection along $\mathrm{id}_Y \times g$ keeps it surjective, because $\otimes$ is right exact, and it preserves flatness over the base and hence the fiberwise Hilbert polynomial, since the fibers of the pulled-back family over points of T are the fibers of the original family over their images in S (proposition 2.1.4 guarantees the polynomial is the same P). So g^* lands in $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(T)$, and functoriality $(g \circ h)^* = h^* \circ g^*$ is the standard compatibility of pullbacks. Second, quotients are recorded up to the coarsest sensible equivalence: identifying $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ with its isomorphic siblings is the same as recording only the kernel subsheaf $\mathcal{K} = \ker(\mathcal{F}_S \to \mathcal{Q}) \subseteq \mathcal{F}_S$, so a $\underline{\mathrm{Quot}}$-point is equally a flat family of *subsheaves* of $\mathcal{F}$ with the complementary Hilbert polynomial. The functor thus captures both quotients and sub-objects at once, which is why it subsumes so many moduli problems.

The single most important special case is the one where $\mathcal{F}$ is the structure sheaf of Y. A surjection $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{Q}$ has for its kernel an ideal sheaf, so its quotient is the structure sheaf of a closed subscheme of Y_S; flatness over S makes this a flat family of closed subschemes. This is the moduli problem of closed subschemes with which the first chapter opened, now made precise.

Definition 2.1.6: The Hilbert Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective and $P \in \mathbb{Q}[t]$. The *Hilbert functor*

$$\underline{\mathrm{Hilb}}^P_Y : \mathbf{Sch}_k^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

is the special case $\mathcal{F} = \mathcal{O}_Y$ of the Quot functor. It assigns to a k-scheme S the set of closed subschemes $Z \subseteq Y_S = Y \times_k S$ that are flat over S and whose fibers $Z_s \subseteq Y_s$ have Hilbert polynomial P; equivalently, the set of surjections $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{O}_Z$ that are flat over S with fiberwise Hilbert polynomial P. To $g\colon T \to S$ it assigns the pullback

$$Z \longmapsto (\mathrm{id}_Y \times g)^{-1}(Z) = Z \times_S T$$

the scheme-theoretic preimage, which is the closed subscheme of Y_T cut out by the pulled-back ideal sheaf.

The Hilbert functor is the moduli of closed subschemes of Y with fixed numerical type P: dimension, degree, and, for curves, genus, all read off P as in example 2.1.2. A $\underline{\mathrm{Hilb}}^P_Y$-point over S is a subscheme $Z \subseteq Y \times S$ whose fiber over each $s \in S$ is a subscheme of Y with the prescribed invariants, varying flatly with s. When $S = \operatorname{Spec} k$ this is a single closed subscheme $Z \subseteq Y$ with Hilbert polynomial P, so the k-points of the eventual Hilbert scheme are exactly the subschemes one wants to parametrize; the content of the next sections is that these k-points assemble into an actual projective scheme, with the flat families over general S recorded as its S-points. That this functor decomposes the "all subschemes" functor by discrete type is the reason a fixed P is imposed: the functor of all closed subschemes of Y, flat over the base, is the disjoint union $\coprod_P \underline{\mathrm{Hilb}}^P_Y$ over all Hilbert polynomials, and each piece is representable while the whole is only a countable coproduct.

Example 2.1.7: The Hilbert Functor of Points on the Line

Take $Y = \mathbb{P}^1_k$ and the constant polynomial $P = d$, a positive integer. A subscheme $Z \subseteq \mathbb{P}^1$ with Hilbert polynomial the constant d is a zero-dimensional subscheme of length d: since P has degree 0, $\dim Z = 0$, and $\chi(\mathcal{O}_Z(m)) = \dim_k H^0(Z, \mathcal{O}_Z) = d$ for all m because Z is affine, so Z is d points counted with multiplicity. Over S, a $\underline{\mathrm{Hilb}}^d_{\mathbb{P}^1}$-point is a closed subscheme $Z \subseteq \mathbb{P}^1 \times S$, flat over S, whose every fiber is length d. Working in the affine chart $\mathbb{A}^1 = \operatorname{Spec} k[x]$, such a Z over an affine base $S = \operatorname{Spec} A$ is cut out by a single monic polynomial

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0 \in A[x]$$

with $a_i \in A$, since flatness of $A[x]/(f)$ over A forces the generator to be monic of degree d so that $A[x]/(f)$ is free of rank d over A. The coefficients $(a_0, \ldots, a_{d-1})$ are d free parameters, so on this chart the functor is represented by affine space $\mathbb{A}^d = \operatorname{Spec} k[a_0, \ldots, a_{d-1}]$: the Hilbert scheme of d points on the line is $\mathbb{A}^d$ over the affine chart, the space of monic degree-d polynomials, and globally $\mathrm{Hilb}^d_{\mathbb{P}^1} \cong \mathbb{P}^d$, the space of binary forms of degree d up to scalar. A fiber over S jumps from d distinct points to a fat point exactly as the discriminant of f vanishes, and flatness is what keeps the length pinned at d throughout.

The example exhibits the entire program in miniature: a moduli functor of flat families with fixed Hilbert polynomial turns out to be represented by an explicit scheme, here $\mathbb{P}^d$, whose points are the objects being classified. The general theorem of section 2.3 promotes this from $\mathbb{P}^1$ and points to arbitrary Y and arbitrary P, and the intervening section 2.2 supplies the boundedness that makes the

Grassmannian into which everything embeds finite dimensional. For the broader picture of Hilbert schemes as the engine of moduli theory, see [12, Hilbert Schemes].

The abstractions above are worth grounding in a case where flatness visibly succeeds and fails, so that the reader can see the Hilbert polynomial jump. The cleanest such case is a plane cubic acquiring an extra embedded point as a parameter degenerates.

Example 2.1.8: A Non-Flat Degeneration

Work over $S = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ and inside $Y = \mathbb{P}^2_k$ with homogeneous coordinates $[x_0 : x_1 : x_2]$; pass to the affine chart $\mathbb{A}^2 = \operatorname{Spec} k[x,y]$ with $x = x_1/x_0$, $y = x_2/x_0$ to do the algebra. Consider the family of subschemes $Z_t \subseteq \mathbb{A}^2$ defined, for $t \in \mathbb{A}^1$, by the ideal

$$I_t = (xy,\ tx)$$

of $k[x,y]$. The total space is $Z \subseteq \mathbb{A}^2 \times \mathbb{A}^1 = \operatorname{Spec} k[x,y,t]$ cut out by $I = (xy, tx) \subseteq k[x,y,t]$, and the family is the projection to $S = \operatorname{Spec} k[t]$.

The general fiber. For $t \neq 0$ the element tx is a unit multiple of x, so $I_t = (xy, x) = (x)$, and $Z_t = \{x = 0\}$ is the line $x = 0$, a copy of $\mathbb{A}^1$ with coordinate y. Compactifying in $\mathbb{P}^2$, the closure is a line, with Hilbert polynomial $P_t(m) = m + 1$: degree 1, dimension 1, and $\chi(\mathcal{O}_{Z_t}) = 1$.

The special fiber. For $t = 0$ the ideal is $I_0 = (xy)$, so $Z_0 = \{xy = 0\}$ is the union of the two coordinate lines $x = 0$ and $y = 0$, the degree-2 plane curve cut out by the single degree-2 form xy. By example 2.1.2 with $d = 2$ its Hilbert polynomial on the projective closure is $P_0(m) = 2m + 1$: degree 2, dimension 1, and $\chi(\mathcal{O}_{Z_0}) = P_0(0) = 1$, so arithmetic genus $p_a = 1 - \chi = 0$. The Hilbert polynomial has jumped from $m + 1$ to the degree-2 polynomial $2m + 1$.

By proposition 2.1.4 the family $Z \to S$ cannot be flat, since $\mathbb{A}^1$ is connected yet the fiberwise Hilbert polynomial is not constant. The failure is visible in the algebra: in $k[x,y,t]$ the element x is annihilated modulo I by both y and t, so $\operatorname{Ann}(x) = (y,t) \bmod I$. Since $I = (xy, tx) = (x) \cap (y,t)$, the associated primes of $k[x,y,t]/I$ are (x) and (y,t), and the second one, (y,t), cuts out the horizontal line $\{y = 0, t = 0\}$ whose image in S is the single point $t = 0$. A component of the total space maps to a proper closed subset of the base, which is precisely the obstruction to flatness: the fiber over $t = 0$ acquires the extra line $y = 0$ that no nearby fiber sees. Concretely, the length of $\mathcal{O}_Z$ localized along $\{t = 0\}$ is not the flat limit of the nearby fibers; a whole extra line materializes at $t = 0$, and the Euler characteristic records it.

The flat repair. The flat limit of the lines $\{x = 0\}$ as $t \to 0$ is just the line $\{x = 0\}$ again: the flat family extending $Z_t = \{x = 0\}$ for $t \neq 0$ over the whole base is the constant family $\{x = 0\} \times \mathbb{A}^1$, cut out by the flat ideal $(x) \subseteq k[x,y,t]$. This constant family has $P_t(m) = m + 1$ for every t, in accordance with proposition 2.1.4, and it is the unique flat family agreeing with Z_t away from $t = 0$. The non-flat Z above differs from it only in the embedded jump at the origin, and the flat limit systematically discards exactly such jumps. This is the sense in which flatness selects the geometrically correct limit: it forbids the fiber from gaining or losing pieces as the parameter specializes.

The two families in the example are the picture to carry through the chapter. A flat family is one whose Hilbert polynomial is a constant of the motion; a non-flat family is one in which the polynomial jumps at special parameters, and the flat families are recovered by taking flat limits that suppress the jumps. The Hilbert and Quot functors are built from flat families precisely so that each of their S-points has a well-defined, constant Hilbert polynomial, and the fixed P in $\underline{\mathrm{Hilb}}^P_Y$ and $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$

is the value of that constant. The task of the rest of the chapter is to show that these functors, so constrained, are representable, and the first step is to prove that the flat families with a given P form a *bounded* collection, which is the subject of section 2.2.

Exercise 2.1.1

1. Let $S \subseteq \mathbb{P}^3_k$ be a smooth surface of degree d, the zero scheme of one homogeneous polynomial of degree d. Using the ideal-sheaf sequence $0 \to \mathcal{O}_{\mathbb{P}^3}(-d) \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_S \to 0$ and the values $\chi(\mathcal{O}_{\mathbb{P}^3}(m)) = \binom{m+3}{3}$, compute the Hilbert polynomial $P_{\mathcal{O}_S}(m)$. Read off from it the dimension and the degree of S, and evaluate $P_{\mathcal{O}_S}(0) = \chi(\mathcal{O}_S)$ for $d = 4$ (a quartic K3 surface), confirming that $\chi(\mathcal{O}_S) = 2$.

2. Consider the family over $S = \operatorname{Spec} k[t]$ given in $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ by the ideal $I_t = (y - tx^2)$ as t varies, a family of parabolas degenerating to the line $y = 0$. Show that the total-space ideal $(y - tx^2) \subseteq k[x, y, t]$ presents a family that *is* flat over $\operatorname{Spec} k[t]$, by exhibiting $k[x, y, t]/(y - tx^2)$ as a free $k[t]$-module, and confirm that every fiber is a copy of $\mathbb{A}^1$ over its residue field. Contrast this with example 2.1.8: why does this affine degeneration stay flat while that one does not? (Caution: taking the projective closure of each affine fiber separately does *not* give a flat family, since the smooth conic $\{yz = tx^2\}$ for $t \neq 0$ acquires an embedded point at infinity in its flat limit; the flatness here is a statement about the affine total space over $\operatorname{Spec} k[t]$.)

3. Let $\mathcal{G}$ be a flat family over a connected base S with fiberwise Hilbert polynomial P. Suppose $\deg P = 0$, so P is a constant d. Prove that every fiber $\mathcal{G}_s$ is a zero-dimensional sheaf of length d, and deduce that the function $s \mapsto \dim_{\kappa(s)} H^0(Y_s, \mathcal{G}_s)$ is constant equal to d (you may use that for a zero-dimensional sheaf all higher cohomology vanishes). Where in the argument is flatness used?

4. Fix $Y = \operatorname{Spec} k$ (the case $Y = \mathbb{P}^0_k$, a single point) and $\mathcal{F} = \mathcal{O}_Y^{\oplus r} = k^{\oplus r}$. Describe the S-points of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ for $S = \operatorname{Spec} k$ in the case P is the constant $r' \leq r$. On a point every coherent sheaf is locally free, so Hilbert polynomial r' means a rank-r' quotient. Show that such a $\operatorname{Spec} k$-point is exactly a rank-r' quotient of $k^{\oplus r}$, and explain in one sentence how this recovers the Grassmannian $\operatorname{Gr}(r', r)$ of section 1.2 as $\underline{\mathrm{Quot}}$ over the point. (The full identification, including the passage from a point to $Y = \mathbb{P}^n$ where the analogous locally free quotients of $\mathcal{O}_{\mathbb{P}^n}^{\oplus r}$ have Hilbert polynomial $r'\binom{m+n}{n}$, is example 2.5.4.)

5. Prove that the Quot functor is a functor, that is, verify the two claims left implicit in definition 2.1.5: that pulling a surjection $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ back along $\mathrm{id}_Y \times g$ for $g\colon T \to S$ yields a surjection with $\mathcal{Q}$-pullback flat over T and fiberwise Hilbert polynomial still P; and that $(g \circ h)^* = h^* \circ g^*$. You may cite proposition 2.1.4 for the Hilbert-polynomial claim and the right-exactness of $\otimes$ for surjectivity.

6. Let $Y = \mathbb{P}^1_k$ and $P = d$ as in example 2.1.7. Give the flat family over $S = \mathbb{A}^1 = \operatorname{Spec} k[t]$ whose fiber over $t \neq 0$ is the d reduced points $\{0, 1, 2, \ldots, d-1\}$ scaled by t, namely $\{0, t, 2t, \ldots, (d-1)t\}$, and whose fiber over $t = 0$ is a single fat point of length d at the origin. Write the monic polynomial $f_t(x) \in k[t][x]$ cutting it out, verify flatness, and identify the limiting ideal at $t = 0$ as (x^d).

2.2 Boundedness and the Flattening Stratification

The Quot and Hilbert functors of section 2.1 package an entire universe of coherent quotients into a single functor of points, and the whole point of the chapter is to show that this functor is representable by a scheme. Representability by a scheme is a finiteness statement: a scheme locally of finite type is cut out, chart by chart, from affine space by finitely many equations, so its functor of points cannot see more than a bounded amount of data at once. The Quot functor, by contrast, is defined with no a priori bound at all. It ranges over *every* coherent quotient of $\mathcal{F}$ with a fixed Hilbert polynomial P, across every base scheme S, and nothing in the definition tells us that these quotients form anything like a set of bounded complexity. Before a construction can even begin, this must be brought under control.

The controlling principle is *boundedness*: the sheaves that can occur as fibers of a flat family with Hilbert polynomial P do not form an unlimited zoo, but a family that a single finite-dimensional object can capture. Concretely, all of them become generated by their global sections, with vanishing higher cohomology, after twisting by one and the same power $\mathcal{O}(m)$ of the hyperplane bundle, once m is large enough (depending only on P). Fixing such an m replaces each quotient by a linear-algebra datum, a quotient of the fixed vector space $H^0(\mathcal{F}(m))$, and this is exactly the datum a Grassmannian parametrizes. Boundedness is what lets a single Grassmannian catch every quotient with a given Hilbert polynomial simultaneously; this section proves it. The measure of "large enough" is Castelnuovo-Mumford regularity, developed first, and the same circle of ideas produces Grothendieck's flattening stratification, the tool that later carves the locus of flat quotients out of the Grassmannian as a locally closed subscheme.

Throughout, $\mathbb{P}^n = \mathbb{P}^n_k$ over a fixed base field k, and $\mathcal{O}(1)$ is the twisting sheaf. For a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n$ and $m \in \mathbb{Z}$ write $\mathcal{F}(m) = \mathcal{F} \otimes \mathcal{O}(m)$, and $H^i(\mathcal{F}) = H^i(\mathbb{P}^n, \mathcal{F})$ with $h^i(\mathcal{F}) = \dim_k H^i(\mathcal{F})$.

Regularity, the numerical invariant that measures "large enough", is where the section begins. Mumford isolated a single number of a coherent sheaf that simultaneously controls global generation, the vanishing of higher cohomology, and the surjectivity of multiplication maps on global sections. The definition looks asymmetric, tracking $H^i(\mathcal{F}(m-i))$ rather than $H^i(\mathcal{F}(m))$, but this shift is precisely what makes regularity propagate upward: if a sheaf is m-regular, it is m'-regular for every $m' \geq m$, so regularity has a well-defined threshold. That threshold is the numerical handle boundedness needs.

Definition 2.2.1: Castelnuovo-Mumford Regularity

Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n_k$ and let $m \in \mathbb{Z}$. The sheaf $\mathcal{F}$ is *m-regular* (in the sense of Castelnuovo and Mumford) if

$$H^i\big(\mathbb{P}^n, \mathcal{F}(m-i)\big) = 0 \qquad \text{for all } i \geq 1$$

The *Castelnuovo-Mumford regularity* of $\mathcal{F}$ is

$$\operatorname{reg}(\mathcal{F}) = \min\{\, m \in \mathbb{Z} \big| \mathcal{F} \text{ is } m\text{-regular} \,\}$$

when the set on the right is nonempty and bounded below, and $\operatorname{reg}(\mathcal{F}) = -\infty$ if $\mathcal{F}$ is m-regular for all m (that is, if $\mathcal{F}$ has finite length).

The single index i runs over $1, 2, \ldots, n$ because H^i of a coherent sheaf on $\mathbb{P}^n$ vanishes for $i > n$. Thus

m-regularity is a finite list of cohomological vanishings: $H^1(\mathcal{F}(m-1)) = 0$, $H^2(\mathcal{F}(m-2)) = 0$, and so on down to $H^n(\mathcal{F}(m-n)) = 0$. The definition asks each higher cohomology group to vanish after a twist that increases with the cohomological degree, and it is this staircase of twists that makes the regularity threshold well behaved. The following theorem records the three properties that make regularity the right invariant; it is the technical engine of the section.

Theorem 2.2.2: Mumford's Regularity Theorem

Let $\mathcal{F}$ be an m-regular coherent sheaf on $\mathbb{P}^n_k$. Then for every $\ell \geq 0$:

1. $\mathcal{F}$ is $(m+\ell)$-regular; in particular $H^i(\mathcal{F}(m+\ell-i)) = 0$ for all $i \geq 1$, and $H^i(\mathcal{F}(j)) = 0$ for all $i \geq 1$ and all $j \geq m-i$.
2. The twisted sheaf $\mathcal{F}(m+\ell)$ is generated by its global sections.
3. The multiplication map
$$H^0(\mathbb{P}^n, \mathcal{F}(m+\ell)) \otimes_k H^0(\mathbb{P}^n, \mathcal{O}(1)) \longrightarrow H^0(\mathbb{P}^n, \mathcal{F}(m+\ell+1))$$
is surjective.

***Proof*:**
The argument is an induction on $n = \dim \mathbb{P}^n$ using a general hyperplane, and the three statements are proved together because part (1) for $\mathbb{P}^{n-1}$ feeds parts (2) and (3) for $\mathbb{P}^n$. Assume first that the base field k is infinite; the general case is recovered at the end by a base extension that does not change any of the cohomology groups involved.

Step 1: Restriction to a hyperplane preserves regularity. Choose a hyperplane $H \subseteq \mathbb{P}^n$ whose defining section $s \in H^0(\mathcal{O}(1))$ is a non-zerodivisor on $\mathcal{F}$, meaning that the multiplication map $\mathcal{F}(-1) \xrightarrow{\cdot s} \mathcal{F}$ is injective. Such an H exists because k is infinite: the associated points of $\mathcal{F}$ are finite in number, and a general hyperplane avoids all of them, so s does not lie in any associated prime and multiplication by s has zero kernel. Let $\mathcal{F}_H = \mathcal{F}|_H = \mathcal{F} \otimes \mathcal{O}_H$ be the restriction, a coherent sheaf on $H \cong \mathbb{P}^{n-1}$. The short exact sequence

$$0 \longrightarrow \mathcal{F}(-1) \xrightarrow{\cdot s} \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0 \tag{2.1}$$

twisted by $\mathcal{O}(m-i)$ gives the long exact cohomology sequence

$$H^i(\mathcal{F}(m-i)) \longrightarrow H^i(\mathcal{F}_H(m-i)) \longrightarrow H^{i+1}(\mathcal{F}(m-1-i))$$

For $i \geq 1$ the left group vanishes because $\mathcal{F}$ is m-regular, and the right group is $H^{i+1}(\mathcal{F}(m-(i+1)))$, which also vanishes because $\mathcal{F}$ is m-regular (the twist $m-1-i$ is exactly the one that m-regularity kills in degree $i+1$). Hence $H^i(\mathcal{F}_H(m-i)) = 0$ for all $i \geq 1$, so $\mathcal{F}_H$ is m-regular as a sheaf on $\mathbb{P}^{n-1}$.

Step 2: The vanishing $H^i(\mathcal{F}(m+\ell-i)) = 0$ for $\ell \geq 0$. Induct on n. When $n = 0$ every higher cohomology group vanishes trivially and there is nothing to prove. For $n \geq 1$, by Step 1 and the inductive hypothesis applied to $\mathcal{F}_H$ on $\mathbb{P}^{n-1}$, the sheaf $\mathcal{F}_H$ satisfies $H^i(\mathcal{F}_H(m+\ell-i)) = 0$ for all $i \geq 1$ and all $\ell \geq 0$. Twist (2.1) by $\mathcal{O}(m+\ell-i)$ and read off the piece

$$H^i(\mathcal{F}_H(m+\ell-i)) \longrightarrow H^{i+1}(\mathcal{F}(m+\ell-1-i)) \xrightarrow{\cdot s} H^{i+1}(\mathcal{F}(m+\ell-i))$$

Fix $i \geq 1$. The first group vanishes by the case just established for $\mathcal{F}_H$, so the multiplication map $\cdot s$ on $H^{i+1}(\mathcal{F}(m+\ell-1-i))$ is injective for every $\ell \geq 0$; equivalently, writing $j = m+\ell-1-i$, multiplication by s is injective on $H^{i+1}(\mathcal{F}(j))$ for all $j \geq m-1-i$. By Serre vanishing $H^{i+1}(\mathcal{F}(j)) = 0$ for $j \gg 0$. An injective endomorphism-up-to-twist of a sequence of groups that eventually vanishes forces the whole sequence to vanish: if $\cdot s\colon H^{i+1}(\mathcal{F}(j)) \hookrightarrow H^{i+1}(\mathcal{F}(j+1))$ is injective for all $j \geq m-1-i$ and the target vanishes for j large, then reading the injections backward, each $H^{i+1}(\mathcal{F}(j))$ injects into a vanishing group and so is itself zero. Reindexing by $i' = i+1 \geq 2$ and $j = m+\ell-i'$ gives $H^{i'}(\mathcal{F}(m+\ell-i')) = 0$ for all $i' \geq 2$ and $\ell \geq 0$. The remaining case $i' = 1$ is the vanishing $H^1(\mathcal{F}(m+\ell-1)) = 0$, and it is proved by a separate induction on ℓ that avoids any appeal to Step 3. The base case $\ell = 0$ is the vanishing $H^1(\mathcal{F}(m-1)) = 0$, which is part of the hypothesis that $\mathcal{F}$ is m-regular. For the inductive step, fix $\ell \geq 1$ and twist (2.1) by $\mathcal{O}(m+\ell-1)$, giving in cohomology the piece

$$H^1(\mathcal{F}(m+\ell-2)) \longrightarrow H^1(\mathcal{F}(m+\ell-1)) \longrightarrow H^1(\mathcal{F}_H(m+\ell-1))$$

The left group is $H^1(\mathcal{F}(m+(\ell-1)-1)) = 0$ by the inductive hypothesis on ℓ. The right group is $H^1(\mathcal{F}_H((m+\ell)-1)) = 0$: by Step 1 the sheaf $\mathcal{F}_H$ is m-regular on $\mathbb{P}^{n-1}$, so by the inductive hypothesis on n it is $(m+\ell)$-regular, and its degree-1 vanishing at level $m+\ell$ is exactly this group. Squeezed between two vanishing groups, $H^1(\mathcal{F}(m+\ell-1)) = 0$, closing the induction on ℓ. This uses only the vanishing of H^1 for $\mathcal{F}_H$ and for $\mathcal{F}$ at the previous level, and in particular no surjectivity of a multiplication map. Combining, $\mathcal{F}$ is $(m+\ell)$-regular for every $\ell \geq 0$, which is part (1); the auxiliary form $H^i(\mathcal{F}(j)) = 0$ for $j \geq m-i$ is the same statement reindexed.

Step 3: Surjectivity of multiplication, part (3). The claim is that

$$\mu_\ell\colon H^0(\mathcal{F}(m+\ell)) \otimes H^0(\mathcal{O}(1)) \longrightarrow H^0(\mathcal{F}(m+\ell+1))$$

is surjective for all $\ell \geq 0$. Restricting to the hyperplane H gives a commutative diagram whose rows come from (2.1). Consider

$$0 \to H^0(\mathcal{F}(m+\ell)) \to H^0(\mathcal{F}(m+\ell+1)) \to H^0(\mathcal{F}_H(m+\ell+1)) \to 0$$

which is exact on the right because $H^1(\mathcal{F}(m+\ell)) = 0$ by part (1). Pick $t \in H^0(\mathcal{F}(m+\ell+1))$. Its restriction $t|_H \in H^0(\mathcal{F}_H(m+\ell+1))$ lies in the image of the multiplication map for $\mathcal{F}_H$, which is surjective by the inductive hypothesis on $\mathbb{P}^{n-1}$ applied to the m-regular sheaf $\mathcal{F}_H$. So $t|_H = \sum_a \bar{u}_a \cdot \bar{v}_a$ with $\bar{u}_a \in H^0(\mathcal{F}_H(m+\ell))$ and $\bar{v}_a \in H^0(\mathcal{O}_H(1))$. Lift each $\bar{u}_a$ to $u_a \in H^0(\mathcal{F}(m+\ell))$ (possible since $H^0(\mathcal{F}(m+\ell)) \to H^0(\mathcal{F}_H(m+\ell))$ is surjective, again because $H^1(\mathcal{F}(m+\ell-1)) = 0$) and each $\bar{v}_a$ to $v_a \in H^0(\mathcal{O}(1))$. Then $t - \sum_a u_a v_a$ restricts to 0 on H, so it is divisible by s: it equals $s \cdot t'$ for a unique $t' \in H^0(\mathcal{F}(m+\ell))$, using the exactness of (2.1) on global sections. Writing $s = \bar{s} \in H^0(\mathcal{O}(1))$, we obtain $t = \sum_a u_a v_a + t' \cdot s$, which exhibits t in the image of μ_ℓ. The base case $n = 0$ is trivial: on a point $\mathcal{F}(m+\ell)$ and $\mathcal{F}(m+\ell+1)$ are the same finite-dimensional vector space, while $H^0(\mathcal{O}(1)) \cong k$ is one-dimensional and its nonzero elements act invertibly, so the multiplication map is surjective.

Step 4: Global generation, part (2). Fix $\ell \geq 0$. By part (3) applied repeatedly, the graded module $\bigoplus_{p\geq 0} H^0(\mathcal{F}(m+\ell+p))$ is generated in degree 0 over the homogeneous coordinate ring $\bigoplus_{p\geq 0} H^0(\mathcal{O}(p))$. Global generation of $\mathcal{F}(m+\ell)$ is a statement about stalks at closed points, and it follows once the evaluation map $H^0(\mathcal{F}(m+\ell)) \otimes \mathcal{O} \to \mathcal{F}(m+\ell)$ is surjective. This surjectivity is local, and at a closed point x the module $\bigoplus_p H^0(\mathcal{F}(m+\ell+p))$ being generated in degree 0 means precisely that sections of $\mathcal{F}(m+\ell)$ generate the stalk after inverting a linear form nonvanishing at x; since such a linear form is a unit at x, the sections already generate $\mathcal{F}(m+\ell)_x$ over $\mathcal{O}_x$ by Nakayama. Hence $\mathcal{F}(m+\ell)$ is globally generated.

Finally, for a base field k that is not infinite, pass to the algebraic closure $\bar{k}$. Cohomology of $\mathcal{F}$ commutes with the flat base change $\mathbb{P}^n_{\bar{k}} \to \mathbb{P}^n_k$, so $\mathcal{F}_{\bar{k}}$ is m-regular, the three conclusions hold for $\mathcal{F}_{\bar{k}}$, and each descends because the maps in (2) and (3) are defined over k and become surjective after a faithfully flat extension, hence were surjective already. This establishes all three parts, completing the proof.

The theorem says that a single number, the regularity, dominates the entire positive-twist behavior of $\mathcal{F}$: past the threshold every higher cohomology group is dead, every twist is globally generated, and the graded module of sections is generated in the lowest surviving degree. The asymmetry in the definition (twisting H^i by $m-i$) is what powers part (1), the propagation upward, and part (1) is what makes "m-regular for $m \gg 0$" a meaningful uniform statement across a family: to bound a whole family it suffices to bound its regularity by a single integer. That is the bridge to boundedness.

A first concrete instance shows the invariant is computable and grounds the definition immediately.

Example 2.2.3: The Regularity of the Ideal Sheaf of a Point

Let $Z = \{x\} \subseteq \mathbb{P}^n$ be a single reduced closed point, with ideal sheaf $\mathcal{I}_Z$ and structure sheaf $\mathcal{O}_Z = k(x)$, a skyscraper. The defining sequence is

$$0 \longrightarrow \mathcal{I}_Z \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_Z \longrightarrow 0$$

Twist by $\mathcal{O}(m-i)$ and take cohomology. The skyscraper $\mathcal{O}_Z$ has $H^0 = k$ and $H^i = 0$ for $i \geq 1$, and $\mathcal{O}_{\mathbb{P}^n}(j)$ has $H^i(\mathcal{O}(j)) = 0$ for $1 \leq i \leq n-1$ (all j), $H^n(\mathcal{O}(j)) = 0$ for $j \geq -n$, and $H^0(\mathcal{O}(j)) \neq 0$ for $j \geq 0$. For $\mathcal{I}_Z$ this yields, for $i \geq 1$ and the twist $m-i$: the group $H^i(\mathcal{I}_Z(m-i))$ sits between $H^{i-1}(\mathcal{O}_Z(m-i))$ and $H^i(\mathcal{O}(m-i))$. For $i \geq 2$ both neighbors vanish once $m-i \geq -n$, so $H^i(\mathcal{I}_Z(m-i)) = 0$ for all $m \geq 1$. For $i = 1$ the relevant sequence is

$$H^0(\mathcal{O}(m-1)) \longrightarrow H^0(\mathcal{O}_Z(m-1)) = k \longrightarrow H^1(\mathcal{I}_Z(m-1)) \longrightarrow H^1(\mathcal{O}(m-1)) = 0$$

The evaluation map $H^0(\mathcal{O}(m-1)) \to k(x)$ is surjective as soon as some degree-$(m-1)$ form is nonzero at x, that is, for $m-1 \geq 0$. Hence $H^1(\mathcal{I}_Z(m-1)) = 0$ exactly for $m \geq 1$, and $H^1(\mathcal{I}_Z(-1)) = k \neq 0$. Therefore $\operatorname{reg}(\mathcal{I}_Z) = 1$: the ideal sheaf of a point is 1-regular but not 0-regular. Consistently, $\mathcal{I}_Z(1)$ is globally generated (the linear forms vanishing at x generate $\mathcal{I}_Z(1)$), while $\mathcal{I}_Z$ itself is not globally generated near x.

Regularity of the structure sheaf of a projective scheme translates directly into regularity of its ideal sheaf, and it is the ideal sheaf that the Hilbert functor sees.

Example 2.2.4: Regularity and the Hilbert Polynomial of a Twisted Cubic

Let $C \subseteq \mathbb{P}^3$ be the twisted cubic, the image of $\mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, $[s:t] \mapsto [s^3 : s^2t : st^2 : t^3]$. Its ideal $\mathcal{I}_C$ is generated by the three quadrics $x_0x_2 - x_1^2$, $x_1x_3 - x_2^2$, $x_0x_3 - x_1x_2$, and $\mathcal{O}_C \cong \mathcal{O}_{\mathbb{P}^1}(3)$, so $H^0(\mathcal{O}_C(m)) = H^0(\mathcal{O}_{\mathbb{P}^1}(3m))$ has dimension $3m+1$ for $m \geq 0$, giving Hilbert polynomial $P_C(m) = 3m+1$. The curve C is projectively normal (arithmetically Cohen-Macaulay), with minimal free resolution

$$0 \longrightarrow \mathcal{O}(-3)^{\oplus 2} \longrightarrow \mathcal{O}(-2)^{\oplus 3} \longrightarrow \mathcal{I}_C \longrightarrow 0$$

Reading regularity off the resolution, the generators sit in degree 2 and the syzygies in degree 3 with a linear shift, so $\mathcal{I}_C$ is 2-regular but not 1-regular: $\operatorname{reg}(\mathcal{I}_C) = 2$, and the quadric generators show $\mathcal{I}_C(2)$ is globally generated. From the defining sequence $0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_C \to 0$ and the 0-regularity of $\mathcal{O}_{\mathbb{P}^3}$, $\operatorname{reg}(\mathcal{O}_C) = \operatorname{reg}(\mathcal{I}_C) - 1 = 1$. The point to retain is that a curve of Hilbert polynomial $3m+1$ in $\mathbb{P}^3$ has ideal-sheaf regularity bounded by an explicit small integer determined by P_C. That is the boundedness the next theorem makes uniform.

With regularity in hand as the measure of "large enough", the family of sheaves that the Quot functor ranges over can now be brought under uniform control. The definition of the Quot functor imposes no size constraint on the quotients it collects, yet a scheme representing it must. Boundedness is the reconciliation: the set of all quotient sheaves that can appear, as fibers over any base and any point, with a fixed Hilbert polynomial P, is uniformly regular. A single integer m_0, depending only on P (and on the ambient $\mathcal{F}$ and $\mathbb{P}^n$), bounds the regularity of every ideal or quotient sheaf in sight. The proof produces an explicit bound by controlling the intermediate cohomology of a sheaf through the coefficients of its Hilbert polynomial.

Theorem 2.2.5: Boundedness of Sheaves with Fixed Hilbert Polynomial

Fix the ambient $\mathbb{P}^n_k$, a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n$ that is a quotient of $\bigoplus_{j=1}^N \mathcal{O}(a_j)$ for fixed integers a_j, and a polynomial $P \in \mathbb{Q}[t]$. There is an integer $m_0 = m_0(P, n, \mathcal{F})$, depending only on P, on n, and on the presentation of $\mathcal{F}$, such that every coherent quotient

$$\mathcal{F} \twoheadrightarrow \mathcal{Q}$$

whose kernel is a subsheaf $\mathcal{G} = \ker(\mathcal{F} \to \mathcal{Q})$ with Hilbert polynomial P is m_0-regular. In particular, the family of all such quotients $\mathcal{Q}$ (equivalently, of all subsheaves $\mathcal{G} \subseteq \mathcal{F}$ with Hilbert polynomial P) is *bounded*: there is a single m_0 with $\mathcal{G}$ and $\mathcal{Q}$ both m_0-regular for every member of the family, so each becomes globally generated with vanishing higher cohomology after the one twist $\mathcal{O}(m_0)$.

***Proof*:**
The heart of the matter is the case of a subsheaf of a fixed sheaf; the case of the ideal sheaves $\mathcal{I}_Z$ relevant to $\underline{\operatorname{Hilb}}^P_Y$ (definition 2.1.6) and of the kernels relevant to $\underline{\operatorname{Quot}}^P_{\mathcal{F}/Y}$ (definition 2.1.5) both reduce to it, since an ideal sheaf is a subsheaf of $\mathcal{O}_Y \subseteq \mathcal{F}$. The argument is an induction on n producing a uniform regularity bound.

Step 1: Reduction to a bound on h^0 and h^1. A coherent sheaf $\mathcal{G}$ on $\mathbb{P}^n$ is m-regular once $H^i(\mathcal{G}(m-i)) = 0$ for all $i \geq 1$. For $i \geq 2$ these groups are controlled by restriction to hyperplanes and the inductive hypothesis, as in theorem 2.2.2; the delicate part is H^1, and the delicate input is a bound on $h^0(\mathcal{G}(j)) = \dim H^0(\mathcal{G}(j))$ in terms of P. The whole proof therefore reduces to producing, from the Hilbert polynomial P alone, an integer past which $H^1(\mathcal{G}(j))$ vanishes, uniformly over the family. Since $\chi(\mathcal{G}(j)) = P(j)$ for all j (the Euler characteristic is the Hilbert polynomial, independent of the member) and $\chi = h^0 - h^1 + \cdots$, a uniform upper bound on h^0 together with the Euler-characteristic identity pins down h^1.

Step 2: The uniform bound on global sections (Grothendieck's lemma). There is an integer-valued function $Q(t)$, depending only on n and the rank and degree data extractable from P, such that every subsheaf $\mathcal{G} \subseteq \mathcal{F}$ with $\mathcal{G}$ having Hilbert polynomial P satisfies

$$h^0(\mathcal{G}(j)) \leq Q(j) \qquad \text{for all } j \tag{2.2}$$

with Q independent of the particular $\mathcal{G}$. This is proved by induction on n. Restrict to a general hyperplane H; the sequence $0 \to \mathcal{G}(j-1) \to \mathcal{G}(j) \to \mathcal{G}_H(j) \to 0$ (using that a general H meets $\mathcal{G}$ in a subsheaf of $\mathcal{F}_H$) gives

$$h^0(\mathcal{G}(j)) \leq h^0(\mathcal{G}(j-1)) + h^0(\mathcal{G}_H(j))$$

On $H \cong \mathbb{P}^{n-1}$ the sheaf $\mathcal{G}_H$ is a subsheaf of the fixed sheaf $\mathcal{F}_H$, and its Hilbert polynomial is $P(t)-P(t-1)$, again fixed. By induction $h^0(\mathcal{G}_H(j)) \leq Q_H(j)$ for a function Q_H depending only on the data on $\mathbb{P}^{n-1}$. Summing the telescoping inequality from a value j_0 where $h^0(\mathcal{G}(j_0)) = 0$ (which exists uniformly: a subsheaf of $\mathcal{F}$ has no sections in sufficiently negative twist, and "sufficiently negative" is controlled by the fixed $\mathcal{F}$) yields the bound (2.2) with $Q(j) = \sum_{j_0<p\leq j} Q_H(p)$, a function of P and n only. The base case $n = 0$ is immediate, where h^0 is the constant length read off from P.

Step 3: From the h^0 bound to a uniform vanishing of H^1. Combine (2.2) with the Euler-characteristic identity. On $\mathbb{P}^n$,

$$h^1(\mathcal{G}(j)) = h^0(\mathcal{G}(j)) - \chi(\mathcal{G}(j)) + \sum_{i\geq 2}(-1)^i h^i(\mathcal{G}(j)) = h^0(\mathcal{G}(j)) - P(j) + \sum_{i\geq 2}(-1)^i h^i(\mathcal{G}(j))$$

For $i \geq 2$ the groups $h^i(\mathcal{G}(j))$ vanish for $j \geq j_1$ where j_1 is a uniform bound coming from the inductive control of higher cohomology in Step 1. This threshold is uniform over the family: twisting the general-hyperplane sequence $0 \to \mathcal{G}(j) \to \mathcal{G}(j+1) \to \mathcal{G}_H(j+1) \to 0$ gives $H^i(\mathcal{G}_H(j+1)) \to H^{i+1}(\mathcal{G}(j)) \xrightarrow{\cdot s} H^{i+1}(\mathcal{G}(j+1))$, so for $i \geq 1$ the multiplication $\cdot s$ on $H^{i+1}(\mathcal{G}(j))$ is injective once $H^i(\mathcal{G}_H(j+1)) = 0$, and by the inductive hypothesis on $\mathbb{P}^{n-1}$ the sheaves $\mathcal{G}_H \subseteq \mathcal{F}_H$ are uniformly regular past a threshold depending only on $P(t) - P(t-1)$ and $n-1$; injectivity feeding into Serre vanishing then forces $h^{i+1}(\mathcal{G}(j)) = 0$ for all j past a threshold set by that same $\mathbb{P}^{n-1}$ datum, uniformly over the family. Reindexing gives the uniform j_1 for all $i \geq 2$. For such j,

$$h^1(\mathcal{G}(j)) = h^0(\mathcal{G}(j)) - P(j) \leq Q(j) - P(j)$$

The right-hand side is a fixed function of j that does not depend on the member $\mathcal{G}$, so $h^1(\mathcal{G}(j))$ is bounded above uniformly. This bound need not be zero on its own; the vanishing comes from combining it with the multiplication-by-s structure past a *uniform* threshold. Twisting the general hyperplane section $0 \to \mathcal{G}(j) \xrightarrow{\cdot s} \mathcal{G}(j+1) \to \mathcal{G}_H(j+1) \to 0$ and taking cohomology gives $H^1(\mathcal{G}(j)) \xrightarrow{\cdot s} H^1(\mathcal{G}(j+1)) \to H^1(\mathcal{G}_H(j+1))$; once $\mathcal{G}_H$ is $(j+1)$-regular on $\mathbb{P}^{n-1}$, the right group vanishes and the multiplication map is surjective. By the inductive control of regularity for the fixed family of subsheaves $\mathcal{G}_H \subseteq \mathcal{F}_H$ on $\mathbb{P}^{n-1}$, this surjectivity holds for all $j \geq j_2$ with j_2 a uniform bound depending only on P and n. Set $j_3 = \max(j_1, j_2)$, again uniform. Surjectivity of $\cdot s$ for all $j \geq j_3$ says that $H^1(\mathcal{G}(j+t))$ is the image of the fixed space $V := H^1(\mathcal{G}(j_3))$ under the iterated map s^t, that is, $H^1(\mathcal{G}(j_3+t)) = s^t V$ for every $t \geq 0$. Consequently the linear map $s^t \colon V \to H^1(\mathcal{G}(j_3+t))$ is surjective, so by rank-nullity $h^1(\mathcal{G}(j_3+t)) = \dim V - \dim \ker(s^t|_V)$. The kernels $\ker(s^t|_V)$ form an increasing chain of subspaces of the finite-dimensional space V, and such a chain stabilizes. Because $\dim V \leq Q(j_3) - P(j_3)$, the chain can strictly grow at most $\dim V$ times, so it has stabilized by $t = \dim V$; for all $t \geq \dim V$ the dimension $h^1(\mathcal{G}(j_3+t))$ is therefore constant. Serre vanishing forces $h^1(\mathcal{G}(j)) = 0$ for $j \gg 0$ for each individual $\mathcal{G}$, so that constant value is 0, and hence $h^1(\mathcal{G}(j_3+t)) = 0$ for all $t \geq \dim V$. Setting $B = Q(j_3) - P(j_3)$, this gives the uniform vanishing $H^1(\mathcal{G}(j)) = 0$ for all $j \geq j_3 + B$. The threshold $m_1 = j_3 + B$ is one and the same for every member, computed from Q, P, and n through the uniform bounds

j_1, j_2, and the a priori bound B; this is the sought uniform vanishing $H^1(\mathcal{G}(j)) = 0$ for all $j \geq m_1$. The surjectivity of $\cdot s$ is what pins $H^1(\mathcal{G}(j_3 + t))$ to a quotient of the single bounded space V, and the stabilizing-kernel count then forces the eventual 0 back to the uniform level m_1; without the a priori bound on $\dim V$ the surjective chain alone would place no ceiling on where the vanishing begins.

Step 4: Assembling the regularity bound. Set $m_0 = \max(m_1 + 1, j_1 + n)$ with j_1 as in Step 3, adjusted so that all of $H^i(\mathcal{G}(m_0 - i)) = 0$ hold: the H^1 vanishing is Step 3, and the H^i vanishings for $2 \leq i \leq n$ are the uniform higher-cohomology bounds. Every $\mathcal{G}$ in the family is then m_0-regular, and m_0 depends only on P, n, and $\mathcal{F}$. The quotient $\mathcal{Q} = \mathcal{F}/\mathcal{G}$ inherits a regularity bound from the regularity of $\mathcal{F}$ (fixed, hence some $m_{\mathcal{F}}$) and of $\mathcal{G}$ (the uniform m_0) through the long exact sequence of $0 \to \mathcal{G} \to \mathcal{F} \to \mathcal{Q} \to 0$: for $i \geq 1$, $H^i(\mathcal{Q}(m - i))$ is squeezed between $H^i(\mathcal{F}(m - i))$ and $H^{i+1}(\mathcal{G}(m - i))$, both of which vanish for $m \geq \max(m_{\mathcal{F}}, m_0)$. Enlarging m_0 to this maximum makes every $\mathcal{Q}$ in the family m_0-regular as well, as we sought to show.

Boundedness is the pivotal reduction of the chapter. It says that although the Quot functor is defined by an open-ended condition, the sheaves it ranges over are all controlled by one integer m_0, and past m_0 each quotient is reconstructed from its degree-m_0 global sections. After the twist by $\mathcal{O}(m_0)$, a flat quotient $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ over any base induces a quotient of the fixed vector space (or fixed free module) $H^0(\mathcal{F}(m_0))$, of fixed rank $P(m_0)$, and this is exactly a point of a Grassmannian. section 2.3 builds the Quot scheme by realizing the functor as a subfunctor of that Grassmannian; boundedness is what guarantees the Grassmannian is large enough to hold every quotient at once. Without it there would be no single finite-dimensional ambient space and no scheme to cut out.

2.2.6 Remark: Boundedness as finiteness of the moduli problem Boundedness is the sheaf-theoretic shadow of a finiteness principle that recurs throughout moduli theory. A moduli functor is representable by a scheme (or by a quasi-compact stack) only if the objects it classifies form a bounded family, that is, are parametrized by a scheme of finite type over k. When boundedness fails, as for the full functor of all coherent sheaves with no fixed Hilbert polynomial, the resulting moduli object is not a scheme and typically not even an algebraic space of finite type; it is an Artin stack that is not quasi-compact. Fixing the Hilbert polynomial P is the standard device that restores boundedness, which is why $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ carries P as a decoration and the Quot scheme is a disjoint union over P of quasi-projective pieces.

Boundedness caught every quotient inside one Grassmannian, but not every point of that Grassmannian gives a *flat* quotient with the right Hilbert polynomial, and flatness is exactly the condition in the definitions of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ and $\underline{\mathrm{Hilb}}^P_Y$. The tool that recognizes where flatness holds is Grothendieck's flattening stratification. Given a coherent sheaf on a product $Y \times S$, it decomposes the base S into locally closed pieces on each of which the sheaf becomes flat, and on each piece the fibers have a single Hilbert polynomial. This turns the flatness-with-Hilbert-polynomial-P condition, which cuts out $\underline{\mathrm{Quot}}^P$ inside the Grassmannian, into a locally closed subscheme, hence a subscheme.

The stratification is governed by a universal property: it is the finest way of splitting S so that the sheaf becomes flat, and any base change that flattens the sheaf factors through it. That universal property is what lets the strata be defined once and for all, functorially in S.

Theorem 2.2.7: Grothendieck's Flattening Stratification

Let S be a Noetherian scheme, let $Y \subseteq \mathbb{P}^n_S$ be a closed subscheme projective over S, and let $\mathcal{F}$ be a coherent sheaf on Y (equivalently, on $\mathbb{P}^n_S$, extended by zero). Then there is a finite set of polynomials $P_1, \ldots, P_r \in \mathbb{Q}[t]$ and a decomposition of S into pairwise disjoint locally closed subschemes

$$S = \coprod_{a=1}^{r} S_a, \qquad S_a = S_{P_a}$$

indexed by the P_a, with the following properties:

1. Each $S_a \hookrightarrow S$ is a locally closed immersion, and $S = \bigcup_a S_a$ set-theoretically; the closure of S_a is contained in $\bigcup_{P_b \geq P_a} S_b$, where $P_b \geq P_a$ means $P_b(m) \geq P_a(m)$ for $m \gg 0$.

2. A point $s \in S$ lies in S_P if and only if the fiber $\mathcal{F}_s = \mathcal{F}|_{Y_s}$ on $Y_s = Y \times_S \operatorname{Spec} k(s)$ has Hilbert polynomial P.

3. The restriction $\mathcal{F}|_{Y \times_S S_a}$ is flat over S_a with all fibers of Hilbert polynomial P_a.

4. *(Universal property.)* A morphism $T \to S$ of Noetherian schemes has the property that the pullback $\mathcal{F}_T$ on $Y_T = Y \times_S T$ is flat over T with locally constant Hilbert polynomial if and only if $T \to S$ factors scheme-theoretically through the disjoint union $\coprod_a S_a \hookrightarrow S$.

***Proof*:**
The construction is local on S, so assume $S = \operatorname{Spec} A$ is affine and Noetherian. The tool is cohomology and base change together with the theorem that a coherent sheaf on a projective scheme over A is flat if and only if a certain graded module is locally free.

Step 1: Reduction to a graded module of finite presentation. Since $Y \subseteq \mathbb{P}^n_S$ is projective over $S = \operatorname{Spec} A$, replace $\mathcal{F}$ by the graded $A[x_0, \ldots, x_n]$-module $M = \bigoplus_{m \geq m_2} H^0(\mathbb{P}^n_S, \mathcal{F}(m))$, taking m_2 large enough (boundedness, theorem 2.2.5, applied to the fibers, or Serre vanishing over A) that for every $s \in S$ and every $m \geq m_2$,

$$H^i(\mathbb{P}^n_{k(s)}, \mathcal{F}_s(m)) = 0 \quad (i \geq 1), \qquad \dim_{k(s)} H^0(\mathbb{P}^n_{k(s)}, \mathcal{F}_s(m)) = P_s(m)$$

where P_s is the Hilbert polynomial of $\mathcal{F}_s$. By cohomology and base change ([12], cohomology-and-base-change over $\operatorname{Spec} A$), for $m \geq m_2$ the A-module $M_m = H^0(\mathbb{P}^n_S, \mathcal{F}(m))$ is finitely generated, and its formation commutes with base change: $M_m \otimes_A k(s) \cong H^0(\mathbb{P}^n_{k(s)}, \mathcal{F}_s(m))$. Thus $\dim_{k(s)}(M_m \otimes_A k(s)) = P_s(m)$ for all s and all $m \geq m_2$.

Step 2: Flatness as local freeness of the M_m. The sheaf $\mathcal{F}$ is flat over A in a neighborhood of s if and only if each M_m ($m \geq m_2$) is locally free of rank $P_s(m)$ near s; this is the graded-flatness criterion, that a coherent sheaf projective over an affine base is flat precisely when its sufficiently-positive-degree cohomology modules are locally free ([12], flatness criterion via H^0 of twists). For a single finitely generated A-module M_m, the function $s \mapsto \dim_{k(s)}(M_m \otimes k(s))$ is upper semicontinuous, and M_m is locally free of rank d exactly on the locus where this function equals d and stays constant, which is a locally closed subscheme of S defined by the vanishing and non-vanishing of appropriate Fitting ideals. Concretely, the d-th Fitting ideal $\operatorname{Fitt}_d(M_m)$ cuts out the closed locus where the rank is $\leq d$, and M_m is locally free of rank d on the locally closed subscheme

$$V(\operatorname{Fitt}_{d-1}(M_m)) \setminus V(\operatorname{Fitt}_d(M_m))$$

equipped with the scheme structure making M_m locally free there.

Step 3: Stratifying by the Hilbert polynomial. For each polynomial P, define $S_P \subseteq S$ to be the locus of points s with $P_s = P$, that is, with $\dim_{k(s)}(M_m \otimes k(s)) = P(m)$ for all $m \geq m_2$. Because $P_s(m) = \dim_{k(s)}(M_m \otimes k(s))$ and each such dimension function is upper semicontinuous with locally closed constancy loci, S_P is the intersection over $m \geq m_2$ of the locally closed loci $\{\dim(M_m \otimes k(s)) = P(m)\}$. By Noetherianity this intersection stabilizes after finitely many m, so S_P is locally closed, and it is nonempty for only finitely many P (the Hilbert polynomials occurring on the Noetherian S form a finite set, since S has finitely many generic points and P_s is constant on a stratification refining the irreducible components). Equip S_P with the Fitting-ideal scheme structure of Step 2, the one making every M_m locally free of rank $P(m)$; then $\mathcal{F}$ restricted to $\mathbb{P}^n \times_S S_P$ is flat over S_P with fibers of Hilbert polynomial P, giving parts (2) and (3). The finitely many nonempty S_P are pairwise disjoint by construction (a point has one Hilbert polynomial) and cover S, giving part (1); the closure relation follows because each fiber-dimension function $s \mapsto \dim_{k(s)}(M_m \otimes k(s)) = P_s(m)$ is upper semicontinuous, so $P_s(m)$ can only jump up under specialization for every $m \geq m_2$, forcing $P_s \geq P_a$ on the closure of S_a and hence $\overline{S_a} \subseteq \bigcup_{P_b \geq P_a} S_b$.

Step 4: The universal property. Let $g\colon T \to S$ be a morphism with T Noetherian. The pullback $\mathcal{F}_T$ is flat over T if and only if each g^*M_m is locally free over T of the correct rank, again by the graded-flatness criterion applied over T and the base-change isomorphism $g^*M_m \cong \bigoplus H^0(\mathbb{P}^n_T, \mathcal{F}_T(m))$ in the stabilized range. Since M_m becomes locally free of rank $P(m)$ precisely after restriction to S_P, and the Fitting-ideal description of local freeness is compatible with base change, g^*M_m is locally free of rank $P(m)$ if and only if g factors through the subscheme S_P cut out by the Fitting conditions. Requiring flatness with locally constant Hilbert polynomial is requiring this for all m simultaneously with a single P per connected component of T, which is exactly factorization of g through $\coprod_P S_P \hookrightarrow S$. This is the asserted universal property, completing the proof.

The flattening stratification says that flatness, though defined pointwise and delicate, is a locally closed condition on the base: the base splits into finitely many locally closed strata, indexed by Hilbert polynomials, over which the sheaf is flat, and any test scheme on which the sheaf flattens factors through this splitting. For the Quot scheme this is decisive. Inside the Grassmannian produced by boundedness sits the universal quotient; applying the flattening stratification to it isolates the single stratum with Hilbert polynomial P as a locally closed subscheme, and the universal property of the stratification is precisely the universal property that identifies this subscheme with the Quot functor. The stratification also explains, geometrically, why the Hilbert polynomial jumps up under specialization: a limit of subschemes can acquire embedded or lower-dimensional components, raising the Euler characteristic in each twist. The next example makes such a jump visible.

Example 2.2.8: The Flattening Stratification of a Jumping Rank

Consider the simplest family in which fiber dimension jumps. Let $S = \mathbb{A}^1 = \operatorname{Spec} k[s]$ and let $Y = \mathbb{P}^0_S = S$ itself, so $Y \times_S S = S$, and take the coherent sheaf

$$\mathcal{F} = \operatorname{coker}\big(\mathcal{O}_S \xrightarrow{\cdot s} \mathcal{O}_S\big) = \mathcal{O}_S/(s) = k[s]/(s)$$

the structure sheaf of the origin, viewed as a sheaf on S over S. The fiber of $\mathcal{F}$ at a point $a \in \mathbb{A}^1$ is

$$\mathcal{F}_a = \mathcal{F} \otimes_{k[s]} k(a) = \big(k[s]/(s)\big) \otimes_{k[s]} k[s]/(s-a) = k[s]/(s, s-a)$$

which is 0 when $a \neq 0$ (the ideal $(s, s-a)$ is the unit ideal) and is k when $a = 0$. So the fiber

Hilbert polynomial, here a constant since Y_s is a point, is

$$P_a = \begin{cases} 0 & a \neq 0 \\ 1 & a = 0 \end{cases}$$

The flattening stratification splits $S = \mathbb{A}^1$ into two strata: the open stratum $S_{P=0} = \mathbb{A}^1 \setminus \{0\}$, where $\mathcal{F}$ restricts to the zero sheaf, flat of fiber length 0; and the closed stratum $S_{P=1} = \{0\}$, where $\mathcal{F}$ restricts to k, flat over the point of fiber length 1. On S itself $\mathcal{F}$ is not flat: flatness over $k[s]$ would force $\mathcal{F}$ to be torsion-free, but $\mathcal{F} = k[s]/(s)$ is all torsion, and indeed $\dim_{k(a)} \mathcal{F}_a$ jumps from 0 to 1 at the origin, violating the constancy that flatness demands. The stratification repairs this by isolating the jump: the closed point where the fiber length rises becomes its own stratum. This models exactly what happens in the Quot scheme when a family of quotients degenerates and an embedded point appears in the special fiber, raising the Hilbert polynomial there.

A second example shows the stratification separating strata of different colength, the closed-subscheme situation of $\underline{\mathrm{Hilb}}^P$ where the Hilbert polynomial is the constant length of a zero-dimensional scheme.

Example 2.2.9: Stratifying a Family of Ideals by Colength

Let $S = \mathbb{A}^1 = \operatorname{Spec} k[s]$ and work in the affine line $\mathbb{A}^1 = \operatorname{Spec} k[x]$ over S, that is, in $\mathbb{A}^1_S = \operatorname{Spec} k[x][s]$; the family of closed subschemes is given by the ideal

$$I_s = (s\,x,\ x^2) \subseteq k[x]$$

with the parameter s ranging over S. The quotient sheaf is $\mathcal{F} = k[x][s]/(sx, x^2)$, viewed as a coherent sheaf on $\mathbb{A}^1_S$ flat-checked over $S = \operatorname{Spec} k[s]$. Compute the fibers over a point $a \in S$: setting $s = a$ gives the ideal $I_a = (ax, x^2) \subseteq k[x]$.

- For $a \neq 0$, the element ax is a times x with a a unit, so $I_a = (x, x^2) = (x)$, and the fiber $Z_a = V(x)$ is a single reduced point, of colength $\dim_k k[x]/(x) = 1$. Its (constant) Hilbert polynomial is $P_a = 1$.
- For $a = 0$, the ideal is $I_0 = (x^2)$, and the fiber $Z_0 = V(x^2)$ is a length-2 non-reduced point (a point with one tangent direction), of colength $\dim_k k[x]/(x^2) = 2$. Its Hilbert polynomial is $P_0 = 2$.

The colength jumps from 1 to 2 at the origin, so the family is not flat over S: flatness forces the fiber length to be locally constant, and here it is not. Concretely, $\mathcal{F} = k[x][s]/(sx, x^2)$ has s-torsion, since $x \cdot s = 0$ while $x \neq 0$ in $\mathcal{F}$, and a torsion module over the domain $k[s]$ is not flat. The flattening stratification of $S = \mathbb{A}^1$ therefore splits into two strata carrying different Hilbert polynomials: the open stratum $S_{P=1} = \mathbb{A}^1 \setminus \{0\}$, over which the family restricts to the flat family of single reduced points (colength 1); and the closed stratum $S_{P=2} = \{0\}$, over which the family is the length-2 fat point (colength 2). Each stratum carries a flat family with constant Hilbert polynomial, while the union does not; the closure relation $\overline{S_{P=1}} = \mathbb{A}^1 \supseteq S_{P=2}$ realizes the general principle that the Hilbert polynomial can only jump up under specialization ($1 \to 2$ as $a \to 0$). This is exactly the picture inside $\underline{\mathrm{Hilb}}^P$: the strata $S_{P=1}$ and $S_{P=2}$ belong to different Hilbert schemes Hilb^1 and Hilb^2, here the Hilbert schemes of 1 and 2 points of the affine line $\mathbb{A}^1 = \operatorname{Spec} k[x]$ (not the projective Hilb^P_Y of the rest of the section, since the ambient is the affine $\mathbb{A}^1_S$), and the flattening stratification is what separates a would-be family straddling both into its flat pieces.

The two engines of this section, boundedness and the flattening stratification, together reduce the construction of the Quot scheme to linear algebra plus a locally closed condition: boundedness embeds every quotient into a fixed Grassmannian, and the flattening stratification carves out the flat, Hilbert-polynomial-P locus as a locally closed subscheme of that Grassmannian. section 2.3 carries this out in full and identifies the resulting scheme with the representing object of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ and, as the special case $\mathcal{F} = \mathcal{O}_Y$, of $\underline{\mathrm{Hilb}}^P_Y$.

Exercise 2.2.1

1. Let $\mathcal{F} = \mathcal{O}_{\mathbb{P}^n}$ be the structure sheaf of projective space. Using the cohomology of the line bundles $\mathcal{O}(j)$, show directly that $\mathcal{O}_{\mathbb{P}^n}$ is 0-regular, that is, $\operatorname{reg}(\mathcal{O}_{\mathbb{P}^n}) = 0$, and deduce that $\mathcal{O}(m)$ is globally generated with vanishing higher cohomology for every $m \geq 0$. Explain how this is consistent with theorem 2.2.2(1).
2. Prove the propagation property of regularity from the definition and theorem 2.2.2: if $\mathcal{F}$ is m-regular then $\mathcal{F}$ is m'-regular for every $m' \geq m$. Then give an example of a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^1$ and an integer m such that $\mathcal{F}$ is m-regular but not $(m-1)$-regular, computing $\operatorname{reg}(\mathcal{F})$ explicitly.
3. Let $Z \subseteq \mathbb{P}^n$ be a hypersurface of degree d, defined by one homogeneous polynomial of degree d, with ideal sheaf $\mathcal{I}_Z \cong \mathcal{O}(-d)$. Compute the Hilbert polynomial of $\mathcal{O}_Z$, compute $\operatorname{reg}(\mathcal{I}_Z)$ and $\operatorname{reg}(\mathcal{O}_Z)$ using the exact sequence $0 \to \mathcal{O}(-d) \to \mathcal{O} \to \mathcal{O}_Z \to 0$, and confirm that the regularity is bounded by an explicit function of d (hence of the Hilbert polynomial), illustrating theorem 2.2.5.
4. Consider the family $\mathcal{F}$ on $\mathbb{P}^1_{\mathbb{A}^1}$ given, in the affine chart $\mathbb{A}^1 \times \mathbb{A}^1 = \operatorname{Spec} k[x][s]$, by the module $M = k[x][s]/(x, s)$ over $k[x][s]$, viewed as a coherent sheaf on $\mathbb{P}^1_S$ over $S = \operatorname{Spec} k[s]$. Determine the fibers $\mathcal{F}_s$ for $s \neq 0$ and $s = 0$, decide whether $\mathcal{F}$ is flat over S, and describe the flattening stratification of S explicitly, identifying each stratum and its Hilbert polynomial.
5. Explain, in the setup of theorem 2.2.5, why fixing the Hilbert polynomial P is necessary for boundedness by exhibiting an infinite family of subschemes of $\mathbb{P}^2$, one for each degree $d \geq 1$, whose ideal-sheaf regularities are unbounded. Then explain in one paragraph how boundedness is used in section 2.3 to reduce the Quot functor to a subfunctor of a single Grassmannian, referencing $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (definition 2.1.5).

2.3 Grothendieck's Existence Theorem

The two preceding sections assembled the raw materials. Section 2.1 phrased the moduli problem: the Quot functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (definition 2.1.5) sends a base S to the set of flat quotients of $\mathcal{F}_S$ with fiberwise Hilbert polynomial P, and the Hilbert functor $\underline{\mathrm{Hilb}}^P_Y$ (definition 2.1.6) is the special case $\mathcal{F} = \mathcal{O}_Y$, parametrizing flat families of closed subschemes. Section 2.2 supplied the two engines: boundedness (theorem 2.2.5), which says the family of quotients with fixed P becomes uniformly m-regular for m large, and the flattening stratification (theorem 2.2.7), which cuts out the locus in a base over which a coherent sheaf is flat with a prescribed Hilbert polynomial. This section fits the pieces together. The outcome is that both functors are representable by projective schemes, the theorem stated without proof in [12, Grothendieck's Existence Theorem] and proved here in full.

The strategy is one the reader has already seen in miniature. In section 1.2 the Grassmannian was built by exhibiting its functor as an explicit gluing of affine charts; here the target is not built from scratch but is cut out of a Grassmannian that already exists. Boundedness is what makes this possible: it collapses the infinite-dimensional datum of a quotient sheaf $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ into the finite-dimensional datum of a single surjection on degree-m global sections, and a surjection of a fixed vector space onto a quotient of fixed dimension is exactly a point of a Grassmannian. The flatness and Hilbert-polynomial conditions of the Quot functor then carve a closed subscheme out of that Grassmannian, and closed subschemes of projective schemes are projective. The whole construction is boundedness plus flattening, packaged through the representability of the Grassmannian from theorem 1.2.4.

Throughout, $Y \subseteq \mathbb{P}^n_k$ is a fixed projective scheme, $\mathcal{F}$ a fixed coherent sheaf on Y, and $P \in \mathbb{Q}[t]$ a fixed Hilbert polynomial; $\mathcal{O}_Y(1)$ is the very ample line bundle pulled back from $\mathbb{P}^n_k$, and for a sheaf $\mathcal{G}$ the twist $\mathcal{G}(m)$ denotes $\mathcal{G} \otimes \mathcal{O}_Y(m)$. For a base scheme S, $\mathcal{F}_S$ denotes the pullback of $\mathcal{F}$ to $Y \times_k S = Y_S$ along the projection $Y_S \to Y$, and $\mathcal{O}_{Y_S}(1)$ is the corresponding relatively very ample bundle. The first task is to record precisely how boundedness lets a quotient be recovered from finitely much data.

The mechanism is that for a fixed flat quotient, taking the degree-m piece of the pushforward is an equivalence between quotient sheaves and quotient vector spaces, once m is past the regularity bound. This is where the m-regularity of definition 2.2.1 does its work: it forces the higher cohomology of the relevant twists to vanish, so that the global-sections functor is exact on the sequences in play and commutes with base change. The following proposition isolates the statement.

Proposition 2.3.1: Recovery of a Quotient from Its Degree-m Sections

Let $Y \subseteq \mathbb{P}^n_k$ be projective, $\mathcal{F}$ coherent on Y, and P a Hilbert polynomial. There is an integer m_0, depending only on Y, $\mathcal{F}$, and P, with the following property. For every k-scheme S and every quotient $q\colon \mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ that is flat over S with fiberwise Hilbert polynomial P, and every $m \geq m_0$:

1. the pushforwards $\pi_{S*}\mathcal{F}_S(m)$ and $\pi_{S*}\mathcal{Q}(m)$ along $\pi_S\colon Y_S \to S$ are locally free of finite rank, the second of rank $P(m)$, and their formation commutes with arbitrary base change $T \to S$;
2. the induced map $H^0(q)(m)\colon \pi_{S*}\mathcal{F}_S(m) \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)$ is surjective; and
3. the quotient q is recovered from $H^0(q)(m)$: the surjection $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ is the sheafification of the surjection of graded modules generated in degree m by $H^0(q)(m)$, so q and $H^0(q)(m)$ determine each other.

***Proof*:**
Step 1: The uniform regularity bound. By boundedness (theorem 2.2.5) there is an integer m_0 such that every quotient $\mathcal{F}_s \twoheadrightarrow \mathcal{Q}_s$ with Hilbert polynomial P, over any field-valued point s, together with its kernel $\mathcal{K}_s$, is m_0-regular. Enlarging m_0 if necessary, arrange also that $\mathcal{F}$ itself is m_0-regular. Fix $m \geq m_0$. For a field-valued point s of S, m-regularity of $\mathcal{Q}_s$ gives $H^i(Y_s, \mathcal{Q}_s(m)) = 0$ for all $i > 0$, and the same for $\mathcal{F}_s$ and for the kernel $\mathcal{K}_s = \ker(\mathcal{F}_s \twoheadrightarrow \mathcal{Q}_s)$.

Step 2: Local freeness and base change. The fiberwise vanishing $H^i(Y_s, \mathcal{Q}_s(m)) = 0$ for $i > 0$, holding at every point of S, is exactly the hypothesis of cohomology and base change ([12], cohomology-and-base-change for a flat family): the higher direct images $R^i\pi_{S*}\mathcal{Q}(m)$ vanish for

$i > 0$, the sheaf $\pi_{S*}\mathcal{Q}(m)$ is locally free, and its formation commutes with any base change $T \to S$. Its rank at s is $h^0(Y_s, \mathcal{Q}_s(m)) = \chi(\mathcal{Q}_s(m)) = P(m)$, the last equality because the higher cohomology vanishes and χ computes P. The same argument applied to $\mathcal{F}_S$ and to the kernel $\mathcal{K} = \ker q$ (which is flat over S because in $0 \to \mathcal{K} \to \mathcal{F}_S \to \mathcal{Q} \to 0$ both $\mathcal{F}_S$ and $\mathcal{Q}$ are S-flat, so $\mathcal{Q}$ flat gives $\mathrm{Tor}_1^{\mathcal{O}_S}(\mathcal{Q}, -) = 0$ and hence $\mathcal{K}$ flat) gives that $\pi_{S*}\mathcal{F}_S(m)$ and $\pi_{S*}\mathcal{K}(m)$ are locally free and compatible with base change, of ranks $\chi(\mathcal{F}_s(m))$ and $\chi(\mathcal{F}_s(m)) - P(m)$.

Step 3: Surjectivity on sections. Twist the short exact sequence $0 \to \mathcal{K} \to \mathcal{F}_S \to \mathcal{Q} \to 0$ by $\mathcal{O}_{Y_S}(m)$ and push forward along π_S. Because $R^1\pi_{S*}\mathcal{K}(m) = 0$ by Step 2, the resulting sequence

$$0 \to \pi_{S*}\mathcal{K}(m) \to \pi_{S*}\mathcal{F}_S(m) \xrightarrow{H^0(q)(m)} \pi_{S*}\mathcal{Q}(m) \to 0$$

is exact; in particular $H^0(q)(m)$ is surjective. This proves (1) and (2).

Step 4: Recovery. For m-regular sheaves the graded module $\bigoplus_{\ell \geq m} \pi_{S*}\mathcal{Q}(\ell)$ is generated in degree m over the section (homogeneous coordinate) ring $\bigoplus_{\ell \geq 0} \pi_{S*}\mathcal{O}_{Y_S}(\ell)$, a further consequence of m-regularity (definition 2.2.1): m-regular sheaves are globally generated in every degree $\geq m$, and the multiplication maps $\pi_{S*}\mathcal{Q}(\ell) \otimes \pi_{S*}\mathcal{O}_{Y_S}(1) \to \pi_{S*}\mathcal{Q}(\ell+1)$ are surjective for $\ell \geq m$. The sheaf $\mathcal{Q}$ is therefore reconstructed as the sheaf associated to this graded module, and the surjection q as the sheafification of the surjection on graded modules determined in degree m by $H^0(q)(m)$ together with the module structure of $\mathcal{F}_S$. Two quotients with the same map $H^0(q)(m)$ on degree-m sections have the same kernel submodule in degree m, hence generate the same graded submodule of $\bigoplus_{\ell \geq m} \pi_{S*}\mathcal{F}_S(\ell)$, and so sheafify to the same quotient q; thus q and $H^0(q)(m)$ determine each other, completing the proof.

This proposition is the technical crux, and its meaning is worth stating plainly before it is used. A flat quotient sheaf on Y_S is a priori an object of infinite type: it is a coherent sheaf, carrying data in every degree. Boundedness says that all this data is forced once the degree-m piece is fixed, for a single m that can be chosen uniformly across the entire moduli problem. So the passage $q \mapsto H^0(q)(m)$ trades a sheaf-theoretic object for a linear-algebra object: a surjection of the locally free sheaf $\pi_{S*}\mathcal{F}_S(m)$ onto the locally free sheaf $\pi_{S*}\mathcal{Q}(m)$ of rank $P(m)$. That surjection is a family of quotient vector spaces, which is the datum a Grassmannian classifies. The construction of the Quot scheme is now the observation that this trade is functorial and that the conditions defining the Quot functor become closed conditions on the Grassmannian side.

There is one subtlety the recovery hides, and it is exactly the one boundedness alone does not settle: not every surjection $\pi_{S*}\mathcal{F}_S(m) \twoheadrightarrow V$ onto a rank-$P(m)$ bundle comes from a flat quotient with the right Hilbert polynomial. A general such surjection sheafifies to *some* quotient of $\mathcal{F}_S$, but this quotient need be neither flat over S nor of fiberwise Hilbert polynomial P. Cutting out the surjections that do arise is the job of the flattening stratification, and it is why the Quot scheme is a proper closed subscheme of the Grassmannian rather than the whole of it.

Before the theorem, it helps to name the Grassmannian in play and the natural transformation into it. Fix $m \geq m_0$ and write $H = \pi_{S*}\mathcal{F}_S(m)$ for the base S; when $S = \operatorname{Spec} k$ this is the vector space $H^0(Y, \mathcal{F}(m))$, and for general S it is a locally free sheaf whose formation commutes with base change (proposition 2.3.1). A quotient $q \in \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(S)$ produces, by the proposition, a rank-$P(m)$ locally free quotient $H \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)$. When $S = \operatorname{Spec} k$ this is a point of the Grassmannian $\mathrm{Gr}(P(m),\, h^0(Y, \mathcal{F}(m)))$ of $P(m)$-dimensional quotients of $H^0(Y, \mathcal{F}(m))$; over a general base it is an S-point of that same Grassmannian, by the functorial description of definition 1.2.1. The assignment $q \mapsto H^0(q)(m)$ is therefore a natural transformation of functors, and everything hinges on identifying its image.

Theorem 2.3.2: Representability of the Quot Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective, $\mathcal{F}$ a coherent sheaf on Y, and $P \in \mathbb{Q}[t]$ a Hilbert polynomial. The functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ of definition 2.1.5 is representable by a projective k-scheme $\mathrm{Quot}^P_{\mathcal{F}/Y}$. Concretely, for $m \gg 0$ the natural transformation $q \mapsto H^0(q)(m)$ embeds $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ as a closed subfunctor of the Grassmannian

$$\underline{\mathrm{Gr}}\big(P(m),\, h^0(Y, \mathcal{F}(m))\big)$$

of $P(m)$-dimensional quotients of $H^0(Y, \mathcal{F}(m))$, and $\mathrm{Quot}^P_{\mathcal{F}/Y}$ is the corresponding closed subscheme, hence projective.

***Proof*:**
Fix $m \geq m_0$ as in proposition 2.3.1, write $N = h^0(Y, \mathcal{F}(m))$ and $G = \mathrm{Gr}(P(m), N)$, the Grassmannian of $P(m)$-dimensional quotients of $H^0(Y, \mathcal{F}(m)) \cong k^N$, which is projective by theorem 1.2.4.

Step 1: The natural transformation to the Grassmannian. The assignment $\tau_S\colon q \mapsto \big(H \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)\big)$, where $H = \pi_{S*}\mathcal{F}_S(m)$, is defined for every S by proposition 2.3.1. To land in $\underline{G} = \underline{\mathrm{Gr}}(P(m), N)$ one must identify H with the constant sheaf $\mathcal{O}_S^N$. Over the point $\operatorname{Spec} k$ the source is $H^0(Y, \mathcal{F}(m)) \cong k^N$; over general S, compatibility with base change (proposition 2.3.1(1)) gives $\pi_{S*}\mathcal{F}_S(m) \cong \mathcal{O}_S \otimes_k H^0(Y, \mathcal{F}(m)) = \mathcal{O}_S^N$ canonically, because $\mathcal{F}_S$ is pulled back from $\mathcal{F}$ and the pushforward commutes with the base change $S \to \operatorname{Spec} k$. Under this identification $\tau_S(q)$ is a rank-$P(m)$ locally free quotient of $\mathcal{O}_S^N$, that is, an element of $\underline{G}(S)$. Naturality in S is the base-change compatibility of proposition 2.3.1(1): for $f\colon T \to S$, $f^*\pi_{S*}\mathcal{Q}(m) = \pi_{T*}(f^*\mathcal{Q})(m)$, so τ commutes with pullback. Hence $\tau\colon \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \to \underline{G}$ is a natural transformation.

Step 2: τ is injective, i.e., a monomorphism of functors. For a fixed S, if two quotients q, q' satisfy $\tau_S(q) = \tau_S(q')$ then they induce the same surjection $H^0(q)(m) = H^0(q')(m)$ on degree-m sections, so by the recovery statement proposition 2.3.1(3) they are equal as quotients of $\mathcal{F}_S$. Thus each τ_S is injective, and τ is a monomorphism.

Step 3: The image is a closed subfunctor via the universal quotient. It remains to characterize which S-points of G lie in the image of τ, and to show this is a closed condition. On G sits the universal quotient of definition 1.2.5, a surjection $\mathcal{O}_G^N \twoheadrightarrow \mathcal{V}$ with $\mathcal{V}$ locally free of rank $P(m)$. Identifying $\mathcal{O}_G^N = \mathcal{O}_G \otimes_k H^0(Y, \mathcal{F}(m))$ and pulling $\mathcal{F}$ back to $Y \times_k G = Y_G$, the composite

$$\mathcal{F}_G(m) \;\longleftarrow\; \pi_G^*\big(\mathcal{O}_G \otimes_k H^0(Y, \mathcal{F}(m))\big) \;\twoheadrightarrow\; \pi_G^*\mathcal{V}$$

is not yet a quotient of $\mathcal{F}_G$; rather, the universal surjection $\mathcal{O}_G^N \twoheadrightarrow \mathcal{V}$ determines a graded quotient of the section module $\bigoplus_{\ell \geq m} \pi_{G*}\mathcal{F}_G(\ell)$, and taking the associated sheaf (Proj-sheafification) of the graded module generated in degree m by $\mathcal{O}_G^N \twoheadrightarrow \mathcal{V}$ produces a universal coherent quotient

$$u_G\colon \mathcal{F}_G \twoheadrightarrow \widetilde{\mathcal{Q}}$$

on Y_G. This $\widetilde{\mathcal{Q}}$ is coherent but in general *not* flat over G: over a general point of G the surjection on sections need not sheafify to a flat quotient with Hilbert polynomial P. Apply the flattening stratification (theorem 2.2.7) to the coherent sheaf $\widetilde{\mathcal{Q}}$ on $Y_G \to G$: there is a locally closed stratification $\{G_\alpha\}$ of G such that the restriction of $\widetilde{\mathcal{Q}}$ to Y_{G_α} is flat over G_α with locally constant Hilbert polynomial. Let $G_P \subseteq G$ be the union of those strata on which the constant Hilbert polynomial equals P.

An S-point $g\colon S \to G$ lifts to an S-point of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ under τ if and only if the pullback $g^*\widetilde{\mathcal{Q}}$ is flat over S with fiberwise Hilbert polynomial P: for then $g^*u_G\colon \mathcal{F}_S \twoheadrightarrow g^*\widetilde{\mathcal{Q}}$ is a Quot-family whose degree-m sections recover g, and conversely $\tau_S(q)$ pulls the universal quotient back to q by the recovery statement. By the universal property of the flattening stratification (theorem 2.2.7), $g^*\widetilde{\mathcal{Q}}$ is flat over S with Hilbert polynomial P if and only if g factors through G_P. Thus the image of τ_S is exactly $h_{G_P}(S) \subseteq h_G(S)$, naturally in S.

Step 4: G_P is closed, and representability. The flattening stratification (theorem 2.2.7) presents G_P as a *locally closed* subscheme of G: it is the union of those strata on which $\widetilde{\mathcal{Q}}$ is flat with constant Hilbert polynomial P, and as a finite union of locally closed sets it is constructible and of finite type over k. The content of this step is that G_P is in fact *closed*. A naive semicontinuity count does not deliver this. For a coherent sheaf $\mathcal{E}$ on G the fiber-dimension function $g \mapsto \dim_{\kappa(g)}(\mathcal{E} \otimes \kappa(g))$ is *upper* semicontinuous, so the individual loci $\{h^0(\widetilde{\mathcal{Q}}(\ell) \otimes \kappa(g)) \geq P(\ell)\}$ are closed; but their intersection is strictly larger than G_P, because a stratum $G_{P'}$ carrying a *larger* polynomial $P' > P$ also satisfies $h^0(\widetilde{\mathcal{Q}}(\ell) \otimes \kappa(g)) = P'(\ell) \geq P(\ell)$ for $\ell \gg 0$ and hence meets every such locus. There is no matching pointwise upper bound $h^0(\widetilde{\mathcal{Q}}(\ell) \otimes \kappa(g)) \leq P(\ell)$ available on all of G: at a point where $\widetilde{\mathcal{Q}}$ is not flat, $\otimes\, \kappa(g)$ does not commute with π_{G*}, and sheafification of the degree-m module can strictly *raise* the dimension of the fiber sections, so no such degree-m inequality controls h^0 at a general point. The closedness of G_P is instead a specialization statement, proved by the valuative criterion, and it is exactly here that boundedness does its final work.

Step 4a: closedness under specialization. Since G_P is constructible and G is Noetherian, G_P is closed if and only if it is stable under specialization: whenever a valuation ring dominates a point of G_P, its closed point again lies in G_P. Concretely ([12], valuative criterion for a constructible set on a Noetherian scheme), it suffices to test discrete valuation rings. Let R be a discrete valuation ring with fraction field K and residue field κ, and let $g\colon \operatorname{Spec} R \to G$ be a morphism whose generic point $g_K\colon \operatorname{Spec} K \to G$ factors through G_P; write $g_0\colon \operatorname{Spec} \kappa \to G$ for the closed point. The claim is that g_0 also factors through G_P.

Pull the universal quotient back along g to obtain a coherent quotient $g^*u_G\colon \mathcal{F}_R \twoheadrightarrow g^*\widetilde{\mathcal{Q}}$ on $Y_R = Y \times_k \operatorname{Spec} R$. Over the generic fiber this restricts to $\mathcal{F}_K \twoheadrightarrow (g^*\widetilde{\mathcal{Q}})_K$, which by hypothesis is flat over K with Hilbert polynomial P. Let $\mathcal{K}_R = \ker(g^*u_G)$ and form the *flat limit*: replace $\mathcal{K}_R$ by the saturation

$$\mathcal{K}_R^\flat := \big(\, j_*\,\mathcal{K}_K \,\big) \cap \mathcal{F}_R$$

where $j\colon Y_K \hookrightarrow Y_R$ is the generic-fiber inclusion, that is, the largest subsheaf of $\mathcal{F}_R$ agreeing with $\mathcal{K}_K$ over K and having no sections supported on the special fiber. Equivalently, $\mathcal{K}_R^\flat$ is the kernel of $\mathcal{F}_R \to j_*(\mathcal{F}_K/\mathcal{K}_K)$; the quotient $\mathcal{Q}_R := \mathcal{F}_R/\mathcal{K}_R^\flat$ is then R-flat, because it is a coherent sheaf on Y_R with no associated points on the special fiber, hence torsion-free and so flat over the discrete valuation ring R ([12], flatness over a Dedekind base is torsion-freeness). Its generic fiber is $\mathcal{F}_K/\mathcal{K}_K = (g^*\widetilde{\mathcal{Q}})_K$, with Hilbert polynomial P. Because $\mathcal{Q}_R$ is flat over the connected base $\operatorname{Spec} R$, its fiberwise Hilbert polynomial is constant (proposition 2.1.4), so the special fiber $(\mathcal{Q}_R)_0 = \mathcal{Q}_R \otimes_R \kappa$ is a flat quotient of $\mathcal{F}_\kappa$ with Hilbert polynomial exactly P. This is the flat limit of the generic Quot-family.

Step 4b: the flat limit is classified by g_0. It remains to see that this flat limit is the family that g_0 names, so that $g_0 \in G_P$. Both the flat limit $\mathcal{Q}_R$ and the original pullback $g^*\widetilde{\mathcal{Q}}$ agree over the generic fiber, so their degree-m section modules agree there; since $m \geq m_0$ is past the regularity bound and $\mathcal{V}$ pulls back to the locally free sheaf $g^*\mathcal{V}$ on $\operatorname{Spec} R$, the degree-m data of $\mathcal{Q}_R$ is a rank-$P(m)$ locally free quotient $g^*\mathcal{O}_G^N = \mathcal{O}_R^N \twoheadrightarrow \pi_{R*}\mathcal{Q}_R(m)$ of $\mathcal{O}_R^N$ extending the generic quotient $g_K^*(\mathcal{O}_G^N \twoheadrightarrow \mathcal{V})$. A locally free quotient of $\mathcal{O}_R^N$ over the discrete valuation ring R is determined by its generic fiber, since $\operatorname{Spec} R$ has $\operatorname{Spec} K$ as a dense open and the Grassmannian G is separated (theorem 1.2.4); the two R-points g and the one classifying $\pi_{R*}\mathcal{Q}_R(m)$ agree on $\operatorname{Spec} K$, hence agree, and in particular have the same closed point. So the closed point g_0 classifies precisely the degree-m data of the flat family $(\mathcal{Q}_R)_0$, and by the recovery statement proposition 2.3.1(3) this degree-m datum is $(\mathcal{Q}_R)_0$ itself. Thus $g_0^*\widetilde{\mathcal{Q}} = (\mathcal{Q}_R)_0$ is flat over κ with Hilbert polynomial P, so g_0 factors through G_P. This proves that G_P is stable under specialization, hence closed in G.

The specialization argument also settles the direction of the earlier semicontinuity, and shows the flat limit cannot escape to a larger polynomial: the flat limit carries polynomial P, not some $P' > P$, precisely because flatness over $\operatorname{Spec} R$ pins the fiberwise polynomial to its generic value P. The strata $G_{P'}$ with $P' > P$ that an unconstrained degeneration might have produced are exactly the non-flat sheafifications the flat limit replaces; they lie in $\overline{G_P}$ only as underlying points of G, never as flat Quot-families with the wrong polynomial. By Steps 2, 3, and 4a, τ restricts to an isomorphism of functors $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \cong h_{G_P}$ over the now-closed subscheme G_P, so $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ is representable by

$$\mathrm{Quot}^P_{\mathcal{F}/Y} := G_P$$

a closed subscheme of the projective scheme $G = \mathrm{Gr}(P(m), N)$. A closed subscheme of a projective k-scheme is projective, so $\mathrm{Quot}^P_{\mathcal{F}/Y}$ is projective over k, as we sought to show.

The theorem is the payoff of the entire chapter, and its architecture repays a second look. Every ingredient entered at a definite point. Boundedness (theorem 2.2.5) chose the degree m and guaranteed that a single Grassmannian, of the fixed finite dimension $\mathrm{Gr}(P(m), h^0(\mathcal{F}(m)))$, receives every family in the moduli problem at once. Cohomology and base change (proposition 2.3.1) turned the assignment $q \mapsto H^0(q)(m)$ into a natural transformation of functors, injective by the recovery of a sheaf from its sections. The flattening stratification (theorem 2.2.7) identified the image as a closed subscheme, isolating those points of the Grassmannian whose associated quotient is flat with Hilbert polynomial P. Representability of the Grassmannian (theorem 1.2.4) provided the ambient projective space in which the Quot scheme sits. The pattern, closed subfunctor of a Grassmannian cut out by a flatness condition, is the template for nearly every construction of a moduli space by hand, and it recurs in the geometric-invariant-theory quotients of chapter 3.

One point of the proof deserves emphasis because it is the source of most subtleties in practice. The universal object $\widetilde{\mathcal{Q}}$ on the full Grassmannian G is coherent but not flat; flatness is imposed, not automatic, and the flattening stratification is precisely the device that imposes it. This is the scheme-theoretic content of the slogan that "the Quot scheme parametrizes flat quotients": flatness is a closed condition, and the Quot scheme is where it holds. The reader should not expect the map to the Grassmannian to be surjective; its image is exactly the flat locus, which is a proper closed subset whenever the moduli problem is nontrivial.

The Hilbert scheme is the special case $\mathcal{F} = \mathcal{O}_Y$, and it requires no new argument. A quotient $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{Q}$ is the same datum as a closed subscheme $Z \subseteq Y_S$ flat over S: the kernel is an ideal sheaf $\mathcal{I}_Z$, and $\mathcal{Q} = \mathcal{O}_Z$ is the structure sheaf of the subscheme it cuts out. Applying theorem 2.3.2 verbatim gives the following.

Corollary 2.3.3: Representability of the Hilbert Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective and $P \in \mathbb{Q}[t]$ a Hilbert polynomial. The Hilbert functor $\underline{\mathrm{Hilb}}^P_Y$ of definition 2.1.6 is representable by a projective k-scheme Hilb^P_Y. It is the closed subscheme $\mathrm{Quot}^P_{\mathcal{O}_Y/Y}$ of the Grassmannian $\mathrm{Gr}\big(P(m),\, h^0(Y, \mathcal{O}_Y(m))\big)$ for $m \gg 0$, and so is projective over k.

***Proof*:**
Take $\mathcal{F} = \mathcal{O}_Y$ in definition 2.1.5. A quotient $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{Q}$ flat over S with fiberwise Hilbert polynomial P has kernel an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_{Y_S}$; the closed subscheme $Z \subseteq Y_S$ it defines is flat over S (its structure sheaf $\mathcal{Q} = \mathcal{O}_Z$ is S-flat by hypothesis) with fiberwise Hilbert polynomial P, and conversely such a Z yields the quotient $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{O}_Z$. This bijection is natural in S (pullback of a subscheme corresponds to pullback of its structure quotient), so $\underline{\mathrm{Hilb}}^P_Y = \underline{\mathrm{Quot}}^P_{\mathcal{O}_Y/Y}$ as functors. By theorem 2.3.2 the right-hand functor is represented by the projective scheme $\mathrm{Quot}^P_{\mathcal{O}_Y/Y}$, which is therefore also a representing object for $\underline{\mathrm{Hilb}}^P_Y$; set $\mathrm{Hilb}^P_Y := \mathrm{Quot}^P_{\mathcal{O}_Y/Y}$, completing the proof.

The corollary is the theorem stated and deferred upstream in [12, Grothendieck's Existence Theorem], now discharged: the Hilbert scheme exists and is projective. The identification of a quotient of $\mathcal{O}_Y$ with a closed subscheme is the reason the Hilbert functor is the geometrically most natural instance of the Quot functor, and it is why Hilb^P_Y, not the more general $\mathrm{Quot}^P_{\mathcal{F}/Y}$, is the object one meets first in practice. Every projective variety with Hilbert polynomial P, embedded in the fixed Y, is a point of Hilb^P_Y; the scheme organizes all such subschemes into a single projective parameter space, and its points can degenerate to non-reduced or reducible subschemes, which is exactly the extra generality flat families buy over classical varieties.

The construction produced not only the representing scheme but, by the very meaning of representability, a universal family living over it. Representability says $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \cong h_{\mathrm{Quot}^P_{\mathcal{F}/Y}}$, and the identity morphism of $\mathrm{Quot}^P_{\mathcal{F}/Y}$ corresponds under this isomorphism to a distinguished quotient over the Quot scheme itself. Naming it is the last piece of the construction, because it is the object from which every family is obtained by pullback.

Definition 2.3.4: The Universal Quotient and Universal Subscheme

Let $\mathrm{Quot}^P_{\mathcal{F}/Y}$ represent $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (theorem 2.3.2), and write $Q = \mathrm{Quot}^P_{\mathcal{F}/Y}$. The *universal quotient* is the surjection

$$u\colon \mathcal{F}_Q \twoheadrightarrow \mathcal{U}$$

of coherent sheaves on $Y_Q = Y \times_k Q$, flat over Q with fiberwise Hilbert polynomial P, that corresponds under the isomorphism $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \cong h_Q$ of theorem 2.3.2 to the identity morphism $\mathrm{id}_Q \in h_Q(Q)$. In the Hilbert case $\mathcal{F} = \mathcal{O}_Y$, the kernel of u is the ideal sheaf of a closed subscheme

$$\mathcal{Z} \subseteq Y \times_k \mathrm{Hilb}^P_Y$$

flat over Hilb^P_Y with fiberwise Hilbert polynomial P, called the *universal subscheme*, and $\mathcal{U} = \mathcal{O}_{\mathcal{Z}}$ is its structure sheaf.

The universal property is what earns the name, and it is worth stating explicitly because it is used constantly. For every k-scheme S and every Quot-family $q\colon \mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ (flat over S, fiberwise Hilbert polynomial P), there is a unique morphism $\varphi_q\colon S \to \mathrm{Quot}^P_{\mathcal{F}/Y}$ such that q is the pullback of the universal quotient:

$$(\mathrm{id}_Y \times \varphi_q)^* u = q$$

as quotients of $\mathcal{F}_S$. Every family is a pullback of the single universal family, in exactly one way. In the Hilbert case this reads: every flat family $Z \subseteq Y_S$ of closed subschemes with Hilbert polynomial P is the pullback $Z = (\mathrm{id}_Y \times \varphi)^{-1}(\mathcal{Z})$ of the universal subscheme along a unique classifying morphism $\varphi\colon S \to \mathrm{Hilb}^P_Y$. This is precisely the statement that Hilb^P_Y is a *fine* moduli space in the sense of definition 1.1.3: the universal subscheme is a single geometric object over Hilb^P_Y from which all families are cut by pullback, exactly as the universal quotient bundle plays that role over the Grassmannian in definition 1.2.5.

The classifying morphism φ_q is not an abstraction; it can be computed. For a family q over S, φ_q is the morphism $S \to G_P \subseteq \mathrm{Gr}(P(m), N)$ classifying the rank-$P(m)$ quotient $\pi_{S*}\mathcal{F}_S(m) \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)$ on degree-m sections, by the proof of theorem 2.3.2. So the universal property is the Grassmannian's universal property (theorem 1.2.4) restricted to the flat locus, and the effective content of "classify a subscheme" is "classify the $P(m)$-dimensional space of its degree-m equations". The following example makes the whole construction concrete for a family of two points on a line.

Example 2.3.5: The Hilbert Scheme of Two Points on $\mathbb{P}^1$

Take $Y = \mathbb{P}^1_k$ and $P(t) = 2$, the constant polynomial: a closed subscheme $Z \subseteq \mathbb{P}^1$ with $\chi(\mathcal{O}_Z(m)) = 2$ for all m is a length-2 subscheme, i.e. two distinct points or one double point. The claim is $\mathrm{Hilb}^2_{\mathbb{P}^1} \cong \mathbb{P}^2$, and the construction of theorem 2.3.2 exhibits the isomorphism.

A length-2 subscheme $Z \subseteq \mathbb{P}^1 = \mathrm{Proj}\, k[x_0, x_1]$ is cut out by a single homogeneous ideal, and its ideal is generated in degree 2 by one quadratic form $f = a_0x_0^2 + a_1x_0x_1 + a_2x_1^2$ (the vanishing of a nonzero binary quadric defines a length-2 subscheme, namely its two roots with multiplicity). Two forms define the same subscheme exactly when they are proportional, so the subschemes are the points $[a_0 : a_1 : a_2]$ of $\mathbb{P}^2 = \mathbb{P}(H^0(\mathbb{P}^1, \mathcal{O}(2)))$. This is the classifying data of the theorem made explicit: $H^0(\mathbb{P}^1, \mathcal{O}(2))$ is 3-dimensional, and a length-2 subscheme is the same as a 1-dimensional subspace of quadrics vanishing on it, equivalently a 2-dimensional quotient

$$H^0(\mathbb{P}^1, \mathcal{O}(2)) \twoheadrightarrow H^0(Z, \mathcal{O}_Z(2))$$

which is a point of $\mathrm{Gr}(2, 3) \cong \mathbb{P}^2$ (a rank-2 quotient of a rank-3 space, dual to the line of equations). Here $m = 2$ already works: $\mathcal{O}_Z(2)$ has no higher cohomology and $h^0(\mathcal{O}_Z(2)) = 2 = P(2)$. The flattening stratification imposes no further condition, because for $Y = \mathbb{P}^1$ every 1-dimensional space of quadrics defines a flat length-2 family, so $\mathrm{Hilb}^2_{\mathbb{P}^1} = \mathrm{Gr}(2, 3) = \mathbb{P}^2$ with no closed subfunctor to cut.

The universal subscheme $\mathcal{Z} \subseteq \mathbb{P}^1 \times \mathbb{P}^2$ is the incidence locus $\{([x], [a]) : a_0x_0^2 + a_1x_0x_1 + a_2x_1^2 = 0\}$, flat of degree 2 over $\mathbb{P}^2$. Its fiber over $[a] \in \mathbb{P}^2$ is the subscheme $Z_{[a]}$ cut by the quadric $[a]$: two distinct points when the discriminant $a_1^2 - 4a_0a_2 \neq 0$, and a single double point (a point with a tangent direction) on the discriminant conic $a_1^2 = 4a_0a_2$. The double points form a conic in $\mathbb{P}^2$, and the support map $Z \mapsto (\text{support of } Z)$ is the Hilbert-Chow morphism $\mathrm{Hilb}^2_{\mathbb{P}^1} = \mathbb{P}^2 \to \mathrm{Sym}^2\mathbb{P}^1$, sending a length-2 subscheme to its support cycle; here $\mathrm{Sym}^2\mathbb{P}^1$ is itself a $\mathbb{P}^2$, presented as the S_2-quotient $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathrm{Sym}^2\mathbb{P}^1 = \mathbb{P}^2$ of the ordered pairs of points, so the Hilbert-Chow map is an isomorphism between these two copies of $\mathbb{P}^2$. This morphism will reappear in section 2.5. So the abstract construction returns, in this case, the classical parameter space of binary quadratic forms.

The example is the pattern in miniature: choose m (here $m = 2$), read the subscheme off its space of degree-m equations, land in a Grassmannian, and check the flatness condition (here vacuous). In nontrivial cases the flatness cut is a proper closed condition, but the shape of the argument does not change. The existence theorem, that the Hilbert and Quot functors are representable by projective schemes carrying universal families, is now established in full.

Exercise 2.3.1

1. Let $Y = \mathbb{P}^n_k$ and let P be the constant polynomial $P(t) = 1$, so $\underline{\mathrm{Hilb}}^1_{\mathbb{P}^n}$ parametrizes length-1 subschemes, i.e. reduced points. Using the construction of theorem 2.3.2, identify $\mathrm{Hilb}^1_{\mathbb{P}^n}$ explicitly as a familiar projective scheme, and describe the universal subscheme $\mathcal{Z} \subseteq \mathbb{P}^n \times \mathrm{Hilb}^1_{\mathbb{P}^n}$. (Hint: a length-1 subscheme is a k-point of $\mathbb{P}^n$; compare with example 2.3.5 and the case $\mathrm{Gr}(1, n+1)$.)

2. In the proof of theorem 2.3.2 the universal coherent quotient $\widetilde{\mathcal{Q}}$ on the full Grassmannian G is asserted to be coherent but not in general flat over G. Explain, using the example $Y = \mathbb{P}^1$, $\mathcal{F} = \mathcal{O}_{\mathbb{P}^1}$, $P(t) = 2$, why some surjection $H^0(\mathcal{O}(2)) \twoheadrightarrow V$ onto a 2-dimensional space might fail to sheafify to a flat length-2 quotient, or argue that in this specific example every such surjection does succeed and so $G_P = G$. Relate your answer to why $\mathrm{Hilb}^2_{\mathbb{P}^1} = \mathbb{P}^2$ with no closed condition imposed.

3. Prove that $\mathrm{Quot}^P_{\mathcal{F}/Y}$ is separated and of finite type over k directly from theorem 2.3.2, without invoking projectivity: deduce both from the fact that it is a closed subscheme of a Grassmannian, and state which property of the Grassmannian each inherits from.

4. Let $Y = \mathbb{P}^n_k$ and $P(t) = \binom{t+n}{n} - \binom{t+n-d}{n}$, the Hilbert polynomial of a degree-d hypersurface in $\mathbb{P}^n$. Show that the universal property of definition 2.3.4 identifies $\mathrm{Hilb}^P_{\mathbb{P}^n}$ with the projective space $\mathbb{P}(H^0(\mathbb{P}^n, \mathcal{O}(d)))$ of degree-d forms, and describe the classifying morphism $\varphi\colon S \to \mathrm{Hilb}^P_{\mathbb{P}^n}$ associated to a flat family of hypersurfaces over S. (You may cite the computation of P for a hypersurface; the content is the identification of the parameter space and the universal family.)

5. The natural transformation $\tau\colon \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \to \underline{\mathrm{Gr}}(P(m), N)$ of theorem 2.3.2 depends on the choice of $m \geq m_0$. Show that the representing scheme $\mathrm{Quot}^P_{\mathcal{F}/Y}$ does not depend on this choice: prove that for two admissible values m, m' the two closed subschemes $G_P \subseteq \mathrm{Gr}(P(m), N)$ and $G'_P \subseteq \mathrm{Gr}(P(m'), N')$ are canonically isomorphic, and explain why this is forced by representability rather than needing a separate verification.

6. Explain why the universal subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}^P_Y$ of definition 2.3.4 being flat over Hilb^P_Y is not an extra hypothesis but a consequence of the construction, and identify where in the proof of theorem 2.3.2 this flatness is arranged. Contrast this with the universal coherent quotient $\widetilde{\mathcal{Q}}$ on the ambient Grassmannian, which is not flat.

2.4 The Tangent Space and Local Structure

The previous section produced the Hilbert scheme Hilb^P_Y as a projective scheme representing the functor $\underline{\mathrm{Hilb}}^P_Y$ (corollary 2.3.3), together with its universal subscheme $\mathcal{Z}$ (definition 2.3.4). Once a moduli problem is represented by a geometric object, the payoff is that all of the object's geometry can be read off functorially. The most immediate piece of geometry is the local structure at a point:

the tangent space at a point $[Z]$, and, one degree deeper, whether the scheme is smooth there and of what dimension. This section carries out the first such computation, and it is the computation that opens the door to all of deformation theory.

The bridge is a single principle already available from the functor-of-points tangent space of definition 1.5.1: because Hilb^P_Y represents a functor, its tangent space at a k-point $[Z]$ is not a differential-geometric object to be constructed by hand but a set of maps out of the dual numbers, computed directly from the functor. A tangent vector at $[Z]$ is a family over $\operatorname{Spec} k[\varepsilon]$ whose special fiber is Z, that is, a way of pushing the subscheme Z infinitesimally inside Y. The content of the section is that these infinitesimal pushes are exactly the global sections of one sheaf on Z, the normal sheaf, and that the obstruction to thickening such a push to a genuine curve of deformations lives in the next cohomology group of the same sheaf. This dictionary between the geometry of Hilb^P_Y and the cohomology of $N_{Z/Y}$ is the prototype for the tangent-obstruction formalism that the deformation-theory Part of this book develops systematically.

The object that governs how Z sits inside Y to first order is the normal sheaf. For a closed subscheme it is built from the ideal sheaf, and the construction is a direct scheme-theoretic analogue of the normal bundle of a submanifold: the conormal sheaf $\mathcal{I}_Z/\mathcal{I}_Z^2$ records first-order functions vanishing on Z, and its dual records first-order directions transverse to Z.

Definition 2.4.1: Normal Sheaf of a Closed Subscheme

Let $Z \subseteq Y$ be a closed subscheme of a scheme Y, with ideal sheaf $\mathcal{I}_Z \subseteq \mathcal{O}_Y$. The *conormal sheaf* of Z in Y is the $\mathcal{O}_Z$-module

$$\mathcal{N}^\vee_{Z/Y} \;=\; \mathcal{I}_Z/\mathcal{I}_Z^2$$

where the $\mathcal{O}_Z = \mathcal{O}_Y/\mathcal{I}_Z$-module structure is the one inherited from the $\mathcal{O}_Y$-module structure on $\mathcal{I}_Z$ (multiplication by $\mathcal{I}_Z$ acts as zero on the quotient, so the action factors through $\mathcal{O}_Z$). The *normal sheaf* of Z in Y is its $\mathcal{O}_Z$-dual,

$$N_{Z/Y} \;=\; \mathcal{H}om_{\mathcal{O}_Z}\big(\mathcal{I}_Z/\mathcal{I}_Z^2,\, \mathcal{O}_Z\big)$$

a coherent $\mathcal{O}_Z$-module.

The conormal sheaf $\mathcal{I}_Z/\mathcal{I}_Z^2$ is the algebraic shadow of the classical conormal bundle: a local defining function $f \in \mathcal{I}_Z$ has a well-defined class modulo $\mathcal{I}_Z^2$, and that class is precisely its first-order behaviour transverse to Z, its "differential in the normal directions." Dualizing turns cotangent data into tangent data, so a section of $N_{Z/Y}$ over an open set assigns to each local equation of Z a function on Z, subject to the Leibniz-type constraint of being $\mathcal{O}_Z$-linear. The interpretation to keep in mind throughout the section is that such a section is a recipe for perturbing each defining equation $f \mapsto f + \varepsilon\,\varphi(f)$ to first order, consistently across overlaps. When Z and Y are both smooth, the normal sheaf is locally free of rank $\dim Y - \dim Z$ and reduces to the normal bundle of differential geometry; in general it can fail to be locally free, and that failure is exactly where the local geometry of Hilb^P_Y becomes interesting.

Two computations make the definition concrete and will be reused below.

Example 2.4.2: The Normal Sheaf of a Hypersurface
Let Y be a scheme and let $Z = V(f) \subseteq Y$ be an effective Cartier divisor, the zero scheme of a single regular function f (or, locally, a nonzerodivisor) generating $\mathcal{I}_Z$. Then $\mathcal{I}_Z \cong \mathcal{O}_Y(-Z)$ is an invertible sheaf, and multiplication by f gives an isomorphism $\mathcal{O}_Z \xrightarrow{\sim} \mathcal{I}_Z/\mathcal{I}_Z^2$ sending $1 \mapsto \bar{f}$. Consequently

$$\mathcal{I}_Z/\mathcal{I}_Z^2 \;\cong\; \mathcal{O}_Y(-Z)\big|_Z, \qquad N_{Z/Y} \;\cong\; \mathcal{O}_Y(Z)\big|_Z$$

the restriction to Z of the line bundle $\mathcal{O}_Y(Z)$. For the concrete case $Y = \mathbb{P}^n$ and $Z \subseteq \mathbb{P}^n$ a hypersurface of degree d, the normal sheaf is $N_{Z/\mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}^n}(d)\big|_Z = \mathcal{O}_Z(d)$. A global section of $N_{Z/Y}$ is then a function on Z scaling the single equation f, matching the intuition that a hypersurface is deformed to first order by perturbing its one defining equation.

Example 2.4.3: The Normal Sheaf of a Reduced Point
Let Y be a smooth n-dimensional variety over k and let $Z = \{p\}$ be a single reduced k-point, so $\mathcal{O}_Z = k$. Choosing a regular system of parameters $t_1, \ldots, t_n$ at p, the ideal $\mathcal{I}_Z = \mathfrak{m}_p$ satisfies $\mathfrak{m}_p/\mathfrak{m}_p^2 \cong k^{\oplus n}$, the Zariski cotangent space at p. Hence

$$N_{Z/Y} \;=\; \mathrm{Hom}_k\big(\mathfrak{m}_p/\mathfrak{m}_p^2,\, k\big) \;\cong\; T_pY$$

the Zariski tangent space of Y at p, an n-dimensional k-vector space. A global section is a single tangent vector at p, matching the intuition that one moves a point inside Y by choosing a direction to move it in.

With the normal sheaf in hand, the tangent space of the Hilbert scheme can be computed. Recall from definition 1.5.1 that for a functor $F\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ with a distinguished point $\xi \in F(\operatorname{Spec} k)$, the tangent space is the fiber

$$T_\xi F \;=\; \big\{\, \eta \in F(\operatorname{Spec} k[\varepsilon]) \;\big|\; \eta \mapsto \xi \text{ under } F(\operatorname{Spec} k[\varepsilon]) \to F(\operatorname{Spec} k) \,\big\}$$

where $k[\varepsilon] = k[t]/(t^2)$ is the ring of dual numbers and the map is induced by $k[\varepsilon] \to k$, $\varepsilon \mapsto 0$; proposition 1.5.2 equips this set with a canonical k-vector space structure. When $F = \underline{\mathrm{Hilb}}_Y^P$ is representable by Hilb_Y^P (corollary 2.3.3), the Yoneda lemma identifies $T_\xi F$ with the Zariski tangent space $T_{[Z]}\,\mathrm{Hilb}_Y^P = (\mathfrak{m}_{[Z]}/\mathfrak{m}_{[Z]}^2)^\vee$, so the functor computation *is* the scheme computation. Everything now reduces to unwinding what an element of $\underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$ restricting to $[Z]$ is.

Theorem 2.4.4: Tangent Space of the Hilbert Scheme

Let $Y \subseteq \mathbb{P}_k^n$ be a projective scheme and let $[Z] \in \mathrm{Hilb}_Y^P(k)$ be the point corresponding to a closed subscheme $Z \subseteq Y$ with Hilbert polynomial P. There is a canonical isomorphism of k-vector spaces

$$T_{[Z]}\,\mathrm{Hilb}_Y^P \;\cong\; H^0\big(Z,\, N_{Z/Y}\big)$$

between the Zariski tangent space of Hilb_Y^P at $[Z]$ and the global sections of the normal sheaf. The same statement holds for the Quot scheme with $N_{Z/Y}$ replaced by $\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{K}, \mathcal{Q})$, where $\mathcal{K}$ is the kernel of the universal quotient $\mathcal{F} \twoheadrightarrow \mathcal{Q}$.

***Proof*:**
By representability (corollary 2.3.3) and the Yoneda identification recalled above, it suffices to compute $\underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$ over $[Z]$ and exhibit a canonical bijection to $H^0(Z, N_{Z/Y})$ that respects the vector space structures.

Step 1: Identify a tangent vector with a flat first-order deformation. By definition of the Hilbert functor (definition 2.1.6), an element of $\underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$ is a closed subscheme

$$\widetilde{Z} \subseteq Y \times_k \operatorname{Spec} k[\varepsilon] = Y_\varepsilon$$

flat over $\operatorname{Spec} k[\varepsilon]$ with fiberwise Hilbert polynomial P. Restricting along the closed immersion $\operatorname{Spec} k \hookrightarrow \operatorname{Spec} k[\varepsilon]$ (cutting by $\varepsilon = 0$) recovers the special fiber; requiring the tangent vector to lie over $[Z]$ is requiring this special fiber to be Z. Thus a tangent vector is exactly a flat family $\widetilde{Z} \subseteq Y_\varepsilon$ restricting to $Z \subseteq Y$ over the closed point, that is, a *first-order deformation of Z inside Y*.

Step 2: Translate flatness into a splitting of ideals. The subscheme $\widetilde{Z}$ is determined by its ideal sheaf $\mathcal{I}_{\widetilde{Z}} \subseteq \mathcal{O}_{Y_\varepsilon}$. Because $\mathcal{O}_{Y_\varepsilon} = \mathcal{O}_Y \otimes_k k[\varepsilon] = \mathcal{O}_Y \oplus \varepsilon\mathcal{O}_Y$ as an $\mathcal{O}_Y$-module, and the special fiber of $\widetilde{Z}$ is Z, the multiplication-by-ε sequence

$$0 \longrightarrow \mathcal{O}_Y \xrightarrow{\cdot\,\varepsilon} \mathcal{O}_{Y_\varepsilon} \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

is exact. The key point is that $\widetilde{Z}$ is flat over $k[\varepsilon]$ if and only if tensoring the structure sequence $0 \to \mathcal{I}_{\widetilde{Z}} \to \mathcal{O}_{Y_\varepsilon} \to \mathcal{O}_{\widetilde{Z}} \to 0$ with the residue field $k = k[\varepsilon]/(\varepsilon)$ keeps it exact on the left, that is, if and only if $\operatorname{Tor}_1^{k[\varepsilon]}(\mathcal{O}_{\widetilde{Z}}, k) = 0$. Unwinding the standard criterion for flatness over the dual numbers (a $k[\varepsilon]$-module M is flat if and only if the multiplication map $\varepsilon\colon M/\varepsilon M \to \varepsilon M$ is an isomorphism), flatness of $\mathcal{O}_{\widetilde{Z}}$ is equivalent to the requirement that

$$\varepsilon \cdot \mathcal{O}_{\widetilde{Z}} \cong \mathcal{O}_{\widetilde{Z}}/\varepsilon\mathcal{O}_{\widetilde{Z}} = \mathcal{O}_Z$$

that is, that reduction mod ε induces an isomorphism $\varepsilon\mathcal{O}_{\widetilde{Z}} \cong \mathcal{O}_Z$. Concretely: locally, if $\mathcal{I}_Z$ is generated by $f_1, \dots, f_r$, a flat lift is generated by elements $f_i + \varepsilon g_i$ with $g_i \in \mathcal{O}_Y$, and flatness forces the ambiguity in the g_i to be exactly by elements of $\mathcal{I}_Z$.

Step 3: Assemble the local data into a section of the normal sheaf. A flat first-order deformation, described locally by generators $f_i + \varepsilon g_i$ of $\mathcal{I}_{\widetilde{Z}}$, defines an $\mathcal{O}_Y$-linear map

$$\varphi\colon \mathcal{I}_Z \longrightarrow \mathcal{O}_Z, \qquad f_i \longmapsto \bar{g}_i$$

as follows. Given $f = \sum a_i f_i \in \mathcal{I}_Z$ with $a_i \in \mathcal{O}_Y$, the corresponding lift is $\sum a_i(f_i + \varepsilon g_i) = f + \varepsilon \sum a_i g_i$, so the "first-order part" is $\sum a_i g_i$, read modulo $\mathcal{I}_Z$ to land in $\mathcal{O}_Z$. Two facts must be checked, and both follow from flatness as pinned down in Step 2. First, φ is well defined: if $\sum a_i f_i = 0$ then the syzygy must be respected to first order, which is exactly the statement that $\sum a_i g_i \in \mathcal{I}_Z$, so φ does not depend on the presentation. Second, φ kills $\mathcal{I}_Z^2$: for $f, f' \in \mathcal{I}_Z$ the product ff' lifts to $(f + \varepsilon(\cdots))(f' + \varepsilon(\cdots)) = ff' + \varepsilon(f \cdot (\cdots) + f' \cdot (\cdots))$, whose first-order part lies in $\mathcal{I}_Z$ and so maps to 0 in $\mathcal{O}_Z$. Therefore φ descends to an $\mathcal{O}_Z$-linear map

$$\varphi\colon \mathcal{I}_Z/\mathcal{I}_Z^2 \longrightarrow \mathcal{O}_Z$$

that is, a global section $\varphi \in H^0(Z, N_{Z/Y})$. Geometrically φ is precisely the "perturb each equation $f \mapsto f + \varepsilon\,\varphi(f)$" recipe anticipated after definition 2.4.1.

Step 4: Invert the construction. Conversely, a global section $\varphi \in H^0(Z, N_{Z/Y}) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z)$ produces a first-order deformation. Lift φ to an $\mathcal{O}_Y$-linear map $\mathcal{I}_Z \to \mathcal{O}_Y$ (choosing local representatives $g_i \in \mathcal{O}_Y$ of $\varphi(\bar{f}_i)$) and set, locally,

$$\mathcal{I}_{\widetilde{Z}} \;=\; \big(f_i + \varepsilon\, g_i \;:\; i = 1, \ldots, r\big) \;\subseteq\; \mathcal{O}_{Y_\varepsilon}$$

That φ vanishes on $\mathcal{I}_Z^2$ guarantees these local ideals patch to a global ideal sheaf $\mathcal{I}_{\widetilde{Z}}$ independent of the lift of φ (a different lift changes each g_i by an element of $\mathcal{I}_Z$, which does not change the ideal $(f_i + \varepsilon g_i)$ since $\varepsilon \mathcal{I}_Z \subseteq \mathcal{I}_{\widetilde{Z}}$ already). The resulting $\widetilde{Z}$ is flat over $k[\varepsilon]$ by the criterion of Step 2 and restricts to Z over $\varepsilon = 0$. Since $\widetilde{Z} \to \operatorname{Spec} k[\varepsilon]$ is flat with special fiber of Hilbert polynomial P, the Hilbert polynomial is preserved (proposition 2.1.4), so $\widetilde{Z} \in \underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$. This is inverse to the construction of Step 3.

Step 5: Compatibility with the vector space structures. The zero section $\varphi = 0$ gives $\mathcal{I}_{\widetilde{Z}} = \mathcal{I}_Z \otimes_k k[\varepsilon]$, the trivial (constant) deformation, which is the base point of the tangent space. Under the $k[\varepsilon]$-module addition used to define the vector space structure in proposition 1.5.2, adding two deformations corresponds to adding the associated maps $\mathcal{I}_Z \to \mathcal{O}_Z$, and scaling $\varepsilon \mapsto c\varepsilon$ for $c \in k$ scales φ by c; both are immediate from the local formula $f_i \mapsto f_i + \varepsilon g_i$. Hence the bijection of Steps 3 and 4 is k-linear, giving the claimed isomorphism $T_{[Z]} \operatorname{Hilb}_Y^P \cong H^0(Z, N_{Z/Y})$.

For the Quot scheme, a point corresponds to a quotient $\mathcal{F} \twoheadrightarrow \mathcal{Q}$ with kernel $\mathcal{K}$, and the identical argument, replacing "ideal sheaf $\mathcal{I}_Z$" by "kernel subsheaf $\mathcal{K}$" and "$\mathcal{O}_Z$" by "$\mathcal{Q}$", identifies first-order deformations of the quotient with $\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{K}, \mathcal{Q})$; the Hilbert case is the special case $\mathcal{F} = \mathcal{O}_Y$, $\mathcal{Q} = \mathcal{O}_Z$, $\mathcal{K} = \mathcal{I}_Z$, where $\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z, \mathcal{O}_Z) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z) = H^0(N_{Z/Y})$ because any $\mathcal{O}_Y$-map to the $\mathcal{O}_Z$-module $\mathcal{O}_Z$ automatically kills $\mathcal{I}_Z \cdot \mathcal{I}_Z$, completing the proof.

The theorem is the first theorem of deformation theory in the book, and it should be read as a template rather than an isolated computation. Its shape recurs throughout the deformation-theory Part: a moduli functor is probed by maps out of $\operatorname{Spec} k[\varepsilon]$, an element of $F(\operatorname{Spec} k[\varepsilon])$ over a fixed point is unwound into linear-algebraic data on the object itself, and that data organizes into the global sections of a coherent sheaf (or, more generally, into a cohomology group). Here the sheaf is $N_{Z/Y}$ and the group is H^0; for deformations of an abstract smooth variety X the sheaf will be the tangent sheaf T_X and the group $H^1(X, T_X)$; for deformations of a coherent sheaf $\mathcal{F}$ it will be $\mathrm{Ext}^1(\mathcal{F}, \mathcal{F})$ ([10]). In the normal-sheaf computation the ambient Y pins everything down, so a first-order deformation is packaged by H^0 of a coherent sheaf alone, with no higher Ext term required.

Two features of the canonical isomorphism deserve emphasis. First, it is entirely local-to-global: whether a first-order deformation exists at all is a purely local question (the Hom sheaf $N_{Z/Y}$ is defined stalk by stalk), whereas the tangent space measures which local perturbations glue to a global one, and that gluing is exactly the passage from the sheaf to its space of global sections. Second, the isomorphism is canonical, so it transports functoriality: a morphism of Hilbert schemes induced by an inclusion $Y \hookrightarrow Y'$ corresponds, on tangent spaces, to the natural map $H^0(N_{Z/Y}) \to H^0(N_{Z/Y'})$ coming from the surjection $\mathcal{I}_{Z/Y'} \twoheadrightarrow \mathcal{I}_{Z/Y}$ of conormal data.

The tangent space measures first-order deformations, but not every first-order deformation extends to an actual curve of subschemes. A tangent vector is a family over $\operatorname{Spec} k[\varepsilon] = \operatorname{Spec} k[t]/(t^2)$; extending it means finding a compatible family over $\operatorname{Spec} k[t]/(t^3)$, then $\operatorname{Spec} k[t]/(t^4)$, and so on, ultimately over a smooth curve or a power series ring. The failure to extend from order n to order $n+1$ is measured by a class in a second cohomology group, the obstruction space. For the Hilbert scheme this space is $H^1(Z, N_{Z/Y})$, and its vanishing forces Hilb^P_Y to be as smooth as its tangent space predicts.

Proposition 2.4.5: The Obstruction Space and Smoothness of the Hilbert Scheme

Let $[Z] \in \mathrm{Hilb}^P_Y(k)$ with $Z \subseteq Y$ as above.

1. Fix $n \geq 2$ and let $\widetilde{Z}_n$ be a flat deformation of Z over $\operatorname{Spec} k[t]/(t^n)$. There is a canonical *obstruction class*
$$\mathrm{ob}(\widetilde{Z}_n) \;\in\; H^1\big(Z, N_{Z/Y}\big)$$
that vanishes if and only if $\widetilde{Z}_n$ extends to a flat deformation over $\operatorname{Spec} k[t]/(t^{n+1})$; when it vanishes, the set of such extensions is a torsor under $H^0(Z, N_{Z/Y})$.

2. If $H^1(Z, N_{Z/Y}) = 0$, then Hilb^P_Y is smooth at $[Z]$ of dimension
$$\dim_{[Z]} \mathrm{Hilb}^P_Y \;=\; h^0\big(Z, N_{Z/Y}\big) \;=\; \dim_k H^0(Z, N_{Z/Y})$$

The obstruction machinery in its full generality, including the precise construction of $\mathrm{ob}(\widetilde{Z}_n)$ as the class of a Čech-type cocycle recording the discrepancy between local lifts, belongs to the deformation theory developed in the next Part of this book; the general obstruction calculus and the proof that vanishing H^1 yields formal smoothness are given there in the setting of arbitrary deformation functors. The statement is recorded here because it is the payoff of the tangent-space computation for the reader's immediate use: it converts a cohomological vanishing into a concrete geometric conclusion about Hilb^P_Y. The mechanism to keep in mind is the "lift-and-compare" picture. Cover Z by opens on which the order-n deformation extends to order $n+1$ (each local piece extends because the local rings are smooth enough, or by the local triviality of the normal-sheaf data); on overlaps two such local extensions differ by a section of $N_{Z/Y}$; the resulting collection of differences is a 1-cocycle whose class in $H^1(Z, N_{Z/Y})$ is ob. The global extension exists precisely when this class is a coboundary, that is, when the local extensions can be adjusted to agree on overlaps. When H^1 vanishes there is no obstruction at any order, the deformation extends to all orders and, by projectivity and the formal criterion, Hilb^P_Y is smooth at $[Z]$; its dimension is then the dimension of the tangent space, $h^0(N_{Z/Y})$.

The proposition also explains why the Hilbert scheme is so often singular. There is no upper bound on $\dim H^1(N_{Z/Y})$ across all subschemes with a fixed Hilbert polynomial, and when it is nonzero the actual dimension of Hilb^P_Y at $[Z]$ can be strictly smaller than $h^0(N_{Z/Y})$, with the tangent space overcounting deformations that do not integrate. This is the algebraic origin of the pathologies of Hilbert schemes: Mumford's example (in his 1962 paper "Further pathologies in algebraic geometry", *Amer. J. Math.* **84**) of a component of the Hilbert scheme of space curves in $\mathbb{P}^3$ that is everywhere non-reduced, and Vakil's "Murphy's law" theorem (from "Murphy's law in algebraic geometry: Badly-behaved deformation spaces", *Invent. Math.* **164** (2006)) that Hilbert schemes realize arbitrarily bad singularity types, both trace to a large obstruction space $H^1(N_{Z/Y})$ obstructing the integration of tangent vectors. The clean cases below are exactly those where H^1 vanishes for dimensional reasons.

The computation becomes concrete when the normal sheaf splits or is a line bundle, so that its cohomology is computable by hand. The following worked case treats a complete intersection, where the conormal sequence is as simple as possible, and specializes to a finite set of points, the running touchstone of this chapter.

Example 2.4.6: Tangent Space of a Complete Intersection and of Points

Let $Y = \mathbb{P}^n_k$ and let $Z = V(f_1, \dots, f_c) \subseteq \mathbb{P}^n$ be a smooth complete intersection of c hypersurfaces of degrees $d_1, \dots, d_c$, so Z has codimension c and dimension $n - c$. The Koszul resolution shows that Z is regularly embedded, so the conormal sheaf is locally free and splits as a direct sum of the individual conormal line bundles:

$$\mathcal{I}_Z/\mathcal{I}_Z^2 \;\cong\; \bigoplus_{i=1}^{c} \mathcal{O}_Z(-d_i)$$

each summand generated by the class of f_i (example 2.4.2 applied to each equation). Dualizing,

$$N_{Z/\mathbb{P}^n} \;\cong\; \bigoplus_{i=1}^{c} \mathcal{O}_Z(d_i)$$

and the tangent space to the Hilbert scheme is the direct sum of the sections of these line bundles:

$$T_{[Z]} \operatorname{Hilb}^P_{\mathbb{P}^n} \;\cong\; H^0\big(Z, N_{Z/\mathbb{P}^n}\big) \;\cong\; \bigoplus_{i=1}^{c} H^0\big(Z, \mathcal{O}_Z(d_i)\big)$$

Each summand $H^0(Z, \mathcal{O}_Z(d_i))$ is the space of degree-d_i perturbations of the i-th equation, restricted to Z: a first-order deformation of a complete intersection is precisely a first-order wiggle of each defining polynomial, and $h^0(\mathcal{O}_Z(d_i))$ counts the independent such wiggles modulo those that do not move Z.

Two specializations make this quantitative.

1. *A single hypersurface.* Take $c = 1$, so $Z \subseteq \mathbb{P}^n$ is a smooth hypersurface of degree d. Then $N_{Z/\mathbb{P}^n} = \mathcal{O}_Z(d)$ and, from the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^n}(d-d) = \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_{\mathbb{P}^n}(d) \to \mathcal{O}_Z(d) \to 0$ together with $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = 0$, one reads off

   $$h^0(N_{Z/\mathbb{P}^n}) \;=\; h^0(\mathbb{P}^n, \mathcal{O}(d)) - 1 \;=\; \binom{n+d}{d} - 1$$

   Moreover $H^1(Z, \mathcal{O}_Z(d)) = 0$ for a smooth hypersurface (again from the same sequence, since $H^1(\mathcal{O}_{\mathbb{P}^n}(d)) = H^2(\mathcal{O}_{\mathbb{P}^n}) = 0$), so proposition 2.4.5 gives that $\operatorname{Hilb}^P_{\mathbb{P}^n}$ is smooth at $[Z]$ of dimension $\binom{n+d}{d} - 1$. This is exactly the projective space $\mathbb{P}\big(H^0(\mathbb{P}^n, \mathcal{O}(d))\big) = \mathbb{P}^{\binom{n+d}{d}-1}$ of degree-d hypersurfaces, the identification developed in full in example 2.5.2.

2. *A finite set of points.* Take $Z = \{p_1, \dots, p_r\} \subseteq \mathbb{P}^n$ a reduced set of r distinct k-points, with constant Hilbert polynomial $P = r$. By example 2.4.3, $N_{Z/\mathbb{P}^n} = \bigoplus_{j=1}^{r} T_{p_j}\mathbb{P}^n$ is a skyscraper sheaf, so

   $$H^0(Z, N_{Z/\mathbb{P}^n}) \;=\; \bigoplus_{j=1}^{r} T_{p_j}\mathbb{P}^n, \qquad H^1(Z, N_{Z/\mathbb{P}^n}) = 0$$

   the latter because Z is 0-dimensional and coherent cohomology of a sheaf on a 0-dimensional scheme vanishes above degree 0. Hence $\operatorname{Hilb}^r_{\mathbb{P}^n} = \operatorname{Hilb}^P_{\mathbb{P}^n}$ is smooth at $[Z]$ of dimension

   $$h^0(N_{Z/\mathbb{P}^n}) \;=\; \sum_{j=1}^{r} \dim T_{p_j}\mathbb{P}^n \;=\; rn$$

This matches the expectation that r distinct points in $\mathbb{P}^n$ move in an rn-dimensional family, each point contributing its n tangent directions independently; the reduced-points locus is a smooth open subscheme of $\mathrm{Hilb}^r_{\mathbb{P}^n}$ of dimension rn. The subtler geometry, where points collide and the tangent space jumps, is the subject of example 2.5.1.

The two clean cases share a structural reason for their smoothness: in both, the obstruction group $H^1(N_{Z/Y})$ vanishes, so the Hilbert scheme is exactly as smooth as the tangent space predicts, and its dimension is computed by a single cohomology count. Vanishing H^1 thus reduces the local geometry of Hilb^P_Y at $[Z]$ to a single H^0 computation, with no correction from higher obstructions. This also isolates the precise input needed to construct many moduli spaces as smooth varieties: control of H^1 of a normal or endomorphism sheaf. The same control, transported to the tangent sheaf T_C of a curve, is what makes the moduli of curves smooth of dimension $3g-3$ in the deformation-theory Part and the capstone, and it is the normal-sheaf computation of theorem 2.4.4 that first exhibits the pattern.

Exercise 2.4.1

1. Let $Z = V(xy) \subseteq \mathbb{A}^2_k$ be the union of the two coordinate axes (the node at the origin, viewed as a subscheme of the plane), with $\mathcal{I}_Z = (xy)$. Compute the normal sheaf $N_{Z/\mathbb{A}^2}$ and describe $H^0(Z, N_{Z/\mathbb{A}^2})$ as a k-vector space. Interpret a first-order deformation of Z that does *not* come from moving the two lines rigidly, and explain in one sentence why the origin is the interesting point.

2. Let $Z \subseteq \mathbb{P}^3_k$ be a smooth conic (a degree-2 curve of arithmetic genus 0) lying in a plane $H \cong \mathbb{P}^2$. Compute $N_{Z/\mathbb{P}^3}$, its global sections $h^0(N_{Z/\mathbb{P}^3})$, and $h^1(N_{Z/\mathbb{P}^3})$, and conclude that the Hilbert scheme of conics in $\mathbb{P}^3$ is smooth at $[Z]$; identify its dimension. (Hint: use the normal-bundle sequence $0 \to N_{Z/H} \to N_{Z/\mathbb{P}^3} \to N_{H/\mathbb{P}^3}|_Z \to 0$, with $N_{Z/H} = \mathcal{O}_H(2)|_Z \cong \mathcal{O}_{\mathbb{P}^1}(4)$ from viewing Z as a degree-2 divisor in H, and $N_{H/\mathbb{P}^3}|_Z = \mathcal{O}_H(1)|_Z \cong \mathcal{O}_{\mathbb{P}^1}(2)$.)

3. Show directly from theorem 2.4.4 that the tangent space to Hilb at the trivial subscheme $Z = Y$ (the whole ambient scheme, with $\mathcal{I}_Z = 0$) and at the empty subscheme $Z = \varnothing$ is zero in each case, and reconcile this with the fact that these are isolated reduced points of the corresponding Hilbert schemes.

4. Let $Z = \operatorname{Spec} k[x]/(x^2) \subseteq \mathbb{A}^1_k$ be the length-2 subscheme supported at the origin (a point with a tangent direction), with $\mathcal{I}_Z = (x^2) \subseteq k[x]$. Compute $\mathcal{I}_Z/\mathcal{I}_Z^2$, the normal sheaf $N_{Z/\mathbb{A}^1}$, and $h^0(N_{Z/\mathbb{A}^1})$. Compare your answer with the dimension of $\mathrm{Hilb}^2(\mathbb{A}^1)$ and explain what the two tangent directions parametrize.

5. Let $Z \subseteq \mathbb{P}^n_k$ be a smooth complete intersection of two quadrics ($c = 2$, $d_1 = d_2 = 2$) with $n \geq 3$. Using example 2.4.6, write down $N_{Z/\mathbb{P}^n}$ and compute $h^0(N_{Z/\mathbb{P}^n})$. For $n = 3$ (so Z is an elliptic curve, a smooth $(2,2)$ curve in $\mathbb{P}^3$), evaluate the dimension explicitly and comment on whether the smoothness criterion of proposition 2.4.5 applies.

2.5 Worked Hilbert Schemes

The previous sections built the Hilbert and Quot schemes in the abstract: $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ is representable by a projective scheme (theorem 2.3.2), the Hilbert functor is the special case $\mathcal{F} = \mathcal{O}_Y$ (corollary 2.3.3), and the tangent space at a point $[Z]$ is $H^0(Z, N_{Z/Y})$ (theorem 2.4.4). What none of this yet supplies is the feel of these schemes as geometric objects one can hold in the hand. A representability theorem tells us Hilb^P_Y exists; it does not tell us that Hilb^n of a smooth surface is itself smooth, or that it maps to the symmetric product, or that the ideal (y, x^2) names a specific length-two subscheme sitting at the origin with a chosen tangent direction. This section closes that gap by working three families of examples all the way down to charts, ideals, and equations.

The three examples are chosen to exercise the whole chapter. The Hilbert scheme of points on a surface is where the local theory of theorem 2.4.4 does the most work: the normal-sheaf tangent space computes to exactly $2n$ everywhere, and the scheme is smooth despite parametrizing objects (fat points, tangent directions) that look singular. The Hilbert scheme of hypersurfaces is the opposite extreme, a case so rigid that the whole scheme is a single projective space read off from one Hilbert polynomial. The Grassmannian as a Quot scheme closes the circle back to section 1.2, showing that the fine moduli space built by hand there is a special case of the machine built here. Together they show the range of the theory, from a space as simple as $\mathbb{P}^N$ to one as intricate as the punctual Hilbert scheme.

The first of the three is the running touchstone of the book: the Hilbert scheme of n points on a surface. A "point" of this scheme is a length-n subscheme, and the subtlety is that length-n subschemes are not just unordered n-tuples of distinct points. When two points collide they can leave behind a tangent direction, and when more collide the bookkeeping of the limiting subscheme becomes intricate, yet the total space of all such subschemes is smooth. This is the phenomenon the example makes concrete.

Fix the Hilbert polynomial to be the constant $P = n$. A closed subscheme $Z \subseteq Y$ with constant Hilbert polynomial n is a zero-dimensional subscheme whose structure sheaf has length n over k, that is, $\dim_k H^0(Z, \mathcal{O}_Z) = n$. The Hilbert scheme $\mathrm{Hilb}^n(Y) := \mathrm{Hilb}^P_Y$ for this constant polynomial is the *Hilbert scheme of n points*. The following worked example states its smoothness and dimension for a smooth surface, gives the tangent-space argument that establishes them, constructs the Hilbert-Chow morphism, and carries the whole picture down to the explicit case $\mathrm{Hilb}^2(\mathbb{A}^2)$.

The governing fact is a theorem of Fogarty, who proved smoothness and irreducibility for a smooth surface over any algebraically closed field; the analogous statement fails in dimension three and higher, where Hilb^n acquires singular and even reducible components once n is large enough.[1] The tangent-space computation rests on the normal-sheaf identification of theorem 2.4.4, and the one input beyond the tools developed here, irreducibility of $\mathrm{Hilb}^n(Y)$, is cited to Fogarty rather than reproved.

Example 2.5.1: The Hilbert Scheme of Points on a Surface

Let Y be a smooth quasi-projective surface over k. The claim is that the Hilbert scheme of n points $\mathrm{Hilb}^n(Y)$ is smooth of dimension $2n$ over k and carries a projective *Hilbert-Chow morphism*

$$\pi\colon \mathrm{Hilb}^n(Y) \longrightarrow \mathrm{Sym}^n Y = Y^n/\mathfrak{S}_n$$

sending a length-n subscheme Z to its support counted with multiplicity, $\sum_p \mathrm{length}_p(\mathcal{O}_Z)\cdot[p]$, which is an isomorphism over the locus of n distinct points and is a resolution of singularities of

[1] Fogarty, *Algebraic families on an algebraic surface* (1968), established smoothness and irreducibility of $\mathrm{Hilb}^n(Y)$ for Y a smooth surface. In dimension ≥ 3 the scheme $\mathrm{Hilb}^n(\mathbb{A}^d)$ acquires singularities and, for n large enough, extra components beyond the closure of the reduced configurations; a tangent-space count at a sufficiently fat point already exceeds the expected dimension dn. The classical example is the length-8 non-smoothable zero-dimensional scheme of Iarrobino and Emsalem in $\mathbb{A}^4$, giving reducibility of $\mathrm{Hilb}^8(\mathbb{A}^4)$; for $\mathbb{A}^3$ irreducibility persists further and the smallest reducible case is larger and was settled only later. `% TODO: verify cite key` (Fogarty 1968, Amer. J. Math. 90).

$\mathrm{Sym}^n Y$.

Step 1: The tangent space at a reduced subscheme. Suppose first that $Z = \{p_1, \ldots, p_n\}$ is n distinct reduced points of Y. By theorem 2.4.4 the tangent space is

$$T_{[Z]} \mathrm{Hilb}^n(Y) = H^0(Z, N_{Z/Y})$$

where $N_{Z/Y} = \mathcal{H}om(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z)$ is the normal sheaf (definition 2.4.1). Since Z is a disjoint union of n reduced points on the smooth surface Y, at each p_i the ideal $\mathcal{I}_Z$ is the maximal ideal $\mathfrak{m}_{p_i}$, and $\mathfrak{m}_{p_i}/\mathfrak{m}_{p_i}^2$ is the two-dimensional cotangent space of the surface. Hence $N_{Z/Y}$ is a skyscraper sheaf whose stalk at each p_i is the two-dimensional tangent space $T_{p_i}Y$, and

$$H^0(Z, N_{Z/Y}) = \bigoplus_{i=1}^{n} T_{p_i} Y$$

has dimension $2n$. Each summand is the freedom to move one point in the two directions of the surface, so the reduced locus already has the expected tangent dimension $2n$.

Step 2: The local dimension is $2n$ everywhere. The locus $U \subseteq \mathrm{Hilb}^n(Y)$ of reduced subschemes (n distinct points) is open and nonempty, and it maps isomorphically to the open subset of $\mathrm{Sym}^n Y$ of distinct points; since $\mathrm{Sym}^n Y$ has dimension $2n$, so does U, and its tangent dimension is $2n$ by Step 1. The two inputs that promote this to smoothness of the whole scheme are irreducibility of $\mathrm{Hilb}^n(Y)$ (so U is dense and $\dim \mathrm{Hilb}^n(Y) = 2n$) and the tangent-dimension count of Step 3; irreducibility is the part beyond the tools here and is cited to Fogarty. Granting it, every point of $\mathrm{Hilb}^n(Y)$ is a limit of reduced configurations and $\dim \mathrm{Hilb}^n(Y) = 2n$.

Step 3: The tangent space has dimension exactly $2n$. Smoothness reduces to the equality $\dim_k H^0(Z, N_{Z/Y}) = 2n$ at every point $[Z]$: combined with $\dim_{[Z]} \mathrm{Hilb}^n(Y) = 2n$ from Step 2 and the general inequality $\dim_{[Z]} \mathrm{Hilb}^n(Y) \leq \dim_k T_{[Z]} \mathrm{Hilb}^n(Y)$, agreement of the tangent dimension with the local dimension forces smoothness at $[Z]$. The tangent count is local: the normal sheaf $N_{Z/Y}$ is supported on the finite set underlying Z, so

$$\dim_k H^0(Z, N_{Z/Y}) = \sum_{p \in Z} \dim_k \mathrm{Hom}_{A_p}(I_p/I_p^2, A_p), \qquad A_p = \mathcal{O}_{Z,p} = \mathcal{O}_{Y,p}/I_p$$

and it suffices to show each local term equals $2\,\mathrm{length}_k(A_p)$. This is where the surface hypothesis enters: for a zero-dimensional subscheme of a smooth surface the local length identity $\dim_k \mathrm{Hom}_{A_p}(I_p/I_p^2, A_p) = 2\,\mathrm{length}_k(A_p)$ holds at every point, whether or not Z is a local complete intersection there, the factor two matching the codimension of Z in the surface. This identity is a nontrivial computation over the regular local ring $\mathcal{O}_{Y,p}$ of dimension two, not a formal consequence of the tangent-normal setup; it is Fogarty's computation, and the argument is cited to Fogarty rather than reproduced here (the same reference that supplies irreducibility). Summing over p and using $\sum_p \mathrm{length}_k(A_p) = n$ gives $\dim_k H^0(Z, N_{Z/Y}) = 2n$. With Step 2 this yields $\dim_k T_{[Z]} = 2n = \dim_{[Z]} \mathrm{Hilb}^n(Y)$ at every point, so $\mathrm{Hilb}^n(Y)$ is smooth of dimension $2n$.

Step 4: The Hilbert-Chow morphism. The universal subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}^n(Y)$ (definition 2.3.4) is finite and flat of degree n over $\mathrm{Hilb}^n(Y)$. Its support, taken with multiplicities, defines for each $[Z]$ the effective zero-cycle $\sum_p \mathrm{length}_p(\mathcal{O}_Z)[p]$ on Y, a point of $\mathrm{Sym}^n Y$. That this assignment is a morphism of schemes, not just of sets, is the content of the construction of $\mathrm{Sym}^n Y$ as a scheme representing the functor of effective relative zero-cycles: the norm of the finite flat family $\mathcal{Z}$ produces a morphism $\pi\colon \mathrm{Hilb}^n(Y) \to \mathrm{Sym}^n Y$. The morphism π is itself projective, hence proper. This is a relative statement over the base $\mathrm{Sym}^n Y$ and holds for quasi-projective Y: it does not require $\mathrm{Hilb}^n(Y)$ to be projective, and indeed for the affine $Y = \mathbb{A}^2$ of Steps 5-6 the scheme $\mathrm{Hilb}^n(\mathbb{A}^2)$ is not projective while π still is, so the properness of π must be read off the morphism, not off an ambient projectivity of the source. It restricts to the identity on the open locus of n distinct points, where a reduced length-n subscheme and its unordered support carry the same information. Since $\mathrm{Hilb}^n(Y)$ is smooth of dimension $2n$ and $\mathrm{Sym}^n Y$ is normal of the same dimension with singularities exactly along the diagonals where points collide, π is a resolution of singularities.

Step 5: The case $\mathrm{Hilb}^2(\mathbb{A}^2)$ *explicitly.* Take $Y = \mathbb{A}^2 = \operatorname{Spec} k[x, y]$ and $n = 2$; a point is an ideal $I \subseteq k[x, y]$ with $\dim_k k[x, y]/I = 2$, and there are two geometric types. If Z is two distinct reduced points $p = (a_1, b_1)$, $q = (a_2, b_2)$, then $I = \mathcal{I}_p \cap \mathcal{I}_q$ and $k[x, y]/I \cong k \times k$ has length two, with tangent space $T_p\mathbb{A}^2 \oplus T_q\mathbb{A}^2$ of dimension $4 = 2 \cdot 2$, and this open locus of distinct pairs maps isomorphically to the corresponding locus of $\mathrm{Sym}^2\mathbb{A}^2$. The subschemes supported at a single point, say the origin, are the interesting ones: a length-two ideal I with $k[x, y]/I$ supported at the origin must contain $\mathfrak{m}^2 = (x^2, xy, y^2)$ (a length-two quotient of the local ring is a quotient of $\mathcal{O}/\mathfrak{m}^2$) and have two-dimensional quotient spanned by 1 and one linear form, so

$$I_{[\lambda:\,\mu]} = (\mathfrak{m}^2,\ \mu x - \lambda y) = (x^2,\ xy,\ y^2,\ \mu x - \lambda y)$$

for a point $[\lambda: \mu] \in \mathbb{P}^1$: the relation $\mu x - \lambda y = 0$ leaves a two-dimensional quotient $k[x, y]/I_{[\lambda:\,\mu]}$ spanned by 1 and the class of any linear form not proportional to $\mu x - \lambda y$. When $\mu \neq 0$ the relation reads $x = (\lambda/\mu)y$, so y survives and $\{1, y\}$ is a basis; when $\lambda \neq 0$ it reads $y = (\mu/\lambda)x$, so x survives and $\{1, x\}$ is a basis. A concrete representative is $[\lambda: \mu] = [1: 0]$, giving

$$I_{[1:\,0]} = (y,\ x^2)$$

since $\mu x - \lambda y = -y$ and modulo y the ideal $\mathfrak{m}^2$ collapses to (x^2). The quotient $k[x, y]/(y, x^2) = k[x]/(x^2)$ has k-basis $\{1, x\}$, length two, supported at the origin with the tangent vector ∂_x singled out. A double point at the origin is therefore not a single subscheme but a $\mathbb{P}^1$ of them, one for each tangent direction $[\lambda: \mu]$.

Step 6: The blow-up of the diagonal. Assembling both types identifies $\mathrm{Hilb}^2(\mathbb{A}^2)$ globally. The symmetric product $\mathrm{Sym}^2\mathbb{A}^2 = \mathbb{A}^4/\mathfrak{S}_2$ is singular exactly along the diagonal $\Delta \cong \mathbb{A}^2$ where the two points coincide. Over a diagonal point the Hilbert-Chow morphism π has fiber the $\mathbb{P}^1$ of tangent directions just found, whereas over a distinct pair the fiber is a single point. A birational morphism from a smooth fourfold to $\mathrm{Sym}^2\mathbb{A}^2$ that replaces each diagonal point by a $\mathbb{P}^1$ of normal directions and is an isomorphism elsewhere is exactly the blow-up of $\mathrm{Sym}^2\mathbb{A}^2$ along its diagonal:

$$\mathrm{Hilb}^2(\mathbb{A}^2) \cong \mathrm{Bl}_\Delta\big(\mathrm{Sym}^2\mathbb{A}^2\big)$$

The exceptional divisor is the locus of nonreduced subschemes, a $\mathbb{P}^1$-bundle over the diagonal $\Delta \cong \mathbb{A}^2$, whose points are the ideals $I_{[\lambda:\,\mu]}$ translated to each point of $\mathbb{A}^2$. The Hilbert scheme of two points is the blow-up that resolves the diagonal of the symmetric square.

The example says the Hilbert scheme repairs the symmetric product. The symmetric product $\mathrm{Sym}^n Y$ is singular precisely along the loci where points come together, and its singularities record only the crude datum of a multiplicity. The Hilbert scheme sits above it, smooth, remembering the finer datum of the subscheme structure that a collision leaves behind, and the Hilbert-Chow morphism forgets that finer datum to recover the cycle. The $\mathrm{Hilb}^2(\mathbb{A}^2)$ computation exhibits exactly what extra information is carried over a diagonal point of $\mathrm{Sym}^2 Y$, namely a projective line of tangent directions.

The tangent-direction $\mathbb{P}^1$ over the origin is the fiber of the Hilbert-Chow morphism at the most singular point of Sym^2, and its appearance is forced by the algebra: a length-two subscheme concentrated at a point is exactly an ideal $I \supseteq \mathfrak{m}^2$ with one-dimensional $\mathfrak{m}/I$, and one-dimensional subspaces of the two-dimensional $\mathfrak{m}/\mathfrak{m}^2$ form a $\mathbb{P}^1$. The passage $(y, x^2) \leftrightarrow$ "origin with tangent ∂_x" is the dictionary to carry forward: an ideal is a subscheme, and the nonreduced structure encodes the limiting geometry of colliding points. Every later appearance of the Hilbert scheme of points, in deformation theory and in the moduli of sheaves, rests on this picture.

The second example swings to the opposite pole. Where the Hilbert scheme of points hides a smooth but intricate space behind a constant Hilbert polynomial, the Hilbert scheme of hypersurfaces of a fixed degree is a single projective space, transparent from the polynomial alone. This is the case where the general construction of theorem 2.3.2 collapses to the linear algebra of homogeneous forms.

A degree-d hypersurface in $\mathbb{P}^n$ is the zero locus $V(f)$ of a nonzero homogeneous polynomial f of degree d, taken up to scalar, so the naive parameter space is the projectivization of the space of such forms. Viewing this as a Hilbert scheme identifies the naive parameter space with a fine moduli space carrying a universal family, and it pins down which Hilbert polynomial to fix. The Hilbert polynomial is read off from the ideal sheaf sequence, and the identification of the scheme with $\mathbb{P}^N$ is a direct application of the representability theorem in a case where the ambient Grassmannian is already a projective space; the example records both.

Example 2.5.2: The Hilbert Scheme of Degree-d Hypersurfaces

Fix $n \geq 1$ and $d \geq 1$. The claim is that the Hilbert scheme of degree-d hypersurfaces in $\mathbb{P}^n$, taken for the Hilbert polynomial

$$P_d(t) = \binom{t+n}{n} - \binom{t-d+n}{n} \tag{2.3}$$

is the projective space

$$\mathrm{Hilb}^{P_d}_{\mathbb{P}^n} \cong \mathbb{P}\big(H^0(\mathbb{P}^n, \mathcal{O}(d))\big) = \mathbb{P}^N, \qquad N = \binom{n+d}{d} - 1$$

of dimension $\binom{n+d}{d} - 1$, with universal family the universal degree-d hypersurface in $\mathbb{P}^n \times \mathbb{P}^N$.

Step 1: The Hilbert polynomial. A degree-d hypersurface $X = V(f) \subseteq \mathbb{P}^n$ has ideal sheaf $\mathcal{I}_X = \mathcal{O}_{\mathbb{P}^n}(-d)$, since f generates the ideal in each chart and cutting by a single degree-d form twists $\mathcal{O}$ down by d. The ideal sheaf sequence

$$0 \to \mathcal{O}_{\mathbb{P}^n}(-d) \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_X \to 0$$

twisted by $\mathcal{O}(t)$ and passed to Euler characteristics gives

$$P_d(t) = \chi(\mathcal{O}_X(t)) = \chi(\mathcal{O}_{\mathbb{P}^n}(t)) - \chi(\mathcal{O}_{\mathbb{P}^n}(t-d)) = \binom{t+n}{n} - \binom{t-d+n}{n}$$

using $\chi(\mathcal{O}_{\mathbb{P}^n}(m)) = \binom{m+n}{n}$ for the projective space. This is (2.3), of degree $n-1$ in t with leading coefficient $d/(n-1)!$, recording that X is an $(n-1)$-dimensional subscheme of degree d.

Step 2: A subscheme with this polynomial is a hypersurface. Conversely, any closed subscheme $X \subseteq \mathbb{P}^n$ with Hilbert polynomial P_d is a degree-d hypersurface. Its ideal sheaf $\mathcal{I}_X$ has $\chi(\mathcal{I}_X(t)) = \binom{t+n}{n} - P_d(t) = \binom{t-d+n}{n} = \chi(\mathcal{O}_{\mathbb{P}^n}(t-d))$, and for $t \geq d$ the higher cohomology $H^{>0}(\mathbb{P}^n, \mathcal{I}_X(t))$ vanishes (Serre vanishing for the twist of a coherent sheaf on $\mathbb{P}^n$, [12]), so the Euler characteristic reduces to h^0: this counts the degree-t part of the saturated homogeneous ideal, forcing $h^0(\mathcal{I}_X(t)) = \binom{t-d+n}{n} = h^0(\mathcal{O}_{\mathbb{P}^n}(t-d))$. In degree $t = d$ this gives $h^0(\mathcal{I}_X(d)) = 1$, a single degree-$d$ form f in the ideal, and matching Hilbert functions in every degree $t \geq d$ shows the ideal is generated by f, so $\mathcal{I}_X = f \cdot \mathcal{O}_{\mathbb{P}^n}(-d) \cong \mathcal{O}_{\mathbb{P}^n}(-d)$. Thus $X = V(f)$ for a unique-up-to-scalar $f \in H^0(\mathbb{P}^n, \mathcal{O}(d))$.

Step 3: Identification with $\mathbb{P}^N$. Steps 1 and 2 show the closed points of $\mathrm{Hilb}^{P_d}_{\mathbb{P}^n}$ are exactly the nonzero degree-d forms up to scalar, that is, the points of $\mathbb{P}(H^0(\mathbb{P}^n, \mathcal{O}(d)))$. To upgrade this bijection of points to an isomorphism of schemes, compare universal families. Over $\mathbb{P}^N = \mathbb{P}(H^0(\mathcal{O}(d)))$ there is a tautological section: the universal form $F \in H^0(\mathbb{P}^n \times \mathbb{P}^N,\ \mathcal{O}_{\mathbb{P}^n}(d) \boxtimes \mathcal{O}_{\mathbb{P}^N}(1))$, whose vanishing locus $\mathcal{X} = V(F) \subseteq \mathbb{P}^n \times \mathbb{P}^N$ is flat over $\mathbb{P}^N$ with every fiber a degree-d hypersurface of Hilbert polynomial P_d. This is a family of subschemes of $\mathbb{P}^n$ with Hilbert polynomial P_d, hence by corollary 2.3.3 it is classified by a morphism $\Phi \colon \mathbb{P}^N \to \mathrm{Hilb}^{P_d}_{\mathbb{P}^n}$. Its inverse sends the universal subscheme over $\mathrm{Hilb}^{P_d}_{\mathbb{P}^n}$, a flat family of hypersurfaces, to the degree-d form cutting it out, a line bundle map $\mathcal{O}_{\mathrm{Hilb}} \to \pi_* \mathcal{O}_{\mathbb{P}^n}(d)$ classifying a morphism back to $\mathbb{P}^N$. The two are mutually inverse because a hypersurface and its defining form determine each other, so Φ is an isomorphism, and the dimension is $\dim H^0(\mathbb{P}^n, \mathcal{O}(d)) - 1 = \binom{n+d}{d} - 1$.

The example is the cleanest possible instance of the fine moduli philosophy. The scheme parametrizing hypersurfaces is not just a set of forms up to scalar, it is a projective space whose universal hypersurface pulls back to every family, and the whole construction of theorem 2.3.2 degenerates here to the linear algebra of the vector space $H^0(\mathbb{P}^n, \mathcal{O}(d))$. This is why classical algebraic geometry could speak of "the space of plane curves of degree d" long before Hilbert schemes: for hypersurfaces the space is a projective space, and no scheme theory beyond $\mathbb{P}^N$ is needed. The next example computes N in the smallest case and connects it to a classical count.

Example 2.5.3: Plane Curves of Degree d

Take $n = 2$, so $\mathbb{P}^2$ and hypersurfaces are plane curves. A degree-d plane curve is $V(f)$ for f a homogeneous form of degree d in three variables x_0, x_1, x_2, and

$$\dim_k H^0(\mathbb{P}^2, \mathcal{O}(d)) = \binom{d+2}{2} = \frac{(d+1)(d+2)}{2}$$

is the number of degree-d monomials in three variables. Hence $\mathrm{Hilb}^{P_d}_{\mathbb{P}^2} \cong \mathbb{P}^N$ with

$$N = \binom{d+2}{2} - 1 = \frac{(d+1)(d+2)}{2} - 1 = \frac{d(d+3)}{2}$$

For $d = 1$, lines in $\mathbb{P}^2$, this gives $N = 2$, so the space of lines is $\mathbb{P}^2$, the dual projective plane: a line in $\mathbb{P}^2$ is a hyperplane, hence a rank-1 quotient of $\mathcal{O}^3$, and the space of these is $\mathrm{Gr}(1,3) = \mathbb{P}^2$ by the identification $\mathrm{Gr}(1,n) = \mathbb{P}^{n-1}$ of theorem 1.2.4. For $d = 2$, conics, $N = 5$: a conic $ax_0^2 + bx_1^2 + cx_2^2 + d'x_0x_1 + ex_0x_2 + fx_1x_2$ has six coefficients up to scalar, so conics form $\mathbb{P}^5$, the classical fact that five general points determine a conic (five linear conditions cut a point in $\mathbb{P}^5$). For $d = 3$, plane cubics, $N = 9$: nine points in general position determine a cubic, the count underlying the group law on an elliptic curve, and the running touchstone $\mathcal{M}_{1,1}$ of the first chapter lives inside this $\mathbb{P}^9$ as the locus of smooth cubics modulo projective equivalence.

The Hilbert polynomial P_d carries exactly the information the moduli problem needs: its degree records the dimension of the hypersurface, and its leading coefficient records the degree d. The dimension count $N = \binom{n+d}{d} - 1$ recovers the classical parameter counts for lines, conics, and cubics, and it is the first invariant one computes for any Hilbert scheme. The point where these hypersurface families connect to the harder theory is that $\mathbb{P}^N$ contains, as locally closed subsets, the loci of smooth, of singular, and of degenerate hypersurfaces, and cutting out the smooth locus and quotienting by PGL_{n+1} is exactly the moduli problem geometric invariant theory solves in the following chapter.

The third example returns to the beginning of the book. The first chapter built the Grassmannian $\mathrm{Gr}(d,n)$ from scratch as a fine moduli space of quotients (section 1.2, definition 1.2.1, theorem 1.2.4). The Quot scheme is a direct generalization of that construction, replacing quotients of a vector space by quotients of a coherent sheaf, so the Grassmannian ought to reappear as a Quot scheme. Confirming this reconciles the hand construction of section 1.2 with the general machine of section 2.3 and shows the two agree on their common ground.

The reconciliation is a matter of matching functors. The Grassmannian functor $\underline{\mathrm{Gr}}(d,n)$ assigns to S the rank-d locally free quotients of $\mathcal{O}_S^n$; the Quot functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ assigns to S the flat quotients of $\mathcal{F}_S$ with fiberwise Hilbert polynomial P. Taking $Y = \operatorname{Spec} k$ a point, $\mathcal{F} = \mathcal{O}^n$ the trivial rank-n sheaf, and $P = d$ the constant polynomial identifies the two.

Example 2.5.4: The Grassmannian as a Quot Scheme

Let $Y = \operatorname{Spec} k$ be a point and $\mathcal{F} = \mathcal{O}^n = \mathcal{O}_Y^{\oplus n}$ the trivial rank-n sheaf, with $P = d$ the constant Hilbert polynomial. The claim is that the Quot functor $\underline{\mathrm{Quot}}^d_{\mathcal{O}^n/\operatorname{Spec} k}$ coincides with the Grassmannian functor $\underline{\mathrm{Gr}}(d,n)$ of definition 1.2.1, and hence

$$\mathrm{Quot}^d_{\mathcal{O}^n/\operatorname{Spec} k} \cong \mathrm{Gr}(d,n)$$

as fine moduli spaces, with the universal quotient $\mathcal{U}$ of definition 2.3.4 equal to the universal quotient bundle $\mathcal{Q}$ of definition 1.2.5.

Over the point $Y = \operatorname{Spec} k$ a scheme S over k has $Y_S = S$, and the pulled-back sheaf is $\mathcal{F}_S = \mathcal{O}_S^n$. A point of $\underline{\mathrm{Quot}}^d_{\mathcal{O}^n/\operatorname{Spec} k}(S)$ is a coherent quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ flat over S and every fiber of constant Hilbert polynomial d. A coherent sheaf on the zero-dimensional fiber $\operatorname{Spec} \kappa(s)$ with constant Hilbert polynomial d is a $\kappa(s)$-vector space of dimension d, so each fiber $\mathcal{Q} \otimes \kappa(s)$ has dimension d; a coherent sheaf on S that is flat with fibers of constant rank d is locally free of rank d. Thus

$$\underline{\mathrm{Quot}}^d_{\mathcal{O}^n/\operatorname{Spec} k}(S) = \left\{ \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q} \;\middle|\; \mathcal{Q} \text{ locally free of rank } d \right\} / \cong$$

which is the definition of $\underline{\mathrm{Gr}}(d,n)(S)$ (definition 1.2.1) verbatim. The two functors are equal, and pullback is the same operation (f^* applied to the surjection) in both. By Yoneda, equal functors have isomorphic representing objects, so $\mathrm{Quot}^d_{\mathcal{O}^n/\operatorname{Spec} k} \cong \mathrm{Gr}(d,n)$. The universal quotient $\mathcal{U}$ over $\mathrm{Quot}^d_{\mathcal{O}^n/\operatorname{Spec} k}$ corresponds under the isomorphism of functors to the universal object over $\mathrm{Gr}(d,n)$, which is the universal quotient bundle $\mathcal{Q}$ of definition 1.2.5.

The example places the Grassmannian inside the theory rather than beside it. The hand construction of section 1.2, with its affine charts and Plücker embedding, and the general existence theorem of section 2.3, with its Grassmannian of degree-m pieces and closed-subfunctor cut, are two routes to the same scheme, and the general route specializes to the specific one when the ambient sheaf is $\mathcal{O}^n$ over a point. This is the reason the construction of theorem 2.3.2 embeds every Quot scheme in a Grassmannian: the Grassmannian is the universal case, the Quot scheme of a trivial sheaf over a point, and every other Quot scheme is cut out of one of these by the flatness and Hilbert-polynomial conditions. The chart-by-chart smoothness of theorem 1.2.4 thus becomes the local model against which the local structure of a general Quot scheme, governed by the tangent-space computation of theorem 2.4.4, is measured.

Taking $Y = \operatorname{Spec} k$ was the degenerate case; allowing Y to be positive-dimensional connects the Quot scheme to maps into the Grassmannian, but two subtleties intervene that are absent over a point. First, the constant polynomial $P = d$ is a trap: over a positive-dimensional Y it selects length-d torsion quotients, whose Quot scheme is a relative Hilbert scheme of points, not a Grassmannian. A rank-d locally free quotient of $\mathcal{O}_Y^n$ has Hilbert polynomial $d \cdot \chi(\mathcal{O}_Y(t))$, a nonconstant polynomial of degree $\dim Y$, so this is the polynomial to fix if one wants locally free quotients in the picture at all. Second, and less obvious, fixing that polynomial does not by itself force a quotient to be locally free, nor does it capture every map $Y \to \mathrm{Gr}(d, n)$: the locally free quotients are only an *open* subscheme of the Quot scheme, and the Hom-scheme into the Grassmannian breaks into components indexed by a discrete pullback invariant, of which the fixed polynomial selects just one. The next example works out exactly what survives of the naive identification.

Example 2.5.5: The Relative Grassmannian as a Quot Scheme

Let Y be a projective scheme over k with a fixed ample $\mathcal{O}_Y(1)$, and $\mathcal{F} = \mathcal{O}_Y^n$ the trivial rank-n sheaf. The Hilbert polynomial of a rank-d locally free quotient $\mathcal{O}_Y^n \twoheadrightarrow \mathcal{E}$ is $\chi(\mathcal{E}(t))$, and this depends on more than the rank: it is governed by the full numerical class (Chern character) of $\mathcal{E}$, not by d alone. Among these, the quotients with *trivial* discrete invariant, those pulled back from a constant map $Y \to \mathrm{Gr}(d, n)$ so that $\mathcal{E} \cong \mathcal{O}_Y^{\oplus d}$, have Hilbert polynomial

$$Q(t) = d \cdot \chi(\mathcal{O}_Y(t))$$

of degree $\dim Y$ in t; this is the polynomial to fix to see the trivial-invariant locally free quotients, whereas the constant $P = d$ selects length-d torsion quotients instead. Fixing Q does *not*, by itself, isolate even these, let alone the whole relative Grassmannian, and the example is a warning against the naive identification $\mathrm{Quot}^Q_{\mathcal{O}_Y^n/Y} = \underline{\mathrm{Hom}}(Y, \mathrm{Gr}(d, n))$, which is false on two counts.

The locally free quotients form only an open locus. A coherent quotient $\mathcal{O}_Y^n \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ flat over $\operatorname{Spec} k$ and Hilbert polynomial Q need not be locally free of rank d: torsion, a rank drop along a divisor, or embedded points all leave $\chi(\mathcal{Q}(t)) = Q(t)$ unchanged while destroying local freeness. So the k-points of $\mathrm{Quot}^Q_{\mathcal{O}_Y^n/Y}$ include quotients that are not rank-d bundles, and the locally free quotients cut out an *open* subscheme

$$\mathrm{Quot}^{Q,\circ}_{\mathcal{O}_Y^n/Y} \subseteq \mathrm{Quot}^Q_{\mathcal{O}_Y^n/Y}$$

(openness of the locally free locus of a coherent sheaf on Y, [12]), not the whole Quot scheme.

The Hilbert polynomial pins one component of the Hom-scheme. A rank-d locally free quotient $\mathcal{O}_Y^n \twoheadrightarrow \mathcal{E}$ is, by the Grassmannian functor (definition 1.2.1), the same datum as a morphism $\varphi\colon Y \to \mathrm{Gr}(d, n)$, the quotient being $\mathcal{E} = \varphi^*\mathcal{Q}$ the pullback of the universal quotient bundle. But $\chi(\varphi^*\mathcal{Q}(t))$ depends on the class of φ: the Hom-scheme $\underline{\mathrm{Hom}}(Y, \mathrm{Gr}(d, n))$ is a disjoint union of pieces indexed by this discrete pullback invariant, and fixing $\chi = Q$ singles out one such piece. For $Y = \mathbb{P}^1$ and $\mathrm{Gr}(1, 2) = \mathbb{P}^1$, a degree-$e$ map pulls the universal quotient back to $\mathcal{O}(e)$, with $\chi(\mathcal{O}(e)(t)) = t + e + 1$; the value $Q(t) = \chi(\mathcal{O}_{\mathbb{P}^1}(t)) = t + 1$ forces $e = 0$, selecting only the constant maps among all $\underline{\mathrm{Hom}}(\mathbb{P}^1, \mathbb{P}^1)$.

Both corrections combine into the true statement: the locally free locus of the Quot scheme, at the fixed Hilbert polynomial Q, is an open subscheme that recovers one component of the Hom-scheme,

$$\mathrm{Quot}^{Q,\circ}_{\mathcal{O}_Y^n/Y} \cong \underline{\mathrm{Hom}}(Y, \mathrm{Gr}(d, n))_Q$$

the subscript Q marking the component on which the pullback invariant matches Q. For Y reduced to a point every quotient of the fixed space k^n of dimension d is automatically locally free and there is a single component, so the open locus is everything and this reduces to $\mathrm{Gr}(d, n)$, recovering example 2.5.4. The new content over a positive-dimensional base is exactly what these two corrections express: the Quot scheme carries non-locally-free quotients its point-model does not see, and even its locally free locus is a component of a Hom-scheme rather than a bare product $\mathrm{Gr}(d, n) \times Y$; that product is the sub-locus of *constant* morphisms $Y \to \mathrm{Gr}(d, n)$, one component among many, not the whole Hom-scheme.

The three examples map the terrain. At one end sits the Hilbert scheme of points, smooth of dimension $2n$ on a surface yet carrying the full subtlety of colliding points and tangent directions; at the other end sit the hypersurface Hilbert schemes, projective spaces read directly from a Hilbert polynomial; and threading them is the Grassmannian, the universal Quot scheme from which the general construction cuts every other. The reader who has followed the ideal (y, x^2) to a tangent direction, the binomial $\binom{n+d}{d} - 1$ to the dimension of a space of forms, and the trivial sheaf over a point back to section 1.2 has seen the whole chapter operate on the objects one meets.

Exercise 2.5.1

1. Compute the length of the subscheme $Z \subseteq \mathbb{A}^2$ cut out by the ideal $I = (x^2, xy, y^2)$, and decide whether $[Z] \in \mathrm{Hilb}^n(\mathbb{A}^2)$ for the appropriate n. Then, using the tangent-direction description in example 2.5.1, explain why (x^2, xy, y^2) does *not* lie in $\mathrm{Hilb}^2(\mathbb{A}^2)$, and identify which Hilb^n it does belong to.

2. For $\mathrm{Hilb}^2(\mathbb{A}^2)$, verify directly that the ideal $I_{[\lambda:\,\mu]} = (\mathfrak{m}^2,\ \mu x - \lambda y)$ has $\dim_k k[x, y]/I_{[\lambda:\,\mu]} = 2$ for every $[\lambda : \mu] \in \mathbb{P}^1$, and show that $I_{[\lambda:\,\mu]} = I_{[\lambda':\,\mu']}$ if and only if $[\lambda : \mu] = [\lambda' : \mu']$. Conclude that the fiber of the Hilbert-Chow morphism over the doubled origin is a $\mathbb{P}^1$.

3. Using example 2.5.2, compute the dimension of the Hilbert scheme of degree-d surfaces in $\mathbb{P}^3$ (quadric surfaces $d = 2$, cubic surfaces $d = 3$). In each case write out $\binom{n+d}{d} - 1$ explicitly, and for quadrics identify the locus of singular quadrics inside $\mathbb{P}^N$ as the zero locus of a single determinant.

4. Prove that the Hilbert polynomial $P_d(t) = \binom{t+n}{n} - \binom{t-d+n}{n}$ of (2.3) has degree $n - 1$ in t with leading coefficient $d/(n - 1)!$, and interpret both facts geometrically for the hypersurface $X = V(f)$.

5. Deduce from example 2.5.4 that $\mathbb{P}^{n-1}$ is a Quot scheme: identify $\underline{\mathrm{Quot}}^1_{\mathcal{O}^n/\mathrm{Spec}\,k}$ with the functor $\underline{\mathrm{Gr}}(1, n)$ and hence with the functor of points of $\mathbb{P}^{n-1}$, matching the universal quotient with $\mathcal{O}(1)$. Relate this to example 1.1.4.

6. Using theorem 2.4.4 and example 2.5.1, compute the dimension of the tangent space to $\mathrm{Hilb}^3(\mathbb{A}^2)$ at the point given by the ideal (x, y^3), a single point of length three concentrated on the y-axis. Confirm the answer equals $2 \cdot 3 = 6$, consistent with smoothness. Then explain where the local argument of Step 3 of example 2.5.1 breaks down for a zero-dimensional subscheme of $\mathbb{A}^3$: contrast the two structural routes to controlling $\mathcal{I}_Z/\mathcal{I}_Z^2$, namely freeness of the conormal module (available exactly at local complete intersection points) and the length-two duality special to a regular local ring of dimension two, and say which route survives on a surface at every point and which is the one that fails once the ambient dimension is three.

3

Geometric Invariant Theory

The Hilbert and Quot schemes of chapter 2 construct moduli spaces of objects that carry an embedding: a closed subscheme of a fixed projective space, or a quotient of a fixed sheaf. Most moduli problems, however, ask for objects only up to isomorphism, with no embedding fixed, and the passage from embedded objects to abstract ones is a quotient by the group that moves the embedding, most often a projective linear group. Geometric invariant theory, Mumford's refinement of Hilbert's nineteenth-century invariant theory, is the machine that forms such quotients. Given a reductive group G acting on a projective scheme together with a choice of linearized ample line bundle, it isolates a locus of *semistable* points on which a good quotient exists and is again projective. This is the second construction engine of the Part: paired with the Hilbert scheme, it produces the moduli spaces of abstract objects, the moduli of curves among them, that the functor-of-points method alone leaves as unrepresentable stacks.

The chapter builds the theory in the order the constructions depend on one another. section 3.1 establishes the affine quotient $X/\!\!/G = \operatorname{Spec} k[X]^G$, which exists as a scheme because the invariant ring is finitely generated for reductive G (Hilbert's finiteness theorem); the reductive groups themselves, and their actions on affine varieties, are taken from [4]. section 3.2 introduces linearizations of line bundles, the data that globalizes the affine quotient, and section 3.3 uses a linearization to cut out the semistable and stable loci and to build the projective quotient $X^{\mathrm{ss}}/\!\!/G$. section 3.4 reduces the stability of a point to a finite computation through the Hilbert–Mumford numerical criterion, and section 3.5 turns the machine on moduli problems, realizing a coarse moduli space as a GIT quotient of a Hilbert scheme and sketching the construction of $\mathcal{M}_g$ that the capstone completes.

3.1 Group Actions and Quotients

The moduli problems of the previous chapters carried their objects already embedded: a point of a Hilbert scheme is a closed subscheme of a fixed $\mathbb{P}^n$, not an abstract variety, and two points that differ only by a projective change of coordinates parametrize the same abstract object. To pass from embedded objects to abstract ones, one must quotient by the group that moves the embedding, typically a projective linear group. The moduli space of abstract objects should be the space of orbits of this group acting on a Hilbert scheme. Geometric invariant theory is the machinery that makes "the space of orbits" into a scheme, and this section builds its foundation: the problem of quotienting a variety by an algebraic group action, and the affine solution to that problem.

The difficulty is that the naive orbit space, the topological quotient X/G with the quotient topology, is almost never a variety. Orbits can fail to be closed, and when the closure of one orbit meets another, any continuous G-invariant function takes the same value on both, so no scheme structure on the set of orbits can separate them. The remedy is to give up on parametrizing all orbits and instead build a scheme that parametrizes the *closed* orbits, identifying two points precisely when their orbit closures meet. This section makes that idea precise for an affine variety, where the construction reduces to a question in invariant theory: the quotient is the spectrum of the ring of invariant functions, and the theorem that makes it a scheme is the finite generation of that ring.

Throughout, k is algebraically closed, G is a reductive linear algebraic group over k, and X is a scheme of finite type over k carrying a (left) algebraic action $\sigma\colon G \times X \to X$. An action is *algebraic* when σ is a morphism of schemes; the basic theory of such actions, their orbits, and their stabilizers is developed in [4], and is used here freely.

Before constructing anything, the target of the construction must be pinned down. [12, Quotients by Group Actions] introduced the two clean notions of quotient a group action can have: a *categorical quotient*, universal among G-invariant morphisms out of X, and a *geometric quotient*, whose points are exactly the orbits and whose structure sheaf is the invariants of the pushforward. These are recalled in [12, Categorical and Geometric Quotients]. The categorical quotient always exists once one has a reasonable ring of invariants but may collapse distinct orbits; the geometric quotient separates all orbits but exists only when every orbit is closed, which fails in the presence of the collapsing phenomenon described above. Geometric invariant theory works with an intermediate notion, tuned so that it exists in wide generality yet retains as much orbit information as the categorical quotient can carry.

That intermediate notion is the good quotient. It asks for a morphism that is affine-local (so it can be built by gluing spectra of invariant rings), whose structure sheaf is exactly the invariant functions (so it is a categorical quotient), and which separates any two disjoint closed invariant subsets (so on the locus of closed orbits it is as good as a geometric quotient). The last condition is the substantive one: it is what forces distinct closed orbits to land at distinct points, and it is exactly the property the topological quotient lacks.

Definition 3.1.1: Good Quotient

Let G act on a scheme X of finite type over k. A morphism $\pi\colon X \to Y$ to a scheme Y is a *good quotient* if:

1. π is G-invariant: $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$ as morphisms $G \times X \to Y$ (equivalently, π is constant on orbits);
2. π is affine and surjective;
3. the natural map $\mathcal{O}_Y \to (\pi_*\mathcal{O}_X)^G$ is an isomorphism, where $(\pi_*\mathcal{O}_X)^G$ denotes the subsheaf of G-invariant sections of $\pi_*\mathcal{O}_X$;
4. π separates disjoint closed invariant subsets: if $W_1, W_2 \subseteq X$ are closed, G-invariant, and disjoint, then $\pi(W_1)$ and $\pi(W_2)$ are closed and disjoint in Y.

A good quotient is a *geometric quotient* if, in addition, the fibers of π are exactly the orbits: $\pi(x) = \pi(x')$ if and only if $Gx = Gx'$.

Conditions (1) and (3) together already make a good quotient a categorical quotient: any G-invariant morphism $f\colon X \to Z$ factors uniquely through π, because on each affine open $\operatorname{Spec} B \subseteq Y$ the map f corresponds to an invariant ring homomorphism, which by (3) lands in $(\pi_*\mathcal{O}_X)^G(\operatorname{Spec} B) = B$. Condition (2) is what lets the whole construction be checked on affine charts and glued, and it is why the affine case treated below is the engine of the entire theory. Condition (4) is the separation property that distinguishes a good quotient from a bare categorical quotient: it guarantees that closed orbits, which are always disjoint from one another and closed, map to distinct points. A good quotient is therefore a geometric quotient precisely on the locus where every orbit is closed; off that locus, several orbits (an orbit and the smaller orbits in its closure) are glued to a single point, and this gluing is not a defect but the only way to obtain a scheme at all.

The affine construction rests on a single input from invariant theory: the ring of invariant functions is again a finitely generated k-algebra, so that its spectrum is a variety. This is Hilbert's finiteness theorem, one of the founding results of the subject. Its proof is an invariant-theory argument whose home in the series is [4], where reductive groups, their representation theory, and the Reynolds operator are developed; the statement is recorded here because it is the theorem that turns definition 3.1.4 into a scheme, and it is used throughout the chapter.

Theorem 3.1.2: Hilbert's Finiteness Theorem

Let G be a linearly reductive group over k acting rationally on a finitely generated k-algebra A by k-algebra automorphisms. Then the subalgebra of invariants

$$A^G = \{\, a \in A \,|\, g \cdot a = a \text{ for all } g \in G \,\}$$

is a finitely generated k-algebra. The same conclusion holds for any reductive group G, without the linear-reductivity hypothesis.

The proof is cited to [4]; the statement above is a recollection of that result, not a claim to reprove it here. The one-paragraph idea is worth carrying, because it explains why reductivity is the operative hypothesis. When G is linearly reductive, every finite-dimensional representation splits as a direct sum of irreducibles, and the trivial isotypic component varies functorially with the representation.

Applied to A, viewed as an increasing union of finite-dimensional G-subspaces, this produces a G-equivariant projection $R\colon A \to A^G$, the *Reynolds operator*, which is a homomorphism of A^G-modules: $R(bc) = b\,R(c)$ whenever $b \in A^G$. From here Hilbert's argument runs as in the graded case of the Hilbert basis theorem. Let A^G_+ be the ideal of A generated by all homogeneous invariants of positive degree; by Noetherianity it is generated by finitely many invariants $f_1, \dots, f_r$, and applying the Reynolds operator to a representation $f = \sum a_i f_i$ of an arbitrary invariant f, using that R is A^G-linear, shows by induction on degree that $f_1, \dots, f_r$ generate A^G as a k-algebra. The Reynolds operator, and with it the projection onto invariants, is exactly what linear reductivity supplies; in positive characteristic linear reductivity can fail for a reductive group, and the finiteness is then recovered through Nagata's theorem, using the weaker *geometric reductivity* that Haboush's theorem establishes for all reductive groups. That general case is the second sentence of the statement and is likewise cited to [4]; it is what makes the construction below available for groups such as SL_n in every characteristic, not only in characteristic zero.

3.1.3 Warning: reductivity is necessary Finiteness fails without a reductivity hypothesis. Nagata's counterexample exhibits an action of a non-reductive group (a suitable additive group $\mathbb{G}_a^n$) on a polynomial ring whose invariant subalgebra is not finitely generated, disproving Hilbert's fourteenth problem in its original generality. The reductivity of G is therefore not a convenience of the proof but a condition the conclusion requires, which is why the quotient theory of this chapter is built for reductive groups.

With finiteness in hand, the affine quotient is immediate to define and carries all four good-quotient properties. Let $X = \operatorname{Spec} A$ be affine with $A = k[X]$ finitely generated, and let G act on X, inducing the dual rational action on A. The ring of invariants $A^G = k[X]^G$ is a finitely generated k-algebra by theorem 3.1.2, so its spectrum is again an affine variety of finite type over k, and the inclusion $A^G \hookrightarrow A$ induces a morphism of schemes.

Definition 3.1.4: Affine GIT Quotient

Let G be a reductive group acting on an affine scheme $X = \operatorname{Spec} A$ of finite type over k. The *affine GIT quotient* of X by G is the affine scheme

$$X/\!\!/G := \operatorname{Spec}\, A^G$$

together with the morphism $\pi\colon X \to X/\!\!/G$ induced by the inclusion $A^G \hookrightarrow A$.

The double-slash notation distinguishes this from a putative orbit space X/G; the two agree precisely when every orbit is closed. The following proposition collects the geometric content of the construction and is the reason the affine quotient deserves the name "quotient" at all.

Proposition 3.1.5: The affine GIT quotient is a good quotient

With notation as in definition 3.1.4, the morphism $\pi\colon X \to X/\!\!/G$ is a good quotient. In particular:

1. π is a categorical quotient of X by G;
2. π is surjective, and its geometric fibers each contain a unique closed orbit;
3. for closed points $x, x' \in X$, one has $\pi(x) = \pi(x')$ if and only if the orbit closures meet, $\overline{Gx} \cap \overline{Gx'} \neq \emptyset$.

***Proof*:**
The four conditions of definition 3.1.1 are verified in turn, and the three numbered consequences are read off from them.

Step 1: π is G-invariant, affine, and surjective. Invariance holds by construction: π is dual to the inclusion $A^G \hookrightarrow A$, whose image consists of functions constant on orbits, so $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$. The morphism π is affine because both source and target are affine. For surjectivity it suffices to show that $\operatorname{Spec} A \to \operatorname{Spec} A^G$ hits every closed point, i.e. that for every maximal ideal $\mathfrak{m} \subset A^G$ the extension $\mathfrak{m}A$ is a proper ideal of A. If $\mathfrak{m}A = A$, write $1 = \sum_i a_i f_i$ with $a_i \in A$ and $f_i \in \mathfrak{m}$, and apply the Reynolds operator R; using its A^G-linearity and $R(1) = 1$ gives $1 = \sum_i R(a_i) f_i \in \mathfrak{m}$, since each $R(a_i) \in A^G$ and $f_i \in \mathfrak{m}$, contradicting that $\mathfrak{m}$ is proper. Hence $\mathfrak{m}A \neq A$, so $\mathfrak{m}$ is the contraction of a maximal ideal of A and lies in the image; the surjectivity of $\operatorname{Spec} A \to \operatorname{Spec} A^G$ for reductive G is established this way in [4]. Thus π is surjective.

Step 2: the structure sheaf condition. Since both schemes are affine, $\pi_*\mathcal{O}_X$ is the sheaf associated to A viewed as an A^G-module, and its G-invariant global sections are A^G by definition. On a basic open $D(f) \subseteq X/\!\!/G$ with $f \in A^G$, the invariant sections of $\mathcal{O}_X(\pi^{-1}D(f)) = A_f = A \otimes_{A^G} A^G_f$ are $(A_f)^G = (A^G)_f$, because inverting an invariant commutes with taking invariants (the Reynolds operator is A^G-linear, so localizing at $f \in A^G$ commutes with R). Hence $\mathcal{O}_{X/\!\!/G} \to (\pi_*\mathcal{O}_X)^G$ is an isomorphism.

Step 3: separation of disjoint closed invariant sets. Let $W_1, W_2 \subseteq X$ be closed, G-invariant, and disjoint, with vanishing ideals $I_1, I_2 \subseteq A$; disjointness means $I_1 + I_2 = A$. Both ideals are G-stable, so applying the Reynolds operator R to a relation $1 = a_1 + a_2$ with $a_i \in I_i$ gives $1 = R(1) = R(a_1) + R(a_2)$ with $R(a_i) \in I_i^G := I_i \cap A^G$, since R maps a G-stable ideal into its invariants. Therefore $I_1^G + I_2^G = A^G$, which says exactly that $V(I_1^G)$ and $V(I_2^G)$ are disjoint closed subsets of $X/\!\!/G$. As $\pi(W_i)$ is contained in $V(I_i^G)$ and is closed because π is a closed map on invariant sets by the same Reynolds argument applied to a single invariant ideal, the images $\pi(W_1)$ and $\pi(W_2)$ are closed and disjoint. This is condition (4), and it is the one place linear reductivity, through the Reynolds operator, does the essential work.

The four conditions establish that π is a good quotient.

Consequence (1). A good quotient is a categorical quotient: given any G-invariant morphism $f\colon X \to Z$, on each affine open $\operatorname{Spec} B \subseteq Z$ the pullback $f^\#\colon B \to A$ has image in A^G by invariance, hence factors through $A^G = \mathcal{O}_{X/\!\!/G}(X/\!\!/G)$, uniquely because $A^G \hookrightarrow A$ is injective. This is the universal property.

Consequence (3), and then (2). For the fiber statement, fix a closed point $y \in X/\!\!/G$ corresponding to a maximal ideal $\mathfrak{m} \subset A^G$. Two closed points $x, x' \in X$ satisfy $\pi(x) = \pi(x') = y$ if and only if every invariant $f \in A^G$ takes the same value on them, namely $f \pmod{\mathfrak{m}}$. Now the orbit closures $\overline{Gx}$ and $\overline{Gx'}$ are disjoint closed invariant sets when they do not meet, so by the separation just proved there is an invariant $f \in A^G$ with $f \equiv 0$ on $\overline{Gx}$ and $f \equiv 1$ on $\overline{Gx'}$, forcing $\pi(x) \neq \pi(x')$. Conversely if the closures meet, any invariant is constant on each closed set and on their common point, hence takes equal values at x and x', giving $\pi(x) = \pi(x')$. This proves (3). Finally each fiber $\pi^{-1}(y)$ is a nonempty closed G-invariant set, so it contains an orbit of minimal dimension, which is therefore closed; by (3) any two closed orbits in the fiber have meeting (hence equal) closures and so coincide, giving the unique closed orbit of (2). This completes the proof.

The proposition's content is worth stating in words. The affine quotient $X/\!\!/G$ does not parametrize all orbits; it parametrizes the *closed* orbits, one per point, and it glues a non-closed orbit to the unique closed orbit sitting in its closure. The invariant functions cannot tell a non-closed orbit apart from the smaller orbit it degenerates to, because a regular invariant is continuous and the two orbits share a limit point, so any quotient built from invariant functions must identify them. Two points map to the same place exactly when their orbit closures collide. This is the precise sense in which $X/\!\!/G$ is the best scheme-theoretic approximation to the orbit space, and the phenomenon it encodes, the collapse of non-closed orbits, is the feature to keep in view for the projective quotients of the following sections.

The following example is the smallest one in which the collapse is visible, and every subsequent GIT computation is a variation on it. It is worth working in complete detail.

Example 3.1.6: Invariant Rings for a Torus and for Binary Forms

1. *The hyperbola action.* Let $G = \mathbb{G}_m$ act on $X = \mathbb{A}^2 = \operatorname{Spec} k[x, y]$ by

$$t \cdot (x, y) = (tx,\, t^{-1}y)$$

so that on functions $t \cdot x = t^{-1}x$ and $t \cdot y = t\,y$ (the coordinate functions transform inversely to the points). A monomial $x^a y^b$ has weight $b - a$ under this action, so it is invariant if and only if $a = b$. Hence the ring of invariants is

$$k[x, y]^{\mathbb{G}_m} = k[xy]$$

the polynomial ring in the single invariant $u := xy$. The affine GIT quotient is therefore

$$\mathbb{A}^2/\!\!/\mathbb{G}_m = \operatorname{Spec} k[u] = \mathbb{A}^1$$

with $\pi(x, y) = xy$. The orbit structure explains the collapse. For $c \neq 0$ the level set $\{xy = c\}$ is a single orbit, closed in $\mathbb{A}^2$, mapping to the point $u = c$. Over $u = 0$ the fiber $\{xy = 0\}$ is the union of the two coordinate axes minus the origin, together with the origin itself, and it decomposes into *three* orbits: the punctured x-axis $\{(x, 0) : x \neq 0\}$, the punctured y-axis $\{(0, y) : y \neq 0\}$, and the fixed point $\{(0, 0)\}$. Only the last is closed; the closures of the two punctured axes each contain the origin. Consistent with proposition 3.1.5, all three orbits map to the single point $u = 0$, and they are identified there precisely because their closures meet at the origin. The two non-closed orbits are glued to the closed orbit in their common closure, and the quotient $\mathbb{A}^1$ sees only the closed orbits: one for each $c \neq 0$ and the origin for $c = 0$.

2. *Binary forms under* SL_2. Let $V_d = \mathrm{Sym}^d(k^2)^\vee$ be the space of binary forms of degree d, on which $G = \mathrm{SL}_2$ acts by linear substitution of variables; write $X = V_d$. For $d = 2$, a binary

quadratic $f = ax^2 + bxy + cy^2$ has the discriminant $\Delta = b^2 - 4ac$, and a direct check shows Δ is SL_2-invariant (the substitution matrices have determinant one). In fact $k[V_2]^{\mathrm{SL}_2} = k[\Delta]$, a polynomial ring in one generator, so

$$V_2 /\!\!/ \mathrm{SL}_2 = \operatorname{Spec} k[\Delta] = \mathbb{A}^1$$

The closed orbits are the nondegenerate conics $\{\Delta = c\}$ for $c \neq 0$; the fiber over $\Delta = 0$ consists of the perfect squares ℓ^2 (a single non-closed orbit) together with the zero form (the closed fixed point), and the two are identified because scaling ℓ^2 by t^2 via the diagonal one-parameter subgroup drives it to 0, so $0 \in \overline{\mathrm{SL}_2 \cdot \ell^2}$. This is the same collapse as in part (1), now realized inside the invariant theory of forms, and the discriminant is the invariant that both detects it and coordinatizes the quotient.

Both computations exhibit the same three-part anatomy. The invariant ring is finitely generated, as theorem 3.1.2 promises; its spectrum is an affine variety, here $\mathbb{A}^1$; and the fiber over the special point collects several orbits, exactly one of which is closed, with the non-closed orbits degenerating into it. The general point of the quotient corresponds to a closed orbit that is a full-dimensional level set of the invariants, while the special point corresponds to the most degenerate closed orbit, on which the invariants that cut out the generic orbits all vanish. This degeneration, an orbit whose closure meets a smaller orbit, is the affine shadow of *instability*, and isolating the points whose closed-orbit collapse can be avoided by a good choice of linearization is precisely the task of the next sections. For the moment the payoff is the construction itself: for a reductive group acting on an affine variety, the ring of invariants produces a scheme that is a categorical quotient, separating exactly the closed orbits.

Exercise 3.1.1

1. Let $\mathbb{G}_m$ act on $\mathbb{A}^n = \operatorname{Spec} k[x_1, \ldots, x_n]$ with weights $w_1, \ldots, w_n \in \mathbb{Z}$, so that $t \cdot x_i = t^{-w_i} x_i$ on functions. Describe the invariant ring $k[x_1, \ldots, x_n]^{\mathbb{G}_m}$ as the span of monomials of weight zero, and compute $\mathbb{A}^n /\!\!/ \mathbb{G}_m$ explicitly when $(w_1, w_2, w_3) = (1, 1, -1)$.

2. Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, ty)$ (weight 1 on both coordinates). Compute $k[x, y]^{\mathbb{G}_m}$ and $\mathbb{A}^2 /\!\!/ \mathbb{G}_m$. Contrast the orbit picture with that of example 3.1.6(1): which orbits are closed, and which single orbit is the unique closed orbit in the fiber over the image of the origin?

3. Show directly from definition 3.1.1 that a good quotient $\pi : X \to Y$ is a categorical quotient. (You may assume X affine; the general case is glued from this one.)

4. Let $\pi : X \to X /\!\!/ G$ be the affine GIT quotient of a reductive group action. Prove that for each closed point $y \in X /\!\!/ G$ the fiber $\pi^{-1}(y)$ contains a *unique* closed orbit, and that this orbit lies in the closure of every orbit in the fiber. (Reduce to the separation property of definition 3.1.1(4).)

5. Give an example of a non-reductive group action for which the invariant ring is not finitely generated, or explain in one paragraph why theorem 3.1.2 cannot simply be extended to all linear algebraic groups, referencing the phenomenon of box 3.1.3.

6. The finite group $G = \mathbb{Z}/2$ acts on $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ (with $\operatorname{char} k \neq 2$) by $(x, y) \mapsto (-x, -y)$. Compute the invariant ring, identify $\mathbb{A}^2 /\!\!/ (\mathbb{Z}/2)$ as an affine variety in $\mathbb{A}^3$, and verify that this quotient is a *geometric* quotient (every orbit is closed).

3.2 Linearizations

The affine quotient of section 3.1 disposes of the affine case completely: when X is affine and G is reductive, the invariant ring $k[X]^G$ is a finitely generated k-algebra, and $X /\!\!/ G = \operatorname{Spec} k[X]^G$ is a categorical quotient whose points are the closed orbits (definition 3.1.4). The moduli spaces this book is after are almost never affine. A moduli of curves, or of subschemes with a fixed Hilbert polynomial, is a quotient of a projective or quasi-projective parameter space, and Proj has no ring of global functions to take invariants of: the only global regular functions on a projective variety are the constants, so the recipe "take Spec of the invariant functions" returns a point and throws away all the geometry.

The way out is to reconstruct the projective variety from its homogeneous coordinate ring, which is not a ring of functions but a ring of *sections* of powers of an ample line bundle. If X is projective and L is ample, then $X = \operatorname{Proj} \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})$, and one recovers the quotient by taking G-invariant sections in each degree and forming $\operatorname{Proj} \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})^G$. For this to make sense G must act not only on X but on the sections of L, and there is no canonical way for a group acting on X to act on L: the action of $g \in G$ carries the fiber L_x to the fiber L_{gx}, but which isomorphism $L_x \to L_{gx}$ to use is extra data. That data, a compatible choice for every g and every x, is a *linearization* of L. It is the choice that glues the local affine quotients over the open sets $X_s = \{s \neq 0\}$ (for s an invariant section) into a single projective quotient, and different linearizations glue them differently. This section defines linearizations, organizes them into a group, and computes, in the one example every reader already knows, how changing the linearization changes the invariant sections and so the eventual quotient.

Throughout, X is a scheme of finite type over the algebraically closed field k, G is a linear algebraic group acting on X by a morphism $\sigma\colon G \times X \to X$, and L is a line bundle on X. Write $\mathrm{pr}_2\colon G \times X \to X$ for the second projection and, for $g \in G$, write $\sigma_g\colon X \to X$ for the automorphism $x \mapsto gx$.

Definition 3.2.1: G-Linearization of a Line Bundle

Let G act on X via $\sigma\colon G \times X \to X$, and let L be a line bundle on X with total space $\pi\colon \mathbb{L} \to X$. A *$G$-linearization* of L is an action $\widetilde{\sigma}\colon G \times \mathbb{L} \to \mathbb{L}$ of G on the total space $\mathbb{L}$ such that:

1. π is equivariant, that is $\pi(\widetilde{\sigma}(g, v)) = \sigma(g, \pi(v))$, so g carries the fiber L_x to the fiber L_{gx}; and
2. $\widetilde{\sigma}$ is *linear on fibers*: for each $g \in G$ and $x \in X$ the induced map $L_x \to L_{gx}$, $v \mapsto \widetilde{\sigma}(g, v)$, is a linear isomorphism of one-dimensional vector spaces.

A line bundle equipped with a G-linearization is called *G-linearized*. Equivalently, a G-linearization is an isomorphism of line bundles on $G \times X$

$$\Phi\colon \sigma^* L \xrightarrow{\sim} \mathrm{pr}_2^* L \tag{3.1}$$

satisfying the *cocycle condition* on $G \times G \times X$: writing $m\colon G \times G \to G$ for multiplication, the two ways of comparing $(m \times \mathrm{id}_X)^*$ and the iterated pullbacks agree, that is $\mathrm{pr}_{23}^* \Phi \circ (\mathrm{id}_G \times \sigma)^* \Phi = (m \times \mathrm{id}_X)^* \Phi$.

The two formulations say the same thing from opposite ends. The total-space description (3.1) in the form of $\widetilde{\sigma}$ is the geometric picture: g moves the line L_x over x to the line L_{gx} over gx by a definite linear map, and doing g then h agrees with doing hg in one step (this is what the cocycle condition encodes). The isomorphism Φ is the same data written on the level of sheaves, which is what one computes with: for a fixed g it restricts to isomorphisms $L_{gx} = (\sigma_g^* L)_x \to L_x$, and the cocycle condition is the associativity of the action. The linearity clause is not automatic and is the crux of the definition: an equivariant map of total spaces that scaled the fibers nonlinearly would not descend to an action on sections.

Proposition 3.2.2: The Induced Action on Sections

A G-linearization of L induces, for every $n \geq 0$, a linear representation of G on the space of global sections $H^0(X, L^{\otimes n})$, by the formula

$$(g \cdot s)(x) = \widetilde{\sigma}\big(g,\, s(g^{-1}x)\big) \tag{3.2}$$

where $\widetilde{\sigma}$ is the linearization of $L^{\otimes n}$ induced by that of L. The subspace of invariants $H^0(X, L^{\otimes n})^G$ consists of the sections s with $g \cdot s = s$ for all $g \in G$, equivalently the sections that are equivariant maps $X \to \mathbb{L}^{\otimes n}$. The representation and the formula (3.2) need nothing beyond the finite-type hypothesis on X; only the final clause uses projectivity, namely that when X is projective $\bigoplus_{n \geq 0} H^0(X, L^{\otimes n})^G$ is a graded subring of the section ring $\bigoplus_{n \geq 0} H^0(X, L^{\otimes n})$.

***Proof*:**
A linearization of L induces one on each tensor power $L^{\otimes n}$: the isomorphism Φ of (3.1) gives $\Phi^{\otimes n} \colon \sigma^* L^{\otimes n} \to \mathrm{pr}_2^* L^{\otimes n}$, and the cocycle condition for Φ passes to $\Phi^{\otimes n}$ because pullback and tensor product are functorial, so each $L^{\otimes n}$ is G-linearized. Fix n and abbreviate $M = L^{\otimes n}$.

Step 1: The formula defines a linear map. For $g \in G$ and a global section $s \in H^0(X, M)$ define $g \cdot s$ by (3.2). Concretely, s is a morphism $X \to \mathbb{M}$ splitting π; the map $g \cdot s$ sends x to $\widetilde{\sigma}(g, s(g^{-1}x))$, which lies in the fiber $M_{g \cdot g^{-1}x} = M_x$ by fiber-preservation, so $g \cdot s$ is again a section. It is regular because it is the composite of the regular maps $x \mapsto g^{-1}x$, then s, then $\widetilde{\sigma}(g, -)$. Linearity in s is the fiberwise linearity of $\widetilde{\sigma}$ (clause (2) of definition 3.2.1): for $a, b \in k$ and sections s, t,

$$\big(g \cdot (as + bt)\big)(x) = \widetilde{\sigma}\big(g, a\, s(g^{-1}x) + b\, t(g^{-1}x)\big) = a\,(g \cdot s)(x) + b\,(g \cdot t)(x)$$

since $\widetilde{\sigma}(g, -) \colon M_{g^{-1}x} \to M_x$ is linear.

Step 2: The action axioms. That $e \in G$ acts as the identity is the identity axiom for $\widetilde{\sigma}$. For $g, h \in G$,

$$\big(g \cdot (h \cdot s)\big)(x) = \widetilde{\sigma}\big(g, (h \cdot s)(g^{-1}x)\big) = \widetilde{\sigma}\big(g, \widetilde{\sigma}(h, s(h^{-1}g^{-1}x))\big) = \widetilde{\sigma}\big(gh, s((gh)^{-1}x)\big) = \big((gh) \cdot s\big)(x)$$

where the third equality is the action axiom $\widetilde{\sigma}(g, \widetilde{\sigma}(h, v)) = \widetilde{\sigma}(gh, v)$ for $\widetilde{\sigma}$ and $(gh)^{-1} = h^{-1}g^{-1}$. So $g \mapsto (s \mapsto g \cdot s)$ is a homomorphism $G \to \mathrm{GL}(H^0(X, M))$; that it is a morphism of algebraic groups, hence a rational representation, follows because (3.2) is given by the morphism σ and the linearization, both algebraic.

Step 3: Invariants form a ring. A section s is invariant iff $s(gx) = \widetilde{\sigma}(g, s(x))$ for all g, x, which is precisely the condition that $s \colon X \to \mathbb{M}$ be G-equivariant. If $s \in H^0(X, L^{\otimes m})^G$ and $t \in$

$H^0(X, L^{\otimes n})^G$ are invariant, their product $st \in H^0(X, L^{\otimes(m+n)})$ satisfies $g\cdot(st) = (g\cdot s)(g\cdot t) = st$, because the linearization on $L^{\otimes(m+n)}$ is the tensor product of those on $L^{\otimes m}$ and $L^{\otimes n}$; and $1 \in H^0(X, \mathcal{O}_X) = H^0(X, L^{\otimes 0})$ is invariant. Hence $\bigoplus_n H^0(X, L^{\otimes n})^G$ is a graded subring of the section ring, completing the proof.

The invariant subring $R(X, L)^G := \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})^G$ is the object from which the projective quotient is built: the next section forms Proj of it, and theorem 3.1.2 of section 3.1 is what guarantees this ring is finitely generated so the Proj is a projective scheme. What proposition 3.2.2 isolates now is that the ring depends on the linearization, not on L alone: the same underlying line bundle carried by two different linearizations produces two different G-actions on the same vector spaces $H^0(X, L^{\otimes n})$, hence two different invariant subrings and two different quotients. The bookkeeping of linearizations is therefore not a formality but the input that selects the quotient. To handle it systematically one collects all linearizations of all line bundles into a single algebraic structure.

The set of linearized line bundles carries the operations one expects of line bundles: two linearizations tensor to a linearization of the tensor product, and the dual of a linearization linearizes the dual bundle. The trivial bundle $\mathcal{O}_X$ has the trivial linearization in which G acts on $\mathcal{O}_X = X \times \mathbb{A}^1$ only through X. These operations make the linearized line bundles into a group, refining the Picard group.

Proposition 3.2.3: The Equivariant Picard Group

Let G act on X. The isomorphism classes of G-linearized line bundles on X form an abelian group $\operatorname{Pic}^G(X)$ under tensor product, with identity the trivially linearized $\mathcal{O}_X$ and inverse given by the dual linearization on $L^\vee$. Forgetting the linearization is a homomorphism

$$\operatorname{Pic}^G(X) \longrightarrow \operatorname{Pic}(X) \tag{3.3}$$

whose kernel is the character group $\widehat{G} = \operatorname{Hom}(G, \mathbb{G}_m)$ of G whenever X is connected and complete with $H^0(X, \mathcal{O}_X) = k$. Moreover:

1. if G is connected and X is normal, then every line bundle is linearizable after passing to a power: for each $L \in \operatorname{Pic}(X)$ there is an integer $N \geq 1$ (depending on L) such that $L^{\otimes N}$ admits a G-linearization. Equivalently, the cokernel of (3.3) is a torsion group.

***Proof*:**
Step 1: The group structure. Given linearizations Φ_L of L and Φ_M of M in the cocycle form (3.1), the isomorphism $\Phi_L \otimes \Phi_M : \sigma^*(L \otimes M) \to \mathrm{pr}_2^*(L \otimes M)$ satisfies the cocycle condition because each factor does and the condition is multiplicative under tensor; this linearizes $L \otimes M$ and is well defined on isomorphism classes. Associativity and commutativity are inherited from tensor product of line bundles. The trivial bundle with $\Phi = \mathrm{id}$ is a two-sided identity. For an inverse, the dual isomorphism $(\Phi_L^\vee)^{-1} : \sigma^* L^\vee \to \mathrm{pr}_2^* L^\vee$ linearizes $L^\vee$, and $L \otimes L^\vee \cong \mathcal{O}_X$ carries the trivial linearization, so $[L^\vee] = [L]^{-1}$. Thus $\operatorname{Pic}^G(X)$ is an abelian group and (3.3) is a homomorphism, since the forgetful map discards Φ and respects tensor product.

Step 2: The kernel. An element of the kernel is a linearization Φ of the *trivial* class in $\operatorname{Pic}(X)$, that is a linearization of $\mathcal{O}_X$ up to isomorphism. A linearization of $\mathcal{O}_X$ is an isomorphism $\Phi : \sigma^*\mathcal{O}_X \to \mathrm{pr}_2^*\mathcal{O}_X$ of the trivial bundle on $G \times X$, hence multiplication by an invertible global function $u \in H^0(G \times X, \mathcal{O})^\times$. Since X is complete and connected with $H^0(X, \mathcal{O}_X) = k$, and G is affine, $H^0(G \times X, \mathcal{O})^\times = H^0(G, \mathcal{O}_G)^\times$: an invertible function on $G \times X$ is constant in the X direction, so $u = u(g)$ is a function of g alone. The cocycle condition on Φ becomes $u(gh) = u(g)u(h)$, that is $u\colon G \to \mathbb{G}_m$ is a homomorphism of algebraic groups, a character. Two such linearizations are isomorphic iff the characters coincide (an isomorphism of linearized trivial bundles is multiplication by a constant, which cancels). Hence the kernel is $\widehat{G} = \mathrm{Hom}(G, \mathbb{G}_m)$.

Step 3: Linearizability of a power. Part (1) is a theorem of Mumford, proved in his *Geometric Invariant Theory* (Chapter 1, §3): if G is a connected linear algebraic group acting on a normal variety X over k, then for every line bundle L there is $N \geq 1$ with $L^{\otimes N}$ G-linearizable. The proof compares $\sigma^* L$ and $\mathrm{pr}_2^* L$ on $G \times X$: their difference is measured by a class in $\mathrm{Pic}\,(G)$ (after restriction to slices), and the obstruction to constructing Φ lies in the finite group $\mathrm{Pic}\,(G)$ for G connected, finiteness of Pic of a connected linear algebraic group being itself a result of Mumford (*GIT* Chapter 1, §3), which is killed by passing to a suitable power $L^{\otimes N}$; the cocycle condition is then arranged using $\mathrm{Pic}\,(G \times G) = \mathrm{Pic}\,(G) \oplus \mathrm{Pic}\,(G)$ and connectedness. The full argument uses the structure of Pic of a connected algebraic group; it is cited rather than reproduced here because it draws on the Picard group of an algebraic group beyond this chapter's scope, and the statement is all that the constructions of this book require. This completes the proof.

The exact sequence packaged by proposition 3.2.3,

$$0 \longrightarrow \widehat{G} \longrightarrow \mathrm{Pic}^G(X) \longrightarrow \mathrm{Pic}\,(X)$$

(valid for X connected complete with only constant global functions) is the organizing principle. It says two things at once. First, whether a given line bundle can be linearized at all is a question about the image of the forgetful map, and part (1) answers it affirmatively up to a power for connected G and normal X, which is exactly the generality of moduli constructions: the ample bundle on a parameter space can always be linearized after replacing it by a power, and replacing L by $L^{\otimes N}$ does not change Proj of the section ring, so no geometry is lost. Second, when a linearization exists it is unique only up to the character group $\widehat{G}$: twisting a linearization by a character $\chi \in \widehat{G}$ produces another linearization of the same bundle, and this twisting is precisely the freedom that will move the semistable locus. For a semisimple group such as SL_{n+1} one has $\widehat{G} = 0$, so the linearization is unique and there is nothing to twist; for a group with characters, such as GL_{n+1} or a torus, the twist is a parameter that changes the invariants. This dichotomy is the whole content of the worked example.

To twist a linearization Φ of L by a character $\chi\colon G \to \mathbb{G}_m$ means to replace the action on the total space by $\widetilde{\sigma}_\chi(g, v) = \chi(g) \cdot \widetilde{\sigma}(g, v)$, scaling each fiber map by $\chi(g)$; equivalently, tensoring the linearized bundle by the trivial bundle carrying the linearization χ. On sections, by (3.2), this multiplies the action on $H^0(X, L^{\otimes n})$ by χ^n, so a section that transformed by a weight ψ under the χ-untwisted action transforms by $\psi + n\chi$ after the twist. Invariance is the vanishing of the total weight, so twisting shifts which sections are invariant. The following example computes this shift in the case of projective space and reads off its effect on the invariant sections.

Example 3.2.4: Linearizations of $\mathcal{O}_{\mathbb{P}^n}(1)$

Let $V = k^{n+1}$ and $\mathbb{P}^n = \mathbb{P}(V)$, and let $\mathrm{GL}(V) = \mathrm{GL}_{n+1}$ act on $\mathbb{P}^n$ in the standard way, $g \cdot [v] = [gv]$. This example carries out three computations: the canonical linearization of $\mathcal{O}(1)$, the resulting action on sections and its invariants, and the effect of twisting by a character.

The canonical linearization. The tautological line bundle $\mathcal{O}_{\mathbb{P}^n}(-1)$ has total space $\{([v], w) \in \mathbb{P}^n \times V \,|\, w \in kv\}$, the union of the lines parametrized by $\mathbb{P}^n$. The group $\mathrm{GL}(V)$ acts on $\mathbb{P}^n \times V$ by $g \cdot ([v], w) = ([gv], gw)$, and this preserves the incidence $w \in kv$ (if $w = \lambda v$ then $gw = \lambda\, gv$), so it restricts to an action on $\mathcal{O}(-1)$ that covers the action on $\mathbb{P}^n$ and is linear on fibers: over $[v]$ the fiber is the line kv, and g maps it isomorphically to $k(gv)$ by $w \mapsto gw$. This is a $\mathrm{GL}(V)$-linearization of $\mathcal{O}(-1)$ in the sense of definition 3.2.1, hence, dualizing (proposition 3.2.3), a canonical linearization of $\mathcal{O}(1) = \mathcal{O}(-1)^\vee$ and of every $\mathcal{O}(d) = \mathcal{O}(1)^{\otimes d}$.

The action on sections. Global sections of $\mathcal{O}(d)$ for $d \geq 0$ are the degree-d homogeneous polynomials on V,

$$H^0(\mathbb{P}^n, \mathcal{O}(d)) = \mathrm{Sym}^d(V^\vee)$$

and the induced action (3.2) is the natural action of $\mathrm{GL}(V)$ on $\mathrm{Sym}^d(V^\vee)$: a linear form $\ell \in V^\vee$ transforms by $(g \cdot \ell)(v) = \ell(g^{-1}v)$, and degree-$d$ forms by extension. To detect invariants it suffices to test the central torus $Z = \{t \cdot \mathrm{id}_V \big| t \in \mathbb{G}_m\} \subseteq \mathrm{GL}(V)$. On V the scalar $t \cdot \mathrm{id}$ acts by t, so on $V^\vee$ by t^{-1}, so on $\mathrm{Sym}^d(V^\vee)$ by t^{-d}: every degree-d form is an eigenvector of weight $-d$ for Z. For $d > 0$ this weight is nonzero, so

$$H^0(\mathbb{P}^n, \mathcal{O}(d))^{\mathrm{GL}_{n+1}} = 0 \qquad (d > 0)$$

already because no nonzero form is Z-invariant. The same computation restricted to $\mathrm{SL}(V) = \mathrm{SL}_{n+1}$ gives $H^0(\mathbb{P}^n, \mathcal{O}(d))^{\mathrm{SL}_{n+1}} = 0$ for $d > 0$: the representation $\mathrm{Sym}^d(V^\vee)$ is irreducible and nontrivial, so it has no nonzero invariants. Since SL_{n+1} acts transitively on $\mathbb{P}^n$, there is a single orbit and no invariant section can separate points; with this linearization no point of $\mathbb{P}^n$ is semistable, which foreshadows the definition of semistability in section 3.3.

Twisting by a character. A character of GL_{n+1} is a power of the determinant, $\chi = \det^a$ for $a \in \mathbb{Z}$ (this is $\widehat{\mathrm{GL}_{n+1}} \cong \mathbb{Z}$). Twisting the linearization of $\mathcal{O}(1)$ by χ multiplies the action on $H^0(\mathbb{P}^n, \mathcal{O}(d))$ by $\chi^d = \det^{ad}$ (proposition 3.2.2 applied to the twist). On the central torus Z the determinant is $\det(t \cdot \mathrm{id}) = t^{n+1}$, so $\det^{ad}$ contributes weight $ad(n+1)$, and the total Z-weight of a degree-d form after the twist is

$$-d + ad(n+1) = d\big(a(n+1) - 1\big)$$

which vanishes exactly when $a(n+1) = 1$. No integer a solves this for $n \geq 1$, so no twist by a GL_{n+1}-character produces invariant sections of $\mathcal{O}(d)$: the central weight can be shifted by multiples of $d(n+1)$ but never landed on 0. This is the arithmetic reason projective space itself has no GL_{n+1}-quotient. The twist does move the weight, which is the phenomenon that matters once the group has a positive-dimensional space of invariants to redistribute, as it does for the torus computation below.

A torus, where the twist changes semistability. Let $T = \mathbb{G}_m \subseteq \mathrm{GL}_{n+1}$ act on $\mathbb{P}^n$ diagonally with distinct weights $a_0 < a_1 < \cdots < a_n$, so $t \cdot [x_0 : \cdots : x_n] = [t^{a_0}x_0 : \cdots : t^{a_n}x_n]$, with the canonical linearization of $\mathcal{O}(1)$ restricted from GL_{n+1}. A monomial $x^{\mathbf{m}} = x_0^{m_0} \cdots x_n^{m_n}$ of degree $d = \sum m_i$ in $H^0(\mathbb{P}^n, \mathcal{O}(d))$ then has T-weight $-\sum_i m_i a_i$ under the untwisted linearization. Twisting by the character $t \mapsto t^r$ shifts every weight by rd, giving twisted weight $rd - \sum_i m_i a_i$. A monomial is invariant when this weight vanishes, that is when $\sum_i m_i a_i = rd$, the average weight equals r. The coordinate point $e_i = [0 : \cdots : 1 : \cdots : 0]$ is a T-fixed point, and the only degree-d monomial nonzero at it is x_i^d, whose twisted weight is $rd - da_i = d(r - a_i)$. That monomial is invariant, and e_i therefore semistable, at the single value $r = a_i$ and at no other: a fixed point of a $\mathbb{G}_m$-action has a one-point weight polytope, so 0 lies in its translated weight set only on the wall $r = a_i$. It is the generic points, those with several nonzero coordinates, whose status changes across the walls $r = a_i$: the semistable locus X^{ss} depends on r, and as r crosses a wall a different set of monomials becomes invariant. Two different twists give two different invariant subrings, hence two different GIT quotients of the same $\mathbb{P}^n$ under the same T-action. This is the concrete meaning of the twisting freedom recorded in proposition 3.2.3, and it is the mechanism the numerical criterion of section 3.4 will quantify.

The example demarcates the two regimes cleanly. For a semisimple group the character group vanishes, the linearization is rigid, and the action on projective space, being transitive, has no invariants and no quotient to speak of; the content of geometric invariant theory in that setting comes from acting on a larger parameter space with many orbits, not on $\mathbb{P}^n$ itself. For a group with characters, the linearization is a parameter, and the invariant sections, and with them the eventual quotient, vary with the chosen character. This is why the definition of semistability in the next section is stated relative to a fixed G-linearized ample line bundle: the choice of linearization is part of the data of the quotient, not a preliminary that can be suppressed. The affine quotient of definition 3.1.4 is the degenerate case where $L = \mathcal{O}_X$ carries the trivial linearization and Proj of the invariant ring is replaced by Spec of the degree-0 invariants; linearizations are what generalize that construction to the projective setting where the interesting moduli live.

Exercise 3.2.1

1. Let G act trivially on X (so $\sigma = \mathrm{pr}_2$). Show that a G-linearization of a line bundle L on X is the same datum as a homomorphism $G \to \mathbb{G}_m$, that is a character, together with the identity isomorphism on L; deduce that $\operatorname{Pic}^G(X) \cong \operatorname{Pic}(X) \times \widehat{G}$ for the trivial action when X is connected complete with $H^0(X, \mathcal{O}_X) = k$, and reconcile this with the kernel computation in proposition 3.2.3.
2. Verify the cocycle condition explicitly for the canonical linearization of $\mathcal{O}_{\mathbb{P}^n}(-1)$ built in example 3.2.4: writing the action on the total space as $g \cdot ([v], w) = ([gv], gw)$, check that the associated isomorphism $\Phi\colon \sigma^*\mathcal{O}(-1) \to \mathrm{pr}_2^*\mathcal{O}(-1)$ on $\mathrm{GL}_{n+1} \times \mathbb{P}^n$ satisfies the identity of (3.1). State where the linearity-on-fibers clause of definition 3.2.1 is used.
3. Let $T = \mathbb{G}_m$ act on $\mathbb{P}^1$ by $t \cdot [x_0 : x_1] = [x_0 : tx_1]$, with the canonical linearization of $\mathcal{O}(1)$. Compute the T-weights of the monomials $x_0^d, x_0^{d-1}x_1, \ldots, x_1^d$ in $H^0(\mathbb{P}^1, \mathcal{O}(d))$. For which twist $t \mapsto t^r$ does the space of invariant sections of $\mathcal{O}(d)$ become nonzero, and what monomials span it? Identify the two T-fixed points $[1 : 0]$ and $[0 : 1]$ as semistable or unstable for that twist.
4. Show that replacing a G-linearized ample line bundle L by its power $L^{\otimes N}$ (with the induced linearization) leaves the graded ring $\bigoplus_n H^0(X, L^{\otimes n})^G$ unchanged up to the Veronese regrading, and hence leaves Proj of the invariant ring unchanged. Explain why this makes the linearizability-of-a-power statement in proposition 3.2.3(1) sufficient for the constructions of this chapter.
5. Let G be a finite group acting on X. Show that $\operatorname{Pic}^G(X)$ is the group of G-equivariant line bundles and that the forgetful map $\operatorname{Pic}^G(X) \to \operatorname{Pic}(X)$ need not be injective by exhibiting, for $G = \mathbb{Z}/2$ acting on $X = \operatorname{Spec} k$ by the trivial action, a line bundle whose distinct linearizations are indexed by $\widehat{\mathbb{Z}/2} = \{\pm 1\}$ (assume $\operatorname{char} k \neq 2$). What does this say about the two linearizations of the trivial bundle?
6. Prove that if L and M are G-linearized and $s \in H^0(X, L)^G$, $t \in H^0(X, M)^G$ are invariant sections, then the open sets $X_s = \{s \neq 0\}$ and $X_t = \{t \neq 0\}$ are G-invariant, and the ring $H^0(X_s, \mathcal{O})^G$ is the degree-0 part of the localized invariant ring $\big(\bigoplus_n H^0(X, L^{\otimes n})^G\big)_{(s)}$. Explain in one sentence how this exhibits the projective quotient of the next section as glued from affine quotients (definition 3.1.4) over the X_s.

3.3 Stability and the Projective Quotient

The affine quotient $X/\!\!/G = \operatorname{Spec} k[X]^G$ of definition 3.1.4 solves the quotient problem for affine X, but it is silent about projective varieties, which carry no nonconstant global functions to be invariant. The way out is already visible in the construction of projective space itself: a projective variety is not built from a ring of functions but from a graded ring of *sections* of an ample line bundle, and Proj of that ring recovers the variety. A G-linearization (definition 3.2.1) endows each space of sections $H^0(X, L^{\otimes n})$ with a linear G-action, so the invariant sections $H^0(X, L^{\otimes n})^G$ form a graded subring, and its Proj is the natural candidate for the quotient.

Two things must be settled before this candidate is usable. First, not every point of X lies in the domain of this rational quotient map: a point where every invariant section vanishes has no invariant coordinate to record its image, and must be discarded. The points that survive, where enough invariant sections are nonzero to place the point in projective space, are the *semistable* points; among them, the ones whose orbits are already as large and as separated as the group allows are the *stable* points. Second, the graded ring of invariants must be finitely generated, so that its Proj is a genuine projective scheme; this is exactly Hilbert's finiteness theorem (theorem 3.1.2) applied to the section ring. This section defines the two loci, proves that their quotient is projective, and works out the picture for configurations of points on the line.

The first task is to define the two loci themselves. The definition selects points by the survival of invariant sections and then refines the selection by orbit geometry. Recall that a section $s \in H^0(X, L^{\otimes n})$ does not have a well-defined value in k at a point x, but the condition $s(x) \neq 0$ (that s does not vanish at x, equivalently that s generates the stalk $L_x^{\otimes n}$) is intrinsic. The nonvanishing locus $X_s = \{x \big| s(x) \neq 0\}$ is open, and if s is G-invariant it is G-stable and affine (it is the complement of the zero divisor of a section of an ample bundle). These invariant affine opens are the charts of the quotient.

Definition 3.3.1: Semistable, Stable, and Unstable Points

Let G be a reductive group acting on a scheme X of finite type over k, and let L be a G-linearized ample line bundle on X. A point $x \in X$ is:

1. *semistable* (with respect to L) if there is an integer $n > 0$ and an invariant section $s \in H^0(X, L^{\otimes n})^G$ with $s(x) \neq 0$;
2. *stable* if it is semistable, some such s can be chosen with X_s affine and the action of G on X_s closed (every orbit in X_s is closed in X_s), the stabilizer G_x is finite (equivalently $\dim Gx = \dim G$), and the orbit Gx is closed in X^{ss};
3. *unstable* if it is not semistable, that is, every invariant section of every positive power of L vanishes at x.

The *semistable locus* $X^{\mathrm{ss}} = X^{\mathrm{ss}}(L)$ is the set of semistable points, and the *stable locus* $X^{\mathrm{s}} = X^{\mathrm{s}}(L)$ the set of stable points.

The three conditions carve X into strata of decreasing pathology. An unstable point is invisible to invariant theory: no invariant survives to record it, so it cannot appear in any quotient built from invariant sections and is thrown away. A semistable point is recorded, but perhaps not faithfully, because invariant sections cannot always tell it apart from other points in the closure of its orbit. A stable point is recorded faithfully: its orbit is closed and its stabilizer finite, so its orbit is exactly one point of the quotient and the fiber over that point is a single orbit. The passage from semistable to stable is the passage from a *good* quotient, which may glue several orbits to a point, to a *geometric* quotient, whose points are exactly the orbits.

The loci are open. If x is semistable, witnessed by $s \in H^0(X, L^{\otimes n})^G$, then every point of the open set X_s is semistable by the same section, so $X^{\mathrm{ss}} = \bigcup_{n,s} X_s$ is a union of opens and hence open. The stable locus is open for a subtler reason, established in the proof of the main theorem below: the conditions of finite stabilizer and closed orbit are open conditions on the semistable locus. Both loci are visibly G-stable, since the defining conditions are phrased entirely in terms of invariant data.

3.3.2 Warning: Stability depends on the linearization The loci X^{ss} and X^{s} depend on L, not only on the underlying line bundle but on the chosen linearization. Two linearizations of the same ample bundle can differ by a character of G, and this shift can move a point across the boundary between stable, strictly semistable, and unstable. The dependence is not a defect but a feature: varying the linearization produces a family of birational models of the quotient, and the study of how the quotient changes as L crosses a wall is the subject of variation of GIT. For a fixed linearization, however, the loci are determined, and the ambiguity of definition 3.2.1 by a character is exactly what makes this variation possible.

A concrete instance grounds the definition before the theory begins. It also previews the extended computation of example 3.3.6.

Example 3.3.3: The Torus Acting on the Projective Line

Let $G = \mathbb{G}_m$ act on $X = \mathbb{P}^1$ by $t \cdot [x_0 : x_1] = [tx_0 : t^{-1}x_1]$, and linearize $L = \mathcal{O}_{\mathbb{P}^1}(2)$ by the natural action on $H^0(\mathbb{P}^1, \mathcal{O}(2))$, the space of binary quadratics in x_0, x_1. A monomial $x_0^a x_1^b$ with $a + b = 2$ is scaled by t^{a-b}, so the invariant sections in $H^0(\mathcal{O}(2))$ are the multiples of x_0x_1, and in $H^0(\mathcal{O}(2)^{\otimes n}) = H^0(\mathcal{O}(2n))$ the invariants are spanned by $(x_0x_1)^n$.

A point $[x_0 : x_1]$ is semistable iff $(x_0x_1)^n$ is nonzero there for some n, that is, iff $x_0x_1 \neq 0$. Thus the two fixed points $0 = [0 : 1]$ and $\infty = [1 : 0]$ are unstable, and every other point is semistable. The orbit of any $[x_0 : x_1]$ with $x_0x_1 \neq 0$ is $\{[tx_0 : t^{-1}x_1]\}$; in the coordinate $\xi = x_0/x_1$ on $\mathbb{P}^1 \setminus \{0, \infty\}$ the action is $\xi \mapsto t^2\xi$, and as t ranges over $\mathbb{G}_m$ the scalar t^2 ranges over all of $\mathbb{G}_m$, so the action is transitive on $X^{\mathrm{ss}} = \mathbb{P}^1 \setminus \{0, \infty\} \cong \mathbb{G}_m$. Each orbit is therefore all of X^{ss}, in particular closed in X^{ss}. The stabilizer is $\mu_2 = \{\pm 1\}$, since $t \cdot [x_0 : x_1] = [x_0 : x_1]$ forces $t = t^{-1}$ in projective coordinates, that is, $t^2 = 1$; this is finite, so the orbit has full dimension $\dim \mathbb{G}_m$ and every semistable point is stable. The invariant section ring is $R = \bigoplus_n H^0(\mathcal{O}(2n))^{\mathbb{G}_m} = k[w]$ with $w = x_0x_1$ a single generator in positive degree, so

$$X^{\mathrm{ss}} /\!\!/ G = X^{\mathrm{s}}/G = \operatorname{Proj} k[w] = \operatorname{Spec} k$$

is a single point. The only $\mathbb{G}_m$-invariant regular functions on X^{ss} are the constants, so there is no modulus: any two semistable points lie in one orbit, and the quotient records nothing beyond the single class.

The two unstable points are exactly the two toric fixed points, where the $\mathbb{G}_m$-action degenerates and no invariant survives. The stable locus is the open orbit-swept complement, a single orbit, and its geometric quotient is a point. This is the smallest nontrivial GIT quotient, and every feature of the general theory is already present in it: an unstable locus to discard, a stable locus with closed orbits, and a quotient recorded by the Proj of the invariant section ring, here a single point because the ring has only one generator in positive degree.

With the loci in hand, the main construction assembles them into the projective quotient itself. The mechanism is a globalization of the affine quotient: over each invariant affine chart X_s the good quotient is the affine GIT quotient of definition 3.1.4, and these affine quotients glue along overlaps to a good quotient of the whole semistable locus, whose total object is Proj of the invariant section ring. Finite generation of that ring, theorem 3.1.2, guarantees the Proj is a projective scheme.

Before stating the theorem, recall the two quotient notions it delivers. A *good quotient* $\pi\colon X^{\mathrm{ss}} \to Y$ (definition 3.1.1) is an affine, G-invariant, surjective map with $\mathcal{O}_Y = (\pi_*\mathcal{O}_{X^{\mathrm{ss}}})^G$ that separates disjoint closed invariant subsets; its fibers are unions of orbits, with exactly one closed orbit apiece. A *geometric quotient* is a good quotient whose fibers are single orbits. The theorem says the good quotient of the semistable locus is projective, and that it restricts to a geometric quotient on the stable locus.

Theorem 3.3.4: The Projective GIT Quotient

Let G be a reductive group acting on a projective scheme X over k, and let L be a G-linearized ample line bundle. Set

$$R = R(X,L)^G = \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})^G$$

the graded ring of invariant sections. Then:

1. R is a finitely generated graded k-algebra, and the quotient
$$X^{\mathrm{ss}}/\!\!/G := \operatorname{Proj} R$$
is a projective scheme over k.
2. The inclusions of invariant affine charts induce a morphism $\pi\colon X^{\mathrm{ss}} \to X^{\mathrm{ss}}/\!\!/G$ that is a good quotient. In particular π is surjective, G-invariant, affine, and its fibers are in bijection with the closed G-orbits in X^{ss}; two semistable points have the same image iff the closures of their orbits meet in X^{ss}.
3. The stable locus X^{s} is open and G-stable, its image $Y^{\mathrm{s}} = \pi(X^{\mathrm{s}})$ is open in $X^{\mathrm{ss}}/\!\!/G$, and the restriction $\pi\colon X^{\mathrm{s}} \to Y^{\mathrm{s}} = X^{\mathrm{s}}/G$ is a geometric quotient. Its fibers are exactly the orbits of stable points.

***Proof*:**
The argument builds the quotient chart by chart, glues, and then identifies the fibers.

Step 1: Finite generation and the projective scheme. Since X is projective and L is ample, the full section ring $R(X,L) = \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})$ is a finitely generated graded k-algebra with $\operatorname{Proj} R(X,L) = X$ (the standard fact that an ample bundle recovers its projective variety as Proj of its section ring). The G-action, made linear on each graded piece by the linearization of definition 3.2.1, makes $R(X,L)$ a graded k-algebra on which G acts by graded algebra automorphisms. Its invariant subring is the graded ring $R = R(X,L)^G$. By Hilbert's finiteness theorem (theorem 3.1.2), the invariant subalgebra of a finitely generated k-algebra under a reductive group is again finitely generated; applied to the graded algebra $R(X,L)$ this gives that R is a finitely generated graded k-algebra. Hence $\operatorname{Proj} R$ is a projective scheme over $\operatorname{Proj} R_0 = \operatorname{Spec} k$, proving (1). Replacing L by a power if necessary, one may assume R is generated in a single degree, which is the standing convenience for the Proj description; this changes neither the loci nor the quotient, since $H^0(X, L^{\otimes nm})^G$ has the same nonvanishing loci as its subring generated by $H^0(X, L^{\otimes m})^G$.

Step 2: The affine charts and the morphism π. For a homogeneous invariant $s \in R_n = H^0(X, L^{\otimes n})^G$, the nonvanishing locus $X_s \subseteq X$ is G-stable, open, and affine (the complement of the support of a section of an ample bundle is affine), and its ring of functions is the degree-zero part of the localization,

$$k[X_s] = (R(X,L)_s)_0 = \left\{ \frac{f}{s^m} \;\middle|\; f \in H^0(X, L^{\otimes nm}),\ m \geq 0 \right\}$$

On this affine open the G-action is an action of a reductive group on an affine scheme, so the affine GIT quotient of definition 3.1.4 applies: $X_s /\!\!/ G = \operatorname{Spec} k[X_s]^G$, and $k[X_s]^G = (R_s)_0$ is the degree-zero part of the localization of the *invariant* ring R at s. But $(R_s)_0$ is exactly the ring of functions on the basic open $D_+(s) \subseteq \operatorname{Proj} R$. Thus the affine quotient of the chart X_s is canonically the affine open $D_+(s)$ of the projective scheme $\operatorname{Proj} R$:

$$X_s /\!\!/ G = \operatorname{Spec} (R_s)_0 = D_+(s) \subseteq \operatorname{Proj} R$$

As s ranges over homogeneous invariants, the X_s cover X^{ss} (by the definition of semistability, every semistable point lies in some X_s) and the $D_+(s)$ cover $\operatorname{Proj} R$. The affine quotient maps $X_s \to D_+(s)$ agree on overlaps $X_s \cap X_{s'} = X_{ss'}$, because on the overlap both restrict to the affine quotient of $X_{ss'}$: localization at s' commutes with taking invariants (invariance is preserved under multiplication by the invariant s' and its inverse, both G-fixed), so $(R_{ss'})_0$ is the common ring of functions computed either way. The maps therefore glue to a single morphism

$$\pi \colon X^{\mathrm{ss}} \longrightarrow \operatorname{Proj} R = X^{\mathrm{ss}} /\!\!/ G$$

Step 3: π is a good quotient. Each of the properties defining a good quotient (definition 3.1.1) is affine-local on the target and holds on each chart $D_+(s)$ by the corresponding property of the affine GIT quotient $X_s \to X_s /\!\!/ G$ (which is a good quotient because G is reductive, the content of the affine theory around definition 3.1.4). Explicitly: π is affine, since $\pi^{-1}(D_+(s)) = X_s$ is affine over $D_+(s)$; π is G-invariant, since each chart map is; π is surjective, since each $X_s \to D_+(s)$ is; and $\mathcal{O}_Y = (\pi_* \mathcal{O}_{X^{\mathrm{ss}}})^G$, since on each chart this reads $k[X_s]^G = (R_s)_0$, which is the invariance statement of the affine quotient. The orbit-separation property, that π separates disjoint closed G-invariant subsets, is again checked chart by chart: on the affine X_s it is exactly the separation property of the affine GIT quotient $X_s \to X_s /\!\!/ G$ established for reductive G in the affine theory around definition 3.1.4 (invariant functions separate disjoint closed invariant sets). This is the affine good-quotient property, so π is a good quotient. The fiber description follows: a good quotient identifies two points iff their orbit closures meet, and each fiber contains a unique closed orbit, proving (2).

Step 4: The stable locus and the geometric quotient. It remains to show that on X^{s} the good quotient π becomes geometric, that is, its fibers are single orbits, and that X^{s} is open with open image. Let $x \in X^{\mathrm{s}}$, so Gx is closed in X^{ss} with finite stabilizer, and x lies in an invariant affine chart X_s on which every orbit is closed. Consider the fiber $\pi^{-1}(\pi(x))$. It is a closed invariant subset of the affine X_s containing the closed orbit Gx. If it contained a second orbit Gy, then Gy would also be closed in X_s (all orbits in X_s are closed), and Gx, Gy would be disjoint closed invariant sets in the same fiber, contradicting the orbit-separation property of Step 3, which forces the closed orbits in a single fiber to coincide. Hence $\pi^{-1}(\pi(x)) = Gx$ is a single orbit, so $\pi|_{X^{\mathrm{s}}}$ is a geometric quotient onto its image.

For openness, work chart by chart on the affine X_s and combine two facts. First, the finite-stabilizer condition is open: the function $x \mapsto \dim Gx$ is lower semicontinuous on X_s (orbit dimension can only jump down under specialization), so its maximal value $\dim G$ is attained on an open set, and $\dim G_x = \dim G - \dim Gx$, so $U_s = \{x \in X_s \big| \dim G_x = 0\} = \{x \big| \dim Gx = \dim G\}$ is open. Finite stabilizer does not by itself force a closed orbit, which is why the second condition must be imposed and its openness argued separately.

Second, the closed-orbit condition is open on U_s. Let $\pi_s \colon X_s \to X_s /\!\!/ G$ be the affine good quotient. For $x \in U_s$ every orbit through U_s has the maximal dimension $\dim G$, so if Gx is not closed in X_s its boundary $\overline{Gx} \setminus Gx$ is a nonempty G-stable closed set consisting of orbits of dimension strictly less than $\dim G$; each such boundary point therefore lies in the strictly-smaller-dimension locus $X_s \setminus U_s$. Consequently, for $x \in U_s$ the orbit Gx is closed in X_s if and only if the fiber $\pi_s^{-1}(\pi_s(x))$ meets U_s in Gx alone: the unique closed orbit in that fiber (which every good-quotient fiber contains) has dimension $\dim G$ and hence lies in U_s, and it equals Gx exactly when Gx is itself closed. The non-closed orbits in U_s are therefore precisely the points of U_s lying in the π_s-saturation of $X_s \setminus U_s$, that is, $U_s \cap \pi_s^{-1}\big(\pi_s(X_s \setminus U_s)\big)$: a maximal-dimension orbit fails to be closed exactly when its closure meets the lower-dimensional locus, which happens exactly when it shares a π_s-fiber with a point of $X_s \setminus U_s$. Since π_s is a good quotient it maps the closed G-stable set $X_s \setminus U_s$ to a closed set (good quotients send closed invariant sets to closed sets), so $\pi_s^{-1}(\pi_s(X_s \setminus U_s))$ is closed, and its complement intersected with U_s,

$$X_s^{\mathrm{s}} := U_s \setminus \pi_s^{-1}\big(\pi_s(X_s \setminus U_s)\big)$$

is open in X_s. This is exactly the set of $x \in X_s$ with finite stabilizer and closed orbit, so the closed-orbit condition is open on U_s. Taking the union over homogeneous invariants s, the stable locus $X^{\mathrm{s}} = \bigcup_s X_s^{\mathrm{s}}$ is open and G-stable. Its image $Y^{\mathrm{s}} = \pi(X^{\mathrm{s}})$ is open because π is a good quotient (open G-saturated sets have open image under a good quotient), and $X^{\mathrm{s}} = \pi^{-1}(Y^{\mathrm{s}})$ is G-saturated by construction. Thus $\pi \colon X^{\mathrm{s}} \to Y^{\mathrm{s}}$ is a geometric quotient X^{s}/G, completing the proof.

The theorem is the projective completion of the affine story. Where the affine quotient $X/\!\!/G$ collapses the failure of orbit separation into a single scheme with no discarded points, the projective quotient must first discard the unstable locus, on which invariant theory is blind, and then perform the same collapse on what remains. Part (2) is the statement that on the semistable locus one can only build a good quotient: strictly semistable orbits, those that are not closed, get glued to the closed orbit in their closure, and the quotient cannot see the difference. Part (3) is the reward for the extra hypotheses defining stability: on the stable locus the orbits are already closed and separated, so nothing is glued, and the quotient is geometric. The moduli interpretation is immediate: if the objects being classified correspond to stable points and their automorphisms to the finite stabilizers, then X^{s}/G is a coarse moduli space, since its points are exactly the isomorphism classes.

3.3.5 Remark: Strictly semistable points and S-equivalence A point that is semistable but not stable is called *strictly semistable*. Its orbit is not closed in X^{ss}, so its closure contains a strictly smaller orbit, and π sends both to the same point. Two semistable points with the same image are called *S-equivalent*: they are identified in the good quotient precisely because their orbit closures meet. The points of $X^{\mathrm{ss}} /\!\!/ G$ are thus in bijection not with orbits but with S-equivalence classes, equivalently with closed orbits in X^{ss}. This is why the semistable quotient is only a good quotient and not geometric: the price of including strictly semistable points is that distinct orbits can share an image. In the moduli of vector bundles, S-equivalence is exactly the relation identifying a bundle with the direct sum of the factors of its Jordan-Holder filtration.

The construction earns its keep on a concrete moduli problem: configurations of points on the line, that is, unordered configurations of n points on $\mathbb{P}^1$ taken up to the automorphisms of the line. This is the projective analogue of the affine example above and the setting in which the classical cross-ratio reappears as the first invariant.

The automorphism group of $\mathbb{P}^1$ is PGL_2, and n unordered points are the roots of a binary form of degree n, an element of $\mathbb{P}H^0(\mathbb{P}^1, \mathcal{O}(n)) = \mathbb{P}^n$ up to scalar. The action of PGL_2 on configurations lifts to the action of SL_2 on binary forms (the $2:1$ cover $\mathrm{SL}_2 \to \mathrm{PGL}_2$ has kernel $\pm I$, which acts on a degree-n form by the scalar $(-1)^n$ and hence trivially on $\mathbb{P}^n$ for every n), so the moduli problem is the GIT quotient of $\mathbb{P}^n = \mathbb{P}(\mathrm{Sym}^n(k^2)^\vee)$ by SL_2, linearized by $\mathcal{O}_{\mathbb{P}^n}(1)$ with its natural SL_2-action. Stability of a configuration is then decided by the invariant sections, which are the classical invariants of binary forms.

Example 3.3.6: n Points on $\mathbb{P}^1$ modulo PGL_2

Let SL_2 act on binary forms of degree n, that is, on $V = \mathrm{Sym}^n(k^2)^\vee$ with coordinates the coefficients of $F(x,y) = \sum_{i=0}^n a_i \binom{n}{i} x^{n-i}y^i$, and consider $X = \mathbb{P}(V) = \mathbb{P}^n$ with $L = \mathcal{O}_{\mathbb{P}^n}(1)$ linearized by the induced SL_2-action. A point of X is a configuration $D = \{p_1, \ldots, p_n\}$ of n points on $\mathbb{P}^1$ (with multiplicity), the zero divisor of F. The stability of D is governed by the multiplicities of its points.

The stability criterion. A configuration D is

1. *stable* iff every point of $\mathbb{P}^1$ occurs in D with multiplicity strictly less than $n/2$;
2. *semistable* iff every point occurs with multiplicity at most $n/2$;
3. *unstable* iff some point occurs with multiplicity strictly greater than $n/2$.

The full derivation of this criterion from one-parameter subgroups is the content of the Hilbert-Mumford analysis of section 3.4; here the picture is read off directly from invariant sections in the small cases. The heuristic is that a form is unstable exactly when it can be pushed to 0 by the group, and a form with a root of multiplicity $> n/2$ can be scaled to concentrate all its mass at that root, sending every invariant to zero.

The case $n = 2$. A binary quadratic $F = a_0x^2 + 2a_1xy + a_2y^2$ has the discriminant $\Delta = a_0a_2 - a_1^2$ as an SL_2-invariant, and Δ spans the invariants in low degree. So D is semistable iff $\Delta \neq 0$, that is, iff the two roots are distinct, each of multiplicity 1, equal to the threshold $2/2 = 1$; since 1 is not strictly less than 1, no configuration of two points is stable, and the semistable ones are the pairs of distinct points. But any two ordered distinct points can be moved to $\{0, \infty\}$ by PGL_2, and the stabilizer of $\{0,\infty\}$ is positive-dimensional (the diagonal torus). The quotient $X^{\mathrm{ss}}/\!\!/\mathrm{SL}_2$ is a single point: all pairs of distinct points are equivalent, and there is nothing to record. This reflects the direct fact that any pair of distinct points can be carried to $\{0,\infty\}$ by an element of PGL_2, so PGL_2 acts transitively on unordered pairs of distinct points.

The case $n=3$. For a binary cubic the analogous invariant is again the discriminant, of degree 4 in the coefficients, nonzero iff the three roots are distinct. Here $n/2 = 3/2$, so a point may appear with multiplicity $1 < 3/2$ but a double point has multiplicity $2 > 3/2$ and is unstable. Since $3/2$ is not an integer, no multiplicity can equal the threshold, so semistable and stable coincide: D is stable = semistable iff its three points are distinct, and unstable as soon as any two collide. Any three distinct points can be moved to $\{0,1,\infty\}$ by PGL_2: for an ordering of the three points the element carrying them to $(0,1,\infty)$ in that order is unique, so the unordered set $\{0,1,\infty\}$ is reached by the six elements corresponding to the six orderings. The stabilizer of the unordered set $\{0,1,\infty\}$ is therefore the anharmonic group S_3 (order 6, generated by $z\mapsto 1-z$ and $z\mapsto 1/z$), which is finite, so the quotient is again a single point. Three unordered points on $\mathbb{P}^1$ have no modulus.

The case $n=4$ and the cross-ratio. Four points are the first configuration with a genuine modulus. Here $n/2 = 2$: D is stable iff all four points are distinct (each multiplicity $1 < 2$), strictly semistable iff its maximum multiplicity is exactly 2 (so all multiplicities are ≤ 2 but not all < 2), and unstable iff some point has multiplicity ≥ 3. The strictly semistable locus therefore comprises both the two-double-point configurations $(2,2)$ and the one-double-point configurations $(2,1,1)$: for instance the quartic with roots $\{0,0,1,-1\}$, $F = x^4 - x^2y^2$, has $a_0 = 1$ and $a_2 = -1/6$, so $I = 3a_2^2 = 1/12 \neq 0$; not every invariant vanishes, and F is semistable but not stable. The ring of SL_2-invariants of binary quartics is generated by two invariants

$$I = a_0a_4 - 4a_1a_3 + 3a_2^2, \qquad J = a_0a_2a_4 - a_0a_3^2 - a_1^2a_4 + 2a_1a_2a_3 - a_2^3$$

of degrees 2 and 3, with the discriminant $\Delta = I^3 - 27J^2$ up to a scalar. The quotient is

$$X^{\mathrm{ss}}/\!\!/\mathrm{SL}_2 = \operatorname{Proj} k[I,J] = \mathbb{P}(2,3) \cong \mathbb{P}^1$$

the weighted projective line $\operatorname{Proj} k[I,J]$ with $\deg I = 2$, $\deg J = 3$, which is abstractly $\mathbb{P}^1$ with coordinate $j = I^3/\Delta$. Four ordered distinct points on $\mathbb{P}^1$ have a cross-ratio $\lambda \in \mathbb{P}^1 \setminus \{0,1,\infty\}$; forgetting the ordering identifies the six values $\lambda, 1-\lambda, 1/\lambda, \ldots$ under the anharmonic action of S_3, and the resulting invariant is exactly the classical

$$j(\lambda) = \frac{(\lambda^2-\lambda+1)^3}{\lambda^2(\lambda-1)^2}$$

up to normalization. Thus the GIT quotient of four points on $\mathbb{P}^1$ is the j-line. Every strictly semistable quartic, whether of type $(2,2)$ or $(2,1,1)$, is S-equivalent to the single closed strictly semistable orbit x^2y^2: the good quotient identifies all of them with that closed orbit, so the entire strictly semistable locus collapses to the one point of $X^{\mathrm{ss}}/\!\!/\mathrm{SL}_2$ (the boundary point $j=\infty$, cross-ratio degenerating to $\{0,1,\infty\}$) where stability fails.

The four-point computation is the first appearance of the j-invariant in this book from the GIT side, and it is the same j-line that arose as the coarse moduli space of elliptic curves in theorem 1.3.6, no accident, since an elliptic curve is a double cover of $\mathbb{P}^1$ branched at four points, and the cross-ratio of those four points determines the curve up to isomorphism. The stable locus (four distinct points) is a fine-enough quotient to be geometric; the strictly semistable locus (all configurations of maximum multiplicity exactly 2) contributes the single boundary point where distinct orbits are S-equivalent. The general pattern, that stability is a bound on how concentrated a configuration may be, is what the numerical criterion of the next section makes precise and computable for arbitrary n and arbitrary reductive G.

Exercise 3.3.1

1. Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, t^{-1}y)$, and extend to $X = \mathbb{P}^2$ (with coordinates $[x : y : z]$) by fixing the extra coordinate, $t \cdot [x : y : z] = [tx : t^{-1}y : z]$. Linearize $L = \mathcal{O}_{\mathbb{P}^2}(1)$ by this action on the coordinates. Compute the invariant sections in each degree, determine the semistable and stable loci, and identify $X^{\mathrm{ss}} /\!\!/ \mathbb{G}_m$ as a scheme.
2. Show directly from definition 3.3.1 that the unstable locus is closed and G-stable, and that the semistable locus is open and G-stable. Deduce that the boundary $X^{\mathrm{ss}} \setminus X^{\mathrm{s}}$ (the strictly semistable locus) is a closed subset of X^{ss}.
3. In the setting of example 3.3.6 with $n = 5$ (binary quintics under SL_2), determine the stability of the following configurations, using the multiplicity criterion: (i) five distinct points; (ii) one double point and three simple points; (iii) one triple point and two simple points; (iv) a point of multiplicity 4 and one simple point.
4. Let $\pi \colon X^{\mathrm{ss}} \to X^{\mathrm{ss}} /\!\!/ G$ be the good quotient of theorem 3.3.4. Prove that a G-invariant morphism $f \colon X^{\mathrm{ss}} \to Z$ to any scheme Z factors uniquely through π, that is, π is a categorical quotient. (Use the good-quotient property that $\mathcal{O}_Y = (\pi_* \mathcal{O}_{X^{\mathrm{ss}}})^G$ and reduce to the affine charts.)
5. Explain why the quotient of two points on $\mathbb{P}^1$ modulo PGL_2 (the case $n = 2$ of example 3.3.6) is a single point, while the quotient of four points is one-dimensional. Precisely: count the dimension of the expected geometric quotient as $\dim X^{\mathrm{s}} - \dim \mathrm{PGL}_2$ in each case, and reconcile the answer with the stability criterion.

3.4 The Hilbert-Mumford Numerical Criterion

The stability notions of the previous section are defined by the existence or non-existence of invariant sections: a point x is semistable when some G-invariant section of a power of the linearization L fails to vanish at x (definition 3.3.1). This is a clean definition, but it is a poor tool for deciding stability in practice. To certify that x is semistable one must exhibit an invariant section not vanishing at x; to certify that x is unstable one must show that *every* invariant section vanishes there. Both require understanding the full invariant ring $\bigoplus_{n \geq 0} H^0(X, L^{\otimes n})^G$, which is exactly the object whose computation is hard. For the action of SL_2 on binary forms of degree d, the invariant ring is a classical and intricate object whose generators were the subject of nineteenth-century invariant theory; reading stability off its structure directly is not feasible beyond small d.

The Hilbert-Mumford criterion resolves this by replacing the group G with its one-parameter subgroups. A one-parameter subgroup is a homomorphism $\lambda \colon \mathbb{G}_m \to G$, and it turns the abstract question of stability into a finite weight computation: for each λ one forms an integer $\mu(x, \lambda)$ measuring how the linearization degenerates as $\lambda(t) \cdot x$ is driven to its limit at $t \to 0$, and x is semistable if and only if $\mu(x, \lambda) \geq 0$ for every λ. The gain is decisive: whether x is unstable is decided by producing a single destabilizing λ, and $\mu(x,\lambda)$ is computed by diagonalizing the λ-action, a linear-algebra task. The criterion is what makes GIT a computational subject.

Fix a reductive group G acting on a projective scheme X over the algebraically closed field k, together with a G-linearized ample line bundle L (definition 3.2.1). The starting object is the simplest nontrivial family of subgroups of G.

Definition 3.4.1: One-Parameter Subgroup and the Numerical Function

A *one-parameter subgroup* (abbreviated *1-PS*) of G is a homomorphism of algebraic groups

$$\lambda\colon \mathbb{G}_m \longrightarrow G$$

It is *trivial* if it is the constant map to the identity. Fix a point $x \in X$ and consider the orbit map $t \mapsto \lambda(t)\cdot x$, a morphism $\mathbb{G}_m \to X$. Since X is projective, hence proper, this morphism extends uniquely to a morphism $\mathbb{A}^1 \to X$, and its value at $t=0$ is a well-defined point

$$x_0 \;=\; \lim_{t\to 0} \lambda(t)\cdot x \;\in\; X$$

The point x_0 is fixed by λ: it satisfies $\lambda(s)\cdot x_0 = x_0$ for all $s \in \mathbb{G}_m$. Because x_0 is λ-fixed, $\mathbb{G}_m$ acts through λ on the one-dimensional fiber L_{x_0}, and every action of $\mathbb{G}_m$ on a one-dimensional vector space is scalar multiplication by a character $t \mapsto t^r$ for a unique integer r. The *numerical function* of the pair (x,λ) is

$$\mu(x,\lambda) \;=\; r$$

where $t \mapsto t^r$ is the character by which λ acts on L_{x_0}. Equivalently, $\mu(x,\lambda) = -r'$ where $t \mapsto t^{r'}$ is the character by which λ acts on the dual line $L_{x_0}^{\vee} = \mathcal{O}(-1)_{x_0}$, the fiber over x_0 of the tautological subbundle; the sign matches the weight formula below because the affine lift $\hat{x}$ lives in $\mathcal{O}(-1)$.

The sign convention is the one that makes the criterion read "semistable iff $\mu \geq 0$" rather than "$\mu \leq 0$"; what μ measures becomes visible in coordinates. Choose a λ-eigenbasis and think of L as embedding X in projective space, so that a point is represented by a nonzero vector $\hat{x}$ in the affine cone over X. Writing $\hat{x} = \sum_i \hat{x}_i$ in eigencoordinates with $\lambda(t)$ acting on the i-th coordinate by t^{w_i}, the limit $\lim_{t\to 0}\lambda(t)\cdot\hat{x}$ is governed by the *smallest* weight w_i appearing with $\hat{x}_i \neq 0$, and one computes

$$\mu(x,\lambda) \;=\; -\min\{\, w_i \,\big|\, \hat{x}_i \neq 0 \,\}$$

Thus $\mu(x,\lambda) > 0$ means the minimal weight is negative: some eigencoordinate of $\hat{x}$ is scaled up (its coordinate blown toward infinity) as $t \to 0$, which is precisely the direction in which the affine lift of the orbit runs away from the origin. A positive μ signals that the orbit resists being pulled into the vertex of the cone, and it is this resistance that stability records.

3.4.2 Note: Computing μ by weights In practice $\mu(x,\lambda)$ is never computed from the abstract fiber L_{x_0}. One fixes a G-equivariant closed embedding $X \hookrightarrow \mathbb{P}(V)$ with $L = \mathcal{O}_{\mathbb{P}(V)}(1)|_X$, where V is a finite-dimensional G-representation. Diagonalizing the $\mathbb{G}_m$-action on V through λ gives a weight decomposition $V = \bigoplus_{w\in\mathbb{Z}} V_w$, where $\lambda(t)$ acts on V_w by t^w. A lift $\hat{x} \in V \setminus \{0\}$ of x decomposes as $\hat{x} = \sum_w \hat{x}_w$, and then

$$\mu(x,\lambda) \;=\; -\min\{\, w \,|\, \hat{x}_w \neq 0 \,\}$$

The value is independent of the chosen lift $\hat{x}$ (rescaling $\hat{x}$ does not change which weights occur) and of the ample power of L used (passing to $L^{\otimes n}$ multiplies every weight by n, hence scales μ by n and preserves its sign).

The numerical function is additive and homogeneous in a way that lets one restrict attention to a manageable supply of one-parameter subgroups. The following elementary properties are used repeatedly.

Proposition 3.4.3: Elementary Properties of the Numerical Function

Let $x \in X$, let λ, λ' be one-parameter subgroups of G, and let $g \in G$. Then:

1. For every integer $n \geq 1$, the 1-PS λ^n defined by $t \mapsto \lambda(t^n)$ satisfies $\mu(x,\lambda^n) = n\,\mu(x,\lambda)$.
2. $\mu(g \cdot x,\, g\lambda g^{-1}) = \mu(x,\lambda)$, where $g\lambda g^{-1}$ is the 1-PS $t \mapsto g\lambda(t)g^{-1}$.
3. If x is itself λ-fixed, then $\mu(x,\lambda)$ is r where $t \mapsto t^r$ is the character of λ on L_x; in particular $\mu(x,\lambda) + \mu(x,\lambda^{-1}) = 0$ in this case, where $\lambda^{-1}(t) = \lambda(t^{-1})$.

***Proof*:**

1. Replacing t by t^n in the orbit map does not change the limit point: $\lim_{t\to 0} \lambda(t^n) \cdot x = x_0$ still, since $t^n \to 0$ as $t \to 0$. The character of λ^n on L_{x_0} is the n-th power $t \mapsto (t^n)^r = t^{nr}$ of the character $t \mapsto t^r$ of λ, so $\mu(x,\lambda^n) = nr = n\,\mu(x,\lambda)$.
2. The element g carries the orbit of x under λ to the orbit of $g \cdot x$ under $g\lambda g^{-1}$: indeed $(g\lambda g^{-1})(t) \cdot (g \cdot x) = g \cdot (\lambda(t) \cdot x)$. Passing to limits, the limit point of $g \cdot x$ under $g\lambda g^{-1}$ is $g \cdot x_0$. The G-linearization identifies the fiber $L_{g\cdot x_0}$ with L_{x_0} compatibly with the two $\mathbb{G}_m$-actions (this is exactly the content of a linearization, definition 3.2.1), so the two actions on the fibers have the same character, giving equal numerical functions.
3. If x is λ-fixed then $x_0 = x$, so the fiber in question is L_x and the first claim is the definition. For the second, λ^{-1} acts on L_x by $t \mapsto (t^{-1})^r = t^{-r}$, hence $\mu(x,\lambda^{-1}) = -r = -\mu(x,\lambda)$, and the two sum to zero.

This establishes all three properties, completing the proof.

The homogeneity in part (1) means the *sign* of $\mu(x,\lambda)$, which is all the criterion asks about, depends only on the ray spanned by λ in the lattice of one-parameter subgroups, not on the particular integral multiple chosen. Part (2) says the numerical function is a G-invariant notion once one moves the point and the subgroup together, so verifying stability may be carried out after replacing λ by any conjugate, which is what allows one to reduce to 1-PS landing in a fixed maximal torus. Part (3) is the source of the fixed-point sign flip that will force strictly positive μ at stable points.

The first concrete instance is the diagonal torus in SL_2, which will carry the binary-forms computation later in the section and already illustrates the entire mechanism.

Example 3.4.4: The Diagonal 1-PS in SL_2

Let $G = \mathrm{SL}_2$ act on $V = k^2$ in the standard way, with basis e_1, e_2, and let $\lambda_a \colon \mathbb{G}_m \to \mathrm{SL}_2$ be the diagonal 1-PS

$$\lambda_a(t) = \begin{pmatrix} t^a & 0 \\ 0 & t^{-a} \end{pmatrix}$$

for a positive integer a; the entries multiply to 1, so this indeed lands in SL_2. On V the weights are $+a$ on e_1 and $-a$ on e_2. Consider the induced action on the projective line $\mathbb{P}(V) = \mathbb{P}^1$, linearized by $L = \mathcal{O}_{\mathbb{P}^1}(1)$, whose global sections $H^0(\mathbb{P}^1, \mathcal{O}(1))$ are spanned by the two coordinate functions dual to e_1, e_2. Take the point $x = [1 : 0]$, represented by $\hat{x} = e_1$, of weight $+a$. The minimal weight occurring in $\hat{x}$ is $+a$, so

$$\mu([1 : 0], \lambda_a) = -a < 0$$

For $x = [0 : 1] = e_2$ the minimal weight is $-a$, giving $\mu([0 : 1], \lambda_a) = a > 0$. For a general point $x = [x_1 : x_2]$ with both $x_1, x_2 \neq 0$, the lift $\hat{x} = x_1e_1 + x_2e_2$ has minimal weight $-a$, so $\mu(x, \lambda_a) = a > 0$ as well. The single point $[1 : 0]$, whose lift e_1 sits at weight $+a$, is the only one on which λ_a takes a negative value. It is the *repelling* fixed point of the projective action: for any $x = [x_1 : x_2]$ with $x_2 \neq 0$ one has $\lambda_a(t) \cdot [x_1 : x_2] = [t^{2a}x_1 : x_2] \to [0 : 1]$ as $t \to 0$, so every other point flows away from $[1 : 0]$ and toward the attracting fixed point $[0 : 1]$. That $[1 : 0]$ is destabilized despite being repelling is exactly the point: its affine lift e_1, of positive weight, is driven to the vertex of the cone as $t \to 0$, and it is the behavior of the *lift*, not of the projective orbit, that μ records.

The computation in example 3.4.4 is the whole idea in miniature: the destabilizing behavior of a 1-PS is concentrated at a fixed point whose affine lift is driven to the vertex of the cone, and the sign of μ there detects whether that fixed point obstructs stability. It remains to prove that these signs, ranged over all λ, capture semistability and stability exactly.

The motivating observation is that an unstable point can always be recognized by a degeneration: if every invariant section vanishes at x, the affine lift of the orbit closure passes through the vertex of the affine cone, and a theorem of Hilbert and Mumford says this degeneration can be realized along a one-parameter subgroup. Conversely, if some 1-PS drives the lift to the vertex, that 1-PS visibly kills every invariant section at x. The criterion packages both directions.

Theorem 3.4.5: The Hilbert-Mumford Numerical Criterion

Let G be a reductive group acting on a projective scheme X over the algebraically closed field k, and let L be a G-linearized ample line bundle. For $x \in X$:

1. x is semistable if and only if $\mu(x, \lambda) \geq 0$ for every one-parameter subgroup λ of G.
2. x is stable if and only if $\mu(x, \lambda) > 0$ for every nontrivial one-parameter subgroup λ of G.

The two directions of the criterion cost very different amounts, and it helps to separate them before the proof. The *easy* direction of (1) is the implication "some $\mu(x,\lambda) < 0 \Rightarrow x$ unstable": a destabilizing 1-PS produces, by hand, a witness that every invariant section vanishes at x. This is elementary and is given in full below. The *hard* direction is its converse: an unstable point must admit a destabilizing 1-PS. This is the Hilbert-Mumford theorem proper, and it rests on a properness input (the statement that the vertex of the cone, if it lies in an orbit closure, is reached along a 1-PS). In the linearly reductive projective setting it is proved in full below; the one step that would otherwise require the semistable-reduction machinery of Mumford's *Geometric Invariant Theory* is isolated and cited there.

***Proof*:**
Fix a G-equivariant closed embedding $X \hookrightarrow \mathbb{P}(V)$ with $L = \mathcal{O}_{\mathbb{P}(V)}(1)|_X$ and V a finite-dimensional G-representation, which exists because L is ample and G-linearized (this is the standard realization of an ample linearization, definition 3.2.1; that some power of L affords such a projective embedding is Serre's theorem on ample line bundles, [12, Ample Line Bundles and Projective Embeddings]). Replacing L by a power multiplies every μ by a positive integer and changes neither stability nor the sign conditions, so this loses nothing. Let $\hat{X} \subset V$ be the affine cone over X, and for $x \in X$ choose a nonzero lift $\hat{x} \in \hat{X}$. Semistability of x is equivalent to the vertex 0 *not* lying in the Zariski closure $\overline{G \cdot \hat{x}}$ of the orbit of $\hat{x}$ in V: this is the affine-cone reformulation of semistability from section 3.3, since an invariant section not vanishing at x is a G-invariant homogeneous function on $\hat{X}$ not vanishing at $\hat{x}$, and a nonzero invariant separates $\hat{x}$ from 0 exactly when $0 \notin \overline{G \cdot \hat{x}}$ (the good quotient separates disjoint closed invariant sets by the Reynolds operator, definition 3.1.1 and definition 3.1.4, which is what supplies the separating invariant). The proof translates each direction through this reformulation.

Step 1: The easy direction of (1). Suppose $\mu(x,\lambda) < 0$ for some 1-PS λ. Decompose $\hat{x} = \sum_w \hat{x}_w$ into weight components for λ. By the weight formula of box 3.4.2, $\mu(x,\lambda) = -\min\{w \big| \hat{x}_w \neq 0\} < 0$ means every weight w with $\hat{x}_w \neq 0$ satisfies $w > 0$. Then

$$\lambda(t) \cdot \hat{x} = \sum_{w>0} t^w \hat{x}_w \xrightarrow{t \to 0} 0$$

since each surviving power t^w has $w > 0$. Thus $0 = \lim_{t\to 0} \lambda(t) \cdot \hat{x}$ lies in $\overline{G \cdot \hat{x}}$, so x is not semistable. This proves that semistability forces $\mu(x,\lambda) \geq 0$ for all λ, which is the easy half of (1).

Step 2: The hard direction of (1). Suppose x is not semistable, so $0 \in \overline{G \cdot \hat{x}}$. The claim is that there is a 1-PS λ with $\lim_{t\to 0} \lambda(t) \cdot \hat{x} = 0$, for then, by the weight formula run in reverse, every weight of $\hat{x}$ under λ is strictly positive and $\mu(x,\lambda) = -\min\{w \big| \hat{x}_w \neq 0\} < 0$. That such a 1-PS exists is the Hilbert-Mumford theorem in the affine setting: if the closure of a G-orbit $G \cdot v$ in a representation V contains a point v_0 in a *closed* orbit, then v_0 can be reached from v along a one-parameter subgroup, that is, there is a 1-PS λ with $\lim_{t\to 0} \lambda(t) \cdot v = g \cdot v_0$ for some $g \in G$. Applied to $v = \hat{x}$ and the closed orbit $\{0\} = G \cdot 0$, this yields a 1-PS with $\lim_{t\to 0} \lambda(t) \cdot \hat{x} = 0$, as needed.

The existence of such a 1-PS is the technical heart of the theorem. For G linearly reductive it can be proved by the following reduction to a torus, which is complete except for one properness input. Since $0 \in \overline{G \cdot \hat{x}}$, there is a curve in the orbit approaching 0: concretely, a map from the spectrum of a discrete valuation ring R with fraction field K, sending the generic point into $G \cdot \hat{x}$ and the closed point to 0. Because G is reductive, the valuative criterion applied to the quotient map $G \to G/P$ (for a suitable parabolic) together with the structure of G over R lets one arrange this curve to factor through the normalizer of a maximal torus after a finite base change; the Cartan decomposition over the local field $G(K) = \coprod_\lambda G(R)\lambda(\pi)G(R)$, where π is a uniformizer of R and λ ranges over 1-PS of a fixed maximal torus, then writes the degenerating curve as $g_1\lambda(\pi)g_2$ with $g_i \in G(R)$. Specializing g_1, g_2 to the closed point and absorbing them into the base point and the conjugate of λ produces a 1-PS of G realizing the degeneration to 0. The single step that this book does not reprove is that the degenerating curve exists and factors through a torus after base change, that is, the properness reduction above; it uses the semistable-reduction machinery for reductive groups and is cited to Mumford's *Geometric Invariant Theory* [25] (Chapter 2), where it is proved in the generality needed. With the 1-PS in hand, $\mu(x, \lambda) < 0$ as computed above, so the contrapositive gives: $\mu(x, \lambda) \geq 0$ for all λ implies x semistable. Together with Step 1 this proves (1).

Step 3: The stable direction. Recall from definition 3.3.1 that x is stable when it is semistable, its stabilizer G_x is finite (equivalently $\dim G \cdot x = \dim G$), and its orbit $G \cdot x$ is closed in the semistable locus. In the affine-cone picture, stability of x is equivalent to the orbit $G \cdot \hat{x}$ being closed in $V \setminus \{0\}$ *and* of dimension $\dim G$ (the stabilizer of $\hat{x}$ being finite modulo the scalars that act on the cone). This is the standard translation from section 3.3.

Suppose first that x is stable, and let λ be a nontrivial 1-PS. Since x is semistable, $\mu(x, \lambda) \geq 0$ by (1). Suppose toward a contradiction that $\mu(x, \lambda) = 0$. Then the minimal weight of $\hat{x}$ under λ is 0, so $\hat{x}_0 := \lim_{t\to 0} \lambda(t) \cdot \hat{x} = \sum_{w=0} \hat{x}_w$ is a nonzero point of $\hat{X}$, and it is a λ-fixed point in the orbit closure $\overline{G \cdot \hat{x}}$. If $\hat{x}_0 \neq \hat{x}$ up to the orbit, that is, if $\hat{x}_0 \notin G \cdot \hat{x}$, then $\overline{G \cdot \hat{x}}$ strictly contains $G \cdot \hat{x}$, so $G \cdot \hat{x}$ is not closed in $V \setminus \{0\}$ (it accumulates on the nonzero point $\hat{x}_0$), contradicting stability. Hence $\hat{x}_0 \in G \cdot \hat{x}$, say $\hat{x}_0 = g \cdot \hat{x}$. Then $g^{-1}\lambda g$ fixes $\hat{x}$, so $g^{-1}\lambda(t)g$ lies in the stabilizer $G_{\hat{x}}$ for all t; but $G_{\hat{x}}$ is finite (stability), and a nontrivial 1-PS has infinite image, a contradiction. Therefore $\mu(x, \lambda) > 0$ for every nontrivial λ.

Conversely, suppose $\mu(x, \lambda) > 0$ for every nontrivial 1-PS λ. Then in particular $\mu(x, \lambda) \geq 0$ for all λ, so x is semistable by (1). It remains to show $G \cdot \hat{x}$ is closed in $V \setminus \{0\}$ and $\dim G \cdot x = \dim G$. If $G \cdot \hat{x}$ were not closed in $V \setminus \{0\}$, its closure would meet a strictly smaller orbit at some nonzero point $\hat{y}$; applying the Hilbert-Mumford theorem of Step 2 to the closed orbit $G \cdot \hat{y}$ inside $\overline{G \cdot \hat{x}}$ produces a nontrivial 1-PS λ with $\lim_{t\to 0} \lambda(t) \cdot \hat{x} = g \cdot \hat{y} \neq 0$. The limit being a nonzero point means the minimal weight of $\hat{x}$ under λ is 0, so $\mu(x, \lambda) = 0$, contradicting the hypothesis. Hence $G \cdot \hat{x}$ is closed. If instead $\dim G \cdot x < \dim G$, the stabilizer G_x is positive-dimensional; moreover, because $G \cdot \hat{x}$ was just shown closed, the stabilizer $G_{\hat{x}}$ of the lift is reductive by Matsushima's criterion (the stabilizer of a point with closed orbit in an affine G-variety is reductive), and G_x differs from $G_{\hat{x}}$ only by the scalars acting on the cone, hence is reductive as well. A positive-dimensional reductive group contains a nontrivial torus, hence a nontrivial 1-PS λ; that λ fixes x, so by part (3) of proposition 3.4.3, $\mu(x, \lambda) + \mu(x, \lambda^{-1}) = 0$, forcing one of the two values to be ≤ 0 and contradicting strict positivity. Hence $\dim G \cdot x = \dim G$ and x is stable. This proves (2), completing the proof.

The criterion converts an existence-of-invariants problem into a sign-of-weights computation. To test semistability one need only bound $\mu(x, \lambda)$ below by 0 across all 1-PS, and by part (2) of proposition 3.4.3 it suffices to check 1-PS in a single fixed maximal torus T, since every 1-PS is conjugate into T and μ is conjugation-invariant once the point is moved along. Inside T the 1-PS form a lattice $\mathbb{Z}^r$ (with $r = \dim T$), and $\mu(x, -)$ is a piecewise linear function on this lattice; deciding a sign across a lattice is a finite combinatorial problem. This is what reduces GIT stability to the linear algebra of weights.

3.4.6 Note: Why one maximal torus suffices The reduction to a single torus is the practical form of the criterion. Every 1-PS λ of G has image in some maximal torus, and any two maximal tori are conjugate; so for a fixed maximal torus T, every 1-PS is of the form $g\lambda' g^{-1}$ with λ' a 1-PS of T and $g \in G$. By proposition 3.4.3(2), $\mu(x, g\lambda' g^{-1}) = \mu(g^{-1} \cdot x, \lambda')$. Thus checking $\mu(x, \lambda) \geq 0$ for all λ is the same as checking $\mu(g \cdot x, \lambda') \geq 0$ for all 1-PS λ' of the fixed torus T and all $g \in G$, that is, over the orbit of x and one torus. When the orbit is understood (as it is for binary forms, where every form is an SL_2-translate of one with a controlled root pattern), this is a finite check.

The archetypal application, and the example that motivated Hilbert's original work, is the action of SL_2 on binary forms. A binary form of degree d is a homogeneous polynomial

$$f(x, y) = \sum_{i=0}^{d} a_i \, x^{d-i} y^i$$

of degree d in two variables; the space of such forms is the SL_2 representation $V_d = \mathrm{Sym}^d(k^2)^\vee$, of dimension $d+1$, and SL_2 acts by substitution on the variables. The projectivization $\mathbb{P}(V_d) = \mathbb{P}^d$ is the space of binary forms up to scalar, and it is linearized by $\mathcal{O}_{\mathbb{P}^d}(1)$ with its natural SL_2-action. Since k is algebraically closed, a nonzero binary form factors into linear factors,

$$f(x, y) = c \prod_{j=1}^{d} (\beta_j x - \alpha_j y)$$

so f is determined up to scalar by its *roots* $[\alpha_j : \beta_j] \in \mathbb{P}^1$ counted with multiplicity; a form of degree d corresponds to an unordered d-tuple of points on $\mathbb{P}^1$. The stability question becomes a question about configurations of d points on the projective line, connecting directly to example 3.3.6.

Example 3.4.7: Stability of Binary Forms
A nonzero binary form f of degree d, viewed as a point of $\mathbb{P}(V_d)$, is

1. *semistable* if and only if every root of f has multiplicity at most $d/2$;
2. *stable* if and only if every root of f has multiplicity strictly less than $d/2$.

Setting up the weight computation. By box 3.4.6, it suffices to evaluate $\mu(f, \lambda)$ for 1-PS λ of the diagonal maximal torus $T \subset \mathrm{SL}_2$, after moving f by SL_2. Every nontrivial 1-PS of T is $\lambda_a(t) = \mathrm{diag}(t^a, t^{-a})$ for some integer $a \neq 0$, and by homogeneity (proposition 3.4.3(1)) the sign of μ is unchanged under $a \mapsto na$, so one may take $a = 1$ and its inverse; write $\lambda = \lambda_1$. A monomial basis of V_d is $\{x^{d-i} y^i\}_{i=0}^{d}$. The 1-PS $\lambda(t)$ sends $x \mapsto t^{-1}x$ and $y \mapsto t\,y$ acting on forms by substitution (the contragredient of the action on k^2), so it scales the monomial $x^{d-i}y^i$ by $t^{-(d-i)} t^i = t^{\,2i-d}$. The weight of the basis vector $x^{d-i}y^i$ is therefore

$$w_i \;=\; 2i - d$$

running through $-d$, $-d+2, \ldots, d-2$, d as i runs from 0 to d.

The weight formula for μ. Write $f = \sum_i a_i \, x^{d-i} y^i$. By box 3.4.2,

$$\mu(f, \lambda) \;=\; -\min\{\, w_i \big| a_i \neq 0 \,\} \;=\; -\min\{\, 2i - d \big| a_i \neq 0 \,\}$$

Let m be the multiplicity of the root $[1:0]$ of f, that is, the largest power of y dividing f (the point $[1:0]$ is the root $y=0$, cut out by y in the coordinates dual to e_1, e_2; here the monomial $x^{d-i}y^i$ is divisible by y^i). Then $a_i = 0$ for $i < m$ and $a_m \neq 0$, so the minimum index i with $a_i \neq 0$ is exactly m, giving

$$\mu(f, \lambda) \;=\; -(2m-d) \;=\; d-2m$$

Thus the numerical function of the standard diagonal 1-PS reads off the multiplicity of the root at $[1:0]$: it is ≥ 0 if and only if $m \leq d/2$, and > 0 if and only if $m < d/2$.

From one 1-PS to all of them. A general 1-PS of SL_2, up to conjugacy and $a \mapsto na$, is $g\lambda g^{-1}$ for some $g \in \mathrm{SL}_2$, and $\mu(f, g\lambda g^{-1}) = \mu(g^{-1} \cdot f, \lambda)$ by proposition 3.4.3(2). The element g moves the root $[1:0]$ to an arbitrary point $p = g \cdot [1:0] \in \mathbb{P}^1$, and the multiplicity of p as a root of f equals the multiplicity of $[1:0]$ as a root of $g^{-1} \cdot f$. So as g ranges over SL_2,

$$\mu(f, g\lambda g^{-1}) \;=\; d - 2\, m_p$$

where $m_p = \mathrm{mult}_p(f)$ is the multiplicity of f at $p = g \cdot [1:0]$. Every point $p \in \mathbb{P}^1$ arises this way (the SL_2-action on $\mathbb{P}^1$ is transitive), and conversely the destabilizing direction detected by any 1-PS λ_a with $a < 0$ swaps the roles of $[1:0]$ and $[0:1]$, which is already covered by ranging over all p. Hence the full family of values of $\mu(f, -)$ over all nontrivial 1-PS is exactly

$$\{\, d - 2\, m_p \big| p \in \mathbb{P}^1 \,\}$$

Reading off stability. By theorem 3.4.5:

- f is semistable iff $d - 2m_p \geq 0$ for all p, that is, $m_p \leq d/2$ for every point p: no root has multiplicity exceeding $d/2$.
- f is stable iff $d - 2m_p > 0$ for all p and every nontrivial 1-PS, that is, $m_p < d/2$ for every point p: every root has multiplicity strictly below $d/2$.

Only the roots of f contribute a nonzero multiplicity, so the condition is vacuous away from the roots, and the two statements are exactly (1) and (2).

The computation makes the geometry transparent. Instability of a binary form is caused by a single root that is "too fat", concentrating more than half the degree; the destabilizing 1-PS is the diagonal torus that fixes that root and, running $t \to 0$, sweeps every other root away from it toward the antipodal fixed point, thinning the configuration everywhere except at the fat root. For $d = 2$ the threshold $d/2 = 1$ says a conic pair of points on $\mathbb{P}^1$ is semistable exactly when the two points are distinct (a double point has multiplicity $2 > 1$, unstable), and is never stable (a distinct pair has each multiplicity $1 = d/2$, so semistable but not stable), matching the elementary fact that SL_2 acts transitively on pairs of distinct points, so the quotient is a single point. For $d = 3$ the threshold $3/2$ says a binary cubic is stable exactly when its three roots are distinct and semistable under the same condition, since a multiplicity of 2 or 3 exceeds $3/2$; a nodal cubic form (two roots, one doubled) is unstable. For $d = 4$ the threshold is 2: a binary quartic is semistable when no root is worse than a double point and stable when all four roots are distinct, and the semistable-but-not-stable locus consists of forms with exactly one double root, which is the source of the strictly semistable boundary in the GIT quotient of four points on $\mathbb{P}^1$.

3.4.8 Remark: Reconciliation with points on $\mathbb{P}^1$ example 3.3.6 analyzed n (unordered) points on $\mathbb{P}^1$ modulo PGL_2 directly through invariant sections, and found a configuration stable when no point is repeated more than $n/2$ times (and semistable with equality allowed). A degree-d binary form is precisely an unordered d-tuple of points on $\mathbb{P}^1$ (its roots with multiplicity), and $\mathrm{PGL}_2 = \mathrm{SL}_2/\{\pm 1\}$ acts on both descriptions compatibly, since the center $\{\pm 1\}$ acts trivially on $\mathbb{P}(V_d)$ and on configurations. The multiplicity of a root is the number of times a point is repeated, so "no root of multiplicity $> d/2$" is "no point repeated more than $d/2$ times", and the two stability criteria coincide with $n = d$. The present derivation via theorem 3.4.5 recovers example 3.3.6 without computing a single invariant section: the weight of a diagonal 1-PS did all the work that a search through the invariant ring would otherwise require.

The reconciliation is the point of the criterion. In section 3.3 the stability of point configurations was extracted by exhibiting invariants, a route that becomes impractical as d grows; here the same answer falls out of a one-line weight count, valid for all d at once. This is the leverage the criterion provides: any GIT stability question with a computable weight decomposition, including the stability of pluricanonically embedded curves that underlies the construction of the moduli of curves, reduces to counting weights of one-parameter subgroups rather than searching an invariant ring.

Exercise 3.4.1

1. Let $\lambda\colon \mathbb{G}_m \to G$ be a 1-PS and $x \in X$ a point that is already fixed by the whole group G. Show that $\mu(x, \lambda)$ can be computed without taking any limit, and that $\mu(x, \lambda) + \mu(x, \lambda^{-1}) = 0$. Deduce that a G-fixed point is semistable if and only if it is a fixed point of trivial weight for every 1-PS, and is never stable when G has a nontrivial 1-PS.

2. Consider SL_2 acting on binary forms of degree $d = 6$. Using example 3.4.7, classify the following forms as stable, strictly semistable (semistable but not stable), or unstable: (a) $f = x^6 + y^6$; (b) $f = x^3y^3$; (c) $f = x^4y^2$; (d) $f = x^5y$. In each case name the destabilizing 1-PS when one exists.

3. Let $\mathbb{G}_m$ act on $V = k^3$ with weights $(2, -1, -1)$, so that $\lambda(t) = \mathrm{diag}(t^2, t^{-1}, t^{-1})$, and consider the induced action on $\mathbb{P}^2$ linearized by $\mathcal{O}(1)$. Compute $\mu(x, \lambda)$ for the points $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[1 : 1 : 1]$. Which of these points is destabilized by λ? Since $G = \mathbb{G}_m$ is a torus, every 1-PS is λ^n; using this, determine the unstable locus in $\mathbb{P}^2$ for this linearization.

4. Prove directly, without invoking the hard direction of theorem 3.4.5, that if $x \in X$ admits a nontrivial 1-PS λ fixing x with $\mu(x, \lambda) = 0$, then x is not stable. (This isolates the argument in Step 3 that a positive-dimensional stabilizer prevents stability.)

5. In the binary-forms computation, the weight of the monomial $x^{d-i}y^i$ under $\lambda(t) = \mathrm{diag}(t, t^{-1})$ was found to be $2i - d$. Verify this directly from the definition of the SL_2 action on $V_d = \mathrm{Sym}^d(k^2)^\vee$, being careful about the contragredient, and explain why replacing λ by λ^{-1} replaces the multiplicity m of the root at $[1 : 0]$ by the multiplicity of the root at $[0 : 1]$ in the formula $\mu(f, \lambda) = d - 2m$.

6. A binary form of degree d is called *polystable* if it is semistable and its SL_2-orbit is closed in the semistable locus. Using example 3.4.7 and the orbit-closure picture from the proof of theorem 3.4.5, show that a strictly semistable binary quartic ($d = 4$) with exactly one double root has a unique polystable form in the closure of its orbit, and identify it. What does this

say about the point of the GIT quotient to which every such form is sent?

3.5 Moduli via GIT

The two great engines for building moduli spaces have now both been assembled. The Hilbert scheme produces, for a fixed projective ambient space and a fixed Hilbert polynomial, a fine moduli space of *embedded* objects: it parametrizes closed subschemes together with the very embedding that pins them down inside $\mathbb{P}^N$ (corollary 2.3.3). Geometric invariant theory, developed across the preceding four sections, produces quotients by group actions, and in particular a projective quotient of the semistable locus. This section closes the circle by combining them. The auxiliary embedding that makes an object into a point of a Hilbert scheme is exactly the extra rigidity that the group PGL_{N+1} acts to forget, and the GIT quotient of the locus of embedded objects by that group is a scheme whose points are the objects taken up to isomorphism. When the numerical stability of GIT matches the geometric class of objects one wishes to parametrize, this quotient is a coarse moduli space.

The strategy is uniform across a wide range of moduli problems, and it is the historically decisive one: it is how Mumford first constructed the moduli space of curves $\mathcal{M}_g$, and the same template governs moduli of polarized varieties, of stable vector bundles, and of abelian varieties with level structure. The section states the recipe as a proposition, proves it, and then walks through the curves case in the detail that its importance warrants, deferring only the one deep stability input to the capstone where the full construction belongs.

The obstruction to fine representability, met already in the first chapter, is automorphisms: an object with a nontrivial automorphism cannot sit as a point of a fine moduli space, because a family with nonconstant isomorphism type but constant fiber can be built by twisting along the automorphisms. The j-line (theorem 1.3.6) is the paradigm. Its objects, elliptic curves, each carry the automorphism $[-1]$ (and the special curves $j = 0, 1728$ carry more), so no fine moduli space exists; yet a coarse moduli space, the affine j-line $\mathbb{A}^1$, does. GIT reproduces exactly this behaviour. On the stable locus the stabilizers are finite, which is precisely the algebraic shadow of "finitely many automorphisms", and the geometric quotient there identifies orbits, that is, identifies embedded objects that differ only by a change of embedding, hence identifies isomorphic objects. The resulting scheme cannot in general be fine (the finite automorphisms survive as stabilizers), but it is coarse.

To state the recipe precisely, fix a moduli problem: a functor

$$F\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$$

whose value $F(S)$ is the set of isomorphism classes of families over S of the objects one wishes to classify (smooth projective curves of genus g, say, or polarized varieties with a fixed Hilbert polynomial). The bridge to the Hilbert scheme is an *auxiliary embedding*: a functorial way to embed each object in a fixed projective space, canonical up to the choice of a basis of the linear system that produces it. This is what the projective linear group acts on.

Proposition 3.5.1: Coarse Moduli via a GIT Quotient

Let $F\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ be a moduli problem for objects that admit a functorial projective embedding of a fixed type. Concretely, suppose there is a fixed projective space $\mathbb{P}^N$, a Hilbert polynomial P, and a locally closed subscheme

$$H \subseteq \mathrm{Hilb}^P(\mathbb{P}^N)$$

of the Hilbert scheme (corollary 2.3.3) whose points are exactly the images of the objects of F under their embeddings, such that:

1. the group PGL_{N+1} acts on $\mathrm{Hilb}^P(\mathbb{P}^N)$ by moving the embedding, and preserves H;
2. two points of H lie in the same PGL_{N+1}-orbit if and only if the objects they represent are isomorphic;
3. for a PGL_{N+1}-linearized ample line bundle on a PGL_{N+1}-invariant projective completion $\overline{H}$ of H, the stable locus $\overline{H}^{\mathrm{s}}$ (definition 3.3.1) equals H, and the stabilizer in PGL_{N+1} of each point of H is finite.

Then the geometric quotient

$$M := H/\mathrm{PGL}_{N+1} = \overline{H}^{\mathrm{s}}/\mathrm{PGL}_{N+1}$$

exists as a quasi-projective scheme (theorem 3.3.4) and is a coarse moduli space for F in the sense of definition 1.3.1.

***Proof*:**
The quotient M is produced by the projective GIT machinery, and the content of the proof is to check that it satisfies the two defining conditions of a coarse moduli space: it corepresents F on geometric points, and it is universal among schemes to which F maps.

Step 1: The quotient exists and is a geometric quotient. By hypothesis (3) the stable locus of $\overline{H}$ for the chosen linearization equals H, and every point of H has finite stabilizer. theorem 3.3.4 constructs the projective GIT quotient $\overline{H}^{\mathrm{ss}} /\!\!/ \mathrm{PGL}_{N+1}$ as a projective scheme and shows that on the stable locus the induced map

$$\overline{H}^{\mathrm{s}} \longrightarrow \overline{H}^{\mathrm{s}}/\mathrm{PGL}_{N+1}$$

is a geometric quotient. Since $\overline{H}^{\mathrm{s}} = H$, the quotient $M = H/\mathrm{PGL}_{N+1}$ exists, and because it is an open subscheme of the projective $\overline{H}^{\mathrm{ss}} /\!\!/ \mathrm{PGL}_{N+1}$ it is quasi-projective. Being a geometric quotient, its closed points are in bijection with the PGL_{N+1}-orbits of closed points of H.

Step 2: The geometric points of M are the isomorphism classes. By hypothesis (2), two closed points of H lie in one orbit exactly when the objects they carry are isomorphic. Combined with Step 1, the closed points of M are in natural bijection with PGL_{N+1}-orbits on H, hence with isomorphism classes of objects of F over k. Since $F(\operatorname{Spec} k)$ is by definition the set of isomorphism classes of objects over k, there is a canonical bijection

$$F(\operatorname{Spec} k) \xrightarrow{\sim} M(k) \tag{3.4}$$

which is the first requirement of definition 1.3.1.

Step 3: The natural transformation $F \to h_M$. A coarse moduli space must carry a morphism of functors $\eta\colon F \to h_M$ inducing (3.4) on k-points; construct it. Let $\xi \in F(S)$ be a family over a scheme S. The functorial embedding turns ξ into a family of embedded objects over S, that is, a closed subscheme $Z \subseteq \mathbb{P}^N_S$ flat over S with fibrewise Hilbert polynomial P, whose fibres land in H. By the universal property of the Hilbert scheme (corollary 2.3.3) this family is classified by a morphism

$$g_\xi\colon S \longrightarrow H$$

The composite $q \circ g_\xi\colon S \to M$ with the quotient map $q\colon H \to M$ depends on ξ only through its isomorphism class: an isomorphism $\xi \cong \xi'$ of families over S changes the embedding by an S-point of PGL_{N+1}, hence changes g_ξ to $g_{\xi'}$ by an element of the group along which q is invariant, so $q \circ g_\xi = q \circ g_{\xi'}$. Setting $\eta_S(\xi) := q \circ g_\xi$ gives a well-defined natural transformation $\eta\colon F \to h_M$. On k-points $\eta_{\operatorname{Spec} k}$ sends an object to the image in M of the point of H carrying its embedding, which is exactly the bijection (3.4).

Step 4: Universality. It remains to show M is initial among schemes receiving a natural transformation from F: given any scheme M' and any $\eta'\colon F \to h_{M'}$, there is a unique morphism $\psi\colon M \to M'$ with $\eta' = h_\psi \circ \eta$. Apply η' to the tautological family over H. The identity of H classifies a family $\zeta \in F(H)$ (the restriction to H of the universal embedded object), and $\eta'_H(\zeta)$ is a morphism $\varphi\colon H \to M'$. Because η' is natural and each object of F over a point is unchanged under the PGL_{N+1}-action on its embedding, φ is constant on PGL_{N+1}-orbits: for $t \in \mathrm{PGL}_{N+1}(S)$ the translated family $t \cdot \zeta_S$ is isomorphic to ζ_S as an element of $F(S)$, so $\varphi \circ (t\cdot) = \varphi$. A PGL_{N+1}-invariant morphism out of H factors uniquely through the geometric quotient $q\colon H \to M$ by the universal property of the categorical quotient underlying M ([12, Categorical and Geometric Quotients]), giving a unique $\psi\colon M \to M'$ with $\varphi = \psi \circ q$. Chasing definitions, $h_\psi \circ \eta = \eta'$, and ψ is the only such morphism because q is an epimorphism of schemes and φ determines ψ on the dense image of q. This is the universal property of definition 1.3.1, completing the proof.

The proposition isolates what has to be checked to run the GIT construction for a given moduli problem, and it makes visible where the difficulty lies. Conditions (1) and (2) are formal: they say only that the embedding is canonical up to the group and that the group's orbits are exactly the isomorphism classes. These are usually easy, once the embedding is chosen well. Condition (3) is the entire mathematical burden, and it splits into two parts. The equality $\overline{H}^{\mathrm{s}} = H$ is a *stability theorem*, an assertion that the objects one wants to parametrize are precisely the GIT-stable ones for a suitable linearization; this is never formal and is proved case by case using the Hilbert-Mumford criterion (theorem 3.4.5). The finiteness of stabilizers is the algebraic incarnation of the objects having finite automorphism groups, and it is what forces M to be coarse rather than fine.

One should read the output correctly. The scheme M is a coarse moduli space, not a fine one, precisely because the stabilizers, though finite, are generally nontrivial. A point of M remembers an object up to isomorphism but forgets its automorphisms, and (3.4) is a bijection on points that does *not* upgrade to an isomorphism of functors $F \cong h_M$. The stack-theoretic repair of exactly this defect, which remembers the automorphism groups as the stabilizers of a quotient stack $[H/\mathrm{PGL}_{N+1}]$, is the subject of the descent-and-stacks part of the book; the present construction is the coarse shadow of that stack.

3.5.2 Why PGL and not GL The group acting is the projective linear group PGL_{N+1}, not GL_{N+1}, because two embeddings of the same object into $\mathbb{P}^N$ differ by a projective change of coordinates, and the scalars $\mathbb{G}_m \subseteq \mathrm{GL}_{N+1}$ act trivially on $\mathbb{P}^N$. Using GL_{N+1} would carry a redundant central $\mathbb{G}_m$ whose orbits are points, inflating stabilizers by a positive-dimensional torus and destroying the stability needed in proposition 3.5.1(3). Because PGL_{N+1} is reductive (it is the quotient of the reductive GL_{N+1} by a central torus), the GIT machinery of theorem 3.3.4 applies unchanged. Linearizations require a little care: an ample line bundle on $\mathrm{Hilb}^P(\mathbb{P}^N)$ is naturally GL_{N+1}-linearized, and one must check the central $\mathbb{G}_m$ acts trivially on the relevant power to descend the linearization to PGL_{N+1}, which is why stability is always stated for a specific power of the natural bundle.

Before the curves example, a small self-contained instance grounds the recipe and shows every hypothesis of proposition 3.5.1 met concretely.

Example 3.5.3: Configurations of Points on the Line

Take the moduli problem $F(S) = \{\text{families of } d \text{ distinct unordered points on } \mathbb{P}^1 \text{ over } S\}$ up to isomorphism, for $d \geq 3$. An unordered d-tuple of distinct points on $\mathbb{P}^1$ is the same as a degree-d divisor with no repeated point, hence a point of the Hilbert scheme $\mathrm{Hilb}^P(\mathbb{P}^1)$ with P the constant polynomial d; this Hilbert scheme is the symmetric product $\mathrm{Sym}^d\mathbb{P}^1 \cong \mathbb{P}^d$, and the locus H of *distinct* configurations is the complement of the discriminant, an open subscheme. The automorphism group of $\mathbb{P}^1$ is PGL_2, which acts on $\mathbb{P}^d$, and two configurations are projectively equivalent exactly when they are isomorphic as abstract d-pointed schemes, so conditions (1) and (2) of proposition 3.5.1 hold.

For condition (3), linearize $\mathcal{O}_{\mathbb{P}^d}(1)$ (equivalently, act on binary forms of degree d) and apply the stability calculation of example 3.3.6: a configuration is GIT-stable precisely when every point has multiplicity strictly less than $d/2$, which for *distinct* points means each multiplicity is 1, automatic once $d \geq 3$. Thus $\overline{H}^{\mathrm{s}}$ is exactly the locus of distinct configurations, and each such configuration has finite PGL_2-stabilizer (a finite group of Mobius transformations permuting the points). All three hypotheses hold, and proposition 3.5.1 yields the coarse moduli space

$$M_{0,d}^{\mathrm{un}} = H/\mathrm{PGL}_2$$

of d distinct unordered points on the line, a quasi-projective variety of dimension $d-3$. For $d = 3$ it is a point (any three distinct points are projectively equivalent to $0, 1, \infty$); for $d = 4$ it is one-dimensional and the invariant is the cross-ratio taken up to the symmetric-group action on the four points, recovering the classical fact that the moduli of four points on $\mathbb{P}^1$ is a j-line-like affine curve.

The reason GIT earns its place beside the Hilbert scheme is the moduli of curves. For a smooth projective curve C of genus $g \geq 2$ there is a canonical projective embedding built from the curve itself, with no choices beyond a basis, and this is exactly the functorial embedding that proposition 3.5.1 requires. The construction that follows is Mumford's, and it is the historical origin of geometric invariant theory: GIT was created to make this construction work.

The embedding comes from pluricanonical linear systems. For a smooth curve C of genus $g \geq 2$ the canonical bundle ω_C has degree $2g-2 > 0$, so for $n \geq 3$ the power $\omega_C^{\otimes n}$ is very ample of degree $n(2g-2)$, and by Riemann-Roch it has

$$N_n + 1 := h^0(C, \omega_C^{\otimes n}) = (2n-1)(g-1)$$

sections. A choice of basis of $H^0(C, \omega_C^{\otimes n})$ embeds C as a curve

$$C \hookrightarrow \mathbb{P}^{N_n}, \qquad N_n = (2n-1)(g-1) - 1$$

whose Hilbert polynomial $P(t) = n(2g-2)t - g + 1$ depends only on g and n. Different bases give embeddings differing by an element of PGL_{N_n+1}, and this is the entire ambiguity: the embedding is canonical up to the projective linear group. This is the setup of proposition 3.5.1, with $\mathbb{P}^N = \mathbb{P}^{N_n}$ and the moduli problem $F = \mathcal{M}_g$ of smooth projective genus-g curves.

Example 3.5.4: The GIT Construction of $\mathcal{M}_g$

Fix $g \geq 2$ and an integer $n \geq 3$ (in fact $n \geq 5$ suffices comfortably, and $n = 3$ works with more care). Set $P(t) = n(2g-2)t - g + 1$ and $N = (2n-1)(g-1) - 1$. The construction proceeds in four movements.

The Hilbert scheme of embedded curves. Inside $\mathrm{Hilb}^P(\mathbb{P}^N)$ consider the subset

$$H_g \;=\; \left\{\, [C \hookrightarrow \mathbb{P}^N] \;:\; C \text{ smooth of genus } g,\ \mathcal{O}_C(1) \cong \omega_C^{\otimes n} \,\right\}$$

of n-canonically embedded smooth genus-g curves. Smoothness is an open condition, and the requirement that the embedding line bundle $\mathcal{O}_C(1)$ be the n-th power of the canonical bundle (rather than some other bundle of the same degree) is a locally closed condition. Hence H_g is a locally closed subscheme of the Hilbert scheme, quasi-projective by corollary 2.3.3. Its points are exactly the smooth genus-g curves *together with* a projective frame for their n-canonical embedding.

The group and its orbits. The group $G = \mathrm{PGL}_{N+1}$ acts on $\mathrm{Hilb}^P(\mathbb{P}^N)$ and preserves H_g, since a projective change of coordinates carries an n-canonical embedding to another n-canonical embedding of the same curve. Two points of H_g lie in one G-orbit if and only if the underlying curves are isomorphic: an isomorphism of curves carries $\omega_C^{\otimes n}$ to $\omega_{C'}^{\otimes n}$ and hence intertwines the two embeddings by a projective transformation, and conversely an element of PGL taking one embedded curve to the other restricts to an abstract isomorphism. This is exactly conditions (1) and (2) of proposition 3.5.1.

Stability. The mathematical heart is that the n-canonically embedded smooth curves are the GIT-stable points, for a suitable linearization on a projective completion of H_g inside $\mathrm{Hilb}^P(\mathbb{P}^N)$. This is theorem 3.5.5 below. Granting it, condition (3) of proposition 3.5.1 holds: $\overline{H_g}^{\mathrm{s}} = H_g$, and each smooth curve of genus $g \geq 2$ has finite automorphism group (by Hurwitz, $|\mathrm{Aut}(C)| \leq 84(g-1)$), so all stabilizers are finite.

The quotient. proposition 3.5.1 now delivers the coarse moduli space

$$\mathcal{M}_g = H_g / \mathrm{PGL}_{N+1}$$

a quasi-projective variety of dimension $\dim H_g - \dim \mathrm{PGL}_{N+1}$. A count confirms the expected value: $\dim H_g = \dim \mathrm{PGL}_{N+1} + (3g-3)$, since the deformations of an n-canonically embedded curve split into the $(N+1)^2 - 1 = \dim \mathrm{PGL}_{N+1}$ directions moving the embedding and the $3g-3$ directions deforming the abstract curve (the deformation-theoretic count $\dim H^1(C, T_C) = 3g-3$ is the subject of the deformation-theory part of the book). Hence

$$\dim \mathcal{M}_g = 3g - 3$$

the classical dimension of the moduli of curves.

The one input taken on faith in example 3.5.4 is the stability theorem, and it is the reason GIT was invented. It is stated here in the form the construction uses.

Theorem 3.5.5: Asymptotic Stability of Pluricanonical Curves

Let $g \geq 2$ and let n be sufficiently large. A smooth projective curve of genus g, embedded in $\mathbb{P}^N$ by the complete linear system $|\omega_C^{\otimes n}|$, is a stable point of the Hilbert scheme $\mathrm{Hilb}^P(\mathbb{P}^N)$ for the natural SL_{N+1}-linearization, in the sense of definition 3.3.1. More generally, a Deligne-Mumford stable curve, embedded by $|\omega_C^{\otimes n}|$, is Hilbert stable (equivalently, Chow stable) for n large.

***Proof Key ideas*:**
The full proof is the technical core of Mumford's geometric invariant theory and occupies the capstone chapter on the moduli of curves; it is deferred there because it requires the compactified problem (stable curves and stable reduction) and a delicate Hilbert-Mumford computation that would be premature before the descent-and-stacks material and the deformation theory of nodes. The strategy is to test stability against every one-parameter subgroup via the numerical criterion (theorem 3.4.5). A 1-PS λ of SL_{N+1} acts on $\mathbb{P}^N$ by weights on the coordinates, and $\mu([C], \lambda)$ is computed from the weighted filtration of the homogeneous coordinate ring of C that λ induces. The estimate that $\mu > 0$ for all nontrivial λ, that is, stability, reduces to a statement about how the sections of $\omega_C^{\otimes n}$ distribute their vanishing across the curve: a destabilizing λ would concentrate too much of the linear system at a subscheme of C, and pluricanonical positivity forbids this once n is large. Mumford's key technical device is to control the leading term of μ using the asymptotic Riemann-Roch expansion of $h^0(C, \omega_C^{\otimes n}(-m\,D))$ for divisors D concentrated by λ, whence the name asymptotic stability. See the capstone construction and Mumford, *Geometric Invariant Theory*, for the complete argument.

The deferral is genuine: the stability theorem is long, its full proof belongs with the compactification and node deformations, and stating it here lets the recipe of proposition 3.5.1 be seen in action without the machinery. What the theorem buys is decisive. It says that the class of curves one wants to parametrize, the smooth ones (and, in the compactified problem, the Deligne-Mumford stable ones), coincides exactly with a GIT-stable locus. This is the non-formal matching demanded by proposition 3.5.1(3), and it is not a coincidence engineered by hand: the notion of *stability* in the moduli of curves, a nodal curve with finite automorphisms, is forced to agree with GIT stability, and this agreement of two a priori unrelated stabilities is what makes the construction succeed.

The genus-one case sits below this construction as its degenerate shadow. For $g = 1$ the canonical bundle ω_C has degree $2g - 2 = 0$, so it is not ample and the pluricanonical embedding of example 3.5.4 collapses: there is no n-canonical embedding, the automorphism group of an elliptic curve is positive-dimensional before rigidifying (translations), and the genus-one moduli problem must be handled with a marked point to obtain $\mathcal{M}_{1,1}$. The coarse space is then the j-line of theorem 1.3.6, constructed there directly through the j-invariant rather than through GIT. The two constructions agree in spirit: the j-invariant is the single invariant function separating isomorphism classes, exactly as the GIT quotient H_g/PGL is Proj of a ring of invariant functions. The j-line is the GIT quotient one would obtain from the moduli of 4 points on $\mathbb{P}^1$ (example 3.5.3 with $d = 4$, since an elliptic curve is a double cover of $\mathbb{P}^1$ branched at four points), and Mumford's theorem for $g \geq 2$ is the higher-genus generalization of the elementary invariant theory that produces j.

3.5.6 Historical Note: GIT and the Moduli of Curves Geometric invariant theory was developed by Mumford in the 1960s expressly to construct $\mathcal{M}_g$ as a quasi-projective variety, published in the first edition of *Geometric Invariant Theory* (1965). The stability of pluricanonical curves (theorem 3.5.5) is the technical result that made the construction possible, and its extension to the Deligne-Mumford stable curves gave the projective coarse space $\overline{M}_g$ of the compactified problem. The stack-theoretic viewpoint, in which $\mathcal{M}_g$ is the moduli stack whose coarse space is this GIT quotient, came later (Deligne-Mumford, 1969) and is the subject of the descent-and-stacks part of the book; historically the GIT construction came first and the stacks were introduced to explain what the coarse space was forgetting.

The GIT quotient is one of the two roads to a coarse moduli space, and comparing it with the direct construction of the j-line clarifies what each accomplishes. The direct construction (theorem 1.3.6) exhibits an explicit invariant and reads off the coarse space as its target; it works when a small ring of invariants can be written down by hand. The GIT construction is systematic: it parametrizes objects with an auxiliary embedding, forgets the embedding by a reductive group, and produces the coarse space as Proj of the full invariant ring, whose finite generation (theorem 3.1.2) guarantees a scheme without ever writing an invariant explicitly. For curves of genus $g \geq 2$, where no analogue of the j-invariant is available, GIT is the only elementary road to $\mathcal{M}_g$, and the descent-and-stacks part of the book will show what is lost along it and how stacks recover it.

Exercise 3.5.1

1. In example 3.5.4 the dimension of H_g was asserted to be $\dim \mathrm{PGL}_{N+1} + (3g-3)$, and $\dim \mathrm{PGL}_{N+1} = (N+1)^2 - 1$. For $g = 2$ and $n = 3$, compute N, $\dim \mathrm{PGL}_{N+1}$, and $\dim H_g$ explicitly, and verify that $\dim \mathcal{M}_g = 3g - 3$ comes out to the expected value.

2. Explain, using proposition 3.5.1, why the coarse moduli space $\mathcal{M}_g$ produced by the GIT construction cannot in general be a fine moduli space. Which specific hypothesis of the proposition is responsible, and what would have to be true of the objects for the space to be fine?

3. In example 3.5.3, take $d = 4$. Show directly that the coarse space $M_{0,4}^{\mathrm{un}}$ of four distinct unordered points on $\mathbb{P}^1$ is one-dimensional, and identify the invariant that coordinatizes it. Relate your answer to the j-line (theorem 1.3.6).

4. The genus-one canonical bundle has degree 0. Explain precisely why this makes the pluricanonical embedding of example 3.5.4 fail for $g = 1$, and why the genus-one moduli problem must be rigidified by a marked point before a GIT-style construction can be attempted. Which step of proposition 3.5.1 breaks without the marking?

5. theorem 3.5.5 asserts that n-canonically embedded smooth curves are GIT-*stable*, not just semistable. Using the stability-versus-semistability distinction of definition 3.3.1 and the role of stabilizers in proposition 3.5.1(3), explain what would go wrong in the construction of $\mathcal{M}_g$ if the curves were only semistable, and connect this to the finiteness of $\mathrm{Aut}(C)$.

6. The GIT construction produces $\mathcal{M}_g$ as a quotient H_g/PGL_{N+1} where H_g is a fine moduli space of embedded curves. Explain why H_g itself *is* fine (as a locally closed subscheme of a Hilbert scheme) while its quotient $\mathcal{M}_g$ is only coarse, and describe in one sentence the object of the descent-and-stacks part of the book that restores the lost information.

Part II

Deformation Theory

Part I built moduli spaces; this Part studies them one point at a time. The question it answers is local: given an object of a moduli problem, in how many directions, and against what obstructions, can it be deformed? Deformation theory linearizes this question. The first-order deformations of an object form a vector space, and that vector space is a cohomology group attached to the object; the obstructions to extending a first-order deformation to higher order live in the next cohomology group. The local geometry of a moduli space, its tangent space and its dimension, its smoothness or singularity at a point, thus becomes a computation in the cohomology of a single object.

The two chapters develop this in increasing depth. chapter 4 treats first-order deformations, identifying the tangent space of a moduli functor with a concrete cohomology group in each of the main cases: the deformations of a subscheme with H^0 of its normal sheaf, of a smooth variety with H^1 of its tangent sheaf, of a coherent sheaf with a first Ext group; from these it reads off dimensions such as the $3g-3$ moduli of a curve of genus $g \geq 2$. The following chapter passes to the obstruction theory governing whether these first-order data integrate: it locates the obstructions in a second cohomology group, develops the naive cotangent complex that packages the tangent and obstruction spaces together, and proves Schlessinger's criteria, which say exactly when a deformation functor has a hull or is pro-representable.

Deformation theory is also the hinge between the two halves of the book. The automorphism obstruction that kept the moduli problems of Part I from being representable reappears here as the group H^0 of infinitesimal automorphisms, now a cohomology group one can compute; and the tangent-obstruction formalism assembled in this Part is precisely the input that Artin's criteria will require, in the descent-and-stacks Part, to certify that a moduli problem is an algebraic stack.

4

First-Order Deformations

The tangent space to a moduli space measures the first-order deformations of the objects it parametrizes, and this chapter computes it. The method is the functor-of-points calculation already run for the Hilbert scheme in chapter 2: a tangent vector at a point $[x]$ is a morphism from the dual numbers $\operatorname{Spec} k[\varepsilon]$ carrying the closed point to $[x]$, which is the same datum as a deformation of the object x over $k[\varepsilon]$. What is new here is that the calculation applies to any moduli functor, whether or not it is representable, and that in each standard case the set of first-order deformations is naturally a cohomology group of the object being deformed.

section 4.1 sets up the general framework, the functor of Artin rings and its tangent space. The next three sections compute that tangent space in the three cases that recur throughout moduli theory: section 4.2 identifies the deformations of a closed subscheme with the global sections of its normal sheaf, recovering the Hilbert-scheme tangent space of chapter 2; section 4.3 classifies the deformations of a smooth variety by the first cohomology of its tangent sheaf, from which follow the rigidity of projective space and the $3g-3$ moduli of a curve; and section 4.4 treats deformations of a coherent sheaf and of a line bundle through Ext groups. section 4.5 closes the chapter by identifying the infinitesimal automorphisms of an object with the zeroth cohomology of the same sheaf, tying the automorphism obstruction of chapter 1 to the cohomological bookkeeping that the next chapter's obstruction theory extends.

4.1 Deformation Functors and the Dual Numbers

Part I built moduli spaces globally: a moduli problem is a functor of points, and when it is representable there is a scheme whose points are the objects being classified. The natural next question is local. Given a specific object, say a smooth curve of genus g or a closed subscheme of $\mathbb{P}^n$, how does the moduli space look in an infinitesimal neighborhood of the corresponding point? The tangent space to a moduli space at a point is the linearization of the classification problem there, and its dimension bounds the number of independent directions in which the object can move. For the Hilbert scheme this tangent space was computed in theorem 2.4.4 as a space of global sections of the normal sheaf; the aim of this chapter is to see that every such computation is an instance of one uniform mechanism.

The mechanism is to probe the moduli problem with the smallest nontrivial thickening of a point, the spectrum of the dual numbers $k[\varepsilon]/(\varepsilon^2)$. A family over this base is an object over the point $\operatorname{Spec} k$ together with exactly one order of infinitesimal wiggle, and the set of such families is the tangent space. To make this precise and to make it apply to schemes, subschemes, and sheaves alike, we package the classification data as a functor on Artinian local k-algebras, evaluate it on the dual numbers, and identify the resulting set with a cohomology group. This section sets up the functor, the dual numbers, and the tangent space; sections 4.2 to 4.4 carry out the three identifications.

Throughout this chapter k is an algebraically closed field, and characteristic 0 is assumed wherever a cohomology-dimension count requires it (this is flagged where it is used). The base category on which deformation problems live is the category of Artinian local k-algebras.

Definition 4.1.1: The Category of Artinian Local k-Algebras

Let k be a field. The category $\mathbf{Art}_k$ has as objects the local Artinian k-algebras A whose residue field $A/\mathfrak{m}_A$ is k, where $\mathfrak{m}_A$ denotes the maximal ideal; a morphism $A \to B$ is a local homomorphism of k-algebras (equivalently, a k-algebra homomorphism, since any k-algebra map between such rings is automatically local). A *small extension* in $\mathbf{Art}_k$ is a surjection

$$\pi \colon A \twoheadrightarrow B$$

in $\mathbf{Art}_k$ whose kernel $I = \ker \pi$ satisfies $\mathfrak{m}_A \cdot I = 0$, that is, I is annihilated by the maximal ideal and so is a k-vector space.

An object of $\mathbf{Art}_k$ is a ring that is finite-dimensional as a k-vector space, local, and with residue field the ground field itself; geometrically, $\operatorname{Spec} A$ is a one-point scheme whose single point is thickened by nilpotents. The prototype is the ring of dual numbers introduced below, whose spectrum is a point carrying a single tangent direction. The requirement $A/\mathfrak{m}_A = k$ says that the fat point sits over the ground field with no residue-field extension, so that everything is measured relative to k; this is the algebraic counterpart of restricting attention to a single closed point of a k-scheme. The structure theory of such rings (that $\mathfrak{m}_A$ is nilpotent, that A is a finite-dimensional local k-algebra, and that A is complete) is developed in [6]; only the surface facts just recorded are needed here.

A small extension is the atomic step by which one fat point is built up from a smaller one. Because $\mathfrak{m}_A I = 0$, the kernel I is a module over $A/\mathfrak{m}_A = k$, hence a k-vector space, and the simplest small extensions have $I \cong k$ one-dimensional. Every surjection in $\mathbf{Art}_k$ factors as a composite of such one-dimensional small extensions, so a property that propagates along small extensions propagates along all of $\mathbf{Art}_k$. This is why small extensions are the correct inductive unit: lifting an object from B to A across a small extension is the elementary move whose iteration builds a deformation over an arbitrary Artinian base, and whose failure to be possible is exactly an obstruction (the subject of the next chapter). The smallest nontrivial small extension of all is the augmentation $k[\varepsilon] \twoheadrightarrow k$ killing ε, with kernel $(\varepsilon) \cong k$.

Definition 4.1.2: Deformation Functor

A *deformation functor* (or *functor of Artin rings*) is a covariant functor

$$D\colon \mathbf{Art}_k \longrightarrow \mathbf{Set}$$

such that $D(k)$ is a single point. The condition $D(k) = \{*\}$ is the *pointing*: it fixes the object being deformed. For A in $\mathbf{Art}_k$, an element of $D(A)$ is a *deformation over* A of the object encoded by that single point; the map $D(\pi)\colon D(A) \to D(k)$ induced by the augmentation $\pi\colon A \to k$ sends every deformation to the unique base point, expressing that a deformation over A restricts over the closed point to the original object.

The functor D records, for each fat point $\operatorname{Spec} A$, the set of ways the fixed object spreads out over that fat point. Covariance is forced by geometry: a k-algebra map $A \to A'$ corresponds to a morphism $\operatorname{Spec} A' \to \operatorname{Spec} A$ of fat points, and pulling a family back along this morphism produces a family over $\operatorname{Spec} A'$, giving a map $D(A) \to D(A')$. The pointing $D(k) = \{*\}$ is what distinguishes a deformation functor from an arbitrary moduli functor: one particular object has been singled out (the point of $D(k)$), and $D(A)$ collects only those families over $\operatorname{Spec} A$ that reduce to that object over the closed point. This is the local, infinitesimal shadow of the global moduli functor $F\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ of Part I: fixing a k-point $x_0 \in F(k)$ and restricting F to fat points supported at x_0 produces a deformation functor D, and D sees exactly the germ of F at x_0.

4.1.3 Warning: a deformation functor is not required to be representable A deformation functor D need not be representable, and indeed the interest lies precisely in the cases where it is not. When D is pro-representable by a complete local k-algebra R, the ring R is the completed local ring of the moduli space at the point, and $D(A) = \operatorname{Hom}_{k\text{-alg}}(R, A)$; the next chapter (Schlessinger's criteria) gives conditions under which this happens. Even when D is not pro-representable, the tangent space $D(k[\varepsilon])$ defined below is a well-behaved invariant. The general non-representability warned about for global moduli in box 1.1.6 has an infinitesimal counterpart here.

Probing D means evaluating it on the smallest Artinian algebra that carries a nonzero tangent direction. This is the ring of dual numbers, whose spectrum is a point with exactly one infinitesimal arrow attached.

Definition 4.1.4: Dual Numbers and First-Order Deformations

The *ring of dual numbers* over k is

$$k[\varepsilon] := k[t]/(t^2)$$

the quotient of the polynomial ring $k[t]$ by the square of the variable, with ε the image of t, so $\varepsilon^2 = 0$. It is an object of $\mathbf{Art}_k$: it is local with maximal ideal (ε), residue field $k[\varepsilon]/(\varepsilon) = k$, and dimension 2 as a k-vector space, with basis $\{1, \varepsilon\}$. A *first-order deformation* of the object classified by a deformation functor D is an element of $D(k[\varepsilon])$: a family over the fat point $\operatorname{Spec} k[\varepsilon]$, flat over $k[\varepsilon]$, whose restriction along the augmentation $k[\varepsilon] \to k$ ($\varepsilon \mapsto 0$) is the given object over $\operatorname{Spec} k$.

The geometric content of $\operatorname{Spec} k[\varepsilon]$ is a point with a single tangent vector: it is the length-two scheme supported at the origin of the affine line, the first infinitesimal neighborhood. A morphism $\operatorname{Spec} k[\varepsilon] \to Y$ into a k-scheme Y is the same as a k-point $y \in Y(k)$ together with a tangent vector to Y at y, because such a morphism is a local k-algebra map $\mathcal{O}_{Y,y} \to k[\varepsilon]$, which sends $\mathfrak{m}_y$ to (ε) and $\mathfrak{m}_y^2$ to $(\varepsilon)^2 = 0$, hence factors through $\mathfrak{m}_y/\mathfrak{m}_y^2$ and is therefore an element of $(\mathfrak{m}_y/\mathfrak{m}_y^2)^\vee = T_yY$. The word *first-order* names exactly this truncation: because $\varepsilon^2 = 0$, a first-order deformation records the linear term of a would-be family and discards all higher-order data. Flatness over $k[\varepsilon]$ is the condition that the family varies rather than degenerating; over the dual numbers it amounts to the concrete requirement that the family be, locally, a free $k[\varepsilon]$-module, which the following sections unwind case by case.

The set $D(k[\varepsilon])$ of first-order deformations is the tangent space to the deformation problem. To justify calling it a tangent space, one needs that under a mild condition it carries the structure of a k-vector space, not just that of a pointed set.

Definition 4.1.5: Tangent Space of a Deformation Functor

Let D be a deformation functor. Its *tangent space* is the set

$$t_D := D(k[\varepsilon])$$

of first-order deformations, pointed by the image of $D(k) = \{*\}$ under the inclusion $k \hookrightarrow k[\varepsilon]$ (the trivial, or product, deformation). Say that D satisfies the *fiber-product condition* (H_ε) if for every pair of maps $A' \to A \leftarrow A''$ in $\mathbf{Art}_k$ with $A'' \to A$ surjective, the natural map

$$D(A' \times_A A'') \longrightarrow D(A') \times_{D(A)} D(A'') \tag{4.1}$$

is a bijection whenever $A = k$. When (H_ε) holds, $t_D = D(k[\varepsilon])$ is a k-vector space.

The point marked in t_D is the trivial deformation: the augmentation $k \to k[\varepsilon]$ is a section of $k[\varepsilon] \to k$, and applying D to it embeds the single point $D(k)$ into $D(k[\varepsilon])$ as the deformation that does not move. Every tangent space has a distinguished zero, and this is it. The condition (H_ε) supplies the arithmetic. Its role is visible already for the specific fiber product

$$k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2 \;\cong\; k[\varepsilon] \times_k k[\varepsilon] \tag{4.2}$$

where the right side is the fiber product over the two augmentations $k[\varepsilon] \to k$. Granting (4.1) for this fiber product, D of the left side is $t_D \times t_D$, and the two k-algebra maps

$$k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2 \longrightarrow k[\varepsilon]$$

given by $\varepsilon_1, \varepsilon_2 \mapsto \varepsilon$ (addition) and by $\varepsilon \mapsto c\varepsilon$ for $c \in k$ (scaling) induce, after applying D, the addition $t_D \times t_D \to t_D$ and scalar multiplication $k \times t_D \to t_D$ making t_D a k-vector space. This is the deformation-theoretic incarnation of the functor tangent space from definition 1.5.1: there the tangent space of a functor F at a k-point was defined as $F(k[\varepsilon])$ probed against the same fiber product, and the vector-space structure was extracted by the same maps (4.2). The deformation functor D is that construction with the base point fixed once and for all, so t_D inherits its linear structure verbatim. The next result records this inheritance and gives the full argument.

Proposition 4.1.6: The Tangent Space is a k-Vector Space

Let D be a deformation functor satisfying the fiber-product condition (H_ε) of definition 4.1.5. Then $t_D = D(k[\varepsilon])$ carries a canonical structure of k-vector space, with zero the trivial deformation, for which every k-algebra map $k[\varepsilon] \to k[\varepsilon]$ over k induces the corresponding linear self-map of t_D. If moreover $D(A' \times_k A'') \to D(A') \times D(A'')$ is a bijection for all A', A'' (the strong form of (H_ε)), the construction is functorial in D.

***Proof*:**
Write $t = D(k[\varepsilon])$, pointed by $0 := D(k \hookrightarrow k[\varepsilon])(*)$, where $* \in D(k)$ is the unique element.

Step 1: Identify D of the double fiber product. Consider the ring $B := k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$, with $\varepsilon_i\varepsilon_j = 0$ for all i, j. The two k-algebra maps $p_1, p_2 \colon B \to k[\varepsilon]$ sending $\varepsilon_1 \mapsto \varepsilon$, $\varepsilon_2 \mapsto 0$ and $\varepsilon_1 \mapsto 0$, $\varepsilon_2 \mapsto \varepsilon$ respectively exhibit B as the fiber product $k[\varepsilon] \times_k k[\varepsilon]$ over the two augmentations, as in (4.2): a k-algebra map into B is a pair of maps into $k[\varepsilon]$ agreeing modulo (ε), and modulo (ε) both land in k. Because $A = k$ here, condition (H_ε) applies and gives a bijection

$$D(B) \xrightarrow{\sim} D(k[\varepsilon]) \times_{D(k)} D(k[\varepsilon]) = t \times t$$

since $D(k)$ is a single point, so the fiber product over $D(k)$ is the full product.

Step 2: Define addition. The map $s \colon B \to k[\varepsilon]$ of k-algebras sending $\varepsilon_1 \mapsto \varepsilon$ and $\varepsilon_2 \mapsto \varepsilon$ is well defined: it carries the relations $(\varepsilon_1, \varepsilon_2)^2 = 0$ to $\varepsilon^2 = 0$, which holds in $k[\varepsilon]$, since each generator $\varepsilon_i\varepsilon_j$ maps to $\varepsilon^2 = 0$. Define addition on t as the composite

$$t \times t \xrightarrow{\sim} D(B) \xrightarrow{D(s)} D(k[\varepsilon]) = t$$

where the first arrow is the inverse of the bijection of Step 1. Concretely, a pair (v_1, v_2) of first-order deformations comes from a unique element of $D(B)$, and its image under $D(s)$ is declared to be $v_1 + v_2$.

Step 3: Define scalar multiplication. For $c \in k$, the k-algebra map $m_c \colon k[\varepsilon] \to k[\varepsilon]$ with $\varepsilon \mapsto c\varepsilon$ is well defined ($\varepsilon^2 \mapsto c^2\varepsilon^2 = 0$). Set $c \cdot v := D(m_c)(v)$ for $v \in t$. In particular $m_1 = \mathrm{id}$ gives $1 \cdot v = v$, and m_0 factors through k, so $0 \cdot v = 0$ is the trivial deformation.

Step 4: Verify the vector-space axioms. Commutativity of addition follows from the symmetry of B under $\varepsilon_1 \leftrightarrow \varepsilon_2$: the swap $\tau \colon B \to B$ interchanges p_1 and p_2, hence interchanges the two factors of $t \times t$, while $s \circ \tau = s$, so $v_1 + v_2 = v_2 + v_1$. Associativity uses the triple fiber product. Set $C := k[\varepsilon_1, \varepsilon_2, \varepsilon_3]/(\varepsilon_1, \varepsilon_2, \varepsilon_3)^2$; iterating (H_ε) over the presentation $C = (k[\varepsilon] \times_k k[\varepsilon]) \times_k k[\varepsilon]$ (each factor a small extension of k, so each application of (H_ε) is legitimate) gives $D(C) \cong t \times t \times t$, and the two ways of bracketing the sum are both computed by the single map $C \to k[\varepsilon]$, $\varepsilon_i \mapsto \varepsilon$, hence agree. The zero 0 is a two-sided identity because the element of $D(B)$ representing the pair $(v, 0)$ pulls back along $p_1 \colon \varepsilon_1 \mapsto \varepsilon$, $\varepsilon_2 \mapsto 0$ to v and along $s \colon \varepsilon_1, \varepsilon_2 \mapsto \varepsilon$ to $v + 0$, and s agrees with p_1 on this element because its second-coordinate deformation is trivial (the map s and the map p_1 differ only in the image of ε_2, whose associated deformation is 0); thus $v + 0 = v$. Distributivity follows from the compatibility of scaling with s: the identity $m_c \circ s = s \circ (m_c \times m_c)$ of k-algebra maps $B \to k[\varepsilon]$ (both send $\varepsilon_1, \varepsilon_2 \mapsto c\varepsilon$) gives $c(v_1 + v_2) = cv_1 + cv_2$, while $(c + c')v = cv + c'v$ is computed by the single map $\varepsilon \mapsto (c + c')\varepsilon = c\varepsilon + c'\varepsilon$ factoring through s applied to the pair $(c \cdot v, c' \cdot v)$, using that D of $\varepsilon \mapsto c\varepsilon$ and $\varepsilon \mapsto c'\varepsilon$ produce cv and $c'v$. Finally, additive inverses are automatic once addition, scaling, and distributivity hold: $v + (-1) \cdot v = (1 + (-1)) \cdot v = 0 \cdot v = 0$ by distributivity and $0 \cdot v = D(m_0)(v) = 0$ (as m_0 factors through k), so $-v = (-1) \cdot v = D(m_{-1})(v)$. All axioms hold, so t is a k-vector space.

Step 5: Functoriality. A natural transformation $\eta \colon D \to D'$ of deformation functors commutes with $D(-)$ applied to s and to each m_c, since these are induced by ring maps and η is natural; hence $\eta_{k[\varepsilon]} \colon t_D \to t_{D'}$ is k-linear. This uses only that η respects the fiber-product bijections, which holds because both D and D' satisfy the strong form of (H_ε) and η is a morphism of functors, completing the proof.

The proposition upgrades t_D from a pointed set to a vector space, and the upgrade is functorial in the deformation functor. This is what makes the tangent space computable: to find $\dim_k t_D$ one need only identify $D(k[\varepsilon])$ with some cohomology group and read off its dimension, and the sections that follow do exactly this in three cases. The condition (H_ε) is the first of Schlessinger's conditions, met by every deformation functor arising from a geometric moduli problem, because pulling families back along the two projections of a fiber product of fat points glues them uniquely; the full list of Schlessinger's conditions, controlling when D has a hull or is pro-representable, is the business of the next chapter. For the present chapter the only fact used is that the three deformation functors below satisfy (H_ε), which will be transparent from their explicit descriptions.

With the framework in place, the three running deformation problems of this chapter can be named. Each is a deformation functor in the sense of definition 4.1.2, and each has a tangent space that the corresponding later section identifies with a cohomology group.

Example 4.1.7: The Three Running Deformation Functors
Fix a smooth projective variety X over k, a closed subscheme $Z \subseteq X$, and a coherent sheaf $\mathcal{F}$ on X. The following are deformation functors, forming the three cases studied in sections 4.2 to 4.4.

1. *Deformations of a scheme.* $\mathrm{Def}_X \colon \mathbf{Art}_k \to \mathbf{Set}$ assigns to A the set of isomorphism classes of *flat families* $\mathcal{X} \to \operatorname{Spec} A$ together with an isomorphism $\mathcal{X} \times_{\operatorname{Spec} A} \operatorname{Spec} k \cong X$ of the closed fiber with X. Two such families are identified if there is an isomorphism over $\operatorname{Spec} A$ restricting to the identity on the closed fiber. Here $\mathrm{Def}_X(k) = \{X\}$ is the single point, the constant family. A first-order deformation is a flat family over $\operatorname{Spec} k[\varepsilon]$ restricting to X; section 4.3 shows $\mathrm{Def}_X(k[\varepsilon]) \cong H^1(X, T_X)$.
2. *Deformations of a closed subscheme.* $\mathrm{Def}_{Z/X} \colon \mathbf{Art}_k \to \mathbf{Set}$ holds the ambient X fixed and assigns to A the set of closed subschemes $\mathcal{Z} \subseteq X \times_k \operatorname{Spec} A$, flat over A, with closed fiber $\mathcal{Z} \times_{\operatorname{Spec} A} \operatorname{Spec} k = Z$. Here $\mathrm{Def}_{Z/X}(k) = \{Z\}$, and $\mathrm{Def}_{Z/X}(k[\varepsilon])$ is the tangent space to the Hilbert scheme at $[Z]$ computed in theorem 2.4.4; section 4.2 identifies it with $H^0(Z, N_{Z/X})$.
3. *Deformations of a coherent sheaf.* $\mathrm{Def}_{\mathcal{F}} \colon \mathbf{Art}_k \to \mathbf{Set}$ assigns to A the set of isomorphism classes of coherent sheaves $\mathcal{F}_A$ on $X \times_k \operatorname{Spec} A$, flat over A, with an isomorphism $\mathcal{F}_A|_{X \times \operatorname{Spec} k} \cong \mathcal{F}$. Here $\mathrm{Def}_{\mathcal{F}}(k) = \{\mathcal{F}\}$, and section 4.4 shows $\mathrm{Def}_{\mathcal{F}}(k[\varepsilon]) \cong \mathrm{Ext}^1_X(\mathcal{F}, \mathcal{F})$.

In each case the functor is covariant (pull back along $\operatorname{Spec} A' \to \operatorname{Spec} A$), the value on k is a single point (the object itself, as the constant family), and the fiber-product condition (H_ε) holds because flat families and flat subschemes and flat sheaves glue uniquely over a fiber product of fat points. So each tangent space is a k-vector space by proposition 4.1.6.

These three functors organize the chapter. Each isolates one kind of geometric object and asks how it deforms while its neighbors stay fixed: the ambient variety deforms in Def_X, a subscheme deforms inside a fixed ambient in $\mathrm{Def}_{Z/X}$, and a sheaf deforms on a fixed variety in $\mathrm{Def}_{\mathcal{F}}$. The pattern is uniform: in every case the tangent space is $D(k[\varepsilon])$, and in every case it turns out to be a cohomology group ($H^1(T_X)$, $H^0(N_{Z/X})$, $\mathrm{Ext}^1(\mathcal{F},\mathcal{F})$) whose dimension is the number of independent first-order deformations. The consistency check between $\mathrm{Def}_{Z/X}$ and the Hilbert scheme, namely that $\mathrm{Def}_{Z/X}(k[\varepsilon])$ recovers exactly the Zariski tangent space $T_{[Z]}\mathrm{Hilb}$ of theorem 2.4.4, is what ties the infinitesimal picture of this chapter to the global moduli spaces of Part I. That check is carried out in section 4.2.

One reading of the trivial deformation deserves emphasis, because it recurs in every case and returns in section 4.5. The zero element of t_D is the constant family $X \times_k \operatorname{Spec} k[\varepsilon]$ (or $Z \times_k \operatorname{Spec} k[\varepsilon]$, or $\mathcal{F} \boxtimes k[\varepsilon]$), the deformation that does not move the object. A nonzero first-order deformation is a family that agrees with the constant one over the closed point but disagrees to first order in ε. Whether such a family is isomorphic to the constant one as a deformation is the question of whether its class in t_D vanishes, and the automorphisms of the constant deformation, which measure the redundancy in this identification, form their own cohomology group $H^0(T_X)$ studied at the end of the chapter.

Exercise 4.1.1

1. Show that $\mathbf{Art}_k$ has k itself as a terminal object and that the polynomial ring $k[t]$ is *not* an object of $\mathbf{Art}_k$. Then show that for any A in $\mathbf{Art}_k$ the maximal ideal $\mathfrak{m}_A$ is nilpotent, and deduce that A is a finite-dimensional k-vector space.
2. Let $\pi\colon A \twoheadrightarrow B$ be a surjection in $\mathbf{Art}_k$ with kernel I. Show that π factors as a finite composite of small extensions. (Filter I by the descending chain $I \supseteq \mathfrak{m}_A I \supseteq \mathfrak{m}_A^2 I \supseteq \cdots$ and observe each successive quotient is killed by $\mathfrak{m}_A$.)
3. Verify directly that a k-algebra homomorphism $\mathcal{O}_{Y,y} \to k[\varepsilon]$ inducing the residue field identity is the same datum as a tangent vector to the k-scheme Y at the k-point y, that is, an element of $(\mathfrak{m}_y/\mathfrak{m}_y^2)^\vee$. Conclude that $Y(k[\varepsilon]) \to Y(k)$ has fiber over y equal to the Zariski tangent space T_yY.
4. Prove the isomorphism $k[\varepsilon_1,\varepsilon_2]/(\varepsilon_1,\varepsilon_2)^2 \cong k[\varepsilon] \times_k k[\varepsilon]$ of (4.2) by exhibiting mutually inverse k-algebra maps, where the fiber product is taken over the two augmentations $k[\varepsilon] \to k$.
5. Let $D = h_R := \mathrm{Hom}_{k\text{-alg}}(R,-)$ be the functor of points of a complete local Noetherian k-algebra $(R,\mathfrak{m}_R)$ with residue field k, restricted to $\mathbf{Art}_k$. Show D is a deformation functor and compute its tangent space $t_D = D(k[\varepsilon])$ as a k-vector space, identifying it with $(\mathfrak{m}_R/\mathfrak{m}_R^2)^\vee$.
6. Give a deformation functor D that does not satisfy the fiber-product condition (H_ε), so that $D(k[\varepsilon])$ fails to be a vector space in the canonical way. (Consider a functor whose value on $k[\varepsilon]$ is a single point but which does not respect the fiber product (4.2), or a set-valued functor built to break additivity.)

4.2 Deformations of a Closed Subscheme

The functor of Artin rings from section 4.1 is a general device, but its first genuine payoff appears in the most concrete deformation problem of all: fixing an ambient variety X and asking how a closed subscheme $Z \subseteq X$ can be moved inside it. This is the infinitesimal version of the question the Hilbert scheme answers globally. In section 4.1 the deformation functor $\mathrm{Def}_{Z/X}$ was introduced as one of the running cases; here it is solved outright. The answer is a single cohomological invariant: first-order deformations of Z are the global sections of a single coherent sheaf on Z, the normal sheaf $N_{Z/X}$. Every deformation is a first-order motion of Z in the directions transverse to itself, and $N_{Z/X}$ is precisely the sheaf of such transverse directions.

The section proves this classification in full, then reads two consequences out of it. For a complete intersection the normal sheaf splits as a sum of line bundles, and the deformation space becomes a cohomology group one can write down explicitly; for a hypersurface or a finite set of points the count reproduces the tangent space to the Hilbert scheme computed in an earlier chapter, closing the circle between the functorial and the geometric pictures. Throughout, X is a smooth projective variety over an algebraically closed field k, and $Z \subseteq X$ is a closed subscheme with ideal sheaf $I_Z \subseteq \mathcal{O}_X$; smoothness hypotheses are stated where the normal sheaf is required to be locally free.

The object that governs the whole section is the normal sheaf, defined in an earlier chapter as the dual of the conormal sheaf I_Z/I_Z^2. Recall its statement so the classification below can be phrased against it.

Recall definition 2.4.1: for a closed subscheme $Z \subseteq X$ with ideal sheaf I_Z, the *conormal sheaf* is the $\mathcal{O}_Z$-module I_Z/I_Z^2, and the *normal sheaf* is its dual

$$N_{Z/X} \;=\; \mathcal{H}om_{\mathcal{O}_Z}(I_Z/I_Z^2,\, \mathcal{O}_Z)$$

When Z and X are both smooth, I_Z/I_Z^2 is locally free of rank equal to the codimension $c = \dim X - \dim Z$, and $N_{Z/X}$ is then a vector bundle of rank c on Z. In general $N_{Z/X}$ is only a coherent sheaf, but the classification below holds regardless: it is the sheaf $\mathcal{H}om_{\mathcal{O}_Z}(I_Z/I_Z^2, \mathcal{O}_Z)$, whose sections are the $\mathcal{O}_Z$-linear maps out of the conormal sheaf.

A homomorphism $I_Z/I_Z^2 \to \mathcal{O}_Z$ is the same datum as an $\mathcal{O}_X$-linear map $I_Z \to \mathcal{O}_Z$, because any such map kills I_Z^2 automatically: if $\varphi\colon I_Z \to \mathcal{O}_Z$ is $\mathcal{O}_X$-linear and $f, g \in I_Z$, then $\varphi(fg) = f\,\varphi(g)$, and $f \in I_Z$ maps to 0 in $\mathcal{O}_Z = \mathcal{O}_X/I_Z$, so $\varphi(fg) = 0$. Thus

$$H^0(Z, N_{Z/X}) \;=\; \mathrm{Hom}_{\mathcal{O}_Z}(I_Z/I_Z^2,\, \mathcal{O}_Z) \;=\; \mathrm{Hom}_{\mathcal{O}_X}(I_Z,\, \mathcal{O}_Z)$$

This identification is the mechanism behind the theorem: a global section of the normal sheaf is an $\mathcal{O}_X$-homomorphism from the ideal to the structure sheaf of Z, and it is precisely such a homomorphism that a first-order deformation produces.

The problem is to understand a flat family over the dual numbers $k[\varepsilon] = k[t]/(t^2)$ that restricts to Z over the closed point. Such a family is a closed subscheme of $X \times_k \operatorname{Spec} k[\varepsilon]$, flat over $\operatorname{Spec} k[\varepsilon]$, whose special fiber is Z (definition 4.1.4). Everything reduces to a statement about ideal sheaves inside $\mathcal{O}_X \otimes_k k[\varepsilon] = \mathcal{O}_X[\varepsilon]$.

Theorem 4.2.1: First-Order Deformations of a Closed Subscheme

Let X be a scheme over k and $Z \subseteq X$ a closed subscheme with ideal sheaf I_Z. The set of first-order deformations of Z inside X, that is, the set $\mathrm{Def}_{Z/X}(k[\varepsilon])$ of closed subschemes $\widetilde{Z} \subseteq X \times_k \operatorname{Spec} k[\varepsilon]$ that are flat over $\operatorname{Spec} k[\varepsilon]$ and restrict to Z over the closed point, is in natural bijection with

$$\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) \;=\; H^0(Z, N_{Z/X})$$

the global sections of the normal sheaf. Under this bijection the trivial deformation $Z \times_k \operatorname{Spec} k[\varepsilon]$ corresponds to the zero section, and the bijection is k-linear, so $\mathrm{Def}_{Z/X}(k[\varepsilon])$ is a k-vector space with $H^0(Z, N_{Z/X})$ as its tangent space.

***Proof*:**

The argument is entirely local: a deformation is a subsheaf $\widetilde{I} \subseteq \mathcal{O}_X[\varepsilon]$, and flatness pins down the shape of $\widetilde{I}$ so tightly that it is determined by a single homomorphism out of I_Z. The proof establishes that shape, extracts the homomorphism, and checks that the construction is reversible.

Step 1: Flat deformations are extensions of I_Z by εI_Z. Write $A = k[\varepsilon]$ and $\mathcal{O}_X[\varepsilon] = \mathcal{O}_X \otimes_k A$, a sheaf of A-algebras on X fitting into the exact sequence

$$0 \longrightarrow \varepsilon\mathcal{O}_X \longrightarrow \mathcal{O}_X[\varepsilon] \xrightarrow{\varepsilon \mapsto 0} \mathcal{O}_X \longrightarrow 0 \tag{4.3}$$

in which multiplication by ε induces an isomorphism $\mathcal{O}_X \xrightarrow{\sim} \varepsilon\mathcal{O}_X$ of $\mathcal{O}_X$-modules. A deformation $\widetilde{Z}$ is given by its ideal sheaf $\widetilde{I} \subseteq \mathcal{O}_X[\varepsilon]$, and the condition that $\widetilde{Z}$ restrict to Z over the closed point is that the image of $\widetilde{I}$ under $\varepsilon \mapsto 0$ be exactly I_Z.

Flatness of $\widetilde{Z}$ over A is equivalent to flatness of $\widetilde{\mathcal{O}} := \mathcal{O}_X[\varepsilon]/\widetilde{I}$ over A. Since $A = k[\varepsilon]$ is a local Artinian ring with maximal ideal (ε) satisfying $\varepsilon^2 = 0$, the local criterion of flatness over the dual numbers takes an elementary form: an A-module M is flat over A if and only if the multiplication map

$$\varepsilon\colon M/\varepsilon M \longrightarrow \varepsilon M \tag{4.4}$$

is an isomorphism, equivalently $\mathrm{Tor}_1^A(M, k) = 0$. Applying this to $M = \widetilde{\mathcal{O}}$: writing $\widetilde{\mathcal{O}}/\varepsilon\widetilde{\mathcal{O}} = \mathcal{O}_Z$ (the special fiber), flatness says that $\varepsilon\widetilde{\mathcal{O}} \cong \mathcal{O}_Z$ via multiplication by ε, so $\widetilde{\mathcal{O}}$ sits in a short exact sequence

$$0 \longrightarrow \varepsilon\mathcal{O}_Z \longrightarrow \widetilde{\mathcal{O}} \longrightarrow \mathcal{O}_Z \longrightarrow 0 \tag{4.5}$$

where $\varepsilon\mathcal{O}_Z$ denotes the copy of $\mathcal{O}_Z$ carried by ε. Dually, on the level of ideals, flatness is equivalent to the requirement that

$$\widetilde{I} \cap \varepsilon\mathcal{O}_X[\varepsilon] \;=\; \varepsilon I_Z \tag{4.6}$$

that is, the part of $\widetilde{I}$ divisible by ε is precisely εI_Z and nothing more, which is the ideal-level form of the iso characterization (4.4).

Step 2: A flat deformation is a lift of the identity, hence a homomorphism $\varphi\colon I_Z \to \mathcal{O}_Z$. Fix a flat deformation with ideal $\widetilde{I}$ satisfying (4.6). Every element of $\widetilde{I}$ has the form $g_0 + \varepsilon g_1$ with $g_0, g_1 \in \mathcal{O}_X$, and reducing modulo ε sends $\widetilde{I}$ onto I_Z; so for $g_0 + \varepsilon g_1 \in \widetilde{I}$ the leading term satisfies $g_0 \in I_Z$. Given a section $f \in I_Z$, choose any lift $f + \varepsilon h \in \widetilde{I}$ of f (one exists because $\widetilde{I} \to I_Z$ is surjective). Two lifts of the same f differ by an element of $\widetilde{I} \cap \varepsilon\mathcal{O}_X[\varepsilon] = \varepsilon I_Z$, so the class of h in $\mathcal{O}_X/I_Z = \mathcal{O}_Z$ is independent of the choice of lift. Define

$$\varphi\colon I_Z \longrightarrow \mathcal{O}_Z, \qquad \varphi(f) \;=\; \overline{h} \;\text{ where } f + \varepsilon h \in \widetilde{I}$$

This is well defined by the previous sentence. It is additive because the sum of two lifts lifts the sum. It is $\mathcal{O}_X$-linear: for $a \in \mathcal{O}_X$ and $f \in I_Z$ with lift $f + \varepsilon h$, the element $a(f + \varepsilon h) = af + \varepsilon(ah)$ lies in $\widetilde{I}$ (an ideal), and it lifts af, so $\varphi(af) = \overline{ah} = \overline{a}\,\overline{h} = a\,\varphi(f)$ in $\mathcal{O}_Z$. Thus $\varphi \in \mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$.

Step 3: The homomorphism recovers the deformation, giving the inverse map. Conversely, let $\varphi \in \mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$ be given, and set

$$\widetilde{I}_\varphi \;=\; \big\{\, f + \varepsilon g \;:\; f \in I_Z,\; g \in \mathcal{O}_X,\; \overline{g} = \varphi(f) \text{ in } \mathcal{O}_Z \,\big\} \;\subseteq\; \mathcal{O}_X[\varepsilon] \tag{4.7}$$

Concretely $\widetilde{I}_\varphi$ consists of $f + \varepsilon g$ where $f \in I_Z$ and g reduces to $\varphi(f)$ modulo I_Z. This is an $\mathcal{O}_X[\varepsilon]$-submodule: for $a + \varepsilon b \in \mathcal{O}_X[\varepsilon]$ and $f + \varepsilon g \in \widetilde{I}_\varphi$,

$$(a + \varepsilon b)(f + \varepsilon g) \;=\; af + \varepsilon(ag + bf)$$

and $af \in I_Z$ with $\overline{ag + bf} = \overline{a}\,\varphi(f) + \overline{b} \cdot 0 = \varphi(af)$ (using $f \in I_Z$, so $\overline{bf} = 0$, and $\mathcal{O}_X$-linearity of φ), so the product lies in $\widetilde{I}_\varphi$. It satisfies (4.6): an element $f + \varepsilon g \in \widetilde{I}_\varphi$ is divisible by ε exactly when $f = 0$, and then $\overline{g} = \varphi(0) = 0$ forces $g \in I_Z$, so $\widetilde{I}_\varphi \cap \varepsilon\mathcal{O}_X[\varepsilon] = \varepsilon I_Z$. Hence $\widetilde{I}_\varphi$ defines a flat deformation. The homomorphism attached to $\widetilde{I}_\varphi$ by Step 2 is φ itself, since $f + \varepsilon g \in \widetilde{I}_\varphi$ has $\overline{g} = \varphi(f)$ by construction.

Step 4: The two constructions are mutually inverse and linear. Starting from a flat deformation $\widetilde{I}$, Step 2 produces φ, and Step 3 applied to φ returns the ideal $\{f + \varepsilon g : \overline{g} = \varphi(f)\}$. This equals $\widetilde{I}$: any $f + \varepsilon g \in \widetilde{I}$ has $\overline{g} = \varphi(f)$ by the definition of φ, and conversely if $\overline{g} = \varphi(f)$ then $f + \varepsilon g$ and the chosen lift $f + \varepsilon h$ of f differ by $\varepsilon(g - h)$ with $\overline{g - h} = 0$, so $g - h \in I_Z$ and $\varepsilon(g - h) \in \varepsilon I_Z \subseteq \widetilde{I}$, giving $f + \varepsilon g \in \widetilde{I}$. Thus the maps of Steps 2 and 3 are inverse bijections. The trivial deformation has $\widetilde{I} = I_Z \cdot \mathcal{O}_X[\varepsilon] = \{f + \varepsilon g : f, g \in I_Z\}$, for which every lift of $f \in I_Z$ can be taken with $h \in I_Z$, so $\varphi = 0$: the trivial deformation is the zero section. Finally, the bijection is k-linear because $\widetilde{I}_{\varphi_1+\varphi_2}$ and $\widetilde{I}_{c\varphi}$ are read off additively and by scaling from the defining condition $\overline{g} = \varphi(f)$ in (4.7). The identification $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) = H^0(Z, N_{Z/X})$ from the discussion preceding the theorem completes the proof.

The theorem is the local model of every tangent-space computation in deformation theory: a deformation of a subscheme is a first-order sliding of its defining equations, encoded by how far each generator of I_Z is allowed to move out of the ideal, measured modulo I_Z. The datum $\varphi(f) = \overline{h}$ records the ε-correction $f \rightsquigarrow f + \varepsilon h$ applied to the equation f, and the constraint that φ be $\mathcal{O}_X$-linear and factor through I_Z/I_Z^2 is exactly the constraint that these corrections assemble into a consistent flat family rather than an arbitrary perturbation of equations. A section of $N_{Z/X}$ is a choice of outward normal direction at every point of Z, and moving Z along it to first order is the deformation. When $H^0(Z, N_{Z/X}) = 0$ the subscheme is rigid inside X: it admits no nontrivial first-order motion, and the corresponding point of the Hilbert scheme is isolated to first order.

The bijection also explains the appearance of the same cohomology group on both sides of the moduli story. Globally, the Hilbert scheme Hilb_X parametrizes closed subschemes of X, and its tangent space at the point $[Z]$ was computed in an earlier chapter to be $H^0(Z, N_{Z/X})$ (theorem 2.4.4). The theorem above is the functor-of-points shadow of that computation: a first-order deformation of Z is exactly a $k[\varepsilon]$-point of Hilb_X lying over $[Z]$, so the two tangent spaces must agree. What the Hilbert scheme sees as its Zariski tangent space, the deformation functor $\mathrm{Def}_{Z/X}$ sees as $\mathrm{Def}_{Z/X}(k[\varepsilon])$, and theorem 4.2.1 identifies both with the sections of the normal sheaf. This is the promised bridge between the geometric and the functorial construction of the tangent space; the remaining examples make the count explicit.

Example 4.2.2: Deformations of a Complete Intersection

Let X be a smooth projective variety and let $Z \subseteq X$ be a smooth complete intersection of codimension c, cut out by a regular sequence of sections $f_1, \ldots, f_c$ of line bundles $\mathcal{L}_1, \ldots, \mathcal{L}_c$ on X; that is, I_Z is generated by the f_i and the associated Koszul complex is a resolution. The conormal sheaf then splits as a direct sum of line bundles restricted to Z:

$$I_Z/I_Z^2 \;\cong\; \bigoplus_{i=1}^{c} \mathcal{L}_i^{-1}\big|_Z$$

with the class of f_i generating the i-th summand. Dualizing gives the normal sheaf as a direct sum of line bundles

$$N_{Z/X} \;\cong\; \bigoplus_{i=1}^{c} \mathcal{L}_i\big|_Z \tag{4.8}$$

so a first-order deformation is a c-tuple $(s_1, \ldots, s_c)$ with $s_i \in H^0(Z, \mathcal{L}_i|_Z)$, and the deformation $f_i \rightsquigarrow f_i + \varepsilon\widetilde{s}_i$ perturbs the i-th equation by a lift $\widetilde{s}_i$ of s_i. The dimension of the space of first-order deformations is therefore

$$h^0(Z, N_{Z/X}) \;=\; \sum_{i=1}^{c} h^0\big(Z,\, \mathcal{L}_i|_Z\big)$$

which (4.8) reduces to a sum of ordinary cohomology computations.

Take the concrete case $X = \mathbb{P}^n$ and Z a complete intersection of hypersurfaces of degrees $d_1, \ldots, d_c$, so $\mathcal{L}_i = \mathcal{O}_{\mathbb{P}^n}(d_i)$ and

$$N_{Z/X} \;\cong\; \bigoplus_{i=1}^{c} \mathcal{O}_Z(d_i)$$

The perturbations of the equation f_i that move Z are the degree-d_i forms restricted to Z; the forms lying in the ideal $(f_1, \ldots, f_c)$ act trivially, since they only reparametrize the same subscheme. For a single smooth surface Z of degree d in $\mathbb{P}^3$ (the case $c = 1$),

$$h^0(Z, N_{Z/X}) \;=\; h^0\big(\mathbb{P}^3, \mathcal{O}(d)\big) - h^0\big(\mathbb{P}^3, \mathcal{O}\big) \;=\; \binom{d+3}{3} - 1$$

the count of degree-d forms modulo the one-dimensional span of f itself, using $H^0(\mathbb{P}^3, \mathcal{O}(d)) \twoheadrightarrow H^0(Z, \mathcal{O}_Z(d))$ (surjective because $H^1(\mathbb{P}^3, \mathcal{O}(d-d)) = H^1(\mathbb{P}^3, \mathcal{O}) = 0$) with kernel $k \cdot f$. This matches the dimension of the projective space of degree-d surfaces through the parametrized family, confirming that the Hilbert scheme of such surfaces is smooth of the expected dimension at $[Z]$.

The complete-intersection case is the one where the abstract normal sheaf becomes a concrete list of line bundles and the deformation count becomes a sum of dimensions of linear systems. More is true: a global complete intersection is *unobstructed*, so $[Z]$ is always a smooth point of the Hilbert scheme, and the first-order count $h^0(Z, N_{Z/X})$ is the actual local dimension there (a standard theorem; see Hartshorne's *Deformation Theory* Ch. 8 or Sernesi 4.6.6). The obstructions to extending a first-order deformation to higher order live in $H^1(Z, N_{Z/X})$, which for the splitting (4.8) is $\bigoplus_i H^1(Z, \mathcal{L}_i|_Z)$; but for a complete intersection every such obstruction vanishes even when this group is nonzero, since the deformations produced by perturbing the defining equations always integrate. Non-vanishing of $H^1(Z, N_{Z/X})$ only says where an obstruction *could* live, not that a genuine obstruction exists, and it is precisely the complete-intersection hypothesis that kills it. The singular and reducible Hilbert schemes of space curves discovered by Mumford arise instead from *non*-complete-intersection curves, where no such splitting is available; complete intersections are among the first examples where the Hilbert scheme is provably smooth, which is the subject the next chapter takes up when it treats obstruction theory in general.

The extreme special cases, a single hypersurface and a finite set of points, are worth isolating because they recover the Hilbert-scheme tangent space of theorem 2.4.4 in its most transparent form.

Example 4.2.3: Hypersurfaces and Points

1. *A degree-d hypersurface in $\mathbb{P}^n$.* Let $Z = V_+(f) \subseteq \mathbb{P}^n$ be a smooth hypersurface of degree d, so $c = 1$, $I_Z = \mathcal{O}_{\mathbb{P}^n}(-d) \cdot f$, and $N_{Z/X} = \mathcal{O}_Z(d)$ by (4.8). The first-order deformations are

$$H^0(Z, N_{Z/X}) \;=\; H^0(Z, \mathcal{O}_Z(d))$$

Restriction from $\mathbb{P}^n$ is surjective with kernel spanned by f, so

$$h^0(Z, N_{Z/X}) \;=\; \binom{n+d}{d} - 1$$

the number of degree-d monomials in $n+1$ variables minus one for the equation f itself. This is exactly $\dim \mathbb{P}\big(H^0(\mathbb{P}^n, \mathcal{O}(d))\big)$, the dimension of the projective space of degree-d hypersurfaces. The Hilbert scheme of degree-d hypersurfaces in $\mathbb{P}^n$ *is* this projective space, smooth of dimension $\binom{n+d}{d} - 1$, and theorem 4.2.1 recovers its tangent space at every point without ever leaving the functor of points. This is the promised recovery of theorem 2.4.4 from the deformation-theoretic side.

2. *A finite set of points.* Let $Z = \{p_1, \ldots, p_r\} \subseteq X$ be r distinct reduced points of a smooth variety X of dimension n. At each p_i the local ring is regular of dimension n, the maximal ideal is generated by a regular system of parameters, and the conormal space $\mathfrak{m}_{p_i}/\mathfrak{m}_{p_i}^2$ is the cotangent space, of dimension n. Hence the normal sheaf is a sum of skyscraper sheaves

$$N_{Z/X} \;=\; \bigoplus_{i=1}^{r} (i_{p_i})_* T_{p_i} X$$

where $i_{p_i} \colon \{p_i\} \hookrightarrow X$ and the stalk at p_i is the tangent space $T_{p_i}X = (\mathfrak{m}_{p_i}/\mathfrak{m}_{p_i}^2)^\vee$, of dimension n. Therefore

$$h^0(Z, N_{Z/X}) \;=\; \sum_{i=1}^{r} \dim_k T_{p_i} X \;=\; rn$$

A first-order deformation of the reduced scheme Z is a choice of tangent vector at each point, moving the r points independently to first order. This is the tangent space to the

Hilbert scheme of r points, $\mathrm{Hilb}^r(X)$, at the reduced configuration $[Z]$: since $\mathrm{Hilb}^r(X)$ has dimension rn near the locus of distinct points, the computation $h^0(Z, N_{Z/X}) = rn$ shows the Hilbert scheme is smooth of the expected dimension there, matching theorem 2.4.4.

3. *A nonreduced point.* The reduced case above conceals the subtlety that makes Hilbert schemes of points interesting. Take $X = \mathbb{A}^2$ and Z the length-2 subscheme defined by $I_Z = (x^2, y)$, supported at the origin. Here Z is not a complete intersection in the splitting sense of c regular parameters cutting a reduced point, but the theorem still applies: $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$ classifies the deformations. Computing this Hom for $I_Z = (x^2, y)$ over $\mathcal{O}_Z = k[x,y]/(x^2, y)$ gives a 4-dimensional space, matching $\dim \mathrm{Hilb}^2(\mathbb{A}^2) = 2 \cdot 2 = 4$; the nonreduced point does not obstruct, and $\mathrm{Hilb}^2(\mathbb{A}^2)$ is smooth. That every $\mathrm{Hilb}^r(\mathbb{A}^2)$ (indeed every Hilb^r of a smooth surface) is smooth of dimension $2r$ is a theorem whose local input is exactly this Hom computation, extended to all finite colength ideals.

The point of these examples is that the single formula $H^0(Z, N_{Z/X})$ specializes to every count one would compute by hand: the dimension of a linear system of hypersurfaces, a tangent vector per point, or a Hom of ideals for a nonreduced scheme. The reduced-points case exhibits the tangent space as a direct sum of ambient tangent spaces, one per point, which is why the Hilbert scheme of distinct points looks locally like a symmetric product; the nonreduced case (part 3) is where the normal-sheaf description earns its keep, since there is no naive count of "directions to move" and the Hom of ideals is the only available handle. In every case the deformation-theoretic tangent space of theorem 4.2.1 agrees with the Zariski tangent space of the Hilbert scheme from theorem 2.4.4, as it must, since a first-order deformation of Z is precisely a $k[\varepsilon]$-point of Hilb_X over $[Z]$.

Exercise 4.2.1

1. Let $Z \subseteq X$ be a closed subscheme with ideal sheaf I_Z, and let $\varphi\colon I_Z \to \mathcal{O}_Z$ be an $\mathcal{O}_X$-linear map. Verify directly from the definition that φ kills I_Z^2, and conclude that $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) = \mathrm{Hom}_{\mathcal{O}_Z}(I_Z/I_Z^2, \mathcal{O}_Z)$. Explain in one sentence why this is the reason the normal sheaf, rather than $\mathcal{H}om_{\mathcal{O}_X}(I_Z, \mathcal{O}_X)$, is the correct object.

2. Let Z be a smooth plane curve of degree d in $\mathbb{P}^2$. Using example 4.2.3, compute $h^0(Z, N_{Z/\mathbb{P}^2})$ and compare it with the dimension of the projective space of degree-d plane curves. For which d is this space positive-dimensional, and what does $d = 1$ say?

3. Let $Z \subseteq \mathbb{P}^3$ be the smooth complete intersection of a quadric and a cubic (a curve of degree 6). Write down $N_{Z/\mathbb{P}^3}$ as a direct sum of line bundles on Z, and set up the computation of $h^0(Z, N_{Z/\mathbb{P}^3})$. You may leave the answer in terms of the degrees $\deg(\mathcal{O}_Z(2))$ and $\deg(\mathcal{O}_Z(3))$ and the genus of Z; identify the genus using the adjunction formula.

4. Consider $X = \mathbb{A}^2$ and the length-2 subscheme Z with $I_Z = (x^2, y)$ supported at the origin, as in example 4.2.3(3). Compute $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$ explicitly by specifying where the generators x^2 and y may be sent, and verify that the dimension is 4. Which of your homomorphisms corresponds to sliding the doubled point along the x-axis?

5. Show that a smooth subvariety $Z \subseteq X$ with $H^0(Z, N_{Z/X}) = 0$ is rigid inside X: every first-order deformation is trivial. Give an example of such a Z (with X specified), and explain in one sentence how this is the subscheme analogue of the infinitesimal rigidity of $\mathbb{P}^n$ studied in the next section.

6. Let $Z \subseteq X$ be a complete intersection of hypersurfaces of degrees $d_1, \ldots, d_c$ in $X = \mathbb{P}^n$. Explain

why the degree-d_i forms lying in the ideal $(f_1, \dots, f_c)$ contribute nothing to $H^0(Z, N_{Z/X})$, and relate this to the fact that $H^0(Z, N_{Z/X})$ is a quotient, not the whole space of equation-perturbations. What geometric operation on Z do the trivial perturbations correspond to?

4.3 Deformations of a Smooth Variety

The previous section deformed a closed subscheme $Z \subseteq X$ inside a fixed ambient space, and the answer, $H^0(Z, N_{Z/X})$, remembered that embedding: it measured the ways the equations cutting out Z could wiggle. The present section drops the ambient space entirely and asks a more intrinsic question. Given a smooth projective variety X, in how many ways can X itself be deformed as an *abstract* variety, with no reference to any surrounding $\mathbb{P}^N$? This is the deformation problem that governs moduli of varieties, and it is where the dimension of a moduli space first becomes visible as a cohomology group. The answer will be $H^1(X, T_X)$, the first cohomology of the tangent sheaf, and the mechanism producing it is the most transparent in deformation theory: a first-order deformation is a way of regluing X from its affine pieces, and the tangent sheaf records the infinitesimal ambiguity in that gluing.

Throughout this section k is algebraically closed of characteristic zero and X is a smooth projective variety over k, of dimension n. The tangent sheaf of X is the dual of the sheaf of Kähler differentials,

$$T_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$$

a locally free sheaf of rank n ([12, Tangent Sheaf]). Its sections are the algebraic vector fields on X; over an affine open $U = \operatorname{Spec} A$, the sections $T_X(U)$ are the k-derivations $\operatorname{Der}_k(A, A)$, the infinitesimal automorphisms of U. This is exactly the data that will glue the affine pieces of a deformation, so the identification of the tangent sheaf with derivations is the hinge of the whole argument.

The passage from the embedded problem of the previous section to the abstract problem here rests on one fact about smooth affine varieties: they cannot be deformed at all. This rigidity is what lets a global deformation be reconstructed purely from gluing data, and it is worth isolating before the main theorem.

Proposition 4.3.1: Rigidity of Smooth Affine Varieties

Let $U = \operatorname{Spec} A$ be a smooth affine variety over k. Then every first-order deformation of U over the dual numbers $k[\varepsilon]$ is trivial: any flat $k[\varepsilon]$-algebra $\tilde{A}$ with $\tilde{A} \otimes_{k[\varepsilon]} k \cong A$ is isomorphic, as a deformation, to $A \otimes_k k[\varepsilon] = A[\varepsilon]$.

***Proof*:**
Let $\tilde{A}$ be a flat $k[\varepsilon]$-algebra reducing to A modulo ε. Flatness over $k[\varepsilon]$ for a module is equivalent to the sequence

$$0 \to A \xrightarrow{\varepsilon} \tilde{A} \to A \to 0$$

being exact, where the outer terms are $\tilde{A}/\varepsilon\tilde{A} \cong A$ and the kernel of reduction is $\varepsilon\tilde{A} \cong A$; concretely, multiplication by ε identifies A with the ideal $\varepsilon\tilde{A} \subseteq \tilde{A}$, and $\tilde{A}$ is an extension of A by this square-zero ideal.

Step 1: Lift generators. Since A is smooth over k, it is in particular formally smooth. Present

$A = P/I$ with $P = k[x_1, \dots, x_m]$ a polynomial ring. Choose lifts $\tilde{x}_i \in \tilde{A}$ of the images of x_i in A, giving a $k[\varepsilon]$-algebra map $\tilde{P} = P[\varepsilon] \to \tilde{A}$. Because $\tilde{A}$ is generated over $k[\varepsilon]$ by the classes reducing to a generating set of A (the reduction is surjective and $\varepsilon\tilde{A}$ is generated by the same classes times ε), this map is surjective.

Step 2: Lift the relations using smoothness. A trivialization is a $k[\varepsilon]$-algebra isomorphism $\tilde{A} \cong A[\varepsilon]$ over A, that is, a ring-homomorphism section $s\colon A \to \tilde{A}$ of the reduction $\tilde{A} \to A$ splitting $0 \to A \to \tilde{A} \to A \to 0$. Producing s is a lifting problem against the square-zero surjection $\tilde{A} \to A$, whose kernel $\varepsilon\tilde{A} \cong A$ is a square-zero ideal. Formal smoothness of A over k is precisely the statement that such lifts exist: given the identity map $A \to A$ and the square-zero surjection $\tilde{A} \to A$, there is a k-algebra homomorphism $s\colon A \to \tilde{A}$ whose reduction modulo ε is the identity.

Step 3: Conclude triviality. Such an s is a ring-theoretic section of $\tilde{A} \to A$, so $\tilde{A} = s(A) \oplus \varepsilon\tilde{A}$ as a k-vector space, and the resulting map $A[\varepsilon] \to \tilde{A}$, $a + \varepsilon b \mapsto s(a) + \varepsilon s(b)$, is a $k[\varepsilon]$-algebra isomorphism restricting to the identity modulo ε. Hence $\tilde{A} \cong A[\varepsilon]$ as a deformation, completing the proof.

The content of proposition 4.3.1 is that all the significant information in a deformation of a global smooth variety is concentrated in how the affine charts are reglued, never in the charts themselves. Over each chart the deformation is trivial; the choices are made on the overlaps, and different trivializations on adjacent charts differ by an infinitesimal automorphism. Since infinitesimal automorphisms of an affine chart are its vector fields, the gluing data is a family of vector fields on overlaps, which is exactly the input to Čech cohomology of the tangent sheaf. This is the entire idea of the theorem; the proof below makes it precise.

Theorem 4.3.2: First-Order Deformations of a Smooth Variety

Let X be a smooth projective variety over an algebraically closed field k of characteristic zero. Then the set of isomorphism classes of first-order deformations of X over $k[\varepsilon]$ is in natural bijection with the cohomology group

$$H^1(X, T_X)$$

the trivial deformation $X \times_k \operatorname{Spec} k[\varepsilon]$ corresponding to the zero class. In particular the deformation functor of X has tangent space $H^1(X, T_X)$ in the sense of definition 4.1.5.

***Proof*:**
Fix a first-order deformation of X: a flat scheme $\mathcal{X}$ over $\operatorname{Spec} k[\varepsilon]$ together with an isomorphism of its special fiber $\mathcal{X} \times_{\operatorname{Spec} k[\varepsilon]} \operatorname{Spec} k$ with X. The strategy is to trivialize $\mathcal{X}$ over an affine cover of X, read off the discrepancy between trivializations on overlaps as a Čech 1-cochain with values in T_X, and check that the cocycle condition holds and that the class is independent of all choices.

Step 1: Trivialize on an affine cover. Choose a finite affine open cover $\mathfrak{U} = \{U_i\}_{i \in I}$ of X, with $U_i = \operatorname{Spec} A_i$. The deformation $\mathcal{X}$ restricts over each U_i to a flat deformation $\mathcal{X}_i = \mathcal{X}|_{U_i}$ of the smooth affine variety U_i. By proposition 4.3.1 each $\mathcal{X}_i$ is trivial: fix an isomorphism of deformations

$$\varphi_i\colon \mathcal{X}_i \xrightarrow{\sim} U_i \times_k \operatorname{Spec} k[\varepsilon]$$

restricting to the identity on the special fiber U_i. Each φ_i is a choice; the cocycle will measure how these choices fail to agree.

Step 2: Extract the transition automorphisms. On a double overlap $U_{ij} = U_i \cap U_j$, the two trivializations φ_i and φ_j both identify $\mathcal{X}|_{U_{ij}}$ with the trivial deformation $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$. Their comparison

$$\theta_{ij} = \varphi_i \circ \varphi_j^{-1} \colon U_{ij} \times_k \operatorname{Spec} k[\varepsilon] \xrightarrow{\sim} U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$$

is an automorphism of the trivial deformation over U_{ij} restricting to the identity modulo ε. An automorphism of $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$ reducing to the identity is, on the coordinate ring $A_{ij}[\varepsilon]$, a $k[\varepsilon]$-algebra map

$$a + \varepsilon b \longmapsto a + \varepsilon\big(b + D_{ij}(a)\big)$$

for a uniquely determined k-linear map $D_{ij} \colon A_{ij} \to A_{ij}$. The requirement that this map be a ring homomorphism forces $D_{ij}(aa') = a\, D_{ij}(a') + a'\, D_{ij}(a)$, so D_{ij} is a k-derivation of A_{ij}: an element of $\operatorname{Der}_k(A_{ij}, A_{ij}) = T_X(U_{ij})$. Thus each overlap carries a vector field $D_{ij} \in T_X(U_{ij})$.

Step 3: Verify the cocycle condition. On a triple overlap $U_{ijk} = U_i \cap U_j \cap U_k$ the three comparisons compose to the identity, $\theta_{ij} \circ \theta_{jk} \circ \theta_{ki} = \mathrm{id}$, since $\theta_{ij}\theta_{jk} = (\varphi_i\varphi_j^{-1})(\varphi_j\varphi_k^{-1}) = \varphi_i\varphi_k^{-1} = \theta_{ik}$. Because each θ reduces to the identity modulo ε, composing two of them adds their derivations: writing out $\theta_{ij}\theta_{jk}$ on $a + \varepsilon b$ gives $a + \varepsilon(b + D_{jk}(a) + D_{ij}(a))$, the cross term $\varepsilon^2 D_{ij}(D_{jk}(a))$ vanishing. Hence the map $\theta \mapsto D$ turns composition into addition, and $\theta_{ij}\theta_{jk} = \theta_{ik}$ becomes

$$D_{ij} + D_{jk} = D_{ik}$$

on U_{ijk}. This is precisely the Čech 1-cocycle condition $D_{ij} - D_{ik} + D_{jk} = 0$ for the cover $\mathfrak{U}$ with values in T_X. The collection (D_{ij}) therefore defines a class

$$\delta(\mathcal{X}) \in \check{H}^1(\mathfrak{U}, T_X)$$

and, passing to the limit over refinements (or using that T_X is coherent on the projective scheme X, so Čech and derived cohomology agree), a class in $H^1(X, T_X)$.

Step 4: Independence of the trivializations. A different choice of trivializations $\varphi_i' = \psi_i \circ \varphi_i$ replaces each φ_i by composing with an automorphism ψ_i of the trivial deformation over U_i reducing to the identity, that is, with a vector field $E_i \in T_X(U_i)$. The new transition maps are $\theta'_{ij} = \psi_i\theta_{ij}\psi_j^{-1}$, and translating to derivations (again composition becomes addition) gives

$$D'_{ij} = D_{ij} + E_i - E_j$$

on U_{ij}. The difference $D' - D$ is the Čech coboundary of the 0-cochain (E_i), so $\delta(\mathcal{X})$ is independent of the chosen trivializations, hence a well-defined element of $H^1(X, T_X)$ attached to the deformation $\mathcal{X}$.

Step 5: The class determines the deformation. If two deformations $\mathcal{X}$ and $\mathcal{X}'$ have the same class in $H^1(X, T_X)$, then after passing to a common refinement of the two covers and choosing trivializations, their cocycles (D_{ij}) and (D'_{ij}) differ by a coboundary $(E_i - E_j)$. Adjusting the trivializations of $\mathcal{X}'$ by the vector fields E_i as in Step 4 makes the two cocycles equal on every overlap. Equal transition data means the glued schemes are isomorphic as deformations: the isomorphisms $\varphi_i^{-1} \circ \varphi_i'$ over the U_i agree on overlaps (their discrepancy is exactly $D'_{ij} - D_{ij} = 0$) and so patch to a global isomorphism $\mathcal{X} \cong \mathcal{X}'$ of deformations. Thus δ is injective on isomorphism classes.

Step 6: Every class is realized. Conversely, given a Čech 1-cocycle (D_{ij}) with values in T_X for the cover $\mathfrak{U}$, form for each overlap the automorphism θ_{ij} of $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$ determined by D_{ij} as in Step 2. The cocycle condition $D_{ij} + D_{jk} = D_{ik}$ is exactly the compatibility $\theta_{ij}\theta_{jk} = \theta_{ik}$ needed to glue the trivial pieces $U_i \times_k \operatorname{Spec} k[\varepsilon]$ along the θ_{ij} into a scheme $\mathcal{X}$ flat over $\operatorname{Spec} k[\varepsilon]$ with special fiber X. Its class under δ is the given cocycle, so δ is surjective. The trivial deformation is glued by the identity transition maps, all $D_{ij} = 0$, and so corresponds to the zero class. Combining Steps 5 and 6, δ is a bijection between isomorphism classes of first-order deformations and $H^1(X, T_X)$, as we sought to show.

The bijection of theorem 4.3.2 is the prototype for every tangent-space computation in the theory of moduli of varieties, and it is worth saying precisely what makes it work. Two features of the smooth global case conspire. First, smoothness makes each affine chart rigid, so the local pieces contribute nothing and the deformation is entirely a gluing phenomenon. Second, the automorphisms doing the gluing reduce to the identity modulo ε, so their linearizations are vector fields and composition of automorphisms becomes addition of vector fields; this linearization is what turns the gluing bookkeeping into a linear cocycle condition, and hence into cohomology. The first cohomology H^1 appears, rather than H^0 or H^2, because the data lives on double overlaps (a 1-cochain) and the coboundaries that trivialize a deformation come from global reparametrizations on single charts (a 0-cochain). The two neighboring groups are not idle: $H^0(X, T_X)$ will reappear in section 4.5 as the infinitesimal automorphisms of X, and $H^2(X, T_X)$ houses the obstructions to lifting a first-order deformation to higher order, the subject of the next chapter.

For a curve the theorem already delivers the number the reader has been promised. Before verifying that, consider the simplest possible outcome: a variety with no first-order deformations at all.

Example 4.3.3: Infinitesimal Rigidity of Projective Space

Let $X = \mathbb{P}^n_k$ with $n \geq 1$. Then
$$H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$$
so by theorem 4.3.2 projective space has no nontrivial first-order deformations: it is infinitesimally rigid.

The computation runs through the Euler sequence ([12, Euler Sequence]), whose dual is the short exact sequence of locally free sheaves
$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$
Taking the long exact sequence in cohomology, the relevant segment is
$$H^1\big(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)\big)^{\oplus(n+1)} \longrightarrow H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) \longrightarrow H^2(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n})$$
The cohomology of line bundles on projective space is known: every twist $\mathcal{O}_{\mathbb{P}^n}(d)$ has cohomology concentrated in degrees 0 and n, so $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) = 0$ whenever $0 < i < n$. For $n \geq 2$ this gives $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) = 0$, since the index 1 lies strictly between 0 and n. It gives $H^2(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = 0$ as well: for $n \geq 3$ the index 2 lies strictly between 0 and n, and for $n = 2$ the group $H^2(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2})$ is Serre dual to $H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3)) = 0$. The middle term is squeezed between two zeros, so $H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$ for $n \geq 2$. For $n = 1$ the tangent sheaf is $T_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(2)$, and $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2)) = 0$ because $\deg = 2 > 2g - 2 = -2$, so the vanishing holds there as well.

The vanishing matches geometric expectation: $\mathbb{P}^n$ is a homogeneous space, so rigid, and the moduli "space" of projective spaces of a fixed dimension is a single reduced point. Contrast this with the deformations of $\mathbb{P}^n$ as a *subvariety* of a larger projective space, which are plentiful; the abstract variety $\mathbb{P}^n$ moves not at all, even though its embeddings move freely.

The rigidity of $\mathbb{P}^n$ is the degenerate case. The first nontrivial count, and the historical origin of the phrase "moduli of curves," comes from applying the same theorem to a curve, where the dimension of $H^1(T_C)$ turns out to be a number Riemann already knew.

Example 4.3.4: Deformations of a Smooth Curve and the Number $3g-3$

Let C be a smooth projective curve of genus g over k. The tangent sheaf T_C is a line bundle, being the dual of the canonical bundle:

$$T_C = \Omega_{C/k}^\vee = \omega_C^{-1}$$

because on a curve $\Omega_{C/k} = \omega_C$ has rank one. By theorem 4.3.2 the first-order deformations of C are classified by $H^1(C, T_C) = H^1(C, \omega_C^{-1})$, and this dimension is computed by Riemann-Roch and Serre duality.

Serre duality. By Serre duality on the curve C ([12, Serre Duality]), for any line bundle $\mathcal{L}$ one has $H^1(C, \mathcal{L}) \cong H^0(C, \omega_C \otimes \mathcal{L}^{-1})^*$. Applying this to $\mathcal{L} = \omega_C^{-1}$ gives

$$H^1(C, \omega_C^{-1}) \cong H^0(C, \omega_C^{\otimes 2})^*$$

so $\dim H^1(C, T_C) = h^0(C, \omega_C^{\otimes 2})$, the dimension of the space of *quadratic differentials* on C.

Riemann-Roch. The degree of the canonical bundle is $\deg \omega_C = 2g-2$ ([12, Riemann-Roch]), so $\deg \omega_C^{\otimes 2} = 4g-4$. Riemann-Roch for the line bundle $\omega_C^{\otimes 2}$ reads

$$h^0(C, \omega_C^{\otimes 2}) - h^1(C, \omega_C^{\otimes 2}) = \deg \omega_C^{\otimes 2} + 1 - g = (4g-4) + 1 - g = 3g-3$$

It remains to show $h^1(C, \omega_C^{\otimes 2}) = 0$ for $g \geq 2$. By Serre duality again, $h^1(C, \omega_C^{\otimes 2}) = h^0(C, \omega_C^{-1})$, and ω_C^{-1} has degree $2-2g < 0$ once $g \geq 2$, so it has no nonzero global sections: a line bundle of negative degree on an integral curve has $h^0 = 0$. Hence for $g \geq 2$

$$\dim H^1(C, T_C) = h^0(C, \omega_C^{\otimes 2}) = 3g-3$$

Low genus. The two small genera behave differently and both are instructive.

- For $g = 0$, the curve is $C = \mathbb{P}^1$ and $T_{\mathbb{P}^1} = \mathcal{O}(2)$, so $H^1(\mathbb{P}^1, \mathcal{O}(2)) = 0$ by example 4.3.3. The rational curve is rigid: $\dim H^1(T_C) = 0$, and the value $3g-3 = -3$ is not the answer (the formula fails because $h^1(\omega^{\otimes 2}) \neq 0$ when the degree $4g-4 = -4$ of $\omega^{\otimes 2}$ is negative).
- For $g = 1$, the curve is an elliptic curve, $\omega_C \cong \mathcal{O}_C$ is trivial, so $T_C \cong \mathcal{O}_C$ and $\dim H^1(C, T_C) = h^1(C, \mathcal{O}_C) = g = 1$. There is a one-dimensional space of first-order deformations, matching the one-dimensional coarse moduli space of elliptic curves recorded by the j-line (theorem 1.3.6). Here $3g-3 = 0$ undercounts because the vanishing $h^1(\omega^{\otimes 2}) = 0$ used above requires $g \geq 2$; when $g = 1$ the line bundle $\omega_C^{\otimes 2} = \mathcal{O}_C$ has $h^1 = 1$, correcting the count from 0 up to 1.

Collecting the three cases,

$$\dim H^1(C, T_C) = \begin{cases} 0 & g = 0 \\ 1 & g = 1 \\ 3g-3 & g \geq 2 \end{cases}$$

For $g \geq 2$ this is exactly the dimension of the moduli space $\mathcal{M}_g$ of smooth curves of genus g, the quantity the capstone will construct as an algebraic stack; the deformation theory computes the answer long before the space itself is built.

The number $3g-3$ is the first place in this book where a moduli dimension falls out of a cohomology computation, and the interpretation of example 4.3.4 deserves to be underlined. The tangent space to the moduli of genus-g curves at a point $[C]$ is $H^1(C, T_C)$, and Serre duality recasts this as the space of quadratic differentials $H^0(C, \omega_C^{\otimes 2})$, a description that will resurface when the cotangent space to $\mathcal{M}_g$ is identified. The genus-one case is the cleanest illustration of the general principle that will organize Part III: an elliptic curve has a one-dimensional family of first-order deformations, so its moduli is one-dimensional, and yet no fine moduli space exists because every elliptic curve carries the automorphism $[-1]$. The infinitesimal deformation count sees the correct dimension; the automorphisms, which will be diagnosed as $H^0(C, T_C)$ in section 4.5, are the obstruction to representability that forces the coarse space and, ultimately, the stack.

A word of caution keeps the bookkeeping straight across the three regimes.

4.3.5 Warning: The Formula $3g-3$ Requires $g \geq 2$ The equality $\dim H^1(C, T_C) = 3g-3$ holds only for $g \geq 2$. The Riemann-Roch computation always gives $h^0(\omega_C^{\otimes 2}) - h^1(\omega_C^{\otimes 2}) = 3g-3$, but this is the dimension of $H^1(T_C)$ only when the correction term $h^1(\omega_C^{\otimes 2}) = h^0(\omega_C^{-1})$ vanishes, which fails for $g \leq 1$. For $g = 0$ and $g = 1$ the correction term is nonzero and the deformation dimension is 0 and 1 respectively, not -3 and 0. The pattern is the general phenomenon that Riemann-Roch computes an Euler characteristic, and reading off a single cohomology dimension from it always requires a vanishing input for the complementary group.

Exercise 4.3.1

1. Let X be a smooth projective variety with $H^1(X, T_X) = 0$. Explain in one or two sentences, citing theorem 4.3.2, what this says about the deformations of X, and give one example of such an X of dimension 2.
2. Verify the vanishing $H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$ for $n \geq 2$ in detail by writing out the full long exact sequence attached to the dual Euler sequence and citing the values of $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$ and $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n})$ used at each step.
3. Compute $\dim H^1(C, T_C)$ for a smooth projective curve of genus $g = 3$, and identify the space of quadratic differentials $H^0(C, \omega_C^{\otimes 2})$ whose dimension it equals.
4. In Step 2 of the proof of theorem 4.3.2, an automorphism of $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$ reducing to the identity was written on coordinate rings as $a + \varepsilon b \mapsto a + \varepsilon(b + D_{ij}(a))$. Show directly that the requirement that this map be a $k[\varepsilon]$-algebra homomorphism forces D_{ij} to be a k-derivation, and hence a section of T_X over U_{ij}.
5. Let A be a smooth projective surface with $H^1(A, \mathcal{O}_A) = g$ and trivial canonical bundle $\omega_A \cong \mathcal{O}_A$ (an abelian surface or a $K3$ surface). Compute $\dim H^1(A, T_A)$ in terms of the Hodge numbers of A, using $T_A \cong \Omega_A$ (which holds because ω_A is trivial and A is a surface) and Serre duality.
6. The proof of theorem 4.3.2 used the rigidity of smooth affine varieties (proposition 4.3.1). Explain what breaks in the argument if X is allowed to be singular, and identify which cohomology group would need to be replaced by a more sophisticated invariant to account for local (as opposed to gluing) deformations.

4.4 Deformations of Sheaves and Line Bundles

The previous sections computed the tangent space of a moduli problem whose objects were *subschemes* of a fixed ambient variety ($H^0(N_{Z/X})$, theorem 4.2.1) and whose objects were *varieties* themselves ($H^1(X, T_X)$, theorem 4.3.2). This section treats the third fundamental moduli problem of algebraic geometry: the moduli of *sheaves*. A coherent sheaf $\mathcal{F}$ on a fixed variety X is a piece of linear-algebraic data spread over X, and one wants a parameter space whose points are such sheaves. Chief among these problems is the moduli of line bundles, whose parameter space is the Picard scheme $\operatorname{Pic}(X)$ already met in section 4.3 through its tangent sheaf. The pattern established in section 4.2 and section 4.3 continues, but with a new cohomological home: the tangent space to a moduli of sheaves is an Ext group, and the degree-one self-extensions $\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{F})$ are exactly the first-order deformations of $\mathcal{F}$.

The mechanism is the same square-zero extension that governed the earlier cases. A first-order deformation of $\mathcal{F}$ is a sheaf on the thickened space $X[\varepsilon] = X \times_k \operatorname{Spec} k[\varepsilon]$, flat over the dual numbers, restricting to $\mathcal{F}$ over the closed point. The whole content of this section is that flatness over $k[\varepsilon]$ turns such a thickened sheaf into a self-extension of $\mathcal{F}$ by $\mathcal{F}$, and conversely. Once that translation is in hand, the moduli of coherent sheaves inherits its tangent space from homological algebra, and the special case of line bundles collapses $\operatorname{Ext}^1(\mathcal{L}, \mathcal{L})$ to the single cohomology group $H^1(X, \mathcal{O}_X)$ that is the tangent space to $\operatorname{Pic}(X)$.

Throughout, k is algebraically closed of characteristic zero and X is a smooth projective k-variety unless stated otherwise; $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ denotes the dual numbers and $X[\varepsilon] = X \times_k \operatorname{Spec} k[\varepsilon]$ the trivial thickening of X over them. A first-order deformation is understood in the sense of definition 4.1.4.

Before stating the classification, a word on the homological tool that carries it. The group Ext^1 measures extensions, and this is precisely the structure a first-order deformation manufactures. The following refresher fixes notation and the one property used below; its proof belongs to homological algebra and is cited, not reproduced.

4.4.1 Note: The Ext^1 Refresher For coherent sheaves $\mathcal{F}, \mathcal{G}$ on a scheme X, the group $\operatorname{Ext}^i_X(\mathcal{F}, \mathcal{G})$ is the i-th right derived functor of $\operatorname{Hom}_X(\mathcal{F}, -)$ evaluated at $\mathcal{G}$, equivalently the i-th hyper-derived functor computed from an injective (or, for $\mathcal{F}$, locally free) resolution. The degree-one group $\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{G})$ classifies *extensions* of $\mathcal{F}$ by $\mathcal{G}$: isomorphism classes of short exact sequences

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0$$

of coherent sheaves, where two extensions are identified when a commuting isomorphism of the middle terms fixing $\mathcal{G}$ and $\mathcal{F}$ exists. The zero class is the split extension $\mathcal{E} = \mathcal{G} \oplus \mathcal{F}$. Addition is the Baer sum, and this bijection is an isomorphism of k-vector spaces. For proofs of the derived-functor description, the extension interpretation, and the Baer sum, see [10].

The extension interpretation is the whole point: a first-order deformation produces a sheaf $\mathcal{E}$ on $X[\varepsilon]$ that, once its $k[\varepsilon]$-structure is unwound, is exactly an extension of $\mathcal{F}$ by $\mathcal{F}$, with the class 0 recording the trivial (product) deformation. The theorem below makes this precise and proves the correspondence is a bijection.

Theorem 4.4.2: First-Order Deformations of a Coherent Sheaf

Let X be a scheme over k and $\mathcal{F}$ a coherent sheaf on X. The set of isomorphism classes of first-order deformations of $\mathcal{F}$, that is, coherent sheaves $\mathcal{F}_\varepsilon$ on $X[\varepsilon]$ that are flat over $k[\varepsilon]$ together with an isomorphism $\mathcal{F}_\varepsilon \otimes_{k[\varepsilon]} k \cong \mathcal{F}$, is in natural bijection with

$$\mathrm{Ext}^1_X(\mathcal{F}, \mathcal{F})$$

The bijection is k-linear, and the zero class corresponds to the trivial deformation $\mathcal{F}_\varepsilon = \mathcal{F} \otimes_k k[\varepsilon]$.

***Proof*:**
Write $R = k[\varepsilon]$ and let $\pi\colon X[\varepsilon] \to X$ be the projection, so that $\mathcal{O}_{X[\varepsilon]} = \mathcal{O}_X \otimes_k R$ as a sheaf of R-algebras on the topological space of X (the closed immersion $X \hookrightarrow X[\varepsilon]$ is a homeomorphism on underlying spaces). Sheaves on $X[\varepsilon]$ are thus sheaves of $\mathcal{O}_X \otimes_k R$-modules on X, and the multiplication by ε is an $\mathcal{O}_X$-linear endomorphism whose square is zero.

Step 1: Flatness over the dual numbers is a square-zero condition. A module M over $R = k[\varepsilon]$ is flat if and only if it is free, since R is a local Artinian ring and finitely generated flat modules over a local ring are free; the same criterion sheafifies. The maximal ideal $(\varepsilon) \subseteq R$ satisfies $(\varepsilon) \cong k$ as an R-module, and the local criterion for flatness reduces to a single condition: an R-module M is flat over R if and only if the multiplication map

$$\varepsilon\colon M \otimes_R k \longrightarrow \varepsilon M$$

is an isomorphism, equivalently $\varepsilon M = \ker(\varepsilon\colon M \to M)$ and the induced map $M/\varepsilon M \to \varepsilon M$ is bijective (see [6] for the local flatness criterion over an Artinian base). Applied to the sheaf $\mathcal{F}_\varepsilon$ on $X[\varepsilon]$, this says: $\mathcal{F}_\varepsilon$ is flat over R if and only if multiplication by ε fits into a short exact sequence of $\mathcal{O}_X$-modules

$$0 \longrightarrow \mathcal{F}_\varepsilon \otimes_R k \xrightarrow{\varepsilon} \mathcal{F}_\varepsilon \longrightarrow \mathcal{F}_\varepsilon \otimes_R k \longrightarrow 0 \tag{4.9}$$

where the right map is reduction modulo ε. By hypothesis $\mathcal{F}_\varepsilon \otimes_R k \cong \mathcal{F}$, so (4.9) reads

$$0 \longrightarrow \mathcal{F} \xrightarrow{\varepsilon} \mathcal{F}_\varepsilon \xrightarrow{\rho} \mathcal{F} \longrightarrow 0 \tag{4.10}$$

a short exact sequence of $\mathcal{O}_X$-modules in which the sub and the quotient are both $\mathcal{F}$. This is an extension of $\mathcal{F}$ by $\mathcal{F}$ in the sense of box 4.4.1, and it determines a class in $\mathrm{Ext}^1_X(\mathcal{F}, \mathcal{F})$.

Step 2: The extension recovers the $\mathcal{O}_{X[\varepsilon]}$-module structure. Conversely, suppose given an extension (4.10) of $\mathcal{O}_X$-modules. Endow $\mathcal{F}_\varepsilon$ with an action of ε by declaring ε to act as the composite

$$\mathcal{F}_\varepsilon \xrightarrow{\rho} \mathcal{F} \xrightarrow{\text{(the inclusion)}} \mathcal{F}_\varepsilon$$

that is, $\varepsilon \cdot x = \iota(\rho(x))$ where $\iota\colon \mathcal{F} \hookrightarrow \mathcal{F}_\varepsilon$ is the given subobject inclusion. This action satisfies $\varepsilon^2 = 0$ because $\rho \circ \iota = 0$ (the sequence is exact at the middle), so it makes $\mathcal{F}_\varepsilon$ a sheaf of $\mathcal{O}_X \otimes_k R = \mathcal{O}_{X[\varepsilon]}$-modules. The exactness of (4.10) says exactly that $\varepsilon\colon \mathcal{F}_\varepsilon \to \mathcal{F}_\varepsilon$ has image equal to its kernel, both being $\iota(\mathcal{F})$, and that $\mathcal{F}_\varepsilon/\varepsilon\mathcal{F}_\varepsilon \cong \mathcal{F}$; by the criterion of Step 1 the resulting $\mathcal{O}_{X[\varepsilon]}$-module is flat over R and restricts to $\mathcal{F}$ over the closed point. The two constructions of Steps 1 and 2 are mutually inverse: passing from a flat $\mathcal{F}_\varepsilon$ to the sequence (4.10) and back returns the same ε-action, since that action was read off from the same inclusion and reduction maps.

Step 3: Isomorphism of deformations matches isomorphism of extensions. An isomorphism of first-order deformations is an $\mathcal{O}_{X[\varepsilon]}$-linear isomorphism $\varphi\colon \mathcal{F}_\varepsilon \to \mathcal{F}'_\varepsilon$ commuting with the identifications of the closed fibers with $\mathcal{F}$. Reducing φ modulo ε gives the identity on $\mathcal{F}$, and φ commutes with multiplication by ε; hence φ carries the sub $\iota(\mathcal{F})$ to $\iota'(\mathcal{F})$ by the identity and induces the identity on the quotient $\mathcal{F}$. This is precisely a morphism of extensions in the sense of box 4.4.1 inducing the identity on sub and quotient, that is, an equivalence of extensions. Conversely an equivalence of extensions is $\mathcal{O}_X$-linear and, commuting with the ε-action reconstructed in Step 2, is $\mathcal{O}_{X[\varepsilon]}$-linear. Isomorphism classes of flat deformations therefore correspond bijectively to equivalence classes of extensions, that is, to elements of $\mathrm{Ext}^1_X(\mathcal{F},\mathcal{F})$.

Step 4: Linearity and the trivial deformation. The trivial deformation $\mathcal{F}_\varepsilon = \mathcal{F}\otimes_k R = \mathcal{F}\oplus\varepsilon\mathcal{F}$ is the split extension $\mathcal{F}\oplus\mathcal{F}$, whose class is 0. That the bijection is k-linear follows because the Baer sum of extensions matches the additive structure on deformations induced by the functoriality of $-\otimes_R k$: scaling ε by $\lambda\in k^\times$ acts on (4.10) by scaling the sub-inclusion, reproducing scalar multiplication in $\mathrm{Ext}^1_X(\mathcal{F},\mathcal{F})$, and the Baer sum of two extensions computes the deformation glued from the two ε-actions. The correspondence is natural in X and in $\mathcal{F}$ because every step used only the flatness criterion and the abelian structure of $\mathcal{O}_X$-modules, both of which are functorial, completing the proof.

The theorem is the sheaf-theoretic member of the family $H^0(N_{Z/X})$, $H^1(T_X)$, $\mathrm{Ext}^1(\mathcal{F},\mathcal{F})$ of tangent-space formulas, and it subsumes the other two conceptually: a subscheme is recorded by its structure sheaf and a smooth variety's deformations are governed by $H^1(T_X) = \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega_X,\mathcal{O}_X)$, so all three are instances of "Ext^1 of the relevant object with itself, or with a partner." The essential mechanism is (4.10): flatness over the dual numbers forces the middle sheaf $\mathcal{F}_\varepsilon$ to be an extension whose kernel and cokernel are both the sheaf being deformed, and the nilpotent ε is precisely the connecting datum that an extension class encodes. A deformation is trivial exactly when this extension splits, that is, when $\mathcal{F}_\varepsilon \cong \mathcal{F}\otimes_k k[\varepsilon]$; the failure to split, measured by the nonzero class in $\mathrm{Ext}^1_X(\mathcal{F},\mathcal{F})$, is the first-order geometry of the moduli of sheaves.

4.4.3 Remark: Automorphisms and Ext^0 The degree-zero group $\mathrm{Ext}^0_X(\mathcal{F},\mathcal{F}) = \mathrm{Hom}_X(\mathcal{F},\mathcal{F})$ is the sheaf-theoretic analogue of the infinitesimal automorphisms $H^0(X,T_X)$ met in section 4.3: it is the space of infinitesimal automorphisms of the trivial deformation. A stable sheaf has $\mathrm{Hom}_X(\mathcal{F},\mathcal{F}) = k$ (only scalars), so its moduli space is fine near $[\mathcal{F}]$; a strictly semistable or decomposable sheaf has larger $\mathrm{Hom}_X(\mathcal{F},\mathcal{F})$, and the extra endomorphisms are the infinitesimal shadow of the automorphism obstruction to fine representability recorded in box 1.1.6. The full $T^0/T^1/T^2$ bookkeeping ($\mathrm{Ext}^0,\mathrm{Ext}^1,\mathrm{Ext}^2$) parallels the tangent-obstruction picture, with Ext^0 the infinitesimal automorphisms treated in the next section (section 4.5) and the obstruction to lifting a first-order deformation living in $\mathrm{Ext}^2_X(\mathcal{F},\mathcal{F})$; that obstruction theory is developed in the next chapter.

The most important sheaves to deform are the line bundles, and here the Ext group collapses to a single cohomology group, recovering the tangent space to the Picard scheme. This is the payoff that ties the section back to $\mathrm{Pic}\,(X)$, whose points parametrize line bundles up to isomorphism and whose identity component $\mathrm{Pic}^0(X)$ is an abelian variety when X is smooth projective.

Proposition 4.4.4: Deformations of a Line Bundle and the Tangent Space to Pic

Let X be a smooth projective variety over k and $\mathcal{L}$ a line bundle on X. Then

$$\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L}) \;\cong\; H^1(X,\mathcal{O}_X)$$

so the first-order deformations of $\mathcal{L}$ form the vector space $H^1(X,\mathcal{O}_X)$. This is the tangent space at $[\mathcal{L}]$ to the Picard scheme $\mathrm{Pic}\,(X)$; since $\mathrm{Pic}\,(X)$ is smooth in characteristic zero with identity component the abelian variety $\mathrm{Pic}^0(X)$, one has

$$T_{[\mathcal{L}]}\mathrm{Pic}\,(X) \;=\; T_0\mathrm{Pic}^0(X) \;\cong\; H^1(X,\mathcal{O}_X)$$

***Proof*:**
Step 1: Reduce $\mathrm{Ext}^1(\mathcal{L},\mathcal{L})$ *to* $H^1(\mathcal{O}_X)$. Because $\mathcal{L}$ is locally free of rank one, the sheaf-Hom $\mathcal{H}om_X(\mathcal{L},\mathcal{L})$ is the trivial line bundle $\mathcal{O}_X$: locally $\mathcal{L}\cong\mathcal{O}_X$ and every $\mathcal{O}_X$-linear endomorphism of $\mathcal{O}_X$ is multiplication by a section, so $\mathcal{H}om_X(\mathcal{L},\mathcal{L})\cong\mathcal{O}_X$ canonically, the isomorphism sending a local endomorphism to its value on 1. For a locally free sheaf $\mathcal{L}$ the local Ext sheaves vanish, $\mathcal{E}xt^i_X(\mathcal{L},\mathcal{L})=0$ for $i>0$, so the local-to-global spectral sequence

$$E_2^{p,q}=H^p\big(X,\mathcal{E}xt^q_X(\mathcal{L},\mathcal{L})\big) \implies \mathrm{Ext}^{p+q}_X(\mathcal{L},\mathcal{L})$$

degenerates to its bottom row, giving $\mathrm{Ext}^i_X(\mathcal{L},\mathcal{L})\cong H^i(X,\mathcal{H}om_X(\mathcal{L},\mathcal{L}))=H^i(X,\mathcal{O}_X)$ for all i (the local-to-global Ext spectral sequence is [10]). In degree one this is the claimed isomorphism

$$\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L})\cong H^1(X,\mathcal{O}_X)$$

Equivalently, tensoring an extension of $\mathcal{L}$ by $\mathcal{L}$ with $\mathcal{L}^{-1}$ gives an extension of $\mathcal{O}_X$ by $\mathcal{O}_X$, so $\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L})\cong\mathrm{Ext}^1_X(\mathcal{O}_X,\mathcal{O}_X)=H^1(X,\mathcal{O}_X)$; every line bundle deforms like the structure sheaf.

Step 2: Identify with the Picard tangent space. By theorem 4.4.2 the first-order deformations of $\mathcal{L}$ as a coherent sheaf are classified by $\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L})$, and Step 1 identifies this with $H^1(X,\mathcal{O}_X)$. Every first-order deformation of a line bundle is again a line bundle, since flatness over $k[\varepsilon]$ preserves the property of being invertible (the restriction to the closed fiber is invertible, and invertibility is an open condition preserved under the square-zero thickening). Thus the deformation functor of $\mathcal{L}$ as a sheaf agrees with its deformation functor as a line bundle, which by definition of $\mathrm{Pic}\,(X)$ as the scheme representing the Picard functor is the tangent space

$$T_{[\mathcal{L}]}\mathrm{Pic}\,(X)=\mathrm{Pic}\,(X)\big(k[\varepsilon]\big)_{[\mathcal{L}]}\cong H^1(X,\mathcal{O}_X)$$

in the sense of the functorial tangent space definition 1.5.1. Finally, translation by $\mathcal{L}$ is an isomorphism of $\mathrm{Pic}\,(X)$ carrying the component of $[\mathcal{L}]$ to $\mathrm{Pic}^0(X)$ and $[\mathcal{L}]$ to the identity $[\mathcal{O}_X]$, so the tangent spaces agree and $T_{[\mathcal{L}]}\mathrm{Pic}\,(X)\cong T_0\mathrm{Pic}^0(X)\cong H^1(X,\mathcal{O}_X)$, as we sought to show.

The proposition is the source of the classical dimension count for the Picard variety. The tangent space to $\mathrm{Pic}\,(X)$ at any point is $H^1(X,\mathcal{O}_X)$, a cohomology group intrinsic to X and independent of the chosen line bundle, so $\mathrm{Pic}^0(X)$ is a smooth group of dimension $h^1(\mathcal{O}_X)$. Over $\mathbb{C}$ this is the abelian variety $H^1(X,\mathcal{O}_X)/H^1(X,\mathbb{Z})$, and the number $h^1(\mathcal{O}_X)$ is the *irregularity* $q(X)$; the proposition says the tangent space to the Picard scheme is the irregularity-dimensional space $H^1(X,\mathcal{O}_X)$. The reason line bundles are so much simpler to deform than general sheaves is Step 1: their local Ext sheaves vanish, so the deformation theory sees only the global cohomology of the trivial line bundle, never any local contribution.

The cleanest instance is a curve, where the Picard variety is the Jacobian and the dimension count reproduces the genus. The example computes the tangent space directly and matches it against the known dimension of $\mathrm{Pic}^0(C)$.

Example 4.4.5: Deformations of a Line Bundle on a Curve

Let C be a smooth projective curve of genus g over k, and let $\mathcal{L}$ be a line bundle on C. By proposition 4.4.4 the first-order deformations of $\mathcal{L}$ form the vector space $H^1(C, \mathcal{O}_C)$, so

$$\dim_k \mathrm{Ext}^1_C(\mathcal{L}, \mathcal{L}) = \dim_k H^1(C, \mathcal{O}_C) = h^1(\mathcal{O}_C)$$

The genus of a smooth projective curve is defined as $g = h^1(C, \mathcal{O}_C) = \dim_k H^1(C, \mathcal{O}_C)$, equivalently $g = h^0(C, \omega_C)$ by Serre duality ([12, Serre Duality] for Serre duality and the genus of a curve). Therefore

$$\dim_k \mathrm{Ext}^1_C(\mathcal{L}, \mathcal{L}) = g$$

Every line bundle on C has a g-dimensional space of first-order deformations, and this dimension is the same for every $\mathcal{L}$.

This matches the geometry of the Picard variety. The degree-zero Picard scheme $\mathrm{Pic}^0(C)$ is the *Jacobian* $\mathrm{Jac}(C)$, an abelian variety of dimension g; over $\mathbb{C}$ it is the complex torus

$$\mathrm{Jac}(C) = H^1(C, \mathcal{O}_C)/H^1(C, \mathbb{Z}) = H^0(C, \omega_C)^\vee/H_1(C, \mathbb{Z})$$

a g-dimensional torus. Its tangent space at the origin is $H^1(C, \mathcal{O}_C)$, of dimension g, in agreement with the Ext-computation above:

$$\dim \mathrm{Jac}(C) = \dim T_0\mathrm{Jac}(C) = h^1(\mathcal{O}_C) = g$$

The deformation theory of a single line bundle on C has recovered the dimension of the entire moduli space of line bundles. For genus $g = 0$, $C = \mathbb{P}^1$ has $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0$, so every line bundle on $\mathbb{P}^1$ is infinitesimally rigid and $\mathrm{Pic}(\mathbb{P}^1) = \mathbb{Z}$ is discrete; for an elliptic curve $g = 1$, the Jacobian is one-dimensional and $\mathrm{Jac}(E) \cong E$ recovers the curve itself (see [7]). For $g \geq 2$ the g-dimensional Jacobian is the first higher-dimensional abelian variety attached to C.

The example closes the loop with section 4.3. There the tangent space to the moduli of curves was $H^1(C, T_C)$ of dimension $3g - 3$; here the tangent space to the moduli of *line bundles on a fixed curve* is $H^1(C, \mathcal{O}_C)$ of dimension g. The two are different moduli problems with different tangent spaces, both computed by the same square-zero mechanism, and both matching the dimensions of the geometric objects ($\mathcal{M}_g$ and $\mathrm{Jac}(C)$) they linearize. The Jacobian is the running touchstone for the abelian-variety chapters that follow this book, where $\mathrm{Jac}(C)$ and its higher analogues $\mathrm{Pic}^0(X)$ become the central objects.

Exercise 4.4.1

1. Let X be a scheme over k and $\mathcal{F}$ a coherent sheaf. Show directly, without invoking theorem 4.4.2, that the trivial deformation $\mathcal{F}_\varepsilon = \mathcal{F} \otimes_k k[\varepsilon]$ is flat over $k[\varepsilon]$ and that as an extension of $\mathcal{F}$ by $\mathcal{F}$ it is the split extension. Conclude that its class in $\mathrm{Ext}^1_X(\mathcal{F}, \mathcal{F})$ is zero.

2. Let X be a smooth projective surface with $h^1(\mathcal{O}_X) = 0$ (for instance $X = \mathbb{P}^2$ or a $K3$ surface). Compute the dimension of the space of first-order deformations of an arbitrary line bundle $\mathcal{L}$ on X, and describe $\operatorname{Pic}^0(X)$ set-theoretically. What does this say about how the line bundles of a fixed numerical class sit inside $\operatorname{Pic}(X)$?

3. Let $\mathcal{F}$ be a coherent sheaf on a smooth projective curve C of genus g, and suppose $\mathcal{F}$ is locally free of rank r (a vector bundle). Using $\operatorname{Ext}^1_C(\mathcal{F},\mathcal{F}) \cong H^1(C, \mathcal{E}nd(\mathcal{F}))$ and Riemann-Roch for the vector bundle $\mathcal{E}nd(\mathcal{F})$ of rank r^2 and degree 0, compute $\dim \operatorname{Ext}^1_C(\mathcal{F},\mathcal{F}) - \dim \operatorname{Hom}_C(\mathcal{F},\mathcal{F})$ and hence show $\dim \operatorname{Ext}^1_C(\mathcal{F},\mathcal{F}) = r^2(g-1) + \dim \operatorname{Hom}_C(\mathcal{F},\mathcal{F})$. For a stable bundle $\mathcal{F}$ (so $\operatorname{Hom}_C(\mathcal{F},\mathcal{F}) = k$), deduce the deformation dimension $r^2(g-1)+1$.

4. Let $\mathcal{L}$ be a line bundle on a smooth projective variety X. Prove the isomorphism $\operatorname{Ext}^1_X(\mathcal{L},\mathcal{L}) \cong \operatorname{Ext}^1_X(\mathcal{O}_X,\mathcal{O}_X)$ used in proposition 4.4.4 directly, by exhibiting the map $\mathcal{E} \mapsto \mathcal{E} \otimes \mathcal{L}^{-1}$ on extensions and checking it is a bijection with inverse $- \otimes \mathcal{L}$. Explain in one sentence why this is the deformation-theoretic statement that “all line bundles deform like $\mathcal{O}_X$.”

5. An extension $0 \to \mathcal{F} \to \mathcal{F}_\varepsilon \to \mathcal{F} \to 0$ realizing a first-order deformation gives a sheaf $\mathcal{F}_\varepsilon$ on $X[\varepsilon]$. Suppose the class in $\operatorname{Ext}^1_X(\mathcal{F},\mathcal{F})$ is nonzero. Show that $\mathcal{F}_\varepsilon$ is nonetheless *not* of the form $\mathcal{G} \otimes_k k[\varepsilon]$ for any $\mathcal{O}_X$-module $\mathcal{G}$, and interpret this as the statement that the deformation is nontrivial. (Hint: compare the ε-action to that on a product.)

4.5 Automorphisms and Infinitesimal Automorphisms

The obstruction that kept the moduli functors of the opening chapter from being representable (box 1.1.6) was never the geometry of any single object; it was the presence of automorphisms. An object with a nontrivial automorphism can be glued to itself along that automorphism in a family, producing a nonconstant family all of whose fibers are isomorphic, and no scheme can separate points a family refuses to separate. This chapter has so far measured the first-order geometry of a deformation problem by a single cohomology group: $H^0(N_{Z/X})$ for a subscheme (theorem 4.2.1), $H^1(X, T_X)$ for a smooth variety (theorem 4.3.2), $\operatorname{Ext}^1_X(\mathcal{F},\mathcal{F})$ for a sheaf (theorem 4.4.2). This closing section measures the automorphisms themselves, and finds that they too are governed by the tangent sheaf, one cohomological degree below the deformations. The infinitesimal automorphisms of a smooth variety X are exactly $H^0(X, T_X)$.

Placing $H^0(X, T_X)$ next to $H^1(X, T_X)$ organizes the local structure of a deformation problem into a graded bookkeeping. The infinitesimal automorphisms sit in degree 0, the first-order deformations in degree 1, and the obstructions to extending a deformation, taken up in the next chapter, sit in degree 2. Writing these as T^0, T^1, and T^2 turns the qualitative dichotomy between fine and coarse moduli established in the opening chapter (theorem 1.3.6) into a computable invariant: a nonzero T^0 is the infinitesimal shadow of the automorphisms that force a coarse space in place of a fine one. This section develops T^0 and the T^0/T^1 bookkeeping; the T^2 layer is the subject of the next chapter.

Throughout, X is a smooth projective variety over an algebraically closed field k of characteristic 0, and $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$ its tangent sheaf. The tangent sheaf, its identification with k-linear derivations of $\mathcal{O}_X$, and the sheaf cohomology $H^i(X, T_X)$ are imported from [12, the tangent sheaf] and are used without re-derivation; the content of this section is the deformation-theoretic meaning of $H^0(X, T_X)$.

Definition 4.5.1: Infinitesimal Automorphism

Let X be a smooth variety over k, and let $X_\varepsilon = X \times_k \operatorname{Spec} k[\varepsilon]$ be the trivial first-order deformation of X over the dual numbers $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ (definition 4.1.4). An *infinitesimal automorphism* of X is an automorphism

$$\varphi\colon X_\varepsilon \xrightarrow{\sim} X_\varepsilon$$

of X_ε over $\operatorname{Spec} k[\varepsilon]$ whose reduction modulo ε is the identity of X; that is, φ restricts to id_X on the closed fiber $X = X_\varepsilon \times_{\operatorname{Spec} k[\varepsilon]} \operatorname{Spec} k$. The set of infinitesimal automorphisms of X is denoted $T^0 = T^0(X)$.

An infinitesimal automorphism is a symmetry of the trivial deformation that is invisible at the level of the original space: it moves points only to first order in ε. The condition that φ reduce to the identity is what makes T^0 an infinitesimal object rather than the full automorphism group $\operatorname{Aut}(X)$; it isolates the part of a symmetry that a first-order neighborhood of the identity can see. The next result identifies this set with global vector fields, which simultaneously gives T^0 the structure of a k-vector space and identifies its dimension with a computable cohomological invariant.

Proposition 4.5.2: Infinitesimal Automorphisms as Global Vector Fields

Let X be a smooth variety over k. There is a canonical bijection

$$T^0(X) \xrightarrow{\sim} H^0(X, T_X)$$

sending an infinitesimal automorphism φ of X_ε to the global vector field D_φ measuring its first-order displacement. Under this bijection the composition of infinitesimal automorphisms corresponds to addition of vector fields, so $T^0(X)$ is a k-vector space and

$$\dim_k T^0(X) = h^0(X, T_X)$$

***Proof*:**

Write $A = k[\varepsilon]$, and let $p\colon X_\varepsilon \to X$ be the projection and $i\colon X \hookrightarrow X_\varepsilon$ the closed immersion of the fiber over $\varepsilon = 0$. An automorphism φ of X_ε over $\operatorname{Spec} A$ is the same datum as a k-algebra automorphism of the sheaf $\mathcal{O}_{X_\varepsilon} = \mathcal{O}_X \otimes_k A = \mathcal{O}_X \oplus \varepsilon\mathcal{O}_X$ that is A-linear and reduces modulo ε to a given automorphism of $\mathcal{O}_X$. The condition in definition 4.5.1 is that this reduction be the identity.

Step 1: A first-order automorphism is $\mathrm{id} + \varepsilon D$ *for a derivation* D. Let $\varphi^\sharp\colon \mathcal{O}_{X_\varepsilon} \to \mathcal{O}_{X_\varepsilon}$ be the comorphism. Because $\varphi^\sharp$ is A-linear and reduces to the identity modulo ε, on a local section $f \in \mathcal{O}_X$ it has the form

$$\varphi^\sharp(f) = f + \varepsilon\, D(f)$$

for a uniquely determined k-linear map $D\colon \mathcal{O}_X \to \mathcal{O}_X$; the value $\varphi^\sharp(\varepsilon) = \varepsilon$ is forced by A-linearity, so $\varphi^\sharp$ is determined by D. That $\varphi^\sharp$ is a ring homomorphism translates into a condition on D. Applying $\varphi^\sharp$ to a product fg and using $\varepsilon^2 = 0$,

$$\varphi^\sharp(fg) = fg + \varepsilon\, D(fg), \qquad \varphi^\sharp(f)\varphi^\sharp(g) = (f + \varepsilon D(f))(g + \varepsilon D(g)) = fg + \varepsilon\big(fD(g) + gD(f)\big)$$

Equating the two, $D(fg) = fD(g) + gD(f)$, so D is a k-linear derivation of $\mathcal{O}_X$. Conversely, for any derivation D the map $f \mapsto f + \varepsilon D(f)$, extended A-linearly with $\varepsilon \mapsto \varepsilon$, is a ring homomorphism reducing to the identity, and it is invertible with inverse $f \mapsto f - \varepsilon D(f)$ (again using $\varepsilon^2 = 0$). Thus $\varphi \mapsto D$ is a bijection between infinitesimal automorphisms of X_ε and k-linear derivations of $\mathcal{O}_X$.

Step 2: Derivations are global vector fields. By definition of the tangent sheaf, $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$, and the universal derivation $d\colon \mathcal{O}_X \to \Omega_{X/k}$ induces for every open $U \subseteq X$ a natural isomorphism between k-linear derivations $\mathcal{O}_X|_U \to \mathcal{O}_X|_U$ and $\mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)(U) = T_X(U)$, sending a derivation D to the $\mathcal{O}_U$-linear map $\omega \mapsto \langle \omega, D\rangle$ determined by $D = \langle d(-), D\rangle$. A derivation defined on all of X is a section over X, so the global k-linear derivations of $\mathcal{O}_X$ are exactly $H^0(X, T_X) = T_X(X)$. Set $D_\varphi \in H^0(X, T_X)$ equal to the section corresponding to the derivation D produced in Step 1. Steps 1 and 2 compose to the asserted bijection $T^0(X) \to H^0(X, T_X)$, $\varphi \mapsto D_\varphi$.

Step 3: Composition corresponds to addition. Let φ, ψ be infinitesimal automorphisms with associated derivations D, E. The composite $\psi \circ \varphi$ has comorphism $\varphi^\sharp \circ \psi^\sharp$, and on a section f,

$$\varphi^\sharp(\psi^\sharp(f)) = \varphi^\sharp(f + \varepsilon E(f)) = f + \varepsilon E(f) + \varepsilon D(f) = f + \varepsilon\,(D + E)(f)$$

where the cross term $\varepsilon D(E(f)) \cdot \varepsilon$ vanishes because it carries a factor $\varepsilon^2 = 0$. Hence the derivation of $\psi \circ \varphi$ is $D + E$, so the group operation of composition is carried to addition of global vector fields. The k-scaling $D \mapsto \lambda D$ likewise corresponds to the infinitesimal automorphism $f \mapsto f + \varepsilon\lambda D(f)$, so the bijection is one of k-vector spaces and $\dim_k T^0(X) = h^0(X, T_X)$, as we sought to show.

The proof exhibits the mechanism behind the whole T^i formalism in miniature. A first-order symmetry is a derivation because $\varepsilon^2 = 0$ kills every term beyond the linear one, so the nonlinear datum of an automorphism collapses to the linear datum of a vector field. That the group law becomes addition is why T^0 is a vector space and not just a group: the commutator of two infinitesimal automorphisms, which would record the Lie bracket $[D, E]$, lives in order ε^2 and is therefore invisible at first order. The Lie-algebra structure on $H^0(X, T_X)$ is real, but it belongs to the second-order theory; the first-order theory sees only the underlying vector space. This is the same reason the first-order deformations $H^1(X, T_X)$ of theorem 4.3.2 form a vector space rather than carrying the finer structure that second-order obstructions will impose in the next chapter.

With T^0 and T^1 both identified as cohomology of T_X, the tangent-obstruction bookkeeping of a smooth variety can be assembled. The following proposition records the assembly and, more importantly, ties T^0 back to the representability failure that opened the book.

Proposition 4.5.3: The T^0/T^1 Bookkeeping

Let X be a smooth projective variety over k. The first-order deformation theory of X is organized by the cohomology of the tangent sheaf:

1. $T^0(X) = H^0(X, T_X)$ is the space of infinitesimal automorphisms of X (proposition 4.5.2).
2. $T^1(X) = H^1(X, T_X)$ is the space of first-order deformations of X (theorem 4.3.2).

Both are finite-dimensional k-vector spaces. Moreover, $T^0(X) \neq 0$ if and only if X carries a nonzero global vector field, and in that case the automorphism group $\mathrm{Aut}(X)$ is positive-dimensional; the presence of such infinitesimal automorphisms is the local, first-order manifestation of the automorphism obstruction to fine representability recorded in box 1.1.6 and realized concretely by the j-line of theorem 1.3.6.

***Proof*:**
Statements (1) and (2) restate proposition 4.5.2 and theorem 4.3.2 respectively. Finite-dimensionality of $H^0(X, T_X)$ and $H^1(X, T_X)$ is the finiteness of sheaf cohomology of a coherent sheaf on a projective variety, imported from [12, finiteness of coherent cohomology].

For the last statement, $T^0(X) = H^0(X, T_X) \neq 0$ means precisely that X admits a nonzero global vector field D. Because X is a projective variety over k, its automorphism functor is represented by a group scheme $\underline{\mathrm{Aut}}(X)$ locally of finite type over k, whose tangent space at the identity is the Lie algebra $H^0(X, T_X)$; a nonzero global vector field is a nonzero tangent vector at the identity, so $\dim \underline{\mathrm{Aut}}(X) = \dim_k H^0(X, T_X) > 0$ and $\mathrm{Aut}(X)$ is positive-dimensional. The identification of $H^0(X, T_X)$ with $\mathrm{Lie}\, \underline{\mathrm{Aut}}(X)$ is exactly the case of proposition 4.5.2 in which the infinitesimal automorphisms are read as the $k[\varepsilon]$-points of the automorphism scheme lying over the identity, completing the proof.

The final clause is the payoff of the section. box 1.1.6 isolated the reason a moduli functor of objects-up-to-isomorphism can fail to be a sheaf, and hence fail to be representable by a scheme: an object with automorphisms admits families that are locally trivial but globally twisted, and the functor cannot see the twist. theorem 1.3.6 made this concrete for elliptic curves, where the extra automorphisms at $j = 0$ and $j = 1728$ force the moduli problem down to the coarse j-line. proposition 4.5.3 says that the same obstruction has an infinitesimal fingerprint: whenever X has a continuous family of automorphisms, $T^0(X) = H^0(X, T_X)$ is nonzero, and this nonvanishing is detectable by a first-order computation with the dual numbers, before any global family is written down. The deformation theory of the next chapter, extending the count to T^2, will diagnose obstructions the same way, turning the qualitative fine-versus-coarse dichotomy into arithmetic on three cohomology groups.

The two extremes of this dictionary are the most instructive: a space whose automorphisms form a positive-dimensional group, so $T^0 \neq 0$, and a space whose automorphism group is finite, so $T^0 = 0$. Projective space and high-genus curves furnish exactly these two poles.

Example 4.5.4: Infinitesimal Automorphisms of $\mathbb{P}^n$ and of Curves

1. *Projective space.* The Euler sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

imported from [12, the Euler sequence], computes $H^0(\mathbb{P}^n, T_{\mathbb{P}^n})$. Taking the long exact sequence in cohomology and using $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = k$, $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) = k^{n+1}$, and $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = 0$,

$$0 \longrightarrow k \longrightarrow k^{(n+1)^2} \longrightarrow H^0(\mathbb{P}^n, T_{\mathbb{P}^n}) \longrightarrow 0$$

so $h^0(\mathbb{P}^n, T_{\mathbb{P}^n}) = (n+1)^2 - 1$. This is exactly $\dim \mathrm{PGL}_{n+1}$, and indeed

$$H^0(\mathbb{P}^n, T_{\mathbb{P}^n}) \cong \mathfrak{pgl}_{n+1} = \mathfrak{sl}_{n+1}$$

the Lie algebra of $\mathrm{PGL}_{n+1} = \mathrm{Aut}(\mathbb{P}^n)$: the global vector fields on $\mathbb{P}^n$ are the infinitesimal projective transformations, the traceless $(n+1) \times (n+1)$ matrices acting by the induced linear flow on homogeneous coordinates, with the scalar matrices (the kernel $\mathcal{O}_{\mathbb{P}^n}$ in the Euler sequence) acting trivially. Thus $T^0(\mathbb{P}^n) = \mathfrak{pgl}_{n+1} \neq 0$: projective space has a positive-dimensional automorphism group, and $\mathbb{P}^n$ is the paradigm of an object with many infinitesimal automorphisms. It is also infinitesimally rigid, $T^1(\mathbb{P}^n) = H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$ (example 4.3.3), so its bookkeeping is $(T^0, T^1) = (\mathfrak{pgl}_{n+1}, 0)$: no deformations, but a large automorphism algebra.

2. *Curves of genus* $g \geq 2$. Let C be a smooth projective curve of genus g. On a curve the tangent sheaf is the dual of the canonical sheaf, $T_C = \Omega_C^\vee = \mathcal{O}_C(-K_C)$, a line bundle of degree $\deg T_C = 2 - 2g$. For $g \geq 2$ this degree is negative, and a line bundle of negative degree on a projective curve has no nonzero global sections, so

$$T^0(C) = H^0(C, T_C) = 0$$

A curve of genus at least 2 has *no* nonzero infinitesimal automorphisms. By proposition 4.5.3, $\mathrm{Aut}(C)$ is therefore 0-dimensional, hence a finite group (it is a closed subgroup scheme of finite type with trivial tangent space at the identity). This is consistent with the Hurwitz bound $|\mathrm{Aut}(C)| \leq 84(g-1)$, which asserts the same finiteness quantitatively. The deformation count sits one degree up: by Serre duality (imported from [12, Serre duality]) and Riemann-Roch,

$$T^1(C) = H^1(C, T_C) \cong H^0(C, \Omega_C^{\otimes 2})^\vee, \qquad \dim_k T^1(C) = 3g - 3$$

for $g \geq 2$ (example 4.3.4), matching the dimension of the moduli space of curves. The bookkeeping is $(T^0, T^1) = (0,\ 3g-3)$: a finite automorphism group and a $(3g-3)$-dimensional family of deformations.

3. *Genus* 0 *and* 1. For contrast, $g = 0$ gives $\mathbb{P}^1$, with $T^0 = H^0(\mathbb{P}^1, T_{\mathbb{P}^1}) \cong \mathfrak{pgl}_2$ of dimension 3 and $T^1 = 0$: the rational curve is rigid with a three-dimensional automorphism group. For $g = 1$, an elliptic curve has $T_C \cong \mathcal{O}_C$ (the canonical bundle is trivial), so $T^0 = H^0(C, \mathcal{O}_C) = k$ is one-dimensional, reflecting the translations, and $T^1 = H^1(C, \mathcal{O}_C) = k$ is one-dimensional, matching the one-dimensional j-line of theorem 1.3.6. The nonvanishing of T^0 at $g = 1$ is the infinitesimal trace of the automorphisms that forced a coarse rather than fine moduli space in theorem 1.3.6.

The two poles read off the same dictionary in opposite directions. For $\mathbb{P}^n$ the bookkeeping is $(T^0, T^1) = (\mathfrak{pgl}_{n+1}, 0)$, all symmetry and no deformation; for a curve of genus $g \geq 2$ it is $(0, 3g-3)$, no symmetry and a large deformation space. The transition happens exactly where $\deg T_X$ changes sign, which on a curve is the transition through genus 1. That T^0 vanishes precisely for $g \geq 2$ is the infinitesimal reason those curves have finite automorphism groups and, in the stack-theoretic framework of the book's third part, why $\mathcal{M}_g$ for $g \geq 2$ is a Deligne-Mumford stack with finite stabilizers rather than an Artin stack with positive-dimensional ones. The vanishing or nonvanishing of a single degree-0 cohomology group thus predicts, before any moduli space is constructed, whether the eventual moduli object will be a scheme, a Deligne-Mumford stack, or something with continuous automorphisms.

Exercise 4.5.1

1. Let X be a smooth projective variety with $H^0(X, T_X) = 0$. Show directly from definition 4.5.1 that the only automorphism of the trivial deformation X_ε over $\operatorname{Spec} k[\varepsilon]$ reducing to id_X is the identity, and explain in one sentence why this says X is "infinitesimally rigid as a symmetric object".

2. Compute T^0, T^1, and $\dim \operatorname{Aut}$ for each of the following, and state whether the automorphism group is finite or positive-dimensional:

    1. $\mathbb{P}^2$;
    2. a smooth plane quartic curve $C \subseteq \mathbb{P}^2$;
    3. an abelian surface A (so $T_A \cong \mathcal{O}_A^{\oplus 2}$).

3. Let C_1 and C_2 be smooth projective curves of genera $g_1, g_2 \geq 2$ and set $X = C_1 \times C_2$. Using $T_X \cong p_1^* T_{C_1} \oplus p_2^* T_{C_2}$ and the Künneth formula, show that $H^0(X, T_X) = 0$, and conclude that $\operatorname{Aut}(X)$ is finite.

4. Show that the Euler characteristic $\chi(T_X) = h^0(X, T_X) - h^1(X, T_X) + \cdots = \dim T^0 - \dim T^1 + \cdots$ is a single number combining the automorphism and deformation counts. For a smooth projective curve C of genus g, compute $\chi(T_C)$ from Riemann-Roch and check that it equals $\dim T^0 - \dim T^1$ in each of the cases $g = 0, 1$ and $g \geq 2$.

5. An elliptic curve E has $T^0(E) = H^0(E, \mathcal{O}_E) = k$. Interpret the nonzero global vector field on E as an infinitesimal automorphism explicitly, and explain how its nonvanishing is compatible with $\operatorname{Aut}(E, O)$ (automorphisms fixing the origin) being finite while $\operatorname{Aut}(E)$ as an abstract variety is positive-dimensional.

6. Suppose X is a smooth projective variety with $H^0(X, T_X) \neq 0$. Argue, using box 1.1.6, that the naive moduli functor parametrizing objects isomorphic to X cannot be represented by a scheme with X as an isolated reduced point, and relate this to the failure diagnosed for the j-line in theorem 1.3.6.

5

Obstructions and Pro-representability

First-order deformations, computed in chapter 4, are the linear approximation to a moduli problem; this chapter asks whether that approximation integrates to an actual family. A tangent vector to a moduli space is a deformation over the dual numbers, but an arc through the point is a deformation over a power-series ring, assembled one order at a time by lifting successively over the small extensions $k[t]/(t^{n+1}) \to k[t]/(t^n)$. At each stage the lift may be blocked, and the block is an element of a second cohomology group; when every such obstruction vanishes the moduli space is smooth at the point, and when they do not, the obstruction theory of this chapter measures the singularity.

section 5.1 locates the obstruction to extending a deformation of a smooth variety in $H^2(X, T_X)$ and defines what it means for a deformation functor to be unobstructed. section 5.2 packages the tangent and obstruction spaces into a single object, the naive cotangent complex, whose cohomology groups T^0, T^1, T^2 are the infinitesimal automorphisms, the first-order deformations, and the obstructions of a k-algebra. section 5.3 proves Schlessinger's criteria, the conditions on a functor of Artin rings under which it admits a hull, a formal versal deformation, or is pro-representable outright. section 5.4 closes Part II with computations: the unobstructedness of curves, abelian varieties, and K3 surfaces, an obstructed example, and the deformation of a node, whose single smoothing parameter is the local shadow of the stable-reduction theorem that the moduli of curves will require.

5.1 Extending Deformations and the Obstruction Space

The previous chapter linearized the deformation problem: a first-order deformation of a smooth projective variety X over the dual numbers $k[\varepsilon]$ is a class in $H^1(X, T_X)$, and this cohomology group is the tangent space to the moduli problem at $[X]$ (theorem 4.3.2). A tangent vector, though, is only the beginning of an arc. Knowing the direction in which X can be bent to first order says nothing about whether that bending can be continued to second order, to third order, and onward to a genuine one-parameter family; a tangent line to a plane curve need not be the start of any branch of the curve when the curve has a singularity at the point. The passage from a tangent vector to an actual deformation is exactly the passage from the linear approximation of the moduli space to the moduli space itself, and it is governed by a second cohomological invariant, one degree up from the tangent space.

This section isolates the mechanism. A deformation over an Artinian base is built by climbing a tower of *small extensions*, extending the deformation one infinitesimal step at a time; at each step the lift either exists or is blocked, and the block is measured by a single class in a *second* cohomology group. When that class vanishes the deformation extends, and the set of extensions is a torsor under the first cohomology group, the same group that measured first-order deformations. When the second cohomology group vanishes identically, every obstruction is forced to be zero and the moduli problem is smooth. This is the dichotomy that separates curves (always unobstructed) from surfaces (which can be obstructed), and it is the computational engine behind every smoothness statement about a moduli space in this book. Throughout, k is algebraically closed of characteristic zero, X is a smooth projective k-variety, and $T_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$ is its tangent sheaf ([12, Tangent Sheaf]).

The infinitesimal tower is built out of the smallest possible steps. Recall from section 4.1 that a *small extension* in $\mathbf{Art}_k$ is a surjection $A' \to A$ of local Artinian k-algebras with residue field k whose kernel (t) is a one-dimensional k-vector space annihilated by the maximal ideal, $\mathfrak{m}_{A'} \cdot t = 0$. Every surjection in $\mathbf{Art}_k$ factors as a finite composite of small extensions, so a deformation over an arbitrary Artinian base is assembled by lifting through a sequence of them. The atom of the whole theory is therefore a single small extension, and the atomic question is whether a deformation already built over A can be pushed one step further, over A'.

Definition 5.1.1: The Small-Extension Lifting Problem

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a deformation functor and let

$$0 \to (t) \to A' \xrightarrow{\pi} A \to 0$$

be a small extension in $\mathbf{Art}_k$, with kernel $(t) \cong k$ satisfying $\mathfrak{m}_{A'} \cdot t = 0$. Given a deformation $\xi \in D(A)$, the *lifting problem* for ξ along π asks whether ξ lies in the image of the map $D(\pi)\colon D(A') \to D(A)$; a preimage $\xi' \in D(A')$ with $D(\pi)(\xi') = \xi$ is called a *lift* (or *extension*) of ξ over A'. The deformation ξ is *obstructed* along π if no lift exists, and *unobstructed* along π if a lift exists.

The definition is deliberately functorial and says nothing yet about how to decide the lifting problem; that is the work of the theorem below. The content of a deformation theory is a recipe that, to each pair (ξ, π), assigns a cohomology class whose vanishing is equivalent to the existence of a lift, and whose ambient group is a fixed invariant of X independent of ξ and π. For the deformation functor Def_X of a smooth projective variety that invariant is $H^2(X, T_X)$, and the assignment is constructed by re-gluing local lifts over an affine cover. The key structural fact, which makes the whole method work, is that the lifting problem is always solvable *locally*: over an affine open where X is smooth there is no obstruction, because a smooth affine scheme deforms trivially. The obstruction is entirely a global gluing phenomenon, and cohomology is the exact bookkeeping for gluing.

Before stating the theorem it is worth fixing what a deformation over an Artinian base concretely is, so that the phrase "re-glue over a cover" has an unambiguous meaning. A deformation of X over A is a flat scheme $\mathcal{X}_A \to \operatorname{Spec} A$ together with an identification of its special fiber $\mathcal{X}_A \times_{\operatorname{Spec} A} \operatorname{Spec} k$ with X. Flatness over A is the algebraic incarnation of "continuously varying," and over an Artinian base it forces the sheaf of functions on $\mathcal{X}_A$ to be, locally, a free A-module deforming $\mathcal{O}_X$. On an affine open $U = \operatorname{Spec} R$ of X a deformation over A is a flat A-algebra R_A with $R_A \otimes_A k \cong R$; smoothness of R over k is what guarantees such a flat lift exists and is unique up to isomorphism. These two facts, local existence and local uniqueness of lifts, are the input to the gluing argument.

Theorem 5.1.2: The Obstruction to Lifting a Deformation

Let X be a smooth projective variety over k, and let

$$0 \to (t) \to A' \xrightarrow{\pi} A \to 0$$

be a small extension in $\mathbf{Art}_k$ with kernel $(t) \cong k$. Let $\mathcal{X}_A \to \operatorname{Spec} A$ be a deformation of X over A. Then there is a canonically associated *obstruction class*

$$\operatorname{ob}(\mathcal{X}_A, \pi) \in H^2(X, T_X) \otimes_k (t)$$

with the following properties.

1. A lift of $\mathcal{X}_A$ to a deformation over A' exists if and only if $\operatorname{ob}(\mathcal{X}_A, \pi) = 0$.
2. When a lift exists, the set of isomorphism classes of lifts is a torsor (a principal homogeneous space) under the group $H^1(X, T_X) \otimes_k (t)$; in particular it is nonempty and any two lifts differ by a unique element of $H^1(X, T_X) \otimes_k (t)$.

In particular, if $H^2(X, T_X) = 0$ then every deformation of X lifts over every small extension.

***Proof*:**
Fix an affine open cover $\mathcal{U} = \{U_i\}$ of X by opens on which X is smooth, and small enough that each U_i, and each pairwise intersection $U_{ij} = U_i \cap U_j$ and triple intersection U_{ijk}, is affine. Write $\mathcal{X}_A|_{U_i}$ for the restriction of the deformation over A to U_i, which is a flat A-algebra deformation of $\mathcal{O}_X(U_i)$. Because a one-step lift and its ambiguity are governed by the same tensor factor (t) throughout, fix once and for all a generator t of the kernel, so that every class produced below is written as (a T_X-valued class) $\otimes\, t$; tensoring with (t) at the end restores canonicity and removes the choice.

Step 1: Local lifts exist. Each U_i is affine and smooth over k, so its ring $R_i = \mathcal{O}_X(U_i)$ is a smooth k-algebra. A smooth algebra is formally smooth: given the surjection $\pi\colon A' \to A$ and the flat A-algebra deformation $\mathcal{X}_A|_{U_i}$ of R_i, formal smoothness produces a flat A'-algebra $\mathcal{X}_i'$ lifting it, that is, a deformation of U_i over A' restricting to $\mathcal{X}_A|_{U_i}$ over A. Concretely, presenting $R_i = P_i/J_i$ over a polynomial ring P_i so that J_i is, locally on U_i, generated by a regular sequence (smoothness gives a local complete-intersection presentation) and lifting those local generators arbitrarily to A'-coefficients produces such an $\mathcal{X}_i'$; flatness over A' holds because the defining sequence stays regular under the nilpotent thickening. So on each chart a lift $\mathcal{X}_i'$ exists. The obstruction is not in producing the local pieces but in matching them.

Step 2: Local lifts are unique up to a T_X-valued isomorphism. On the overlap U_{ij} the two local lifts $\mathcal{X}_i'$ and $\mathcal{X}_j'$ both restrict to the same deformation $\mathcal{X}_A|_{U_{ij}}$ over A. Two lifts of the same object over A to the same small extension of A differ by an automorphism that is the identity over A; the group of such automorphisms of a smooth affine over the kernel (t) is exactly the module of k-derivations $\mathrm{Der}_k(R_{ij}, R_{ij}) \otimes_k (t)$, that is, the vector fields $T_X(U_{ij}) \otimes_k (t)$. Concretely, an isomorphism $\theta_{ij} : \mathcal{X}'_j|_{U_{ij}} \xrightarrow{\sim} \mathcal{X}'_i|_{U_{ij}}$ restricting to the identity over A acts on functions by $f \mapsto f + D_{ij}(\bar{f})\, t$ for a unique derivation $D_{ij} \in T_X(U_{ij})$, where $\bar{f}$ is the reduction of f modulo $\mathfrak{m}_{A'}$; the term $D_{ij}(\bar{f})t$ is well defined because $\mathfrak{m}_{A'}t = 0$, so only the residue $\bar{f}$ enters. Choose such an isomorphism θ_{ij} on each overlap. This is the gluing datum; whether it is a genuine gluing depends on whether it is consistent on triple overlaps.

Step 3: The cocycle. On a triple overlap U_{ijk} compare the two ways of passing from $\mathcal{X}'_k$ to $\mathcal{X}'_i$: directly via θ_{ik}, or around through $\mathcal{X}'_j$ via $\theta_{ij} \circ \theta_{jk}$. Both are isomorphisms $\mathcal{X}'_k|_{U_{ijk}} \xrightarrow{\sim} \mathcal{X}'_i|_{U_{ijk}}$ restricting to the identity over A, so their difference is again a T_X-valued datum: set

$$c_{ijk} \;=\; \theta_{ik}^{-1} \circ \theta_{ij} \circ \theta_{jk} \;\in\; T_X(U_{ijk})$$

using Step 2 to identify an automorphism of $\mathcal{X}'_k|_{U_{ijk}}$ over A with a vector field. Because each θ is additive in its derivation to first order in t (the products $\mathfrak{m}_{A'}t = 0$ kill all higher terms), the failure of associativity is exactly the additive combination

$$c_{ijk} \;=\; D_{jk} - D_{ik} + D_{ij}$$

in $T_X(U_{ijk})$. The collection (c_{ijk}) is a Čech 2-cochain for $\mathcal{U}$ with values in T_X. It is a *cocycle*: on a quadruple overlap U_{ijkl} the alternating sum $c_{jkl} - c_{ikl} + c_{ijl} - c_{ijk}$ vanishes, because it is the difference of two ways of reassociating the four-fold composite θ_{il}, and reassociation of composable isomorphisms is associative. Thus (c_{ijk}) defines a Čech cohomology class

$$\mathrm{ob}(\mathcal{X}_A, \pi) \;=\; [(c_{ijk})] \otimes t \;\in\; \check{H}^2(\mathcal{U}, T_X) \otimes_k (t) \;=\; H^2(X, T_X) \otimes_k (t)$$

where the last identification uses that $\mathcal{U}$ is an affine (hence Leray) cover for the coherent sheaf T_X, so Čech and derived-functor cohomology agree.

Step 4: The class is independent of the choices. Two choices enter: the local lifts $\mathcal{X}'_i$ and the overlap isomorphisms θ_{ij}. Changing θ_{ij} to another gluing isomorphism θ'_{ij} over A changes D_{ij} by a vector field $E_{ij} \in T_X(U_{ij})$, and replaces c_{ijk} by $c_{ijk} + (E_{jk} - E_{ik} + E_{ij})$, that is, by a Čech coboundary; the class is unchanged. Changing the local lift $\mathcal{X}'_i$ to another local lift $\mathcal{X}''_i$ over A' changes it by a vector field $F_i \in T_X(U_i)$ (Step 2, now over the affine U_i itself, where a lift is unique up to $T_X(U_i) \otimes (t)$), and adjusts D_{ij} by $F_j - F_i$; this again changes c_{ijk} by the coboundary of (F_i). In both cases the cohomology class is invariant. So $\mathrm{ob}(\mathcal{X}_A, \pi) \in H^2(X, T_X) \otimes_k (t)$ depends only on $\mathcal{X}_A$ and π.

Step 5: Vanishing is equivalent to liftability. If a global lift $\mathcal{X}_{A'}$ of $\mathcal{X}_A$ over A' exists, take $\mathcal{X}'_i = \mathcal{X}_{A'}|_{U_i}$ to be its restrictions; these already glue (they are restrictions of a single object), so one may take $\theta_{ij} = \mathrm{id}$, whence $c_{ijk} = 0$ and $\mathrm{ob}(\mathcal{X}_A, \pi) = 0$. Conversely, suppose $\mathrm{ob}(\mathcal{X}_A, \pi) = 0$. Then (c_{ijk}) is a Čech coboundary, so there are vector fields $(G_{ij}) \in T_X(U_{ij})$ with $c_{ijk} = G_{jk} - G_{ik} + G_{ij}$ on every triple overlap. Replacing each θ_{ij} by $\theta_{ij} \circ$ (the automorphism $-G_{ij}$) modifies D_{ij} to $D_{ij} - G_{ij}$ and hence replaces c_{ijk} by $c_{ijk} - (G_{jk} - G_{ik} + G_{ij}) = 0$. The corrected gluing data θ'_{ij} now satisfy $\theta'_{ik} = \theta'_{ij} \circ \theta'_{jk}$ on triple overlaps, so the local lifts $\mathcal{X}'_i$ glue, along the θ'_{ij}, to a scheme $\mathcal{X}_{A'}$ over A'; flatness is local, so $\mathcal{X}_{A'}$ is flat over A', and its restriction over A is $\mathcal{X}_A$ because each θ'_{ij} is the identity over A. Thus a lift exists, proving (1).

Step 6: The torsor structure. Suppose a lift exists; fix one, say $\mathcal{X}_{A'}$, presented by local pieces $\mathcal{X}'_i$ glued by θ_{ij} with $c_{ijk} = 0$. Any other lift is presented, over the same local pieces (replacing

$\mathcal{X}_i'$ by an isomorphic lift changes nothing), by gluing data θ_{ij}' differing from θ_{ij} by vector fields $\eta_{ij} \in T_X(U_{ij})$, subject to the cocycle condition $\eta_{jk} - \eta_{ik} + \eta_{ij} = 0$ that both glue. So (η_{ij}) is a Čech 1-cocycle, defining a class in $H^1(X, T_X)$; and (η_{ij}) a coboundary means precisely that the two lifts are isomorphic rel $\mathcal{X}_A$. Adding a class of $H^1(X, T_X) \otimes (t)$ to the gluing data sends one lift to another and acts simply transitively on isomorphism classes of lifts. This is the torsor structure of (2); when $H^1(X, T_X) = 0$ the lift, when it exists, is unique. This establishes both claims and, in the case $H^2(X, T_X) = 0$, the final assertion, completing the proof.

The theorem is the local structure theory of a moduli space packaged in two cohomology groups. The first, $H^1(X, T_X)$, is the tangent space: it measures the directions in which X can move to first order, and, by the torsor statement, the ambiguity in continuing a deformation once continuation is possible. The second, $H^2(X, T_X)$, is the *obstruction space*: it is the receptacle for the classes that block continuation. The dimension of the moduli space at $[X]$ is at most $\dim H^1(X, T_X)$, and equals it exactly when the obstructions all vanish; the gap between the tangent-space dimension and the actual dimension is cut out by the image of the obstruction map into $H^2(X, T_X)$. This is the precise sense in which H^1 is the linear approximation and H^2 measures the failure of the linear approximation to be exact. A reader who has met the analogous statement in another guise, the obstruction to extending a section or a class living one degree up in a long exact sequence, is seeing the same homological phenomenon: extending a structure over a thickening is a lifting problem, and lifting problems are controlled by the next cohomology group.

The proof also explains *why* the obstruction is a global object even though the lifting problem is posed globally from the start. Locally there is never any obstruction, because smooth affines are unobstructed (Step 1); the class $\mathrm{ob}(\mathcal{X}_A, \pi)$ measures only the failure of the local solutions to be compatible on overlaps, and that failure is a Čech 2-cocycle. This is the same mechanism by which a line bundle can be locally trivial yet globally nontrivial, one dimension up. The obstruction is a purely cohomological shadow of a gluing that cannot be carried out, not a local pathology.

One consequence deserves to be named, because it will organize every smoothness result in the sequel: unobstructedness is the property that turns the tangent space into the whole local picture.

Definition 5.1.3: Unobstructed Deformation Functor

A deformation functor $D\colon \mathbf{Art}_k \to \mathbf{Set}$ is *unobstructed*, or *formally smooth*, if for every small extension $A' \to A$ in $\mathbf{Art}_k$ the map $D(A') \to D(A)$ is surjective; that is, every deformation over A lifts over every small extension. Equivalently, a hull R of D (a complete local k-algebra pro-representing D up to the tangent space, in the sense of the next sections) is a formal power-series ring $R \cong k[\![x_1, \ldots, x_n]\!]$ with $n = \dim_k D(k[\varepsilon])$, and the corresponding formal moduli space is smooth of dimension n at the point $[X]$.

The word "formally smooth" is chosen because the surjectivity condition on D is exactly the infinitesimal lifting criterion for smoothness of a scheme: a scheme M is smooth at a point m if and only if its functor of points lifts along every small extension near m. Applied to the local model of a moduli space, unobstructedness says that the moduli space, near $[X]$, looks formally like affine space of dimension $\dim H^1(X, T_X)$, with no equations cutting it down. When $D = \mathrm{Def}_X$ and $H^2(X, T_X) = 0$, theorem 5.1.2 shows Def_X is unobstructed and the moduli space is smooth of dimension $\dim H^1(X, T_X)$ at $[X]$; the two cohomology groups then give both the smoothness and the dimension in one stroke. The equivalence with a power-series hull is proved once Schlessinger's criteria are in place (see section 5.3); the definition records it now so that "unobstructed" and "power-series hull" are known to be the same condition wherever either appears.

The definition is only as useful as the vanishing statement $H^2(X, T_X) = 0$ is checkable. The following example is the whole payoff of the section: it makes the vanishing checkable for curves, deducing the smoothness of the moduli of curves from a one-line cohomological fact, and it exhibits the opposite phenomenon for surfaces, where $H^2(X, T_X)$ can be nonzero and a deformation can be obstructed.

Example 5.1.4: Curves are Unobstructed, Surfaces Need Not Be

1. *Curves.* Let C be a smooth projective curve of genus $g \geq 2$ over k. The tangent sheaf $T_C = \Omega_C^\vee$ is a line bundle, and C is a one-dimensional variety, so the coherent-sheaf cohomology $H^i(C, \mathcal{F})$ vanishes for all $i > 1$ and every coherent sheaf $\mathcal{F}$; this is Grothendieck's vanishing theorem for a Noetherian space of dimension 1. In particular
$$H^2(C, T_C) \ = \ 0$$
so by theorem 5.1.2 the deformation functor of C is unobstructed and the moduli space is smooth at $[C]$ of dimension $\dim H^1(C, T_C)$. The dimension is computed from Riemann-Roch. Since $T_C = \Omega_C^\vee$ has degree $-\deg \Omega_C = -(2g-2) = 2-2g < 0$, the space $H^0(C, T_C)$ of global vector fields vanishes for $g \geq 2$ (a nonzero global section of a negative-degree line bundle on a curve cannot exist), so C has no infinitesimal automorphisms in the sense of definition 4.5.1. Riemann-Roch on the curve ([12, Riemann-Roch]) then gives
$$h^0(C, T_C) - h^1(C, T_C) \ = \ \deg T_C + 1 - g \ = \ (2-2g) + 1 - g \ = \ 3 - 3g$$
and with $h^0(C, T_C) = 0$ this yields $h^1(C, T_C) = 3g - 3$. Thus the tangent space to the moduli of curves at $[C]$ has dimension $3g-3$, and since $H^2 = 0$ forces unobstructedness, $\mathcal{M}_g$ is smooth of dimension $3g-3$ at every point coming from a curve with no automorphisms. The absence of obstructions is the reason the count $3g-3$, first a tangent-space dimension, is the actual dimension of the moduli space and not just an upper bound.

2. *Surfaces.* For a smooth projective surface X the tangent sheaf T_X is a rank-2 bundle on a two-dimensional variety, and $H^2(X, T_X)$ need not vanish. By Serre duality on the surface ([12, Serre Duality]),
$$H^2(X, T_X) \ \cong \ H^0(X, \Omega_X \otimes \omega_X)^\vee$$
where $\omega_X = \Omega_X^2$ is the canonical bundle, and the right-hand side is a space of global sections that can be nonzero. A surface of general type with ample canonical bundle can have $H^0(X, \Omega_X \otimes \omega_X) \neq 0$, hence $H^2(X, T_X) \neq 0$, so the receptacle for obstructions is nonzero and the vanishing-room argument that made curves unobstructed is no longer available. Nonzero $H^2(X, T_X)$ is only a *necessary* condition for an obstructed deformation, not a sufficient one: the obstruction map $\xi \mapsto \frac{1}{2}[\xi, \xi] \colon H^1(X, T_X) \to H^2(X, T_X)$ (exercise 5.1.1 (5)) can be identically zero even when its target does not vanish. This is precisely the phenomenon behind the unobstructedness of higher-dimensional Calabi-Yau manifolds (of dimension ≥ 3), where $H^2(X, T_X) \neq 0$ yet every obstruction still vanishes (the Bogomolov-Tian-Todorov theorem, section 5.4). An abelian surface exhibits the same nonzero target already in dimension two: its tangent bundle $T_X \cong \mathcal{O}_X^{\oplus 2}$ is trivial, so $H^2(X, T_X) \cong H^2(X, \mathcal{O}_X)^{\oplus 2}$ is two-dimensional, yet abelian surfaces are unobstructed.

Whether obstructions occur is therefore a question about the obstruction map, not about the dimension of its target, and it must be settled by a construction rather than by a vanishing count.

Obstructed surfaces do exist. Kas and Horikawa exhibited surfaces of general type with obstructed deformations, whose moduli spaces are singular at the corresponding points, with local dimension there strictly less than $\dim H^1(X, T_X)$ (see [18] and [28] for the constructions and computations). The dichotomy with curves is dimensional at its root: on a curve, cohomology stops at degree 1 and there is no room for an obstruction to live, so unobstructedness is automatic; on a surface, degree 2 is available and an obstruction *can* be present, so smoothness of the moduli space is no longer free and must be established case by case.

The contrast in the example is the organizing principle of the rest of the chapter. For a curve, $H^2(C, T_C) = 0$ is automatic from dimension, so the moduli of curves is smooth and its dimension is read off from a single Riemann-Roch computation; this is the cohomological source of the number $3g - 3$ that reappears throughout the study of $\mathcal{M}_g$. For a surface, the obstruction space is a nontrivial invariant that can be nonzero, and deciding smoothness of the moduli space requires either computing $H^2(X, T_X)$ directly or invoking a structural reason for the obstructions to vanish. When $H^2(X, T_X) = 0$ outright, as for a K3 surface (where $\omega_X \cong \mathcal{O}_X$ forces $H^2(X, T_X) \cong H^0(X, \Omega_X)^\vee = 0$), theorem 5.1.2 already gives unobstructedness with no further input; the harder case is a higher-dimensional Calabi-Yau, where $H^2(X, T_X) \neq 0$ yet the obstruction map still vanishes, which is the content of the Bogomolov-Tian-Todorov theorem (section 5.4). The next section replaces the sheafy groups $H^i(X, T_X)$ by the algebraic cotangent functors $T^i(A/k, -)$, which compute the same obstruction theory for singular objects, where no tangent sheaf is available and the naive picture of "vector fields on overlaps" breaks down.

Exercise 5.1.1

1. Let X be a smooth projective variety with $H^2(X, T_X) = 0$, and let $A' \to A$ be an arbitrary surjection in $\mathbf{Art}_k$ (not necessarily small). Show that every deformation of X over A lifts to a deformation over A'. (Reduce to the small-extension case using that every surjection in $\mathbf{Art}_k$ factors as a finite composite of small extensions.)

2. Compute $H^1(\mathbb{P}^n, T_{\mathbb{P}^n})$ and $H^2(\mathbb{P}^n, T_{\mathbb{P}^n})$ for $n \geq 1$, and conclude that $\mathbb{P}^n$ is both rigid (no nontrivial first-order deformations) and unobstructed. (Use the Euler sequence $0 \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \to T_{\mathbb{P}^n} \to 0$, from [12, the Euler Sequence], and the cohomology of line bundles on $\mathbb{P}^n$.)

3. Let C be a smooth projective curve of genus g. Using example 5.1.4, determine $h^0(C, T_C)$ and $h^1(C, T_C)$ for $g = 0$ and $g = 1$, and interpret the results in terms of infinitesimal automorphisms and the dimension of the moduli of curves in those two genera.

4. In the setting of theorem 5.1.2, suppose $\mathcal{X}_A$ is a deformation of X over A that admits a lift over A', and suppose $H^1(X, T_X) = 0$. Show that the lift is unique up to isomorphism, and explain why the hypothesis $H^1 = 0$ says the deformation is rigid at every infinitesimal order.

5. Let $A = k[\varepsilon] = k[t]/(t^2)$ and $A' = k[t]/(t^3)$, so that $A' \to A$ is a small extension with kernel $(t^2) \cong k$. Interpret a lift of a first-order deformation $\xi \in H^1(X, T_X)$ over A to a deformation over A' as a "second-order deformation," and show directly from theorem 5.1.2 that the obstruction to the existence of such a lift is a class in $H^2(X, T_X)$ depending on ξ. Argue that this class is the quadratic term of the obstruction, so that unobstructedness at first order is exactly the vanishing of this quadratic obstruction for every ξ.

5.2 The Naive Cotangent Complex and the T^i Functors

The previous section measured deformations and obstructions of a *smooth* projective variety through the cohomology groups $H^i(X, T_X)$ of its tangent sheaf: first-order deformations sit in $H^1(X, T_X)$, obstructions in $H^2(X, T_X)$, and infinitesimal automorphisms in $H^0(X, T_X)$. That bookkeeping used smoothness at every turn, because the tangent sheaf $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$ only sees a variety's deformations correctly when the variety is smooth. Yet moduli problems force singular objects on us at once. The capstone stack $\overline{\mathcal{M}}_g$ parametrizes *nodal* curves; the deformation theory of a node, and of every other singularity that appears at the boundary of a moduli space, cannot be read off from $H^i(T_X)$ alone. What is needed is an invariant that specializes to $H^i(X, T_X)$ when X is smooth but continues to compute deformations and obstructions when X is singular.

That invariant is the cotangent complex, and its cohomology groups are the T^i functors of a k-algebra. This section constructs the version adequate for our purposes, the naive or Lichtenbaum-Schlessinger cotangent complex of a presentation $A = P/I$, and extracts from it the three functors T^0, T^1, T^2. The reader should read them as the algebraic, local counterparts of the geometric groups $H^0(T_X), H^1(T_X), H^2(T_X)$: T^0 collects infinitesimal automorphisms, T^1 classifies first-order deformations, and T^2 receives their obstructions. The section closes by reconciling the two languages: for a smooth affine $\operatorname{Spec} A$ the higher T^i vanish, and for any smooth X the local T^i sheafify to recover exactly the groups $H^i(X, T_X)$ of section 5.1.

The full theory of Illusie assigns to any morphism of schemes a cotangent *complex* $L_{A/k}$ in the derived category, whose homotopy groups compute deformations and obstructions in all degrees and whose construction requires simplicial resolutions. That machinery is out of scope for this book (BOOK.md §6): it belongs to derived algebraic geometry, and the naive two-term complex below suffices for every deformation problem treated here. The naive version agrees with the full cotangent complex in the degrees -1 and 0 that produce T^0, T^1, T^2, which is all the present chapter uses.

The construction begins with a presentation. Fix a finitely generated k-algebra A and write it as a quotient $A = P/I$ of a polynomial ring $P = k[x_1, \ldots, x_n]$ by an ideal I. Such a presentation always exists and is far from unique, but the functors extracted from it will turn out to depend only on A. Two modules over A organize the first-order geometry of the presentation. The first is the module of Kähler differentials $\Omega_{P/k}$, a free P-module on the symbols $dx_1, \ldots, dx_n$, restricted to A by base change to $\Omega_{P/k} \otimes_P A \cong A^n$. The second is the *conormal module* I/I^2, an A-module recording the equations cutting out $\operatorname{Spec} A$ inside affine space $\mathbb{A}^n = \operatorname{Spec} P$ to first order. They are linked by the universal derivation: the map sending a class $\bar{f} \in I/I^2$ to its total differential $df = \sum_i (\partial f/\partial x_i)\, dx_i$ lands in $\Omega_{P/k} \otimes_P A$ because differentiating a relation records how that relation constrains the tangent directions.

Definition 5.2.1: Naive Cotangent Complex

Let $A = P/I$ with $P = k[x_1, \ldots, x_n]$ a polynomial ring and I an ideal. The *naive* (Lichtenbaum-Schlessinger) *cotangent complex* of A over k is the two-term complex of A-modules

$$L_{A/k}\colon \quad I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P A$$

placed in cohomological degrees -1 and 0, where $d(\bar{f}) = df \otimes 1 = \sum_{i=1}^n \frac{\partial f}{\partial x_i}\, dx_i$.
For the second-order term, choose generators $f_1, \ldots, f_r$ of I, so that I is the image of a surjection $P^r \twoheadrightarrow I$ from a free module, and let $\mathrm{Rel} \subseteq P^r$ be the module of relations (first syzygies) among the f_j. The submodule $\mathrm{Kos} \subseteq \mathrm{Rel}$ of *Koszul* (trivial) relations is generated by the elements $f_j e_k - f_k e_j$. The naive cotangent complex is extended to a three-term complex by the module of *nontrivial second syzygies*

$$L^{-2}_{A/k} \;:=\; (\mathrm{Rel}\,/\,\mathrm{Kos}) \otimes_P A$$

in cohomological degree -2, mapping to I/I^2 through the presentation of I.

The complex is short by design. Its degree-0 term is the restriction to A of the cotangent space of the ambient affine space; its degree-(-1) term is the conormal module, the equations; and its degree-(-2) term records the relations *among* the equations modulo the tautological Koszul ones. Reading the complex geometrically, $\Omega_{P/k} \otimes_P A$ is the cotangent bundle of $\mathbb{A}^n$ restricted to $\operatorname{Spec} A$, and d tells one which of its directions are killed by the defining equations. When $\operatorname{Spec} A$ is smooth of dimension d, the map d is injective with locally free cokernel $\Omega_{A/k}$ of rank d, and the whole complex collapses to $\Omega_{A/k}$ in degree 0; every correction term measures a failure of smoothness. The dependence on the presentation is genuine at the level of the complex but not at the level of the invariants below: the T^i are independent of all choices, a fact whose proof, a homotopy-invariance argument comparing two presentations through a common refinement, is the homological content this section imports and is carried out in full in [10].

The three functors of interest are the cohomology of this complex after dualizing into a coefficient module. Fixing an A-module M, apply $\mathrm{Hom}_A(-, M)$ to $L_{A/k}$ and read off cohomology. Concretely, the dualized complex is

$$\mathrm{Hom}_A(\Omega_{P/k} \otimes_P A, M) \xrightarrow{d^\vee} \mathrm{Hom}_A(I/I^2, M) \longrightarrow \mathrm{Hom}_A(L^{-2}_{A/k}, M)$$

sitting in cohomological degrees $0, 1, 2$, and the T^i are its cohomology in those degrees.

Definition 5.2.2: The T^i Functors

Let $A = P/I$ be a finitely generated k-algebra and M an A-module. The *cotangent functors* $T^i(A/k, M)$ for $i = 0, 1, 2$ are the cohomology groups of the complex $\mathrm{Hom}_A(L_{A/k}, M)$:

1. $T^0(A/k, M) = \ker\big(d^\vee \colon \mathrm{Hom}_A(\Omega_{P/k} \otimes_P A, M) \to \mathrm{Hom}_A(I/I^2, M)\big)$, the derivations that annihilate the equations;
2. $T^1(A/k, M) = \ker\big(\mathrm{Hom}_A(I/I^2, M) \to \mathrm{Hom}_A(L^{-2}_{A/k}, M)\big) \big/ \mathrm{im}(d^\vee)$;
3. $T^2(A/k, M) = \mathrm{coker}\big(\mathrm{Hom}_A(I/I^2, M) \to \mathrm{Hom}_A(L^{-2}_{A/k}, M)\big)$.

When $M = A$ one writes $T^i(A/k) := T^i(A/k, A)$.

The degree-zero functor admits a description that owes nothing to the presentation. A k-derivation of A into M is a k-linear map $\delta \colon A \to M$ satisfying the Leibniz rule $\delta(ab) = a\,\delta(b) + b\,\delta(a)$, and the collection of these is the module $\mathrm{Der}_k(A, M)$. A derivation of A is the same datum as a derivation of P that kills the ideal I, which is exactly an element of the kernel defining T^0. This gives the clean identification

$$T^0(A/k, M) = \mathrm{Der}_k(A, M) = \mathrm{Hom}_A(\Omega_{A/k}, M)$$

so that $T^0(A/k, A) = \mathrm{Der}_k(A, A)$ is the module of vector fields on $\mathrm{Spec}\, A$, the algebraic incarnation of $H^0(X, T_X)$.

The other two functors carry the deformation-theoretic content. A first-order deformation of the affine scheme $\mathrm{Spec}\, A$ over the dual numbers $k[\varepsilon]$ is a flat $k[\varepsilon]$-algebra $\tilde{A}$ reducing to A modulo ε, and such deformations up to equivalence are classified by $T^1(A/k, A)$. Equivalently, $T^1(A/k, M)$ classifies the square-zero extensions of A by M as k-algebras: the short exact sequences $0 \to M \to B \to A \to 0$ with $M^2 = 0$ in B, modulo isomorphism fixing the ends. The functor $T^2(A/k, A)$ is the home of obstructions: given a first-order deformation and a small extension of the base to lift it along, the obstruction to the lift is a class in $T^2(A/k, A)$ that vanishes exactly when a lift exists. These readings are the content of the next theorem, and they mirror term by term the smooth-case bookkeeping of section 5.1.

Before stating that theorem, one interpretive point deserves emphasis. The passage from the geometric groups $H^i(X, T_X)$ to the algebraic groups $T^i(A/k, A)$ is not a change of subject but a change of vantage. For a smooth variety the tangent sheaf already captures the deformation theory, and its cohomology is the answer. For a singular scheme the tangent sheaf is the wrong object, and the cotangent complex is the correction: T^1 and T^2 are what $H^1(T_X)$ and $H^2(T_X)$ *ought* to have been, computed through a resolution that remembers the equations and their relations rather than through a sheaf that has forgotten them.

Theorem 5.2.3: Interpretation of the Cotangent Functors

Let A be a finitely generated k-algebra.

1. $T^1(A/k, A)$ is in natural bijection with the set of first-order deformations of $\operatorname{Spec} A$ over $k[\varepsilon]$, up to equivalence; the trivial deformation corresponds to 0.

2. Given a first-order deformation of $\operatorname{Spec} A$ and a small extension $0 \to (t) \to A' \to k[\varepsilon] \to 0$ in $\mathbf{Art}_k$, there is a canonical obstruction class in $T^2(A/k, A) \otimes_k (t)$ whose vanishing is necessary and sufficient for a lift to exist; when a lift exists, the lifts form a torsor under $T^1(A/k, A) \otimes_k (t)$.

3. If A is smooth over k, then $T^i(A/k, M) = 0$ for all $i \geq 1$ and all M.

4. If X is a smooth k-scheme covered by affine opens $\operatorname{Spec} A_\alpha$, the local functors $T^i(A_\alpha/k, A_\alpha)$ sheafify (by (3), only the degree-zero sheaf survives, and it is the tangent sheaf T_X), and the local-to-global spectral sequence assembles them, converging to the geometric groups $H^i(X, T_X)$ of section 5.1. For X affine, or in low degree, this recovers $T^i(A/k, A) = H^i(X, T_X)$ directly.

***Proof*:**
The homological input, that $T^i(A/k, M)$ is independent of the presentation and computes the derived functors of derivations in degrees ≤ 2, is proved in [10]; the deformation-theoretic statements are deduced from the complex directly.

Step 1: First-order deformations and T^1. A first-order deformation of $\operatorname{Spec} A$ is a flat $k[\varepsilon]$-algebra $\tilde{A}$ with $\tilde{A} \otimes_{k[\varepsilon]} k = A$. Present it by lifting the presentation of A: since P is smooth, the surjection $P \to A$ lifts to a surjection $\tilde{P} := P[\varepsilon] \to \tilde{A}$, and flatness over $k[\varepsilon]$ forces $\tilde{A} = \tilde{P}/\tilde{I}$ where $\tilde{I}$ is generated by lifts $\tilde{f}_j = f_j + \varepsilon g_j$ of the generators f_j of I, with $g_j \in P$. Reducing modulo ε recovers A, so the data of the deformation is the tuple $(g_j) \in P^r$ modulo two indeterminacies. First, the g_j are only well-defined modulo I once reduced to A, and modulo the ambiguity in the lifts $\tilde{f}_j$; both amount to changing (g_j) by an element of the image of $d^\vee$, that is, by the derivation coming from an infinitesimal change of coordinates in $\tilde{P}$. Second, flatness requires that every relation $\sum_j p_j f_j = 0$ among the f_j lift to a relation $\sum_j \tilde{p}_j \tilde{f}_j = 0$, and the obstruction to lifting the relation is precisely the requirement that the induced functional on I/I^2 annihilate the second syzygies, that is, that (g_j) define a class in $\ker(\operatorname{Hom}_A(I/I^2, A) \to \operatorname{Hom}_A(L^{-2}_{A/k}, A))$. The two conditions together identify the equivalence classes of first-order deformations with

$$\frac{\ker\big(\operatorname{Hom}_A(I/I^2, A) \to \operatorname{Hom}_A(L^{-2}_{A/k}, A)\big)}{\operatorname{im}(d^\vee)} = T^1(A/k, A)$$

and the trivial deformation $\tilde{A} = A[\varepsilon]$, given by $g_j = 0$, maps to the zero class. This proves (1).

Step 2: Lifting over a small extension and T^2. Let $0 \to (t) \to A' \to k[\varepsilon] \to 0$ be a small extension, so $\mathfrak{m}_{A'} \cdot t = 0$ and $(t) \cong k$ as an A'-module through the residue field. A deformation $\tilde{A}$ over $k[\varepsilon]$ is presented as above; to lift it over A' is to choose lifts $\tilde{f}'_j \in P \otimes_k A'$ of the $\tilde{f}_j$ so that the relations among the $\tilde{f}_j$ continue to hold modulo t. Choose arbitrary set-theoretic lifts $\tilde{f}'_j$. For each relation $\sum_j \tilde{p}_j \tilde{f}_j = 0$ holding over $k[\varepsilon]$, the same combination $\sum_j \tilde{p}'_j \tilde{f}'_j$ of the lifted generators need not vanish, but it lies in $(t) \cdot (P \otimes_k A') = t \cdot P$ because it vanishes modulo t;

writing it as $t \cdot c_{(\mathrm{rel})}$ produces, as the relation ranges over the second syzygies, a well-defined element
$$\mathrm{ob} \in \mathrm{Hom}_A\big(L^{-2}_{A/k}, A\big) \otimes_k (t)$$
because the Koszul relations impose no condition (they lift tautologically) and the ambiguity in the set-theoretic lifts changes ob by the image of $\mathrm{Hom}_A(I/I^2, A) \otimes_k (t)$. Its class
$$\mathrm{ob}(\tilde{A}, A') \ \in \ T^2(A/k, A) \otimes_k (t)$$
is therefore canonical, and by construction the lifted generators can be adjusted to satisfy all relations exactly, that is, a lift $\tilde{A}'$ exists, if and only if ob vanishes. When it vanishes, two lifts differ by an element of $\mathrm{Hom}_A(I/I^2, A) \otimes_k (t)$ modulo those coming from coordinate changes, which is $T^1(A/k, A) \otimes_k (t)$; hence the lifts form a torsor under this group. This proves (2).

Step 3: Vanishing for smooth A. Suppose A is smooth over k of dimension d. By the Jacobian criterion the map $d\colon I/I^2 \to \Omega_{P/k} \otimes_P A$ is a split injection of A-modules with locally free cokernel $\Omega_{A/k}$ of rank d; the conormal sequence
$$0 \to I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P A \to \Omega_{A/k} \to 0$$
is exact and split, and I is generated by a regular sequence, so its first syzygies are Koszul and the nontrivial second syzygies vanish: $L^{-2}_{A/k} = 0$. Applying $\mathrm{Hom}_A(-, M)$ to the split sequence keeps it exact, so $d^\vee$ is surjective, forcing $T^1(A/k, M) = 0$; and $L^{-2}_{A/k} = 0$ gives $T^2(A/k, M) = 0$. The same vanishing propagates to higher T^i in the full theory, but the two computed here are all the chapter requires. This proves (3).

Step 4: Sheafification and reconciliation. For X smooth, Step 3 gives $T^0(A_\alpha/k, A_\alpha) = \mathrm{Der}_k(A_\alpha, A_\alpha) = \Gamma(\operatorname{Spec} A_\alpha, T_X)$ and $T^{\geq 1}(A_\alpha/k, A_\alpha) = 0$ on each affine chart. The presheaf $U \mapsto T^0(\mathcal{O}_X(U)/k, \mathcal{O}_X(U))$ sheafifies to the tangent sheaf $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$, and the local-to-global passage from the affine-local groups to the global ones is governed by a spectral sequence whose E_2 page has the sheaf cohomology $H^p(X, \mathcal{T}^q_X)$ of the sheafified cotangent functors $\mathcal{T}^q_X$. Since $\mathcal{T}^0_X = T_X$ and $\mathcal{T}^q_X = 0$ for $q \geq 1$ by the local vanishing, the spectral sequence degenerates to the edge isomorphism
$$T^i(X/k, \mathcal{O}_X) \ \cong \ H^i(X, T_X)$$
identifying the global algebraic groups with the geometric groups of section 5.1. For X affine both sides are the derivations in degree 0 and vanish above, giving the claimed direct equality, which proves (4) and completes the proof.

The theorem closes the circle opened in the previous section. There the tangent and obstruction spaces of a smooth variety were the cohomology groups $H^1(T_X)$ and $H^2(T_X)$; here they are revealed as shadows of the cotangent functors T^1 and T^2, which continue to compute the same information for singular schemes on which the tangent sheaf fails. The notion of unobstructedness transfers verbatim: a deformation functor is unobstructed in the sense of definition 5.1.3 exactly when the relevant T^2 vanishes, so that every obstruction class in part (2) of theorem 5.2.3 is automatically zero and every lift exists. The class of schemes for which T^2 vanishes therefore deserves a name, and the next example identifies a large supply of them.

The most important such class is the local complete intersections, and the cleanest instance is a hypersurface. A hypersurface has a single defining equation, so its ideal is generated by a regular sequence of length one, its syzygies are all Koszul, and T^2 vanishes for the same reason it did in the smooth case: there are no nontrivial relations among the equations. What survives in T^1 is a finite-dimensional algebra built from the equation and its partial derivatives, the Jacobian or Tjurina algebra, which measures the singularity directly.

Example 5.2.4: The T^i of a Hypersurface

Let $A = P/(f)$ with $P = k[x_1, \dots, x_n]$ and f a nonzero nonunit, so $\operatorname{Spec} A$ is a hypersurface, and suppose $\operatorname{Spec} A$ has an isolated singularity (over a field of characteristic 0). The ideal $I = (f)$ is generated by the single regular element f, so the naive cotangent complex of definition 5.2.1 is short:

- $I/I^2 = (f)/(f^2) \cong A$, free of rank 1, generated by the class $\bar{f}$;
- $\Omega_{P/k} \otimes_P A \cong A^n$, free on $dx_1, \dots, dx_n$;
- the differential $d(\bar{f}) = \sum_{i=1}^n \partial_i f \, dx_i$, where $\partial_i f := \partial f / \partial x_i$;
- $L^{-2}_{A/k} = 0$, because I is generated by a regular sequence, so every relation among the (single) generator is Koszul.

Vanishing of T^2. Since $L^{-2}_{A/k} = 0$, the target $\operatorname{Hom}_A(L^{-2}_{A/k}, A) = 0$, and definition 5.2.2 gives

$$T^2(A/k, A) = \operatorname{coker}\big(\operatorname{Hom}_A(I/I^2, A) \to 0\big) = 0$$

The hypersurface is unobstructed. The same computation applies to any local complete intersection $A = P/(f_1, \dots, f_c)$ with $f_1, \dots, f_c$ a regular sequence: the first syzygies are Koszul, $L^{-2}_{A/k} = 0$, and $T^2(A/k, A) = 0$. Complete intersections are unobstructed, which is why they deform freely and why the boundary singularities appearing in moduli of curves, all of them nodes and hence hypersurface singularities, carry no obstruction to smoothing.

Computation of T^1. By definition 5.2.2, with $L^{-2}_{A/k} = 0$,

$$T^1(A/k, A) = \operatorname{coker}\big(d^\vee \colon \operatorname{Hom}_A(A^n, A) \to \operatorname{Hom}_A(A, A)\big)$$

Under the identifications $\operatorname{Hom}_A(A^n, A) \cong A^n$ and $\operatorname{Hom}_A(A, A) \cong A$, the dual map $d^\vee$ sends a functional $\varphi \colon A^n \to A$, given by $\varphi(dx_i) = a_i$, to the composite $\varphi \circ d$, whose value on the generator $\bar{f}$ is

$$(\varphi \circ d)(\bar{f}) = \varphi\Big(\sum_i \partial_i f \, dx_i\Big) = \sum_{i=1}^n a_i \, \partial_i f$$

so $d^\vee \colon A^n \to A$ is the map $(a_1, \dots, a_n) \mapsto \sum_i a_i \, \partial_i f$, whose image is the Jacobian ideal $(\partial_1 f, \dots, \partial_n f) \cdot A$. Therefore

$$T^1(A/k, A) = A \big/ (\partial_1 f, \dots, \partial_n f) = \frac{P}{(f, \partial_1 f, \dots, \partial_n f)}$$

the *Jacobian* or *Tjurina algebra* of f. Because the singularity is isolated, the vanishing locus of f together with all its partials is the singular point alone, so this algebra is a finite-dimensional k-vector space; its dimension, the *Tjurina number* $\tau(f) = \dim_k T^1(A/k, A)$, counts the parameters in the versal deformation of the singularity.

The node. Take $n = 2$ and $f = xy$, the node $A = k[x, y]/(xy)$. Then $\partial_x f = y$, $\partial_y f = x$, and

$$T^1(A/k, A) = \frac{k[x, y]}{(xy, y, x)} = \frac{k[x, y]}{(x, y)} \cong k$$

so the node has a one-dimensional space of first-order deformations and $\tau = 1$. The single parameter is realized by the family $xy = t$, which smooths the node for $t \neq 0$; this is the local model that reappears in section 5.4 and, later, in the deformation theory of stable curves at the boundary of the moduli space.

The hypersurface computation isolates the mechanism behind unobstructedness. Obstructions live in T^2, and T^2 is built from the nontrivial relations among the defining equations; when the equations form a regular sequence there are no such relations, so T^2 vanishes and the scheme deforms without obstruction. The surviving invariant T^1, the Tjurina algebra, is a direct algebraic measurement of the singularity, and its dimension counts deformation parameters. This is the pattern the applications section exploits: to prove a moduli space smooth at a point, show that the T^2 of the object it parametrizes vanishes, and to compute its dimension there, compute the corresponding T^1.

Exercise 5.2.1

1. Let $A = k[x, y]/(y^2 - x^3)$ be the cuspidal cubic curve over a field of characteristic 0. Compute $T^1(A/k, A)$ and $T^2(A/k, A)$, and give the Tjurina number τ. Compare with the node of example 5.2.4: which singularity has more deformation parameters?

2. Show that a smooth affine k-scheme $\operatorname{Spec} A$ has $T^1(A/k, A) = 0$ directly from definition 5.2.2, and interpret the vanishing in terms of first-order deformations. Why does theorem 5.2.3(1) then say $\operatorname{Spec} A$ has no nontrivial first-order deformations?

3. Let $A = k[x, y, z]/(xy, xz, yz)$, the union of the three coordinate axes in $\mathbb{A}^3$. This is not a complete intersection: three equations cut out a one-dimensional scheme in three-space. Exhibit a nontrivial relation among the three generators that is not Koszul, and explain why this forces $L^{-2}_{A/k} \neq 0$, so that $T^2(A/k, A)$ can be nonzero. (You need not compute T^2; the point is to see the obstruction module come alive when the codimension exceeds the number of equations only by failing to be a complete intersection.)

4. Verify that $T^0(A/k, A) = \operatorname{Der}_k(A, A)$ is compatible with the two usages of section 5.1: for $A = k[x]$ (so $\operatorname{Spec} A = \mathbb{A}^1$ is smooth), compute $\operatorname{Der}_k(A, A)$ explicitly and confirm it agrees with the global vector fields $H^0(\mathbb{A}^1, T_{\mathbb{A}^1})$.

5. Let $A = P/(f)$ be a hypersurface as in example 5.2.4, and suppose $f = f_2 + f_3 + \cdots$ is the sum of its homogeneous pieces with $f_2 \neq 0$ a nondegenerate quadratic form (an ordinary double point, A_1 singularity, in n variables over characteristic 0). Compute $\dim_k T^1(A/k, A)$. *(Hint: after a linear change of coordinates the leading term is $x_1^2 + \cdots + x_n^2$; work out the Jacobian ideal of the pure quadric first and argue that the higher-order terms do not change the dimension.)*

5.3 Schlessinger's Criteria

The previous two sections assembled the local anatomy of a deformation problem: a tangent space of first-order deformations and an obstruction space receiving the failure to lift across a small extension. What is still missing is the global verdict. Given a deformation functor, does there exist a single formal object, living over a complete local ring, from which every deformation is induced? For a smooth projective variety with a fine moduli space, the answer is a completed local ring of that space, and the deformation functor is the functor of points of that ring restricted to Artinian algebras. But most deformation functors are not representable at all, precisely because the objects being deformed carry automorphisms, and automorphisms destroy the uniqueness that a universal family demands. This is the same obstruction that produced the j-line rather than a fine moduli space of elliptic curves in theorem 1.3.6.

Schlessinger's theorem cuts exactly along this fault line. It isolates a short list of conditions on a functor of Artin rings, phrased entirely in terms of how the functor behaves on fiber products, and it extracts two verdicts. The weaker three conditions guarantee a *hull*: a formal object that is versal, so every deformation is induced from it, and minimal, so its tangent space is as small as possible. The fourth condition upgrades versality to genuine pro-representability, producing a universal formal deformation that is unique with its inducing map. Everything a deformation theorist needs to know about whether a moduli problem has a well-behaved local model is encoded in these four conditions, and the proof of hull existence is a concrete order-by-order construction that this section carries out in full.

Throughout, k is an algebraically closed field, $\mathbf{Art}_k$ is the category of local Artinian k-algebras with residue field k, and $D\colon \mathbf{Art}_k \to \mathbf{Set}$ is a functor with $D(k)$ a single point. Recall from definition 4.1.5 that a *small extension* is a surjection $\pi\colon A' \to A$ in $\mathbf{Art}_k$ whose kernel (t) is a one-dimensional k-vector space annihilated by the maximal ideal $\mathfrak{m}_{A'}$, so that $\mathfrak{m}_{A'} \cdot t = 0$ and $t^2 = 0$.

The right notion of a solution to a deformation problem is not always a representing ring. When automorphisms are present, the best one can hope for is a formal object that induces all others but is not required to do so uniquely. To state this precisely, one works with $\widehat{\mathbf{Art}}_k$, the category of complete local Noetherian k-algebras with residue field k; for the theory of complete local rings, their construction as limits of Artinian quotients, and the Cohen structure theorem, see [6]. A complete local k-algebra R with maximal ideal $\mathfrak{m}_R$ defines a functor on $\mathbf{Art}_k$ by

$$h_R(A) = \mathrm{Hom}_{k\text{-alg}}(R, A) = \varinjlim_n \mathrm{Hom}_{k\text{-alg}}(R/\mathfrak{m}_R^n, A)$$

where the colimit is over n and identifies a homomorphism out of the complete ring R with a compatible system of homomorphisms out of its Artinian quotients, each of which factors through some $R/\mathfrak{m}_R^n$ because A is Artinian. The functor h_R is the object of study: it is the algebraic shadow cast by a formal deformation.

Definition 5.3.1: Hull of a Functor of Artin Rings

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a functor with $D(k) = \{*\}$. A *formal element* of D is a complete local k-algebra $R \in \widehat{\mathbf{Art}}_k$ together with a compatible system $\xi = (\xi_n)$ of elements $\xi_n \in D(R/\mathfrak{m}_R^n)$, equivalently a morphism of functors $h_R \to D$. Such a ξ is *versal* if for every small extension $\pi\colon A' \to A$ in $\mathbf{Art}_k$ the map

$$h_R(A') \longrightarrow h_R(A) \times_{D(A)} D(A')$$

is surjective, that is, every deformation over A' that agrees over A with a deformation induced by ξ is itself induced by ξ. A versal element is a *hull* (or a *miniversal formal deformation*) if in addition the induced map on tangent spaces

$$d\xi\colon \mathrm{Hom}_{k\text{-alg}}(R, k[\varepsilon]) \longrightarrow D(k[\varepsilon])$$

is a bijection, so the formal object is as small as versality allows.

The versality condition says that ξ can be extended along any small extension whenever the obstruction to extending the underlying deformation vanishes: no deformation over A' compatible with ξ escapes being induced by ξ. The minimality condition pins down the tangent direction: a hull has exactly the tangent space of D and no spurious first-order directions. Versality alone does not force uniqueness of the map $h_R \to A'$ lifting a given map $h_R \to A$; it guarantees existence of such a lift but permits several. This is exactly what one must give up when the objects have automorphisms, and it is the precise sense in which a hull is weaker than a representing ring.

Definition 5.3.2: Pro-representable Functor

A functor $D\colon \mathbf{Art}_k \to \mathbf{Set}$ with $D(k) = \{*\}$ is *pro-representable* if there exists $R \in \widehat{\mathbf{Art}}_k$ and an isomorphism of functors

$$h_R \xrightarrow{\sim} D$$

on $\mathbf{Art}_k$. Equivalently, D admits a formal element (R, ξ) that is *universal*: for every small extension $\pi\colon A' \to A$ the map $h_R(A') \to h_R(A) \times_{D(A)} D(A')$ is a bijection, not only a surjection.

Pro-representability is the strongest possible outcome: the deformation functor is literally the functor of points of a formal parameter space R, and every deformation over an Artinian base is induced by a unique map out of R. The prefix "pro" records that R is a limit of the Artinian quotients through which the deformations factor, so D is representable in the pro-category of $\mathbf{Art}_k$ even when it is not representable by a single object of $\mathbf{Art}_k$. A hull is a pro-representable functor with the uniqueness relaxed to existence, and the entire content of Schlessinger's theorem is to say which functors reach each of these two levels.

Before quantifying the conditions, it is worth seeing the tangent space enter as a vector space rather than a bare set. The following is the mechanism by which the finiteness hypothesis (H3) below feeds into the construction.

Proposition 5.3.3: The Tangent Space as a Vector Space

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ satisfy $D(k) = \{*\}$ and the condition that the natural map

$$D(A' \times_A A'') \longrightarrow D(A') \times_{D(A)} D(A'')$$

is a bijection whenever $A = k$ and $A' = A'' = k[\varepsilon]$. Then the set $t_D = D(k[\varepsilon])$ carries a canonical structure of a k-vector space, natural in D, for which the bijection above is an isomorphism of the underlying additive groups $D(k[\varepsilon] \times_k k[\varepsilon]) \cong t_D \oplus t_D$.

***Proof*:**
The ring $k[\varepsilon] \times_k k[\varepsilon]$ is isomorphic to $k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_1\varepsilon_2, \varepsilon_2^2)$, and under the stated bijection $D(k[\varepsilon] \times_k k[\varepsilon]) \cong t_D \times t_D$. *Addition.* The addition map on $k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$ sending both ε_1 and ε_2 to ε is a k-algebra homomorphism $\sigma\colon k[\varepsilon] \times_k k[\varepsilon] \to k[\varepsilon]$, well-defined because it carries the relations $(\varepsilon_1, \varepsilon_2)^2 = 0$ to $\varepsilon^2 = 0$, which holds in $k[\varepsilon]$. Applying D and precomposing with the bijection defines a map $t_D \times t_D \to t_D$, which is the vector addition. *Scalar multiplication.* For $\lambda \in k$, the map $\varepsilon \mapsto \lambda\varepsilon$ is a k-algebra endomorphism of $k[\varepsilon]$, and D applied to it gives scalar multiplication by λ on t_D. The zero element is the image of $D(k) \to D(k[\varepsilon])$ induced by $k \to k[\varepsilon]$, which is a single point since $D(k) = \{*\}$. The vector-space axioms follow from functoriality applied to the corresponding identities among k-algebra homomorphisms of $k[\varepsilon]$ and $k[\varepsilon] \times_k k[\varepsilon]$; associativity and commutativity of addition, for instance, come from the associativity and symmetry of σ up to the automorphisms permuting the two copies of ε. Naturality in D is immediate since every ingredient is the image under D of a fixed homomorphism of test rings, completing the proof.

The proposition shows that once a deformation functor is even slightly compatible with fiber products, its set of first-order deformations is automatically a vector space, recovering the tangent space of definition 4.1.5 from an abstract categorical hypothesis rather than a hands-on cocycle description. This is the first payoff of phrasing everything through the fiber-product map, and it is the template for the conditions to come.

With the two target notions in place, everything reduces to a small list of axioms on a single natural map. Schlessinger's conditions all concern the map

$$\eta\colon D(A' \times_A A'') \longrightarrow D(A') \times_{D(A)} D(A'') \tag{5.1}$$

induced by the two projections $A' \times_A A'' \to A'$ and $A' \times_A A'' \to A''$, defined whenever $A' \to A$ and $A'' \to A$ are morphisms in $\mathbf{Art}_k$. The map η compares gluing on the ring side with gluing on the deformation side. It is always defined; the conditions ask how close it is to a bijection under progressively weaker hypotheses on the two morphisms. A functor of this shape, with $D(k) = \{*\}$, is often called a *deformation functor* or a *functor of Artin rings*, and the conditions below are its structural axioms.

Definition 5.3.4: Schlessinger's Conditions (H1)-(H4)

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a functor with $D(k) = \{*\}$. Consider morphisms $A' \to A$ and $A'' \to A$ in $\mathbf{Art}_k$ and the associated map η of (5.1). The conditions are:

- *(H1)* η is surjective whenever $A'' \to A$ is a small extension.
- *(H2)* η is bijective when $A = k$ and $A'' = k[\varepsilon]$, so that $D(A' \times_k k[\varepsilon])$ maps isomorphically onto $D(A') \times D(k[\varepsilon])$.
- *(H3)* the tangent space $t_D = D(k[\varepsilon])$ is a finite-dimensional k-vector space.
- *(H4)* η is bijective whenever $A' = A''$ and $A' \to A$ is the *same* small extension, that is, whenever $A' \to A$ is a small extension and $A'' = A'$ with the identical map.

The four conditions are graded in strength and in what they control. Condition (H1) is a lifting axiom: it says two deformations that agree over A can be combined into a deformation over the fiber product $A' \times_A A''$ whenever one of the two sides is a small extension, so gluing along small extensions never fails for lack of an object. Condition (H2) makes η a bijection in the single case that governs the tangent space, and combined with (H1) it is exactly the hypothesis of proposition 5.3.3, so under (H1) and (H2) the tangent space t_D is a k-vector space. Condition (H3) makes that vector space finite-dimensional, which is what allows the construction to terminate in Noetherian rings at each stage. Condition (H4) is the uniqueness upgrade: it demands that η be an actual bijection, not just a surjection, in the self-fiber-product case that detects whether distinct liftings can be told apart, and it is precisely the condition that distinguishes a universal formal deformation from a versal one.

Before the theorem it helps to see how (H4) fails in the presence of automorphisms, since that failure is the entire reason pro-representability is the exception rather than the rule. Consider deforming an object E with a nontrivial automorphism group. Two liftings of a given deformation over A to the fiber product $A' \times_A A'$ can differ by the action of an automorphism that is the identity over A but nontrivial over A'; such an automorphism produces two different elements of $D(A' \times_A A')$ mapping to the same element of $D(A') \times_{D(A)} D(A')$, so η is not injective and (H4) fails. This is the same phenomenon that obstructs fine representability in theorem 1.3.6: the automorphisms of an elliptic curve are what force the j-line to be a coarse rather than fine space.

Everything now assembles into the central result. The theorem is stated for a functor D on $\mathbf{Art}_k$ with $D(k) = \{*\}$, and its two clauses give the two levels of solution introduced above.

Theorem 5.3.5: Schlessinger's Theorem

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a functor with $D(k) = \{*\}$.

1. D admits a hull if and only if it satisfies conditions (H1), (H2), and (H3) of definition 5.3.4.
2. D is pro-representable if and only if it satisfies (H1), (H2), (H3), and (H4).

Moreover, when a hull exists it is unique up to (non-canonical) isomorphism, and its tangent space is canonically $t_D = D(k[\varepsilon])$.

The theorem is due to Schlessinger [27]. On the necessity side, if D has a hull (R, ξ), then h_R satisfies all four conditions (part (a) of the exercises), and (H1) for D follows because the versal map $h_R \to D$ is smooth: given elements of $D(A')$ and $D(A'')$ agreeing over $D(A)$, versality lifts each into h_R, and (H1) for h_R then produces the required element of $D(A' \times_A A'')$. Finiteness of $\mathrm{Hom}(R, k[\varepsilon]) = (\mathfrak{m}_R/\mathfrak{m}_R^2)^\vee$ forces (H3). The full necessity of (H1) and (H2), whose (H2) half asserts *bijectivity* of η in the tangent case and so requires the minimality of the tangent map rather than versality alone, uses a separate minimality argument and is deferred to [27] and the exercises; the necessity is not needed for the hull-existence construction below. The substance is the converse for part (1): the construction of a hull from (H1)-(H3), carried out below in full. The upgrade to part (2) under (H4) is then a bookkeeping refinement of the same construction, tracking uniqueness at each stage.

***Proof*:**
Necessity of the conditions was discussed above and is deferred in part to [27]; a versal formal element makes the maps in definition 5.3.4 surjective and (H3) is the finiteness of the cotangent space of a Noetherian complete local ring, while the bijectivity in (H2) and (H4) uses the minimality of a hull. The content is the construction of a hull assuming (H1), (H2), and (H3); this occupies the remainder of the proof. Uniqueness of the hull and the upgrade to pro-representability under (H4) are addressed at the end.

Write $t = t_D = D(k[\varepsilon])$, a finite-dimensional k-vector space by (H1)-(H3) and proposition 5.3.3, and set $d = \dim_k t$. Let $S = k[[x_1, \ldots, x_d]]$ be the power-series ring in d variables with maximal ideal $\mathfrak{n}$, so that $\mathrm{Hom}_{k\text{-alg}}(S, k[\varepsilon]) = (\mathfrak{n}/\mathfrak{n}^2)^\vee$ is d-dimensional and canonically dual to t. The strategy is to build the hull R as a quotient S/J for an ideal J obtained as the limit of a decreasing chain of ideals, constructing at each stage the smallest quotient of S through which a versal family extends.

Step 1: The tower of quotients. Fix a k-basis of t and let $x_1, \ldots, x_d$ correspond to the dual basis, so that the tangent element of any quotient S/J with $J \subseteq \mathfrak{n}^2$ is identified with t itself. The hull will be produced as a limit $R = \varprojlim_q R_q$ of a tower

$$S/\mathfrak{n}^2 = R_1 \twoheadleftarrow R_2 \twoheadleftarrow R_3 \twoheadleftarrow \cdots$$

in which each R_q is a quotient S/J_q with $\mathfrak{n}^{q+1} \subseteq J_q \subseteq \mathfrak{n}^2$, so that R_q is Artinian, together with a compatible system of elements $\xi_q \in D(R_q)$. At each stage the ideal J_q is chosen as small as possible subject to ξ_{q-1} extending to R_q; smallness of J_q is what forces the resulting formal element to have the correct tangent space and to be versal.

Step 2: The base of the tower. Set $R_1 = S/\mathfrak{n}^2$. The tangent space of R_1 is $\mathrm{Hom}(R_1, k[\varepsilon]) = (\mathfrak{n}/\mathfrak{n}^2)^\vee \cong t$. Since $D(k[\varepsilon]) = t$ and R_1 has been built with tangent space canonically t, the identity of t corresponds under the duality to a distinguished element $\xi_1 \in D(R_1)$: concretely, $R_1 \cong k[\varepsilon] \times_k \cdots \times_k k[\varepsilon]$ (d factors), and by iterating (H2) the map $D(R_1) \to D(k[\varepsilon])^d = t^d$ is a bijection, so ξ_1 is the unique element mapping to the tuple of basis vectors. This ξ_1 induces the identity on tangent spaces by construction.

Step 3: The inductive extension. Suppose $R_q = S/J_q$ and $\xi_q \in D(R_q)$ have been constructed with $\mathfrak{n}^{q+1} \subseteq J_q \subseteq \mathfrak{n}^2$ and ξ_q inducing the fixed identification on tangent spaces. To pass to stage $q+1$, consider the collection of ideals

$$\mathcal{J} = \{J \subseteq S \mid \mathfrak{n}J_q \subseteq J \subseteq J_q \text{ and } \xi_q \text{ lifts to } D(S/J)\}$$

where ξ_q *lifts to* $D(S/J)$ means that $\xi_q \in D(S/J_q)$ lies in the image of $D(S/J) \to D(S/J_q)$ induced by the quotient $S/J \twoheadrightarrow S/J_q$. Because $\mathfrak{n}J_q \subseteq J \subseteq J_q$ and $J_q \supseteq \mathfrak{n}^{q+1}$, every such quotient

$S/J \to S/J_q$ is a composite of small extensions, so lifting is controlled by (H1). Here and below (H1), stated for a single small extension, is applied to a composite of small extensions by factoring the surjection into a chain of small extensions and lifting one stage at a time; each stage's kernel is a one-dimensional subspace of the total kernel, which is finite-dimensional and annihilated by $\mathfrak{n}$. The claim is that $\mathcal{J}$ has a smallest element J_{q+1}, and this smallest element, with any chosen lift ξ_{q+1}, gives the next stage.

Step 4: $\mathcal{J}$ is closed under intersection. Let $J', J'' \in \mathcal{J}$. To show $J' \cap J'' \in \mathcal{J}$, first reduce to $\mathfrak{n}J_q \subseteq J' \cap J''$, which is immediate, and $J' \cap J'' \subseteq J_q$, also immediate. It remains to show ξ_q lifts to $D(S/(J' \cap J''))$. There is a fiber-product presentation

$$S/(J' \cap J'') \cong S/J' \times_{S/(J'+J'')} S/J''$$

since the kernel of $S \to S/J' \times S/J''$ is exactly $J' \cap J''$ and the image is the fiber product over $S/(J' + J'')$. The projection $S/J'' \to S/(J' + J'')$ has kernel $(J' + J'')/J''$, which is annihilated by $\mathfrak{n}$ because $\mathfrak{n}(J' + J'') \subseteq \mathfrak{n}J_q \subseteq J' \cap J'' \subseteq J''$; a finitely generated module annihilated by $\mathfrak{n}$ is a finite-dimensional k-vector space, so this projection is a composite of small extensions, filtering the kernel by one-dimensional subspaces. By (H1) the map

$$D\big(S/(J' \cap J'')\big) \longrightarrow D(S/J') \times_{D(S/(J'+J''))} D(S/J'')$$

is surjective. Now ξ_q lifts to elements $\eta' \in D(S/J')$ and $\eta'' \in D(S/J'')$ by hypothesis, and both η' and η'' restrict to the same element of $D(S/(J' + J''))$, namely the further pushforward of ξ_q, because $S/(J' + J'')$ is a common quotient of S/J_q through which both restrictions factor. Thus (η', η'') is an element of the target fiber product, and by surjectivity it is the image of some $\zeta \in D(S/(J' \cap J''))$. Pushing ζ forward to $D(S/J_q)$ recovers ξ_q, so ξ_q lifts to $D(S/(J' \cap J''))$ and $J' \cap J'' \in \mathcal{J}$.

Step 5: Existence of a smallest element. Every $J \in \mathcal{J}$ satisfies $\mathfrak{n}J_q \subseteq J \subseteq J_q$, and $J_q/\mathfrak{n}J_q$ is a finite-dimensional k-vector space because S is Noetherian and J_q is finitely generated. The ideals in $\mathcal{J}$ correspond to k-subspaces of $J_q/\mathfrak{n}J_q$, and by Step 4 this collection of subspaces is closed under finite intersection. A collection of subspaces of a finite-dimensional vector space that is closed under intersection has a smallest member, namely the intersection of all of them, which is a finite intersection since the dimension is finite. Let J_{q+1} be this smallest ideal and set $R_{q+1} = S/J_{q+1}$. By definition ξ_q lifts to some $\xi_{q+1} \in D(R_{q+1})$; fix one such lift. This completes the inductive step, and the minimality of J_{q+1} is the "smallest quotient through which the versal family extends" that drives the construction.

Step 6: Passage to the limit. The ideals J_q form a decreasing chain with $\bigcap_q J_q =: J$, and setting $R = S/J = \varprojlim_q R_q$ produces a complete local Noetherian k-algebra; completeness holds because $\mathfrak{n}^{q+1} \subseteq J_q$ forces the $\mathfrak{m}_R$-adic topology to agree with the tower. The compatible system (ξ_q) defines a formal element $\xi = (\xi_q) \in \varprojlim_q D(R_q)$, that is, a morphism $h_R \to D$. Its tangent map is the identification of t with $D(k[\varepsilon])$ from Step 2, hence a bijection, so ξ satisfies the minimality requirement of definition 5.3.1 at once.

Step 7: Versality. It remains to show ξ is versal, meaning that for every small extension $\pi\colon A' \to A$ the map $h_R(A') \to h_R(A) \times_{D(A)} D(A')$ is surjective. Fix such a π and an element (u, ζ) of the target, where $u\colon R \to A$ is a k-algebra map and $\zeta \in D(A')$ restricts along π to $u_*\xi \in D(A)$. Since A is Artinian, u factors through some $R_q = S/J_q$, giving $\bar{u}\colon R_q \to A$ with $\bar{u}_*\xi_q = u_*\xi$. Choose a lift of $\bar{u}$ to a k-algebra homomorphism $v\colon S \to A'$: this is possible because S is a power-series ring, hence (formally) smooth, so any map to the quotient A lifts to the small extension A'. The composite $S \xrightarrow{v} A'$ need not kill J_{q+1}, but consider the ideal $J' = J_q \cap \ker\bigl(v \bmod (t)\bigr)$ inside S; more directly, form the fiber product

$$B = R_q \times_A A'$$

which sits in a small extension $B \to R_q$ with the same kernel (t) as π, since π is small and the fiber product with the surjection $\bar{u}$ preserves the kernel. By (H1) applied to the small extension $A' \to A$ and the map $R_q \to A$, the element $\xi_q \in D(R_q)$ and $\zeta \in D(A')$, agreeing over A, lift to an element $\beta \in D(B)$. Now $B = R_q \times_A A'$ is a quotient of S by an ideal J_B with $\mathfrak{n}J_q \subseteq J_B \subseteq J_q$: the surjection $S \to R_q \to A$ lifts to $S \to A'$ as above, giving $S \to B$, and its kernel J_B contains $\mathfrak{n}J_q$ because $B \to R_q$ is a small extension with $\mathfrak{m}_B \cdot (t) = 0$. The element β shows ξ_q lifts to $D(S/J_B)$, so $J_B \in \mathcal{J}$ and therefore $J_{q+1} \subseteq J_B$ by minimality. Hence $S \to B$ factors through $R_{q+1} = S/J_{q+1}$, yielding a map $w\colon R_{q+1} \to B \to A'$ whose composite $R_{q+1} \to A' \to A$ is $\bar{u}$ precomposed with $R_{q+1} \to R_q$, and pulling ξ_{q+1} back along w recovers (u, ζ) up to the lift chosen. Composing w with the projection $h_R \to h_{R_{q+1}}$ gives the required element of $h_R(A')$ mapping to (u, ζ). This establishes surjectivity, so ξ is versal, and (R, ξ) is a hull.

Step 8: Uniqueness of the hull. If (R, ξ) and (R', ξ') are two hulls, versality of each produces maps $h_{R'} \to h_R$ and $h_R \to h_{R'}$ compatible with the formal elements. The composite $h_R \to h_R$ induces the identity on tangent spaces (both hulls have tangent space t), and a map of complete local rings inducing an isomorphism on cotangent spaces $\mathfrak{m}/\mathfrak{m}^2$ is surjective by the complete Nakayama lemma; a surjective endomorphism of a Noetherian complete local ring is an isomorphism. Hence both composites are isomorphisms, and $R \cong R'$, though the isomorphism depends on the choices of lift and is not canonical. This proves clause (1).

Step 9: Pro-representability under (H4). Assume in addition (H4). The only place the construction used surjectivity rather than bijectivity of η was in the lifting steps; (H4) makes η bijective in the self-fiber-product case $A' = A''$, which is exactly the case that measures whether two lifts of a map $h_R \to A$ to $h_R \to A'$ can differ. Concretely, given $u\colon R \to A$ and a small extension $A' \to A$, two lifts $R \to A'$ of u differ by a k-derivation $R \to (t)$ that kills $\ker u$ and $\mathfrak{m}_R^2$, since (t) is annihilated by $\mathfrak{m}_{A'}$; hence the set of lifts of u, when nonempty, is a torsor under $\mathrm{Hom}(\mathfrak{m}_R/(\mathfrak{m}_R^2 + \ker u), (t))$, and the set of lifts of $u_*\xi$ in $D(A')$ is correspondingly a torsor under $t_D \otimes_k (t)$, which is $\cong t_D$ since (t) is one-dimensional, via the tangent-space action; (H4) forces the natural map between these two torsors, $h_R(A') \to h_R(A) \times_{D(A)} D(A')$, to be injective. Injectivity together with the surjectivity from Step 7 makes η for $h_R \to D$ a bijection on every small extension, so $h_R \to D$ is an isomorphism on all of $\mathbf{Art}_k$ by induction on the length of the Artinian ring, factoring any A as a tower of small extensions of k. Thus $D = h_R$ is pro-representable. Conversely, pro-representability makes every η a bijection, in particular (H4), completing the proof of clause (2) and the theorem, as we sought to show.

The proof follows a single template: build the parameter ring one infinitesimal order at a time, and at each order divide out by the smallest ideal that still allows the versal family to extend. Smallness is what makes the tangent space come out correctly, since a larger ideal would kill tangent directions that D possesses, and it is also what makes the hull unique. The single categorical input at every stage is condition (H1), which guarantees that deformations agreeing over a common base glue over the fiber product; (H2) and (H3) enter only to make the tangent space a finite-dimensional vector space, fixing the number of variables in the power-series ring. The passage from a hull to full pro-representability is entirely about injectivity, and injectivity is exactly what automorphisms destroy, which is why (H4) is the automorphism-free condition.

One structural remark clarifies the role of the completion. The hull R is a formal object: it lives in $\widehat{\mathbf{Art}}_k$, not in $\mathbf{Art}_k$, and represents D only in the pro-sense of a limit over the finite-length quotients. Whether this formal deformation *algebraizes* to an actual family over a ring of finite type, or over a Henselian ring, is a separate question answered by Artin's approximation and algebraization theorems; those provide the bridge from the formal local structure produced here to genuine algebraic moduli, and they are the subject of the algebraic-stacks development later in the book. For now the hull is the sharpest local invariant available, and Schlessinger's criteria are the precise test for its existence.

The conditions (H1)-(H3) are easy to verify for the deformation functors that arise in geometry, so the existence of a hull is the generic situation. Pro-representability is the exceptional case, gated entirely by (H4), and (H4) holds exactly when the objects being deformed have no infinitesimal automorphisms. The following example makes both statements concrete, tying the abstract criteria back to the cohomology groups of definition 4.1.5.

Example 5.3.6: Hulls and Universal Deformations in Geometry

Let X be a smooth projective variety over k and let $D = \mathrm{Def}_X$ be its deformation functor, sending $A \in \mathbf{Art}_k$ to the set of isomorphism classes of flat families $\mathcal{X} \to \operatorname{Spec} A$ with special fiber X.

1. Def_X *has a hull.* Conditions (H1) and (H2) follow from the gluing property of flat families: given flat deformations $\mathcal{X}' \to \operatorname{Spec} A'$ and $\mathcal{X}'' \to \operatorname{Spec} A''$ agreeing over $\operatorname{Spec} A$, with $A'' \to A$ a small extension, the flat family over $A' \times_A A''$ is obtained by gluing $\mathcal{X}'$ and $\mathcal{X}''$ along their common restriction, and flatness over the fiber product is checked on the local rings, where it reduces to the fact that $A' \times_A A''$ is the fiber product and a module flat over both factors and agreeing over A is flat over the fiber product (the standard flatness-descent lemma for a fiber square of rings, see [6]). This gives (H1); taking $A = k$, $A'' = k[\varepsilon]$ makes the gluing an equivalence, giving (H2). Condition (H3) is the finiteness of the tangent space, which for the smooth X under consideration is
$$t_{\mathrm{Def}_X} = D(k[\varepsilon]) = \mathrm{Ext}^1_X(\Omega_{X/k}, \mathcal{O}_X) = H^1(X, T_X)$$
the group of theorem 4.3.2, the second equality holding because $\Omega_{X/k}$ is locally free on the smooth X, so $\mathrm{Ext}^1_X(\Omega_{X/k}, \mathcal{O}_X) = H^1(X, \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)) = H^1(X, T_X)$. This is a finite-dimensional k-vector space because X is projective and $H^1(X, T_X)$ is coherent cohomology on a projective variety, hence finite over k. (For a singular scheme the first-order deformation space is instead $T^1_X = \mathrm{Ext}^1_X(\mathbf{L}_{X/k}, \mathcal{O}_X)$, which differs from $\mathrm{Ext}^1_X(\Omega_{X/k}, \mathcal{O}_X)$ once X is not lci; the smooth case sidesteps this correction.) By theorem 5.3.5(1), Def_X admits a hull. The hull is a complete local k-algebra whose tangent space is $H^1(X, T_X)$ and whose relations are cut out by the obstruction map into $H^2(X, T_X)$; when X is unobstructed the hull is the power-series ring $k[[H^1(X, T_X)]]$, by which is meant the completed symmetric algebra on $H^1(X, T_X)$, that is, the power-series ring on a chosen k-basis of $H^1(X, T_X)$.
2. *No infinitesimal automorphisms gives pro-representability.* The functor Def_X fails (H4) exactly when X carries infinitesimal automorphisms, that is, when $T^0 = H^0(X, T_X) \neq 0$: a nonzero global vector field integrates to a one-parameter family of automorphisms over the dual numbers, and this automorphism produces two distinct elements of $\mathrm{Def}_X(A' \times_A A')$ with the same image under η, exactly the failure of injectivity described after definition 5.3.4. When $H^0(X, T_X) = 0$ the object X is infinitesimally rigid as an automorphism problem, every lifting is unique, η is injective in the self-fiber-product case, and (H4) holds. By theorem 5.3.5(2), Def_X is then pro-representable: there is a complete local k-algebra R and a universal formal deformation $\mathcal{X}_{\mathrm{univ}} \to \operatorname{Spec} R$ from which every deformation of X over

an Artinian base is induced by a unique map. A curve C of genus $g \geq 2$ is the standard instance: $H^0(C, T_C) = 0$ because $\deg T_C = 2 - 2g < 0$, so Def_C is pro-representable with hull $k[[t_1, \ldots, t_{3g-3}]]$, since $\dim H^1(C, T_C) = 3g - 3$ and $H^2(C, T_C) = 0$ on a curve.

The two clauses of the example separate cleanly. The existence of a hull is a soft consequence of the geometry of flat families and the finiteness of coherent cohomology on a projective scheme, so essentially every geometric deformation functor clears (H1)-(H3). Whether that hull is universal is a question about automorphisms alone: $H^0(X, T_X)$ measures the infinitesimal automorphisms, and it vanishing is both necessary and sufficient for the hull to become a genuine universal formal deformation. The genus $g \geq 2$ curves are pro-representable and their hull is a smooth $(3g-3)$-dimensional formal disk, which is the local model of the moduli stack $\mathcal{M}_g$ and the reason $\mathcal{M}_g$ is smooth of dimension $3g - 3$. For an elliptic curve, by contrast, $H^0(E, T_E) = k \neq 0$, the functor is not pro-representable, and one is forced into the coarse space of theorem 1.3.6 or the stack-theoretic correction of Part III.

5.3.7 Historical Note: Schlessinger's Functors of Artin Rings The criteria and the hull-existence theorem are the main results of Schlessinger's 1968 paper *Functors of Artin Rings* [27], which grew out of his 1964 Harvard thesis under John Tate. The framework recast deformation theory, previously a collection of cohomological computations, as the study of functors on $\mathbf{Art}_k$, and it made the hull the correct universal object in the presence of automorphisms, before the language of stacks made the automorphisms themselves into geometry. The four conditions have been the standard entry point to formal deformation theory ever since.

Exercise 5.3.1

1. Let $R \in \widehat{\mathbf{Art}}_k$ be a complete local k-algebra with residue field k. Show directly that the functor $h_R = \mathrm{Hom}_{k\text{-alg}}(R, -)$ satisfies conditions (H1), (H2), and (H4) of definition 5.3.4, and that it satisfies (H3) if and only if R is Noetherian. Conclude that a pro-representable functor satisfies all four conditions, recovering one direction of theorem 5.3.5(2).

2. Using proposition 5.3.3, show that for $R \in \widehat{\mathbf{Art}}_k$ Noetherian the tangent space of h_R is canonically $t_{h_R} = (\mathfrak{m}_R/\mathfrak{m}_R^2)^\vee$, the k-linear dual of the cotangent space. Deduce that if h_R is a hull of a functor D, then $\dim_k D(k[\varepsilon])$ equals the minimal number of generators of $\mathfrak{m}_R$.

3. Give an example of a functor $D\colon \mathbf{Art}_k \to \mathbf{Set}$ with $D(k) = \{*\}$ that satisfies (H1) and (H2) but not (H3), and explain in one sentence why the hull-existence construction of theorem 5.3.5 breaks down for it. (Suggestion: take the deformation functor of an infinite-dimensional object, or a functor whose tangent space is $k^{\oplus \mathbb{N}}$.)

4. Let X be a smooth projective variety with $H^0(X, T_X) = 0$ and $H^2(X, T_X) = 0$. Show that Def_X is pro-representable and that its hull is a power-series ring $k[[t_1, \ldots, t_n]]$ with $n = \dim_k H^1(X, T_X)$. Identify n for a smooth hypersurface of degree $d \geq 5$ in $\mathbb{P}^3$ (so that X is of general type and $H^0(T_X) = 0$), given the standard fact that for such a surface every deformation of X is realized by varying the defining degree-d form, so that $H^1(X, T_X)$ is computed by the space of degree-d forms modulo scaling and the projective linear group. State what geometric family this hull parametrizes, and explain why the same count *fails* to give $\dim_k H^1(X, T_X)$ for the quartic $d = 4$.

5. Using the failure of injectivity of η described after definition 5.3.4, explain precisely which

automorphisms obstruct (H4) for Def_X, defined as isomorphism classes of flat families with special fiber X. Show that the relevant automorphisms are the *infinitesimal* ones, measured by $H^0(X, T_X)$, and explain why a nontrivial *finite* automorphism group (for instance the hyperelliptic involution of a genus-2 curve C, for which $H^0(C, T_C) = 0$) does *not* obstruct (H4), so that Def_C is pro-representable. Where, then, do the finite automorphisms make themselves felt, and why does this concern the global moduli stack rather than the local functor Def_X?

5.4 Applications and Computations

The three sections before this one built an apparatus and left it idling: the obstruction space $H^2(X, T_X)$ and its algebraic avatar $T^2(A/k, A)$ (section 5.1, section 5.2), the criterion by which a deformation functor acquires a hull (section 5.3). This section puts the apparatus to work. The payoff of an obstruction theory is that it decides, for a given object, whether the deformations one can write down to first order extend to families over larger bases, and when they do, it pins down the local structure of the moduli space. Three questions organize the section: which classical objects are *unobstructed*, so that their moduli spaces are smooth; what the deformations of the simplest singular curve, the node, look like; and what an obstructed deformation, one that exists to first order but dies at second order, looks like when written out.

The first question is answered for the objects this book most cares about: curves, abelian varieties, and K3 surfaces are all unobstructed, and their moduli spaces are correspondingly smooth of the expected dimension. The second question produces the single number 1, the dimension of the space of smoothings of a node, and that number is the local shadow of the stable-reduction theorem that the moduli of curves in chapter 1's eventual sequel will need: a node deforms in exactly one way that separates its two branches, and this one-parameter smoothing $xy = t$ is the local model for how a nodal degeneration is resolved. The third question is settled by an example, the cone over a projective variety chosen so that a first-order deformation of the cone fails to lift, exhibiting $T^2 \neq 0$ in the flesh rather than as an abstract receptacle.

We begin with the unobstructedness results, since they are what make the moduli spaces of Part III's capstone smooth.

Proposition 5.4.1: Unobstructedness of Curves, Abelian Varieties, and K3 Surfaces

Let k be an algebraically closed field of characteristic 0.

1. Every smooth projective curve C over k is unobstructed: $H^2(C, T_C) = 0$, so Def_C is formally smooth and its hull is a power-series ring in $h^1(C, T_C)$ variables. For $g = \mathrm{genus}(C) \geq 2$ this dimension is $3g - 3$, and $\mathcal{M}_g$ is smooth of dimension $3g - 3$.

2. Every abelian variety A of dimension g over k is unobstructed: $T_A \cong \mathcal{O}_A^{\oplus g}$ is trivial and $H^*(A, \mathcal{O}_A) \cong \bigwedge^* H^1(A, \mathcal{O}_A)$ is an exterior algebra, so $H^i(A, T_A) \cong \left(\bigwedge^i H^1(A, \mathcal{O}_A)\right) \otimes_k \mathfrak{a}$ with $\mathfrak{a} = T_{A,0}$; the deformation space $H^1(A, T_A)$ has dimension g^2, and the moduli space $\mathcal{A}_g$ of principally polarized abelian varieties is smooth of dimension $\binom{g+1}{2}$.

3. Every K3 surface S over k is unobstructed: $H^2(S, T_S) = 0$ (via the Calabi-Yau isomorphism $T_S \cong \Omega^1_S$ and $h^{1,0}(S) = 0$), so Def_S is formally smooth with hull a power-series ring in $h^1(S, T_S) = 20$ variables, and the moduli space of K3 surfaces is smooth of dimension 20 (respectively 19 for a fixed polarization). For a Calabi-Yau variety of dimension ≥ 3 the group $H^2(X, T_X)$ need not vanish, yet unobstructedness still holds: this is the Bogomolov-Tian-Todorov theorem, stated below with its proof cited.

***Proof*:**
The obstruction to lifting a deformation of a smooth projective X over a small extension lies in $H^2(X, T_X)$ by theorem 5.1.2, and unobstructedness in the sense of definition 5.1.3 follows as soon as this group vanishes; parts (1) and (3) proceed by exhibiting the vanishing, while part (2) has $H^2(A, T_A) \neq 0$ for $g \geq 2$ and requires a substitute argument. Throughout, the first-order deformation space is $H^1(X, T_X)$ by theorem 4.3.2, so once formal smoothness is established the hull is a power-series ring on $h^1(X, T_X)$ generators by definition 5.1.3.

Part (1): curves. A smooth projective curve C has dimension 1, so T_C is a coherent sheaf on a 1-dimensional scheme and $H^2(C, T_C) = 0$ by Grothendieck vanishing, since cohomology of a coherent sheaf vanishes above the dimension of the space ([12, Grothendieck vanishing]). Thus C is unobstructed. The tangent space is $H^1(C, T_C)$ with $T_C = \omega_C^{-1}$ the anticanonical bundle, of degree $2 - 2g$; by Serre duality $H^1(C, T_C) \cong H^0(C, \omega_C \otimes T_C^{-1})^\vee = H^0(C, \omega_C^{\otimes 2})^\vee$, and $\omega_C^{\otimes 2}$ has degree $4g - 4 > 2g - 2$ for $g \geq 2$, so $H^1(C, \omega_C^{\otimes 2}) = 0$ and Riemann-Roch gives

$$h^1(C, T_C) = h^0(C, \omega_C^{\otimes 2}) = \deg(\omega_C^{\otimes 2}) - g + 1 = (4g - 4) - g + 1 = 3g - 3$$

Hence the hull of Def_C is $k[[t_1, \ldots, t_{3g-3}]]$ and, since the deformation theory of $\mathcal{M}_g$ at $[C]$ is governed by Def_C, the coarse moduli space is smooth of dimension $3g - 3$ at a point with no extra automorphisms.

Part (2): abelian varieties. An abelian variety is a smooth projective group scheme, and translations act transitively, so the tangent bundle is globally generated by the g translation-invariant vector fields forming a basis of the Lie algebra $\mathfrak{a} = T_{A,0}$. This trivializes T_A:

$$T_A \cong \mathcal{O}_A \otimes_k \mathfrak{a} \cong \mathcal{O}_A^{\oplus g}$$

Consequently $H^i(A, T_A) \cong H^i(A, \mathcal{O}_A) \otimes_k \mathfrak{a}$, and the cohomology of the structure sheaf of an abelian variety is the exterior algebra on $H^1(A, \mathcal{O}_A)$, a g-dimensional space: $H^i(A, \mathcal{O}_A) \cong$

$\bigwedge^i H^1(A, \mathcal{O}_A)$, so $h^i(A, \mathcal{O}_A) = \binom{g}{i}$. In particular $H^2(A, T_A) \cong \bigwedge^2 H^1(A, \mathcal{O}_A) \otimes \mathfrak{a}$ is nonzero for $g \geq 2$, so the vanishing route fails, yet unobstructedness holds for a different reason. In characteristic 0 the polarized deformations of A are realized by a single explicit family: varying the period lattice, equivalently the point of the Siegel upper half-space, produces a family of abelian varieties over a smooth base whose Kodaira-Spencer map at $[A]$ is an isomorphism onto the polarized first-order deformations. Every polarized first-order deformation is therefore tangent to an actual family over a smooth base and so extends to all orders, which is exactly unobstructedness. Concretely, the local moduli of A together with a principal polarization is the analytic germ of the symmetric space $\mathrm{Sp}(2g, \mathbb{R})/U(g)$, which is smooth of dimension $\binom{g+1}{2}$; algebraically, Def_A has tangent space $H^1(A, T_A) \cong \bigwedge^1 H^1(A, \mathcal{O}_A) \otimes \mathfrak{a}$ of dimension $g \cdot g = g^2$ (all deformations of the abelian variety forgetting the polarization), cut down to $\binom{g+1}{2}$ by the symmetry condition the polarization imposes, and the germ is smooth. The full argument that the obstruction map vanishes on abelian varieties, via the triviality of T_A and the Hodge-theoretic splitting, is [28, Ch. 3]; here the vanishing of the obstruction is the content of the classical construction of $\mathcal{A}_g$ as a quotient of Siegel space.

Part (3): K3 surfaces. A K3 surface S is a smooth projective surface with $\omega_S \cong \mathcal{O}_S$ and $H^1(S, \mathcal{O}_S) = 0$. Since $\omega_S \cong \mathcal{O}_S$ and $\dim S = 2$, the tangent bundle satisfies $T_S = (\Omega^1_S)^\vee \cong \Omega^1_S \otimes \omega_S^{-1} \cong \Omega^1_S$, using that on a surface $(\Omega^1_S)^\vee \cong \Omega^1_S \otimes (\det \Omega^1_S)^{-1} = \Omega^1_S \otimes \omega_S^{-1}$. Serre duality then gives

$$H^2(S, T_S) \cong H^0(S, T_S^\vee \otimes \omega_S)^\vee \cong H^0(S, \Omega^1_S)^\vee$$

and $h^0(S, \Omega^1_S) = h^{1,0}(S) = 0$ for a K3, so $H^2(S, T_S) = 0$. By theorem 5.1.2 the K3 is unobstructed, with hull $k[[t_1, \ldots, t_{20}]]$, where $h^1(S, T_S) = h^1(S, \Omega^1_S) = h^{1,1}(S) = 20$ by the same Calabi-Yau isomorphism and Hodge theory. For a Calabi-Yau variety X of dimension ≥ 3 the group $H^2(X, T_X)$ need not vanish, so the vanishing route fails and unobstructedness is the substantive Bogomolov-Tian-Todorov theorem, whose proof uses the Tian-Todorov lemma on the vanishing of a certain bracket in the differential-graded Lie algebra controlling deformations. That proof requires Hodge-theoretic input beyond the scope of this book, so it is cited rather than reproduced ([28, Thm. 14.10], [18, §6]), consistent with the deferral policy for results whose machinery lies outside the present development. In every case the moduli space is smooth of the stated dimension, completing the proof.

The three parts illustrate the two routes to unobstructedness that recur throughout the subject. The first route, taken by curves, is to kill the obstruction space outright: when $H^2(X, T_X) = 0$ there is nowhere for an obstruction to live, and formal smoothness is automatic. This is the cleanest situation and is exactly why $\mathcal{M}_g$ is smooth, a fact the moduli of curves relies on completely. The second route, taken by abelian varieties and by general Calabi-Yau manifolds, is subtler: the obstruction space is nonzero, yet the obstruction map into it is identically zero, so deformations lift despite having somewhere to fail. Abelian varieties see this through the triviality of the tangent bundle and the explicit construction via period lattices; Calabi-Yau manifolds see it through the Bogomolov-Tian-Todorov theorem. The lesson is that a nonzero H^2 is a warning, not a verdict: it flags where obstructions could appear, but only the vanishing of the actual obstruction cocycle, computed in theorem 5.1.2, decides whether the moduli space is smooth. For a K3 the warning is empty because $H^2(S, T_S)$ vanishes; for a Calabi-Yau threefold it is a nonzero group whose obstruction map is nonetheless zero, and the vanishing of that map is the content of Bogomolov-Tian-Todorov.

With smoothness in hand for the objects Part III will compactify, the focus turns from smooth objects to singular ones, where the geometric $H^i(T_X)$ picture no longer suffices and the algebraic cotangent functors $T^i(A/k, A)$ of section 5.2 take over. The governing example is the node, both because it is the simplest singular point of a curve and because its deformation theory is the local input to the stable-reduction theorem. The computation is short enough to carry out in full from the naive cotangent complex.

Example 5.4.2: Deforming the Node

Let $A = k[x, y]/(xy)$, the coordinate ring of a node: two lines meeting transversally at the origin of $\mathbb{A}^2$, an A_1 (equivalently ordinary double point) singularity. Up to completion of the local ring, this is the local model of a nodal point on a curve. Its cotangent functors $T^i(A/k, A)$ are computed from the presentation $A = P/I$ with $P = k[x, y]$ and $I = (xy)$, a hypersurface, hence a complete intersection.

The naive cotangent complex. By definition 5.2.1 the two-term complex $L_{A/k}$ attached to this presentation is

$$I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P A$$

placed in degrees 1 and 0. Here I/I^2 is free of rank 1 over A on the class of $f = xy$ (a single relation, no syzygies among a single generator, so the second-syzygy term vanishing gives $T^2 = 0$ immediately for a hypersurface, as recorded in example 5.2.4), and $\Omega_{P/k} \otimes_P A = A\,dx \oplus A\,dy$ is free of rank 2. The differential sends the generator $f = xy$ to

$$d(xy) = y\,dx + x\,dy \in A\,dx \oplus A\,dy$$

Computing T^0, T^1, T^2. The functors $T^i(A/k, A)$ are the cohomology of $\mathrm{Hom}_A(L_{A/k}, A)$ in the appropriate degrees, following definition 5.2.2 and theorem 5.2.3. Dualizing the two-term complex against A gives

$$\mathrm{Hom}_A(\Omega_{P/k} \otimes_P A,\ A) \xrightarrow{d^\vee} \mathrm{Hom}_A(I/I^2,\ A)$$

that is, $A^2 \xrightarrow{d^\vee} A$, where a pair $(a, b) \in A^2$ (representing the derivation $a\,\partial_x + b\,\partial_y$) maps to

$$d^\vee(a, b) = a \cdot y + b \cdot x \in A$$

the pairing of the derivation with the relation $d(xy) = y\,dx + x\,dy$. Then $T^0(A/k, A) = \ker d^\vee = \mathrm{Der}_k(A, A)$ is the module of vector fields, $T^1(A/k, A) = \mathrm{coker} d^\vee$ is the deformation space, and $T^2(A/k, A)$ is the obstruction space.

First, $T^2(A/k, A) = 0$: the node is a hypersurface, hence a complete intersection, hence l.c.i., and l.c.i. singularities have no second syzygies among a single defining equation, so they are unobstructed by example 5.2.4. There is no obstruction to deforming a node.

Next, $T^1(A/k, A) = \mathrm{coker}(d^\vee) = A/(x, y)$. The image of $d^\vee$ is the ideal generated by the two coordinates $d^\vee(1, 0) = y$ and $d^\vee(0, 1) = x$, so it is $(x, y) \subseteq A$, and the cokernel is

$$T^1(A/k, A) = A/(x, y) = k$$

a one-dimensional k-vector space. This is the Tjurina algebra $A/(f, \partial_x f, \partial_y f) = k[x, y]/(xy,\ y,\ x) = k$ of the isolated hypersurface singularity, computed as the Jacobian ring predicted by example 5.2.4. The node has exactly one first-order deformation up to isomorphism.

The versal deformation. A cocycle representing the generator of T^1 is the perturbation $f = xy \rightsquigarrow xy - t$ of the defining equation, since modulo (x, y) the constant -1 (the coefficient of t) spans $A/(x, y) = k$. Because $T^2 = 0$, this first-order deformation extends to all orders without obstruction, and because $\dim_k T^1 = 1$, one parameter suffices: the versal deformation of the node is

$$\operatorname{Spec} k[x, y, t]/(xy - t) \longrightarrow \operatorname{Spec} k[t]$$

with the node sitting over $t = 0$. For $t \neq 0$ the fiber is $\{xy = t\}$, a smooth conic (a hyperbola), while the fiber over $t = 0$ is the node itself. The single parameter t smooths the singularity: the two branches $x = 0$ and $y = 0$ that met at the origin are pulled apart into a single smooth curve as t moves off 0.

Reading the smoothing. The one-dimensionality of T^1 is the local input to the stable-reduction theorem in the moduli of curves. When a family of smooth curves degenerates to a singular limit, the limit can be taken to be a nodal curve, and each node contributes exactly this one-parameter smoothing $xy = t$: locally, the total space of a one-parameter degeneration to a nodal curve looks like $\{xy = t\}$ near each node, a surface smooth in the total space even though the special fiber is singular. The versal deformation $\{xy = t\}$ is therefore the local model for the boundary of the Deligne-Mumford compactification $\overline{\mathcal{M}}_g$: a node is a codimension-one degeneration, its single smoothing parameter is the normal direction to the boundary divisor, and the boundary of $\overline{\mathcal{M}}_g$ is where the nodes live. That the smoothing is unobstructed ($T^2 = 0$) is why the boundary is a divisor with normal crossings rather than a more singular locus, and why $\overline{\mathcal{M}}_g$ is smooth as a stack even where its coarse space is not. This connection is developed when the moduli of curves is constructed.

The node is the model unobstructed singularity, and its computation shows the mechanism by which a singular point deforms: the tangent space T^1 measures the independent smoothing directions, and the vanishing of T^2 guarantees each of them integrates to an actual family. Two features generalize. First, for any isolated hypersurface singularity $\{f = 0\}$, the same computation gives $T^1 = k[x_1, \ldots, x_n]/(f, \partial_1 f, \ldots, \partial_n f)$, the Tjurina algebra, whose dimension counts the smoothing parameters and whose vanishing detects smoothness; and $T^2 = 0$ always, so hypersurface singularities are always unobstructed. Second, the passage from the algebraic $T^i(A/k, A)$ to the geometric moduli is exactly the gluing of theorem 5.2.3: the local T^1 at each singular point assembles, together with the global $H^1(T)$ of the smooth part, into the deformation space of the whole curve, and this is what makes the boundary of $\overline{\mathcal{M}}_g$ a normal-crossings divisor whose components record which nodes are being smoothed.

Unobstructed examples are the well-behaved case; to see the machinery earn its keep, one needs a deformation that exists to first order but fails to extend, so that $T^2 \neq 0$ receives a nonzero obstruction. Such examples are less elementary than the node, because l.c.i. singularities are always unobstructed and one must leave the l.c.i. world to find $T^2 \neq 0$. The standard source is the affine cone over a projective variety embedded so that its second cotangent functor is nonzero, and the cleanest instance is the cone over a rational normal curve of high enough degree, or over a Segre-embedded product.

Example 5.4.3: An Obstructed Cone

Let $Y = \mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$ be the Segre embedding of the quadric surface, whose image is the smooth quadric $\{X_0X_3 - X_1X_2 = 0\} \subseteq \mathbb{P}^3$, and let A be the homogeneous coordinate ring of the corresponding *affine cone*

$$C(Y) = \operatorname{Spec} A, \qquad A = k[X_0, X_1, X_2, X_3]/(X_0X_3 - X_1X_2)$$

the affine quadric cone over Y, a three-dimensional hypersurface with an isolated singularity at the vertex. This particular cone is l.c.i. (it is a hypersurface), so it is unobstructed and would not serve; the point of naming it is to contrast it with the obstructed cone obtained by a non-complete-intersection embedding, taken up next.

Take instead $Y = \mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$ under the Segre embedding, whose homogeneous ideal is generated by the 2×2 minors of the 2×3 matrix (Z_{ij}) of coordinates, three quadrics

$$q_1 = Z_{00}Z_{11} - Z_{01}Z_{10}, \quad q_2 = Z_{00}Z_{12} - Z_{02}Z_{10}, \quad q_3 = Z_{01}Z_{12} - Z_{02}Z_{11}$$

cutting out a codimension-2 variety in $\mathbb{P}^5$, so the cone $C(Y) = \operatorname{Spec} A$ with $A = k[Z_{ij}]/(q_1, q_2, q_3)$ is a four-dimensional variety defined by three equations, hence *not* a complete intersection (three equations, codimension two). It is precisely the failure of the complete-intersection condition that lets T^2 be nonzero.

$T^2 \neq 0$. For this determinantal cone the second cotangent functor $T^2(A/k, A)$ is nonzero. The naive cotangent complex of definition 5.2.1 now has a nontrivial syzygy term. The three minors q_1, q_2, q_3 do not form a regular sequence (they cut out a codimension-2 locus with three equations), so there are syzygies among them beyond the trivial Koszul ones: the classical Hilbert-Burch resolution of a codimension-2 determinantal ideal presents (q_1, q_2, q_3) with two linear syzygies, one for each row of the matrix (Z_{ij}), and the departure of these from the Koszul syzygies is what feeds the second-syzygy term of $L_{A/k}$. The resulting computation of the second cotangent functor gives $T^2(A/k, A) \neq 0$ ([18, §3.1], and [28, Ch. 3] for deformations of determinantal schemes); this is the standard example of an obstructed singularity, cited here because the full determinantal T^2 computation is longer than the point it illustrates. By theorem 5.2.3 the cone can carry an obstructed deformation.

A first-order deformation that fails to lift. Perturb the three defining quadrics to first order,

$$q_i \rightsquigarrow q_i + \varepsilon\, g_i, \qquad \varepsilon^2 = 0$$

with g_i chosen quadrics. At first order the deformation is flat over $k[\varepsilon]$ exactly when the syzygies among the q_i lift modulo ε^2: each linear relation $\sum_i r_{ji}(Z)\, q_i = 0$ among the quadrics must extend to $\sum_i r_{ji}(Z)\,(q_i + \varepsilon g_i) = \varepsilon\, h_j$ with $h_j \in (q_1, q_2, q_3)$, and such a first-order lift can always be arranged, so every (g_1, g_2, g_3) satisfying this condition is a valid element of T^1. The obstruction appears one order up: lifting the first-order deformation to second order over $k[\varepsilon']/(\varepsilon'^3)$ requires quadratic corrections to the equations that keep the syzygies satisfied modulo ε'^3, and the class in $T^2(A/k, A)$ measures whether these corrections exist. For a class in T^1 pointing off the versal base, they do not: the second-order syzygy conditions cannot all be met simultaneously, and their failure lands in the nonzero group T^2. Concretely, the versal base space of $C(Y)$ is not smooth: it is a cone (a quadric cone in the relevant example) cut out by the obstruction equations in the T^1-directions, so a tangent direction pointing off the versal cone is a first-order deformation with a nonzero obstruction, and it does not lift. This exhibits the phenomenon the obstruction theory was built to detect: a class in T^1 that is not tangent to any actual family, whose obstruction in T^2 is nonzero.

The obstructed cone is where the abstract T^2 of section 5.2 becomes visible. The versal deformation of the node was a smooth base, Spec $k[t]$, one parameter for one first-order deformation, because $T^2 = 0$; the versal base of the obstructed cone is singular, a cone inside the affine space T^1 cut out by the obstruction equations valued in T^2. The dimension of T^1 still counts first-order deformations, but now not every first-order deformation integrates: the ones that do are exactly those tangent to the versal base, and the obstruction map $T^1 \to T^2$ (more precisely, the quadratic and higher terms of the Kuranishi map) records which do not. This is the general shape of a moduli space at an obstructed point: the tangent space is T^1, the ambient obstruction space is T^2, and the moduli space is locally the zero locus of the obstruction map inside T^1, smooth of dimension $\dim T^1$ exactly when $T^2 = 0$ or the obstruction map vanishes. The unobstructedness results of proposition 5.4.1 are precisely the statement that for curves, abelian varieties, and K3 surfaces this obstruction map vanishes, so their moduli spaces are smooth, while the cone shows what smoothness would fail to hold for a determinantal singularity carrying a nonzero obstruction.

Exercise 5.4.1

1. Let C be a smooth projective curve of genus g. Using proposition 5.4.1(1), compute $\dim_k H^1(C, T_C)$ for $g = 0$ and $g = 1$, and reconcile the answers with the geometry: explain why the $g = 0$ curve $\mathbb{P}^1$ is rigid (no moduli) and why the $g = 1$ answer is the 1-dimensional $\mathcal{M}_{1,1}$ deformation computed in chapter 1. (Recall $T_C = \omega_C^{-1}$ has degree $2 - 2g$, and treat the cases $g = 0, 1$ where the Riemann-Roch shortcut of the proposition's proof, valid for $g \geq 2$, does not directly apply.)

2. For the node $A = k[x, y]/(xy)$, verify by hand that the Tjurina algebra $A/(f, \partial_x f, \partial_y f)$ with $f = xy$ equals k, and check that the cusp $A' = k[x, y]/(y^2 - x^3)$ has $\dim_k T^1(A'/k, A') = 2$ by the same Jacobian-ring computation. Interpret the difference between the numbers 1 and 2 in terms of how many parameters are needed to smooth each singularity.

3. Write down the versal deformation $\operatorname{Spec} k[x, y, t]/(xy - t) \to \operatorname{Spec} k[t]$ of the node explicitly as a family, and prove that the total space $\operatorname{Spec} k[x, y, t]/(xy - t)$ is smooth (regular) at the origin even though the special fiber over $t = 0$ is singular there. Explain in one sentence why this is the reason $\overline{\mathcal{M}}_g$ can be smooth as a stack while parametrizing singular (nodal) curves.

4. Let S be a K3 surface. Using $\omega_S \cong \mathcal{O}_S$, $H^1(S, \mathcal{O}_S) = 0$, and Serre duality, verify all three of $h^0(S, T_S) = 0$, $h^1(S, T_S) = 20$, and $h^2(S, T_S) = 0$ from proposition 5.4.1(3). (You may use the Hodge numbers of a K3: $h^{0,0} = h^{2,0} = h^{0,2} = 1$, $h^{1,0} = h^{0,1} = 0$, $h^{1,1} = 20$.) Which of these three vanishings is the one that makes S unobstructed, and which makes Def_S pro-representable rather than only having a hull?

5. Let $A = k[x, y, z]/(xy, xz, yz)$, the coordinate ring of the three coordinate axes in $\mathbb{A}^3$ (a non-l.c.i. singularity: three equations, codimension two). Show that A is not a complete intersection by comparing the number of equations with the codimension, and explain why this makes it a candidate for $T^2(A/k, A) \neq 0$, in contrast to the node of example 5.4.2. (You need not compute T^2; identify the structural reason a non-complete-intersection can be obstructed.)

6. Let A be an abelian variety of dimension g. Using the triviality $T_A \cong \mathcal{O}_A^{\oplus g}$ and $H^i(A, \mathcal{O}_A) \cong \bigwedge^i H^1(A, \mathcal{O}_A)$ from proposition 5.4.1(2), compute $\dim_k H^i(A, T_A)$ for all i as a function of g, and identify $\dim_k H^1(A, T_A)$ as the dimension of the unpolarized deformation space. For $g = 2$, compare this with the dimension $\binom{g+1}{2}$ of $\mathcal{A}_g$ and explain the discrepancy in terms of the polarization constraint.

Part III

Descent and Algebraic Stacks

The moduli problems of Part I were functors, and the recurring verdict was that they are not representable by schemes. Part II diagnosed the obstruction one point at a time: the automorphisms of an object, visible as the infinitesimal automorphism group H^0, are what a scheme cannot record. This Part builds the geometry that can. It replaces the scheme, an object glued from affine pieces in the Zariski topology, with objects glued in finer topologies and remembering their automorphisms, and it shows that in this larger world the moduli problems finally acquire the spaces they were asking for.

The construction proceeds in three stages, and this Part devotes a chapter to each. chapter 6 develops descent theory: the recognition that the sheaf condition, and with it the notion of gluing, makes sense in any Grothendieck topology, and that quasi-coherent sheaves, morphisms, and (with a polarization) schemes descend along faithfully flat covers. The failure of a bare descent datum for schemes to be effective is the precise reason schemes are too rigid, and torsors and their classification by H^1 are the first sign of the automorphisms about to be geometrized. The following chapters pass to categories fibered in groupoids and to stacks, the bookkeeping for objects-with-automorphisms glued in a topology, and then to algebraic spaces and algebraic stacks, the geometric objects that represent the moduli functors schemes could not; Artin's criteria, fed by the deformation theory of Part II, certify when a moduli problem is such a stack. A final chapter treats gerbes and twisted sheaves, the 2-categorical layer above torsors.

The capstone is the moduli of curves. There $\mathcal{M}_g$ is the stack whose deformation theory is the $H^1(T_C)$ of Part II, whose coarse space is the geometric invariant theory quotient of Part I, and whose Deligne–Mumford compactification $\overline{\mathcal{M}}_g$ by stable curves is proper and smooth as a stack even where its coarse space is singular. The three Parts of the book meet in that single object.

6

Descent Theory

A scheme is glued from affine pieces, and the glue is the Zariski topology: a morphism, a sheaf, or a subscheme is specified by giving it on an open cover together with the condition that the pieces agree on overlaps. The moduli problems of the earlier Parts strain against this. A family of curves can be locally trivial in a topology finer than the Zariski one and yet globally nontrivial; an object can have automorphisms, so that two local descriptions agree not on the nose but up to an isomorphism that must itself be tracked. To build the spaces these problems require, the first move is to enlarge the notion of an open cover, and then to ask which structures still glue.

That enlargement is Grothendieck's, and this chapter develops it. A Grothendieck topology axiomatizes the covering families for which gluing should work, and the étale, smooth, and faithfully flat topologies it produces are the ones in which moduli objects are locally trivial. The sheaf condition, rewritten as an equalizer over such a covering family, then makes sense on any site, and the central theorem, faithfully flat descent, says that quasi-coherent sheaves, morphisms, and, when a polarization is present, schemes are recovered from their restrictions to a flat cover together with gluing data on the overlaps. The point where this recovery fails, a descent datum for schemes with no descended polarization, is exactly the point where schemes must give way to the algebraic spaces and stacks of the chapters to follow. The chapter closes with torsors and their classification by the first cohomology of a site, the one-categorical first appearance of the automorphisms that stacks will geometrize and of the gerbes that Chapter 9 will build on them.

6.1 Grothendieck Topologies and Sites

Every construction in the first two Parts of this book rested on a single tacit assumption: that the geometric objects of a moduli problem glue in the Zariski topology. A family of curves over a base S was assembled from its restrictions to a Zariski-open cover; a coherent sheaf was recovered from its restrictions to open sets by the sheaf axiom. This assumption is where representability broke down.

The functor $\mathcal{M}_{1,1}$ of elliptic curves is not a Zariski sheaf in the sense that would let one build a fine moduli space: two families that agree Zariski-locally can differ globally, because an elliptic curve has automorphisms, and gluing along those automorphisms is invisible to the Zariski topology. The repair begins here. The Zariski topology is too coarse for descent because its covers are too large, too few isomorphisms are captured, and a morphism that is Zariski-locally an isomorphism is already an isomorphism. What Part III needs is a notion of "cover" fine enough that faithfully flat maps count as covers, so that objects glued along them descend. That is exactly what a Grothendieck topology provides.

This section axiomatizes the word "cover." The insight, due to Grothendieck, is that gluing never used the points of a topological space, only the lattice of open sets and the relation "these opens cover that one." Abstracting that relation to an arbitrary category produces a *site*, and on a site one can speak of sheaves, cohomology, and descent with no topological space in sight. The étale, smooth, and fppf topologies on schemes are the payoff: covers there are surjective families of étale, smooth, or flat morphisms, and each strictly refines the Zariski topology. This chapter develops descent in that language; the present section builds the language itself, grounding each topology in a concrete cover as soon as the machinery permits.

The starting point is to isolate what a family of open inclusions $\{U_i \hookrightarrow X\}$ does in ordinary topology, stripped of any reference to points. Three closure properties do all the work in every gluing argument. A cover of X restricts to a cover of any open subset (a cover is stable under passing to a smaller open); covers compose (a cover of each member of a cover assembles to a cover of the whole); and the one-element family consisting of the identity, or any isomorphism, is a cover. In a general category the role of "restriction to a smaller open" is played by fiber product, and this substitution is the one nontrivial conceptual step. The following definition records these three axioms in the generality of an arbitrary category with fiber products, following [30] and the categorical foundations of [5].

Definition 6.1.1: Grothendieck Topology and Site

Let $\mathcal{C}$ be a category admitting fiber products. A *Grothendieck pretopology* on $\mathcal{C}$ assigns to each object X of $\mathcal{C}$ a collection of *covering families* $\{U_i \xrightarrow{u_i} X\}_{i\in I}$ of morphisms with target X, subject to three axioms.

1. *(Isomorphisms.)* If $u\colon U \to X$ is an isomorphism, then the one-element family $\{u\colon U \to X\}$ is a covering family.
2. *(Base change.)* If $\{U_i \to X\}_{i\in I}$ is a covering family and $V \to X$ is any morphism, then the fiber products $U_i \times_X V$ exist, and the family of projections $\{U_i \times_X V \to V\}_{i\in I}$ is a covering family of V.
3. *(Composition.)* If $\{U_i \to X\}_{i\in I}$ is a covering family and, for each i, $\{V_{ij} \to U_i\}_{j\in J_i}$ is a covering family, then the composite family $\{V_{ij} \to U_i \to X\}_{i\in I,\, j\in J_i}$ is a covering family of X.

A category $\mathcal{C}$ equipped with a Grothendieck pretopology is called a *site*.

The three axioms are precisely the three moves a gluing argument performs. Isomorphisms as covers guarantee that anything glued from a single copy of itself is unchanged, so the trivial cover trivializes; without it a sheaf could fail to satisfy $F(X) = F(X)$. Base change is the substitute for "restrict the cover to a smaller open": it says a cover pulls back to a cover, which is what lets one localize a descent problem. Composition says a cover of a cover is a cover, the property that makes "locally" a transitive relation, so that a statement true locally on a cover, and locally on that cover, is true locally. The axioms make no mention of surjectivity, injectivity, or any point-set notion; they are entirely about the arrows.

The word *pretopology* rather than *topology* flags a redundancy in the data. Two covering families of the same object can generate the same notion of "locally" even when they are not equal as families, just as in ordinary topology two different open covers of a space refine each other and so give the same sheaves. The invariant that a pretopology determines is not the set of covering families but the set of *sieves* they generate. A sieve on X is a collection S of morphisms into X closed under precomposition: if $(V \to X) \in S$ and $W \to V$ is any morphism, then the composite $(W \to X) \in S$. A covering family $\{U_i \to X\}$ generates the sieve of all morphisms $V \to X$ that factor through some u_i, and the object a Grothendieck topology assigns to X is a distinguished collection of *covering sieves* satisfying axioms equivalent to those above. The sieve formulation is cleaner for proving general theorems, since it quotients out the choice of indexing, but the pretopology formulation is what one computes with, and it is the one used throughout this book. Passing between the two is a bookkeeping exercise carried out in [30, Definition 2.24]; nothing below depends on which formulation is adopted, and "topology" will mean a pretopology given by explicit covering families.

The definition is abstract by necessity, but its content is already visible in the one topology the reader has used for years.

Example 6.1.2: The Zariski Site of a Scheme

Let X be a scheme and let $\mathcal{C} = \mathrm{Open}(X)$ be the category whose objects are the Zariski-open subsets of X and whose morphisms are the inclusions. Fiber products in this category are intersections: for opens $U, V \subseteq W$ the fiber product $U \times_W V$ is the intersection $U \cap V$. Declare a family of inclusions $\{U_i \hookrightarrow X\}$ to be a covering family exactly when $\bigcup_i U_i = X$ set-theoretically. The three axioms hold transparently.

1. The identity $X = X$ is a surjective one-element family, and any isomorphism of opens is an equality, so isomorphisms cover.
2. If $\bigcup_i U_i = X$ and $V \subseteq X$ is open, then $U_i \times_X V = U_i \cap V$ and $\bigcup_i (U_i \cap V) = V \cap \bigcup_i U_i = V$, so the pullback family covers V.
3. If $\bigcup_i U_i = X$ and $\bigcup_j V_{ij} = U_i$ for each i, then $\bigcup_{i,j} V_{ij} = \bigcup_i U_i = X$, so the composite family covers.

This is the site whose sheaves are exactly the sheaves on the topological space X: the abstract sheaf condition of section 6.2 specializes to the ordinary one. The Zariski topology is thus a Grothendieck topology on a category whose only morphisms are inclusions, and the leap of the next definition is to allow morphisms that are not inclusions.

The Zariski site is a Grothendieck topology in which the covering morphisms are open immersions and the underlying category is a poset. To capture descent along faithfully flat maps one enlarges the category to allow arbitrary scheme morphisms as members of a cover, and enlarges the supply of covering families to include surjective families of morphisms of a prescribed type. The prescription is the topology; changing the class of allowed morphisms changes the topology.

Fix a base scheme S and work in the category $\mathbf{Sch}/S$ of S-schemes. A topology on this category is specified by naming which surjective families count as covers, and there is a standard tower of four choices, each obtained by enlarging the class of covering morphisms from the last. The requirement common to all four is joint surjectivity: a family $\{U_i \to X\}$ covers only if the images of the U_i cover X as a set. What distinguishes the topologies is the type of morphism permitted. Recall from [12, Étale Morphisms] that an étale morphism is a smooth morphism of relative dimension zero, equivalently a flat and unramified morphism locally of finite presentation; that a smooth morphism is flat, locally of finite presentation, with smooth fibers; and from [12, Faithfully Flat Morphisms] that a morphism is faithfully flat when it is flat and surjective. These are the four morphism classes that define the four topologies.

Definition 6.1.3: The Standard Topologies on Sch/S

Let τ be one of the labels *Zariski*, *étale*, *smooth*, or *fppf*. A family of morphisms $\{u_i\colon U_i \to X\}_{i\in I}$ of S-schemes is a *τ-covering family* if it is jointly surjective, that is $\bigcup_i u_i(U_i) = X$, and each u_i is a morphism of the corresponding type:

- *Zariski:* each u_i is an open immersion;
- *étale:* each u_i is étale;
- *smooth:* each u_i is smooth;
- *fppf:* each u_i is flat and locally of finite presentation (*fidèlement plat de présentation finie*).

Each choice satisfies the axioms of definition 6.1.1 and so makes **Sch**/S a site, written $(\mathbf{Sch}/S)_\tau$ and called the *big τ-site*. The *small étale site* $X_{\text{ét}}$ of a scheme X is the full subcategory of étale X-schemes $U \to X$, with the étale covering families; the small Zariski site is the site Open(X) of example 6.1.2.

The distinction between the big site and the small site is a matter of how much of the category one carries. The big τ-site $(\mathbf{Sch}/S)_\tau$ contains *all* S-schemes as objects and admits τ-covers between any of them; it is the natural home for a moduli functor, which is defined on all test schemes. The small étale site $X_{\text{ét}}$ retains only the schemes étale over a fixed X, and it is the natural home for the étale cohomology of X itself, because a sheaf on $X_{\text{ét}}$ is a local invariant of X that varies only in the étale direction. This book works predominantly with big sites, since the objects of a moduli problem are tested against arbitrary schemes; the small site returns when a fixed scheme's own cohomology is at issue. The fppf topology is the widest of the four that this book uses, and it is the one in which representable functors become sheaves, the reason it governs descent.

The four topologies are nested, and the nesting is the reason a hierarchy of descent statements exists. A cover in a coarser topology is automatically a cover in a finer one, so a sheaf for the finer topology is in particular a sheaf for the coarser, and a descent statement proved for the finer topology subsumes the coarser cases. Establishing the nesting is a matter of tracking the implications among the morphism classes: an open immersion is étale, an étale morphism is smooth, and a smooth morphism is flat and locally of finite presentation. The one non-formal input is that a smooth cover, though finer than an étale cover as a family, generates the *same* notion of sheaf on the small site, because smooth morphisms admit sections étale-locally.

Proposition 6.1.4: Comparison of the Standard Topologies

Let S be a scheme. Every Zariski-covering family is an étale-covering family, every étale-covering family is a smooth-covering family, and every smooth-covering family is an fppf-covering family. Consequently there are inclusions of topologies

$$\text{Zariski} \subseteq \text{étale} \subseteq \text{smooth} \subseteq \text{fppf}$$

in the sense that every τ-sheaf is a τ'-sheaf whenever τ is finer than τ'. Moreover, every smooth-covering family $\{U_i \to X\}$ admits, after further étale base change, a section: there is an étale cover $\{X_a \to X\}$ such that each $X_a \to X$ factors through some U_i. In particular the smooth and étale topologies define the same sheaves.

***Proof*:**
Step 1: The inclusions of morphism classes. Each inclusion of covering families follows from an implication between the defining properties of the morphisms, the joint-surjectivity requirement being identical across the four topologies.

An open immersion $u\colon U \to X$ is flat, since localization is flat, and unramified with trivial relative differentials, and locally of finite presentation; a morphism that is flat, unramified, and locally of finite presentation is étale by [12, Étale Morphisms]. Hence every open immersion is étale, and a jointly surjective family of open immersions is a jointly surjective family of étale morphisms: every Zariski cover is an étale cover.

An étale morphism is by definition smooth of relative dimension zero, so in particular smooth. Hence a jointly surjective family of étale morphisms is a jointly surjective family of smooth morphisms: every étale cover is a smooth cover.

A smooth morphism is flat and locally of finite presentation by definition, these being two of the three conditions (the third being geometric regularity of the fibers). Hence a jointly surjective family of smooth morphisms is a jointly surjective family of flat, locally-finitely-presented morphisms: every smooth cover is an fppf cover.

Step 2: Finer topologies give more sheaves. Suppose τ is finer than τ', meaning every τ'-cover is a τ-cover, which is the content of Step 1 for adjacent pairs and follows by transitivity in general. If F is a τ-sheaf, then F satisfies the sheaf condition of section 6.2 for every τ-covering family, hence in particular for every τ'-covering family, since these form a subclass. Thus F is a τ'-sheaf. This is the asserted inclusion of topologies: finer topology, more covers to satisfy, fewer sheaves, and the sheaves for the finer topology are among those for the coarser.

Step 3: Smooth covers admit étale-local sections. Let $u\colon U \to X$ be a smooth surjective morphism. The structure theorem for smooth morphisms states that u is étale-locally on U a composite of an étale morphism with a projection from affine space: for each point of U there is an open neighborhood $U' \subseteq U$ and an étale morphism $U' \to \mathbb{A}^n_X$ over X. The projection $\mathbb{A}^n_X \to X$ has the zero section $X \to \mathbb{A}^n_X$, and pulling this section back along the étale map produces, over an étale cover of X, a section of u. The full statement is that a smooth surjection admits sections étale-locally on the target, proved via the local structure of smooth morphisms in [12, Local Structure of Smooth Morphisms]; see also [29, Tag 054L]. Concretely, there is an étale cover $\{X_a \to X\}$ and, for each a, a lift $X_a \to U$ of $X_a \to X$ through u. Applied to a smooth-covering family $\{U_i \to X\}$, choosing for each member such an étale refinement produces an étale cover of

X factoring through the U_i, which is the claimed étale refinement.

Step 4: Smooth and étale sheaves coincide. A τ-sheaf is determined by its behavior on a cofinal system of covers. By Step 3 every smooth cover is refined by an étale cover, and by Step 1 every étale cover is a smooth cover; the two classes of covers are therefore mutually cofinal. A presheaf satisfies the sheaf condition for a cover if and only if it satisfies it for any refinement, since refinement composes covers (definition 6.1.1(3)) and the sheaf condition passes to refinements. Hence a presheaf is a smooth sheaf if and only if it is an étale sheaf, completing the proof.

The comparison proposition organizes Part III's descent statements into a single ascending ladder. Descent along an fppf cover, once proved, descends along smooth, étale, and Zariski covers as special cases, because each of those is an fppf cover; and a functor that is an fppf sheaf is a fortiori a sheaf for the three coarser topologies. This is why faithfully flat descent, the subject of section 6.3, is stated and proved at the fppf level: it is the strongest statement, and the others follow. The coincidence of smooth and étale sheaves in the last clause is the reason algebraic stacks can be defined with either a smooth atlas or, in the Deligne-Mumford case, an étale atlas without changing the resulting sheaf theory, a point that returns when stacks are constructed in Chapter 7. On the small site of a fixed scheme the smooth and étale topologies are interchangeable; the distinction between them is felt only through the morphisms allowed in an atlas, not through the sheaves.

One further topology sits above the fppf topology and completes the tower, the fpqc topology, whose covers are the faithfully flat quasi-compact families. Descent holds in the fpqc topology as well, and it is the coarsest topology for which quasi-coherent descent is available, so it is the correct maximal setting for the descent theorems. There is a technical cost: the fpqc topology is not a small site in the naive sense, because the class of fpqc covers of a fixed scheme is not a set but a proper class, and the resulting sheaf theory requires a set-theoretic device (a universe, or a bound on the cardinality of covers) to be well-founded. This subtlety is inessential for the constructions of this book, all of which can be carried out at the fppf level, so the fpqc topology is invoked only when a descent statement is strictly stronger there, and its foundational treatment is deferred to [29, Tags 022A, 03NV]. The working hierarchy is

$$\text{Zariski} \subseteq \text{étale} \subseteq \text{smooth} \subseteq \text{fppf} \subseteq \text{fpqc}$$

with the last inclusion carrying the set-theoretic caveat just noted.

The definitions are now in place, and the promise of the section is to make them concrete. The following worked covers exhibit the difference between the Zariski and finer topologies in the smallest possible examples and introduce the cover that will power the theory of μ_n-torsors in section 6.5. Each is a family of morphisms; the question in each case is whether it is jointly surjective and of the required type.

Example 6.1.5: Three Model Covers

Work over a field k; write $\mathbb{A}^1 = \operatorname{Spec} k[x]$ and $\mathbb{G}_m = \operatorname{Spec} k[x, x^{-1}]$.

1. *A non-cover.* Consider the single-morphism family $\{\operatorname{Spec} k[x, x^{-1}] \to \operatorname{Spec} k[x]\}$, the inclusion of the punctured line $\mathbb{G}_m$ into $\mathbb{A}^1$ dual to the localization $k[x] \to k[x, x^{-1}]$. This morphism is an open immersion, hence étale, smooth, and flat of finite presentation. It is nonetheless *not* a covering family in any of the four topologies, because it is not jointly surjective: its image is the open set $x \neq 0$, missing the origin. Joint surjectivity is a genuine constraint, not an afterthought, and it is what fails here. Adjoining the origin, say via the second chart $\{\operatorname{Spec} k[x, (x-a)^{-1}] \to \operatorname{Spec} k[x]\}$ for a scalar $a \neq 0$, or more simply the identity chart at 0, restores a cover; a single localization does not.

2. *An fppf and étale cover of a point.* Let L/k be a finite field extension of degree n and consider the one-morphism family $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ dual to the inclusion $k \hookrightarrow L$. A finite extension is a finite, flat (indeed free of rank n as a k-module) morphism of finite presentation, and it is surjective since $\operatorname{Spec} L$ and $\operatorname{Spec} k$ are both single points and the map hits it. Hence $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ is an *fppf cover of* $\operatorname{Spec} k$. If moreover L/k is separable, then $k \to L$ is unramified (the trace form is nondegenerate, equivalently $\Omega_{L/k} = 0$), so the morphism is étale and the family is an *étale cover.* This single-point cover is the model on which Galois descent runs: a field extension is a cover in the étale or fppf topology precisely when the Zariski topology, which sees $\operatorname{Spec} k$ as an irreducible point with no proper open cover, is blind. Over $k = \mathbb{R}$ the extension $\mathbb{C}/\mathbb{R}$ gives the étale cover $\{\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} \mathbb{R}\}$ underlying the descent of real forms.

3. *The Kummer cover.* Fix an integer $n \geq 1$ and consider the n-th power map on the multiplicative group,
$$\nu_n \colon \mathbb{G}_m \to \mathbb{G}_m, \qquad x \mapsto x^n$$
dual to the k-algebra map $k[t, t^{-1}] \to k[x, x^{-1}]$ sending $t \mapsto x^n$. As a module map, $k[x, x^{-1}]$ is free of rank n over $k[t, t^{-1}]$ with basis $1, x, \ldots, x^{n-1}$, so ν_n is finite, flat of rank n, and of finite presentation; it is surjective because every nonzero scalar has an n-th root in an extension, so on points it hits every point of the target. Thus $\{\nu_n \colon \mathbb{G}_m \to \mathbb{G}_m\}$ is an *fppf cover.* Its ramification is controlled by the derivative: Ω of the extension is generated by dx with relation $nx^{n-1}\, dx = dt \mapsto 0$, which vanishes identically iff n is invertible in k. Hence ν_n is *étale exactly when* $\operatorname{char} k \nmid n$, and in that case the family is an étale cover. The fiber of ν_n over the identity is $\operatorname{Spec} k[x, x^{-1}]/(x^n - 1) = \operatorname{Spec} k[\mu_n]$, the group scheme μ_n of n-th roots of unity, and ν_n is a μ_n-torsor (section 6.5): $\mathbb{G}_m$ acts on itself covering ν_n, freely and transitively on fibers over the fppf cover. This cover is the geometric source of the Kummer sequence and of every μ_n-torsor computation to come.

The three covers isolate the three phenomena that make the fppf topology the right setting for descent. The first shows that joint surjectivity cannot be dropped: a topology is not a class of "nice" morphisms but a class of *surjective* families of them, and forgetting surjectivity would make every open immersion a cover of its target and collapse the theory. The second shows that the étale and fppf topologies see structure the Zariski topology cannot: a field, Zariski-irreducible and uncoverable, acquires genuine covers by finite extensions, and this is the precise sense in which Galois theory is a descent theory, developed in section 6.4. The third exhibits the cover from which μ_n-torsors are built, and it already displays the characteristic subtlety of the fppf topology: the map ν_n is a perfectly good fppf cover in every characteristic, but it is étale only away from n, so a torsor theory that insisted on étale covers would lose the interesting phenomena in characteristic dividing n. The fppf topology retains them. These covers recur throughout Part III as the concrete instances on which the abstract descent machinery is tested.

Exercise 6.1.1

1. Verify directly from definition 6.1.1 that the base-change axiom is consistent with the composition axiom in the following sense: if $\{U_i \to X\}$ is a covering family and $V \to X$ is any morphism, and if $\{W_{ij} \to U_i \times_X V\}$ is a covering family of each pullback, show that the composite $\{W_{ij} \to V\}$ is a covering family of V. Which two axioms did you use, and in what order?

2. Let k be a field. Show that the one-morphism family $\{\operatorname{Spec} k \xrightarrow{\mathrm{id}} \operatorname{Spec} k\}$ is a covering family

of $\operatorname{Spec} k$ in every one of the four topologies of definition 6.1.3, and deduce that any presheaf on $(\mathbf{Sch}/k)_\tau$ automatically satisfies the sheaf condition for this particular cover. What does this cover fail to tell you about the sheaf condition?

3. Consider the family $\{\operatorname{Spec} k[x, x^{-1}] \to \mathbb{A}^1,\ \operatorname{Spec} k[(x-1)^{-1}] \to \mathbb{A}^1\}$ of two open immersions into $\mathbb{A}^1 = \operatorname{Spec} k[x]$. Is it a Zariski cover? Is it an étale cover? Justify your answer using proposition 6.1.4, and identify the point of $\mathbb{A}^1$, if any, that is not in the image.

4. Let $L = k(\alpha)$ with $\alpha^p = a$ for a prime p equal to $\operatorname{char} k$ and $a \in k$ not a p-th power (a purely inseparable extension). Show that $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ is an fppf cover but *not* an étale cover. Contrast this with example 6.1.5(2), and explain in one sentence why the fppf topology is indispensable in positive characteristic.

5. For the Kummer cover $\nu_n \colon \mathbb{G}_m \to \mathbb{G}_m$ of example 6.1.5(3), compute the fiber product $\mathbb{G}_m \times_{\nu_n, \mathbb{G}_m, \nu_n} \mathbb{G}_m$ as an explicit affine scheme, and identify it with $\mathbb{G}_m \times \mu_n$. Interpret the two projections to $\mathbb{G}_m$ in terms of the group action; this fiber product is the object on which a descent datum for ν_n lives.

6. Explain why the sieve generated by a covering family $\{U_i \to X\}$ determines the sheaf condition, so that two covering families generating the same sieve impose the same condition on a presheaf. Illustrate with the two Zariski covers $\{U_1, U_2\}$ and $\{U_1, U_2, U_1 \cap U_2\}$ of a space $X = U_1 \cup U_2$: show they generate the same sieve.

6.2 Sheaves on a Site

The previous section built the covers. A Grothendieck topology on a category $\mathcal{C}$ selects, for each object, the families $\{U_i \to X\}$ that count as covers of it, and definition 6.1.3 supplied the four topologies this book uses on $(\mathbf{Sch}/S)$: Zariski, étale, smooth, and fppf. A covering family is machinery with a purpose, and the purpose is gluing. On an ordinary topological space the data that glues is a sheaf: an assignment of sets to open sets that is determined by its restrictions to any open cover, subject to a compatibility condition on overlaps. This section transports that notion to an arbitrary site, where overlaps become fiber products $U_i \times_X U_j$ and the compatibility condition becomes an equalizer diagram.

Three results organize the section. First, the sheaf condition itself, recast as an equalizer, together with the weaker separatedness condition that a presheaf may satisfy on the way to being a sheaf. Second, the sheafification functor, which forces any presheaf to become a sheaf in the universal way; it is the left adjoint to the inclusion of sheaves into presheaves, and the plus construction builds it explicitly, two applications sufficing. Third, and the payoff for moduli theory, Grothendieck's theorem that every representable functor $h_T = \operatorname{Hom}_S(-, T)$ is already a sheaf in the fppf topology. That theorem is why a moduli functor built out of representable pieces has a chance of being a sheaf, and its proof is faithfully flat descent of morphisms in disguise. The final example computes that $\mathbb{G}_m$, μ_n, and GL_n are fppf sheaves while a naive quotient presheaf is not, isolating the exact place where sheafification becomes unavoidable.

Throughout, $\mathcal{C}$ denotes a site: a category equipped with a Grothendieck topology, its covering families written $\{U_i \to X\}$. The reader should keep two examples in view, the small étale site $X_{\text{ét}}$ and the big site $(\mathbf{Sch}/S)_{\text{fppf}}$, and should read every abstract statement against them.

A presheaf is the raw assignment of data to objects, with no gluing required; it is a functor, nothing more. The sheaf condition is what turns local data into global data. On a space, a section over an open set U is recovered from its restrictions to the members of a cover of U, provided those restrictions agree on pairwise intersections. On a site the pairwise intersection $U_i \cap U_j$ has no meaning, but its correct replacement is the fiber product $U_i \times_X U_j$, and the correct formulation of "the restrictions agree" and "recovered from its restrictions" is a single equalizer diagram.

Definition 6.2.1: Presheaf and Sheaf on a Site

Let $\mathcal{C}$ be a site. A *presheaf* on $\mathcal{C}$ (of sets) is a functor $F\colon \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$; the image of a morphism $g\colon V \to U$ is the *restriction map* $F(g)\colon F(U) \to F(V)$, written $s \mapsto s|_V$. A presheaf of abelian groups is a functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Ab}$.

A presheaf F is a *sheaf* if for every covering family $\{U_i \to X\}$ in $\mathcal{C}$ the diagram

$$F(X) \xrightarrow{e} \prod_i F(U_i) \overset{\mathrm{pr}_1^*}{\underset{\mathrm{pr}_2^*}{\rightrightarrows}} \prod_{i,j} F(U_i \times_X U_j) \tag{6.1}$$

is an equalizer of sets: e sends $s \in F(X)$ to the family $(s|_{U_i})_i$, the two parallel maps send $(s_i)_i$ to $(s_i|_{U_i\times_X U_j})_{i,j}$ and $(s_j|_{U_i\times_X U_j})_{i,j}$ respectively, and e identifies $F(X)$ with the subset of $\prod_i F(U_i)$ on which the two parallel maps agree.

A presheaf F is *separated* if for every covering family the map e in (6.1) is injective (equivalently, a section is determined by its restrictions to a cover, even if not every compatible family arises from one).

Unwinding the equalizer restores the two familiar sheaf axioms. That e is injective is *separatedness*: two global sections with the same restrictions to a cover are equal. That the image of e is exactly the equalizer subset is the *gluing* axiom: any family (s_i) of local sections that agrees on overlaps, meaning $s_i|_{U_i\times_X U_j} = s_j|_{U_i\times_X U_j}$ for all i, j, comes from a unique global section. A sheaf is a separated presheaf in which every compatible family glues; a separated presheaf need only enforce uniqueness, not existence. The single diagram (6.1) packages both axioms because an equalizer is by definition the universal solution to "equalize the two maps," and the universal solution being $F(X)$ itself is precisely the demand that global sections biject with compatible families.

The fiber products play the role intersections play on a space, and the difference is the entire content of descent theory. On a space, $U_i \cap U_j$ is the set-theoretic intersection, and a point of it is a point lying in both U_i and U_j. On a site with a cover $\{U_i \to X\}$ by, say, étale or flat maps, the map $U_i \to X$ need not be injective, so $U_i \times_X U_j$ can be strictly larger than any naive intersection; over a Galois cover $\operatorname{Spec} L \to \operatorname{Spec} k$ it is $\operatorname{Spec}(L \otimes_k L)$, which for L/k Galois of degree n is a product of n copies of L. The extra components are exactly what records the automorphisms, and they are why a sheaf on the étale or fppf site sees more than a Zariski sheaf does.

Example 6.2.2: The Structure Presheaf and a Constant Presheaf

Two presheaves on the big Zariski or étale site $(\mathbf{Sch}/S)_\tau$ illustrate the definition.

1. *The structure presheaf.* Send a scheme $T \to S$ to the ring $\mathcal{O}(T) = \Gamma(T, \mathcal{O}_T)$ of global functions, and a morphism to the induced pullback of functions. This is the functor $\mathbb{A}^1_S = \operatorname{Hom}_S(-, \mathbb{A}^1_S)$, the affine line viewed as a functor of points, since $\operatorname{Hom}_S(T, \mathbb{A}^1_S) = \mathcal{O}(T)$. It is a sheaf in every topology considered here, a fact that is the base case of theorem 6.2.4 below: global functions on T are recovered from their restrictions to any flat cover, and a compatible family of functions on the cover that agrees on the double overlaps glues to a function on T.

2. *A constant presheaf that is not a sheaf.* Fix a set A with at least two elements and send every nonempty scheme to A and the empty scheme to a point, with all restriction maps the identity. This presheaf is separated but fails gluing already in the Zariski topology: if $T = T_1 \sqcup T_2$ is a disjoint union of two nonempty schemes, the cover $\{T_1 \to T,\ T_2 \to T\}$ has $T_1 \times_T T_2 = \emptyset$, so *every* pair $(a_1, a_2) \in A \times A$ is compatible (the overlap condition is vacuous). The map $e\colon F(T) = A \to F(T_1) \times F(T_2) = A \times A$ is the diagonal $a \mapsto (a, a)$, which is injective, so F is separated; but its image is only the diagonal, so $F(T) = A$ cannot surject onto the equalizer $A \times A$, and gluing fails. The sheafification of this presheaf is the sheaf $T \mapsto A^{\pi_0(T)}$ of locally constant A-valued functions, recording one value of A per connected component; this is the constant sheaf $\underline{A}$.

The second item is the prototype of everything that follows: a presheaf can carry the right data on connected schemes yet fail to glue across a disjoint union, and the cure is to enlarge $F(T)$ so that it records a value per component. That enlargement is sheafification, and its universal character is what the next theorem establishes.

Not every presheaf is a sheaf, but every presheaf has a best sheaf approximation. The construction that produces it must be universal: any map from F to a sheaf should factor uniquely through the approximation, so that no information is lost and none is invented beyond what gluing forces. This universal property is an adjunction, the sheafification functor being left adjoint to the forgetful inclusion of sheaves into presheaves. The adjunction is not an abstract certificate alone; it is what lets one compute in the category of sheaves by lifting constructions from presheaves and sheafifying the result, and it is the reason the category of sheaves inherits limits and (after sheafification) colimits from presheaves. The explicit construction is the *plus construction*, which replaces sections by compatible families modulo refinement; applied once it separates, applied twice it sheafifies.

Theorem 6.2.3: Sheafification

Let $\mathcal{C}$ be a site and let $\mathrm{PSh}(\mathcal{C})$ and $\mathrm{Sh}(\mathcal{C})$ denote the categories of presheaves and sheaves of sets on $\mathcal{C}$. The inclusion $\iota\colon \mathrm{Sh}(\mathcal{C}) \hookrightarrow \mathrm{PSh}(\mathcal{C})$ admits a left adjoint

$$a\colon \mathrm{PSh}(\mathcal{C}) \longrightarrow \mathrm{Sh}(\mathcal{C}) \qquad \text{(sheafification)}$$

so that for every presheaf F and every sheaf G there is a natural bijection

$$\mathrm{Hom}_{\mathrm{Sh}(\mathcal{C})}(aF, G) \;\cong\; \mathrm{Hom}_{\mathrm{PSh}(\mathcal{C})}(F, \iota G) \tag{6.2}$$

The sheafification aF is constructed as F^{++}, where $(-)^{+}$ is the plus construction: F^{+} is always separated, and F^{+} is a sheaf whenever F is separated, so two applications yield a sheaf. There is a canonical map $\theta\colon F \to aF$, universal among maps from F to sheaves, and a preserves finite limits.

***Proof*:**
The plus construction and the two lemmas about it carry the whole argument; the adjunction (6.2) is then a formal consequence of the universal property of θ, which we record by citing the general adjoint-from-universal-arrows bookkeeping of [5].

Step 1: The plus construction. For an object $X \in \mathcal{C}$ and a covering family $\mathcal{U} = \{U_i \to X\}$, call a family $(s_i) \in \prod_i F(U_i)$ *compatible* if $s_i|_{U_i \times_X U_j} = s_j|_{U_i \times_X U_j}$ for all i, j, and write $\check{H}^0(\mathcal{U}, F) \subseteq \prod_i F(U_i)$ for the set of compatible families, which is the equalizer of the two parallel maps in

(6.1). If $\mathcal{V} = \{V_a \to X\}$ refines $\mathcal{U}$, meaning each $V_a \to X$ factors through some $U_{i(a)} \to X$, restriction along the refinement induces a map $\check{H}^0(\mathcal{U}, F) \to \check{H}^0(\mathcal{V}, F)$ that is independent of the chosen factorizations (two factorizations of $V_a \to X$ through U_i agree after pulling back the compatible-family condition to $V_a \times_X V_b$). The covering families of X form a filtered system under refinement, and one sets

$$F^+(X) \;:=\; \varinjlim_{\mathcal{U}} \check{H}^0(\mathcal{U}, F)$$

the colimit over that system. A morphism $g\colon Y \to X$ pulls a cover of X back to a cover of Y and so induces $F^+(g)\colon F^+(X) \to F^+(Y)$, making F^+ a presheaf. The map $F(X) \to \check{H}^0(\mathcal{U}, F)$ sending $s \mapsto (s|_{U_i})$ is compatible with refinement, so it induces a natural transformation $F \to F^+$.

Step 2: F^+ is separated. Let $\{U_i \to X\}$ be a cover and let $t, t' \in F^+(X)$ have equal restrictions to each U_i. Represent t, t' by compatible families over covers $\mathcal{U}_t, \mathcal{U}_{t'}$ of X; passing to a common refinement $\mathcal{V} = \{V_a \to X\}$ (the family of all $V_a \to X$ that factor through both a member of $\mathcal{U}_t$ and a member of $\mathcal{U}_{t'}$, a cover because covers are stable under the base-change and composition axioms of definition 6.1.3), represent both t and t' by compatible families $(s_a), (s'_a) \in \check{H}^0(\mathcal{V}, F)$. The hypothesis $t|_{U_i} = t'|_{U_i}$ means that after refining $\mathcal{V}$ further so each V_a factors through some U_i, the representatives agree: $s_a = s'_a$ for every a in this common refinement. Hence $t = t'$ in the colimit $F^+(X)$, so F^+ is separated. This uses only that any two covers have a common refinement, which the site axioms guarantee.

Step 3: F^+ is a sheaf when F is separated. Assume F separated. Separatedness of F^+ is Step 2, so it remains to glue. Let $\{U_i \to X\}$ be a cover and $(t_i) \in \prod_i F^+(U_i)$ a compatible family. Each t_i is represented by a compatible family over some cover $\{U_{ij} \to U_i\}$; the composite family $\{U_{ij} \to X\}$ is a cover of X by the composition axiom. Because F is separated, the representatives of t_i and $t_{i'}$ agree on the overlaps $U_{ij} \times_X U_{i'j'}$ (their images in F^+ agree by the compatibility of the t_i, and separatedness of F upgrades the F^+-agreement to equality of the underlying F-sections after a further common refinement). The resulting family over $\{U_{ij} \to X\}$ is therefore compatible and defines an element $t \in F^+(X)$ whose restriction to U_i is t_i; separatedness of F^+ makes t unique. Thus F^+ satisfies the sheaf condition.

Step 4: Sheafification and the universal property. For any presheaf F, apply the plus construction twice. By Step 2 the presheaf F^+ is separated; by Step 3, $F^{++} = (F^+)^+$ is a sheaf. Set $aF := F^{++}$ and let $\theta\colon F \to F^+ \to F^{++} = aF$ be the composite of the two canonical maps. Given any sheaf G and any morphism of presheaves $\varphi\colon F \to G$, the plus construction is functorial and $G^+ \cong G$ canonically because G is a sheaf (its compatible families glue uniquely, so the colimit defining $G^+(X)$ collapses to $G(X)$). Hence φ induces $F^{++} \to G^{++} \cong G$, giving a factorization $\varphi = \psi \circ \theta$ with $\psi\colon aF \to G$. Uniqueness of ψ follows because θ has dense image in the sense that any two maps $aF \to G$ agreeing after precomposition with θ agree on every compatible family, hence on aF, using separatedness of G. This is exactly the universal arrow from F to ι, so by the standard criterion ([5]) the assignment $F \mapsto aF$ is left adjoint to ι, establishing the natural bijection (6.2).

Step 5: Left exactness. Left exactness is checked objectwise, and finite limits of presheaves are computed objectwise, so it suffices to show that each $F \mapsto F^+(X)$ preserves finite limits. Fix a finite diagram of presheaves. For a cover $\mathcal{U}$ the assignment $F \mapsto \check{H}^0(\mathcal{U}, F)$ is a limit (an equalizer of two maps between products), hence commutes with the finite limit; it is not itself a *finite* limit when $\mathcal{U}$ has infinitely many members, but limits commute with limits, so this poses no obstruction. Passing to the colimit, $F^+(X) = \varinjlim_{\mathcal{U}} \check{H}^0(\mathcal{U}, F)$ is a filtered colimit by Step 2's common-refinement fact, and filtered colimits commute with finite limits in **Set**. Composing the two, $(-)^+$ preserves finite limits; iterating, $a = (-)^{++}$ preserves finite limits. Therefore a is exact on the left, completing the proof.

The adjunction is the working characterization of sheafification: aF is the initial sheaf receiving a map from F, so it adds exactly the gluing F was missing and nothing more. The plus construction makes this concrete by replacing each $F(X)$ with the colimit of compatible families over finer and finer covers, which is the operation of "allowing sections that are only locally defined." Left exactness of a matters downstream: it is why kernels of sheaf maps are computed presheaf-wise while cokernels require sheafification, and it is the technical fact that makes $\mathrm{Sh}(\mathcal{C})$ an abelian category when $\mathcal{C}$ carries abelian-group-valued sheaves, the setting in which the cohomology $H^i(X_\tau, -)$ of the final two sections lives. The étale and fppf topologies produce genuinely different sheafifications of the same presheaf, and the gap between them is where the arithmetic of a scheme resides.

The plus construction also settles a subtlety flagged by the disjoint-union example. The presheaf F of example 6.2.2(2) is separated (its map e is the injective diagonal), so by Step 3 a *single* plus already sheafifies it: F^+ is the sheaf $T \mapsto A^{\pi_0(T)}$ of locally constant functions, which records one value per connected component and glues across disjoint unions. Two applications are necessary only when the presheaf is not already separated: the first plus then enforces uniqueness and the second enforces existence, and neither can be skipped for a presheaf that fails both axioms. The general principle that $(-)^{++}$, not $(-)^+$, is what always yields a sheaf is thus not decorative, even though the separated example above needs only the one application.

The point of the whole apparatus, for moduli theory, is that the functors one wants to represent geometric objects should be sheaves. A moduli functor $F\colon (\mathbf{Sch}/S)^{\mathrm{op}} \to \mathbf{Set}$ that is not a sheaf cannot be represented by a scheme, because a scheme's functor of points *is* a sheaf. The next theorem is the source of that last fact, and it is Grothendieck's: the functor of points of any scheme is a sheaf not only in the Zariski topology, where it is elementary, but in the fppf topology, where it is the descent theorem of theorem 6.4.1 rephrased. It is the single most-used sheaf statement in the book, because it certifies that every representable piece of a moduli problem already satisfies the gluing every later construction needs.

Theorem 6.2.4: Representable Functors Are fppf Sheaves

Let S be a scheme and T an S-scheme. The representable presheaf

$$h_T = \mathrm{Hom}_S(-, T)\colon (\mathbf{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

is a sheaf in the fppf topology on $(\mathbf{Sch}/S)$. In particular it is a sheaf in the smaller étale and Zariski topologies. The same statement holds in the fpqc topology.

***Proof*:**
By the definition of the fppf topology (definition 6.1.3) it suffices to check the sheaf condition for a covering family, and because h_T carries finite disjoint unions to products, it suffices to treat a *single* surjective, flat, locally-finitely-presented morphism $U \to X$ of S-schemes; a general fppf cover $\{U_i \to X\}$ reduces to the single cover $U = \coprod_i U_i \to X$, which is again fppf. The fpqc strengthening is addressed at the end.

Step 1: Reduction to the equalizer for one cover. For the single cover $U \to X$, the sheaf condition

(6.1) reads: the sequence

$$h_T(X) \xrightarrow{e} h_T(U) \overset{\mathrm{pr}_1^*}{\underset{\mathrm{pr}_2^*}{\rightrightarrows}} h_T(U \times_X U) \tag{6.3}$$

is an equalizer of sets, where $\mathrm{pr}_1, \mathrm{pr}_2 \colon U \times_X U \to U$ are the two projections. Concretely, an element of $h_T(U)$ is an S-morphism $f \colon U \to T$, and the equalizer condition splits into two claims:

1. *Separatedness* (e injective): if $g, g' \colon X \to T$ satisfy $g \circ (U \to X) = g' \circ (U \to X)$, then $g = g'$.
2. *Gluing* (image of e is the equalizer): if $f \colon U \to T$ satisfies $f \circ \mathrm{pr}_1 = f \circ \mathrm{pr}_2$ as morphisms $U \times_X U \to T$, then f descends, meaning $f = g \circ (U \to X)$ for a unique S-morphism $g \colon X \to T$.

Step 2: Separatedness. Since $U \to X$ is fppf it is in particular faithfully flat and (being surjective) an epimorphism in the category of schemes for the purpose at hand: two morphisms $g, g' \colon X \to T$ agreeing after composition with a faithfully flat surjection are equal. This is the geometric shadow of the algebraic fact that a faithfully flat ring map $R \to R'$ is injective, so $g^\sharp, g'^\sharp \colon \mathcal{O}_T \to g_*\mathcal{O}_X$ are equal once they become equal after the injection $\mathcal{O}_X \hookrightarrow (U \to X)_*\mathcal{O}_U$; faithful flatness gives the injection ([6]). Hence e is injective and claim (1) holds. This separatedness is in fact already contained in the full descent equalizer invoked in Step 3, which yields injectivity and surjectivity of e onto the equalizer at once; Step 2 records the injective half separately because it has the transparent geometric reading above and isolates the one place faithful flatness enters directly.

Step 3: Gluing is descent of morphisms. Claim (2) is exactly the assertion that morphisms to T descend along the fppf cover $U \to X$. This is the content of theorem 6.4.1, proved in full in section 6.4: for S-schemes Y, Z the presheaf $R \mapsto \mathrm{Hom}_R(Y_R, Z_R)$ on the fppf site is a sheaf, equivalently an X-morphism is the same datum as a U-morphism whose two pullbacks to $U \times_X U$ coincide. Applying that theorem with $Y = X$ (as an X-scheme via the identity) and $Z = T \times_S X$ (so that $\mathrm{Hom}_X(X, Z) = \mathrm{Hom}_S(X, T) = h_T(X)$ and $\mathrm{Hom}_U(U, Z_U) = \mathrm{Hom}_S(U, T) = h_T(U)$), the descent equalizer for morphisms is precisely (6.3). A morphism $f \colon U \to T$ with $f \circ \mathrm{pr}_1 = f \circ \mathrm{pr}_2$ is a U-point of T whose two pullbacks to $U \times_X U$ agree, so it descends uniquely to an X-point $g \colon X \to T$ with $g \circ (U \to X) = f$. This establishes claim (2), and with Step 2 the sequence (6.3) is an equalizer. Therefore h_T is an fppf sheaf.

Step 4: The dependency and the fpqc case. The proof rests entirely on theorem 6.4.1: the gluing half of the sheaf condition for h_T *is* descent of morphisms, and no independent argument is needed. We record this dependency explicitly because the two theorems are stated in different sections and it is descent of morphisms, not representability, that supplies the mathematical work. The Zariski and étale cases follow because every Zariski or étale cover is an fppf cover (definition 6.1.3), so a sheaf for the finer fppf topology is a sheaf for the coarser ones. The fpqc strengthening replaces "locally of finite presentation" by "quasi-compact," and its proof requires the same descent of morphisms along a faithfully flat quasi-compact cover together with a limit argument managing the loss of finite presentation; the set-theoretic subtleties of the fpqc topology (it is not a small site in the naive sense) place its full treatment beyond this book's framework, and we cite [29] for the fpqc statement in complete generality, completing the proof.

The theorem is the load-bearing beam of everything that follows. Every moduli functor this book studies is assembled from representable functors h_T by limits, quotients, and gluing; theorem 6.2.4 certifies that the raw material is already fppf-local, so the only obstacle to a moduli functor being a sheaf is what the assembly does to it. When a quotient or a colimit destroys the sheaf property, the repair is sheafification (theorem 6.2.3), and the resulting sheaf is the first candidate for a representing object. This is the precise sense in which descent is the technical foundation of Part III: to represent a moduli problem one first makes it a sheaf, and representable functors are where the sheaf property begins.

The reduction in the proof is worth isolating. Descent of morphisms (theorem 6.4.1) is proved by faithfully flat descent of the graph of the morphism, so theorem 6.2.4 is downstream of the module-level descent of section 6.3. The logical order is: modules descend, therefore morphisms descend, therefore representable functors are sheaves. Grothendieck's insight was that the representability question and the sheaf question are governed by the same faithfully flat descent, and this single flow is what makes the fppf topology, rather than the Zariski, the natural home of moduli.

The abstract theorems become concrete on the group schemes that recur throughout the book. The multiplicative group $\mathbb{G}_m$, the group of n-th roots of unity μ_n, and the general linear group GL_n are all representable, so theorem 6.2.4 makes each an fppf sheaf without further work; the value of computing them explicitly is to see the equalizer condition satisfied by hand and, by contrast, to watch a naive quotient presheaf fail it. The failure is not a pathology but the mechanism: quotients of sheaves of groups are the source of torsors and cohomology, and the gap between the presheaf quotient and its sheafification is exactly the first cohomology group that the next sections compute.

Example 6.2.5: $\mathbb{G}_m$, μ_n, GL_n, and a Quotient That Fails
Work over a base scheme S in the fppf topology on $(\mathbf{Sch}/S)$.

1. *The multiplicative group.* $\mathbb{G}_m = \operatorname{Spec} \mathbb{Z}[t, t^{-1}] \times_{\operatorname{Spec} \mathbb{Z}} S$ represents the functor $T \mapsto \mathcal{O}(T)^\times$, the units of the global function ring. It is representable, hence an fppf sheaf by theorem 6.2.4. Directly: for a cover $U \to X$, a unit $u \in \mathcal{O}(U)^\times$ with $\mathrm{pr}_1^* u = \mathrm{pr}_2^* u$ in $\mathcal{O}(U \times_X U)^\times$ descends to a unit on X because $\mathcal{O}(X)^\times$ is the equalizer of $\mathcal{O}(U)^\times \rightrightarrows \mathcal{O}(U \times_X U)^\times$, which is faithfully flat descent of the structure sheaf restricted to units.

2. *Roots of unity.* $\mu_n = \operatorname{Spec} \mathbb{Z}[t]/(t^n - 1) \times S$ represents $T \mapsto \{u \in \mathcal{O}(T)^\times \,|\, u^n = 1\}$, the n-th roots of unity in $\mathcal{O}(T)$. It is the kernel of the n-th power map $\mathbb{G}_m \xrightarrow{x \mapsto x^n} \mathbb{G}_m$, a representable subfunctor, hence an fppf sheaf. Kernels of maps of sheaves of groups are computed pointwise and remain sheaves, which confirms the equalizer condition without a separate check.

3. *The general linear group.* $\mathrm{GL}_n = \operatorname{Spec} \mathbb{Z}[x_{ij}, \det(x_{ij})^{-1}] \times S$ represents $T \mapsto \mathrm{GL}_n(\mathcal{O}(T))$, the invertible $n \times n$ matrices over $\mathcal{O}(T)$. Representable, hence an fppf sheaf; the case $n = 1$ recovers $\mathbb{G}_m$.

4. *A quotient presheaf that is not a sheaf.* Consider the presheaf Q on $(\mathbf{Sch}/S)_{\mathrm{fppf}}$ sending T to the naive quotient
$$Q(T) = \mathcal{O}(T)^\times / \big(\mathcal{O}(T)^\times\big)^n$$
the cokernel presheaf of $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$, formed group-by-group. This Q is a presheaf of abelian groups but is *not* an fppf sheaf. Concretely, take $S = \operatorname{Spec} k$ with k a field containing a primitive n-th root of unity and $\operatorname{char} k \nmid n$, and take $X = \mathbb{G}_{m,k} = \operatorname{Spec} k[x, x^{-1}]$. The class

of the coordinate $x \in \mathcal{O}(X)^\times$ is nonzero in $Q(X)$ because x is not an n-th power in $k[x, x^{-1}]$. Pull back along the fppf cover $U = \operatorname{Spec} k[y, y^{-1}] \to X$, $x \mapsto y^n$ (the n-th power map, finite flat of degree n): on U the pullback of x is y^n, which *is* an n-th power, so the image of x in $Q(U)$ is 0. A nonzero section of $Q(X)$ restricting to 0 on a cover violates separatedness, so Q fails the sheaf condition of definition 6.2.1.

The last item is the entire moral of the section in miniature. The presheaf cokernel of $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$ is not a sheaf because an element can become an n-th power fppf-locally without being one globally, which is the failure of separatedness observed on $x = y^n$. The sheafification of Q is the fppf sheaf $\mathbb{G}_m/\mathbb{G}_m^{[n]}$, and the discrepancy between an element being locally an n-th power and globally an n-th power is measured by a cohomology class: the connecting map in the long exact sequence of the Kummer sequence $1 \to \mu_n \to \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \to 1$ sends the class of x to a nontrivial element of $H^1(X_{\text{fppf}}, \mu_n)$, and that class is the μ_n-torsor $U = \operatorname{Spec} k[y, y^{-1}] \to X$ with $y^n = x$. The failure of the naive quotient to be a sheaf, the need for sheafification, and the appearance of a torsor are three faces of one phenomenon, and the sections that follow develop each face in turn: faithfully flat descent (section 6.3) proves the sheaf statements used here, and torsors and H^1 (section 6.5) identify the sheafification gap with a cohomology group.

A closing contrast sharpens the Zariski-versus-fppf distinction. Every functor above is already a Zariski sheaf, so passing to the fppf topology does not change whether $\mathbb{G}_m$, μ_n, or GL_n is a sheaf; it changes the *covers* against which the quotient Q is tested. Over the Zariski topology the cover $U \to X$ of item (4) is not available (the n-th power map is not a Zariski cover, being finite of degree n rather than an open immersion), so Q is closer to a sheaf Zariski-locally; it is precisely the finer fppf covers that detect the local n-th power and force the sheafification. This is why moduli problems are posed in the fppf or étale topology and not the Zariski: the finer topology sees the identifications, the automorphisms, and the local trivializations that the Zariski topology cannot.

Exercise 6.2.1

1. Let F be a presheaf of sets on a site $\mathcal{C}$. Show that F is separated in the sense of definition 6.2.1 if and only if the map $\theta\colon F \to F^+$ of the plus construction is injective on sections over every object, and that F is a sheaf if and only if θ is an isomorphism. Deduce that a sheaf G satisfies $G^+ \cong G$, the fact used in Step 4 of the proof of theorem 6.2.3.

2. Fix a finite Galois extension L/k with group $\Gamma = \operatorname{Gal}(L/k)$ and work on the small étale site of $\operatorname{Spec} k$. Compute the fiber product $\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L$ as a scheme, and show that it is isomorphic to $\coprod_{\gamma \in \Gamma} \operatorname{Spec} L$, a disjoint union of $[L : k]$ copies of $\operatorname{Spec} L$ indexed by Γ. Explain, in one or two sentences, why this makes the sheaf condition (6.1) for the cover $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ into a condition involving the Galois action, and relate it to the extra components mentioned after definition 6.2.1.

3. Let $\{U_i \to X\}$ be a covering family in which one member $U_{i_0} \to X$ is an isomorphism. Show directly from definition 6.2.1 that every presheaf F satisfies the sheaf condition for this particular cover, so that a cover containing an isomorphism imposes no condition. (This is why the topology only needs to be tested on "genuine" covers.)

4. Give a presheaf on the big Zariski site of $\operatorname{Spec} \mathbb{Z}$ that is separated but not a sheaf. *(Hint: take a subpresheaf of a sheaf whose sections satisfy a global condition that is not local, for example the presheaf $T \mapsto \{f \in \mathcal{O}(T) \,|\, f$ is constant on $T\}$ of globally constant functions, and exhibit a cover on which local constants fail to glue to a global constant.)*

5. Using theorem 6.2.4, prove that the fiber product of two representable functors $h_T \times_{h_S} h_{T'}$ is a sheaf in the fppf topology, first by identifying it with a representable functor and second by a general argument that limits of sheaves are sheaves. Which of the two arguments also shows that an arbitrary (not necessarily finite) product of fppf sheaves is a sheaf?

6. Return to the quotient presheaf $Q(T) = \mathcal{O}(T)^\times/(\mathcal{O}(T)^\times)^n$ of example 6.2.5(4), and pin down exactly why the fppf cover $U = \operatorname{Spec} k[y, y^{-1}] \to X = \operatorname{Spec} k[x, x^{-1}]$, $x \mapsto y^n$, detects the failure of separatedness while no Zariski open cover of X can. First, for a Zariski open cover $\{U_i \hookrightarrow X\}$ of an integral scheme X, verify that the class of a unit $u \in \mathcal{O}(X)^\times$ restricting to 0 in every $Q(U_i)$ need not itself be 0 in $Q(X)$: the local n-th roots v_i with $u|_{U_i} = v_i^n$ satisfy $(v_i/v_j)^n = 1$ on $U_i \cap U_j$, so they define a Čech 1-cocycle valued in μ_n, and u is a global n-th power exactly when this cocycle is a coboundary, that is, when its class in $\check{H}^1(\{U_i\}, \mu_n)$ vanishes. Conclude that Zariski separatedness of Q is equivalent to the vanishing of $\check{H}^1_{\mathrm{Zar}}(X, \mu_n)$ and is therefore *not* automatic. Second, contrast this with the fppf cover above: show that the two pullbacks of y to $U \times_X U$ differ by the μ_n-action $y \mapsto \zeta y$, so y defines a nontrivial descent datum and does not descend to an n-th root of x on X; identify this descent datum with the μ_n-torsor of example 6.2.5. Pinpoint the step that would use Zariski covers being open immersions, and explain why it is unavailable for the finite-flat cover $x \mapsto y^n$.

6.3 Faithfully Flat Descent

The previous two sections built the language: a Grothendieck topology axiomatizes what it means to cover a scheme, and a sheaf on a site is a functor whose values glue along covers. What remains is to make gluing *do work*. The Zariski topology lets one build a sheaf on X from sheaves on an open cover $\{U_i\}$ together with agreement on the overlaps $U_i \cap U_j$. Descent theory is the assertion that the same reconstruction succeeds for the finer topologies of definition 6.1.3, where the "overlaps" are not intersections inside X but fiber products $U_i \times_X U_j$ that need not embed in X at all. The engine that makes this work is faithful flatness, and the section proves the foundational theorem of the whole chapter: a quasi-coherent sheaf on X is the same datum as a quasi-coherent sheaf on a faithfully flat cover $U \to X$ equipped with a compatible identification of its two pullbacks to $U \times_X U$.

This theorem is the reason Part III can exist. Every later construction in the book depends on it. Grothendieck's theorem that representable functors are fppf sheaves (theorem 6.2.4) reduces to descent; the stack condition of the next chapter is descent for objects with automorphisms; and the very definition of an algebraic space or an algebraic stack presupposes that quasi-coherent sheaves, morphisms, and suitable schemes can be reconstructed from local data in the fppf or étale topology. The heart of the matter is a single exactness statement in commutative algebra, the exactness of the Amitsur complex of a faithfully flat ring map, which the section proves in full and then transports to schemes. The theorem is due to Grothendieck, who developed it in the FGA seminar [15]; the treatment here follows the modern account in [30].

Throughout, $U \to X$ denotes a single covering morphism (rather than a family $\{U_i \to X\}$; a family is recovered by taking $U = \coprod_i U_i$, and the two formulations are equivalent). The two projections $U \times_X U \to U$ are pr_1 and pr_2, and the three projections $U \times_X U \times_X U \to U \times_X U$ are $\mathrm{pr}_{12}, \mathrm{pr}_{13}, \mathrm{pr}_{23}$, where pr_{ij} forgets the factor not named. The first task is to say precisely what data on U should reconstruct a sheaf on X.

The naive hope is that a quasi-coherent sheaf on X is the same as a quasi-coherent sheaf on U. This is far too much to ask: pulling a sheaf back to U loses information, because $U \to X$ is not an isomorphism. What one gains on U must be paid for by a gluing datum recording how the sheaf on U is to be reassembled. Over the Zariski cover the gluing datum is the requirement that the restrictions to U agree on double overlaps; over a faithfully flat cover the double overlap becomes $U \times_X U$, and "agreement" becomes an isomorphism between the two pullbacks $\mathrm{pr}_1^* \mathcal{F}_U$ and $\mathrm{pr}_2^* \mathcal{F}_U$. Compatibility of this isomorphism on triple overlaps is the cocycle condition. This package is a descent datum.

Definition 6.3.1: Descent Datum for Quasi-Coherent Sheaves

Let $U \to X$ be a morphism of schemes. A *descent datum* for quasi-coherent sheaves on $U \to X$ is a pair $(\mathcal{F}, \varphi)$ consisting of a quasi-coherent sheaf $\mathcal{F}$ on U and an isomorphism of $\mathcal{O}_{U\times_X U}$-modules

$$\varphi\colon\ \mathrm{pr}_1^* \mathcal{F} \xrightarrow{\sim} \mathrm{pr}_2^* \mathcal{F}$$

on $U \times_X U$, subject to the *cocycle condition*: on $U \times_X U \times_X U$ the diagram of pullbacks

$$\mathrm{pr}_{13}^* \varphi = \mathrm{pr}_{23}^* \varphi \circ \mathrm{pr}_{12}^* \varphi$$

holds as an equality of isomorphisms $\mathrm{pr}_1^* \mathcal{F} \to \mathrm{pr}_3^* \mathcal{F}$, where $\mathrm{pr}_i\colon U \times_X U \times_X U \to U$ is the projection to the i-th factor and each $\mathrm{pr}_{ij}^* \varphi$ is interpreted through the canonical identifications $\mathrm{pr}_{ij}^* \mathrm{pr}_a^* \cong \mathrm{pr}_b^*$.
A *morphism of descent data* $(\mathcal{F}, \varphi) \to (\mathcal{G}, \psi)$ is a morphism $f\colon \mathcal{F} \to \mathcal{G}$ of $\mathcal{O}_U$-modules commuting with the gluing isomorphisms, that is, $\psi \circ \mathrm{pr}_1^* f = \mathrm{pr}_2^* f \circ \varphi$. Descent data on $U \to X$ and their morphisms form a category, written $\mathrm{Desc}(U \to X)$.

The cocycle condition is the precise sense in which the gluing is consistent. To read it, note that $\mathrm{pr}_{12}^* \varphi$ glues the first pullback to the second, $\mathrm{pr}_{23}^* \varphi$ glues the second to the third, and $\mathrm{pr}_{13}^* \varphi$ glues the first to the third directly; the condition demands that gluing first-to-second and then second-to-third agrees with gluing first-to-third in one step. Setting the second and third factors equal (pulling back along the diagonal $U \times_X U \to U \times_X U \times_X U$) and using the condition once forces φ to restrict to the identity on the diagonal $\Delta\colon U \to U \times_X U$, so a normalization $\Delta^*\varphi = \mathrm{id}_{\mathcal{F}}$ is automatic rather than an extra axiom. This is exactly the structure of transition functions for a vector bundle, promoted from a Zariski cover to an arbitrary cover: φ plays the role of the transition isomorphisms g_{ij}, and the cocycle condition is the familiar $g_{ik} = g_{jk} g_{ij}$.

There is a canonical source of descent data. Any quasi-coherent sheaf on X can be pulled back to U, and its two pullbacks to $U \times_X U$ are canonically isomorphic because both projections $U \times_X U \to U \to X$ agree after composition with $U \to X$. This produces the pullback functor into descent data, whose essential surjectivity, the statement that *every* descent datum arises this way, is the theorem the section is built to prove.

Definition 6.3.2: The Pullback Functor

Let $p\colon U \to X$ be a morphism. The *pullback functor*

$$p^*\colon \mathrm{QCoh}(X) \longrightarrow \mathrm{Desc}(U \to X)$$

sends a quasi-coherent sheaf $\mathcal{E}$ on X to the descent datum $(p^*\mathcal{E}, \varphi_{\mathrm{can}})$, where $\varphi_{\mathrm{can}}\colon\ \mathrm{pr}_1^* p^*\mathcal{E} \to \mathrm{pr}_2^* p^*\mathcal{E}$ is the canonical isomorphism arising from the equality $p \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2\colon U \times_X U \to X$. It sends a morphism $\mathcal{E} \to \mathcal{E}'$ to its pullback $p^*\mathcal{E} \to p^*\mathcal{E}'$, which automatically commutes with the canonical gluings.

The canonical isomorphism φ_{can} satisfies the cocycle condition because all three faces of the triple product agree after composition to X: on $U \times_X U \times_X U$ the three morphisms $\mathrm{pr}_1, \mathrm{pr}_2, \mathrm{pr}_3$ to U all become the same morphism to X, so the canonical identifications compose canonically. Thus $p^*\mathcal{E}$ is indeed a descent datum. The content of the theorem below is that, when p is faithfully flat and quasi-compact, this functor is an equivalence: no information is lost by passing to U with its gluing, and no descent datum is exotic.

Before the general theorem, the case that carries every idea is the affine one. If $X = \operatorname{Spec} B$ and $U = \operatorname{Spec} A$ with $B \to A$ a ring map, then $U \times_X U = \operatorname{Spec}(A \otimes_B A)$ and $U \times_X U \times_X U = \operatorname{Spec}(A \otimes_B A \otimes_B A)$. A quasi-coherent sheaf on U is an A-module N; the two pullbacks to $U \times_X U$ are $A \otimes_B A \otimes_A N = A \otimes_B N$ (via pr_1, tensoring on the left) and $N \otimes_A A \otimes_B A = N \otimes_B A$ (via pr_2, tensoring on the right); and a descent datum is an isomorphism of $(A \otimes_B A)$-modules $\varphi\colon A \otimes_B N \to N \otimes_B A$ satisfying the cocycle condition on $A \otimes_B A \otimes_B A$. The whole theorem is the statement that, for $B \to A$ faithfully flat, such an N with its φ comes from a unique B-module M by $N = A \otimes_B M$, with φ the tautological swap. The bridge to this statement is one exact sequence.

The algebraic core is the exactness of the Amitsur complex. A faithfully flat ring map $B \to A$ makes every B-module M recoverable from $A \otimes_B M$ together with a coequalizer datum, and the precise form of this recovery is that the augmented complex below is exact. Recall from [6] that a ring map $B \to A$ is *faithfully flat* if it is flat and the induced map $\operatorname{Spec} A \to \operatorname{Spec} B$ is surjective; equivalently, A is flat over B and $M \otimes_B A = 0$ implies $M = 0$ for every B-module M. The two facts imported from there are that a faithfully flat map is injective (so $B \hookrightarrow A$) and that a sequence of B-modules is exact if and only if it becomes exact after $- \otimes_B A$.

Lemma 6.3.3: Exactness of the Amitsur Complex

Let $B \to A$ be a faithfully flat ring map and M a B-module. Write $A^{\otimes n}$ for the n-fold tensor power $A \otimes_B \cdots \otimes_B A$. Then the augmented *Amitsur complex*

$$0 \longrightarrow M \xrightarrow{\eta} M \otimes_B A \xrightarrow{d^0} M \otimes_B A \otimes_B A \xrightarrow{d^1} M \otimes_B A^{\otimes 3} \longrightarrow \cdots$$

is exact, where $\eta(m) = m \otimes 1$ and $d^n = \sum_{i=0}^{n+1} (-1)^i \delta_i$ with δ_i inserting a factor 1 in the i-th slot of the A-factors. In particular the first three terms form an exact sequence

$$0 \longrightarrow M \xrightarrow{\eta} M \otimes_B A \xrightarrow{d^0} M \otimes_B A \otimes_B A$$

with $d^0(m \otimes a) = m \otimes 1 \otimes a - m \otimes a \otimes 1$.

***Proof*:**
The strategy is the standard reduction that gives the complex its power: a faithfully flat map is exact-detecting, so it suffices to prove exactness after applying $- \otimes_B A$, and after that base change the complex acquires a contracting homotopy for a purely formal reason, namely that $B \to A$ acquires a section.

Step 1: Reduction to the split case. By faithful flatness ([6]), a complex of B-modules is exact if and only if it stays exact after applying $- \otimes_B A$. Tensor the Amitsur complex with A over B. Since $A \otimes_B (M \otimes_B A^{\otimes n}) = M \otimes_B A^{\otimes(n+1)}$, the base-changed complex is the Amitsur complex of M regarded over A for the ring map $A \to A \otimes_B A =: A'$ given by $a \mapsto 1 \otimes a$, computed with the A-module $M \otimes_B A$. Concretely, after base change the map $B \to A$ is replaced by the map

$A \to A'$, and this map is *split*: the multiplication map $\mu\colon A' = A \otimes_B A \to A$, $a \otimes a' \mapsto aa'$, is a ring map that is a retraction, $\mu \circ (a \mapsto 1 \otimes a) = \mathrm{id}_A$. Thus it suffices to prove that the Amitsur complex of a ring map $A \to A'$ possessing an A-algebra retraction $\mu\colon A' \to A$ is exact for every A'-module (here $M \otimes_B A$).

Step 2: The contracting homotopy from a section. Rename the split situation: let $R \to S$ be a ring map with an R-algebra retraction $\sigma\colon S \to R$ (so σ restricts to the identity on R), and let L be a module over the base. The complex to kill is

$$0 \to L \xrightarrow{\eta} L \otimes_R S \xrightarrow{d^0} L \otimes_R S \otimes_R S \xrightarrow{d^1} \cdots$$

Define R-linear maps $h^n\colon L \otimes_R S^{\otimes(n+1)} \to L \otimes_R S^{\otimes n}$ by applying σ to the last tensor factor with an alternating sign,

$$h^n\big(\ell \otimes s_1 \otimes \cdots \otimes s_{n+1}\big) = (-1)^n\, \sigma(s_{n+1}) \cdot \big(\ell \otimes s_1 \otimes \cdots \otimes s_n\big)$$

with $h^0(\ell \otimes s) = \sigma(s) \cdot \ell$. The sign is what makes the signs in $d^n = \sum_i (-1)^i \delta_i$ telescope; without it the two terms of the homotopy relation would reinforce rather than cancel. A direct check on the face maps δ_i shows the cosimplicial identity

$$h^{n+1} \circ d^n + d^{n-1} \circ h^n = \mathrm{id}$$

in each degree $n \geq 0$, where in degree 0 the second term is replaced by $\eta \circ$ (augmentation): for the first three terms, $h^0 \circ \eta = \mathrm{id}_L$ (because $\sigma(1) = 1$), and $h^1 \circ d^0 + \eta \circ h^0 = \mathrm{id}$ on $L \otimes_R S$. Verifying the middle identity explicitly, on a simple tensor $\ell \otimes s$, the degree-1 homotopy carries its sign $h^1(\ell' \otimes s_1' \otimes s_2') = -\sigma(s_2')(\ell' \otimes s_1')$, so

$$h^1(d^0(\ell \otimes s)) = h^1(\ell \otimes 1 \otimes s - \ell \otimes s \otimes 1) = -\sigma(s)(\ell \otimes 1) + \sigma(1)(\ell \otimes s) = \ell \otimes s - \ell \otimes \sigma(s)$$

while $\eta(h^0(\ell \otimes s)) = \eta(\sigma(s)\ell) = \sigma(s)\ell \otimes 1 = \ell \otimes \sigma(s)$ after moving the scalar $\sigma(s) \in R$ across the tensor product. The two therefore sum to $\ell \otimes s$, as required. A homotopy h with $hd + dh = \mathrm{id}$ forces every cohomology group to vanish, so the split complex is exact.

Step 3: Conclusion. The complex of Step 1 is exactly of the form treated in Step 2 (with $R = A$, $S = A' = A \otimes_B A$, $\sigma = \mu$, $L = M \otimes_B A$), hence exact. By the faithful-flatness reduction of Step 1, the original Amitsur complex over B is exact, which is what we sought to show.

The proof is worth pausing over, because its two-step shape, reduce to a split map by faithful flatness, then contract by hand, recurs everywhere in descent theory. Faithful flatness never lets one see the base B directly; it only guarantees that what is true after base change to A was already true over B. And after base change the situation trivializes completely, because the map $A \to A \otimes_B A$ always splits (via multiplication), so the complex acquires a formal contracting homotopy. This is the mechanism of splitting off a faithfully flat base. The exact sequence in low degrees is the one that matters for descent: M is the equalizer of the two maps $\eta_1, \eta_2\colon A \rightrightarrows A \otimes_B A$ applied to $M \otimes_B A$, precisely the sheaf condition for the cover $\operatorname{Spec} A \to \operatorname{Spec} B$.

Everything now assembles into the descent theorem. The module M is recovered from $M \otimes_B A$ as the equalizer of η_1 and η_2, and a descent datum supplies exactly the extra structure that identifies an A-module with $M \otimes_B A$ for a unique M.

Theorem 6.3.4: Faithfully Flat Descent for Quasi-Coherent Sheaves

Let $U \to X$ be a faithfully flat quasi-compact morphism of schemes. The pullback functor of definition 6.3.2

$$p^* \colon \mathrm{QCoh}(X) \longrightarrow \mathrm{Desc}(U \to X)$$

is an equivalence of categories. In the affine case $X = \operatorname{Spec} B$, $U = \operatorname{Spec} A$ with $B \to A$ faithfully flat, this says: the functor sending a B-module M to the A-module $A \otimes_B M$ with its canonical descent datum is an equivalence from B-modules to the category of A-modules equipped with a descent datum, with quasi-inverse $(N, \varphi) \mapsto M$, where

$$M = \left\{ n \in N \;\middle|\; \varphi(1 \otimes n) = n \otimes 1 \text{ in } N \otimes_B A \right\}$$

is the submodule of descent-invariant elements.

***Proof*:**
Both the source and target of p^* are local on X in the Zariski topology, and a faithfully flat quasi-compact morphism is, Zariski-locally on X, a faithfully flat morphism of affines (cover X by affine opens and, over each, cover the preimage by finitely many affine opens whose union is faithfully flat, using quasi-compactness). Descent data and quasi-coherent sheaves both glue along a Zariski cover of X, so it suffices to prove the theorem when $X = \operatorname{Spec} B$ and $U = \operatorname{Spec} A$ are affine with $B \to A$ faithfully flat. The affine statement is proved by constructing a quasi-inverse and checking the two composites are the identity.

Step 1: The descended module. Let (N, φ) be a descent datum, so N is an A-module and $\varphi \colon A \otimes_B N \xrightarrow{\sim} N \otimes_B A$ is an isomorphism of $(A \otimes_B A)$-modules satisfying the cocycle condition. Define

$$M = \ker\big(N \xrightarrow{\psi} N \otimes_B A\big), \qquad \psi(n) = \varphi(1 \otimes n) - n \otimes 1$$

the B-submodule of N on which the two structure maps agree. The claim is that the natural map $A \otimes_B M \to N$, $a \otimes m \mapsto am$, is an isomorphism of A-modules, and that under it φ becomes the canonical descent datum on $A \otimes_B M$.

Step 2: Comparison with the Amitsur equalizer. The cocycle condition makes φ compatible with the two ways of mapping N into $N \otimes_B A \otimes_B A$, which is precisely the statement that ψ fits into the Amitsur complex of $B \to A$ with coefficients twisted by φ. Concretely, transport the standard Amitsur complex of a B-module along φ: the isomorphism φ identifies the complex

$$0 \to M \to N \xrightarrow{\psi} N \otimes_B A \longrightarrow N \otimes_B A \otimes_B A$$

with the augmented Amitsur complex of the underlying B-module of N computed via the descent isomorphism, where the higher differentials are built from φ and its pullbacks and the cocycle condition is exactly what makes the composite of consecutive differentials vanish. Applying lemma 6.3.3 to the flat base change, the sequence

$$0 \to M \to N \xrightarrow{\psi} N \otimes_B A$$

is exact, so $M = \ker \psi$ is the equalizer.

Step 3: The comparison map is an isomorphism. Consider the map $\alpha \colon A \otimes_B M \to N$, $a \otimes m \mapsto am$. To prove α is an isomorphism it suffices, by faithful flatness ([6]), to prove that $A \otimes_B \alpha$ is an isomorphism of A-modules, since $-\otimes_B A$ reflects isomorphisms. Base-changing the exact sequence of Step 2 along the flat map $B \to A$ keeps it exact:

$$0 \to A \otimes_B M \to A \otimes_B N \xrightarrow{\mathrm{id}\otimes\psi} A \otimes_B N \otimes_B A$$

so $A \otimes_B M = \ker(\mathrm{id} \otimes \psi)$. On the other hand, the descent isomorphism $\varphi\colon A \otimes_B N \xrightarrow{\sim} N \otimes_B A$ carries this kernel isomorphically onto the kernel of the corresponding map out of $N \otimes_B A$, which by the cocycle condition is the equalizer computing N back again; unwinding the identifications, $A \otimes_B \alpha$ is carried to the identity of N. Hence $A \otimes_B \alpha$ is an isomorphism, so α is an isomorphism. This shows $N = A \otimes_B M$ with its canonical descent datum, so p^* is essentially surjective.

Step 4: Full faithfulness. For B-modules M, M' a morphism of descent data $p^*M \to p^*M'$ is an A-linear map $g\colon A \otimes_B M \to A \otimes_B M'$ commuting with the canonical gluing isomorphisms φ_{can}, that is, satisfying $\mathrm{pr}_2^* g \circ \varphi_{\mathrm{can}} = \varphi_{\mathrm{can}} \circ \mathrm{pr}_1^* g$ on $A \otimes_B A$. Under the canonical identifications this equation says that the two pullbacks $\mathrm{pr}_1^* g$ and $\mathrm{pr}_2^* g$ of g to $A \otimes_B A$ agree, that is, g is descent-invariant. The descent-invariant A-linear maps $A \otimes_B M \to A \otimes_B M'$ are then exactly the base changes $\mathrm{id}_A \otimes h$ of B-linear maps $h\colon M \to M'$. This is computed as an equalizer directly, avoiding any finite-presentation hypothesis on the modules. By Step 3 the target module M' is the equalizer of $\eta_1, \eta_2\colon A \otimes_B M' \rightrightarrows A \otimes_B A \otimes_B M'$; applying the left-exact functor $\mathrm{Hom}_B(M, -)$ to the resulting exact sequence $0 \to M' \to A \otimes_B M' \rightrightarrows A \otimes_B A \otimes_B M'$ presents $\mathrm{Hom}_B(M, M')$ as the equalizer

$$0 \to \mathrm{Hom}_B(M, M') \to \mathrm{Hom}_B(M, A \otimes_B M') \rightrightarrows \mathrm{Hom}_B(M, A \otimes_B A \otimes_B M')$$

whose middle and right terms are the A- and $(A \otimes_B A)$-linear map modules $\mathrm{Hom}_A(A \otimes_B M, A \otimes_B M')$ and $\mathrm{Hom}_{A\otimes_B A}(\cdots)$ by adjunction. Thus the invariant maps are $\mathrm{Hom}_B(M, M')$ outright, with no comparison of Hom with a tensor product needed. Hence every morphism of descent data descends uniquely to a B-linear map $M \to M'$, and p^* is fully faithful. An essentially surjective, fully faithful functor is an equivalence, completing the proof.

The theorem is the technical center of the chapter, and its meaning is worth restating in plain terms. A quasi-coherent sheaf on X can be handed over entirely by handing over its pullback to a faithfully flat cover U, together with the single isomorphism φ on $U \times_X U$ that says how the two pullbacks match. Nothing is lost. This is what makes it legitimate to *define* a sheaf on a base by working over a cover, the maneuver on which all of stack theory rests. The invariant submodule $M = \{n : \varphi(1 \otimes n) = n \otimes 1\}$ is the concrete recipe: it is the "G-invariants" of the descent, and in the Galois case below it will literally be a fixed-point set. The quasi-compactness hypothesis is what reduces the general statement to finitely many affines; for the coarser fpqc topology, where covers need not be quasi-compact, the same theorem holds but requires care with set-theoretic size, and the full fpqc statement is cited to [29].

Descent of quasi-coherent sheaves is the master case, and it immediately descends more than sheaves. An affine morphism $Y \to X$ is the same datum as a quasi-coherent sheaf of $\mathcal{O}_X$-algebras, namely $\mathcal{A} = f_*\mathcal{O}_Y$, with $Y = \underline{\mathrm{Spec}}_X \mathcal{A}$; and descent of sheaves upgrades to descent of sheaves of algebras with no new work, because the algebra structure is extra data that the sheaf equivalence carries along. The consequence is that a descent datum of affine schemes over U always glues to an affine scheme over X. This is the first *effectivity* result: not only sheaves but geometric objects, provided they are affine, can be reconstructed from a faithfully flat cover.

Theorem 6.3.5: Effectivity of Descent for Affine Morphisms

Let $U \to X$ be a faithfully flat quasi-compact morphism. The category of affine X-schemes is equivalent to the category of descent data of affine U-schemes, equivalently of quasi-coherent $\mathcal{O}_U$-algebras equipped with a descent datum compatible with the algebra structure. In the affine case $X = \operatorname{Spec} B$, $U = \operatorname{Spec} A$: a descent datum consisting of an A-algebra R together with an isomorphism of $(A \otimes_B A)$-algebras $\varphi\colon A \otimes_B R \xrightarrow{\sim} R \otimes_B A$ satisfying the cocycle condition descends to the B-algebra

$$R_0 = \left\{ r \in R \;\middle|\; \varphi(1 \otimes r) = r \otimes 1 \right\}$$

and $R = A \otimes_B R_0$ as A-algebras. Every descent datum of affine schemes is effective.

***Proof*:**
The strategy is to run the sheaf equivalence of theorem 6.3.4 and check that it respects algebra structures on both sides.

Step 1: Descend the underlying module. By theorem 6.3.4, the descent datum on the underlying quasi-coherent sheaf $\mathcal{A}_U = f_*\mathcal{O}_Y$ (in the affine case, the underlying A-module of R) descends uniquely to a quasi-coherent sheaf $\mathcal{A}$ on X (the B-module R_0), with $\mathcal{A}_U = p^*\mathcal{A}$ (that is, $R = A \otimes_B R_0$). The descended module R_0 is computed as the descent-invariant submodule, exactly as in the theorem.

Step 2: Descend the algebra structure. The multiplication and unit of R are morphisms of A-modules

$$\mu_R\colon R \otimes_A R \to R, \qquad u_R\colon A \to R$$

The hypothesis that φ is an isomorphism of $(A \otimes_B A)$-algebras says precisely that μ_R and u_R are morphisms of descent data, when $R \otimes_A R$ is given its induced descent datum from $\varphi \otimes \varphi$. Because p^* is an equivalence of categories (theorem 6.3.4) it is compatible with tensor products over the structure sheaf, so $(R \otimes_A R, \varphi \otimes \varphi)$ descends to $R_0 \otimes_B R_0$. A morphism of descent data descends uniquely to a morphism over X; applying this to μ_R and u_R produces B-linear maps

$$\mu_{R_0}\colon R_0 \otimes_B R_0 \to R_0, \qquad u_{R_0}\colon B \to R_0$$

whose pullback to A recovers μ_R and u_R.

Step 3: The descended structure is an algebra. The associativity, unit, and commutativity axioms for $(R_0, \mu_{R_0}, u_{R_0})$ are equalities of B-linear maps between finite tensor powers of R_0. Each such equality holds after applying the faithfully flat base change $- \otimes_B A$, where it becomes the corresponding axiom for R, which holds by hypothesis. By faithful flatness ([6]) a map of B-modules that vanishes after $- \otimes_B A$ already vanishes, so each axiom holds over B. Hence R_0 is a commutative B-algebra with $A \otimes_B R_0 = R$ as A-algebras.

Step 4: Equivalence. Conversely, any B-algebra R_0 yields the affine X-scheme $\operatorname{Spec} R_0$ whose pullback to U carries the canonical descent datum, and the two constructions are mutually inverse by theorem 6.3.4 applied to the underlying modules together with the uniqueness of descended morphisms in Steps 2 and 3. Passing from rings to sheaves of algebras and gluing over a Zariski cover of X, exactly as in the proof of theorem 6.3.4, gives the general statement for affine morphisms, completing the proof.

Effectivity for affine morphisms is the sharp boundary of what descends for free. The proof used nothing beyond the sheaf equivalence and the fact that an affine scheme is its algebra of functions; there was no geometry to lose. For non-affine schemes the situation changes: a descent datum of schemes along an fppf cover need not be effective, because the scheme structure can fail to glue even when the underlying sheaves do. The repair, requiring either a descending polarization or the passage to algebraic spaces, is the subject of the next section, and its failure is the reason Part III must eventually leave schemes behind. For now, the affine case is enough to power the descent of morphisms and Grothendieck's sheaf theorem.

The example that grounds the entire theory, and that connects it to material the reader already knows, is descent along a field extension. Here the cover is $\operatorname{Spec} L \to \operatorname{Spec} k$ for a finite Galois extension L/k, which is faithfully flat (it is finite free of rank $[L : k]$, and surjective on spectra) and even étale when L/k is separable. The fiber product $\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L$ is $\operatorname{Spec}(L \otimes_k L)$, and for L/k Galois with group $G = \operatorname{Gal}(L/k)$ there is the classical decomposition $L \otimes_k L \cong \prod_{g \in G} L$, sending $a \otimes b \mapsto (a \cdot g(b))_{g \in G}$. This turns the abstract descent datum into something completely concrete: a semilinear action of the Galois group.

Example 6.3.6: Descent Along a Finite Galois Extension

Let L/k be a finite Galois extension with group $G = \operatorname{Gal}(L/k)$, and consider the faithfully flat cover $\operatorname{Spec} L \to \operatorname{Spec} k$. A quasi-coherent sheaf on $\operatorname{Spec} L$ is an L-vector space N. Under the isomorphism $L \otimes_k L \cong \prod_{g \in G} L$, the two pullbacks of N become

$$L \otimes_k N \cong \prod_{g \in G} N, \qquad N \otimes_k L \cong \prod_{g \in G} N$$

and an isomorphism $\varphi \colon L \otimes_k N \to N \otimes_k L$ of $(L \otimes_k L)$-modules is the same as a family of k-linear isomorphisms $\rho_g \colon N \to N$, one for each $g \in G$, each g-*semilinear*: $\rho_g(\lambda n) = g(\lambda)\rho_g(n)$ for $\lambda \in L$, $n \in N$. The cocycle condition on $\operatorname{Spec}(L \otimes_k L \otimes_k L) \cong \prod_{g,h} \operatorname{Spec} L$ becomes the composition law $\rho_{gh} = \rho_g \circ \rho_h$. Thus:

> A descent datum on $\operatorname{Spec} L \to \operatorname{Spec} k$ is exactly a *semilinear G-action* on the L-vector space N.

Theorem 6.3.4 then reads: the functor $V \mapsto L \otimes_k V$ from k-vector spaces to L-vector spaces with semilinear G-action is an equivalence, with quasi-inverse $N \mapsto N^G = \{n \in N : \rho_g(n) = n \ \forall g\}$, the fixed-point space. An L-vector space with a compatible semilinear Galois action is thus the base change of a unique k-vector space, recovered as the G-invariants. This is the Galois-descent dictionary.

The example is the concrete face of theorem 6.3.4: the descent-invariant submodule $M = \{n : \varphi(1 \otimes n) = n \otimes 1\}$ of the theorem is here literally the fixed-point space N^G, because the condition $\varphi(1 \otimes n) = n \otimes 1$ unwinds, factor by factor over G, into $\rho_g(n) = n$ for every g. Faithfully flat descent thus contains classical Galois descent as the special case of the one-point étale cover $\operatorname{Spec} L \to \operatorname{Spec} k$. The same statement, run with a k-algebra in place of a vector space (invoking theorem 6.3.5), says that a k-algebra is the same as an L-algebra with semilinear G-action: forms of an algebra over k are classified by descent data over L.

The point at which this becomes a computational tool rather than a tautology is when one asks for the *forms* of a fixed object. Given a k-object V_0 (a vector space, an algebra, later a variety), a *twisted* *form* of V_0 is a k-object V that becomes isomorphic to V_0 after base change to L. Every twisted form gives a semilinear action on $L \otimes_k V_0$ that differs from the standard one by an automorphism $c_g \in \mathrm{Aut}_L(L \otimes_k V_0)$ for each g, and the cocycle condition $\rho_{gh} = \rho_g \rho_h$ forces the relation

$$c_{gh} = c_g \cdot {}^g c_h$$

This is exactly the condition for $(c_g)_{g \in G}$ to be a 1-*cocycle* of G with values in the automorphism group, and two forms are isomorphic precisely when their cocycles differ by a coboundary. Thus twisted forms of V_0 are classified by the pointed cohomology set $H^1(G, \mathrm{Aut}(L \otimes_k V_0))$. This is the first appearance of nonabelian Galois cohomology in the book; its formalism, the pointed sets $H^1(G, -)$ and their long exact sequences, is developed in [9] and returned to in the next section (section 6.4), where descent of schemes makes the twisted forms geometric.

Exercise 6.3.1

1. Let $B \to A$ be faithfully flat. Using only lemma 6.3.3 in low degrees, prove directly that $B \to A$ is injective, and that a sequence $0 \to M' \to M \to M'' \to 0$ of B-modules is exact if and only if it is exact after applying $- \otimes_B A$. (These are the two commutative- algebra inputs the section imports from [6]; the point of the exercise is an *alternative* derivation, showing they can equally be read off the Amitsur complex and so are of a piece with descent, rather than the route by which the lemma's own proof obtains them.)

2. Take $B = \mathbb{R}$ and $A = \mathbb{C}$, so $L = \mathbb{C}$, $k = \mathbb{R}$, and $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, \sigma\}$ with σ complex conjugation. Describe explicitly the semilinear action on $\mathbb{C}^n = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n$ coming from the standard descent datum, and verify that its fixed-point space is $\mathbb{R}^n$. Then exhibit a *different* semilinear action on $\mathbb{C}^2$ (a nonstandard ρ_σ) and compute its fixed-point space; confirm it is again a 2-dimensional $\mathbb{R}$-vector space, as theorem 6.3.4 predicts.

3. Verify the cocycle identity for the canonical descent datum of definition 6.3.2. That is, let $\mathcal{E}$ be a quasi-coherent sheaf on X, $p\colon U \to X$ a morphism, and φ_{can} the canonical isomorphism $\mathrm{pr}_1^* p^* \mathcal{E} \to \mathrm{pr}_2^* p^* \mathcal{E}$; show $\mathrm{pr}_{13}^* \varphi_{\mathrm{can}} = \mathrm{pr}_{23}^* \varphi_{\mathrm{can}} \circ \mathrm{pr}_{12}^* \varphi_{\mathrm{can}}$ on $U \times_X U \times_X U$.

4. Show that the cocycle condition forces the normalization $\Delta^* \varphi = \mathrm{id}_{\mathcal{F}}$, where $\Delta\colon U \to U \times_X U$ is the diagonal. (Hint: pull the cocycle identity back along a suitable map $U \times_X U \to U \times_X U \times_X U$ so that two of the three factors coincide, and use that φ is invertible.)

5. Let L/k be finite Galois with group G, and let A_0 be the L-algebra $A_0 = L \times L$ (two copies of L) with the semilinear G-action in which ρ_g acts on the factors both by g *and* by swapping them according to some homomorphism $\chi\colon G \to \mathbb{Z}/2$. Determine, in terms of χ, the descended k-algebra A_0^G, and identify it as a quadratic étale k-algebra. Which χ gives $k \times k$, and which gives a field?

6. Explain why the quasi-compactness hypothesis in theorem 6.3.4 is needed for the reduction to the affine case, and give an example of a faithfully flat but not quasi-compact morphism (e.g. an infinite disjoint union of open immersions covering X redundantly). What goes wrong with the finiteness argument in Step 1 of the proof if quasi-compactness is dropped?

6.4 Descent of Morphisms and of Schemes

The previous section proved that quasi-coherent sheaves descend along a faithfully flat cover: the category $\mathrm{QCoh}(X)$ is equivalent to the category of descent data on $U \to X$, and this equivalence is effective for affine morphisms (theorem 6.3.4, theorem 6.3.5). That result answers one half of what a moduli problem needs. To glue objects in the fppf topology one must descend not only the objects but the *morphisms* between them, and, one level up, the geometric spaces themselves. This section addresses both. The first result, that morphisms descend, is the exact input the sheaf-theoretic backbone of the whole chapter rests on: it is what makes a representable functor an fppf sheaf, closing the gap left open in section 6.2. The second circle of results concerns properties of morphisms, and the third, effectivity of descent for schemes.

The third result records where descent breaks down. Descent data for schemes are *not* always effective: a compatible family of schemes over the pieces of a cover may fail to glue to a scheme over the base. Effectivity holds when a relatively ample line bundle descends alongside the schemes, which is the condition that keeps the descended object projective and hence a scheme. The failure of effectivity in general is not a technical nuisance. It is the precise reason the remainder of Part III abandons schemes in favor of algebraic spaces and stacks: the category of schemes is too small to be closed under the gluing that moduli problems demand. Galois descent, the classical theory of forms of a variety over a field, is the étale-of-a-point special case of everything here, and it is where the abstract machinery first pays a concrete dividend.

Throughout, $U \to X$ is a faithfully flat cover, quasi-compact and locally of finite presentation where the fppf topology is invoked, with $U \times_X U$ carrying the two projections $\mathrm{pr}_1, \mathrm{pr}_2$ and $U \times_X U \times_X U$ the three projections $\mathrm{pr}_{12}, \mathrm{pr}_{13}, \mathrm{pr}_{23}$. For an X-scheme Y and a morphism $T \to X$ write $Y_T = Y \times_X T$ for the base change.

The starting point is the observation that a morphism between two X-schemes is determined by, and can be reconstructed from, its restriction to a cover, provided the restriction is compatible with the two ways of pulling back to the double overlap. This is the morphism analogue of the sheaf axiom, and its proof is a direct application of faithfully flat descent for sheaves to the sheaf of homomorphisms. The mechanism is worth stating cleanly before the theorem, because it recurs at every level of the descent hierarchy: a morphism is a section of a sheaf, and sections of a sheaf are exactly the data that satisfy the equalizer condition.

Theorem 6.4.1: Descent of Morphisms

Let X be a scheme and $U \to X$ a faithfully flat cover, quasi-compact and locally of finite presentation. Let Y, Z be X-schemes. Then the presheaf on the big fppf site $(\mathbf{Sch}/X)_{\mathrm{fppf}}$

$$\mathcal{F}\colon \quad (T \to X) \longmapsto \mathrm{Hom}_X(Y_T, Z_T)$$

is an fppf sheaf. Concretely, giving an X-morphism $g\colon Y \to Z$ is equivalent to giving a U-morphism $g_U\colon Y_U \to Z_U$ whose two pullbacks to $U \times_X U$ agree:

$$\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U \quad \text{in} \quad \mathrm{Hom}_{U\times_X U}\big(Y_{U\times_X U}, Z_{U\times_X U}\big)$$

That is, the sequence

$$\mathrm{Hom}_X(Y,Z) \longrightarrow \mathrm{Hom}_U(Y_U, Z_U) \rightrightarrows \mathrm{Hom}_{U\times_X U}\big(Y_{U\times_X U}, Z_{U\times_X U}\big)$$

is an equalizer.

***Proof*:**
The proof reduces the descent of morphisms to the descent of quasi-coherent sheaves already established, by exhibiting a morphism as a section of an affine scheme over the base and applying theorem 6.3.4. It is enough to prove the equalizer statement for the single cover $U \to X$; the general sheaf condition for the fppf topology follows because every fppf covering family is refined by such a single quasi-compact faithfully flat morphism, and the equalizer condition passes to the refinement.

Step 1: Injectivity of the restriction map. Suppose $g, g'\colon Y \to Z$ are two X-morphisms with $g_U = g'_U$ after base change to U; the assertion is that $g = g'$. Because $U \to X$ is faithfully flat, so is the base change $Y_U \to Y$; a morphism out of Y is determined by its composite with $Y_U \to Y$ precisely when the latter is an epimorphism of schemes, and a faithfully flat quasi-compact morphism is an epimorphism in the category of schemes. Concretely, g and g' agree if and only if they agree on the affine pieces, where equality of two ring maps $B \to A$ can be tested after the faithfully flat base change $A \to A'$, since $A \to A'$ is injective. Thus $g_U = g'_U$ forces $g = g'$, and the restriction map $\mathrm{Hom}_X(Y,Z) \to \mathrm{Hom}_U(Y_U, Z_U)$ is injective.

Step 2: The gluing datum defines a morphism, affine target case. Assume first that $Z \to X$ is affine, say $Z = \underline{\mathrm{Spec}}_X \mathcal{B}$ for a quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{B}$. Then for any X-scheme W,

$$\mathrm{Hom}_X(W,Z) = \mathrm{Hom}_{\mathcal{O}_X\text{-alg}}\big(\mathcal{B}, (f_W)_*\mathcal{O}_W\big)$$

where $f_W\colon W \to X$ is the structure map, by the universal property of the relative $\underline{\mathrm{Spec}}$. Suppose given $g_U\colon Y_U \to Z_U$ with $\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U$. Translating through the adjunction, g_U is a map of $\mathcal{O}_U$-algebras $\mathcal{B}_U \to (f_{Y_U})_*\mathcal{O}_{Y_U}$, and the cocycle condition says exactly that this map of algebras is compatible with the two pullbacks to $U \times_X U$. The sheaf $\mathcal{B}$ and the pushforward $(f_Y)_*\mathcal{O}_Y$ are quasi-coherent on X (the pushforward because f_Y is quasi-compact and quasi-separated, which holds under the book's standing finite-type-over-k convention), and by theorem 6.3.4 the category of quasi-coherent sheaves on X is equivalent to the category of descent data on $U \to X$. A homomorphism of descent data is a homomorphism of the descended sheaves; hence the compatible algebra map $\mathcal{B}_U \to (f_{Y_U})_*\mathcal{O}_{Y_U}$ descends to a unique algebra map $\mathcal{B} \to (f_Y)_*\mathcal{O}_Y$ over X. This algebra map is the sought morphism $g\colon Y \to Z$, and it restricts to g_U by uniqueness in the descent equivalence. This proves the equalizer statement when $Z \to X$ is affine.

Step 3: Reduction of the general target to the affine case. For general Z, the point is that the morphism condition is local on the source Y and on the target Z, and locality is a Zariski (hence fppf) condition to which the sheaf property already proved applies. Cover Z by affine opens $Z = \bigcup_\alpha Z_\alpha$. A morphism $g_U : Y_U \to Z_U$ with matching pullbacks determines, over each preimage $g_U^{-1}(Z_{\alpha,U}) \subseteq Y_U$, a morphism into the affine Z_α. The preimages $g_U^{-1}(Z_{\alpha,U})$ are open in Y_U and, because the compatibility $\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U$ identifies them across the two pullbacks, they descend to open subschemes $Y_\alpha \subseteq Y$ with $(Y_\alpha)_U = g_U^{-1}(Z_{\alpha,U})$: an open subscheme of Y_U stable under the descent datum is the pullback of a unique open subscheme of Y, which is descent for open immersions, itself the affine-target case applied to the structure sheaf. On each Y_α the restricted datum $g_U|_{(Y_\alpha)_U} : (Y_\alpha)_U \to (Z_\alpha)_U$ has affine target, so Step 2 produces a morphism $g_\alpha : Y_\alpha \to Z_\alpha \hookrightarrow Z$. On overlaps $Y_\alpha \cap Y_\beta$ the two morphisms g_α, g_β agree, because they agree after the faithfully flat base change to U (both restrict to g_U) and Step 1 promotes that to equality on the base. The g_α therefore glue to a single X-morphism $g : Y \to Z$ restricting to g_U. This establishes the equalizer for arbitrary Z, and with it the sheaf property, completing the proof.

The theorem is the keystone the chapter's sheaf theory needs. In section 6.2 the claim that a representable presheaf $h_T = \mathrm{Hom}_X(-, T)$ is an fppf sheaf (Grothendieck's theorem, theorem 6.2.4) was stated with its proof deferred to the descent of morphisms. That debt is now paid. Taking $Y = X$ and $Z = T$ fixed, so that for a test scheme $S \to X$ one has $Y_S = S$ and $Z_S = T_S$, the sheaf $\mathcal{F}$ of the theorem evaluates to $\mathcal{F}(S) = \mathrm{Hom}_X(S, T_S) = h_T(S)$; it is exactly the representable functor h_T. An fppf-local morphism into T that agrees on double overlaps glues to a genuine morphism into T, which is precisely the sheaf condition for h_T. Thus theorem 6.2.4 is a corollary of theorem 6.4.1, and the fppf topology is fine enough that representable functors remain sheaves in it. The strengthening to the fpqc topology, where the same conclusion holds for arbitrary faithfully flat quasi-compact covers without the finite-presentation hypothesis, requires a more delicate set-theoretic treatment of the site and is treated in full in [29]; the fppf statement above is what the moduli constructions of this book use.

A first concrete consequence is that morphisms between varieties over a field can be constructed after passing to a field extension, provided they are compatible with the two pullbacks. This is the mechanism behind the descent of forms in the Galois case below, and it deserves an explicit worked instance.

Example 6.4.2: Descending a Morphism from a Field Extension

Let k be a field, L/k a finite Galois extension with Galois group $\Gamma = \mathrm{Gal}(L/k)$, and let Y, Z be k-schemes. The cover is $\operatorname{Spec} L \to \operatorname{Spec} k$, and the double fiber product decomposes as a product over the Galois group,

$$\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L \;\cong\; \coprod_{\sigma \in \Gamma} \operatorname{Spec} L$$

one copy for each $\sigma \in \Gamma$, the σ-th copy embedded by $(\mathrm{id}, \sigma) : L \otimes_k L \to L$. Under this identification the two projections $\mathrm{pr}_1, \mathrm{pr}_2$ send the σ-th copy to the source and target factors, and the compatibility condition $\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U$ of theorem 6.4.1 becomes the statement that the base-changed morphism $g_L : Y_L \to Z_L$ commutes with the semilinear Γ-action on both sides: for every $\sigma \in \Gamma$,

$$g_L \circ \sigma_Y = \sigma_Z \circ g_L$$

where σ_Y, σ_Z are the automorphisms of Y_L, Z_L induced by σ acting on L. Concretely, take

$Y = Z = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ and $k = \mathbb{R}$, $L = \mathbb{C}$, $\Gamma = \{1, c\}$ with c complex conjugation. An L-morphism $g_L \colon \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$ is a polynomial $g_L(t) = \sum a_i t^i$ with $a_i \in \mathbb{C}$. The conjugation c acts on $\mathbb{A}^1_{\mathbb{C}}$ by $t \mapsto t$ on the coordinate and by $a_i \mapsto \bar{a}_i$ on the coefficients read as functions, so g_L is Γ-equivariant precisely when $\bar{a}_i = a_i$ for all i, that is, when every coefficient is real. The descended morphism is then the real polynomial map $g \colon \mathbb{A}^1_{\mathbb{R}} \to \mathbb{A}^1_{\mathbb{R}}$ with those coefficients. A polynomial with a complex coefficient, such as $g_L(t) = it$, fails equivariance and does not descend: it defines a morphism only over $\mathbb{C}$, not over $\mathbb{R}$.

Once objects and morphisms descend, the next question is which *properties* of a morphism can be detected after a faithfully flat base change. A property P of morphisms is called *fppf-local on the base* when, for every faithfully flat cover $U \to X$ locally of finite presentation, a morphism $f \colon Y \to X$ has property P if and only if the base change $f_U \colon Y_U \to U$ has property P. This is what one wants for constructions built by gluing: to check that a descended morphism is flat, or smooth, or proper, it suffices to check the property on the pieces of a cover, where the geometry is often simpler. The list of such properties is long and central to the theory, so the pattern of proof matters more than any single entry. The argument for flatness and for smoothness is the model; the rest follow the same faithfully flat descent of the defining local condition.

Theorem 6.4.3: Descent of Properties of Morphisms

Let $U \to X$ be a faithfully flat cover, quasi-compact and locally of finite presentation, and let $f \colon Y \to X$ be a morphism of schemes. For each property P in the list

$$\{\text{flat},\ \text{smooth},\ \text{étale},\ \text{proper},\ \text{finite},\ \text{an immersion},\ \text{surjective},\ \text{quasi-compact}\}$$

the morphism f has property P if and only if the base change $f_U \colon Y_U \to U$ has property P. In other words, each of these properties is fppf-local on the base.

***Proof*:**
Necessity in each case is immediate, since every property in the list is stable under arbitrary base change: if f has property P then so does f_U, being a base change of f. The content is sufficiency, that property P on f_U descends to f. The full table is verified property by property in [29], where each entry is reduced to a local statement in commutative algebra; the two representative cases are proved here in full, and the remaining entries follow the same pattern of descending a local condition along a faithfully flat ring map.

Step 1: Flatness descends. Flatness of a morphism is a condition on stalks: f is flat if and only if for every point $y \in Y$ the local ring $\mathcal{O}_{Y,y}$ is flat over $\mathcal{O}_{X,f(y)}$. Reduce to the affine situation $X = \operatorname{Spec} R$, $U = \operatorname{Spec} R'$ with $R \to R'$ faithfully flat, and Y affine over X near y, say $Y = \operatorname{Spec} A$ with A an R-algebra; flatness of f is flatness of A as an R-module, and flatness of f_U is flatness of $A' := A \otimes_R R'$ as an R'-module. Suppose A' is flat over R'. Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of R-modules; applying $- \otimes_R A$ gives a complex whose exactness is to be shown. Base change to R': the complex $(- \otimes_R A) \otimes_R R'$ is

$$M_i \otimes_R A \otimes_R R' \;\cong\; (M_i \otimes_R R') \otimes_{R'} A'$$

by associativity of tensor product, and this is the sequence $M_i \otimes_R R'$ tensored over R' with the flat R'-module A'. Since $R \to R'$ is flat, the sequence $M_i \otimes_R R'$ is exact, and since A' is R'-flat, tensoring it with A' keeps it exact. Thus the complex $(M_\bullet \otimes_R A) \otimes_R R'$ is exact. But $R \to R'$ is *faithfully* flat, so a complex of R-modules is exact if and only if it becomes exact after $- \otimes_R R'$; therefore $M_\bullet \otimes_R A$ is itself exact. Hence A is flat over R, and f is flat.

Step 2: Smoothness descends. A morphism is smooth if and only if it is flat, locally of finite presentation, and has geometrically regular fibers of the expected dimension. Under flatness and local finite presentation, the geometric regularity of the fibers is encoded by the sheaf of relative differentials $\Omega_{Y/X}$ being locally free of rank equal to the fiber dimension: for a flat, finitely presented morphism the fiber over a point is regular of dimension n exactly when $\Omega_{Y/X}$ is locally free of rank n there. Thus smoothness is equivalent to flatness and local finite presentation together with local freeness of $\Omega_{Y/X}$ of the expected rank. Flatness of f descends by Step 1. Local finite presentation descends because $R \to R'$ is faithfully flat: an R-algebra A with $A \otimes_R R'$ of finite presentation over R' is of finite presentation over R, a faithfully flat descent statement for the finiteness of a presentation carried out in [29] and resting on [6]. It remains to descend the freeness and rank of $\Omega_{Y/X}$. Relative differentials commute with base change, so $\Omega_{Y_U/U}$ is the pullback of $\Omega_{Y/X}$ along the projection $Y_U \to Y$. If $\Omega_{Y_U/U}$ is locally free of rank n, then the pullback of $\Omega_{Y/X}$ along the faithfully flat morphism $Y_U \to Y$ is locally free of rank n; and local freeness of a coherent sheaf descends along a faithfully flat morphism, because a finitely presented module N over $\mathcal{O}_{Y,y}$ is free precisely when $N \otimes \mathcal{O}_{Y_U,y'}$ is free over the faithfully flat local extension, the rank being read off either side. Hence $\Omega_{Y/X}$ is locally free of rank n, and f is smooth.

Step 3: The remaining properties. Each of étale, finite, immersion, surjective, and quasi-compact is a combination of conditions already treated or a local condition of the same form. Étale is smooth of relative dimension 0, so it descends by Step 2 together with the descent of fiber dimension, the relative dimension at a point being unchanged under the faithfully flat base change $Y_U \to Y$ since fibers of f_U are base changes of fibers of f to field extensions. Surjectivity descends because $U \to X$ is surjective on points: f_U surjective forces f surjective by a diagram chase on underlying topological spaces, since $Y_U \to Y$ and $U \to X$ are surjective. Being an immersion, finite, or proper is checked by descending the corresponding property of the associated affine or the corresponding valuative or topological criterion, each reduced to a faithfully flat descent of a commutative-algebra condition in the manner of Steps 1 and 2; the full verifications, uniform in shape, are in [29]. This exhausts the list and completes the proof.

The descent of properties is what makes the atlas philosophy of Part III work. An algebraic space or stack is presented by a cover from a scheme, and every geometric property one wants to attribute to the space, that it is smooth, that a morphism between two of them is proper, is defined by checking the property on the cover. theorem 6.4.3 is the guarantee that such definitions are well posed: the property read off the atlas does not depend on the atlas, because it descends. The same theorem is what lets the Hilbert and Quot schemes, and the GIT quotients built earlier in this book, be assembled by gluing over a cover while retaining flatness or properness. Flatness in particular is the most-used entry, because it is the technical hypothesis under which cohomology and base change behave, and its fppf-locality on the base is invoked silently throughout the moduli literature.

The final and most consequential result concerns descent of the schemes themselves, not their morphisms or their properties alone. A descent datum for schemes along a cover $U \to X$ is a scheme V over U together with an isomorphism $\varphi\colon \mathrm{pr}_1^* V \xrightarrow{\sim} \mathrm{pr}_2^* V$ over $U \times_X U$ satisfying the cocycle condition on the triple overlap. The datum is *effective* when it arises from base change of a single scheme over X: there exists a scheme $Y \to X$ and an isomorphism $Y_U \cong V$ over U carrying the canonical descent datum of Y_U to φ. For quasi-coherent sheaves, descent is always effective (theorem 6.3.4); for affine morphisms it is again always effective (theorem 6.3.5). For general schemes it is not. The obstruction is that gluing the affine pieces of V across the cover can produce an object that is a scheme locally over the cover but has no global scheme structure, because it cannot be covered by affine opens compatibly. The remedy, and the boundary of what schemes can do, is a polarization.

Theorem 6.4.4: Effective Descent for Polarized Schemes

Let $U \to X$ be a faithfully flat cover, quasi-compact and locally of finite presentation, and let (V, φ) be a descent datum for schemes along $U \to X$ with $V \to U$ quasi-projective. Suppose there is a line bundle $\mathcal{L}$ on V, relatively ample for $V \to U$, together with an isomorphism $\mathrm{pr}_1^* \mathcal{L} \xrightarrow{\sim} \mathrm{pr}_2^* \mathcal{L}$ over $\mathrm{pr}_1^* V \cong \mathrm{pr}_2^* V$ compatible with φ and satisfying the cocycle condition, that is, the polarization itself carries a descent datum. Then the descent datum (V, φ) is effective: there is a scheme $Y \to X$, quasi-projective, with $Y_U \cong V$ compatibly with φ, and $\mathcal{L}$ descends to a relatively ample line bundle on Y.

***Proof*:**
The strategy is to trade the scheme V for a quasi-coherent sheaf of graded algebras, descend the sheaf by theorem 6.3.4, and recover the scheme as a relative $\underline{\mathrm{Proj}}$. This converts an effectivity question for schemes, where descent can fail, into an effectivity question for quasi-coherent sheaves, where it never does.

Step 1: Reconstruct V from its polarization. Because $\mathcal{L}$ is relatively ample for the quasi-projective morphism $g \colon V \to U$, the scheme V is recovered from the graded $\mathcal{O}_U$-algebra of its sections,

$$\mathcal{S}_U \ := \bigoplus_{n \geq 0} g_* \mathcal{L}^{\otimes n}$$

as the relative Proj, $V \cong \underline{\mathrm{Proj}}_U \mathcal{S}_U$, with $\mathcal{L}$ recovered as the relative $\mathcal{O}(1)$ (for $\mathcal{L}$ relatively ample rather than very ample one first replaces it by a Veronese power $\mathcal{L}^{\otimes d}$ generating $\mathcal{S}_U$ in degree one, which does not affect the descent argument since the polarization datum passes to the power). Each graded piece $g_* \mathcal{L}^{\otimes n}$ is a quasi-coherent $\mathcal{O}_U$-module, so $\mathcal{S}_U$ is a quasi-coherent sheaf of graded $\mathcal{O}_U$-algebras.

Step 2: The graded algebra carries a descent datum. The isomorphism $\mathrm{pr}_1^* \mathcal{L} \cong \mathrm{pr}_2^* \mathcal{L}$ over the double overlap, being compatible with φ and satisfying the cocycle condition, induces for each n an isomorphism of pushforwards

$$\mathrm{pr}_1^* \big(g_* \mathcal{L}^{\otimes n} \big) \ \xrightarrow{\sim} \ \mathrm{pr}_2^* \big(g_* \mathcal{L}^{\otimes n} \big)$$

compatible with the grading and the algebra structure, because pushforward commutes with the flat base changes $\mathrm{pr}_1, \mathrm{pr}_2$ (flat base change for quasi-coherent cohomology applies since pr_i are flat, being base changes of the flat $U \to X$). Assembling over n, the graded algebra $\mathcal{S}_U$ acquires a descent datum $\psi \colon \mathrm{pr}_1^* \mathcal{S}_U \xrightarrow{\sim} \mathrm{pr}_2^* \mathcal{S}_U$ of quasi-coherent sheaves of graded algebras, satisfying the cocycle condition because φ and the polarization datum do.

Step 3: Descend the algebra and take Proj. By theorem 6.3.4, applied to each graded piece and assembled, the descent datum $(\mathcal{S}_U, \psi)$ is effective as a quasi-coherent sheaf: there is a quasi-coherent sheaf of graded $\mathcal{O}_X$-algebras $\mathcal{S}$ on X with $\mathrm{pr}^* \mathcal{S} \cong \mathcal{S}_U$ compatibly with ψ, where $\mathrm{pr} \colon U \to X$. The algebra structure descends because it is a morphism of quasi-coherent sheaves $\mathcal{S}_U \otimes \mathcal{S}_U \to \mathcal{S}_U$ compatible with the descent data, and morphisms of quasi-coherent sheaves descend by the same theorem. Now set

$$Y \ := \ \underline{\mathrm{Proj}}_X \mathcal{S}$$

Formation of relative Proj commutes with the flat base change $U \to X$, so $Y_U = \underline{\mathrm{Proj}}_U(\mathrm{pr}^* \mathcal{S}) = \underline{\mathrm{Proj}}_U \mathcal{S}_U \cong V$, and this isomorphism carries the canonical descent datum of Y_U to φ because it does so on the defining graded algebras. The relative $\mathcal{O}(1)$ on Y is a relatively ample line bundle descending $\mathcal{L}$. Thus $Y \to X$ is a quasi-projective scheme representing the descent datum, and $\mathcal{L}$ descends to a relatively ample bundle on it, as we sought to show.

The hypothesis that the polarization descends is exactly what cannot be dropped. Without a compatible relatively ample line bundle there is in general no scheme over X realizing the datum, and this failure is genuine, not an artifact of the proof. The standard obstruction is that the descended object, while locally a scheme, may be proper but non-projective, and a proper non-projective object has no ample line bundle to descend. The next result records the classical counterexample.

Theorem 6.4.5: Non-Effective Descent

There exist an algebraic space X, a faithfully flat (indeed étale) cover $U \to X$ by a scheme U, and a descent datum (V, φ) for schemes along $U \to X$ whose descended object over X would be proper, that is not effective in the category of schemes: no scheme $Y \to X$ has $Y_U \cong V$ compatibly with φ. The cover U is a Hironaka-type proper non-projective threefold carrying a free $\mathbb{Z}/2$-action, X is its free quotient, and $V = U$ with the tautological descent datum recording the action. The obstruction is the absence of a descending polarization, the free action exchanging two curves that no $(\mathbb{Z}/2)$-invariant ample bundle can distinguish, so the free quotient exists as an algebraic space but not as a scheme.

***Proof Key ideas*:**
The full construction, with the geometry of the proper non-projective threefold and the verification that the resulting descent datum admits no scheme representing it, is carried out in [30] and, in the setting of Grothendieck's original descent talks, in [15]; the argument uses the non-projective threefold geometry that is beyond the present chapter's scope, so the mechanism is described here and the detailed verification is cited. The idea is as follows. Hironaka constructed a smooth proper threefold W that is not projective, with a free action of a group of order two, $\mathbb{Z}/2$, whose two orbits of a certain curve are exchanged in a way no ample class can respect: the quotient $W/(\mathbb{Z}/2)$ exists as an algebraic space but not as a scheme, because a scheme quotient would require a $(\mathbb{Z}/2)$-invariant ample bundle, which does not exist.

Encoded as descent, take $U = W$ with the free $\mathbb{Z}/2$-action, let $X = W/(\mathbb{Z}/2)$ as an algebraic space so that $U \to X$ is an étale (hence fppf) cover with $U \times_X U = U \times \mathbb{Z}/2 = U \sqcup U$, and let $V = U$ with the tautological descent datum φ recording the $\mathbb{Z}/2$-action. Effectivity of this datum in schemes would produce a scheme Y with $Y_U \cong U$, that is, a scheme quotient $W/(\mathbb{Z}/2)$; its nonexistence as a scheme is exactly Hironaka's non-projectivity obstruction. Thus (V, φ) is a descent datum for schemes whose descended object over X would be proper that is not effective in the category of schemes, though it is effective in the larger category of algebraic spaces. This is the phenomenon claimed.

This is the fault line that Part III is built to cross. Descent for quasi-coherent sheaves is always effective; descent for affine morphisms is always effective; but descent for schemes fails, and it fails for a structural reason, the possible absence of a polarization, not a repairable technical one. The moment a moduli problem forces a gluing of the second kind, a compatible family of proper objects with no globally coherent ample bundle, the glued object exists as a geometric space but not as a scheme. The response is to enlarge the category. Algebraic spaces, introduced in the chapter on algebraic spaces and stacks below, are precisely the objects obtained by allowing étale descent data for schemes to be effective by fiat, and algebraic stacks go one level further, allowing descent data with automorphisms. Hironaka's non-effective datum is a scheme-theoretic ghost that becomes a genuine algebraic space, and this is the first concrete evidence that the category of schemes is the wrong home for moduli. The counterexample is not a pathology to be avoided but the signpost pointing from schemes to spaces and stacks.

Descent theory has a classical ancestor that predates the language of sites and covers by decades: Galois descent, the theory of forms of an algebraic object over a field. When the cover is a Galois field extension $\operatorname{Spec} L \to \operatorname{Spec} k$, the machinery specializes exactly, and the abstract notion of a descent datum reduces to the concrete notion of a semilinear Galois action. The classification of forms by a nonabelian cohomology set $H^1(\mathrm{Gal}, \mathrm{Aut})$ is the shadow, at the level of a single point, of the torsor classification developed in the next section. Galois descent is where the theory of this chapter first meets objects a reader has seen before, and it supplies the running dictionary between cocycles and twisted forms.

Example 6.4.6: Galois Descent and Forms of a Variety

Let k be a field with separable closure k^{sep} and absolute Galois group $\Gamma = \mathrm{Gal}(k^{\mathrm{sep}}/k)$. Fix a k-scheme (or a k-vector space, or a k-algebra) X_0. A *form* of X_0 is a k-object X that becomes isomorphic to X_0 after base change to k^{sep}: an isomorphism $X_{k^{\mathrm{sep}}} \cong (X_0)_{k^{\mathrm{sep}}}$ exists, though no k-isomorphism $X \cong X_0$ need. Two forms are equivalent when they are k-isomorphic. The content of Galois descent is that the set of forms of X_0 up to equivalence is classified by nonabelian Galois cohomology.

The dictionary. By theorem 6.4.1 applied to the cover $\operatorname{Spec} k^{\mathrm{sep}} \to \operatorname{Spec} k$, and its extension to descent of schemes, a k-form of X_0 is the same datum as a descent datum on $(X_0)_{k^{\mathrm{sep}}}$, which by the Galois decomposition of the double fiber product is a continuous system of isomorphisms

$$c_\sigma \colon (X_0)_{k^{\mathrm{sep}}} \xrightarrow{\sim} {}^{\sigma}(X_0)_{k^{\mathrm{sep}}}, \qquad \sigma \in \Gamma$$

one for each element of the Galois group, satisfying the cocycle condition $c_{\sigma\tau} = {}^{\sigma}c_\tau \circ c_\sigma$. Composing with the tautological semilinear action lets one record the datum as a 1-cocycle $\sigma \mapsto a_\sigma \in \mathrm{Aut}\big((X_0)_{k^{\mathrm{sep}}}\big)$ valued in the automorphism group, and changing the chosen k^{sep}-isomorphism $X_{k^{\mathrm{sep}}} \cong (X_0)_{k^{\mathrm{sep}}}$ replaces a_σ by a coboundary. The forms of X_0 are therefore in bijection with the pointed cohomology set

$$\{k\text{-forms of } X_0\}/\cong \;\longleftrightarrow\; H^1\big(\Gamma, \mathrm{Aut}((X_0)_{k^{\mathrm{sep}}})\big)$$

the base point (trivial cocycle) corresponding to X_0 itself. The nonabelian H^1 here is a pointed set, not a group, and the formalism of cocycles, coboundaries, and twisting for a nonabelian coefficient group is developed in [9].

Worked form: the conics as forms of $\mathbb{P}^1$. Take $k = \mathbb{R}$, $k^{\mathrm{sep}} = \mathbb{C}$, $\Gamma = \{1, c\}$ with c complex conjugation, and $X_0 = \mathbb{P}^1_{\mathbb{R}}$. A smooth conic $C \subseteq \mathbb{P}^2_{\mathbb{R}}$ with no real point, such as the anisotropic conic

$$C\colon \quad x^2 + y^2 + z^2 = 0$$

becomes isomorphic to $\mathbb{P}^1_{\mathbb{C}}$ over $\mathbb{C}$ (every smooth conic over an algebraically closed field is a rational curve isomorphic to $\mathbb{P}^1$), so C is a form of $\mathbb{P}^1_{\mathbb{R}}$. It is not $\mathbb{R}$-isomorphic to $\mathbb{P}^1_{\mathbb{R}}$, because $\mathbb{P}^1_{\mathbb{R}}$ has real points and C does not. The automorphism group is $\mathrm{Aut}(\mathbb{P}^1_{\mathbb{C}}) = \mathrm{PGL}_2(\mathbb{C})$, and

$$H^1\big(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathrm{PGL}_2(\mathbb{C})\big)$$

has exactly two classes: the trivial class, giving $\mathbb{P}^1_{\mathbb{R}}$, and one nontrivial class, giving the anisotropic conic C. The two forms of $\mathbb{P}^1$ over $\mathbb{R}$ are the split line and the pointless conic, and the nontrivial cohomology class measures the failure of C to have a real point. Equivalently, quaternion algebras over $\mathbb{R}$ enter: the conic C is the Severi-Brauer variety of Hamilton's quaternions $\mathbb{H}$, and the two elements of H^1 correspond to the two $\mathbb{R}$-quaternion algebras $M_2(\mathbb{R})$ and $\mathbb{H}$, the split and division cases.

A second form: the two forms of $\mathbb{G}_m$. Over a field k the one-dimensional split torus $\mathbb{G}_m$ has automorphism group $\mathrm{Aut}(\mathbb{G}_{m,k^{\mathrm{sep}}}) = \{\pm 1\}$, generated by inversion $t \mapsto t^{-1}$. Its forms are classified by

$$H^1(\Gamma, \mathbb{Z}/2) = \mathrm{Hom}_{\mathrm{cont}}(\Gamma, \mathbb{Z}/2)$$

the continuous homomorphisms $\Gamma \to \mathbb{Z}/2$, which by Galois theory are the quadratic étale k-algebras: the trivial homomorphism gives the split torus $\mathbb{G}_m$, and a surjection $\Gamma \to \mathrm{Gal}(L/k) = \mathbb{Z}/2$ for a separable quadratic extension L/k gives the nonsplit norm-one torus

$$R^{(1)}_{L/k}\mathbb{G}_m \;=\; \ker\big(\mathrm{Nm}\colon R_{L/k}\mathbb{G}_m \to \mathbb{G}_m\big)$$

whose k-points are the norm-one elements of $L^\times$. For $k = \mathbb{R}$, $L = \mathbb{C}$, this nonsplit form is the circle group $\mathrm{SO}(2)$, the unit circle $\{z \in \mathbb{C} : |z| = 1\}$, which becomes $\mathbb{G}_m$ over $\mathbb{C}$ through $z \mapsto z$ but is anisotropic over $\mathbb{R}$. The two forms of $\mathbb{G}_m$ over $\mathbb{R}$, the multiplicative group and the circle, are the two elements of $H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{Z}/2)$, matching the two quadratic étale $\mathbb{R}$-algebras $\mathbb{R} \times \mathbb{R}$ and $\mathbb{C}$.

Galois descent shows the machinery of this chapter operating at its smallest nontrivial scale, a cover by a single point $\mathrm{Spec}\, L \to \mathrm{Spec}\, k$, and yields a classification a reader can compute by hand. The passage from the general descent datum to a 1-cocycle valued in an automorphism group is the same passage that, in the next section, turns a torsor into a class in $\check{H}^1$; the forms of X_0 classified by $H^1(\Gamma, \mathrm{Aut}(X_0))$ are the point-level instance of torsors classified by H^1 of a site. The automorphism group $\mathrm{Aut}(X_0)$ that appears here as the coefficient of the cohomology is the same automorphism obstruction that made $\mathcal{M}_{1,1}$ fail to be a fine moduli space in example 1.3.4: forms, torsors, and the automorphisms that obstruct fine representability are three faces of one phenomenon, and the sheaf-theoretic language of theorem 6.4.1 is what unifies them. The next section develops the torsor picture directly.

Exercise 6.4.1

1. Let $U \to X$ be a faithfully flat quasi-compact morphism and $g, g'\colon Y \to Z$ two X-morphisms of X-schemes. Using theorem 6.4.1, show directly that if $g_U = g'_U$ then $g = g'$, by reducing to the affine case and using that a faithfully flat ring map $A \to A'$ is injective. Explain where faithful flatness (as opposed to plain flatness) is used.
2. Let $k = \mathbb{R}$, $L = \mathbb{C}$, and consider the $\mathbb{C}$-morphism $g_L\colon \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$ given by $t \mapsto (2+3i)t^2 + 5t$. Does g_L descend to an $\mathbb{R}$-morphism $\mathbb{A}^1_{\mathbb{R}} \to \mathbb{A}^1_{\mathbb{R}}$? If not, exhibit the failure of the compatibility condition of theorem 6.4.1 explicitly. Then find the condition on the coefficients of a general $g_L(t) = \sum a_i t^i$ under which it descends.
3. Show that "being an open immersion" is fppf-local on the base, in the pattern of theorem 6.4.3:

for $U \to X$ faithfully flat and $f\colon Y \to X$, if $f_U\colon Y_U \to U$ is an open immersion then so is f. (*Hint: an open immersion is a flat monomorphism locally of finite presentation; descend flatness by the theorem, descend the monomorphism property using injectivity of* Hom*-restriction, and use that $U \to X$ is surjective on points to descend openness of the image.*)

4. Explain why theorem 6.4.4 does not contradict theorem 6.4.5. Specifically, identify which hypothesis of the polarized-descent theorem fails for Hironaka's example, and state in one sentence why the same descent datum *is* effective in the category of algebraic spaces.

5. Compute $H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathrm{Aut})$ for the two forms of $\mathbb{G}_m$ over $\mathbb{R}$ from example 6.4.6, and identify the nonsplit form explicitly as the circle group. Then compute the forms of the two-dimensional split torus $\mathbb{G}_m^2$ over $\mathbb{R}$: what is $\mathrm{Aut}(\mathbb{G}_{m,\mathbb{C}}^2)$ as a Γ-module, and how many forms result? (*Hint:* $\mathrm{Aut}(\mathbb{G}_{m,\mathbb{C}}^2) = \mathrm{GL}_2(\mathbb{Z})$*; the forms are classified by conjugacy classes of order-dividing-*2 *elements* $\Gamma \to \mathrm{GL}_2(\mathbb{Z})$*, i.e. by* $\mathrm{GL}_2(\mathbb{Z})$*-conjugacy classes of involutions.*)

6. Let A/k be a finite-dimensional k-algebra, and let $A_0 = M_n(k)$ be the $n \times n$ matrix algebra. Explain why the forms of $M_n(k)$ over k, classified by $H^1(\Gamma, \mathrm{PGL}_n)$ (since $\mathrm{Aut}(M_n) = \mathrm{PGL}_n$ by the Skolem-Noether theorem), are exactly the central simple k-algebras of degree n, and connect this to the conic example: what is the form of $M_2(\mathbb{R})$ corresponding to the nontrivial class, and which conic is its Severi-Brauer variety?

6.5 Torsors and H^1

Faithfully flat descent, developed across the previous sections, answers the question of when an object defined locally on a cover glues to a global object. It does not by itself count the ways in which gluing can fail, or succeed nontrivially. That count is the content of the present section. The objects that measure it are torsors, and their classifying invariant is the first cohomology H^1 of the site. A torsor is the sheaf-theoretic shadow of a familiar geometric object: a line bundle is a torsor under the multiplicative group, a rank-n vector bundle is a torsor under GL_n, and in each case the set of isomorphism classes is a first cohomology group. The section makes this dictionary precise, proves that isomorphism classes of G-torsors are classified by $\check{H}^1$ of the site, and identifies the concrete geometry behind three fundamental groups of coefficients: $\mathbb{G}_m$, μ_n, and GL_n.

The torsor is also the first place in the book where the automorphism obstruction of example 1.3.4 acquires a positive counterpart. In Part I an object with nontrivial automorphisms resisted representation by a fine moduli space; the sheaf of trivializations of that obstruction is a torsor, and the failure of the moduli functor to be representable is measured by an H^1. Pushing the same idea one categorical level higher produces a gerbe, the two-dimensional analogue of a torsor, whose classifying invariant is H^2. The gerbe is named here and its full development deferred to Chapter 9; what this section supplies is the one-dimensional theory on which that development rests, together with the concrete operation, twisting a quasi-coherent sheaf by a torsor, that turns a cohomology class into geometry.

Throughout, τ denotes one of the topologies of definition 6.1.3 (Zariski, étale, smooth, or fppf), X a scheme over the base S, and G a sheaf of groups on the site $(\mathbf{Sch}/X)_\tau$ or on the small site X_τ. The multiplicative group $\mathbb{G}_m$, the group of n-th roots of unity μ_n, and the general linear group GL_n are the sheaves of groups $T \mapsto \mathbb{G}_m(T) = \Gamma(T, \mathcal{O}_T)^\times$, $T \mapsto \mu_n(T) = \{u \in \mathcal{O}_T(T)^\times \,|\, u^n = 1\}$, and $T \mapsto \mathrm{GL}_n(\mathcal{O}_T(T))$ respectively, each a representable sheaf in the fppf topology by theorem 6.2.4.

Definition 6.5.1: G-Torsor

Let G be a sheaf of groups on the site $(\mathbf{Sch}/X)_\tau$. A *G-torsor* (or *principal homogeneous space* under G) is a τ-sheaf P on X equipped with a left action $G \times P \to P$ such that

1. P is τ-locally nonempty: there is a covering family $\{U_i \to X\}$ with $P(U_i) \neq \varnothing$ for every i; and
2. the action is simply transitive: the map
$$G \times P \longrightarrow P \times P, \qquad (g, p) \longmapsto (g \cdot p,\, p)$$
is an isomorphism of sheaves.

Equivalently, P is τ-locally isomorphic to G acting on itself by left translation: over each U_i of a covering family there is a $G|_{U_i}$-equivariant isomorphism $P|_{U_i} \cong G|_{U_i}$. A *morphism of torsors* $P \to P'$ is a G-equivariant morphism of sheaves; every such morphism is an isomorphism. The torsor P is *trivial* if it has a global section, that is, if $P(X) \neq \varnothing$.

The two formulations of the action condition are equivalent, and the equivalence is worth spelling out because it isolates what a torsor supplies over what a mere G-set supplies. Condition (2), that $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism, says that once a point p of P is fixed, every other point is $g \cdot p$ for a unique g: the fiber of P is a copy of G, but with no distinguished element playing the role of the identity. A global section would supply such a distinguished element and thereby an equivariant isomorphism $P \cong G$; this is exactly why the existence of a global section is the same as triviality. The content of a nontrivial torsor is precisely that it looks like G from every local vantage but admits no global identification with G.

The claim that every morphism of torsors is an isomorphism is the first thing to verify, and it explains why the moduli of torsors is a *set* of isomorphism classes rather than a category with interesting morphisms.

Proposition 6.5.2: Torsor Morphisms Are Isomorphisms

Let P and P' be G-torsors on X in the topology τ. Any G-equivariant morphism of sheaves $f\colon P \to P'$ is an isomorphism. Consequently the category of G-torsors on X is a groupoid (every arrow is invertible), and the isomorphism classes of G-torsors form a set. The automorphism sheaf $\underline{\mathrm{Aut}}_G(P)$ of a torsor is the inner form ${}^{P}G$ of G, the twist of G by its conjugation action along P (the right-translation model of G glued by conjugation $h_j = g_{ij}\, h_i\, g_{ij}^{-1}$ along the transition cocycle); it recovers G exactly when P is trivial or G is abelian.

***Proof*:**
Being an isomorphism of sheaves is a τ-local property: a morphism of sheaves is an isomorphism if and only if it is one after restriction to each member of a covering family, since the sheaf condition recovers global sections from local ones. Choose a covering family $\{U_i \to X\}$ over which both P and P' trivialize, and refine so that a single family trivializes both. Over U_i, fix equivariant isomorphisms $P|_{U_i} \cong G|_{U_i}$ and $P'|_{U_i} \cong G|_{U_i}$; under these, the restriction $f|_{U_i}$ becomes a G-equivariant endomorphism of $G|_{U_i}$ acting on itself by left translation. Every

left-equivariant endomorphism of G is right translation by the fixed element $h_i = f(e) \in G(U_i)$, since $f(g) = f(g \cdot e) = g \cdot f(e) = gh_i$; this map is a bijection with inverse $g \mapsto gh_i^{-1}$. Hence $f|_{U_i}$ is an isomorphism for every i, and therefore f is an isomorphism. Every arrow of G-torsors is thus invertible, so the category is a groupoid; and since every torsor is glued from copies of a fixed local model G along a cover, the isomorphism classes form a set.

For the automorphism sheaf, apply the same computation with $P = P'$: over U_i an automorphism f restricts to right translation R_{h_i} by some $h_i \in G(U_i)$. The local models are glued along U_{ij} by the transition right translation $R_{g_{ij}}$, and for the restrictions of f to define a single global endomorphism they must commute with this gluing: $R_{h_i} \circ R_{g_{ij}} = R_{g_{ij}} \circ R_{h_j}$ on U_{ij}. Evaluating, $xg_{ij}h_i = xh_jg_{ij}$ for all x, so $g_{ij}h_i = h_jg_{ij}$, that is $h_j = g_{ij}\, h_i\, g_{ij}^{-1}$. The local elements h_i therefore do not agree on overlaps; they glue only after conjugation by the transition cocycle, so the family (h_i) is a section not of the constant sheaf G but of G twisted by the inner automorphisms $x \mapsto g_{ij}\, x\, g_{ij}^{-1}$. Sending f to this twisted section identifies $\underline{\mathrm{Aut}}_G(P)$ with G acting on itself by right translation, glued by inner automorphisms of the transition cocycle; this is the inner form of G cut out by P, and it recovers the constant G exactly when P is trivial. Thus $\underline{\mathrm{Aut}}_G(P)$ is a form of G, as we sought to show.

That every morphism of torsors is an isomorphism is the fact that keeps torsor theory one-dimensional. Torsors form a groupoid, not a category with genuinely different objects linked by non-invertible maps, so their classifying data are just the gluing isomorphisms on overlaps modulo the freedom to change local trivializations, with no higher coherence to track. The automorphism sheaf is a form of G, but it contributes no new isomorphism classes: it records the internal symmetry of a single torsor, not a way of building new ones. The moment the coefficient object acquires automorphisms whose own automorphisms matter, the classification jumps a categorical level, and the classifying invariant becomes H^2 rather than H^1; that is the gerbe, and it is the two-dimensional refinement that proposition 6.5.2 rules out for ordinary torsors.

Before the classification theorem, the mechanism by which a torsor produces a cohomology class must be made explicit, because the proof is nothing but that mechanism run forwards and backwards. A trivialized torsor over a cover is a copy of G over each cover-piece; the failure of the local trivializations to agree on overlaps is a family of elements of G indexed by pairs of cover-pieces, and the compatibility of these elements on triple overlaps is exactly the cocycle condition. This is the same descent datum of definition 6.3.1 in the special case where the fibered object is a torsor, and it is the reason the classification is a theorem about descent rather than a new construction.

Theorem 6.5.3: Classification of Torsors by H^1

Let G be a sheaf of groups on the site $(\mathbf{Sch}/X)_\tau$. There is a natural bijection

$$\{\text{isomorphism classes of } G\text{-torsors on } X \text{ in } \tau\} \xrightarrow{\sim} \check{H}^1(X_\tau, G)$$

between isomorphism classes of G-torsors and the first Čech cohomology of the site with coefficients in G, under which the trivial torsor corresponds to the distinguished (base-point) class. When G is nonabelian, both sides are pointed sets and the bijection is a bijection of pointed sets; when G is abelian, $\check{H}^1(X_\tau, G)$ is an abelian group, the bijection is a group isomorphism, and it agrees with the derived-functor cohomology $H^1(X_\tau, G)$.

***Proof*:**
Recall the Čech description of H^1. For a covering family $\mathcal{U} = \{U_i \to X\}$, write $U_{ij} = U_i \times_X U_j$ and $U_{ijk} = U_i \times_X U_j \times_X U_k$. A 1-*cocycle* on $\mathcal{U}$ with values in G is a family $(g_{ij}) \in \prod_{i,j} G(U_{ij})$ satisfying the cocycle condition

$$\mathrm{pr}_{13}^* g_{ik} = \mathrm{pr}_{12}^* g_{ij} \cdot \mathrm{pr}_{23}^* g_{jk} \quad \text{on } U_{ijk} \tag{6.4}$$

where pr_{ab} are the projections to the double overlaps. Two cocycles (g_{ij}) and (g'_{ij}) are *cohomologous* if there is a family $(h_i) \in \prod_i G(U_i)$ with $g'_{ij} = (\mathrm{pr}_1^* h_i)\, g_{ij}\, (\mathrm{pr}_2^* h_j)^{-1}$ on U_{ij}. The pointed set of cocycles modulo this equivalence is $\check{H}^1(\mathcal{U}, G)$, and $\check{H}^1(X_\tau, G)$ is the colimit over refinements of covering families. The base point is the class of the trivial cocycle $g_{ij} = e$.

Step 1: From a torsor to a cocycle. Let P be a G-torsor. Choose a covering family $\mathcal{U} = \{U_i \to X\}$ and trivializations, that is, equivariant isomorphisms $t_i \colon G|_{U_i} \xrightarrow{\sim} P|_{U_i}$; equivalently, by condition (2) of definition 6.5.1, sections $s_i = t_i(e) \in P(U_i)$, which exist by τ-local nonemptiness. On the overlap U_{ij} the two sections $\mathrm{pr}_1^* s_i$ and $\mathrm{pr}_2^* s_j$ are both points of $P(U_{ij})$, and by simple transitivity there is a unique $g_{ij} \in G(U_{ij})$ with

$$\mathrm{pr}_1^* s_i = g_{ij} \cdot \mathrm{pr}_2^* s_j$$

Comparing the three sections $\mathrm{pr}_1^* s_i$, $\mathrm{pr}_2^* s_j$, $\mathrm{pr}_3^* s_k$ over the triple overlap U_{ijk} and using uniqueness of the transition element gives, after pulling back along the appropriate projections, the identity (6.4): the composite of the transition from k to j with the transition from j to i is the transition from k to i. Thus (g_{ij}) is a 1-cocycle. A different choice of sections $s'_i = h_i \cdot s_i$ with $h_i \in G(U_i)$ replaces g_{ij} by the cohomologous cocycle $\mathrm{pr}_1^* h_i \cdot g_{ij} \cdot (\mathrm{pr}_2^* h_j)^{-1}$, and a refinement of the cover restricts the cocycle to the refinement. Hence the class $[(g_{ij})] \in \check{H}^1(X_\tau, G)$ is independent of all choices and depends only on the isomorphism class of P.

Step 2: From a cocycle to a torsor. Conversely, let (g_{ij}) be a 1-cocycle on $\mathcal{U}$. Build a sheaf P by descent: on each U_i take the trivial torsor $G|_{U_i}$, and glue $G|_{U_j}$ to $G|_{U_i}$ over U_{ij} by the isomorphism $\varphi_{ij} \colon \mathrm{pr}_2^*(G|_{U_j}) \to \mathrm{pr}_1^*(G|_{U_i})$ given by *right* translation by g_{ij}^{-1}, that is $x \mapsto x \cdot g_{ij}^{-1}$. Right translation is forced: the torsor G-action to be constructed is *left* translation, and only right translations commute with left translations, so only a right-translation gluing descends to a G-equivariant sheaf. Composing the transitions $k \to j \to i$ gives $\varphi_{ij} \circ \varphi_{jk} = R_{g_{ij}^{-1}} \circ R_{g_{jk}^{-1}}$, the map $x \mapsto x\, g_{jk}^{-1} g_{ij}^{-1} = x\,(g_{ij} g_{jk})^{-1}$; this equals $\varphi_{ik} = R_{g_{ik}^{-1}}$ precisely when $g_{ik} = g_{ij} g_{jk}$, so the descent cocycle condition of definition 6.3.1 for this gluing datum is equivalent to the 1-cocycle condition (6.4). Since G is a sheaf of groups rather than a quasi-coherent module, the relevant descent is that sheaves form a stack for τ: a gluing datum of τ-sheaves satisfying the cocycle condition is effective, and the local pieces glue to a τ-sheaf P on X (this is the sheaf-theoretic content underlying the faithfully flat descent of theorem 6.3.4, specialized to the sheaf of sets underlying G). The torsor G-action on P is the one induced by *left* translation on each $G|_{U_i}$: left translation commutes with the right translations φ_{ij} that glue the pieces, so it descends to a well-defined left action on P, and it is simply transitive because left translation is so on each $G|_{U_i}$. Thus P is a G-torsor trivialized over $\mathcal{U}$, with tautological section s_i on U_i the identity of the i-th piece $G|_{U_i}$. It remains to check that Step 1 applied to P returns the class $[(g_{ij})]$ with no inversion. Over U_{ij} the transition isomorphism φ_{ij} identifies the section $s_j = e$ of the j-th piece with the section $\varphi_{ij}(e) = e \cdot g_{ij}^{-1} = g_{ij}^{-1}$ of the i-th piece; call this common section $\mathrm{pr}_2^* s_j$. Applying left translation, the Step 1 recipe reads off the unique element $\gamma_{ij} \in G(U_{ij})$ with $\mathrm{pr}_1^* s_i = \gamma_{ij} \cdot \mathrm{pr}_2^* s_j$, that is $e = \gamma_{ij}\, g_{ij}^{-1}$, whence $\gamma_{ij} = g_{ij}$. Thus the recovered cocycle is (g_{ij}) itself, and no inversion arises. Hence Step 1 applied to this P returns the class $[(g_{ij})]$.

Step 3: The two constructions are mutually inverse. Step 1 sends a torsor to a class; Step 2 sends a cocycle to a torsor whose class is the given one, and cohomologous cocycles produce isomorphic torsors because a coboundary (h_i) is precisely a τ-local change of trivialization, which by proposition 6.5.2 induces an isomorphism of the glued torsors. Conversely a torsor P, trivialized over $\mathcal{U}$ with cocycle (g_{ij}), is isomorphic to the torsor glued from (g_{ij}) in Step 2 via the map that sends the local section s_i to the tautological section of the i-th piece. The two assignments are therefore mutually inverse bijections of pointed sets, and the trivial torsor, having a global section, has all transition elements equal to e after a suitable refinement, hence maps to the base-point class.

Step 4: The abelian case. When G is abelian, the cocycle condition (6.4) is additive and the set of cocycles is an abelian group under pointwise multiplication, with coboundaries a subgroup; thus $\check{H}^1(\mathcal{U}, G)$ and its colimit $\check{H}^1(X_\tau, G)$ are abelian groups. The group law transports to torsors as the *contracted product*: given torsors P, P' with cocycles (g_{ij}), (g'_{ij}), the torsor $P \wedge^G P' := (P \times P')/G$, quotient by the antidiagonal action $g \cdot (p, p') = (gp, g^{-1}p')$, has cocycle $(g_{ij}g'_{ij})$, the inverse of P is the torsor with cocycle (g_{ij}^{-1}), and the trivial torsor is the identity. That the Čech group $\check{H}^1$ agrees with the derived-functor cohomology $H^1(X_\tau, G)$ for abelian G is a standard comparison: $\check{H}^1 \to H^1$ is always an isomorphism in degree 1 on any site, since a class in either group is a torsor and the two descriptions of a torsor, gluing data versus a sheaf that is locally G, coincide. The derived statement, that $H^1(X_\tau, G) = R^1\Gamma(X, G)$ classifies the same torsors, is proved in [10]through the machinery of right-derived functors of global sections and holds for any abelian sheaf on a site. This gives the group isomorphism and completes the proof.

The classification is the organizing principle of the entire chapter turned into a single statement. Descent, in the form of theorem 6.3.4, manufactures a global object from local data glued by a cocycle; the classification recognizes that when the object being glued is a torsor, the gluing data are exactly a 1-cocycle, and the ambiguity in the gluing is exactly a coboundary. Every geometric object that is locally trivial in some topology, a line bundle, a vector bundle, a covering with structure group, a form of a fixed object, is a torsor under its automorphism sheaf, and its isomorphism classes are therefore an H^1. The theorem is the bridge that carries every such classification problem into cohomology, where exact sequences and long exact sequences become available. The point of Galois descent in example 6.4.6, that forms of an object over k are classified by H^1 of the Galois group with values in the automorphism group, is the special case of this theorem for the étale topology on $\operatorname{Spec} k$, where the site is the category of finite separable extensions and Čech H^1 is Galois cohomology; the point/field version is developed in full in [9].

The nonabelian case deserves a word, because the loss of a group structure on H^1 is not a defect but a feature of the geometry. When G is nonabelian, $\check{H}^1(X_\tau, G)$ is only a pointed set: there is no natural way to multiply two G-torsors, because the contracted product of Step 4 requires the antidiagonal action to be well-defined, which needs commutativity. What survives is the base point, the trivial torsor, and the distinguished role it plays in exact sequences of pointed sets. This is precisely the structure needed to classify forms and vector bundles, neither of which forms a group under any natural operation, and it is why GL_n- and PGL_n-torsors are handled with pointed-set cohomology rather than group cohomology.

The torsor is the one-dimensional shadow of a phenomenon that recurs one level up. To see the next level, consider not a single obstruction but the obstruction to trivializing an obstruction. When a moduli problem fails to be representable because its objects carry automorphisms, as in example 1.3.4, the sheaf recording local trivializations of the obstruction is a torsor, and its class in H^1 measures the failure. But the objects of the moduli problem themselves form a sheaf of categories, not a sheaf of sets, and the two-dimensional analogue of a torsor, a sheaf of categories that is locally equivalent to the classifying stack of G, carries a class in H^2. That object is a gerbe. The full theory of gerbes, their classification by H^2, and the Brauer group $H^2(X, \mathbb{G}_m)$ they compute, is the subject of Chapter 9; the present section supplies only its one-dimensional foundation and the name.

Definition 6.5.4: Twisting by a Torsor, and Gerbes

Let P be a G-torsor on X and let F be a quasi-coherent sheaf on X carrying a left action of G by $\mathcal{O}_X$-linear automorphisms (a G-equivariant sheaf). The *twist of F by P*, or *associated bundle*, is the quasi-coherent sheaf

$$P \times^G F := (P \times F)/G$$

the quotient of $P \times F$ by the diagonal action $g \cdot (p, v) = (g \cdot p,\ g \cdot v)$; it is τ-locally isomorphic to F and glued by the image of the transition cocycle (g_{ij}) of P under the action $G \to \mathrm{Aut}_{\mathcal{O}_X}(F)$. A *gerbe banded by G* (for G a sheaf of abelian groups) is the two-dimensional analogue: a stack of groupoids over X_τ that is τ-locally equivalent to the classifying stack BG and whose automorphism band is identified with G. Gerbes are classified by $H^2(X_\tau, G)$, generalizing the classification of torsors by H^1; their theory is developed in Chapter 9, and only their name is fixed here.

Twisting is the operation that turns a cohomology class into a new geometric object, and it is the concrete payoff of the classification theorem. Given a G-torsor P with class in H^1 and any representation of G on a sheaf F, the twist $P \times^G F$ is a sheaf that agrees with F locally but is reglued by the transition functions of P. The construction is functorial in F and multiplicative in P: the whole category of G-equivariant sheaves is transported by a single torsor, which is why a torsor is the correct notion of "structure group data". When $G = \mathbb{G}_m$ acts on $F = \mathcal{O}_X$ by scaling, the twist is a line bundle, and the following example makes this the first of three fundamental computations.

The passage from torsors to gerbes is the passage from H^1 to H^2, and the automorphism gerbe of a point of a moduli stack, the two-dimensional refinement of the automorphism obstruction of example 1.3.4, is what a stack sees where a scheme sees only a coarse point. This is the thread that ties example 1.3.4 and the j-line of theorem 1.3.6 to the gerbe chapter.

With twisting in hand, the three anchor examples identify the geometry behind the three coefficient sheaves that recur throughout algebraic geometry. Each is an instance of the classification theorem, and together they justify the claim that H^1 is where line bundles, roots of unity, and vector bundles live.

Example 6.5.5: $\mathbb{G}_m$-, μ_n-, and GL_n-Torsors

The three fundamental coefficient sheaves each classify a familiar class of geometric objects.

1. *$\mathbb{G}_m$-torsors are line bundles.* A line bundle L on X is a locally free $\mathcal{O}_X$-module of rank 1; its sheaf of local trivializations, the sheaf $\underline{\mathrm{Isom}}(\mathcal{O}_X, L)$ of nowhere-vanishing local generators, is a $\mathbb{G}_m$-torsor, since two local generators differ by a unit. Conversely, twisting $\mathcal{O}_X$ by a $\mathbb{G}_m$-torsor P via the scaling action produces a line bundle $P \times^{\mathbb{G}_m} \mathcal{O}_X$. By theorem 6.5.3 the isomorphism classes of line bundles are

$$\mathrm{Pic}\,(X) = H^1(X_{\mathrm{Zar}}, \mathbb{G}_m) = H^1(X_{\text{ét}}, \mathbb{G}_m)$$

the equality of the Zariski and étale computations being Hilbert's Theorem 90 ([9]): a $\mathbb{G}_m$-torsor that is étale-locally trivial is already Zariski-locally trivial, so no new line bundles appear on passing to the finer topology. The group law on $H^1(X, \mathbb{G}_m)$ is the tensor product of line bundles, and the abelian structure of Step 4 of theorem 6.5.3 is exactly $L \otimes L'$.

2. *μ_n-torsors and the Kummer sequence.* The n-th power map on units gives, in the fppf topology (and in the étale topology when n is invertible on X), a short exact sequence of sheaves of abelian groups

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\cdot)^n} \mathbb{G}_m \longrightarrow 1 \tag{6.5}$$

the *Kummer sequence*, exact because every unit fppf-locally admits an n-th root (adjoin a root of $T^n - u$). Its long exact cohomology sequence reads

$$\mathcal{O}_X(X)^\times \xrightarrow{n} \mathcal{O}_X(X)^\times \to H^1(X, \mu_n) \to \operatorname{Pic}(X) \xrightarrow{n} \operatorname{Pic}(X)$$

which exhibits $H^1(X, \mu_n)$ as an extension

$$0 \to \mathcal{O}_X(X)^\times/(\mathcal{O}_X(X)^\times)^n \to H^1(X, \mu_n) \to \operatorname{Pic}(X)[n] \to 0$$

of the n-torsion $\operatorname{Pic}(X)[n]$ by the global units modulo n-th powers. A μ_n-torsor is thus the data of a line bundle together with a trivialization of its n-th power, the geometric object underlying cyclic covers and the Kummer theory of [9].

3. GL_n*-torsors are vector bundles.* A locally free sheaf E of rank n has, as its sheaf of local frames $\underline{\mathrm{Isom}}(\mathcal{O}_X^n, E)$, a GL_n-torsor, and twisting the standard representation $\mathcal{O}_X^n$ by a GL_n-torsor recovers a rank-n vector bundle. By theorem 6.5.3, isomorphism classes of rank-n vector bundles on X are the pointed set $H^1(X_{\mathrm{Zar}}, \mathrm{GL}_n) = H^1(X_{\text{ét}}, \mathrm{GL}_n)$, again with Zariski and étale agreeing by the generalized Hilbert 90 for GL_n ([9]). Since GL_n is nonabelian for $n \geq 2$, this H^1 is only a pointed set: vector bundles do not form a group, matching the general nonabelian phenomenon.

The three computations exhibit the entire range of the classification. For $\mathbb{G}_m$ the coefficient is abelian and H^1 is the Picard group, a genuine group under tensor product. For μ_n the same abelian formalism, run through the long exact sequence of the Kummer sequence (6.5), extracts the n-torsion of the Picard group and the units modulo n-th powers, a computation that recurs whenever a variety is studied through its n-torsion. For GL_n the coefficient is nonabelian and H^1 is a pointed set, the moduli of vector bundles up to isomorphism, with no group structure because there is no natural product of vector bundles of fixed rank. Each is an application of the single theorem theorem 6.5.3, and each identifies a cohomology group the reader has met before with a class of geometric objects.

The automorphism thread of Part I now closes into this picture. In example 1.3.4 the elliptic curves with extra automorphisms obstructed the moduli functor $\mathcal{M}_{1,1}$ from being representable by a fine moduli space, and the coarse space, the j-line of theorem 1.3.6, forgot the automorphisms entirely. The torsor theory of this section explains why: over the j-line, the family of elliptic curves with a given j-invariant is not a single curve but a $\mathbb{G}_m$- or μ_n-torsor's worth of curves, twisted by the automorphism group, and the fibers over the special points $j = 0$ and $j = 1728$, where in characteristic $\neq 2, 3$ the automorphism group jumps to μ_6 and μ_4 respectively (in characteristics 2 and 3 these loci merge and the group can be larger), carry nontrivial H^1 with values in those groups. The correct object over the j-line is not a scheme but a stack whose points carry these automorphism groups, and the two-dimensional refinement of the obstruction, the automorphism gerbe of a point of $\mathcal{M}_{1,1}$, is the H^2-shadow that Chapter 9 develops. The torsor and its H^1 are the first one-dimensional trace of the stack that Part III is built to construct.

Exercise 6.5.1

1. Let P be a G-torsor on X in a topology τ. Prove that P is trivial if and only if it has a global section, and that a global section, when it exists, determines an equivariant isomorphism $P \cong G$; describe the set of all such isomorphisms and show it is a torsor under $G(X)$ acting by right translation.

2. Compute $H^1(\operatorname{Spec} k, \mathbb{G}_m)$ and $H^1(\operatorname{Spec} k, \mathrm{GL}_n)$ for a field k, and interpret each geometrically. (Hint: a line bundle on $\operatorname{Spec} k$ is a one-dimensional k-vector space; a rank-n vector bundle is an n-dimensional k-vector space. What is the classification up to isomorphism?)

3. Using the Kummer sequence (6.5), compute $H^1(\operatorname{Spec} k, \mu_n)$ for a field k with n invertible in k, and identify it with a familiar group from Kummer theory. Then compute $H^1(\mathbb{P}^1_k, \mu_n)$ using $\operatorname{Pic}(\mathbb{P}^1_k) = \mathbb{Z}$ and $\Gamma(\mathbb{P}^1_k, \mathcal{O})^\times = k^\times$.

4. Let L/k be a finite Galois extension with group Γ, and let V be a finite-dimensional k-vector space. Show that the twists of V by $\mathrm{GL}(V)$-torsors trivialized by $\operatorname{Spec} L \to \operatorname{Spec} k$ are classified by the Galois cohomology $H^1(\Gamma, \mathrm{GL}(V)(L))$, and deduce from example 6.4.6 that this pointed set is trivial: every such twist is isomorphic to V. (This is the vector-space case of Hilbert's Theorem 90; compare [9].)

5. Let P be a $\mathbb{G}_m$-torsor with class $[P] \in H^1(X, \mathbb{G}_m) = \operatorname{Pic}(X)$ corresponding to a line bundle L. Show that the twist $P \times^{\mathbb{G}_m} \mathcal{O}_X$ (scaling action) is isomorphic to L, and that the twist $P \times^{\mathbb{G}_m} \mathcal{O}_X$ by the action of weight d (where $\lambda \in \mathbb{G}_m$ acts by λ^d) is $L^{\otimes d}$. Conclude that twisting realizes the group law on $\operatorname{Pic}(X)$.

6. Explain why the classification theorem 6.5.3 produces a group when G is abelian but only a pointed set when G is nonabelian, by locating the exact step in the proof where commutativity is used. Then give an example of two GL_2-torsors (rank-2 vector bundles) on a curve for which no natural product exists, illustrating the absence of a group structure.

7

Fibered Categories and Stacks

Descent theory taught the objects of a moduli problem to glue in a Grothendieck topology, but it treated them as elements of a set: two local objects either agree on an overlap or they do not. The moduli problems that resisted representability throughout Part I fail exactly this dichotomy. An elliptic curve has automorphisms, so two families that restrict to isomorphic curves over an overlap are identified not on the nose but through a chosen isomorphism, and different choices give different global objects. A structure that records objects together with the isomorphisms between them is not a sheaf of sets but a sheaf of groupoids, and the geometric incarnation of a sheaf of groupoids is a category fibered in groupoids.

This chapter builds the resulting language. The first section defines fibered categories and categories fibered in groupoids and proves the 2-Yoneda lemma, which embeds schemes among them and fixes the meaning of representability. The second imposes the descent condition from Chapter 6 one categorical level higher: a category fibered in groupoids whose morphisms and whose objects both glue in the topology is a stack, and the faithfully flat descent already proved becomes the statement that quasi-coherent sheaves and torsors form stacks, while stackification manufactures a stack from any fibered category. The third section constructs the first geometric stacks, the quotient stack $[X/G]$ and the classifying stack BG, whose points are torsors with an equivariant map; here $\mathcal{M}_{1,1}$ finally becomes an object in its own right, a quotient stack whose coarse space is the j-line and whose points carry the automorphisms the j-line forgot. The last section explains how geometry is imposed on a stack through representable morphisms and the diagonal, isolating the two conditions that Chapter 8 will promote to the definition of an algebraic stack.

7.1 Fibered Categories

Chapter 6 taught how data glue along a faithfully flat cover: a quasi-coherent sheaf on X is the same as a sheaf on a cover $U \to X$ with a gluing isomorphism satisfying the cocycle condition, and a torsor on X is classified by a first cohomology set. The data descended there were sets, sheaves, and morphisms between them. A moduli problem asks for something richer. The elliptic curves parametrized by $\mathcal{M}_{1,1}$ carry automorphisms, and any object with a nontrivial automorphism refuses to be recorded by a functor to sets: the value $F(T)$ of a moduli functor at a scheme T is a *set* of isomorphism classes, and the moment two families become isomorphic without being equal, the isomorphism, and with it the automorphisms, is discarded. This section builds the categorical vessel that keeps that information. Instead of a set-valued functor, a moduli problem is organized as a category $\mathcal{F}$ mapping to the category $\mathcal{C}$ of test schemes, whose fiber over each T is not a set but a groupoid of families and their isomorphisms.

The organizing structure is that of a *fibered category*: a functor $p\colon \mathcal{F} \to \mathcal{C}$ in which objects can be pulled back along morphisms of the base, coherently and essentially uniquely. When every fiber is a groupoid, this is a *category fibered in groupoids*, the correct home for a moduli problem with automorphisms. The section proves the 2-categorical Yoneda lemma, which shows that a scheme is faithfully recorded as such a fibered category and pins down what it means for a fibered category to be representable, and it grounds the abstraction in the three objects that thread the chapter: the fibered category of quasi-coherent sheaves, the classifying object BG of G-torsors, and $\mathcal{M}_{1,1}$. The 2-categorical language, groupoids, pseudofunctors, and the ordinary Yoneda lemma, is imported from [5]; the treatment of fibered categories follows the canonical reference [30].

Throughout, fix a base scheme S and a topology τ (default: the fppf topology, definition 6.1.3), and let $\mathcal{C} = (\mathbf{Sch}/S)_\tau$ be the category of S-schemes equipped with τ. A moduli problem is a rule assigning to each test scheme T a category of families over T, together with pullback along morphisms $T' \to T$. Recording this as a single category over $\mathcal{C}$ requires a mechanism for pullback that is intrinsic to the projection p, and that is what a cartesian arrow provides.

The definition of a cartesian arrow copies the universal property of a fiber product, stripped to the part that survives when there is no base change to speak of, only a projection p. An arrow $\varphi\colon x \to y$ in $\mathcal{F}$ lying over $f\colon U \to V$ in $\mathcal{C}$ deserves to be called the pullback of y along f if every arrow into y that factors through f on the base factors uniquely through φ upstairs.

Definition 7.1.1: Cartesian Arrow and Fibered Category

Let $p\colon \mathcal{F} \to \mathcal{C}$ be a functor. An arrow $\varphi\colon x \to y$ in $\mathcal{F}$ is *cartesian* (over $f = p(\varphi)\colon U \to V$) if for every arrow $\psi\colon z \to y$ in $\mathcal{F}$ and every factorization $p(\psi) = f \circ h$ of $p(\psi)$ through f in $\mathcal{C}$, there is a unique arrow $\theta\colon z \to x$ in $\mathcal{F}$ with $p(\theta) = h$ and $\varphi \circ \theta = \psi$. When such a cartesian φ exists, x is a *pullback* of y along f, written f^*y.
The functor p is a *fibered category* over $\mathcal{C}$ (and $\mathcal{C}$ is the *base*) if:

1. for every arrow $f\colon U \to V$ in $\mathcal{C}$ and every object y of $\mathcal{F}$ with $p(y) = V$, there is a cartesian arrow $\varphi\colon f^*y \to y$ over f; and
2. the composite of two cartesian arrows is cartesian.

For an object U of $\mathcal{C}$, the *fiber* $\mathcal{F}(U)$ is the subcategory of $\mathcal{F}$ whose objects are the x with $p(x) = U$ and whose arrows are those α with $p(\alpha) = \mathrm{id}_U$; such an α is called *vertical*. An object x with $p(x) = U$ is written x/U.

The universal property in the definition of cartesian is exactly the one that makes a pullback well-defined up to unique isomorphism. If both $\varphi\colon x \to y$ and $\varphi'\colon x' \to y$ are cartesian over the same f, applying the universal property of φ to φ' (with $h = \mathrm{id}_U$) and of φ' to φ produces mutually inverse vertical isomorphisms $x \cong x'$ over U. So the pullback f^*y is an object of the fiber $\mathcal{F}(U)$ determined up to a unique isomorphism compatible with the maps to y. This is the precise sense in which "pullback" makes sense in a fibered category even though no actual choice of pullback has been fixed: the fibered-category axioms guarantee existence, and the universal property guarantees uniqueness up to unique isomorphism, but a concrete pullback is a further choice.

Making that choice, once and for all, is a *cleavage*, and it turns the intrinsic pullback into a well-defined functor between fibers. A cleavage is the device that converts the geometry of $\mathcal{F}$ into the bookkeeping of a pseudofunctor.

Definition 7.1.2: Cleavage and Pullback Functors

A *cleavage* of a fibered category $p\colon \mathcal{F} \to \mathcal{C}$ is a choice, for each arrow $f\colon U \to V$ in $\mathcal{C}$ and each object y/V, of a cartesian arrow $\alpha_f(y)\colon f^*y \to y$ over f. Given a cleavage, each $f\colon U \to V$ induces a *pullback functor*

$$f^*\colon \mathcal{F}(V) \longrightarrow \mathcal{F}(U)$$

sending y/V to the chosen f^*y and sending a vertical arrow $\beta\colon y \to y'$ to the unique vertical arrow $f^*\beta\colon f^*y \to f^*y'$ with $\alpha_f(y') \circ f^*\beta = \beta \circ \alpha_f(y)$ (which exists and is unique because $\alpha_f(y')$ is cartesian). For composable $U \xrightarrow{f} V \xrightarrow{g} W$ there are canonical isomorphisms of functors

$$\epsilon_{f,g}\colon f^*g^* \xrightarrow{\sim} (g \circ f)^*, \qquad \mathrm{id}_U^* \cong \mathrm{id}_{\mathcal{F}(U)}$$

coming from the uniqueness of cartesian lifts, and these satisfy a compatibility (cocycle) condition for triple composites.

The isomorphisms $\epsilon_{f,g}$ are the reason a fibered category is *not* literally a contravariant functor from $\mathcal{C}$ to categories: pulling back along g and then along f agrees with pulling back along $g \circ f$ only up to the canonical isomorphism $\epsilon_{f,g}$, not on the nose, because the cleavage chose the two sides independently. The assignment $U \mapsto \mathcal{F}(U)$, $f \mapsto f^*$, $\epsilon_{f,g}$ is precisely the data of a *pseudofunctor* $\mathcal{C}^{\mathrm{op}} \to \mathbf{Cat}$, where the coherence condition on triple composites is the pentagon-type axiom for pseudofunctors (see [5] for the general notion, and [30] for the equivalence in this setting). The converse also holds: from a pseudofunctor one reconstructs a fibered category by the Grothendieck construction, taking objects to be pairs (U, x) with $x \in \mathcal{F}(U)$ and arrows $(U, x) \to (V, y)$ to be pairs (f, β) with $f\colon U \to V$ and $\beta\colon x \to f^*y$ vertical. The two viewpoints are equivalent, and the section moves freely between them: the fibered category $p\colon \mathcal{F} \to \mathcal{C}$ when the intrinsic geometry is cleaner, the pseudofunctor $U \mapsto \mathcal{F}(U)$ when an explicit rule for the fibers is at hand.

7.1.3 Convention: Fibered Categories versus Pseudofunctors A cleavage always exists (it is a choice of one cartesian arrow per (f, y), an application of choice at the level of the class of arrows), so every fibered category yields a pseudofunctor and conversely. The two formulations carry the same information, and no result below depends on a particular cleavage. When a moduli problem is presented by a rule "over T, the objects are such-and-such families", it is being described as a pseudofunctor; when it is presented as a category of all families at once with a projection to the base, it is being described as a fibered category. The word "pullback" means the intrinsic operation of definition 7.1.1, canonical up to unique isomorphism, whether or not a cleavage has been named.

A moduli problem with automorphisms lives in a fibered category of a special kind: one whose fibers are groupoids, so that every family has a well-defined group of automorphisms and every morphism of families is invertible. The two conditions, all arrows cartesian and all fibers groupoids, turn out to be the same.

Definition 7.1.4: Category Fibered in Groupoids

A fibered category $p\colon \mathcal{F} \to \mathcal{C}$ is a *category fibered in groupoids* (a *CFG*) if every arrow of $\mathcal{F}$ is cartesian. Equivalently (see proposition 7.1.5), $p\colon \mathcal{F} \to \mathcal{C}$ is a CFG if and only if:

1. for every arrow $f\colon U \to V$ in $\mathcal{C}$ and every y/V there is an arrow $x \to y$ over f; and
2. for every pair of arrows $\varphi\colon x \to z$, $\psi\colon y \to z$ in $\mathcal{F}$ and every arrow $h\colon p(x) \to p(y)$ in $\mathcal{C}$ with $p(\psi) \circ h = p(\varphi)$, there is a unique arrow $\theta\colon x \to y$ over h with $\psi \circ \theta = \varphi$.

Under a cleavage, the associated pseudofunctor takes values in the 2-category **Grpd** of groupoids: each fiber $\mathcal{F}(U)$ is a groupoid.

The equivalence of the "all arrows cartesian" description with the two-condition description, and with the requirement that every fiber be a groupoid, is worth proving in full, because it is the criterion one checks in practice: to verify that a rule $T \mapsto$ (a category of families) is a CFG one confirms that families pull back and that morphisms of families glue uniquely.

Proposition 7.1.5: Characterizations of a CFG

Let $p\colon \mathcal{F} \to \mathcal{C}$ be a functor. The following are equivalent:

1. p is a fibered category and every arrow of $\mathcal{F}$ is cartesian;
2. conditions (1) and (2) of definition 7.1.4 hold;
3. p is a fibered category and every fiber $\mathcal{F}(U)$ is a groupoid.

***Proof*:**
The argument runs (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

Step 1: (1) $\Rightarrow$ (2). Condition (2)(1) is the existence half of the fibered-category axiom, so it holds. For (2)(2), let $\varphi\colon x \to z$, $\psi\colon y \to z$ and $h\colon p(x) \to p(y)$ satisfy $p(\psi) \circ h = p(\varphi)$. Since ψ is cartesian (all arrows are, by hypothesis) and $p(\varphi) = p(\psi) \circ h$ factors $p(\varphi)$ through $p(\psi)$, the universal property of the cartesian arrow ψ yields a unique $\theta\colon x \to y$ with $p(\theta) = h$ and $\psi \circ \theta = \varphi$. This is exactly (2)(2).

Step 2: (2) $\Rightarrow$ (3). First, p is a fibered category. Existence of cartesian lifts: given $f\colon U \to V$ and y/V, condition (2)(1) supplies an arrow $\varphi\colon x \to y$ over f; this φ is cartesian, for given $\psi\colon z \to y$ with $p(\psi) = f \circ h$, apply (2)(2) to $\varphi\colon x \to y$, $\psi\colon z \to y$ and h (noting $p(\varphi) \circ h = f \circ h = p(\psi)$) to get the unique $\theta\colon z \to x$ over h with $\varphi \circ \theta = \psi$. Stability of cartesian arrows under composition follows directly from the universal property. Now each fiber is a groupoid: let $\alpha\colon x \to y$ be an arrow vertical over U (so $p(\alpha) = \mathrm{id}_U$). Apply condition (2)(2) to the pair $\varphi = \mathrm{id}_y\colon y \to y$, $\psi = \alpha\colon x \to y$ with target y and $h = \mathrm{id}_U$ (valid since $p(\mathrm{id}_y) \circ \mathrm{id}_U = \mathrm{id}_U = p(\alpha)$): this yields a unique arrow $\beta\colon y \to x$ over id_U with $\alpha \circ \beta = \mathrm{id}_y$, a vertical right inverse of α. Applying (2)(2) once more, this time to $\varphi = \alpha$, $\psi = \alpha$ with $h = \mathrm{id}_U$, shows that $\beta \circ \alpha = \mathrm{id}_x$: both $\beta \circ \alpha$ and id_x are vertical arrows $x \to x$ solving the same factorization $\alpha \circ (-) = \alpha$, and (2)(2) forces them equal. Hence β is a two-sided inverse and α is an isomorphism, so $\mathcal{F}(U)$ is a groupoid.

Step 3: (3) $\Rightarrow$ (1). Let $\varphi\colon x \to y$ be any arrow of $\mathcal{F}$, over $f = p(\varphi)\colon U \to V$. Since p is a fibered category there is a cartesian arrow $\varphi_0\colon x_0 \to y$ over f, and by its universal property (with $h = \mathrm{id}_U$, $\psi = \varphi$) there is a vertical $\theta\colon x \to x_0$ over id_U with $\varphi_0 \circ \theta = \varphi$. By hypothesis (3) the fiber $\mathcal{F}(U)$ is a groupoid, so θ is an isomorphism. An arrow equal to a cartesian arrow precomposed with a vertical isomorphism is again cartesian: the universal property transports across the isomorphism θ. Therefore φ is cartesian, and every arrow of $\mathcal{F}$ is cartesian, completing the proof.

The proposition is the working definition. A category over $\mathcal{C}$ is a CFG precisely when families pull back (condition (2)(1)) and morphisms of families are determined by their behavior on the base together with a compatible target (condition (2)(2)); the payoff is that every fiber is automatically a groupoid, so automorphisms are built in. In the pseudofunctor picture, a CFG is a pseudofunctor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Grpd}$, and this is the definitive form of a moduli problem: to each test scheme T, a groupoid of families over T; to each $T' \to T$, a pullback functor; and coherence isomorphisms for composites.

Morphisms between CFGs and the 2-morphisms between those are what make CFGs into a 2-category, the ambient world in which stacks live.

Definition 7.1.6: Morphisms and 2-Morphisms of CFGs

A *morphism* of CFGs $F\colon \mathcal{F} \to \mathcal{G}$ over $\mathcal{C}$ is a functor F with $p_{\mathcal{G}} \circ F = p_{\mathcal{F}}$ (so F carries $\mathcal{F}(U)$ into $\mathcal{G}(U)$ for each U). Because every arrow of a CFG is cartesian, such an F automatically preserves cartesian arrows. A 2-*morphism* $\eta\colon F \Rightarrow F'$ between two such morphisms is a natural transformation that is the identity on the base, meaning $p_{\mathcal{G}}(\eta_x) = \mathrm{id}_{p_{\mathcal{F}}(x)}$ for every object x of $\mathcal{F}$; equivalently, each component $\eta_x\colon F(x) \to F'(x)$ is a vertical arrow of $\mathcal{G}$. Since fibers are groupoids, every such η is invertible, so it is a natural isomorphism. CFGs over $\mathcal{C}$, their morphisms, and their 2-morphisms form a 2-category, in which the Hom between two objects is itself a groupoid. An *equivalence* of CFGs is an equivalence in this 2-category: an F admitting G with $F \circ G$ and $G \circ F$ 2-isomorphic to the identities.

Two CFGs are considered the same when they are equivalent, never when they are isomorphic: an equivalence of categories is the correct notion of sameness for groupoid-valued data, and demanding an isomorphism would be as unreasonable as demanding that two equivalent categories have equal objects. The groupoid $\underline{\mathrm{Hom}}_{\mathcal{C}}(\mathcal{F}, \mathcal{G})$ of morphisms and 2-morphisms is the internal Hom of the 2-category, and its appearance in the Yoneda statement below is what makes that statement a 2-categorical refinement of the ordinary one.

The first and most basic examples are the schemes themselves. A scheme X over S becomes a CFG in a canonical way, and the construction is the 2-categorical shadow of the functor of points [12, the Functor of Points].

Definition 7.1.7: The CFG Represented by a Scheme

For an S-scheme X, the *representable CFG* $\underline{X}$ is the category whose objects are the S-morphisms $T \to X$ (for T ranging over $\mathcal{C}$) and whose arrows $(T' \xrightarrow{a} X) \to (T \xrightarrow{b} X)$ are the S-morphisms $g\colon T' \to T$ with $b \circ g = a$. The projection $p\colon \underline{X} \to \mathcal{C}$ sends $(T \to X)$ to T and g to g. A CFG $\mathcal{F}$ over $\mathcal{C}$ is *representable* (by a scheme) if there is an equivalence $\mathcal{F} \simeq \underline{X}$ for some S-scheme X.

The category $\underline{X}$ is fibered in groupoids because its fibers are discrete: the fiber $\underline{X}(T)$ is the *set* $\mathrm{Hom}_S(T, X)$ regarded as a groupoid with only identity arrows, since a vertical arrow is a $g = \mathrm{id}_T$ commuting with the structure maps, forcing $a = b$. Thus $\underline{X}$ has no nontrivial automorphisms anywhere; a scheme, viewed as a moduli problem, parametrizes objects with no automorphisms. This is the structural reason a moduli problem with automorphisms cannot be represented by a scheme, and the reason stacks are needed: representability by a scheme forces every fiber to be a set. The pseudofunctor associated with $\underline{X}$ is exactly the functor of points $h_X = \mathrm{Hom}_S(-, X)$, now landing in discrete groupoids rather than sets.

Representing X this way loses no information about X, and CFG-morphisms out of $\underline{X}$ are computed by evaluation. This is the content of the 2-Yoneda lemma, which lifts the ordinary Yoneda lemma of [5] to the groupoid-valued setting. The ordinary Yoneda lemma identifies natural transformations $h_X \Rightarrow F$ with elements of $F(X)$; the 2-categorical version identifies the *groupoid* of CFG-morphisms $\underline{X} \to \mathcal{F}$ with the fiber groupoid $\mathcal{F}(X)$, morphisms and 2-morphisms included.

Theorem 7.1.8: The 2-Yoneda Lemma

Let $\mathcal{F}$ be a CFG over $\mathcal{C}$ and X an object of $\mathcal{C}$. Evaluation at the object $\mathrm{id}_X \in \underline{X}(X)$ defines a functor

$$\mathrm{ev}\colon \underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{X}, \mathcal{F}) \longrightarrow \mathcal{F}(X), \qquad F \longmapsto F(\mathrm{id}_X)$$

from the groupoid of morphisms $\underline{X} \to \mathcal{F}$ (and their 2-morphisms) to the fiber groupoid $\mathcal{F}(X)$, and this functor is an equivalence of groupoids. Consequently the assignment $X \mapsto \underline{X}$ embeds $\mathcal{C}$ 2-fully-faithfully into the 2-category of CFGs: for $X, Y \in \mathcal{C}$ the evaluation $\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{X}, \underline{Y}) \simeq \underline{Y}(X) = \mathrm{Hom}_S(X, Y)$ is an equivalence onto a discrete groupoid. A CFG is representable if and only if it is equivalent to $\underline{X}$ for some X.

***Proof*:**
A choice of cleavage on $\mathcal{F}$ is fixed throughout; for $g\colon T \to X$ and $\xi \in \mathcal{F}(X)$ write $g^*\xi \in \mathcal{F}(T)$ for the chosen pullback. The proof constructs a quasi-inverse to ev from this cleavage and checks that the two composites are 2-isomorphic to the identities.

Step 1: The quasi-inverse. Given $\xi \in \mathcal{F}(X)$, define a morphism $F_\xi\colon \underline{X} \to \mathcal{F}$ as follows. An object of $\underline{X}$ is a morphism $g\colon T \to X$; set $F_\xi(g) = g^*\xi \in \mathcal{F}(T)$. An arrow in $\underline{X}$ from $g'\colon T' \to X$ to $g\colon T \to X$ is a morphism $h\colon T' \to T$ with $g \circ h = g'$; the cleavage supplies canonical cartesian arrows $g^*\xi \to \xi$ and $(g')^*\xi \to \xi$ over g and $g' = g \circ h$, and the composite isomorphism ϵ of definition 7.1.2 identifies $(g')^*\xi = (g \circ h)^*\xi \cong h^*g^*\xi$. Define $F_\xi(h)\colon (g')^*\xi \to g^*\xi$ to be the cartesian arrow over h provided by the cleavage of $\mathcal{F}$ (the unique arrow over h lying above the identity of ξ). Functoriality of F_ξ up to the coherence isomorphisms ϵ makes F_ξ a morphism of CFGs; this is where the pseudofunctor coherence of definition 7.1.2 is used, and it is verified in [30]. By construction $p_{\mathcal{F}}(F_\xi(g)) = T = p_{\underline{X}}(g)$, so F_ξ is a morphism over $\mathcal{C}$.

Step 2: $\mathrm{ev} \circ (\xi \mapsto F_\xi)$ *is the identity.* Evaluating F_ξ at $\mathrm{id}_X \in \underline{X}(X)$ gives $F_\xi(\mathrm{id}_X) = \mathrm{id}_X^*\xi$, and the canonical isomorphism $\mathrm{id}_X^*\xi \cong \xi$ (the unit part of definition 7.1.2) is natural in ξ. Hence $\mathrm{ev}(F_\xi) \cong \xi$ naturally, so the composite $\mathrm{ev} \circ (\xi \mapsto F_\xi)$ is 2-isomorphic to $\mathrm{id}_{\mathcal{F}(X)}$.

Step 3: $(\xi \mapsto F_\xi) \circ \mathrm{ev}$ *is the identity.* Let $F\colon \underline{X} \to \mathcal{F}$ be a morphism of CFGs and put $\xi = F(\mathrm{id}_X) = \mathrm{ev}(F)$. A natural isomorphism $F \cong F_\xi$ must be produced. For $g\colon T \to X$, note that g is, as an arrow of $\underline{X}$, a morphism from $g\colon T \to X$ to $\mathrm{id}_X\colon X \to X$ (since $\mathrm{id}_X \circ g = g$). Write $\eta_g^{\mathrm{src}} := F(g\colon T \to X) \in \mathcal{F}(T)$ for the image of the *object* g, to keep it distinct from the image of the *arrow* g. Applying F to the arrow g gives an arrow $F(g)\colon \eta_g^{\mathrm{src}} \to F(\mathrm{id}_X) = \xi$ over g, and since every arrow of a CFG is cartesian, this $F(g)$ is a cartesian arrow over g. But $F_\xi(g) = g^*\xi$ is also equipped with a cartesian arrow to ξ over g, namely the cleavage arrow. Two cartesian arrows over the same g with the same target ξ differ by a unique vertical isomorphism (as shown following definition 7.1.1), yielding a unique isomorphism $\eta_g\colon \eta_g^{\mathrm{src}} \xrightarrow{\sim} F_\xi(g)$ in $\mathcal{F}(T)$ over id_T. The uniqueness makes $\eta = (\eta_g)_g$ natural in g: for $h\colon T' \to T$ with $g' = g \circ h$, both $\eta_{g'}$ and the mediating arrows are forced by the same universal property, so the naturality square commutes. Each η_g is vertical, so $\eta\colon F \Rightarrow F_\xi$ is a 2-morphism, and being a fiberwise isomorphism it is a 2-isomorphism. Hence $(\xi \mapsto F_\xi) \circ \mathrm{ev} \cong \mathrm{id}$, and ev is an equivalence.

Step 4: The embedding and representability. Taking $\mathcal{F} = \underline{Y}$ in the equivalence gives $\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{X}, \underline{Y}) \simeq \underline{Y}(X) = \mathrm{Hom}_S(X, Y)$, a discrete groupoid, so the 2-functor $X \mapsto \underline{X}$ induces equivalences on Hom-groupoids: it is a 2-fully-faithful embedding of $\mathcal{C}$ into CFGs. Finally, a CFG $\mathcal{F}$ is representable if and only if $\mathcal{F} \simeq \underline{X}$ for some X, which is the definition (definition 7.1.7); the equivalence just proved shows that when this holds the equivalence is recorded by an object $\xi \in \mathcal{F}(X)$ that is universal, the image of id_X, as we sought to show.

The lemma has two payoffs used constantly in the sequel. First, it lets one compute the groupoid of maps into a CFG by evaluation: a map $\underline{T} \to \mathcal{F}$ from a scheme is the same as an object $\xi \in \mathcal{F}(T)$, and a 2-morphism between two such maps is the same as an isomorphism between the corresponding objects. This is why the "T-points of $\mathcal{F}$" can be defined as the objects of $\mathcal{F}(T)$: the 2-Yoneda lemma guarantees that the categorical Hom recovers exactly this groupoid. Second, it makes "representable" a checkable condition: $\mathcal{F}$ is a scheme precisely when its fibers are discrete and glue to a representable functor of points, and the failure of representability for a moduli problem is detected in the fiber $\mathcal{F}(T)$ carrying nontrivial automorphisms. The embedding $\mathcal{C} \hookrightarrow \{\mathrm{CFGs}\}$ means that no generality is lost by working with CFGs from the outset: schemes sit inside as the CFGs with discrete fibers.

With the framework in place, the recurring objects of the chapter can be introduced. Each is a CFG; the difference among them is entirely in how much automorphism their fibers carry.

Example 7.1.9: Quasi-Coherent Sheaves, BG, and $\mathcal{M}_{1,1}$ as CFGs

Each of the following is a CFG over $\mathcal{C} = (\mathbf{Sch}/S)_\tau$; the automorphisms in the fiber are identified in each case.

1. *Quasi-coherent sheaves.* Let $\underline{\mathrm{QCoh}}$ have as objects the pairs $(T, \mathcal{E})$ with $T \in \mathcal{C}$ and $\mathcal{E}$ a quasi-coherent sheaf on T, and as arrows $(T', \mathcal{E}') \to (T, \mathcal{E})$ the pairs (f, φ) with $f\colon T' \to T$ and $\varphi\colon \mathcal{E}' \xrightarrow{\sim} f^*\mathcal{E}$ an isomorphism of $\mathcal{O}_{T'}$-modules. The projection sends $(T, \mathcal{E})$ to T. Pullback along $f\colon T' \to T$ is f^* of sheaves, and this is cartesian; every arrow is an isomorphism onto a pullback, so $\underline{\mathrm{QCoh}}$ is a CFG. The fiber over T is the groupoid $\mathrm{QCoh}(T)^\times$ of quasi-coherent sheaves on T with isomorphisms as arrows; the automorphisms of $(T, \mathcal{E})$ are $\mathrm{Aut}_{\mathcal{O}_T}(\mathcal{E})$, the sheaf automorphisms of $\mathcal{E}$. Restricting to locally free sheaves of rank n gives the CFG $\underline{\mathrm{Vect}}_n$, whose fiber over T is rank-n vector bundles and whose automorphism group at $(T, \mathcal{E})$ is $\mathrm{GL}(\mathcal{E}) = \mathrm{GL}_n(\Gamma(T, \mathcal{O}_T))$ when $\mathcal{E}$ is trivial.

2. *The classifying CFG BG.* Let G be a flat, finitely presented S-group scheme. Let BG have as objects the pairs (T, P) with $T \in \mathcal{C}$ and $P \to T$ a G-torsor for the topology τ (definition 6.5.1), and as arrows $(T', P') \to (T, P)$ the pairs (f, φ) with $f\colon T' \to T$ and $\varphi\colon P' \xrightarrow{\sim} P \times_T T' = f^*P$ an isomorphism of G-torsors over T'. Pullback of a torsor is a torsor, and every arrow is an isomorphism of torsors over a base change, so BG is a CFG. The fiber over T is the groupoid of G-torsors on T; its set of isomorphism classes is $H^1_\tau(T, G)$

(theorem 6.5.3). The automorphisms of (T, P) are the G-equivariant automorphisms of the torsor P, which for the trivial torsor $P = G \times_S T$ is the group $G(T)$ acting by translation. Thus BG has fibers with automorphism group G: it is the CFG that "remembers" the group.

3. *The moduli CFG of elliptic curves.* Let $\mathcal{M}_{1,1}$ have as objects the families $(T, E \to T)$ of elliptic curves, that is, smooth proper morphisms $E \to T$ with a section whose geometric fibers are genus-1 curves, and as arrows $(T', E') \to (T, E)$ the pairs (f, φ) with $f: T' \to T$ and $\varphi: E' \xrightarrow{\sim} E \times_T T'$ an isomorphism of elliptic curves over T' (respecting the sections). Pullback of a family along f is the fiber product $E \times_T T'$, which is again a family, and it is cartesian; every arrow is a fiberwise isomorphism, so $\mathcal{M}_{1,1}$ is a CFG. The fiber over T is the groupoid of elliptic curves over T; over an algebraically closed field of characteristic 0 (or of characteristic $\neq 2, 3$), the automorphisms of a single elliptic curve are $\{\pm 1\}$ generically, growing to a group of order 4 or 6 at $j = 1728$ or $j = 0$ (example 1.3.4). The presence of these automorphisms in every fiber is exactly why $\mathcal{M}_{1,1}$ is not representable by a scheme: by theorem 7.1.8 a representable CFG has discrete fibers.

4. *A scheme.* For an S-scheme X, the representable CFG $\underline{X}$ of definition 7.1.7 has fiber over T the discrete groupoid $\mathrm{Hom}_S(T, X)$: every automorphism is the identity. A scheme is the extreme case of a moduli problem, one with no automorphisms.

The four examples arrange themselves by the size of their automorphisms. A scheme $\underline{X}$ has trivial automorphisms and is representable; $\mathcal{M}_{1,1}$ carries the automorphisms $\{\pm 1\}$ and more at special j, and is not; BG has automorphism group G uniformly; and $\underline{\mathrm{QCoh}}$ has the full sheaf automorphism group in each fiber. In every case the fiber is a genuine groupoid, and the arrows of that groupoid, the isomorphisms discarded by a set-valued functor, are the data that a CFG keeps. That $\underline{\mathrm{QCoh}}$, BG, and $\mathcal{M}_{1,1}$ are not only CFGs but *stacks*, meaning their objects and morphisms glue in the topology τ, is the subject of section 7.2; that BG and $\mathcal{M}_{1,1}$ are quotient stacks built from group actions is the subject of section 7.3. For now the reader should hold onto the picture that a moduli problem with automorphisms is a CFG, and that the automorphisms live in the fibers where a functor of points would have flattened them to a set.

Exercise 7.1.1

1. Let $p: \mathcal{F} \to \mathcal{C}$ be a fibered category and $\varphi: x \to y$, $\varphi': x' \to y$ two cartesian arrows over the same $f: U \to V$. Prove directly from definition 7.1.1 that there is a unique vertical isomorphism $u: x \to x'$ over U with $\varphi' \circ u = \varphi$. (This is the "pullback unique up to unique isomorphism" statement.)

2. Show that an identity arrow $\mathrm{id}_y: y \to y$ is always cartesian, and deduce that in the representable CFG $\underline{X}$ of definition 7.1.7, every arrow is cartesian, so $\underline{X}$ is a CFG. Identify the fiber $\underline{X}(T)$ as a groupoid and confirm it is discrete.

3. Let G be an S-group scheme and BG the classifying CFG of example 7.1.9(2). Using the 2-Yoneda lemma (theorem 7.1.8) as a template, compute the groupoid $\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{T}, BG)$ of morphisms from a scheme T to BG, and show it is equivalent to the groupoid of G-torsors on T. Conclude that the isomorphism classes of maps $\underline{T} \to BG$ are classified by $H^1_\tau(T, G)$ (theorem 6.5.3).

4. A morphism of CFGs $F: \mathcal{F} \to \mathcal{G}$ over $\mathcal{C}$ is defined to be a functor with $p_{\mathcal{G}} \circ F = p_{\mathcal{F}}$. Prove that such an F automatically preserves cartesian arrows, so the requirement "preserves cartesian

arrows" is redundant for CFGs (though not for general fibered categories). Where does the argument use that all arrows of $\mathcal{F}$ are cartesian?

5. Give the pseudofunctor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Grpd}$ associated with $\underline{\mathrm{Vect}}_1$ (line bundles), specifying the value at each T, the pullback functors, and the automorphism group of an object. Explain why the isomorphism classes of objects in the fiber over T form the Picard group $\mathrm{Pic}(T)$, and identify the automorphism group of a line bundle $\mathcal{L}$ on T with $\Gamma(T, \mathcal{O}_T)^\times$.
6. Explain, using theorem 7.1.8, why a CFG whose fiber $\mathcal{F}(T)$ contains an object with a nontrivial automorphism for some T cannot be representable by a scheme. Apply this to $\mathcal{M}_{1,1}$ (example 7.1.9(3)) and to $B\mathbb{G}_m$ to conclude that neither is a scheme.

7.2 Prestacks and Stacks

The previous section built the container. A category fibered in groupoids over $\mathcal{C} = (\mathbf{Sch}/S)_\tau$ records objects with automorphisms varying over the base, exactly the shape of a moduli problem in which isomorphic objects must not be silently identified. What that section did *not* do is ask whether the objects and morphisms of a CFG can be reconstructed from local data. This is the same question the previous chapter answered for sheaves of sets and for quasi-coherent sheaves, now posed one categorical level up: given a covering family $\{U_i \to U\}$ and objects $x_i \in \mathcal{F}(U_i)$ together with isomorphisms of their restrictions on the overlaps, is there an object over U from which they all descend? A CFG for which the answer is always yes is a *stack*, and the passage from a fibered category to the descent condition is the step that turns a bookkeeping device into a geometric object.

The condition splits cleanly into two halves, and it is worth separating them from the start. Morphisms between objects are themselves a local datum: two objects over U are identified through their pullbacks to a cover, and an isomorphism between them should be constructible from compatible isomorphisms over the pieces. Objects are the harder datum: a family of objects on a cover, glued by isomorphisms on the overlaps satisfying a cocycle condition, should assemble into a single object over the base. A CFG in which morphisms glue is a *prestack*; a prestack in which objects also glue is a *stack*. The bulk of the work in verifying that a concrete moduli problem is a stack is already discharged by faithfully flat descent, and this section makes that reduction precise: $\underline{\mathrm{QCoh}}$, BG, and the CFG of G-torsors are stacks because theorem 6.3.4 and theorem 6.5.3 say exactly that their objects and morphisms descend. The section closes with stackification, the universal repair that forces a CFG to become a stack, and with the running example of a prestack that is not one.

Throughout, $\mathcal{F} \to \mathcal{C}$ is a CFG in the sense of definition 7.1.4, and τ is the fixed topology on $\mathcal{C}$ (fppf unless stated otherwise). For an object $U \in \mathcal{C}$ and two objects $x, y \in \mathcal{F}(U)$, the assignment sending an arrow $f\colon V \to U$ to the set $\mathrm{Isom}_{\mathcal{F}(V)}(f^*x, f^*y)$ of isomorphisms between the pullbacks in the fiber groupoid over V is a presheaf on the slice category $\mathcal{C}/U$: a morphism $g\colon W \to V$ over U acts by pulling back isomorphisms along g, using the pseudofunctor structure to identify $g^*(f^*x) \cong (fg)^*x$. This presheaf is the central object of the whole section, and it deserves a name.

Definition 7.2.1: The Isom Presheaf

Let $\mathcal{F} \to \mathcal{C}$ be a category fibered in groupoids, $U \in \mathcal{C}$, and $x, y \in \mathcal{F}(U)$. The *Isom presheaf* $\underline{\mathrm{Isom}}_U(x, y)$ is the presheaf of sets on the slice category $\mathcal{C}/U$ defined on objects by

$$\underline{\mathrm{Isom}}_U(x, y)(f : V \to U) = \mathrm{Isom}_{\mathcal{F}(V)}(f^*x, f^*y)$$

the set of isomorphisms in the fiber groupoid $\mathcal{F}(V)$ between the pullbacks f^*x and f^*y, and on a morphism $g\colon (W \to U) \to (V \to U)$ over U by the pullback map $\varphi \mapsto g^*\varphi$, formed through the canonical isomorphisms $g^*f^*x \cong (fg)^*x$ of the cleavage. The presheaf $\underline{\mathrm{Aut}}_U(x) = \underline{\mathrm{Isom}}_U(x, x)$ is the *Aut presheaf*; it takes values in groups.

The Isom presheaf is the fibered-category incarnation of the set of maps between two families. When $\mathcal{F} = \underline{\mathrm{QCoh}}$ and x, y are quasi-coherent sheaves on U, $\underline{\mathrm{Isom}}_U(x, y)(V \to U)$ is the set of $\mathcal{O}_V$-linear isomorphisms $x|_V \to y|_V$, so this presheaf is the sheaf-Hom of the previous chapter read as a presheaf on the site $\mathcal{C}/U$. When $\mathcal{F} = BG$ and x, y are G-torsors on U, $\underline{\mathrm{Isom}}_U(x, y)$ is the presheaf of torsor isomorphisms, which is itself a torsor under $\underline{\mathrm{Aut}}(x)$, the inner form ${}^xG$ of G (equal to G when G is abelian, as for $\mathbb{G}_m$ and μ_n). Whether these presheaves are sheaves is precisely the prestack condition, and the reader has already met the reason they are: a G-equivariant isomorphism of torsors, or an isomorphism of quasi-coherent sheaves, is a local datum that glues.

The two-tier definition that organizes the rest of the chapter can now be stated. The first tier asks that maps glue; the second asks that objects glue along a descent datum, and the descent datum is the CFG version of the gluing package of definition 6.3.1: instead of an isomorphism between the two pullbacks of a single sheaf, one has objects x_i over the pieces U_i of a cover and isomorphisms φ_{ij} over the double overlaps, subject to a cocycle condition on the triple overlaps.

Definition 7.2.2: Prestack and Stack

Let $(\mathcal{C}, \tau)$ be a site and $\mathcal{F} \to \mathcal{C}$ a category fibered in groupoids.
A *descent datum* for $\mathcal{F}$ on a covering family $\{f_i \colon U_i \to U\}$ of τ is a pair $(\{x_i\}, \{\varphi_{ij}\})$ consisting of an object $x_i \in \mathcal{F}(U_i)$ for each i and, writing $U_{ij} = U_i \times_U U_j$ with projections $\mathrm{pr}_i \colon U_{ij} \to U_i$ and $\mathrm{pr}_j \colon U_{ij} \to U_j$, an isomorphism

$$\varphi_{ij} \colon \mathrm{pr}_j^* x_j \xrightarrow{\sim} \mathrm{pr}_i^* x_i \quad \text{in } \mathcal{F}(U_{ij})$$

for each pair (i, j), subject to the *cocycle condition*: on the triple overlap $U_{ijk} = U_i \times_U U_j \times_U U_k$, with $\mathrm{pr}_{ab} \colon U_{ijk} \to U_{ab}$ the projections,

$$\mathrm{pr}_{ik}^* \varphi_{ik} = \mathrm{pr}_{ij}^* \varphi_{ij} \circ \mathrm{pr}_{jk}^* \varphi_{jk}$$

as isomorphisms $\mathrm{pr}_k^* x_k \to \mathrm{pr}_i^* x_i$ in $\mathcal{F}(U_{ijk})$, formed through the canonical cleavage identifications. A *morphism of descent data* $(\{x_i\}, \{\varphi_{ij}\}) \to (\{x_i'\}, \{\varphi_{ij}'\})$ is a family of isomorphisms $\psi_i \colon x_i \to x_i'$ in $\mathcal{F}(U_i)$ with $\varphi_{ij}' \circ \mathrm{pr}_j^* \psi_j = \mathrm{pr}_i^* \psi_i \circ \varphi_{ij}$ over each U_{ij}.
Every object $x \in \mathcal{F}(U)$ induces a *canonical* descent datum on $\{f_i \colon U_i \to U\}$: take $x_i = f_i^* x$ and let φ_{ij} be the canonical isomorphism $\mathrm{pr}_j^* f_j^* x \cong \mathrm{pr}_i^* f_i^* x$ arising from $f_i \circ \mathrm{pr}_i = f_j \circ \mathrm{pr}_j \colon U_{ij} \to U$.
A descent datum is *effective* if it is isomorphic to a canonical one.
The CFG $\mathcal{F}$ is a *prestack* (over $(\mathcal{C}, \tau)$) if for all $U \in \mathcal{C}$ and all $x, y \in \mathcal{F}(U)$ the Isom presheaf $\underline{\mathrm{Isom}}_U(x, y)$ of definition 7.2.1 is a τ-sheaf on $\mathcal{C}/U$. The CFG $\mathcal{F}$ is a *stack* if it is a prestack and, in addition, every descent datum on every covering family of τ is effective.

The definition packs two ideas that are worth prying apart. The prestack condition is descent for *morphisms*. Unwinding the sheaf axiom for $\underline{\mathrm{Isom}}_U(x, y)$ on a cover $\{U_i \to U\}$: an isomorphism $x \to y$ over U is the same as a family of isomorphisms $f_i^* x \to f_i^* y$ over the pieces that agree on the overlaps U_{ij}. The separated half of the sheaf axiom says a morphism over U is determined by its restrictions (no two distinct maps agree everywhere locally); the gluing half says compatible local maps come from a global one. A prestack is therefore a CFG in which Hom-sets are recovered from local data. The stack condition adds descent for *objects*: the objects x_i on the pieces, glued by the φ_{ij} satisfying the cocycle condition, assemble into a single x over U inducing them. The cocycle condition $\varphi_{ik} = \varphi_{ij} \circ \varphi_{jk}$ is the transitivity of the gluing, identical in form to the one in definition 6.3.1, and it is what guarantees the local objects can be consistently overlapped.

The two conditions are separate, and the split is not cosmetic. Once $\mathcal{F}$ is a prestack, the effective descent data are unique up to unique isomorphism when they exist, because the gluing isomorphisms between two solutions are themselves controlled by the (now sheafy) Isom presheaves; the stack condition is then the pure *existence* statement that every descent datum has a solution. This is why one first checks that Isom is a sheaf and only then checks effectivity, and why the failure of a moduli problem to be a stack is always a failure of gluing objects, never a failure of gluing maps once the prestack condition holds.

Before the theorem, a first concrete instance grounds the descent datum in an object the reader already controls: a single quasi-coherent sheaf. This is the CFG $\underline{\mathrm{QCoh}}$ from example 7.1.9, whose fiber over T is the groupoid of quasi-coherent sheaves and their isomorphisms, and its stack condition is faithfully flat descent verbatim.

Example 7.2.3: The Descent Datum of a Quasi-Coherent Sheaf

Let $\mathcal{F} = \underline{\mathrm{QCoh}}$ and let $\{U_i \to U\}$ be a single-morphism cover $U' \to U$ (recover a family by $U' = \coprod_i U_i$). A descent datum for $\underline{\mathrm{QCoh}}$ on $U' \to U$ is a quasi-coherent sheaf $\mathcal{E}'$ on U' together with an isomorphism

$$\varphi\colon \mathrm{pr}_2^* \mathcal{E}' \xrightarrow{\sim} \mathrm{pr}_1^* \mathcal{E}' \quad \text{on } U' \times_U U'$$

satisfying the cocycle condition on $U' \times_U U' \times_U U'$. This is exactly the descent datum of definition 6.3.1, transcribed into the fibered-category language: the objects x_i are the single sheaf $\mathcal{E}'$ and the gluing isomorphism φ_{12} is the map φ. The prestack condition is that $\underline{\mathrm{Isom}}(\mathcal{E}, \mathcal{G})$ is an fppf sheaf for quasi-coherent $\mathcal{E}, \mathcal{G}$ on U, which holds because morphisms of quasi-coherent sheaves satisfy fppf descent: a section of $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$ over an object is recovered from its restrictions to any fppf cover. The effectivity of every descent datum is the essential surjectivity of the pullback functor in theorem 6.3.4. Thus every ingredient of "$\underline{\mathrm{QCoh}}$ is a stack" is already proved; the theorem below only records the packaging.

The example carries the content of the theorem in the affine, single-cover case; the theorem states the conclusion in general and adds the two group theoretic companions BG and the CFG of torsors, whose stack property is the torsor classification of theorem 6.5.3 read in the same way. The motivating point is that no new descent is needed: the entire content of this chapter's first stack examples was established in the previous chapter, and recognizing that is the payoff of separating the fibered category from its descent condition.

Theorem 7.2.4: The Stack of Quasi-Coherent Sheaves and Classifying Stacks

Over the site $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$:

1. The CFG $\underline{\mathrm{QCoh}}$, whose fiber over T is the groupoid of quasi-coherent $\mathcal{O}_T$-modules, is a stack. The same holds for the CFGs $\underline{\mathrm{Vect}}_n$ of rank-n vector bundles and $\underline{\mathrm{Coh}}$ of coherent sheaves.
2. For a flat, finitely presented, affine group scheme G over S, the CFG of G-torsors, whose fiber over T is the groupoid of G-torsors on T, is a stack. In particular the classifying CFG BG is a stack.

***Proof*:**

Each part is a prestack condition (Isom is a sheaf) followed by an effectivity condition (descent data glue), and both are the descent theorems of the previous chapter transcribed. The pseudofunctor coherence, the verification that the chosen cleavage makes the cocycle conditions strictly compatible with the associativity constraints of the 2-category of groupoids, is a formal bookkeeping check carried out in full in [30, Ch. 4]; it is cited rather than reproduced, as it is the same coherence for every fibered category and adds nothing to the geometry.

Step 1: $\underline{\mathrm{QCoh}}$ is a prestack. Fix U and quasi-coherent sheaves $\mathcal{E}, \mathcal{G}$ on U. The Isom presheaf $\underline{\mathrm{Isom}}_U(\mathcal{E}, \mathcal{G})$ is the presheaf of $\mathcal{O}$-linear isomorphisms, the subpresheaf of isomorphisms in $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$. The presheaf $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$ sends $V \to U$ to $\mathrm{Hom}_{\mathcal{O}_V}(\mathcal{E}|_V, \mathcal{G}|_V)$; that this is an fppf sheaf is faithfully flat descent for morphisms of quasi-coherent sheaves, which is the full-faithfulness half of theorem 6.3.4 applied to the sheaf $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$: a morphism $\mathcal{E}|_V \to \mathcal{G}|_V$ is determined by, and can be glued from, its restrictions to an fppf cover of V, because theorem 6.3.4 identifies $\mathrm{Hom}_{\mathcal{O}_V}(\mathcal{E}|_V, \mathcal{G}|_V)$ with the equalizer of the two pullbacks of Hom over the cover, which is exactly the sheaf condition for $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$. The subpresheaf of isomorphisms is a sheaf because being an isomorphism is fppf-local on the base (a morphism of quasi-coherent sheaves is an isomorphism iff it is so after faithfully flat base change, again by theorem 6.3.4). Hence $\underline{\mathrm{Isom}}_U(\mathcal{E}, \mathcal{G})$ is an fppf sheaf and $\underline{\mathrm{QCoh}}$ is a prestack.

Step 2: Descent data for $\underline{\mathrm{QCoh}}$ *are effective.* Effectivity is local on U, so it suffices to treat U affine, hence quasi-compact. A descent datum on a cover $\{U_i \to U\}$ is, by example 7.2.3, a quasi-coherent sheaf on $U' = \coprod_i U_i$ with a gluing isomorphism on $U' \times_U U'$ satisfying the cocycle condition. Because U is quasi-compact, the surjection $\coprod_i U_i \to U$ admits a finite subcover $U_{i_1}, \dots, U_{i_r}$; replacing U' by the finite disjoint union $U'' = U_{i_1} \sqcup \cdots \sqcup U_{i_r}$ (and restricting the descent datum, which loses nothing since a descent datum for a cover restricts to one for any subcover realizing the same object) yields a morphism $U'' \to U$ that is faithfully flat and quasi-compact: a finite disjoint union of fppf morphisms is fppf, and finiteness of the union supplies the quasi-compactness that theorem 6.3.4 requires. That theorem then applies: the pullback functor $\mathrm{QCoh}(U) \to \mathrm{Desc}(U'' \to U)$ is an equivalence, hence essentially surjective, so every such descent datum is isomorphic to a canonical one $\mathrm{pr}^* \mathcal{E}$. This is exactly effectivity. Together with Step 1, $\underline{\mathrm{QCoh}}$ is a stack. The variants $\underline{\mathrm{Vect}}_n$ and $\underline{\mathrm{Coh}}$ are the full subcategories cut out by the fppf-local conditions “locally free of rank n” and “coherent”; both conditions descend along a faithfully flat quasi-compact cover ([6]), so the descended sheaf lies in the subcategory, and the same two steps apply.

Step 3: The CFG of G-torsors is a prestack. Fix U and two G-torsors P, Q on U. An arrow $V \to U$ pulls them back to torsors on V, and $\underline{\mathrm{Isom}}_U(P, Q)(V) = \mathrm{Isom}_G(P|_V, Q|_V)$ is the set of G-equivariant isomorphisms. This is a sheaf: a G-equivariant morphism of torsors is a morphism of the underlying fppf sheaves commuting with the G-action, and both the sheaf $\underline{\mathrm{Hom}}$ and the equivariance condition are fppf-local, so an equivariant isomorphism over V is determined by and glued from its restrictions to an fppf cover. Concretely $\underline{\mathrm{Isom}}_U(P, Q)$ is a torsor under the automorphism group scheme $\underline{\mathrm{Aut}}(P)$ when $P \cong Q$ locally; this is the inner form ${}^P G$, which equals G on the nose when G is abelian (as for the running anchors $\mathbb{G}_m$ and μ_n). Hence the CFG of torsors is a prestack.

Step 4: Descent data for G-torsors are effective. A descent datum on $\{U_i \to U\}$ is a torsor P_i on each U_i with gluing isomorphisms φ_{ij} over the overlaps satisfying the cocycle condition. Passing to $U' = \coprod_i U_i$, this is a G-torsor P' on U' with an isomorphism $\varphi\colon \mathrm{pr}_2^* P' \to \mathrm{pr}_1^* P'$ of G-torsors over $U' \times_U U'$ satisfying the cocycle condition. A G-torsor is, in particular, a quasi-affine (indeed affine, as G is affine) faithfully flat U'-scheme with G-action; its descent along the fppf cover $U' \to U$ is effective by faithfully flat descent for affine morphisms, the effectivity of affine descent that upgrades theorem 6.3.4 from quasi-coherent modules to their quasi-coherent sheaves of algebras ([30, Ch. 4]), and the descended scheme $P \to U$ inherits a G-action making it a torsor because the torsor axioms (the action map $G \times_S P \to P \times_U P$, $(g, p) \mapsto (gp, p)$, is an isomorphism and $P \to U$ is an fppf cover) are fppf-local on U and hold after pullback to U'. Thus every descent datum of torsors is effective, and the CFG of torsors, in particular BG, is a stack. The classification $H^1_{\mathrm{fppf}}(U, G) = \{G\text{-torsors on } U\}/\cong$ of theorem 6.5.3 is the set of isomorphism classes in the fiber $BG(U)$, so the stack BG is the groupoid-valued refinement of the cohomology functor $U \mapsto H^1_{\mathrm{fppf}}(U, G)$, completing the proof.

The theorem is the first payoff of the entire descent apparatus: the two principal geometric stacks of the book, $\underline{\mathrm{QCoh}}$ and BG, are stacks for a reason the reader has already proved. The Isom presheaf being a sheaf is descent for maps of sheaves or torsors, and the effectivity of descent data is descent for the objects themselves; there is nothing to check beyond re-reading theorem 6.3.4 and theorem 6.5.3

through the fibered-category dictionary. This is the sense in which a stack is descent theory for objects with automorphisms: the automorphisms are the reason a set-valued functor will not do, and descent is the reason the groupoid-valued replacement still glues. The classifying stack BG deserves particular emphasis. Its T-points are G-torsors on T, and its isomorphism classes over T are $H^1_{\text{fppf}}(T, G)$; BG is the object that remembers not only the cohomology set but the automorphisms $\underline{\text{Aut}}(P) = G$ that made the moduli problem non-representable in the first place. The quotient $[X/G]$ of the next section will be the relative version, with $BG = [S/G]$ the case $X = S$.

The reader should hold in mind that both parts used the fppf topology because both descent theorems of the previous chapter were proved there, but the stack property of these two CFGs is not special to fppf. Both $\underline{\text{QCoh}}$ and the CFG of G-torsors are stacks in every subcanonical topology, including Zariski: quasi-coherent sheaves glue along a Zariski cover by the classical gluing of sheaves, and G-torsors glue along a Zariski cover as schemes-with-action, so their descent data are effective there too. What changes with the topology is not effectivity but the *local structure* of the objects: a nontrivial G-torsor need not be Zariski-locally trivial (it becomes trivial only after passing to a finer fppf cover), so fppf descent sees a local triviality that Zariski descent cannot. The choice of topology is not incidental. A CFG is a stack *for a given topology*, and the same CFG can be a stack for one τ and fail for another; the example of such topology dependence is not $\underline{\text{QCoh}}$ or BG but a CFG of *framed* objects, where the framing itself does not descend, as in the naive quotient constructed next. There the descent data whose gluing cocycle defines a nontrivial torsor are exactly the ineffective ones, so the CFG is a stack for a coarser topology in which those torsors have not yet appeared and fails for a finer one in which they have.

Not every CFG is a stack, and the ones that fail are the ones that need repair. Before constructing the repair, it helps to see the failure mechanism cleanly. A prestack can fail to be a stack because descent data fail to glue, and the simplest source of such failure is a moduli problem whose objects have been rigidified by extra structure that does not itself descend. The following is the running counterexample of the chapter, and it is the naive quotient that stackification will correct into $[X/G]$.

Example 7.2.5: The Naive Presheaf Quotient $\underline{X}/G$

Let G be a flat affine group scheme over S acting on an S-scheme X. Define a CFG $\mathcal{F} = [X/G]^{\text{pre}}$, the *naive presheaf quotient*, whose fiber over T is the groupoid whose objects are morphisms $T \to X$ and whose morphisms $u \to u'$ are elements $g \in G(T)$ with $g \cdot u = u'$. Equivalently $\mathcal{F}$ is the CFG associated to the action groupoid $[X(T)/G(T)]$ pointwise, before sheafification. This is a CFG: pullback along $S' \to S$ acts on maps to X and on group elements, and every arrow is invertible because $G(T)$ is a group.

$\mathcal{F}$ is a prestack. For $u, u' \in \mathcal{F}(T)$ the Isom presheaf sends $V \to T$ to $\{g \in G(V) : g \cdot u|_V = u'|_V\}$, a subsheaf of the representable sheaf $\underline{G}$ cut out by the equalizer condition $g \cdot u = u'$; since $\underline{G}$ is an fppf sheaf (it is representable) and the condition is fppf-local, $\underline{\text{Isom}}_T(u, u')$ is an fppf sheaf. So morphisms glue.

$\mathcal{F}$ is not a stack. The objects of $\mathcal{F}(T)$ are maps $T \to X$, which record only the *trivial* torsor $G \times T \to T$ carrying its tautological equivariant map $(g, t) \mapsto g \cdot u(t)$ to X; a nontrivial torsor with an equivariant map to X is invisible to $\mathcal{F}$. A descent datum on a cover $\{U_i \to T\}$ consists of maps $u_i \colon U_i \to X$ and elements $g_{ij} \in G(U_{ij})$ with $g_{ij} \cdot u_j = u_i$ on U_{ij} satisfying the cocycle condition $g_{ik} = g_{ij} g_{jk}$. Such a datum glues to an object of $\mathcal{F}(T)$, a single map $T \to X$, exactly when the cocycle (g_{ij}) is a coboundary, that is, when the associated G-torsor is trivial. But (g_{ij}) is precisely the transition cocycle of a G-torsor on T (theorem 6.5.3), and there exist nontrivial such torsors on many T as soon as $H^1_{\text{fppf}}(T, G) \neq 0$. For such a datum there is no map $T \to X$ inducing it:

the descent datum encodes a nontrivial twisted family, which the naive quotient cannot represent. Hence $\mathcal{F}$ is a prestack that is not a stack, and the obstruction to gluing is measured by $H^1(T, G)$.

The failure is diagnostic. The naive presheaf quotient records only untwisted families, the ones built on the trivial torsor, so a descent datum whose gluing cocycle defines a nontrivial torsor has nothing to glue to. The information the CFG is missing is exactly a G-torsor: the correct quotient must allow objects to be G-torsors $P \to T$ with an equivariant map $P \to X$, not only maps $T \to X$. This is the definition of $[X/G]$ in the next section, and the example makes the definition inevitable rather than ad hoc: one adjoins to $\mathcal{F}$ precisely the twisted objects that its descent data demand. The general procedure for performing this repair, turning any CFG into the closest stack, is stackification.

Stackification is the reflection of the 2-category of CFGs onto its sub-2-category of stacks. The motivating demand is universal: given any CFG $\mathcal{F}$, one wants a stack $\mathcal{F}^{\mathrm{a}}$ and a morphism $\mathcal{F} \to \mathcal{F}^{\mathrm{a}}$ through which every morphism from $\mathcal{F}$ to a stack factors uniquely up to unique 2-isomorphism. The construction proceeds in the two steps the definition of a stack suggests: first force the Isom presheaves to become sheaves, producing a prestack; then adjoin the missing effective descent data, producing a stack. The example above shows both steps at work: $[X/G]^{\mathrm{pre}}$ is already a prestack, so the first step is vacuous, and the second step adjoins the twisted objects, producing $[X/G]$.

Theorem 7.2.6: Stackification

Let $(\mathcal{C}, \tau)$ be a site. The inclusion of the 2-category of stacks on $(\mathcal{C}, \tau)$ into the 2-category of categories fibered in groupoids over $\mathcal{C}$ admits a left 2-adjoint

$$(-)^{\mathrm{a}} \colon \{\mathrm{CFGs}\} \longrightarrow \{\mathrm{stacks}\}$$

the *stackification*. For each CFG $\mathcal{F}$ there is a morphism $\iota\colon \mathcal{F} \to \mathcal{F}^{\mathrm{a}}$ to a stack such that, for every stack $\mathcal{G}$, composition with ι induces an equivalence of groupoids

$$\underline{\mathrm{Hom}}(\mathcal{F}^{\mathrm{a}}, \mathcal{G}) \xrightarrow{\sim} \underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$$

Consequently $\mathcal{F}^{\mathrm{a}}$ is unique up to equivalence, and ι is an equivalence iff $\mathcal{F}$ is already a stack. The morphism ι is fully faithful on Isom sheaves and locally essentially surjective: every object of $\mathcal{F}^{\mathrm{a}}(U)$ is, on a cover of U, in the essential image of $\mathcal{F}$.

***Proof*:**
The construction is in two steps mirroring the two clauses of definition 7.2.2: sheafify morphisms to reach a prestack, then adjoin descent data to reach a stack. The set-theoretic size control (that the sheafifications and the glued objects form a set, not a proper class, so $\mathcal{F}^{\mathrm{a}}$ is a legitimate CFG) and the strict 2-categorical coherence of the resulting pseudofunctor are the bookkeeping carried out in full in [29, Tag 02ZM]; they are cited, and the argument below gives the construction and the universal property.

Step 1: The associated prestack. Define a new CFG $\mathcal{F}^{\mathrm{pre}}$ with the same objects as $\mathcal{F}$ over each U, but with morphism sheaves replaced by their sheafifications. Precisely, for $x, y \in \mathcal{F}(U)$ let $\underline{\mathrm{Isom}}_U(x, y)^{\#}$ be the τ-sheafification (theorem 6.2.3) of the Isom presheaf, and declare an isomorphism $x \to y$ in $\mathcal{F}^{\mathrm{pre}}(U)$ to be a global section of $\underline{\mathrm{Isom}}_U(x, y)^{\#}$. Composition is induced by the composition of Isom presheaves and passes to sheafifications because sheafification is a left-exact functor preserving finite products (theorem 6.2.3). The unit map $\underline{\mathrm{Isom}}_U(x, y) \to \underline{\mathrm{Isom}}_U(x, y)^{\#}$ gives a morphism of CFGs $\mathcal{F} \to \mathcal{F}^{\mathrm{pre}}$, the identity on objects. By construction the Isom presheaves of $\mathcal{F}^{\mathrm{pre}}$ are sheaves, so $\mathcal{F}^{\mathrm{pre}}$ is a prestack; and any morphism from $\mathcal{F}$ to a prestack $\mathcal{G}$ factors uniquely through $\mathcal{F}^{\mathrm{pre}}$, because a morphism to a prestack sends the Isom presheaves into sheaves and so factors through their sheafification by the universal property of theorem 6.2.3. This proves the claim of the theorem for the sub-2-category of prestacks.

Step 2: Adjoining effective descent data. Now let $\mathcal{F}$ be a prestack; construct a stack $\mathcal{F}^{\mathrm{a}}$ by adjoining the missing objects. Define $\mathcal{F}^{\mathrm{a}}(U)$ to be the groupoid whose objects are descent data on τ-covers of U: an object is a pair $(\{U_i \to U\}, (\{x_i\}, \{\varphi_{ij}\}))$ of a cover and a descent datum for $\mathcal{F}$ on it, and a morphism to $(\{V_a \to U\}, (\{y_a\}, \{\psi_{ab}\}))$ is, after passing to a common refinement of the two covers, a morphism of descent data as in definition 7.2.2. Two descent data related by refinement are declared isomorphic, so $\mathcal{F}^{\mathrm{a}}(U)$ is the (filtered) 2-colimit of the descent-data groupoids over all covers of U. Pullback along $V \to U$ pulls a cover of U back to a cover of V and a descent datum to a descent datum, making $\mathcal{F}^{\mathrm{a}} \to \mathcal{C}$ a CFG, and the tautological "one-element cover" embedding gives $\iota\colon \mathcal{F} \to \mathcal{F}^{\mathrm{a}}$, $x \mapsto (\{U \xrightarrow{\mathrm{id}} U\}, (x, \mathrm{id}))$.

Step 3: $\mathcal{F}^{\mathrm{a}}$ is a stack. That $\mathcal{F}^{\mathrm{a}}$ is a prestack follows because $\mathcal{F}$ is: an isomorphism of two descent data on a common refinement is a compatible family of isomorphisms of the underlying objects, whose gluing is governed by the (sheafy) Isom presheaves of $\mathcal{F}$, so $\underline{\mathrm{Isom}}$ for $\mathcal{F}^{\mathrm{a}}$ is a sheaf. For effectivity, a descent datum for $\mathcal{F}^{\mathrm{a}}$ on a cover $\{U_i \to U\}$ assigns to each U_i a descent datum for $\mathcal{F}$ on a cover of U_i, glued over the overlaps. Refining all these covers to a single cover $\{W_\alpha \to U\}$ of U (the composite of the two layers of covers, using that τ-covers compose), the data assemble into a single descent datum for $\mathcal{F}$ on $\{W_\alpha \to U\}$, which is by construction an object of $\mathcal{F}^{\mathrm{a}}(U)$ inducing the given one. Hence every descent datum for $\mathcal{F}^{\mathrm{a}}$ is effective, and $\mathcal{F}^{\mathrm{a}}$ is a stack.

Step 4: The universal property. Let $\mathcal{G}$ be a stack and $F\colon \mathcal{F} \to \mathcal{G}$ a morphism. Define $\widehat{F}\colon \mathcal{F}^{\mathrm{a}} \to \mathcal{G}$ on an object $(\{U_i \to U\}, (\{x_i\}, \{\varphi_{ij}\}))$ of $\mathcal{F}^{\mathrm{a}}(U)$ as follows: the images $F(x_i) \in \mathcal{G}(U_i)$ with the isomorphisms $F(\varphi_{ij})$ form a descent datum for $\mathcal{G}$ on $\{U_i \to U\}$, which is effective because $\mathcal{G}$ is a stack, so it glues to an object $z \in \mathcal{G}(U)$, well-defined up to unique isomorphism by the prestack property of $\mathcal{G}$. Set $\widehat{F}$ of the object to be z. This is functorial and satisfies $\widehat{F} \circ \iota \cong F$ because ι sends x to the trivial descent datum, which glues back to $F(x)$. Uniqueness up to unique 2-isomorphism: any $H\colon \mathcal{F}^{\mathrm{a}} \to \mathcal{G}$ with $H \circ \iota \cong F$ must send a descent datum to the gluing of $\{F(x_i)\}$, since H commutes with pullback and the descent datum is, locally on U, the image under ι of the x_i; the prestack property of $\mathcal{G}$ then makes the comparison 2-isomorphism $\widehat{F} \cong H$ unique. Therefore composition with ι induces the asserted equivalence $\underline{\mathrm{Hom}}(\mathcal{F}^{\mathrm{a}}, \mathcal{G}) \xrightarrow{\sim} \underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$, and $(-)^{\mathrm{a}}$ is left 2-adjoint to the inclusion.

Combining Steps 1 through 4, an arbitrary CFG is stackified by first applying Step 1 to reach a prestack and then Steps 2 and 3 to reach a stack, and the composite has the universal property of Step 4. That ι is an equivalence exactly when $\mathcal{F}$ is already a stack follows from the universal property applied to $\mathcal{G} = \mathcal{F}$, completing the proof.

Stackification is the free construction that makes descent data glue, and its universal property is what justifies calling $\mathcal{F}^{\mathrm{a}}$ "the" stack associated to $\mathcal{F}$: any construction on stacks applied to $\mathcal{F}^{\mathrm{a}}$ is determined by its restriction to $\mathcal{F}$, so passing to the stackification loses no mapping-out information. The two steps are worth holding separately in mind, because they diagnose two different pathologies. Step 1 repairs a CFG whose maps do not glue, replacing the Isom presheaves by their sheafifications; this is invisible when the objects already have sheafy automorphisms, as in $[X/G]^{\mathrm{pre}}$. Step 2 repairs a prestack whose objects do not glue, formally adjoining the missing twisted objects; this is the step that turns $[X/G]^{\mathrm{pre}}$ into $[X/G]$ by supplying the nontrivial-torsor families that example 7.2.5 found missing. The construction here is the concrete two-step one; the size-theoretic guarantee that the adjoined objects form a set and the strict coherence of the resulting pseudofunctor are the parts deferred to [29], because they are formal and identical for every site, contributing nothing to the geometry.

The contrast with $\underline{\mathrm{QCoh}}$ closes the loop. A stack is a CFG that needs no repair: $\iota\colon \mathcal{F} \to \mathcal{F}^{\mathrm{a}}$ is an equivalence, and $\underline{\mathrm{QCoh}}$ is such a CFG by theorem 7.2.4. The naive quotient $[X/G]^{\mathrm{pre}}$ is not, and its stackification is $[X/G]$; the discrepancy between them is measured by the nontrivial G-torsors, which is to say by $H^1(-, G)$, which is to say by exactly the automorphism data that made a set-valued moduli functor inadequate. Stackification is the machine that converts that inadequacy into a geometric object, and the quotient stack $[X/G]$ is the first geometric output of that machine.

Exercise 7.2.1

1. Let $\mathcal{F} \to \mathcal{C}$ be a CFG over a site $(\mathcal{C}, \tau)$. Show that $\mathcal{F}$ is a prestack if and only if, for every cover $\{U_i \to U\}$ and every pair $x, y \in \mathcal{F}(U)$, the map
$$\mathrm{Isom}_{\mathcal{F}(U)}(x, y) \longrightarrow \left\{(\psi_i) : \psi_i \in \mathrm{Isom}_{\mathcal{F}(U_i)}(x|_{U_i}, y|_{U_i}),\ \psi_i|_{U_{ij}} = \psi_j|_{U_{ij}}\right\}$$
is a bijection. Explain how injectivity is the "separated" half and surjectivity is the "gluing" half of the sheaf axiom for $\underline{\mathrm{Isom}}_U(x, y)$.

2. Prove that in any prestack, an effective descent datum, when it exists, is unique up to a unique isomorphism. (Hint: if $x, x' \in \mathcal{F}(U)$ both induce the descent datum $(\{x_i\}, \{\varphi_{ij}\})$, the local isomorphisms $x|_{U_i} \cong x_i \cong x'|_{U_i}$ are compatible on overlaps; glue them using the prestack condition.)

3. Consider the CFG $\underline{\mathrm{Vect}}_1$ of line bundles over $(\mathbf{Sch}/S)_{\mathrm{fppf}}$. Describe a descent datum on a cover $\{U_i \to U\}$ explicitly as a collection of line bundles L_i on U_i with transition isomorphisms φ_{ij}, and identify the cocycle condition with the usual 1-cocycle condition on transition functions. Conclude from theorem 7.2.4 that a line bundle on U is the same as a descent datum, and relate this to the identification $\mathrm{Pic}(U) = H^1_{\mathrm{fppf}}(U, \mathbb{G}_m)$.

4. Let $\mathcal{F} = \underline{\mathrm{Set}}$ be the CFG whose fiber over T is the groupoid of *constant* finite sets (sets with the trivial pullback maps), regarded over the small Zariski site of a fixed scheme X. Show that for X disconnected $\mathcal{F}$ is *not* even a prestack: exhibit a pair of objects whose Isom presheaf, the constant presheaf $\mathrm{Isom}(x, y)$, fails the gluing axiom on the cover by the connected components. Identify the associated prestack $\mathcal{F}^{\mathrm{pre}}$ (Isom presheaves sheafified) and the stackification $\mathcal{F}^{\mathrm{a}}$ (locally constant sheaves of finite sets). This is the "sheafify morphisms" step of stackification in its simplest form.

5. For the naive presheaf quotient $[X/G]^{\mathrm{pre}}$ of example 7.2.5, take $S = \mathrm{Spec}\, k$, $X = \mathrm{Spec}\, k$ a point, and $G = \mathbb{G}_m$. Identify $[X/G]^{\mathrm{pre}}$ concretely (its fiber over T has one object and automorphism group $\mathbb{G}_m(T)$), show it is a prestack but not a stack, and identify a nontrivial descent datum with a nontrivial line bundle. What is the stackification?

6. Show that the stack condition depends on the topology by exhibiting a single CFG that is a stack for one topology and fails to be a stack for a finer one. Take the naive presheaf quotient $\mathcal{F} = [\mathrm{pt}/\mu_n]^{\mathrm{pre}}$ of example 7.2.5 with $S = X = \mathrm{Spec}\, k$, n invertible in k, and μ_n in place of $\mathbb{G}_m$; its objects are *framed* (trivialized) μ_n-torsors and its descent data glue exactly when the gluing cocycle is a coboundary. Deduce that $\mathcal{F}$ is a τ-stack if and only if $H^1_\tau(T, \mu_n) = 0$ for every T. Then explain why this makes $\mathcal{F}$ a stack for a topology in which every μ_n-torsor over

the relevant base is trivial but not for the fppf topology, where nontrivial μ_n-torsors exist. Contrast this with the *full* CFG of μ_n-torsors, which (by theorem 7.2.4) is a stack for every subcanonical topology, and explain in one sentence why the framed CFG, and not the full torsor CFG, is where the topology dependence lives.

7.3 Quotient Stacks and Classifying Stacks

The previous two sections built the abstract apparatus: a moduli problem with automorphisms is a category fibered in groupoids (definition 7.1.4), and when its objects and morphisms glue in a topology it is a stack (definition 7.2.2). What has been missing is a supply of stacks one can hold in the hand. This section produces the first geometric ones. Given a group scheme G acting on a scheme X, there is a stack $[X/G]$, the *quotient stack*, whose points over a test scheme T are not orbits of T-points, but G-torsors on T carrying an equivariant map to X. The special case $X = \mathrm{pt}$ produces the *classifying stack* BG, whose T-points are precisely the G-torsors on T. These two constructions are the workhorses of the theory: every quotient stack is built from a torsor and an equivariant map, and BG is where the torsor classification of theorem 6.5.3 becomes a geometric object rather than a cohomology set.

The quotient stack repairs the central defect diagnosed in chapter 7's opening and in the $\mathcal{M}_{1,1}$ thread of Chapter 1: when G acts with stabilizers, the naive orbit space X/G forgets the automorphisms of the orbits and fails to represent the moduli problem. The stack $[X/G]$ keeps that information. Its coarse space, when one exists, is the good quotient $X/\!\!/G$ of geometric invariant theory (definition 3.1.1), and the two agree exactly where the action is free. The section closes by exhibiting $\mathcal{M}_{1,1}$ as a quotient stack in the Weierstrass presentation, recovering the j-line (theorem 1.3.6) as its coarse space and locating the generic μ_2 and special μ_4, μ_6 automorphisms of Chapter 1 inside the stack.

Throughout, $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ and G is a flat, finitely presented, affine group scheme over S acting on an S-scheme X by a morphism $\sigma\colon G \times_S X \to X$. A G-torsor is as in definition 6.5.1: an S-scheme P with a G-action and a G-invariant map $P \to T$ that is fppf-locally on T isomorphic to the trivial torsor $G \times_S T$ with its translation action. The action being fixed, all torsors are for this same G.

The definition of $[X/G]$ is dictated by descent. If X/G were a fine moduli space, a T-point of it would be a map $T \to X/G$, that is, an orbit varying over T. Locally on T such an orbit is cut out by choosing a point of X in it, but the choice is ambiguous up to the G-action, and globally the ambiguities are recorded by a torsor. A T-point of the quotient stack is therefore the intrinsic version of "a G-orbit's worth of maps to X": a principal G-bundle $P \to T$ together with a G-equivariant map $P \to X$. When the torsor is trivial, $P = G \times_S T$, an equivariant map $G \times_S T \to X$ is the same as an ordinary map $T \to X$ (evaluate at the identity section), so the trivial-torsor points recover ordinary T-points of X; the general points twist this by a torsor.

Definition 7.3.1: The Quotient Stack $[X/G]$

Let G be a flat, finitely presented, affine group scheme over S acting on an S-scheme X. The *quotient stack* $[X/G]$ is the category fibered in groupoids over $\mathcal{C} = (\mathbf{Sch}/S)_{\text{fppf}}$ whose fiber over an S-scheme T is the groupoid $[X/G](T)$ with:

- *objects* the pairs $(P \xrightarrow{\pi} T,\ P \xrightarrow{f} X)$ where $\pi\colon P \to T$ is a G-torsor and $f\colon P \to X$ is a G-equivariant S-morphism (so $f(g \cdot p) = g \cdot f(p)$);
- *morphisms* $(P, f) \to (P', f')$ the isomorphisms of G-torsors $\alpha\colon P \xrightarrow{\sim} P'$ over T that commute with the maps to X, that is, $f' \circ \alpha = f$.

A morphism $h\colon T' \to T$ in $\mathcal{C}$ acts by pullback: the object (P, f) over T is sent to $(h^*P, f \circ \mathrm{pr}_P)$ over T', where $h^*P = P \times_T T'$ is the pulled-back torsor and $\mathrm{pr}_P\colon h^*P \to P$ the projection. This assignment makes $[X/G]$ fibered in groupoids, with the pullback square exhibiting each such h^*-arrow as cartesian.

The morphisms are worth dwelling on, because they are the entire reason $[X/G]$ is a stack and not a space. An object (P, f) over T has automorphisms: an automorphism is a torsor automorphism $\alpha\colon P \to P$ over T with $f \circ \alpha = f$. When $P = G \times_S T$ is trivial and f corresponds to a point $x\colon T \to X$, such an α is right-translation by a T-point g of G, and the condition $f \circ \alpha = f$ says g fixes x, that is, g lies in the stabilizer $\mathrm{Stab}_G(x)$. Thus the automorphism group of a point of $[X/G]$ is its stabilizer under the action. This is exactly the datum the orbit space X/G discards, and its retention is what lets $[X/G]$ see the difference between a free action (all stabilizers trivial) and one with fixed points.

The pullback of torsors is the cartesian structure, and it is functorial because fiber products are: $(h \circ h')^*P \cong h'^*(h^*P)$ canonically, so the pullbacks assemble into a pseudofunctor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Grpd}$, $T \mapsto [X/G](T)$, in the sense of definition 7.1.4. That $[X/G]$ is fibered in groupoids is immediate from the definition of its morphisms: every arrow is a torsor isomorphism, hence invertible, so each fiber $[X/G](T)$ is a groupoid.

The first worked instance is the one that names the construction: taking X to be a single point produces a stack whose points over T are torsors with no further structure, since an equivariant map to a point is no constraint at all.

Definition 7.3.2: The Classifying Stack BG

Let G be a flat, finitely presented, affine group scheme over S. The *classifying stack* of G is the quotient stack

$$BG = [S/G] = [\mathrm{pt}/G]$$

of G acting (necessarily trivially) on the terminal object $S = \mathrm{pt}$. Concretely, the fiber $BG(T)$ is the groupoid of G-torsors on T: an object over T is a G-torsor $P \to T$, a morphism is an isomorphism of G-torsors over T, and a map $h\colon T' \to T$ acts by pullback of torsors. There is a distinguished object over S, the *trivial* torsor $G \to S$; the induced map $\mathrm{pt} = S \to BG$ it represents is the *universal torsor*.

The classifying stack is the geometric shadow of the torsor classification proved in theorem 6.5.3. There, G-torsors on T (up to isomorphism) were shown to be classified by the pointed cohomology set $H^1_{\text{fppf}}(T, G)$, or $H^1_{\text{ét}}(T, G)$ for smooth G. The stack BG upgrades this classification from a *set* of isomorphism classes to a *groupoid* that remembers the isomorphisms themselves. The 2-Yoneda lemma (theorem 7.1.8) makes this precise: for any S-scheme T,

$$\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{T}, BG) \simeq BG(T) = \{G\text{-torsors on } T\}$$

an equivalence of groupoids. Passing to isomorphism classes on both sides recovers $\pi_0\big(\underline{\mathrm{Hom}}(\underline{T}, BG)\big) \cong H^1(T, G)$: the set of T-points of BG, up to 2-isomorphism, is the first cohomology of T with coefficients in G. The classifying stack is the object that represents the functor $T \mapsto H^1(T, G)$, refined to a groupoid so that the automorphisms $\underline{\mathrm{Aut}}(P) = G$ of the trivial torsor are visible; the automorphism group of the universal point $\mathrm{pt} \to BG$ is G itself. This is the sense in which BG deserves the name "classifying": it is the moduli stack of G-torsors, and a map into it is a torsor.

Before proving that these constructions are stacks, it is worth naming the universal torsor in the general case, because it is the structure that makes $[X/G]$ deserve to be called a quotient. Over X itself there is a tautological object of $[X/G]$: the trivial torsor $G \times_S X \to X$ equipped with the action map $\sigma\colon G \times_S X \to X$ as its equivariant map to X. This object is classified by a morphism $q\colon X \to [X/G]$, and the theorem below records that q is itself a G-torsor, the universal one from which every object of $[X/G]$ is pulled back.

Everything now rests on a single assertion, that $[X/G]$ satisfies descent, and that assertion is not new work: it is the descent already proved in Chapter 6, repackaged. A quotient-stack object over T is a torsor with an equivariant map, and both torsors and morphisms of torsors were shown to glue in the fppf topology. The stack axioms for $[X/G]$ are the sheaf condition for isomorphisms and the effectivity of descent for objects; the first is the statement that torsor isomorphisms glue, the second that torsor-with-map data glue, and both follow from theorems 6.3.4 and 6.5.3. The strict verification of the pseudofunctor coherence (that the chosen pullbacks compose associatively up to the coherence 2-isomorphisms) is bookkeeping, and is cited to [30].

Theorem 7.3.3: The Quotient Stack and Its Universal Torsor

Let G be a flat, finitely presented, affine group scheme over S acting on an S-scheme X. Then:

1. $[X/G]$ is a stack for the fppf topology on $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$.

2. The tautological object $(G \times_S X \to X,\ \sigma)$ defines a morphism $q\colon X \to [X/G]$, and q is a G-torsor: for every T and every object $(P, f) \in [X/G](T)$, the fiber product $X \times_{[X/G]} T$ is canonically isomorphic to P as a G-torsor over T. In particular q is the *universal G-torsor*: giving a G-torsor $P \to T$ together with a G-equivariant map $P \to X$ is the same as giving a morphism $T \to [X/G]$.

3. If a good quotient $X/\!\!/G$ in the sense of definition 3.1.1 exists, there is a canonical morphism
$$\pi\colon [X/G] \longrightarrow X/\!\!/G$$
which is universal among maps from $[X/G]$ to schemes, and which is an isomorphism precisely over the open locus of $X/\!\!/G$ above which G acts with trivial stabilizers.

***Proof*:**
The three parts are proved in turn, the first by reducing to Chapter 6 descent, the second by unwinding the 2-fiber product against the 2-Yoneda lemma, and the third by the universal property of the good quotient.

Step 1: $[X/G]$ *is a stack.* By definition 7.2.2, two conditions must hold: for objects $(P, f), (P', f') \in [X/G](T)$ the presheaf $\underline{\mathrm{Isom}}\big((P, f), (P', f')\big)$ on $\mathcal{C}/T$ is an fppf sheaf, and every descent datum on every fppf cover is effective.

For the isomorphism sheaf, fix a cover $\{T_i \to T\}$. An isomorphism $(P, f) \to (P', f')$ is a torsor isomorphism $\alpha\colon P \to P'$ over T with $f' \circ \alpha = f$. Torsor isomorphisms form an fppf sheaf: the functor $T' \mapsto \mathrm{Isom}_{T'}(P|_{T'}, P'|_{T'})$ is a torsor under the sheaf $\underline{\mathrm{Aut}}(P) \cong G$ (twisted), and torsors are fppf sheaves by theorem 6.5.3; equivalently, an isomorphism of torsors is a section of the fppf-sheaf $\underline{\mathrm{Isom}}(P, P')$, so compatible local isomorphisms α_i agreeing on overlaps glue to a global α. The compatibility with f, f' is a closed condition that glues because it holds on the members of the cover and X is a sheaf. Hence $\underline{\mathrm{Isom}}$ is a sheaf and $[X/G]$ is a prestack.

For effectivity, let $\{T_i \to T\}$ be an fppf cover and let a descent datum be given: objects $(P_i, f_i) \in [X/G](T_i)$ together with isomorphisms $\varphi_{ij}\colon (P_i, f_i)|_{T_{ij}} \to (P_j, f_j)|_{T_{ij}}$ over the double overlaps $T_{ij} = T_i \times_T T_j$, satisfying the cocycle condition on triple overlaps. Forgetting the maps f_i, the torsors P_i with their φ_{ij} form a descent datum for G-torsors, which is effective: G-torsors form a stack because they are locally trivial and torsor isomorphisms glue, so the P_i glue to a G-torsor $P \to T$ with $P|_{T_i} \cong P_i$. This is the content of theorem 6.5.3 read as descent (a torsor is an fppf-sheaf locally isomorphic to G, and such sheaves glue). The equivariant maps $f_i\colon P_i \to X$ agree on overlaps under the φ_{ij} by hypothesis, so they define compatible maps $P|_{T_i} \to X$; since X is a sheaf and $\{P|_{T_i} \to P\}$ is an fppf cover of P, these glue to a G-equivariant map $f\colon P \to X$ (equivariance is checked after pulling back to the cover, where it holds). The pair (P, f) is an object of $[X/G](T)$ restricting to (P_i, f_i), so the descent datum is effective. The coherence of the pseudofunctor structure is verified in [30]. Thus $[X/G]$ is a stack.

Step 2: q *is the universal torsor.* The tautological object $t_X = (G \times_S X \xrightarrow{\mathrm{pr}_X} X,\ \sigma)$ over X is a G-torsor (the trivial one on X) with equivariant map σ, hence an object of $[X/G](X)$; by 2-Yoneda (theorem 7.1.8) it corresponds to a morphism $q\colon X \to [X/G]$. To identify the fiber product, fix an object $(P, f) \in [X/G](T)$, corresponding to a morphism $\xi\colon T \to [X/G]$. By the universal property of the 2-fiber product, an object of $\big(X \times_{[X/G]} T\big)(T')$ over a T-scheme T' is a triple consisting of a T'-point $a\colon T' \to X$, the given point $\xi|_{T'}$, and a 2-isomorphism $q \circ a \cong \xi|_{T'}$ in $[X/G](T')$. Unwinding the definitions, $q \circ a$ is the trivial torsor $G \times_S T' \to T'$ with equivariant map $g \mapsto g \cdot a$, and a 2-isomorphism to $(P, f)|_{T'}$ is an isomorphism of torsors $G \times_S T' \xrightarrow{\sim} P|_{T'}$ commuting with the maps to X. Such an isomorphism is the choice of a section of $P|_{T'} \to T'$ (the image of the identity), and it determines a uniquely by $a = f \circ (\text{section})$. Therefore

$$\big(X \times_{[X/G]} T\big)(T') = \{\text{sections of } P|_{T'} \to T'\} = \mathrm{Hom}_T(T', P)$$

naturally in T', which says $X \times_{[X/G]} T \cong P$ as T-schemes, and the identification is G-equivariant. Hence the base change of q along any $\xi\colon T \to [X/G]$ is the torsor P; taking ξ to be the identity of $[X/G]$ (formally, working over the stack itself) exhibits q as a G-torsor over $[X/G]$. That every object of $[X/G]$ arises as a pullback of t_X is the same computation read backwards: an object (P, f) over T is $\xi^* t_X$ for the classifying map $\xi = \xi_{(P,f)}$, since its fiber product with q returns P with f. This is the universal property claimed.

Step 3: The map to the good quotient. Suppose $Y = X /\!\!/ G$ is a good quotient, so Y carries a G-invariant morphism $p\colon X \to Y$ that is universal among G-invariant maps to schemes and identifies Y with the ring of invariants locally (definition 3.1.1). To produce $\pi\colon [X/G] \to Y$ it suffices, by 2-Yoneda, to give for each object $(P, f) \in [X/G](T)$ a morphism $T \to Y$, functorially. The composite $p \circ f\colon P \to Y$ is G-invariant, because f is equivariant and p is invariant: $p(f(g \cdot x)) = p(g \cdot f(x)) = p(f(x))$. A G-invariant map out of a G-torsor $P \to T$ factors uniquely through T, since $P \to T$ is an fppf-local trivial torsor and a G-invariant map out of $G \times_S T$ factors through the projection to T; the factorization glues by descent (theorem 6.3.4). This produces $\bar{f}\colon T \to Y$ with $\bar{f} \circ \pi = p \circ f$, and the assignment $(P, f) \mapsto \bar{f}$ is functorial in T, hence defines $\pi\colon [X/G] \to Y$. The tautological object over X is sent to $p\colon X \to Y$, so $\pi \circ q = p$.

Universality: given any scheme Z and a morphism $[X/G] \to Z$, precompose with $q\colon X \to [X/G]$ to get a G-invariant map $X \to Z$ (invariant because $q \circ \sigma$ agrees with $q \circ \mathrm{pr}_X$ up to the tautological 2-isomorphism, so the two ways of mapping $G \times_S X$ to Z coincide), which factors uniquely through $Y = X/\!\!/G$ by the universal property of the good quotient; the induced $Y \to Z$ composes with π to give back the original map, and uniqueness follows because q is an fppf cover and maps out of a stack are determined by their pullback along an fppf cover. Thus π is universal among maps from $[X/G]$ to schemes, that is, Y is the coarse space of $[X/G]$ in the categorical sense.

Finally, π is an isomorphism exactly over the free locus. On the open $U \subseteq Y$ over which G acts with trivial stabilizers, the action on $p^{-1}(U)$ is free with quotient a scheme (a good quotient of a free action is geometric, and $p^{-1}(U) \to U$ is then a G-torsor by definition 3.1.1), so $[p^{-1}(U)/G] \cong U$: a torsor $P \to T$ with equivariant map to $p^{-1}(U)$ is, since the action downstairs is free, the pullback of $p^{-1}(U) \to U$ along the induced $T \to U$, giving an equivalence $[X/G]|_U \simeq U$. Conversely, at a point of X with nontrivial stabilizer H, the fiber of π over its image contains a copy of BH (the automorphisms of that point form the group H, as computed after definition 7.3.1), which is not a scheme unless H is trivial, so π is not an isomorphism there. Hence π is an isomorphism precisely over the trivial-stabilizer locus, completing the proof.

The theorem is the structural payoff of the section, and its three parts say three complementary things. Part (1) is the promise kept: $[X/G]$ is a genuine stack, and its proof consumed no new descent, only the descent of Chapter 6 applied to torsors-with-a-map. Part (2) is the universal property that justifies the notation: $[X/G]$ is the target that turns a G-equivariant family into a map, exactly as a fine moduli space turns a family into a map to it, with the torsor absorbing the ambiguity that a scheme could not. The map $q\colon X \to [X/G]$ is a G-torsor, so X is a "cover" of the stack in the same sense that $\operatorname{Spec} L$ covers $\operatorname{Spec} k$ in Galois descent; this presentation $X \to [X/G]$ is what will make $[X/G]$ algebraic in Chapter 8, where an atlas is exactly such a smooth cover by a scheme. Part (3) situates the stack above its coarse space: the good quotient $X/\!\!/G$ of geometric invariant theory (proposition 1.4.6) is the best scheme approximation to $[X/G]$, and the two coincide only where the action is free. Where stabilizers jump, the stack carries a residual BH that the coarse space cannot see. This is the precise mechanism by which the automorphism obstruction of Chapter 1 survives into the stack and is resolved there rather than lost.

One reading of part (3) deserves emphasis, because it reorganizes the whole of geometric invariant theory. GIT constructs $X/\!\!/G$ by hand, through invariant sections of a linearized line bundle (definition 3.2.1), and the construction is delicate precisely because it must decide which orbits to keep (the semistable ones) and how to separate them. The quotient stack $[X^{\mathrm{ss}}/G]$ is the same data assembled without those choices: it exists for any action, and GIT is the statement that its coarse space is a projective scheme when the linearization is ample. The stack is the primary object and the projective GIT quotient is its shadow, obtained by descending to schemes and paying the price of forgetting stabilizers.

The construction acquires its full meaning through examples, and the richest cluster is the classifying stacks BG for the groups that appear in algebraic geometry. Each BG is a moduli stack of a familiar structure, because a G-torsor is a familiar structure in disguise: for $G = \mathrm{GL}_n$ a torsor is a rank-n vector bundle, for $G = \mathbb{G}_m$ a line bundle, for $G = \mu_n$ an n-torsion line bundle with a chosen trivialization of its n-th power. The classifying stacks are therefore the universal moduli of these bundles, and computing them concretely turns the abstract H^1-classification into geometry.

Example 7.3.4: Classifying Stacks of Line and Vector Bundles
Fix the base $S = \operatorname{Spec} k$ and the fppf topology.

1. *$B\mathbb{G}_m$ and line bundles.* A $\mathbb{G}_m$-torsor on T is the complement of the zero section of a line bundle, and the assignment (line bundle) $\mapsto$ (its associated $\mathbb{G}_m$-torsor of frames) is an equivalence between line bundles on T and $\mathbb{G}_m$-torsors on T. Hence

   $$B\mathbb{G}_m(T) \simeq \{\text{line bundles on } T\}$$

   as groupoids, with morphisms the bundle isomorphisms. The isomorphism classes are $\operatorname{Pic}(T) = H^1(T, \mathbb{G}_m)$, matching theorem 6.5.3. The automorphism group of a line bundle is $\mathbb{G}_m$ (multiplication by a unit), so every point of $B\mathbb{G}_m$ has automorphism group $\mathbb{G}_m$; the stack is a gerbe over a point in the sense to be developed in Chapter 9. The universal object is the tautological line bundle on $B\mathbb{G}_m$.

2. *BGL_n and vector bundles.* A GL_n-torsor on T is the frame bundle of a rank-n vector bundle, and again the associated-bundle construction gives an equivalence

   $$BGL_n(T) \simeq \{\text{rank-}n \text{ vector bundles on } T\}$$

   so BGL_n is the moduli stack of rank-n vector bundles, with a point's automorphism group the automorphisms GL_n of the trivial bundle. The determinant homomorphism $\det\colon \mathrm{GL}_n \to \mathbb{G}_m$ induces a morphism $BGL_n \to B\mathbb{G}_m$ sending a bundle to its determinant line bundle.

3. *$B\mu_n$ and torsion line bundles.* A μ_n-torsor on T is, by the Kummer sequence $1 \to \mu_n \to \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \to 1$, a line bundle L together with a trivialization of $L^{\otimes n}$. Thus $B\mu_n(T)$ is the groupoid of such pairs, and its isomorphism classes fit into

   $$0 \to \mathcal{O}(T)^\times/(\mathcal{O}(T)^\times)^n \to H^1(T, \mu_n) \to \operatorname{Pic}(T)[n] \to 0$$

   the Kummer sequence in cohomology. The automorphism group of the trivial μ_n-torsor is μ_n.

The three classifying stacks are the same construction viewed through three groups, and the pattern is uniform: BG is the moduli stack of the geometric objects that G-torsors encode, its isomorphism classes are $H^1(-, G)$, and the automorphism group of every point is G. The maps between them ($\det\colon BGL_n \to B\mathbb{G}_m$, or the Kummer inclusion $B\mu_n \to B\mathbb{G}_m$) are induced by homomorphisms of groups, which is the general principle that a group homomorphism $G \to H$ induces a morphism of classifying stacks $BG \to BH$ by pushing torsors forward. These stacks are not schemes, and not even algebraic spaces, precisely because their points have nontrivial automorphisms; they are the simplest stacks that are not spaces, and every quotient stack is built by mixing a BG of this kind into the geometry of X.

The example the book has been building toward is the moduli of elliptic curves. In Chapter 1 the functor $\mathcal{M}_{1,1}$ failed to be representable by a scheme because every elliptic curve has the automorphism $[-1]$, and special curves have more (example 1.3.4); the best a scheme could do was the coarse j-line (theorem 1.3.6). The quotient-stack construction resolves this: $\mathcal{M}_{1,1}$ is a quotient stack, and its coarse space is exactly the j-line, with the automorphisms sitting in the fibers.

Example 7.3.5: $\mathcal{M}_{1,1}$ as a Quotient Stack

Work over $S = \operatorname{Spec} k$ with $\operatorname{char} k \neq 2, 3$. Every elliptic curve over a k-scheme T admits, fppf-locally on T, a Weierstrass equation

$$y^2 = x^3 + ax + b$$

with discriminant $\Delta = -16(4a^3 + 27b^2)$ invertible. Two such equations define isomorphic curves if and only if they are related by the change of variables $(x, y) \mapsto (\lambda^2 x, \lambda^3 y)$ for a unit $\lambda \in \mathbb{G}_m$, under which $(a, b) \mapsto (\lambda^4 a, \lambda^6 b)$. Hence the moduli stack is the quotient of the parameter space by this $\mathbb{G}_m$-action with weights $(4, 6)$:

$$\mathcal{M}_{1,1} \simeq \left[\left(\mathbb{A}^2_{a,b} \setminus \{\Delta = 0\}\right)/\mathbb{G}_m\right]$$

where $\lambda \cdot (a, b) = (\lambda^4 a, \lambda^6 b)$. An object over T is a $\mathbb{G}_m$-torsor $P \to T$ (a line bundle minus its zero section) together with a $\mathbb{G}_m$-equivariant map $P \to \mathbb{A}^2 \setminus \{\Delta = 0\}$; unwinding the weights, this is exactly a line bundle $\mathcal{L}$ on T (the "Hodge line bundle") with sections $a \in \mathcal{L}^{\otimes 4}$, $b \in \mathcal{L}^{\otimes 6}$ whose discriminant is nowhere zero, which is the twisted Weierstrass data of an elliptic curve over T.

The coarse space is the good quotient. After inverting Δ, the ring of invariants of the $\mathbb{G}_m$-action on $k[a, b]$ (weights $4, 6$) is generated by the single weight-0 element $j = 1728 \cdot \frac{4a^3}{4a^3+27b^2}$ (equivalently by a^3/Δ and b^2/Δ with one relation), and the good quotient of $\mathbb{A}^2 \setminus \{\Delta = 0\}$ is the affine j-line $\mathbb{A}^1_j$. By theorem 7.3.3(3) and proposition 1.4.6 the coarse space of $\mathcal{M}_{1,1}$ is therefore the j-line $\mathbb{A}^1_j = \operatorname{Spec} k[j]$, recovering theorem 1.3.6.

The automorphisms are the stabilizers. A point (a, b) has $\mathbb{G}_m$-stabilizer $\{\lambda : \lambda^4 a = a,\ \lambda^6 b = b\}$:

- generically (a, b both nonzero, so $j \neq 0, 1728$) the stabilizer is $\mu_2 = \{\pm 1\}$, matching the automorphism $[-1]$ of a generic elliptic curve;
- at $b = 0$ ($j = 1728$) the condition is $\lambda^4 = 1$, so the stabilizer is μ_4, matching the extra automorphism of the curve $y^2 = x^3 + ax$;
- at $a = 0$ ($j = 0$) the condition is $\lambda^6 = 1$, so the stabilizer is μ_6, matching the curve $y^2 = x^3 + b$.

These are precisely the automorphism groups recorded in example 1.3.4. Because μ_2 acts trivially on the whole parameter space (every (a, b) has ± 1 in its stabilizer), $\mathcal{M}_{1,1}$ carries a generic μ_2 in the automorphism group of every point: it is a μ_2-gerbe over its coarse space away from $j = 0, 1728$, and the residual gerbe of the stack point over $j = 1728$ (resp. $j = 0$) is $B\mu_4$ (resp. $B\mu_6$). The systematic study of these residual gerbes and of the gerbe structure of $\mathcal{M}_{1,1}$ over the j-line is the subject of Chapter 9.

The elliptic-curve example closes the circle opened in Chapter 1. There, the automorphism $[-1]$ was an obstruction to fine representability, and the j-line was a lossy consolation prize that identified curves related by an isomorphism but forgot that each carries automorphisms. The quotient stack $\mathcal{M}_{1,1} = [(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ loses nothing: it is a fine moduli object in the 2-categorical sense, its

T-points are exactly families of elliptic curves over T, and the automorphisms that obstructed a scheme now live visibly in the stabilizers. The j-line reappears as its coarse space, and the special automorphisms at $j = 0$ and $j = 1728$ are the points where the stabilizer jumps from μ_2 to μ_6 and μ_4. What was a defect of the theory is now a feature of the object: the stack is the moduli space, and the automorphisms are part of its geometry rather than an obstruction to its existence.

The construction is uniform across the running examples of the book. The Hilbert scheme of Chapter 2 was a fine moduli space because its objects (closed subschemes) have no automorphisms; there the stack $\underline{\mathrm{Hilb}}$ is representable, a scheme viewed as a stack with trivial stabilizers. The moduli of curves is the opposite extreme, where every object has automorphisms and only a stack will do. The quotient stack is the general mechanism spanning the two: it is representable exactly where the action is free, and it acquires nontrivial stabilizers exactly where the objects it parametrizes acquire automorphisms.

Exercise 7.3.1

1. Let G act on X, and let $(P, f) \in [X/G](T)$ be an object over T. Show that its automorphism group in the groupoid $[X/G](T)$ is the group of T-points of the stabilizer group scheme along f: when $P = G \times_S T$ is trivial and f corresponds to $x: T \to X$, this automorphism group is $\mathrm{Stab}_G(x)(T)$. Deduce that $[X/G] \to X/\!\!/G$ is an isomorphism near a point if and only if the stabilizer there is trivial.

2. Verify from definition 7.3.2 that a group homomorphism $\rho: G \to H$ of S-group schemes induces a morphism of stacks $B\rho: BG \to BH$, given on T-points by pushing a G-torsor P forward to the H-torsor $P \times^G H = (P \times_S H)/G$. Check that $B\rho$ sends the universal G-torsor to the pushforward of the trivial torsor, and that $B(\rho' \circ \rho) = B\rho' \circ B\rho$. Identify the determinant morphism $BGL_n \to B\mathbb{G}_m$ of example 7.3.4 as $B(\det)$.

3. Compute the ring of $\mathbb{G}_m$-invariants of $k[a, b]$ under the weight-$(4, 6)$ action $\lambda \cdot (a, b) = (\lambda^4 a, \lambda^6 b)$, and confirm that inverting $\Delta = -16(4a^3 + 27b^2)$ produces a ring whose spectrum is the affine j-line, with $j = 1728 \cdot \frac{4a^3}{4a^3+27b^2}$. (Compute the invariant monomials $a^i b^j$ with $4i + 6j = 0$, note there are none of positive degree, then pass to the localization at Δ where negative-weight invariants appear.) Explain in one sentence why this makes the coarse space of $\mathcal{M}_{1,1}$ the j-line.

4. Show that when G acts *freely* on X (trivial stabilizers everywhere) and a geometric quotient X/G exists as a scheme with $X \to X/G$ a G-torsor, the quotient stack $[X/G]$ is representable by the scheme X/G, that is, the canonical map $[X/G] \to X/G$ is an equivalence. (Use part (2) of theorem 7.3.3: a T-point of $[X/G]$ is a torsor with equivariant map to X, and freeness lets you descend it to a map $T \to X/G$.)

5. Let G be a finite group (constant group scheme) acting on a scheme X over k. Describe the objects and morphisms of $[X/G](T)$ concretely for $T = \operatorname{Spec} k$ with k algebraically closed: show that a k-point of $[X/G]$ is a point $x \in X(k)$ and that its automorphism group is the set-theoretic stabilizer $\mathrm{Stab}_G(x) \subseteq G$. How many isomorphism classes of k-points lie over a given G-orbit, and how does this compare with the single point that orbit contributes to X/G?

6. The naive presheaf quotient $(\underline{X}/G)^{\mathrm{pre}}$ assigns to T the set $X(T)/G(T)$ of orbits of T-points. Explain why the stackification of $\underline{X}/G$ (as a prestack of sets, viewed as a CFG with trivial automorphisms) is *not* in general $[X/G]$, and identify what extra data the passage to $[X/G]$

records that the set-theoretic sheafification of $X(T)/G(T)$ cannot. (Contrast the case of a free action, where they agree, with $BG = [\mathrm{pt}/G]$, where the naive quotient of $\mathrm{pt}(T)/G(T) = \mathrm{pt}$ is a point but BG is not.)

7.4 Representable Morphisms of Stacks

The previous sections built stacks as geometric objects in their own right: the quotient stack $[X/G]$ and the classifying stack BG are stacks in the fppf topology, and a scheme X sits among them through its representable stack $\underline{X}$. What is missing is a supply of *adjectives*. For an ordinary morphism of schemes one says it is flat, smooth, proper, or surjective, and these words carry the geometry. A morphism of stacks $f\colon \mathcal{F} \to \mathcal{G}$ has no underlying continuous map of topological spaces to test, so none of these adjectives has an immediate meaning. The device that repairs this, and that turns out to be the organizing principle for the whole geometric theory of stacks, is representability: a morphism f is understood by pulling it back along every map from a scheme and asking what happens over that scheme, where the classical adjectives *do* make sense.

This section isolates the one class of morphisms for which the transfer of geometric language is unproblematic, the representable morphisms, and shows how a property P of scheme morphisms is transported to them by base change. The payoff is the diagonal. A single morphism, the diagonal $\Delta_{\mathcal{F}}\colon \mathcal{F} \to \mathcal{F} \times \mathcal{F}$, packages all the automorphism data of a stack into the Isom-sheaves of section 7.2, and its representability is the first of the two conditions that will make a stack *algebraic* in the next chapter. Everything here is bookkeeping in the 2-category of stacks, but it is the bookkeeping without which no geometry on a stack can even be stated.

Throughout, $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ and all stacks are stacks over $\mathcal{C}$; "scheme" means a scheme over S, identified with its representable stack $\underline{T}$ via the 2-Yoneda lemma (theorem 7.1.8). The 2-fiber product $\mathcal{F} \times_{\mathcal{G}} \mathcal{H}$ of stacks is the 2-categorical fiber product: its objects over T are triples (x, y, α) with $x \in \mathcal{F}(T)$, $y \in \mathcal{H}(T)$, and α an isomorphism in $\mathcal{G}(T)$ between the images of x and y. It is again a stack, and it enjoys the universal property of a fiber product up to canonical 2-isomorphism; the construction is imported from [5] and used freely.

The definition begins with the observation that some morphisms of stacks have schematic fibers. When they do, every geometric question about the morphism can be pushed down to a question about schemes.

Definition 7.4.1: Representable Morphism of Stacks

Let $f\colon \mathcal{F} \to \mathcal{G}$ be a morphism of stacks over $\mathcal{C}$. The morphism f is *representable by schemes* (or simply *representable*) if for every scheme T and every morphism $t\colon \underline{T} \to \mathcal{G}$ the 2-fiber product

$$\mathcal{F} \times_{\mathcal{G}} \underline{T}$$

is representable by a scheme, that is, equivalent as a stack to $\underline{T'}$ for some scheme T'. The morphism f is *representable by algebraic spaces* if, under the same hypotheses, $\mathcal{F} \times_{\mathcal{G}} \underline{T}$ is (equivalent to) an algebraic space.

The definition tests f against all schemes mapping to the target $\mathcal{G}$ and demands that each fiber over a scheme be a scheme. The word "fiber" is meant in the 2-categorical sense: the objects of $\mathcal{F} \times_{\mathcal{G}} \underline{T}$ over a T-scheme U are pairs consisting of an object $x \in \mathcal{F}(U)$ and an isomorphism in $\mathcal{G}(U)$ between

$f(x)$ and the image of $U \to T$. When this stack is a scheme, it has no nontrivial automorphisms in any fiber, so a representable morphism is exactly one whose fibers over schemes have trivialized their automorphisms. This is the precise sense in which representable morphisms are the "schematic" part of stack theory: they are the morphisms along which a scheme, pulled back, stays a scheme. The two versions of the definition differ only in what one is willing to accept as a fiber. Representability by algebraic spaces is the more flexible notion and is the one that survives in full generality, because algebraic spaces are closed under the quotient operations that produce stacks, whereas schemes are not (this is the lesson of the failure of effective descent for non-affine schemes, section 6.3); until algebraic spaces are constructed in the next chapter, the reader may keep the schematic version in mind, as every example below is representable by schemes.

A first sanity check: a morphism $\underline{X} \to \underline{Y}$ of representable stacks is representable, because for any $T \to \underline{Y}$ the fiber product $\underline{X} \times_{\underline{Y}} \underline{T}$ is $\underline{X \times_Y T}$, the representable stack of the scheme fiber product $X \times_Y T$, which exists as a scheme. So the notion extends the category of schemes without disturbing it. The content of the definition is entirely in the morphisms with stacky source or target, where it is a real restriction.

Example 7.4.2: A Morphism to BG from a Scheme

Let G be an affine group scheme over S and $BG = [S/G]$ the classifying stack (definition 7.3.2). Consider the structure morphism $\pi\colon BG \to \underline{S}$, and test its representability against a scheme $T \to \underline{S}$, that is, an S-scheme T. By definition of the 2-fiber product,

$$BG \times_{\underline{S}} \underline{T}$$

has, over a T-scheme U, the objects (P, θ) where P is a G-torsor on U (an object of $BG(U)$) and θ is an isomorphism in $\underline{S}(U)$ between $\pi(P)$ and the given point, which is no data at all since $\underline{S}(U)$ is discrete. Thus the fiber is again BG restricted to T-schemes, that is, $BG_T = [T/G]$, and this is a scheme only when it has no automorphisms, that is, only when G is trivial. For G nontrivial, $\pi\colon BG \to \underline{S}$ is *not* representable: the torsor P carries its automorphism group $\mathrm{Aut}(P) = G(U)$, which no scheme fiber can absorb. This is the prototype of a non-representable morphism, and it will be diagnosed precisely by the diagonal below: the automorphisms that obstruct representability of π are exactly the ones the diagonal of BG records.

The example already shows the pattern. Non-representability of a morphism of stacks is caused by automorphisms in the fibers, and the same automorphisms are what a stack carries at each of its points. Before turning them into a single object, the section records the reason representable morphisms matter: they are precisely the morphisms to which one can attach a geometric adjective.

Suppose $f\colon \mathcal{F} \to \mathcal{G}$ is representable and P is a property of morphisms of schemes. To say "f has P" one wants to say that every scheme-fiber of f, that is, every morphism $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$, has P. Two things must hold for this to be well-posed. First, the property must make sense on the fibers: since each $\mathcal{F} \times_{\mathcal{G}} \underline{T}$ is a scheme, the map to $\underline{T}$ is a morphism of schemes, so any P of scheme morphisms applies. Second, and this is the subtle point, the answer must not depend on *which* scheme $T \to \mathcal{G}$ one tests against. Two different maps $T \to \mathcal{G}$ and $T' \to \mathcal{G}$ can present overlapping information, and the fibers must give a consistent verdict. Consistency is guaranteed exactly when P descends, and this is where Chapter 6 enters.

Definition 7.4.3: Property of a Representable Morphism via Base Change

Let P be a property of morphisms of schemes that is *stable under base change* (if $g\colon Y \to Z$ has P, so does every base change $Y \times_Z Z' \to Z'$) and *fppf-local on the target* (a morphism $g\colon Y \to Z$ has P if and only if, for some fppf cover $\{Z_i \to Z\}$, each base change $Y \times_Z Z_i \to Z_i$ has P). Standard examples are: flat, smooth, étale, unramified, surjective, (locally) of finite presentation, proper, separated, quasi-compact, affine, and a closed (or open) immersion. A representable morphism $f\colon \mathcal{F} \to \mathcal{G}$ of stacks *has property* P if for every scheme T and every morphism $t\colon \underline{T} \to \mathcal{G}$ the base change

$$\mathcal{F} \times_{\mathcal{G}} \underline{T} \longrightarrow \underline{T}$$

of schemes has P.

The two hypotheses on P are exactly what make the definition consistent, and the mechanism is descent. The next proposition is where the whole apparatus of Chapter 6 pays off: it says that stability under base change plus fppf-locality is not only convenient but is the precise condition under which the base-change definition is independent of choices and can be checked on a single well-chosen test scheme rather than on all of them.

Proposition 7.4.4: Well-Definedness of Properties via Descent

Let P be a property of scheme morphisms that is stable under base change and fppf-local on the target, and let $f\colon \mathcal{F} \to \mathcal{G}$ be a representable morphism of stacks. Then:

1. If $\{U_i \to \underline{T}\}$ is any fppf cover refining a test $t\colon \underline{T} \to \mathcal{G}$, then $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ has P if and only if each $\mathcal{F} \times_{\mathcal{G}} U_i \to U_i$ has P. In particular the verdict "f has P" can be checked fppf-locally on each test scheme.

2. If $\mathcal{G} = \underline{Y}$ is representable, so that f is a morphism of schemes $X \to Y$, then f has P in the sense of definition 7.4.3 if and only if $X \to Y$ has P as a morphism of schemes.

3. If $\mathcal{G}$ admits a representable morphism $\pi\colon \underline{U} \to \mathcal{G}$ from a scheme that is an fppf cover after base change to any test scheme (an *atlas*), then f has P if and only if the single base change $\mathcal{F} \times_{\mathcal{G}} \underline{U} \to \underline{U}$ has P.

***Proof*:**
The three statements share one engine, the fppf-locality of P on the target, applied to the descent of properties from Chapter 6 (theorem 6.4.3).

1. *Fppf-local check on a test scheme.* Fix $t\colon \underline{T} \to \mathcal{G}$ and write $Z = \mathcal{F} \times_{\mathcal{G}} \underline{T}$, a scheme by representability, with structure map $g\colon Z \to T$. For an fppf cover $\{U_i \to \underline{T}\}$, the compatibility of 2-fiber products gives canonical equivalences

$$\mathcal{F} \times_{\mathcal{G}} U_i \;\simeq\; Z \times_T U_i$$

because $U_i \to \underline{T} \to \mathcal{G}$ presents U_i as a T-scheme and fiber products compose. Hence $\mathcal{F} \times_{\mathcal{G}} U_i \to U_i$ is the base change $g_i\colon Z \times_T U_i \to U_i$ of g along $U_i \to T$. Since P is fppf-local on the target, g has P if and only if each g_i has P. This is the assertion.

2. *Agreement with the scheme-theoretic property.* Suppose $\mathcal{G} = \underline{Y}$ and $\mathcal{F} = \underline{X}$, so f is (the stack associated to) a morphism $h\colon X \to Y$ of schemes. For a test $t\colon \underline{T} \to \underline{Y}$, that is, a morphism $T \to Y$, the 2-fiber product is the representable stack of the scheme fiber product,
$$\underline{X} \times_{\underline{Y}} \underline{T} \;\simeq\; \underline{X \times_Y T}$$
so its structure map to $\underline{T}$ is h base-changed along $T \to Y$. If h has P, then every such base change has P because P is stable under base change, so f has P in the sense of definition 7.4.3. Conversely, taking the tautological test $t = \mathrm{id}_Y\colon \underline{Y} \to \underline{Y}$ (with $T = Y$) gives the base change $\underline{X} \times_{\underline{Y}} \underline{Y} \simeq \underline{X}$ with structure map h itself, so if f has P then h has P. The two notions coincide.

3. *Checking on an atlas.* Let $\pi\colon \underline{U} \to \mathcal{G}$ be representable and fppf-surjective, and suppose the base change $g_U\colon \mathcal{F} \times_{\mathcal{G}} \underline{U} \to \underline{U}$ has P; the claim is that $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ has P for every test $t\colon \underline{T} \to \mathcal{G}$. Form the fiber product $V = \underline{U} \times_{\mathcal{G}} \underline{T}$, which is a scheme because π is representable, and let $q\colon V \to T$ be the projection. Then q is an fppf cover of T: it is a base change of the fppf cover π, and the three properties defining an fppf cover, flat, locally of finite presentation, and surjective, are each stable under base change. Two applications of the composition of fiber products give a canonical equivalence
$$\big(\mathcal{F} \times_{\mathcal{G}} \underline{T}\big) \times_T V \;\simeq\; \mathcal{F} \times_{\mathcal{G}} V \;\simeq\; \big(\mathcal{F} \times_{\mathcal{G}} \underline{U}\big) \times_U V$$
where the last step uses $V = \underline{U} \times_{\mathcal{G}} \underline{T}$ once more to view V as a U-scheme. Under this equivalence the base change of $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ along the fppf cover $q\colon V \to T$ is identified with the base change of g_U along $V \to U$. Since g_U has P and P is stable under base change, this base change has P; since P is fppf-local on the target and q is an fppf cover, $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ has P, as we sought to show. The converse is immediate: if f has P then in particular the base change along the test π has P, completing the proof.

The proposition is the reason the definition is usable in practice. Part (2) confirms that the base-change definition does not conflict with the classical one on schemes: a smooth morphism of schemes is smooth as a morphism of stacks, with no shift in meaning. Part (3) is the working tool. Testing a property against *all* schemes over $\mathcal{G}$ is impossible to carry out, but once $\mathcal{G}$ has an atlas, a single representable fppf-surjective $\underline{U} \to \mathcal{G}$, the property is decided by one base change $\mathcal{F} \times_{\mathcal{G}} \underline{U} \to \underline{U}$. This is precisely how "smooth", "proper", and the rest will be defined for morphisms between algebraic stacks in the next chapter: pull back along an atlas of the target and read the property off the resulting morphism of schemes. The transport of geometric adjectives to stacks is thus not an ad hoc extension of vocabulary but a theorem about descent.

Example 7.4.5: Smoothness of the Universal Torsor $X \to [X/G]$

Let G be a smooth affine group scheme over S acting on an S-scheme X, and let $q\colon X \to [X/G]$ be the projection to the quotient stack (definition 7.3.1). The morphism q is representable: for a test $t\colon \underline{T} \to [X/G]$, corresponding to a pair $(P \to T, P \to X)$ with P a G-torsor and $P \to X$ equivariant, the fiber product $\underline{X} \times_{[X/G]} \underline{T}$ is exactly the torsor P itself, which is a scheme (an fppf-locally trivial G-torsor over a scheme is a scheme, being fppf-locally $G \times T$; when G is affine, $P \to T$ is even affine, so P is a scheme). Thus every scheme-fiber of q is a G-torsor $P \to T$, and to decide whether q is smooth, flat, or surjective it suffices to decide these for $P \to T$. A G-torsor $P \to T$ is fppf-locally on T isomorphic to the projection $G \times T \to T$ (theorem 6.5.3), so it inherits every fppf-local property of the structure map $G \to S$. Because G is smooth over S, the projection $G \times T \to T$ is smooth and surjective, hence so is every G-torsor $P \to T$, and therefore, by proposition 7.4.4(1), the morphism $q\colon X \to [X/G]$ is smooth and surjective. In particular q is an atlas for $[X/G]$: a representable, smooth, surjective morphism from a scheme. This is the concrete meaning of the slogan that $[X/G]$ is "X modulo a smooth group": the quotient map is a smooth cover in the stack sense.

The example does more than illustrate the definition; it produces the first atlas. The map $X \to [X/G]$ is representable and smooth-surjective, which is exactly the data an algebraic stack will be required to possess. For $[X/G]$ this atlas is given by construction; the definition of a general algebraic stack in the next chapter axiomatizes the existence of such an atlas, and definition 7.4.3 is what lets one read the geometry of the stack off it.

There remains one representability question that no atlas resolves on its own, and it is the one that controls whether an atlas can be used at all. When is a morphism $\underline{T} \to \mathcal{F}$ from a *scheme* representable? Equivalently, when is the fiber product of two schemes over $\mathcal{F}$ again a scheme? The answer is encoded in a single morphism of $\mathcal{F}$, its diagonal, and this is the central result of the section.

The diagonal $\Delta_{\mathcal{F}}\colon \mathcal{F} \to \mathcal{F} \times \mathcal{F}$ is the morphism whose component in each factor is the identity; on objects over T it sends $x \in \mathcal{F}(T)$ to the pair (x, x). Its role is to detect isomorphisms. Given two objects $x, y \in \mathcal{F}(T)$, presented as a single morphism $(x, y)\colon \underline{T} \to \mathcal{F} \times \mathcal{F}$, the fiber product of $\Delta_{\mathcal{F}}$ against (x, y) measures how x and y can be identified. This measurement is exactly the Isom-sheaf of section 7.2, which is why the diagonal is the object that carries all automorphism information of $\mathcal{F}$.

Theorem 7.4.6: Representability of the Diagonal

Let $\mathcal{F}$ be a stack over $\mathcal{C}$ and $\Delta_{\mathcal{F}}\colon \mathcal{F} \to \mathcal{F} \times \mathcal{F}$ its diagonal. The following are equivalent.

1. The diagonal $\Delta_{\mathcal{F}}$ is representable (by schemes, respectively by algebraic spaces).
2. For every scheme T and all objects $x, y \in \mathcal{F}(T)$, the presheaf $\underline{\mathrm{Isom}}_T(x, y)$ on $(\mathbf{Sch}/T)$, sending $U \xrightarrow{u} T$ to the set of isomorphisms $u^*x \to u^*y$ in $\mathcal{F}(U)$, is representable by a T-scheme (respectively by an algebraic space over T).

When these hold, every morphism $\underline{T} \to \mathcal{F}$ from a scheme T is representable, and for any two morphisms $a\colon \underline{T_1} \to \mathcal{F}$, $b\colon \underline{T_2} \to \mathcal{F}$ from schemes the 2-fiber product $\underline{T_1} \times_{\mathcal{F}} \underline{T_2}$ is a scheme (respectively an algebraic space).

***Proof*:**
The proof turns on a single computation: the fiber product of two objects over $\mathcal{F}$ along the diagonal is the Isom-sheaf. Everything then follows by recognizing each stated fiber product as such an Isom-sheaf.

Step 1: The diagonal fiber product is the Isom-sheaf. Fix a scheme T and two objects $x, y \in \mathcal{F}(T)$, assembled into the morphism $(x, y)\colon \underline{T} \to \mathcal{F} \times \mathcal{F}$. Form the 2-fiber product

$$I \;=\; \underline{T} \times_{(\mathcal{F}\times\mathcal{F})} \mathcal{F}$$

taken against $\Delta_{\mathcal{F}}$. By the definition of the 2-fiber product, an object of I over a T-scheme $u\colon U \to T$ is a triple $\big(u,\ z \in \mathcal{F}(U),\ \alpha\big)$ where α is an isomorphism in $(\mathcal{F} \times \mathcal{F})(U)$ between (u^*x, u^*y) and $\Delta_{\mathcal{F}}(z) = (z, z)$. Such an α is a pair of isomorphisms $\alpha_1\colon u^*x \to z$ and $\alpha_2\colon u^*y \to z$ in $\mathcal{F}(U)$. The datum (z, α_1, α_2) is equivalent, up to unique isomorphism, to the single isomorphism $\beta = \alpha_2^{-1} \circ \alpha_1\colon u^*x \to u^*y$, obtained by taking $z = u^*y$ and $\alpha_2 = \mathrm{id}$. Hence the groupoid $I(U)$

is equivalent to the *set* of isomorphisms $u^*x \to u^*y$, with no nontrivial automorphisms, so I is (equivalent to) the presheaf $\underline{\mathrm{Isom}}_T(x,y)$. Because $\mathcal{F}$ is a stack, this presheaf is an fppf sheaf: it is precisely the Isom-sheaf whose sheaf property is the prestack half of the stack condition (definition 7.2.2). Thus

$$\underline{T} \times_{(\mathcal{F}\times\mathcal{F})} \mathcal{F} \;\simeq\; \underline{\mathrm{Isom}}_T(x,y)$$

as sheaves on $(\mathbf{Sch}/T)$.

Step 2: (1) ⇔ (2). By definition, $\Delta_{\mathcal{F}}$ is representable if and only if for every scheme T and every morphism $\underline{T} \to \mathcal{F}\times\mathcal{F}$ the fiber product $\underline{T} \times_{(\mathcal{F}\times\mathcal{F})} \mathcal{F}$ is a scheme (respectively an algebraic space). A morphism $\underline{T} \to \mathcal{F}\times\mathcal{F}$ is, by the universal property of the product, the same as a pair of objects $x, y \in \mathcal{F}(T)$. By Step 1 this fiber product is $\underline{\mathrm{Isom}}_T(x,y)$. Therefore $\Delta_{\mathcal{F}}$ is representable if and only if every $\underline{\mathrm{Isom}}_T(x,y)$ is representable by a scheme (respectively an algebraic space), which is precisely condition (2).

Step 3: Representability of morphisms from a scheme. Assume (1) holds and let $a\colon \underline{T_1} \to \mathcal{F}$ and $b\colon \underline{T_2} \to \mathcal{F}$ be two morphisms from schemes, corresponding by 2-Yoneda (theorem 7.1.8) to objects $x_1 \in \mathcal{F}(T_1)$ and $x_2 \in \mathcal{F}(T_2)$. Consider the 2-fiber product $\underline{T_1} \times_{\mathcal{F}} \underline{T_2}$. Its objects over a scheme U are triples $(U \to T_1,\ U \to T_2,\ \gamma)$ with γ an isomorphism in $\mathcal{F}(U)$ between the two pullbacks of x_1 and x_2. Writing $T = T_1 \times_S T_2$ and pulling x_1, x_2 back to objects $\tilde{x}_1, \tilde{x}_2 \in \mathcal{F}(T)$ along the two projections, a morphism $U \to T_1$ and a morphism $U \to T_2$ are the same as a single morphism $U \to T$, and the isomorphism γ is an element of $\underline{\mathrm{Isom}}_T(\tilde{x}_1, \tilde{x}_2)(U)$. Hence there is a Cartesian square

$$\underline{T_1} \times_{\mathcal{F}} \underline{T_2} \;\simeq\; \underline{\mathrm{Isom}}_T(\tilde{x}_1, \tilde{x}_2)$$

of sheaves on $(\mathbf{Sch}/T)$, and the right side is representable by a scheme (respectively an algebraic space) by (2). This is exactly the assertion that the 2-fiber product $\underline{T_1} \times_{\mathcal{F}} \underline{T_2}$ is a scheme (respectively an algebraic space) for every pair of morphisms from schemes. In particular, holding a fixed, every base change of $a\colon \underline{T_1} \to \mathcal{F}$ along a morphism $\underline{T_2} \to \mathcal{F}$ from a scheme is a scheme, which is precisely representability of a, completing the proof.

The equivalence is the technical heart of stack geometry, and its meaning is worth stating plainly. The diagonal of $\mathcal{F}$ is a single morphism, but it secretly is the whole family of Isom-sheaves $\underline{\mathrm{Isom}}_T(x,y)$ ranging over all schemes and all pairs of objects. Asking that the diagonal be representable is asking that isomorphisms between objects of $\mathcal{F}$ be parametrized by a scheme (or algebraic space): the automorphisms of $\mathcal{F}$ are themselves geometric. The second conclusion of the theorem is the payoff that makes atlases work. Once the diagonal is representable, any two maps from schemes to $\mathcal{F}$ have a schematic fiber product, so a scheme mapping to $\mathcal{F}$ is a representable morphism, and the notion of an atlas $\underline{U} \to \mathcal{F}$ acquires content: one can pull the atlas back against itself to get a scheme $\underline{U} \times_{\mathcal{F}} \underline{U}$, the “relation” that presents $\mathcal{F}$ as a quotient. This is why the definition of an algebraic stack, taken up in the next chapter, has exactly two clauses: the diagonal is representable, and there exists a smooth surjective representable morphism $\underline{U} \to \mathcal{F}$ from a scheme. The first clause is theorem 7.4.6; the second is the atlas of example 7.4.5. Representability of the diagonal is logically prior, because without it the atlas cannot even be tested for the properties (smooth, surjective) that its definition demands.

7.4.7 Note: The Two Diagonals and Separatedness The diagonal $\Delta_{\mathcal{F}}$ plays for stacks the role the scheme-theoretic diagonal plays for schemes, where the diagonal $\Delta_X\colon X \to X \times_S X$ being a closed immersion is separatedness. For a stack whose diagonal is representable, the adjectives of definition 7.4.3 apply to $\Delta_{\mathcal{F}}$: the stack $\mathcal{F}$ is called *separated* when $\Delta_{\mathcal{F}}$ is proper, *quasi-separated* when $\Delta_{\mathcal{F}}$ is quasi-compact, and the automorphism groups are unramified, that is, finite étale in the finite-type setting (a Deligne-Mumford condition met in the next chapter), exactly when $\Delta_{\mathcal{F}}$ is unramified. This is strictly stronger than finiteness of the automorphism groups: $B\mu_p$ in characteristic p has finite automorphism groups yet a ramified diagonal (example 7.4.8(4)), so it is not Deligne-Mumford. The single morphism $\Delta_{\mathcal{F}}$ thus simultaneously encodes the separation properties of $\mathcal{F}$ and the size of its automorphism groups, because both are properties of the Isom-sheaves it represents.

The theorem is now applied to the running stacks. In each case the diagonal is computed explicitly and its fibers are identified with stabilizer or automorphism groups, making concrete the claim that the diagonal records automorphisms.

Example 7.4.8: Diagonals of BG, $[X/G]$, $B\mathbb{G}_m$, and $B\mu_n$

Each computation identifies the diagonal Δ as a morphism of stacks and reads off its fibers.

1. *The classifying stack BG.* Let G be an affine group scheme over S. Over a scheme T the objects of BG are G-torsors, and for two torsors P, Q on T the sheaf $\underline{\mathrm{Isom}}_T(P,Q)$ is the sheaf of G-equivariant isomorphisms $P \to Q$. When $P = Q$ this is the sheaf of automorphisms of a torsor, which is representable: for the trivial torsor $P = G \times T$ one has $\underline{\mathrm{Aut}}(G \times T) \cong \underline{G}_T$, the group scheme G over T acting by right translation, and for a general P the automorphism sheaf is the inner form ${}^{P}G = P \times^{G} G$ (the twist of G by P under conjugation), again a scheme because G is affine and torsors are fppf-locally trivial. Hence $\underline{\mathrm{Isom}}_T(P,Q)$ is representable (it is a torsor under $\underline{\mathrm{Aut}}(P)$, empty or an $\underline{\mathrm{Aut}}(P)$-torsor according to whether P and Q are isomorphic), so by theorem 7.4.6 the diagonal $\Delta_{BG}\colon BG \to BG \times BG$ is representable. In fact the diagonal fiber over the point of $BG \times BG$ given by (P,P) is exactly $\underline{\mathrm{Aut}}(P)$; over the trivial torsor this is $\underline{G}$. One summarizes: the diagonal of BG "is" G. By contrast, the structure morphism $BG \to \underline{S}$ is representable only when G is trivial (example 7.4.2); the diagonal is representable for every affine G, which is the content that makes BG an algebraic stack while $BG \to \underline{S}$ remains non-representable.

2. *The quotient stack $[X/G]$.* Let G act on X. Over T an object of $[X/G]$ is a pair $(P \to T,\ f\colon P \to X)$ with P a G-torsor and f equivariant, and an isomorphism of two such objects is a torsor isomorphism commuting with the maps to X. Computing the diagonal $\Delta_{[X/G]}$ as in theorem 7.4.6, its fiber over a pair of objects is the sheaf of such compatible isomorphisms, which for the two objects coming from points $x_1, x_2 \in X(T)$ (that is, the trivial torsors with the equivariant maps determined by x_1, x_2) is the transporter
$$\mathrm{Transp}_G(x_1, x_2) = \{g \in G : g \cdot x_1 = x_2\}$$
a closed subscheme of $G \times T$ cut out by the equation $g \cdot x_1 = x_2$. In particular the diagonal is represented by the morphism
$$G \times X \longrightarrow X \times X, \qquad (g,x) \longmapsto (x,\ g \cdot x)$$
the "action-and-projection" morphism: its fiber over $(x_1, x_2) \in X \times X$ is exactly $\{g : g \cdot x_1 = x_2\}$, and over the diagonal (x,x) it is the *stabilizer* $\mathrm{Stab}_G(x) = \{g : g \cdot x = x\}$. Because

$G \times X \to X \times X$ is a morphism of schemes, it is representable, so theorem 7.4.6 makes $\Delta_{[X/G]}$ representable. The stabilizers are the automorphism groups of the points of $[X/G]$, exactly as expected from the moduli reading of a quotient stack. Setting $X = S$ recovers item (1): the action-and-projection morphism $G \times S \to S \times S$ collapses to $G \to S$ (both projections trivial), so Δ_{BG} is G.

3. *The classifying stack of line bundles $B\mathbb{G}_m$.* Take $G = \mathbb{G}_m$, so $B\mathbb{G}_m$ is the stack of line bundles: its fiber over T is the groupoid of invertible $\mathcal{O}_T$-modules and their isomorphisms. For a line bundle L on T the automorphism sheaf is $\underline{\mathrm{Aut}}(L) = \underline{\mathbb{G}_m}$, since an automorphism of an invertible module is multiplication by a unit, and this holds functorially in T: $\underline{\mathrm{Aut}}_T(L)(U) = \mathcal{O}_U(U)^\times = \mathbb{G}_m(U)$. Thus every point of $B\mathbb{G}_m$ has automorphism group $\mathbb{G}_m$, the diagonal fiber is $\underline{\mathbb{G}_m}$, and the diagonal
$$\Delta_{B\mathbb{G}_m} : B\mathbb{G}_m \longrightarrow B\mathbb{G}_m \times B\mathbb{G}_m$$
is representable, being $\mathbb{G}_m$ over each point. The Isom-sheaf $\underline{\mathrm{Isom}}_T(L, M)$ of two line bundles is the $\mathbb{G}_m$-torsor of trivializations of $L^{-1} \otimes M$, representable as the complement of the zero section in the total space of $L^{-1} \otimes M$; it is nonempty exactly when $L \cong M$. Every line bundle carries the scalar automorphisms $\mathbb{G}_m$, so no point of $B\mathbb{G}_m$ has trivial automorphisms, and $B\mathbb{G}_m \to \underline{S}$ is nowhere representable, consistently with item (1).

4. *The stack $B\mu_n$.* Take $G = \mu_n$, the group scheme of n-th roots of unity, $\mu_n = \mathrm{Spec}\big(\mathbb{Z}[t]/(t^n - 1)\big)$, an affine (finite, flat) group scheme over S. Then $B\mu_n$ is the stack of μ_n-torsors, and for a torsor P the automorphism sheaf is $\underline{\mathrm{Aut}}(P) \cong \underline{\mu_n}$ (finite flat of degree n), so the diagonal $\Delta_{B\mu_n}$ is represented by μ_n, a finite flat group scheme. Since μ_n is unramified over S exactly when n is invertible on S (equivalently μ_n is étale), the diagonal $\Delta_{B\mu_n}$ is unramified over such a base, so $B\mu_n$ has finite unramified automorphism groups and is a Deligne-Mumford stack there; when n is not invertible (for instance μ_p in characteristic p) the automorphisms are finite but not unramified, and $B\mu_n$ is an Artin stack that is not Deligne-Mumford. This is the smallest example in which the distinction between the two kinds of algebraic stack is visible on the diagonal alone.

In every case the fiber of the diagonal over a point is the automorphism group of that point: G for BG, the stabilizer $\mathrm{Stab}_G(x)$ for $[X/G]$, $\mathbb{G}_m$ for $B\mathbb{G}_m$, and μ_n for $B\mu_n$. The representability of the diagonal is the assertion that these automorphism groups vary in a scheme. This automorphism group at a point is what the next chapters will call the *residual gerbe* of the point, a copy of $B(\mathrm{Aut})$ sitting inside the stack over the point. The generic elliptic curve has automorphism group $\{\pm 1\}$, giving the residual gerbe $B\mu_2$; but at the two special j-invariants the automorphism group is larger (example 1.3.4). The curve $y^2 = x^3 + b$ with j-invariant 0 has the order-6 automorphism $(x, y) \mapsto (\zeta_3 x, -y)$, so $\mathrm{Aut} = \mathbb{Z}/6$ and the residual gerbe is $B(\mathbb{Z}/6)$; the curve $y^2 = x^3 + ax$ with j-invariant 1728 has the order-4 automorphism $(x, y) \mapsto (-x, iy)$, so $\mathrm{Aut} = \mathbb{Z}/4$ and the residual gerbe is $B(\mathbb{Z}/4)$. The gerbe structure is developed systematically when gerbes are introduced.

The four computations exhibit the mechanism promised at the start. A morphism of stacks is understood by its fibers over schemes; the fibers of the diagonal are the automorphism groups; and representability of the diagonal is the precise condition that these automorphism groups be geometric objects. The passage from BG (diagonal $= G$) to $[X/G]$ (diagonal = action-and-projection, fibers = stabilizers) to $B\mathbb{G}_m$ and $B\mu_n$ traces the full range of what the diagonal records, from a fixed group to a family of stabilizers to the arithmetic distinction between étale and non-étale automorphism schemes. With representable morphisms and the diagonal in hand, the geometry of an algebraic stack, smoothness, properness, dimension, can be defined by pulling back to an atlas, which is the task of the next chapter.

Exercise 7.4.1

1. Let $f\colon \mathcal{F} \to \mathcal{G}$ and $g\colon \mathcal{G} \to \mathcal{H}$ be morphisms of stacks. Show that if f and g are representable then so is the composite $g \circ f$, and that if $g \circ f$ is representable and the diagonal of $\mathcal{G}$ is representable then f is representable. (These are the "two out of three" stability properties that make representability behave like a class of geometric morphisms.)
2. Show that representability of morphisms is stable under base change: if $f\colon \mathcal{F} \to \mathcal{G}$ is representable and $\mathcal{G}' \to \mathcal{G}$ is any morphism of stacks, then the projection $\mathcal{F} \times_{\mathcal{G}} \mathcal{G}' \to \mathcal{G}'$ is representable.
3. Let G be a finite group (a constant finite group scheme) over a field k, acting on a k-scheme X. Compute the diagonal of $[X/G]$ explicitly as a morphism of schemes, and describe its fiber over a point $(x_1, x_2) \in X \times X$. When is $[X/G]$ a scheme? When does every point have trivial stabilizer, so that $[X/G]$ is representable by the ordinary quotient?
4. For the classifying stack $B\mathbb{G}_m$ over a field k, exhibit an atlas $\underline{U} \to B\mathbb{G}_m$ from a scheme U: find a representable, smooth, surjective morphism from a scheme. (Hint: the universal torsor $\mathrm{pt} \to B\mathbb{G}_m$ of definition 7.3.2, i.e. $\operatorname{Spec} k \to B\mathbb{G}_m$, classifies the trivial line bundle; verify it is a $\mathbb{G}_m$-torsor and hence smooth surjective. What is $\underline{U} \times_{B\mathbb{G}_m} \underline{U}$?)
5. Show that a stack $\mathcal{F}$ whose diagonal is representable and which admits an atlas $\pi\colon \underline{U} \to \mathcal{F}$ (a representable smooth surjective morphism from a scheme) has the property that $R = \underline{U} \times_{\mathcal{F}} \underline{U}$ is a scheme, and that the two projections $R \rightrightarrows \underline{U}$ are smooth. Interpret R as a groupoid in schemes presenting $\mathcal{F}$. (This is the "groupoid presentation" that the next chapter takes as an alternative definition of an algebraic stack.)
6. Determine for which base schemes S the diagonal of $B\mu_n$ is unramified, and deduce for which S the stack $B\mu_n$ is Deligne-Mumford. Contrast $B\mu_p$ with $B(\mathbb{Z}/p)$ (the constant group) in characteristic p: both have automorphism group of order p at every point, yet one is Deligne-Mumford and the other is not. Explain the discrepancy in terms of the diagonal.

8

Algebraic Spaces and Algebraic Stacks

Stacks solved the bookkeeping problem: a category fibered in groupoids that satisfies descent records objects with their automorphisms and glues them in a topology, and $\mathcal{M}_{1,1}$, BG, and $[X/G]$ are such objects. What stacks did not yet come with is geometry. A scheme carries a topology, a structure sheaf, a dimension, and a notion of smoothness; a bare stack carries none of these. The gap is closed by presenting a stack the way a scheme is presented by an affine cover: through an atlas, a surjection from a scheme that is well behaved enough to transport geometric notions from its source to the stack. This chapter carries out that construction and, with it, finally produces the geometric objects that represent the moduli problems the earlier Parts could not.

The construction is stratified by how singular the automorphisms are allowed to be. When a moduli problem has no automorphisms but still fails to be a scheme, the right object is an algebraic space, a sheaf that is étale-locally a scheme, and the first section builds these from étale equivalence relations. When the automorphisms are finite, the atlas can be taken étale and the object is a Deligne–Mumford stack; the second section defines these, characterizes them by the unramifiedness of their diagonal, and records the Keel–Mori theorem that extracts a coarse moduli space. When the automorphisms are positive dimensional, as for the stack of all coherent sheaves or of principal bundles, only a smooth atlas is available and the object is an Artin stack; the third section defines these and computes the dimension of a stack from a presentation.

The fourth section is the keystone of the book. Artin's criteria answer the question the whole subject has been circling: given a moduli functor, when is it an algebraic stack? The answer is a checklist, and its central entries are precisely the deformation theory of Part II. A moduli problem is algebraic when it is a limit-preserving stack whose deformations are governed by a tangent space and an obstruction space satisfying Schlessinger's conditions and whose formal deformations algebraize. The criteria convert the local, infinitesimal data computed in Part II into the global existence of a smooth atlas, and applying them shows that the moduli of curves and the Hilbert and Quot functors are algebraic. The final section imposes the remaining geometry, quasi-coherent sheaves, the Picard group, cohomology, and the properties smooth, separated, and proper, on an algebraic stack through its atlas and its diagonal, closing with the fact that $\overline{\mathcal{M}}_g$ is smooth and proper as a stack even where its coarse space is singular, the statement the final chapter will prove.

8.1 Algebraic Spaces

The previous two chapters supplied two closely related repairs for the failure of a moduli functor to be a scheme. Descent theory recognized that the objects of a moduli problem glue in the étale or fppf topology, not the Zariski one (theorem 6.4.3), and the theory of fibered categories (chapter 7) built the categorical bookkeeping that keeps track of objects together with their automorphisms. This chapter turns those repairs into geometry. The first and gentlest step is to enlarge the category of schemes just far enough to include every quotient that ought to exist but does not: an *algebraic space* is a sheaf that looks like a scheme in the étale topology, and nothing more. It sits strictly between schemes and Deligne-Mumford stacks, and it is the object that finally represents moduli problems whose objects have no automorphisms but whose parameter space cannot be glued from affines in the Zariski topology.

The guiding image is the one that already governed Galois descent and the quotient stack. A scheme U mapping to an object X by an étale surjection presents X as a quotient, and the fibers of $U \times_X U \rightrightarrows U$ record which points of U have been identified. When X is itself a scheme, this is nothing new. The definition of an algebraic space is obtained by inverting the arrow of logic: rather than starting with X and forming the presentation, one starts with the presentation, an étale equivalence relation $R \rightrightarrows U$ on schemes, and *defines* X to be its quotient sheaf. The gain is that such a quotient sheaf need not be a scheme, and the objects that arise this way are exactly what is needed to run the moduli program one rung below the stacks of the coming sections.

Throughout, the base site is $\mathcal{C} = (\mathbf{Sch}/S)_{\text{et}}$, the category of S-schemes with the étale topology, and S is a fixed base scheme (often $\operatorname{Spec} k$ or a Noetherian scheme). A sheaf means an étale sheaf of sets on $\mathcal{C}$; a scheme U is identified with the sheaf it represents, $T \mapsto \operatorname{Hom}_S(T, U)$, which is an étale sheaf because étale descent is effective for morphisms to a scheme (theorem 6.4.3). All spaces below are quasi-separated unless noted.

The starting datum is an equivalence relation on a scheme, realized not set-theoretically but by an actual subscheme of the product whose two projections are étale. The étale hypothesis is what makes the quotient behave like a scheme locally; without it one would land in a wilder category still. The definition below packages the relation, its two structure maps, and the equivalence axioms into a single object.

Definition 8.1.1: Étale Equivalence Relation

Let U be an S-scheme. An *étale equivalence relation* on U is a scheme R together with a morphism

$$j = (s,t)\colon R \longrightarrow U \times_S U$$

such that:

1. j is a monomorphism (so R is, functorially, a relation: for every S-scheme T, the image of $R(T) \hookrightarrow U(T) \times U(T)$ is a subset);
2. that subset $R(T) \subseteq U(T) \times U(T)$ is an equivalence relation on $U(T)$, functorially in T (reflexive, symmetric, and transitive);
3. both projections $s = \mathrm{pr}_1 \circ j$ and $t = \mathrm{pr}_2 \circ j$ from R to U are étale morphisms in the sense of [12, Étale Morphisms].

The two maps are written as a parallel pair $s, t\colon R \rightrightarrows U$, the *source* and *target*. When the context demands the ambient monomorphism, the relation is written $R \hookrightarrow U \times_S U$.

The three conditions unwind to a picture that recurs throughout the chapter. Condition (1) says R is a subfunctor of $U \times_S U$, so a T-point of R is literally a pair of T-points of U that are declared equivalent; there is at most one way to be equivalent. Condition (2) is the reflexive, symmetric, transitive package, and it is what distinguishes an equivalence relation from a more general groupoid: reflexivity gives a diagonal section $e\colon U \to R$ with $j \circ e = \Delta_U$, symmetry an involution $\iota\colon R \to R$ swapping s and t, and transitivity a composition $R \times_{s,U,t} R \to R$. Condition (3), the étale hypothesis on both projections, is the geometric heart: the two ways of viewing R as living over U are both local isomorphisms, so passing between the "upstairs" scheme U and its quotient neither creates nor destroys infinitesimal information. It is exactly this that will let the quotient inherit the local structure of a scheme.

The elementary source of étale equivalence relations, and the one that shows the definition subsumes the classical case, is a free action of a finite (or étale) group scheme.

Example 8.1.2: Quotient by a Free Finite Group Action

Let a finite group G act freely on an S-scheme U, with the action morphism $a\colon G \times_S U \to U$, $(g,u) \mapsto g \cdot u$. Take $R = G \times_S U$ and

$$j = (s,t) = (\mathrm{pr}_U,\ a)\colon G \times_S U \longrightarrow U \times_S U$$

sending (g,u) to $(u, g\cdot u)$. On T-points, $R(T) = G \times U(T)$ maps to the pairs $(u, g \cdot u)$, whose image is the relation "$u \sim u'$ iff $u' = g \cdot u$ for some g": the orbit equivalence relation. It is reflexive ($g = e$), symmetric ($g \mapsto g^{-1}$), and transitive (multiply the group elements), so condition (2) holds. Freeness of the action is precisely the statement that j is a monomorphism: if $(u, g\cdot u) = (u, g' \cdot u)$ then $g \cdot u = g' \cdot u$ forces $g = g'$ because the stabilizer of u is trivial, so a T-point of the image determines (g,u) uniquely, giving condition (1). Finally G finite over S makes $G \times_S U \to U$ (the first projection) finite étale, so $s = \mathrm{pr}_U$ is finite étale. For $t = a$, introduce the shear isomorphism

$$\varphi\colon G \times_S U \xrightarrow{\ \sim\ } G \times_S U, \qquad (g,u) \longmapsto (g,\ g \cdot u)$$

whose inverse is $(g,u) \mapsto (g, g^{-1} \cdot u)$; then $t = a$ factors as $\mathrm{pr}_U \circ \varphi$, the composite of an isomorphism with the finite étale projection pr_U, so t is finite étale as well. Thus condition (3) holds and

$R = G \times_S U \rightrightarrows U$ is an étale equivalence relation. Its quotient, constructed below, is the sheaf U/G.

The quotient of a scheme by such a relation is defined by a universal property in the category of étale sheaves, exactly as one would quotient a set by an equivalence relation. When the resulting sheaf happens to be a scheme (as it does, for instance, when U is quasi-projective and G acts freely) nothing new has been built. The point of the theory is that in general it is not, and the objects one gains deserve a name.

Definition 8.1.3: Algebraic Space

An *algebraic space* over S is an étale sheaf $X \colon \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ on $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$ satisfying:

1. *(representable diagonal)* the diagonal morphism $\Delta_X \colon X \to X \times_S X$ is representable by schemes: for every S-scheme T and every morphism $T \to X \times_S X$, the fiber product $T \times_{X \times_S X} X$ is (representable by) a scheme;
2. *(étale atlas)* there is a scheme U and a morphism $p \colon U \to X$ that is surjective and étale (both properties are representable and hold after every base change $T \to X$, using (1)).

The morphism $p \colon U \to X$ is called an *étale atlas* or *étale presentation* of X.

Two features of this definition need immediate comment, since together they are the whole content. Condition (1), representability of the diagonal, is what makes the words "surjective" and "étale" in condition (2) meaningful: a property of a morphism $U \to X$ to a sheaf is defined by requiring it after base change to every scheme mapping to X, and the diagonal being representable is exactly what guarantees those base changes are schemes, so the property can be tested there. Condition (2) then asserts that X is étale-locally a scheme. The diagonal condition and the atlas condition are the two halves of algebraicity that will reappear, verbatim in spirit, in the definitions of Deligne-Mumford and Artin stacks in the next two sections, where a stack replaces a sheaf and "étale" is weakened to "smooth".

The definition by an atlas and the definition by a relation are two descriptions of the same objects, and moving between them is the basic fluency the chapter requires. Given an atlas $p \colon U \to X$, the fiber product

$$R = U \times_X U$$

carries its two projections $s, t \colon R \rightrightarrows U$, and because p is representable étale these projections are étale; the monomorphism $R \hookrightarrow U \times_S U$ (which lands in $U \times_S U$ because a point of R is a pair of points of U with the same image in X) is an equivalence relation because "$p(u) = p(u')$" is one. Thus every algebraic space with a chosen atlas produces an étale equivalence relation. Conversely, the quotient sheaf reconstructs the space.

8.1.4 Note: Algebraic Spaces as Quotients U/R Let $R \rightrightarrows U$ be an étale equivalence relation (definition 8.1.1). Its *quotient sheaf* U/R is the sheafification, in the étale topology, of the presheaf $T \mapsto U(T)/R(T)$ (orbits of T-points under the equivalence relation). Then U/R is an algebraic space, with $U \to U/R$ an étale atlas and $R = U \times_{U/R} U$; and every algebraic space arises this way from any of its atlases. Thus

$$\{\text{algebraic spaces}\} \longleftrightarrow \{\text{étale equivalence relations } R \rightrightarrows U\}/(\text{refinement of atlas})$$

is the working dictionary. The equivalence "étale-sheaf quotient of an étale equivalence relation = étale sheaf with representable diagonal and an étale atlas" is one of the foundational structural theorems; the direction that U/R has a representable diagonal (so that $U \to U/R$ is a bona fide étale surjection of the kind in definition 8.1.3) requires the machinery of [20] and is recorded in [29] (Tag 02WW and following). The easy direction, that an algebraic space with an atlas yields the relation $R = U \times_X U$, is the computation above.

The dictionary makes the first nontrivial fact free of charge: a scheme is an algebraic space, in the most economical way possible. This is the sanity check that the enlargement really is an enlargement and loses no old objects.

The claim that a scheme is an algebraic space is the base of every comparison that follows, and it is proved by exhibiting the identity as an atlas. The equivalence relation is then the trivial one, in which nothing is identified.

Theorem 8.1.5: Basic Properties of Algebraic Spaces

Let S be a base scheme and work over $\mathcal{C} = (\mathbf{Sch}/S)_{\text{et}}$. Then:

1. Every scheme is an algebraic space, presented by the identity atlas $U \xrightarrow{\text{id}} U$, whose associated étale equivalence relation is the diagonal $\Delta_U : U \hookrightarrow U \times_S U$.
2. The diagonal $\Delta_X : X \to X \times_S X$ of an algebraic space is representable, and X is quasi-separated (resp. separated) iff Δ_X is quasi-compact (resp. a closed immersion).
3. Algebraic spaces are closed under fiber products: if $X \to Z$ and $Y \to Z$ are morphisms of algebraic spaces, then $X \times_Z Y$ is an algebraic space.
4. A morphism $T \to X$ from a scheme T to an algebraic space X is representable: for every scheme T' and morphism $T' \to X$, the fiber product $T \times_X T'$ is a scheme.
5. Every quasi-separated algebraic space X contains a scheme as a dense open subspace; that is, there is a dense open $V \subseteq X$ which is representable by a scheme.

***Proof*:**
Parts (1) to (4) are short and are proved here; part (5) is one of the structural theorems whose proof is cited.

Part (1): a scheme is an algebraic space. Let U be an S-scheme, viewed as the sheaf $T \mapsto \mathrm{Hom}_S(T, U)$, which is an étale sheaf by effectivity of étale descent for morphisms to a scheme (theorem 6.4.3). Its diagonal $\Delta_U : U \to U \times_S U$ is a morphism of schemes, hence representable, and for a scheme mapping to $U \times_S U$ the fiber product $T \times_{U\times_S U} U$ is a scheme (it is the locus in T where the two maps $T \to U$ agree), so condition (1) of definition 8.1.3 holds. The identity $\mathrm{id}\colon U \to U$ is an isomorphism, hence surjective and étale, giving an atlas and condition (2). The associated relation is $R = U \times_U U = U$ with $j = \Delta_U$, the trivial equivalence relation identifying nothing. Thus U is an algebraic space.

Part (2): the diagonal. Representability of Δ_X is condition (1) of definition 8.1.3, so there is nothing to prove for the first clause. For the separation clauses, recall that quasi-compactness and being a closed immersion are representable properties of morphisms, so they may be tested after the base change of Δ_X along any scheme mapping to $X \times_S X$; a quasi-separated (resp. separated) algebraic space is defined precisely by requiring the representable morphism Δ_X to be quasi-compact (resp. a closed immersion), which is the stated equivalence.

Part (3): fiber products. Write $W := X \times_Z Y$ for the sheaf fiber product, which is an étale sheaf because sheaves are closed under limits. Two things must be checked: that Δ_W is representable, and that W has an étale atlas.

For the diagonal, fix a scheme T and a morphism $T \to W \times_S W$; the task is to show the base change $T \times_{W\times_S W} W$ (the fiber of Δ_W over T) is a scheme. The projection $W = X \times_Z Y \to X \times_S Y$ identifies $W \times_S W$ with a subsheaf of $(X \times_S X) \times_S (Y \times_S Y)$, and under this identification Δ_W is the restriction to W of the product $\Delta_X \times_S \Delta_Y \colon X \times_S Y \to (X \times_S X) \times_S (Y \times_S Y)$. Hence the base change of Δ_W along $T \to W \times_S W$ fits in a cartesian square with the base change of $\Delta_X \times_S \Delta_Y$ along the composite $T \to W \times_S W \to (X \times_S X) \times_S (Y \times_S Y)$:

$$T \times_{W\times_S W} W = T \times_{(X\times_S X)\times_S(Y\times_S Y)} (X \times_S Y)$$

The right-hand side is obtained by base-changing Δ_X (a scheme by (2)) along $T \to X \times_S X$ and then Δ_Y (a scheme by (2)) along the resulting scheme mapping to $Y \times_S Y$; each step yields a scheme because Δ_X and Δ_Y are representable. Thus $T \times_{W\times_S W} W$ is a scheme for every T, so Δ_W is representable.

For the atlas, choose étale atlases $a\colon U \to X$ and $b\colon V \to Y$. Pulling the sheaf $W = X \times_Z Y$ back along a on the X-side and along b on the Y-side replaces both X and Y by their atlas schemes, giving the sheaf

$$P = U \times_X (X \times_Z Y) \times_Y V = U \times_Z V,$$

the fiber product of the two scheme morphisms $U \to Z$ and $V \to Z$ over the algebraic space Z. This P is a scheme: since $U \to Z$ is a morphism from a scheme it is representable by (4), so its base change $U \times_Z V \to V$ along the scheme V is a scheme, whence $P = U \times_Z V$ is a scheme. The projection $P = U \times_Z V \to X \times_Z Y = W$ is the fiber product over Z of the two atlas maps a and b; it factors as the composite

$$U \times_Z V \xrightarrow{a\times_Z \mathrm{id}} X \times_Z V \xrightarrow{\mathrm{id}\times_Z b} X \times_Z Y = W,$$

each factor being a base change of an étale surjection (a, resp. b) along a morphism to X, resp. Y, hence itself étale and surjective; the composite is therefore étale and surjective. Thus $P \to W$ is an étale atlas by a scheme, and $W = X \times_Z Y$ has representable diagonal and an étale atlas, so it is an algebraic space, completing the proof of this part.

Part (4): a map from a scheme is representable. Let $T \to X$ be a morphism from a scheme, and let $T' \to X$ be any morphism from a scheme. Then $T \times_X T'$ is the fiber product of

$(T, T') \to X \times_S X$ against the diagonal Δ_X:

$$T \times_X T' = (T \times_S T') \times_{X \times_S X} X$$

and the right-hand side is a scheme because Δ_X is representable by condition (1) of definition 8.1.3 and $T \times_S T'$ is a scheme. Thus every map from a scheme is representable, which is what was to be shown.

Part (5): the dense open subscheme. Let X be a quasi-separated algebraic space with étale atlas $p\colon U \to X$. Over the generic points of U the étale map p is, after shrinking, an isomorphism onto its image, so X contains a scheme-theoretic dense open; the full statement (that a quasi-separated algebraic space has a dense open subscheme, and even generic separatedness) is one of the structural theorems whose proof uses Noetherian induction and the theory of étale-local structure of spaces. It is beyond the elementary framework here and is cited to [20] and [29] (Tag 06NH). This completes the proof of the parts proved in full and cites the remainder.

The four proved parts already organize the theory. Part (1) confirms that schemes sit inside algebraic spaces, part (2) records that the diagonal is the seat of separatedness, part (3) closes the category under the operation moduli theory uses constantly, and part (4) is what makes the notion workable: it says a scheme mapping to an algebraic space behaves like a scheme mapping to a scheme, so all the usual notions (open, closed, étale, smooth, flat, proper) transport to morphisms of algebraic spaces by testing on schemes. This last point is what makes algebraic spaces usable rather than only definable. Part (5), the dense open subscheme, is the precise sense in which an algebraic space is a scheme with only a small "non-scheme" locus: the pathology is always concentrated away from a dense open, so an algebraic space is birationally a scheme.

The representability of the diagonal deserves a second look, because it is the exact place where the stack theory of chapter 7 feeds in. There, a morphism of stacks was called representable when its base change to any scheme is a scheme (or algebraic space), and the representability of the diagonal (theorem 7.4.6) was identified as the condition encoding the Isom sheaves. For an algebraic space the diagonal is representable by schemes rather than by algebraic spaces, and the Isom sheaf of two objects is a subscheme of a product; this is the degenerate case of the stack story in which the automorphism sheaves are trivial. An algebraic space is thus exactly a stack whose objects have no nontrivial automorphisms, presented by an étale rather than a smooth atlas. The sections to come loosen each of these two constraints in turn.

The theory would be idle if every algebraic space were secretly a scheme. The example that follows is the one that justifies the whole enlargement: a free étale equivalence relation whose quotient is a genuine algebraic space and not a scheme, because the identification cannot be carried out inside any single affine chart.

Example 8.1.6: A Free Involution Quotient That Is Not a Scheme

Work over $S = \operatorname{Spec} k$ with k algebraically closed of characteristic $\neq 2$. A free finite group action on a *quasi-projective* scheme has a scheme quotient, because every orbit then lies in a common affine open (the exercises establish this). To produce a non-scheme algebraic space the atlas must therefore be non-quasi-projective, and the canonical source of such atlases is Hironaka's example of a smooth complete non-projective threefold.

Hironaka constructed a smooth complete (proper) threefold U over k, non-projective, carrying a free involution $\sigma\colon U \to U$ (an action of $\mathbb{Z}/2$ with no fixed points) whose orbit through a certain point is not contained in any affine open of U. Because σ is a free action of the finite étale group

$\mathbb{Z}/2$, it defines an étale equivalence relation $R = U \times (\mathbb{Z}/2) \rightrightarrows U$ in the sense of example 8.1.2: both projections $s, t\colon R \to U$ are finite étale (the constant group $\mathbb{Z}/2$ is finite étale over k in every characteristic, being $\operatorname{Spec}(k \times k)$; the hypothesis $\operatorname{char} k \neq 2$ is what makes the involution act freely), and freeness makes $(s,t)\colon R \hookrightarrow U \times_k U$ a monomorphism. The quotient sheaf

$$X = U/R = U/(\mathbb{Z}/2)$$

is therefore an algebraic space (definition 8.1.3), with étale atlas $U \to X$. It is a smooth proper algebraic space of dimension 3.

Why X is not a scheme. Suppose X were a scheme. Then the image $* \in X$ of the distinguished orbit would have an affine open neighborhood $\operatorname{Spec} A \subseteq X$, and its preimage $W \subseteq U$ under the étale double cover $U \to X$ would be a $\mathbb{Z}/2$-invariant open with $\operatorname{Spec} A = W/(\mathbb{Z}/2)$; since $\operatorname{Spec} A$ is affine and $\mathbb{Z}/2$ finite, the quotient map $W \to \operatorname{Spec} A$ would be a finite (hence affine) morphism, forcing W to be affine (an affine morphism over an affine scheme has affine total space), and W would contain the whole orbit of the distinguished point in a single affine open. This contradicts Hironaka's construction, in which that orbit lies in no affine open of U. Hence X has no affine open neighborhood of $*$ and is not a scheme. The verification that Hironaka's threefold has these properties, and that the resulting X is an algebraic space but not a scheme, is a standard structural computation carried out in [20] and [29] (the non-scheme algebraic space, Tag 02Z1 and the Hironaka example); its full details use resolution-of-singularities-era birational geometry beyond the present scope, so the construction of U itself is cited.

The obstruction is the one advertised in the outline of the theory: the distinguished orbit fails to be contained in a single affine open of the atlas, so the construction of a quotient scheme by patching affine invariant-ring quotients breaks down there, while the étale-sheaf quotient exists without difficulty. The involution is free and étale, so no stacky automorphism appears; X has trivial automorphism sheaves and is a bona fide space, one rung below the stacks of the next sections.

The example is worth holding onto because it isolates the single phenomenon that separates algebraic spaces from schemes. A scheme is glued from affines in the Zariski topology, and the gluing of a quotient by a group action requires each orbit to fit inside a stable affine chart. When some orbit refuses to fit, the Zariski gluing fails but the étale gluing does not, and the resulting object is an algebraic space with no scheme structure. Every non-scheme algebraic space is, at bottom, a variation on this theme: a free étale relation whose orbits are not uniformly affine.

This is exactly the shape of the failure encountered at the very start of the book, and naming the connection closes a thread carried since its beginning. There, $\mathcal{M}_{1,1}$ failed to be a fine moduli scheme because elliptic curves carry the automorphism $[-1]$, and the quotient presentation $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ (example 1.3.4 recorded the automorphisms; theorem 1.3.6 the j-line) exhibited it as a quotient. The distinction is that $\mathcal{M}_{1,1}$ has nontrivial automorphisms, so its correct home is a stack rather than a space. But the geometric mechanism, that a quotient by an étale (or smooth) presentation need not descend to a scheme, is identical: an algebraic space is the automorphism-free instance of that mechanism, a Deligne-Mumford stack the finite-automorphism instance.

Exercise 8.1.1

1. Let $R \rightrightarrows U$ be an étale equivalence relation with quotient sheaf $X = U/R$. Using theorem 8.1.5(4) and the description $R = U \times_X U$, prove that the source and target maps $s, t\colon R \to U$ are étale directly from the representability of the atlas $U \to X$, without invoking

the definition of R. (Base change $U \to X$ along $U \to X$.)

2. Show that a morphism $f: X \to Y$ of algebraic spaces has representable diagonal, i.e. $\Delta_f: X \to X \times_Y X$ is representable by schemes. Deduce that for any scheme T and morphism $T \to Y$, the fiber product $X \times_Y T$ is an algebraic space. (Use theorem 8.1.5(2),(3),(4).)

3. Consider the $\mathbb{Z}/2$-action $x \mapsto -x$ on $U = \mathbb{A}^1 = \operatorname{Spec} k[x]$ ($\operatorname{char} k \neq 2$), and let $R' \hookrightarrow \mathbb{A}^1 \times_k \mathbb{A}^1$ be the *closed* subscheme $\{(y-x)(y+x) = 0\}$, the full graph of the action. Show that R' is an equivalence relation but is *not* étale (the origin is a fixed point, so the projections ramify there), and that the quotient is nevertheless the scheme $\operatorname{Spec} k[x^2] \cong \mathbb{A}^1$. Explain in one sentence why this shows that a *non*-étale relation can still have a scheme quotient, so the étale hypothesis in definition 8.1.1 is a hypothesis about the presentation, not about the quotient.

4. Prove that an algebraic space X is a scheme if and only if it admits an étale atlas $U \to X$ that is a *Zariski* covering after a suitable choice, more precisely: X is a scheme iff there is an open subspace covering $\{X_i \hookrightarrow X\}$ by algebraic spaces each of which is a scheme. (One direction is trivial; for the other, glue the schemes X_i along $X_i \times_X X_j$, using that these fiber products are schemes by theorem 8.1.5(4).)

5. Let G be a finite group acting freely on a quasi-projective scheme U over k. Show that the quotient sheaf U/G is representable by a scheme. Explain why this is consistent with example 8.1.6: identify the single hypothesis of this exercise that Hironaka's atlas violates, and confirm that it is exactly the property whose failure lets the orbit escape every affine open. (Recall that on a quasi-projective scheme every finite set of points lies in a common affine open; apply this to each orbit.)

6. Explain the sense in which theorem 8.1.5(5) says an algebraic space is "birationally a scheme," and reconcile this with example 8.1.6: describe a dense open subscheme $V \subseteq X$ of Hironaka's quotient $X = U/(\mathbb{Z}/2)$ (take the image of a $\mathbb{Z}/2$-invariant affine open of U on which the action is free with a scheme quotient), and explain why the non-scheme locus is contained in the nowhere-dense closed complement.

8.2 Deligne-Mumford Stacks

The previous section built the first geometric objects beyond schemes: algebraic spaces, the étale-sheaf quotients of étale equivalence relations (definition 8.1.3). An algebraic space is a sheaf that is étale-locally a scheme, and it repairs the failures of representability that stem from gluing, not from automorphisms. The moduli problems this book exists to serve have a second, deeper source of non-representability: their objects carry automorphisms, and no sheaf, algebraic space or otherwise, can record them. Chapter 7 built the categorical object that does record them, the stack, and produced a supply of them as quotient stacks $[X/G]$ (definition 7.3.1) and classifying stacks BG (definition 7.3.2). This chapter imposes geometry on stacks. The present section treats the case where the automorphisms are as small as they can be while still being present: finite. A stack whose objects have finite, reduced automorphism groups admits an *étale* atlas, and this is exactly the class of *Deligne-Mumford stacks.* They are the stacks closest to schemes, and they are the stacks the moduli of curves lives in.

The organizing principle is a dictionary between the automorphisms of the objects and the geometry of the diagonal. A stack $\mathcal{X}$ has a diagonal $\Delta_{\mathcal{X}}: \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ whose fibers are the $\underline{\operatorname{Isom}}$ sheaves of pairs of objects, and whose "fiber over the diagonal point" at an object x is the automorphism group scheme $\underline{\mathrm{Aut}}(x)$. Representability of the diagonal (Chapter 7, theorem 7.4.6) says these $\underline{\mathrm{Isom}}$ sheaves are algebraic spaces, which is half of what it means for a stack to be geometric. The other half is the atlas. The theorem of this section is that the atlas can be taken étale, rather than only smooth, exactly when the diagonal is unramified, which happens exactly when every automorphism group scheme is finite and reduced. The three conditions, an étale atlas, unramified diagonal, finite reduced automorphisms, are one condition wearing three faces, and Deligne-Mumford stacks are the stacks satisfying it.

Definition 8.2.1: Deligne-Mumford Stack

Let S be a base scheme and $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$ the big étale site. A stack $\mathcal{X}$ over $\mathcal{C}$ is a *Deligne-Mumford stack* (a *DM stack*) if:

1. the diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is representable (by algebraic spaces), quasi-compact, and separated; and
2. there exist a scheme U and a morphism $u \colon U \to \mathcal{X}$ that is *representable*, *étale*, and *surjective*.

Such a $u \colon U \to \mathcal{X}$ is called an *étale atlas* (or an *étale presentation*) of $\mathcal{X}$. Given an atlas, the fiber product $R = U \times_{\mathcal{X}} U$ is an algebraic space, both projections $s, t \colon R \to U$ are étale (being base changes of u), and the induced groupoid

$$R = U \times_{\mathcal{X}} U \rightrightarrows U$$

is an étale groupoid presenting $\mathcal{X}$ as its stack quotient $\mathcal{X} \simeq [U/R]$. When R is a scheme, this exhibits $\mathcal{X}$ as $[U/R]$ for an étale equivalence relation in the sense of definition 8.1.1, except that $R \hookrightarrow U \times_S U$ need not be a monomorphism (the groupoid $R \rightrightarrows U$ records automorphisms, so (s,t) can have positive-dimensional fibers only if automorphisms are positive dimensional, which the DM condition will forbid).

The definition is the stack-level analogue of the algebraic-space definition of the previous section, with two changes. First, the atlas is a map to a stack, so the requirement that it be *representable* (Chapter 7, definition 7.4.1) is not automatic and must be imposed: representability of u says that for every scheme T and every map $T \to \mathcal{X}$, the fiber product $U \times_{\mathcal{X}} T$ is a scheme (or algebraic space), which is what makes it meaningful to ask that u be étale, since "étale" is a property of morphisms of schemes and algebraic spaces. Second, the equivalence relation $R \rightrightarrows U$ is replaced by a groupoid, because two points of U over the same object of $\mathcal{X}$ can be identified in more than one way, once for each isomorphism between the objects they represent. The groupoid $R \rightrightarrows U$ is the geometric incarnation of the $\underline{\mathrm{Isom}}$ sheaves: $R = U \times_{\mathcal{X}} U$ parametrizes triples (a, b, φ) with $a, b \in U$ mapping to isomorphic objects and φ an isomorphism between them, so the fiber of $(s,t) \colon R \to U \times_S U$ over a diagonal point (a, a) is precisely $\underline{\mathrm{Aut}}$ of the object $u(a)$.

The requirement that the diagonal be representable is not redundant given the atlas; rather, the two together are what "geometric" means for a stack. A useful reformulation, which the following remark records for use throughout, is that representability of the diagonal is equivalent to the algebraicity of all $\underline{\mathrm{Isom}}$ sheaves.

8.2.2 Note: The Diagonal and the Isom Sheaves For a stack $\mathcal{X}$ over $\mathcal{C}$, the diagonal $\Delta_{\mathcal{X}}$ is representable by algebraic spaces if and only if, for every scheme T and every pair of objects $x, y \in \mathcal{X}(T)$, the sheaf

$$\underline{\mathrm{Isom}}_T(x, y) \colon (T' \to T) \longmapsto \mathrm{Isom}_{\mathcal{X}(T')}(x|_{T'}, y|_{T'})$$

on $\mathcal{C}/T$ is an algebraic space over T (Chapter 7, theorem 7.4.6). Taking $y = x$ gives the automorphism group algebraic space $\underline{\mathrm{Aut}}_T(x) = \underline{\mathrm{Isom}}_T(x, x)$, which is a group object over T; taking $T = \operatorname{Spec} K$ for a field gives the automorphism group scheme $\underline{\mathrm{Aut}}(x)$ of a single object. The diagonal being quasi-compact and separated translates into these $\underline{\mathrm{Isom}}$ sheaves being quasi-compact and separated over the base, which for the automorphism groups means, in particular, that they are separated group schemes.

The heart of the section is the equivalence that gives the class its name. Deligne and Mumford isolated the DM stacks in their proof of the irreducibility of the moduli of curves precisely because these are the stacks one can study with the étale-local methods of schemes, and the criterion for recognizing them is intrinsic: it reads off the automorphism groups, which are exactly the data a moduli problem hands to the geometer. The theorem below is the bridge between the two ways of looking at a stack, from above (an atlas one can take étale) and from within (objects whose automorphisms are finite).

Theorem 8.2.3: DM Stacks and the Unramified Diagonal

Let $\mathcal{X}$ be a stack over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$ whose diagonal $\Delta_{\mathcal{X}}$ is representable, quasi-compact, and separated, and which admits a *smooth* representable surjection $V \to \mathcal{X}$ from a scheme (so $\mathcal{X}$ is an algebraic stack in the sense of the next section). The following are equivalent:

1. $\mathcal{X}$ is a Deligne-Mumford stack; that is, $\mathcal{X}$ admits an *étale* atlas $U \to \mathcal{X}$.
2. The diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is *unramified*.
3. For every scheme T and every object $x \in \mathcal{X}(T)$, the automorphism group algebraic space $\underline{\mathrm{Aut}}_T(x) \to T$ is *unramified*; equivalently, for every field K and every $x \in \mathcal{X}(K)$, the automorphism group scheme $\underline{\mathrm{Aut}}(x)$ is *finite and reduced* over K.

When S lies over $\mathbb{Q}$ (or more generally when the relevant automorphism groups have order prime to the residue characteristics), the reducedness in (3) is automatic, and the condition is simply that every $\underline{\mathrm{Aut}}(x)$ is *finite*.

***Proof*:**
The argument threads three reformulations of a single geometric fact: a morphism to a stack can be taken étale exactly when the smooth atlas one already has fails to be étale only along the automorphisms, and those automorphisms are the fibers of the diagonal.

Step 1: (2) ⇔ (3), the diagonal and the automorphisms. The diagonal $\Delta_{\mathcal{X}}$ is a representable morphism, so "unramified" is defined by base change: $\Delta_{\mathcal{X}}$ is unramified if and only if for every scheme T and every map $T \to \mathcal{X} \times_S \mathcal{X}$, the base change $T \times_{\mathcal{X} \times_S \mathcal{X}} \mathcal{X} \to T$ is an unramified morphism of algebraic spaces (Chapter 7, definition 7.4.3). A map $T \to \mathcal{X} \times_S \mathcal{X}$ is a pair of objects $x, y \in \mathcal{X}(T)$, and the base change is exactly the $\underline{\mathrm{Isom}}$ sheaf: $T \times_{\mathcal{X} \times_S \mathcal{X}} \mathcal{X} = \underline{\mathrm{Isom}}_T(x, y)$

by theorem 7.4.6. Hence $\Delta_{\mathcal{X}}$ is unramified if and only if every $\underline{\mathrm{Isom}}_T(x,y) \to T$ is unramified. Since unramifiedness is preserved and detected under base change, and the automorphism groups $\underline{\mathrm{Aut}}_T(x) = \underline{\mathrm{Isom}}_T(x,x)$ are the restrictions to the diagonal $x = y$, it is enough to test on the automorphism groups: the general $\underline{\mathrm{Isom}}_T(x,y)$ is either empty or, after passing to a cover trivializing the isomorphism, a torsor under $\underline{\mathrm{Aut}}_T(x)$, and a torsor under an unramified group space is unramified. This gives (2) $\Leftrightarrow$ (3) in the form "every $\underline{\mathrm{Aut}}_T(x)$ is unramified over T."

For the pointwise reformulation, a group scheme G of finite type over a field K is unramified over K if and only if $\Omega_{G/K} = 0$, which by translation invariance holds if and only if the cotangent space at the identity vanishes, that is, if and only if G is étale over K. An étale K-group scheme of finite type is finite (over a field, étale plus finite type, hence quasi-compact, forces finite, since it becomes a finite disjoint union of copies of $\operatorname{Spec} \bar{K}$ after base change) and reduced (étale schemes over a field are reduced). Conversely a finite reduced group scheme over K is a finite étale K-scheme, hence unramified. When $\operatorname{char} K = 0$, Cartier's theorem makes every finite group scheme automatically reduced; when $|G|$ is prime to $\operatorname{char} K$, reducedness follows from the separate (and easier) fact that the order is invertible, so the group scheme is étale by order count. Either way "finite" alone suffices. This establishes the equivalence of (3)'s two formulations and its equivalence to (2).

Step 2: (1) $\Rightarrow$ (2), an étale atlas forces an unramified diagonal. Suppose $u \colon U \to \mathcal{X}$ is an étale atlas. Form the fiber product $R = U \times_{\mathcal{X}} U$; by representability of u and of the diagonal, R is an algebraic space, and both projections $s, t \colon R \to U$ are base changes of u, hence étale. The diagonal of $\mathcal{X}$ fits into a cartesian square whose top row is the map $R \to U \times_S U$, in the sense that after base change along the étale surjection $U \times_S U \to \mathcal{X} \times_S \mathcal{X}$, the diagonal $\Delta_{\mathcal{X}}$ pulls back to $(s,t) \colon R \to U \times_S U$. Unramifiedness is étale-local on the target, since it is the vanishing of the relative cotangent sheaf and Ω commutes with the étale base change $U \times_S U \to \mathcal{X} \times_S \mathcal{X}$ (theorem 6.4.3 for the descent of étale-local properties along a cover). So $\Delta_{\mathcal{X}}$ is unramified if and only if $(s,t) \colon R \to U \times_S U$ is. Now $s \colon R \to U$ is étale, and the structure map $U \to S$ composed appropriately shows (s,t) factors as $R \xrightarrow{(s,t)} U \times_S U \xrightarrow{\mathrm{pr}_1} U$ with $\mathrm{pr}_1 \circ (s,t) = s$ étale. The morphism (s,t) is then unramified because $\Omega_{R/U \times_S U} = 0$: the surjection $\Omega_{R/U} \twoheadrightarrow \Omega_{R/U \times_S U}$ from the factorization has $\Omega_{R/U} = 0$ (as s is étale), so its quotient vanishes. Hence $\Delta_{\mathcal{X}}$ is unramified.

Step 3: (2) $\Rightarrow$ (1), an unramified diagonal produces an étale atlas. This is the substance. Start from a smooth atlas $p \colon V \to \mathcal{X}$, which exists by hypothesis. The goal is to replace V by a scheme U mapping to $\mathcal{X}$ étale-ly and surjectively. The obstruction to p being étale is its positive relative dimension, and this dimension is accounted for entirely by the automorphisms: the fiber of p over an object x is the automorphism space $\underline{\mathrm{Aut}}(x)$ (up to torsor twisting), whose dimension is the relative dimension of p minus the "geometric" dimension. When the diagonal is unramified, Step 1 makes $\underline{\mathrm{Aut}}(x)$ unramified, hence zero-dimensional, so the automorphisms contribute nothing to the relative dimension and p is smooth of the "correct" dimension, with the excess directions absent.

To extract an étale atlas, work locally on V and slice. Fix a point $v \in V$ with image $x = p(v)$, and let $n = \dim_v(V/\mathcal{X})$ be the relative dimension of the smooth morphism p at v. Because $\Delta_{\mathcal{X}}$ is unramified, the diagonal $V \to V \times_{\mathcal{X}} V$ is an open immersion: indeed $V \times_{\mathcal{X}} V = V \times_{\mathcal{X}} V$ carries the two étale-locally-defined projections, and its diagonal $V \to V \times_{\mathcal{X}} V$ is a section of the unramified separated morphism $V \times_{\mathcal{X}} V \to V$, hence an open immersion (a section of an unramified separated morphism is an open immersion, being a monomorphism that is étale). Now choose a locally closed subscheme $W \hookrightarrow V$ through v that is transverse to the fibers of p

and of relative dimension 0 over $\mathcal{X}$: since p is smooth of relative dimension n at v, the fiber $p^{-1}(x)$ is smooth of dimension n near v in the appropriate sense, and one selects W by cutting V with n functions $f_1, \ldots, f_n$ vanishing at v whose differentials $df_1, \ldots, df_n$ restrict to a basis of the fiber cotangent space $\Omega_{p^{-1}(x)/\kappa(x)} \otimes \kappa(v)$. The restriction $p|_W : W \to \mathcal{X}$ is then unramified because its relative cotangent sheaf vanishes at v: the df_i span the fiber directions of p, so cutting them out kills $\Omega_{W/\mathcal{X}}$ at v. It is flat at v by a regular-sequence argument: $f_1, \ldots, f_n$ form a regular sequence on the smooth (hence Cohen-Macaulay) fibers of p near v, cutting the relative dimension from n down to 0 without dropping the fiber dimension by more than one at each step, so, checked étale-locally on the smooth atlas where the target is a scheme, miracle flatness applies to the Cohen-Macaulay source W over the regular target and $p|_W$ is flat at v. Unramified and flat at v makes $p|_W$ étale at v. Étaleness is an open condition, so after shrinking W around v the map $p|_W : W \to \mathcal{X}$ is étale.

Doing this at every point of $\mathcal{X}$ and taking the disjoint union of the resulting slices produces a scheme $U = \bigsqcup_\alpha W_\alpha$ with an étale map $u = \sqcup p|_{W_\alpha} : U \to \mathcal{X}$. This map is surjective because every point x of $\mathcal{X}$ is hit: x lifts to some $v \in V$ under the surjection p, and the slice W_α through v contains v, so u hits x. Thus $u : U \to \mathcal{X}$ is a representable, étale, surjective map from a scheme, that is, an étale atlas, and $\mathcal{X}$ is a Deligne-Mumford stack. This completes the cycle (1) $\Rightarrow$ (2) $\Rightarrow$ (1), and together with Step 1 the equivalence of all three conditions, as we sought to show.

The theorem is the recognition principle for the class. It says that to check a moduli stack is Deligne-Mumford, one never needs to construct an étale atlas by hand: it suffices to compute the automorphism group of a single object over each field and confirm it is finite (in characteristic 0) or finite and reduced (in general). This is a computation entirely internal to the moduli problem, made before any geometry is built, and it is why the DM condition is so usable. The deformation theory developed earlier is the tool that turns this into a smooth atlas: Schlessinger's hull (theorem 5.3.5) produces the formal neighborhood of a point, the tangent space T^1 (definition 5.2.2) measures its dimension, and $\underline{\mathrm{Aut}}$ finite means the extra dimensions that would force a smooth but non-étale atlas are absent, so the hull algebraizes to an étale slice. The passage from "finite automorphisms" to "étale atlas" in Step 3 is the geometric shadow of this deformation-theoretic fact.

The other reason the DM class matters is that it is the largest class of stacks for which the classical scheme-theoretic invariants transport cleanly. Because an étale atlas is a local isomorphism for the étale topology, the étale-local geometry of a DM stack is the geometry of a scheme, and notions that are étale-local, smoothness, normality, the cotangent complex, coherent cohomology, descend from the atlas to the stack without correction. The one place a DM stack visibly departs from a scheme is at its points, where the automorphism group survives as a finite "residual gerbe," a copy of $B\underline{\mathrm{Aut}}(x)$ that the coarse space of the next theorem will collapse.

The construction of a coarse space is the tool that connects a DM stack to classical algebraic geometry: it produces the best possible algebraic space whose points are the isomorphism classes of objects, forgetting the automorphisms. For the moduli of elliptic curves this is the j-line (theorem 1.3.6); for the moduli of curves it is the classical M_g that predates the stack. The existence of the coarse space in general is the theorem of Keel and Mori, and it is the last general result of this section.

Theorem 8.2.4: Keel-Mori

Let S be a locally Noetherian scheme and $\mathcal{X}$ a Deligne-Mumford stack, separated and of finite type over S, with *finite inertia*: the inertia stack $I_{\mathcal{X}} = \mathcal{X} \times_{\Delta, \mathcal{X} \times_S \mathcal{X}, \Delta} \mathcal{X} \to \mathcal{X}$ (whose fiber over an object x is the automorphism group $\underline{\mathrm{Aut}}(x)$) is finite over $\mathcal{X}$. Then there exists a *coarse moduli space*: an algebraic space X and a morphism

$$\pi \colon \mathcal{X} \longrightarrow X$$

such that

1. π is *initial* among morphisms from $\mathcal{X}$ to algebraic spaces: every morphism $\mathcal{X} \to Y$ with Y an algebraic space factors uniquely through π;
2. π induces a *bijection* on geometric points: for every algebraically closed field K over S, the map $\pi \colon \mathcal{X}(K)/\cong \to X(K)$ from isomorphism classes of objects to K-points is a bijection; and
3. π is *proper* and *quasi-finite*, and the structure map $\mathcal{O}_X \to \pi_* \mathcal{O}_{\mathcal{X}}$ is an isomorphism.

Moreover X is of finite type and separated over S, its formation commutes with flat base change on X, and X is a *scheme* whenever $\mathcal{X}$ admits a quasi-projective coarse space, in particular whenever $\mathcal{X}$ is the quotient of a quasi-projective scheme by a finite group or, more generally, carries an ample line bundle descending to X.

***Proof Key ideas*:**
The full proof constructs X étale-locally by taking, on a suitable étale chart $\operatorname{Spec} A \to \mathcal{X}$ where the finite inertia acts by a finite group G on A, the invariant ring $\operatorname{Spec} A^G$, and then glues these local invariant quotients along the groupoid. The delicate part is that the local groups G vary and the gluing must be shown to be effective in the étale topology; the verification that the local quotients patch to a global algebraic space, and that π has the stated properties, uses the theory of finite flat groupoids and is carried out in full in [19] and, in the modern stack-theoretic language, in [26]. The argument is beyond the scope of the present treatment, which is why it is cited rather than reproduced; the universal property (1) and the bijection (2) are the load-bearing facts for this book and are used below without further comment. That X becomes a scheme under the quasi-projective hypotheses is the classical statement, going back to the GIT construction of coarse moduli (proposition 1.4.6), where the coarse space is cut out as a projective quotient and is manifestly a scheme.

The universal property is the precise sense in which X is the "best scheme approximation" to $\mathcal{X}$: it is the terminal object under $\mathcal{X}$ in the category of algebraic spaces, so any classical invariant one might compute by mapping $\mathcal{X}$ to a space factors through X. What π forgets is exactly the automorphisms: it records only the isomorphism class of an object, discarding its automorphism group, and the residual gerbe $B\underline{\mathrm{Aut}}(x)$ over a point of X is precisely the fiber of π that the coarse space cannot resolve. The condition $\mathcal{O}_X \cong \pi_* \mathcal{O}_{\mathcal{X}}$ records that π is a coarse space and not a mere continuous map: functions on $\mathcal{X}$ are the same as functions on X, so nothing is lost at the level of the structure sheaf, only at the level of automorphisms.

Two subtleties are worth flagging. First, the coarse space is an *algebraic space*, not always a scheme; the passage from the previous section's algebraic spaces to this section's stacks is what makes the non-scheme case unavoidable, since the coarse space of a DM stack can be a non-scheme algebraic space of exactly the shape produced in example 8.1.6. It becomes a scheme under the projective hypotheses that hold for the moduli of curves, which is why the classical M_g and $\overline{M}_g$ are quasi-projective schemes even though the general Keel-Mori coarse space need not be. Second, the finiteness of the inertia is essential and is exactly the DM (indeed the separated-DM) condition made quantitative: a stack with positive-dimensional automorphisms, such as $B\mathbb{G}_m$, has no coarse space in this sense, because collapsing a $\mathbb{G}_m$ to a point destroys too much, and the theory of the next section (Artin stacks) is where such stacks are handled.

The section's examples are the ones the whole book points toward. The moduli of elliptic curves, the moduli of curves of higher genus, and its Deligne-Mumford compactification are the canonical Deligne-Mumford stacks, and in each case the DM property is verified by the automorphism computation of theorem 8.2.3 rather than by constructing an atlas directly. The automorphisms that obstructed a fine moduli *scheme* in Chapter 1 (example 1.3.4) are precisely the finite groups that make these stacks Deligne-Mumford rather than schemes.

Example 8.2.5: The Moduli of Curves as DM Stacks
Work over a base S, and impose in each item the characteristic condition noted.

1. $\mathcal{M}_{1,1}$*, the moduli of elliptic curves.* Over $S = \operatorname{Spec} k$ with $\operatorname{char} k \neq 2, 3$, the stack $\mathcal{M}_{1,1}$ is the quotient stack $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ of example 7.3.5. The automorphism group of an elliptic curve E over an algebraically closed field is its group of automorphisms fixing the origin: generically $\mu_2 = \{\pm 1\}$ (the inversion $[-1]$), enlarging to μ_4 at $j = 1728$ and μ_6 at $j = 0$, as recorded in example 1.3.4. Each of these groups is finite, and each is reduced because $\operatorname{char} k \neq 2, 3$ keeps their orders prime to the characteristic, so the group schemes are étale by order count. By theorem 8.2.3, $\mathcal{M}_{1,1}$ is a Deligne-Mumford stack. Its coarse space is the j-line $\mathbb{A}^1_j = \operatorname{Spec} k[j]$ (theorem 1.3.6), recovered here as the Keel-Mori coarse space $\pi\colon \mathcal{M}_{1,1} \to \mathbb{A}^1_j$: the map π sends a curve to its j-invariant, is a bijection on geometric points (elliptic curves over $\bar{k}$ are classified up to isomorphism by j), and forgets the automorphism groups, which survive as the residual gerbes $B\mu_2$ generically, $B\mu_4$ over $j = 1728$, and $B\mu_6$ over $j = 0$.

2. $\mathcal{M}_g$*, the moduli of smooth curves of genus $g \geq 2$.* A smooth projective curve C of genus $g \geq 2$ over an algebraically closed field has a *finite* automorphism group: by the theorem of Hurwitz its order is bounded by $84(g-1)$, and in particular $\underline{\operatorname{Aut}}(C)$ is finite. Over a base where the relevant orders are prime to the residue characteristics (automatically over $\mathbb{Q}$), these groups are also reduced, so by theorem 8.2.3 the stack $\mathcal{M}_g$ is a Deligne-Mumford stack. Its Keel-Mori coarse space is the classical coarse moduli space M_g, a quasi-projective scheme by the GIT construction (proposition 1.4.6) that presents M_g as a projective quotient of a locus in a Hilbert scheme of pluricanonically embedded curves. The automorphisms are exactly what prevents M_g from being a fine moduli space: a curve with automorphisms cannot admit a universal family in a Zariski neighborhood, so no scheme carries the universal curve, and the stack is the object that does.

3. $\overline{\mathcal{M}}_g$*, the Deligne-Mumford compactification.* The compactification $\overline{\mathcal{M}}_g$ parametrizes *stable curves* of genus g: connected nodal projective curves C with at worst nodal singularities, arithmetic genus g, and finite automorphism group, the last condition being equivalent (for $g \geq 2$) to the ampleness of the dualizing sheaf ω_C, that is, to every smooth rational component meeting the rest of C in at least three points. Deligne and Mumford introduced this notion

in [13] precisely so that the compactified moduli space would remain a Deligne-Mumford stack: the stability condition is exactly the demand that $\underline{\mathrm{Aut}}(C)$ be finite, so theorem 8.2.3 applies verbatim and $\overline{\mathcal{M}}_g$ is a DM stack (over $\mathbb{Q}$, or with the usual characteristic caveats). Its coarse space is the projective scheme $\overline{M}_g$, and $\overline{\mathcal{M}}_g$ is smooth and proper as a stack even though $\overline{M}_g$ has quotient singularities, a point taken up in section 8.5 and in the capstone chapter.

In every case the DM property is a statement about the finiteness of automorphism groups, checked object by object, and the stack is the correct moduli object precisely because those finite automorphisms, which obstruct a fine moduli scheme, are recorded rather than discarded.

The three examples are one phenomenon at three levels of difficulty. The moduli of elliptic curves is the smallest case, where the whole stack is a single explicit quotient $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ and every automorphism group can be read off a Weierstrass equation. The moduli of smooth curves of higher genus is the case the theory was built for, where the finiteness of automorphisms is a nontrivial input (Hurwitz's bound) but the DM property still follows from a single application of the recognition theorem. The compactification is where the definition of the objects is engineered to preserve the DM property: stability is not a geometric afterthought but the exact condition that keeps automorphism groups finite, and this is why the Deligne-Mumford boundary is the right one. The coarse spaces (the j-line, M_g, and $\overline{M}_g$) are the classical objects, and the stacks are the refinements that see the automorphisms the coarse spaces forget.

The passage from finite to positive-dimensional automorphism groups is the subject of the next section. When a moduli problem has objects whose automorphism groups are positive-dimensional, such as the coherent sheaves whose automorphisms contain $\mathbb{G}_m$, no étale atlas exists and the DM condition fails; the atlas can then only be taken smooth, and the resulting class of stacks is the Artin (algebraic) stacks, of which Deligne-Mumford stacks are the special case of an unramified atlas.

Exercise 8.2.1

1. Let $\mathcal{X}$ be a stack over $(\mathbf{Sch}/S)_{\mathrm{et}}$ with an étale atlas $u\colon U \to \mathcal{X}$, and let $R = U \times_{\mathcal{X}} U$. Show that the two projections $s, t\colon R \to U$ are étale, and that the fiber of $(s,t)\colon R \to U \times_S U$ over a point (a,a) on the diagonal is the automorphism group scheme $\underline{\mathrm{Aut}}(u(a))$ of the object $u(a)$. Conclude that if $\mathcal{X}$ is DM then every such automorphism group is a finite étale scheme over the residue field.
2. Verify directly that a scheme X, viewed as a stack (a category fibered in sets), is a Deligne-Mumford stack. Identify its étale atlas, describe the groupoid $R \rightrightarrows U$, and confirm that every automorphism group is trivial, so the unramified-diagonal criterion of theorem 8.2.3 holds vacuously. What is its Keel-Mori coarse space?
3. For G a finite group (a finite constant group scheme) acting on a scheme X over a field k, show that the quotient stack $[X/G]$ is a Deligne-Mumford stack by exhibiting an étale atlas and computing the automorphism groups of its points. When is it separated with finite inertia, so that Keel-Mori applies, and what is the coarse space $[X/G] \to X/G$ then? Contrast this with $B\mathbb{G}_m$, and explain in one sentence why the latter is *not* a DM stack.
4. Let k be a field of characteristic $p > 0$ and consider $\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$ as a group scheme over k. Show that μ_p is finite but *not* reduced (not étale) over k. Deduce that a stack having an object whose automorphism group scheme is μ_p fails the reducedness

condition of theorem 8.2.3(3) and so is *not* Deligne-Mumford in characteristic p, even though the automorphism group is finite. Relate this to the caveat "$\operatorname{char} k \neq 2, 3$" in the $\mathcal{M}_{1,1}$ computation of example 8.2.5.

5. Using the Hurwitz bound $|\underline{\mathrm{Aut}}(C)| \leq 84(g-1)$ for a smooth projective curve C of genus $g \geq 2$ over $\bar{k}$ with $\operatorname{char} \bar{k} = 0$, together with theorem 8.2.3, give the complete argument that $\mathcal{M}_g$ is a Deligne-Mumford stack. Which hypothesis of theorem 8.2.3 does the Hurwitz bound supply, and which hypotheses (representable diagonal, existence of a smooth atlas) are supplied elsewhere?

6. Let $\mathcal{X}$ be a separated DM stack of finite type over a Noetherian base with finite inertia, and let $\pi \colon \mathcal{X} \to X$ be its Keel-Mori coarse space. Show that a morphism $\mathcal{X} \to Y$ to a scheme Y factors uniquely through π, and deduce that two DM stacks with the same objects up to isomorphism (the same groupoid of geometric points) but different automorphism groups can have the *same* coarse space. Illustrate with $\mathcal{M}_{1,1}$ and its rigidification $\mathcal{M}_{1,1} /\!\!/ \mu_2$ (the μ_2-gerbe removed), which share the coarse j-line.

8.3 Algebraic Stacks

The previous section imposed geometry on a stack by demanding an *étale* atlas: a Deligne-Mumford stack (definition 8.2.1) is one presented by a scheme mapping étale-surjectively onto it, and the price of the étale hypothesis is that automorphism groups must be finite and reduced. That hypothesis is exactly what fails for the moduli problems this book has been circling around. A vector bundle has the automorphism group GL_n; a coherent sheaf has at least the scalars $\mathbb{G}_m$ inside its automorphisms; a principal G-bundle has the group G itself. These automorphism groups are positive dimensional, so no étale atlas can exist, and the moduli stacks of such objects lie outside the Deligne-Mumford world. The remedy is to relax the atlas from étale to *smooth*. A smooth morphism is submersive with smooth fibers of arbitrary dimension, so a smooth atlas can absorb the positive-dimensional automorphisms that an étale atlas cannot. The resulting objects are the *algebraic stacks* of Artin, and they are the widest class of geometric objects in this book: every Deligne-Mumford stack, every algebraic space, and every scheme is one, and the moduli of coherent sheaves and of principal bundles finally find their home here.

This section makes the definition, proves that the quotient stacks $[X/G]$ built in chapter 7 are algebraic (with $X \to [X/G]$ the smooth atlas), records how dimension and points are read off a presentation, and exhibits $\mathfrak{Coh}_X$ and $\mathfrak{Bun}_G$ as the motivating Artin, non-Deligne-Mumford examples. The passage from étale to smooth is the last enlargement of the geometric category the book needs; section 8.4 then gives the criterion that decides which stacks are algebraic, and the capstone chapter builds $\mathcal{M}_g$ inside this framework.

Throughout, the base site is $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ (the fppf topology is the natural one for smooth atlases, since a smooth surjection is an fppf cover), and all stacks are quasi-separated and locally of finite presentation over a locally Noetherian base S unless stated otherwise. The two conditions that make a stack algebraic are a representable diagonal, encoding that its $\underline{\mathrm{Isom}}$ sheaves are geometric objects, and the existence of a smooth atlas.

Definition 8.3.1: Algebraic (Artin) Stack

Let $\mathcal{X}$ be a stack in groupoids over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$. Then $\mathcal{X}$ is an *algebraic stack* (or *Artin stack*) if:

1. the diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is representable, quasi-compact, and separated (equivalently, for all schemes T and all objects $x, y \in \mathcal{X}(T)$, the sheaf $\underline{\mathrm{Isom}}_T(x, y)$ on $\mathcal{C}/T$ is an algebraic space, quasi-compact and separated over T); and

2. there exist a scheme U and a morphism $u \colon U \to \mathcal{X}$ that is representable, smooth, and surjective. Such a u is called a *smooth atlas* (or *presentation*) of $\mathcal{X}$.

A *Deligne-Mumford stack* (definition 8.2.1) is the special case in which the atlas u may be taken *étale*; the étale hypothesis is a strengthening of "smooth", so every Deligne-Mumford stack is algebraic.

The two clauses of the definition play complementary roles, and it is worth separating them. Clause (1), representability of the diagonal, is the condition that makes the notion of a *representable* morphism $u \colon U \to \mathcal{X}$ meaningful: by theorem 7.4.6, a morphism from a scheme T to $\mathcal{X}$ has representable fibers exactly when $\Delta_{\mathcal{X}}$ is representable, so without clause (1) one could not even ask that the atlas u in clause (2) be smooth (a property that is tested on the fibers $U \times_{\mathcal{X}} T$, which must be schemes or algebraic spaces for the test to make sense). Clause (1) is therefore logically prior: it is what lets the automorphisms of objects be geometric objects in their own right, and it is half of algebraicity all by itself. Clause (2) is the geometric input proper: it says $\mathcal{X}$ has "enough charts", a scheme U covering it smoothly, so that any question about $\mathcal{X}$ can be pulled back to a question about the scheme U and its self-intersection groupoid.

That self-intersection is the structure to keep in view. Given a smooth atlas $u \colon U \to \mathcal{X}$, form the fiber product

$$R = U \times_{\mathcal{X}} U$$

which, by clause (1), is an algebraic space (in fact a scheme when $\Delta_{\mathcal{X}}$ is representable by schemes), together with the two projections $s, t \colon R \rightrightarrows U$. Both projections are smooth, being base changes of the smooth u, and $R \rightrightarrows U$ is a groupoid in algebraic spaces: it has an identity $U \to R$ (the diagonal of u), an inverse $R \to R$ (swapping the two factors), and a composition $R \times_{s,U,t} R \to R$. The stack $\mathcal{X}$ is recovered as the stackification of this groupoid, exactly as $[X/G]$ was recovered from the groupoid $G \times_S X \rightrightarrows X$ in chapter 7. This is the precise sense in which an algebraic stack is "a scheme up to a smooth groupoid": the pair (U, R) is the presentation, and $\mathcal{X} = [U/R]$ is the quotient. The smooth-groupoid presentation is the working object for every computation on a stack, from dimension to quasi-coherent sheaves.

The definition would be inert without a supply of examples, and the richest supply is the one already constructed. Chapter 7 built the quotient stack $[X/G]$ for a group scheme G acting on a scheme X, and produced the morphism $q \colon X \to [X/G]$ classifying the tautological torsor. That morphism is the atlas: the theorem below shows q is smooth and surjective, so $[X/G]$ is algebraic whenever G is a smooth affine group scheme. This is the mechanism that makes BG, $B\mathbb{G}_m$, BGL_n, and $\mathcal{M}_{1,1}$ into geometric objects, and it is the source of essentially every algebraic stack met in practice, since a stack with a smooth atlas is, locally, a quotient stack.

Theorem 8.3.2: Quotient Stacks Are Algebraic

Let G be a smooth affine group scheme of finite presentation over S acting on an S-scheme X. Then:

1. the quotient stack $[X/G]$ is an algebraic stack over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$;
2. the tautological morphism $q\colon X \to [X/G]$ of theorem 7.3.3 is a G-torsor, hence representable, smooth, and surjective; that is, q is a smooth atlas, and $R = X \times_{[X/G]} X \cong G \times_S X$ is the presentation groupoid;
3. in particular the classifying stack $BG = [\mathrm{pt}/G]$ is algebraic, with atlas $\mathrm{pt} = S \to BG$ and presentation groupoid $G \rightrightarrows \mathrm{pt}$; conversely, any algebraic stack with a smooth atlas $u\colon U \to \mathcal{X}$ is, Zariski-locally on U, of the form $[U'/R']$ for a smooth groupoid $R' \rightrightarrows U'$, and if moreover $\mathcal{X}$ has affine stabilizers (affine diagonal) it is étale-locally on the base a quotient stack $[V/\mathrm{GL}_N]$.

***Proof*:**
The three parts are proved in turn. Part (2) does the geometric work (identifying q and its base change R), part (1) then reads off algebraicity from the diagonal and the atlas, and part (3) specializes and reverses the construction.

Step 1: The atlas q is a G-torsor. By theorem 7.3.3(2), the tautological morphism $q\colon X \to [X/G]$ has the property that for every object $(P, f) \in [X/G](T)$, corresponding to a morphism $\xi\colon T \to [X/G]$, the fiber product satisfies

$$X \times_{[X/G],\xi} T \cong P$$

as G-torsors over T. In particular q is *representable*: its base change along any morphism $T \to [X/G]$ from a scheme is the scheme (indeed G-torsor) P. Moreover this base change $P \to T$ is a G-torsor, so it is faithfully flat and of finite presentation, and, because G is smooth over S, the torsor $P \to T$ is smooth (fppf-locally on T it is the projection $G \times_S T \to T$, which is smooth as a base change of the smooth $G \to S$, and smoothness is fppf-local on the base by theorem 6.4.3). A G-torsor is surjective (again fppf-locally it is $G \times_S T \to T$, which is surjective). Smoothness and surjectivity of q are properties of a representable morphism tested after base change to a scheme (definition 7.4.3), and they hold on every such base change, so q is smooth and surjective.

For the presentation groupoid, base-change q along itself. Taking $T = X$ and $\xi = q$ (which classifies the trivial torsor $G \times_S X \to X$ with equivariant map σ), the identity above gives

$$R = X \times_{[X/G]} X \cong G \times_S X$$

the trivial torsor pulled back, with the two projections $s, t\colon G \times_S X \rightrightarrows X$ being the action $\sigma\colon (g, x) \mapsto g \cdot x$ and the projection $\mathrm{pr}_X\colon (g, x) \mapsto x$. This is exactly the action groupoid of G on X.

Step 2: $[X/G]$ is algebraic. Two conditions must be checked: representability of the diagonal, and existence of a smooth atlas. The atlas is $q\colon X \to [X/G]$ from Step 1, which is a smooth surjection from a scheme. It remains to check that $\Delta_{[X/G]}$ is representable, quasi-compact, and separated.

The diagonal of $[X/G]$ is computed by the $\underline{\mathrm{Isom}}$ sheaves. For a scheme T and two objects $(P, f), (P', f') \in [X/G](T)$, the sheaf $\underline{\mathrm{Isom}}_T((P, f), (P', f'))$ assigns to a T-scheme T' the set of G-torsor isomorphisms $P|_{T'} \to P'|_{T'}$ commuting with the maps to X. This sheaf is representable by a scheme affine over T: locally on T where both torsors trivialize, an isomorphism of trivial (left) G-torsors $G \times_S T' \to G \times_S T'$ is right-translation $g \mapsto g\gamma$ by a section $\gamma \in G(T')$ (this is the general form of an automorphism of the trivial torsor), and compatibility with the equivariant maps $g \mapsto g \cdot x$ and $g \mapsto g \cdot x'$ to X forces $g\gamma \cdot x' = g \cdot x$ for all g, that is, $\gamma \cdot x' = x$. Thus $\underline{\mathrm{Isom}}$ is cut out inside the transporter

$$\mathrm{Transp}_G(x', x) = \{\gamma \in G : \gamma \cdot x' = x\}$$

a closed subscheme of $G \times_S T$ (the preimage of the diagonal under $(\gamma, t) \mapsto (\gamma \cdot x'(t), x(t))$), hence affine and of finite presentation over T. Since G is separated (affine over S) and quasi-compact, the transporter is separated and quasi-compact over T. Gluing these local descriptions along the trivializing cover (the identifications are torsor isomorphisms, which glue by descent, theorem 6.3.4), the sheaf $\underline{\mathrm{Isom}}$ is representable by a scheme quasi-compact and separated over T. By theorem 7.4.6, representability of all these $\underline{\mathrm{Isom}}$ sheaves is precisely representability of the diagonal $\Delta_{[X/G]}$, and the quasi-compactness and separatedness of the sheaves give the corresponding properties of Δ. Thus $\Delta_{[X/G]}$ is representable, quasi-compact, and separated, and with the smooth atlas q this establishes that $[X/G]$ is algebraic.

Step 3: BG, and the converse. Applying parts (1) and (2) with $X = S = \mathrm{pt}$ (the terminal object, with G acting trivially) gives that $BG = [\mathrm{pt}/G]$ is algebraic. Its atlas is $q\colon \mathrm{pt} = S \to BG$ (classifying the trivial torsor $G \to S$), and the presentation groupoid is $R = \mathrm{pt} \times_{BG} \mathrm{pt} \cong G \times_S \mathrm{pt} = G$, with both projections the structure map $G \to S$; that is, $G \rightrightarrows \mathrm{pt}$ with the multiplication as composition. This is the group G regarded as a one-object groupoid, whose stackification is BG.

For the converse, let $\mathcal{X}$ be algebraic with smooth atlas $u\colon U \to \mathcal{X}$ and groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$. Working Zariski-locally on U, choose an open $U' \subseteq U$ and set $R' = R \times_{U \times U} (U' \times U')$; then $R' \rightrightarrows U'$ is a smooth groupoid in algebraic spaces and $\mathcal{X}|_{u(U')} \cong [U'/R']$ is its stackification, which is the assertion that $\mathcal{X}$ is locally a groupoid quotient. To upgrade this to a genuine quotient stack $[V/G]$ one needs R' to be an action groupoid; this holds étale-locally after a further construction. Where $\mathcal{X}$ has affine (or quasi-affine) diagonal and admits a locally free sheaf trivializing the atlas, the frame bundle of that sheaf is a GL_N-torsor $V \to \mathcal{X}$ from a scheme, and $\mathcal{X} \cong [V/\mathrm{GL}_N]$; the general local-quotient statement (that every algebraic stack with affine stabilizers is étale-locally a quotient stack $[V/\mathrm{GL}_N]$) is a structural theorem whose full proof uses the local structure of algebraic stacks and is beyond the present scope, cited to [26] and [29]. The smooth-groupoid presentation $\mathcal{X} = [U/R]$, on the other hand, is immediate from the atlas and holds for every algebraic stack, as we sought to show.

The theorem is the engine of the whole chapter: it converts the categorical construction $[X/G]$ of chapter 7 into a geometric object with charts, and it does so by identifying the atlas $X \to [X/G]$ as a torsor, hence smooth. Two features deserve comment. First, the smoothness of the atlas is inherited directly from the smoothness of the group G: a G-torsor is smooth exactly when G is, which is why the theorem requires G smooth and why the positive-dimensional automorphism groups of coherent sheaves and bundles pose no obstruction to algebraicity, only to Deligne-Mumfordness. Second, the presentation groupoid $R \cong G \times_S X$ is the action groupoid, so the dimension of the stack is legible from the group: R is larger than $U = X$ by exactly $\dim G$, and the next definition turns this into the formula $\dim[X/G] = \dim X - \dim G$. The converse in part (3) is the reason quotient stacks are not a special case but the general picture: every algebraic stack with affine stabilizers is locally of the form $[V/\mathrm{GL}_N]$, so understanding quotient stacks is, locally, understanding all algebraic stacks with affine diagonal (the abelian-variety classifying stacks BA, whose stabilizers are proper, fall outside this local model).

Before the general examples, the smallest instances anchor the definition. The classifying stacks $B\mathbb{G}_m$ and BGL_n, built as groupoids in chapter 7, are now algebraic stacks by the theorem, and they are the simplest algebraic stacks that are *not* Deligne-Mumford.

Example 8.3.3: $B\mathbb{G}_m$ and BGL_n as Algebraic Stacks

Over $S = \operatorname{Spec} k$, the group $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ is smooth affine of dimension 1, so by theorem 8.3.2 the classifying stack $B\mathbb{G}_m$ is algebraic. Its atlas is the point $\mathrm{pt} \to B\mathbb{G}_m$ and its presentation groupoid is $\mathbb{G}_m \rightrightarrows \mathrm{pt}$. The atlas $\mathrm{pt} \to B\mathbb{G}_m$ is smooth (a $\mathbb{G}_m$-torsor, and $\mathbb{G}_m$ is smooth) and surjective, but it is not unramified: the automorphism group of the unique geometric point is $\mathbb{G}_m$ itself, which is positive dimensional, so the diagonal

$$\Delta\colon B\mathbb{G}_m \to B\mathbb{G}_m \times_k B\mathbb{G}_m$$

is not unramified, and $B\mathbb{G}_m$ is not a Deligne-Mumford stack (by theorem 8.2.3). No étale atlas exists: an étale morphism has discrete (finite) fibers, but a $\mathbb{G}_m$-torsor has one-dimensional fibers.

The same holds for BGL_n: the group GL_n is smooth affine of dimension n^2, so BGL_n is algebraic with atlas $\mathrm{pt} \to BGL_n$ and groupoid $GL_n \rightrightarrows \mathrm{pt}$, and the automorphism group of its point is GL_n, of dimension $n^2 > 0$, so BGL_n is not Deligne-Mumford. By contrast, for a *finite* group G (a smooth group scheme of dimension 0 in characteristic not dividing $|G|$) the atlas $\mathrm{pt} \to BG$ is étale, and BG is Deligne-Mumford: the distinction between Deligne-Mumford and Artin is exactly the distinction between $\dim G = 0$ and $\dim G > 0$ for the automorphism groups.

The example isolates the mechanism cleanly: BG is Deligne-Mumford precisely when G is finite, and Artin (but not Deligne-Mumford) when G is positive dimensional. The dimension of the automorphism group is the sole obstruction to the étale atlas, and the smooth atlas ignores it. This suggests that dimension is measured not by the atlas alone but by the atlas relative to the groupoid, and the following definition makes that precise.

Definition 8.3.4: Dimension and Points of an Algebraic Stack

Let $\mathcal{X}$ be an algebraic stack over S with smooth atlas $u\colon U \to \mathcal{X}$ and presentation groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$.

1. The *dimension* of $\mathcal{X}$ at a point is

$$\dim_x \mathcal{X} = \dim_{\tilde{x}} U - \dim_{\tilde{x}}(u\text{-fiber})$$

where $\tilde{x} \in U$ lies over x and the u-fiber is the fiber of u through $\tilde{x}$; equivalently, since $s\colon R \to U$ is smooth of relative dimension equal to the fiber dimension of u,

$$\dim \mathcal{X} = \dim U - (\dim R - \dim U) = 2\dim U - \dim R$$

the last equality holding when U, R are equidimensional. For a quotient stack this specializes (see the remark below) to

$$\dim[X/G] = \dim X - \dim G$$

and in particular $\dim BG = -\dim G$.

2. The *underlying topological space* $|\mathcal{X}|$ is the set of isomorphism classes of points $x\colon \operatorname{Spec} K \to \mathcal{X}$ over fields K, where $x\colon \operatorname{Spec} K \to \mathcal{X}$ and $x'\colon \operatorname{Spec} K' \to \mathcal{X}$ are identified when there is a common field extension over which they become 2-isomorphic; equivalently $|\mathcal{X}| = |U|/|R|$ is the quotient of $|U|$ by the equivalence relation cut out by the image of $|R| \to |U| \times |U|$. It carries the quotient topology, for which $|u|\colon |U| \to |\mathcal{X}|$ is open.

3. The *residual gerbe* at a point $x \in |\mathcal{X}|$ with field of definition K is the reduced locally closed substack $\mathcal{G}_x \hookrightarrow \mathcal{X}$ through x; it is a gerbe over $\operatorname{Spec} K$ (banded by the automorphism group scheme $\underline{\operatorname{Aut}}(x)$), and it is the intrinsic remnant of the automorphisms at x. A point of $\mathcal{X}$ is thus not a bare point but a point together with its residual gerbe $B\underline{\operatorname{Aut}}(x)$.

Three points about the definition are worth drawing out. First, the dimension formula. The atlas u is smooth, and its fibers have the dimension of the automorphism groups of the objects being covered; for a quotient stack $[X/G]$ the atlas is $q\colon X \to [X/G]$ with fibers G-torsors, so the fiber dimension is $\dim G$, and $\dim[X/G] = \dim X - \dim G$. The subtraction is the point: a stack can have *negative* dimension, and BG does, with $\dim BG = -\dim G$. This is not a pathology but the correct bookkeeping: BG has a single point with a $\dim G$-dimensional automorphism group, and the automorphisms count against the dimension exactly as the parameters count for it. The Hilbert scheme, whose objects have no automorphisms, has atlas fibers of dimension 0 and so $\dim \operatorname{Hilb}$ equals the naive parameter count; the moduli stack $\mathcal{M}_g$, whose objects have finite automorphism groups (dimension 0), has $\dim \mathcal{M}_g = 3g - 3$ equal to the deformation count, with the automorphisms invisible to dimension but visible to the stack structure.

Second, the topological space $|\mathcal{X}|$. It is the set of R-equivalence classes of points of the atlas, and it forgets everything about the automorphisms, retaining only which points are identified. For BG it is a single point; for $[X/G]$ it is the orbit space $|X|/|G|$ with its quotient topology; for $\mathcal{M}_{1,1}$ it is the underlying set of the j-line. The space $|\mathcal{X}|$ is where a stack looks like its coarse space, and it is the object on which one defines irreducibility, connectedness, and the dimension of an irreducible component.

Third, the residual gerbe. This is the datum that a point of a stack carries beyond its location in $|\mathcal{X}|$: at each point sits the classifying stack $B\underline{\mathrm{Aut}}(x)$ of its automorphism group, embedded in $\mathcal{X}$ as the residual gerbe. For $\mathcal{M}_{1,1}$ over a generic j-value the residual gerbe is $B\mu_2$ (the automorphism $[-1]$), and over $j = 0, 1728$ it jumps to $B\mu_6, B\mu_4$, recovering the picture of chapter 7. The residual gerbe is the local model for the stack near a point, and it is what the coarse space $|\mathcal{X}|$ discards. The systematic study of these gerbes is the subject of the next chapter.

Example 8.3.5: Dimensions of BG, $[X/G]$, and $\mathcal{M}_{1,1}$

The dimension formula is read off the presentation in each case.

1. $\dim B\mathbb{G}_m = -\dim \mathbb{G}_m = -1$. The atlas $\mathrm{pt} \to B\mathbb{G}_m$ has $\dim \mathrm{pt} = 0$ and the atlas fiber is a $\mathbb{G}_m$-torsor over pt, of dimension 1, so $\dim B\mathbb{G}_m = 0 - 1 = -1$. Similarly $\dim BGL_n = -n^2$ and $\dim BG = 0$ for a finite group G.

2. For $\mathcal{M}_{1,1} = [(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ with the weight-$(4,6)$ action (example 7.3.5), the atlas is $X = \mathbb{A}^2 \setminus \{\Delta = 0\}$ of dimension 2 and $G = \mathbb{G}_m$ of dimension 1, so

$$\dim \mathcal{M}_{1,1} = \dim X - \dim \mathbb{G}_m = 2 - 1 = 1$$

matching the one-parameter family of elliptic curves recorded by the j-line. The single generic automorphism μ_2 is 0-dimensional and does not affect the dimension; it is finite, consistent with $\mathcal{M}_{1,1}$ being Deligne-Mumford of dimension 1.

3. For the affine plane modulo GL_2, $[\mathbb{A}^2/\mathrm{GL}_2]$ has $\dim = 2 - 4 = -2$, reflecting that the GL_2-action on $\mathbb{A}^2$ has a dense orbit (the nonzero vectors) with two-dimensional stabilizer, so the stack is concentrated at essentially one point with a large automorphism group.

The dimension examples show the formula behaving as bookkeeping should: parameters add, automorphisms subtract, and the balance is the dimension of the stack. With dimension and points in hand, the section reaches the examples that motivated the enlargement from étale to smooth atlases in the first place. These are the moduli stacks whose objects have positive-dimensional automorphism groups: the stack of coherent sheaves and the stack of principal bundles. They are the prototypical Artin, non-Deligne-Mumford stacks, and their construction is the reason algebraic stacks in Artin's sense exist.

The stack $\mathfrak{Coh}_X$ of coherent sheaves on a projective scheme X has, as its T-points, the flat families of coherent sheaves on $X \times T$ over T. A single coherent sheaf $\mathcal{F}$ has automorphism group $\underline{\mathrm{Aut}}(\mathcal{F}) = \mathrm{End}_{\mathcal{O}_X}(\mathcal{F})^\times$, the units of the endomorphism algebra, an open subscheme of the affine space $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F})$, which always contains the scalars $\mathbb{G}_m$ (multiplication by a unit) and is positive dimensional. The presence of $\mathbb{G}_m$ in every automorphism group is the structural reason $\mathfrak{Coh}_X$ can never be Deligne-Mumford: its diagonal is not unramified. Yet it is algebraic, because it is covered by Quot schemes.

Example 8.3.6: $\mathfrak{Coh}_X$ and $\mathfrak{Bun}_G$: the Motivating Artin Stacks

Let X be a projective scheme over $S = \operatorname{Spec} k$ and let G be a smooth affine group scheme over k.

The stack of coherent sheaves. The stack $\mathfrak{Coh}_X$ has $\mathfrak{Coh}_X(T)$ the groupoid of T-flat coherent sheaves on $X_T = X \times_k T$, with morphisms the $\mathcal{O}_{X_T}$-isomorphisms. This is a stack for the fppf topology: quasi-coherent sheaves satisfy descent (theorem 6.3.4), and flatness and coherence are fppf-local. Its diagonal is representable: for two families $\mathcal{F}, \mathcal{G}$ over T, the sheaf $\underline{\mathrm{Isom}}_T(\mathcal{F}, \mathcal{G})$ is an open subscheme of the linear scheme $\underline{\mathrm{Hom}}_T(\mathcal{F}, \mathcal{G}) = \mathrm{Spec}\,_T\mathrm{Sym}\,\mathcal{H}om(\mathcal{F}, \mathcal{G})^\vee$ (the automorphisms being the invertible homomorphisms), hence affine of finite type over T; so Δ is representable, quasi-compact, and separated.

For the atlas, fix a numerical type (a Hilbert polynomial P) and an integer m. By boundedness (theorem 2.2.5), the family of coherent sheaves with Hilbert polynomial P is bounded, so for $m \gg 0$ every such sheaf $\mathcal{F}$ is m-regular, meaning $\mathcal{F}(m)$ is globally generated with $H^i(X, \mathcal{F}(m)) = 0$ for $i > 0$ and $h^0(X, \mathcal{F}(m)) = P(m)$. A choice of surjection $\mathcal{O}_X(-m)^{\oplus P(m)} \twoheadrightarrow \mathcal{F}$ is then a point of the Quot scheme $\mathrm{Quot}(\mathcal{O}_X(-m)^{\oplus P(m)}, P)$, and forgetting the choice of surjection gives a morphism

$$\mathrm{Quot}\big(\mathcal{O}_X(-m)^{\oplus P(m)}, P\big) \longrightarrow \mathfrak{Coh}_X^{\leq P,m}$$

onto the open substack of m-regular sheaves with Hilbert polynomial P. This morphism is a $\mathrm{GL}_{P(m)}$-torsor (the fiber over $\mathcal{F}$ is the choice of basis of $H^0(X, \mathcal{F}(m))$, a $\mathrm{GL}_{P(m)}$-torsor), hence smooth and surjective; ranging over P and m, these Quot schemes cover $\mathfrak{Coh}_X$ by smooth morphisms from schemes. Thus $\mathfrak{Coh}_X$ is an algebraic stack, presented as

$$\mathfrak{Coh}_X^{\leq P,m} \cong \big[\mathrm{Quot}(\mathcal{O}_X(-m)^{\oplus P(m)}, P)/\mathrm{GL}_{P(m)}\big]$$

a quotient stack on each numerical piece. Its automorphism groups contain $\mathbb{G}_m$, so it is Artin and not Deligne-Mumford. The full verification that these charts glue into a global algebraic stack is carried out via Artin's criteria (section 8.4); a complete treatment is in [21] and [26].

The stack of G-bundles. The stack $\mathfrak{Bun}_G$ has $\mathfrak{Bun}_G(T)$ the groupoid of G-torsors (principal G-bundles) on X_T, flat and of finite presentation over T, with morphisms the isomorphisms of G-torsors. When $G = GL_n$ this is the stack $\mathfrak{Bun}_{GL_n}$ of rank-n vector bundles on X. Its T-points have automorphism groups the global automorphisms of a G-bundle, which contain the center $Z(G) \supseteq \mathbb{G}_m$ (for $G = GL_n$) and are positive dimensional, so $\mathfrak{Bun}_G$ is Artin and not Deligne-Mumford. It is algebraic: for a bounded family of bundles one obtains a smooth atlas from a Quot-scheme parametrization of the bundle together with a rigidifying frame, presenting $\mathfrak{Bun}_G$ locally as a quotient stack $[V/\mathrm{GL}_N]$ exactly as above. The full proof is via Artin's criteria, cited to [21], [26], and [29].

These two stacks are the reason the Artin theory exists. The moduli of coherent sheaves and of principal bundles are among the central objects of algebraic geometry, and their automorphism groups are irreducibly positive dimensional: every coherent sheaf has its scalar automorphisms $\mathbb{G}_m$, every vector bundle its GL_n-worth of frames. There is no way to rigidify these away while retaining the moduli problem, because the scalars are intrinsic to the linear-algebraic structure of sheaves. The Deligne-Mumford framework, which requires finite automorphism groups, simply cannot accommodate them, and the smooth atlas of the Artin framework is the minimal enlargement that does. The presentation as a quotient of a Quot scheme by GL_N is the concrete form the abstract atlas takes, and it is the recurring shape of every algebraic stack in practice: a bounded family of objects, parametrized by a scheme (Quot or Hilbert), modulo the group of choices (GL_N) that the parametrization introduced.

The pattern that emerges across the section is uniform. An algebraic stack is a scheme up to a smooth groupoid; the smooth atlas is a scheme U covering it, the presentation groupoid $R \rightrightarrows U$ records the automorphisms and identifications, and every geometric question about the stack is answered by pulling back to U and checking compatibility over R. The quotient stack $[X/G]$ is the model case, and the local-quotient theorem says it is not only a case but the general picture. What separates the Artin world from the Deligne-Mumford world is a single number, the dimension of the automorphism groups: finite for Deligne-Mumford, positive dimensional for Artin, and the smooth atlas is the device that makes the positive-dimensional case geometric.

Exercise 8.3.1

1. Show directly from definition 8.3.1 that every scheme U, regarded as a stack via its functor of points $\underline{U}$, is an algebraic stack, and that every algebraic space is an algebraic stack. (For the scheme case: the identity $U \to \underline{U}$ is an atlas, and the diagonal of a scheme is a closed immersion, hence representable, quasi-compact, and separated. For an algebraic space X, use its étale atlas $U \to X$ from definition 8.1.3, noting étale implies smooth.) Conclude the inclusions: schemes $\subseteq$ algebraic spaces $\subseteq$ Deligne-Mumford stacks $\subseteq$ algebraic stacks.

2. Let G be a smooth affine group scheme over k acting on X. Using theorem 8.3.2, identify the presentation groupoid $R \rightrightarrows U$ of $[X/G]$ explicitly, and verify that the two projections $s, t\colon R \to U$ are both smooth of relative dimension $\dim G$. Deduce the dimension formula $\dim[X/G] = \dim X - \dim G$ from definition 8.3.4, and check it against $\dim BG = -\dim G$ and $\dim \mathcal{M}_{1,1} = 1$.

3. The stack $B\mathbb{G}_m$ over $\operatorname{Spec} k$ has a single geometric point with automorphism group $\mathbb{G}_m$. Show that its diagonal $\Delta\colon B\mathbb{G}_m \to B\mathbb{G}_m \times_k B\mathbb{G}_m$ is representable by computing the fiber product $\mathrm{pt} \times_{B\mathbb{G}_m \times B\mathbb{G}_m} \mathrm{pt}$ over the two maps $\mathrm{pt} \to B\mathbb{G}_m$ classifying the trivial torsor, and identify this fiber product with $\mathbb{G}_m$. Explain why the positive dimension of this $\underline{\mathrm{Isom}}$ scheme is exactly the failure of Δ to be unramified, so that $B\mathbb{G}_m$ is algebraic but not Deligne-Mumford (contrast with $B\mu_n$).

4. Let X be a projective scheme and fix a Hilbert polynomial P. Using boundedness (theorem 2.2.5), explain why the open substack $\mathfrak{Coh}_X^{\leq P,m}$ of m-regular sheaves with Hilbert polynomial P (for $m \gg 0$) is a quotient stack $[\mathrm{Quot}/\mathrm{GL}_{P(m)}]$ as in example 8.3.6. Identify the automorphism group of the point of $\mathfrak{Coh}_X$ corresponding to a stable sheaf $\mathcal{F}$ (one with $\mathrm{End}(\mathcal{F}) = k$), and confirm it is $\mathbb{G}_m$, so that even the "best-behaved" sheaves keep a positive-dimensional automorphism group.

5. Show that a group homomorphism $\rho\colon G \to H$ of smooth affine group schemes induces a morphism of algebraic stacks $B\rho\colon BG \to BH$, and compute the effect on dimensions and on atlases: the atlas $\mathrm{pt} \to BG$ maps to the atlas $\mathrm{pt} \to BH$, and $\dim BG - \dim BH = \dim H - \dim G = -\dim\ker\rho + \dim\operatorname{coker}$ (interpret suitably for the determinant $\det\colon GL_n \to \mathbb{G}_m$, where $\ker = SL_n$). Verify $\dim BGL_n = -n^2$ and $\dim B\mathbb{G}_m = -1$ are consistent with $\dim BSL_n = -(n^2 - 1)$ via the fibration $BSL_n \to BGL_n \to B\mathbb{G}_m$.

6. Explain why $[\mathbb{A}^1/\mathbb{G}_m]$ (with $\mathbb{G}_m$ acting by scaling) is an algebraic stack, compute its dimension, and describe its two points (the open orbit $\mathbb{A}^1 \setminus 0$ and the origin) together with their automorphism groups and residual gerbes. Show that $|[\mathbb{A}^1/\mathbb{G}_m]|$ is the two-point Sierpiński space (a generic point whose closure is the whole space, plus a closed point $B\mathbb{G}_m$), and contrast this with the coarse space, which is a single point. Why does this stack fail to be separated?

8.4 Artin's Representability Criteria

The whole book converges here. Part I posed moduli problems as functors of points and learned that the best of them, the Hilbert and Quot schemes, are representable, while the most natural of them, the moduli of curves, are not. Part II opened the object being classified and read its infinitesimal structure: a first-order deformation is a class in a tangent cohomology group, and the obstruction to continuing it lives one degree up. Part III built the geometric objects, algebraic spaces and algebraic stacks, capacious enough to represent moduli problems that no scheme could. What is still missing is the bridge. Given a moduli problem, one can compute its deformation theory by the methods of Chapter 5, but nothing so far turns that formal, order-by-order infinitesimal data into the statement that the moduli problem is an algebraic stack in the sense of definition 8.3.1. Artin's criteria are that bridge: a checklist, phrased entirely in terms of the functor and its deformation theory, whose satisfaction is equivalent to algebraicity.

The difficulty the criteria overcome is the gap between the formal and the algebraic. Schlessinger's theorem (theorem 5.3.5) produces, at each point of a moduli problem, a formal versal deformation living over a complete local ring, the hull of definition 5.3.1. A hull is a power series neighborhood of the point, but a power series ring is not a scheme of finite type, and a formal deformation is not a family over an actual scheme; it is the limit of families over Artinian thickenings. To build an atlas, a smooth surjection from a scheme, one must *algebraize* the hull: find an actual finite-type scheme, or at least a finitely presented one, whose completion at a point recovers the formal versal deformation, and which maps to the stack. This is the analytic heart of the theory, and it is supplied by Artin approximation, whose proof rests on Néron-Popescu desingularization. That proof is beyond the present scope and is cited; everything else, the precise statement of the criteria, the deformation-theoretic mechanism by which they assemble an atlas, and the verification of the criteria for $\mathcal{M}_g$ and for the Hilbert and Quot functors, is carried out in full.

The first task is to name the deformation-theoretic input to the criteria as a structure on the stack itself. Chapter 5 developed deformation and obstruction theory for a fixed variety or a fixed singular scheme, through the cotangent functors $T^i(A/k, -)$ (definition 5.2.2) and their interpretation (theorem 5.2.3). For a stack the same data must be attached uniformly to every point, and it must vary well enough that a formal versal deformation at one point remains versal nearby. The following packages this.

Definition 8.4.1: Deformation and Obstruction Theory for a Stack

Let $\mathcal{X}$ be a stack over $(\mathbf{Sch}/S)_{\text{fppf}}$ with S locally Noetherian, and let $x \in \mathcal{X}(k)$ be an object over a field k that is a residue field of S. A *deformation and obstruction theory* for $\mathcal{X}$ at x consists of:

1. a finite-dimensional k-vector space $T^1_{\mathcal{X}}(x)$, the *tangent space*, in natural bijection with the set of first-order deformations of x, that is, the lifts of x to an object over $k[\varepsilon]$ up to isomorphism, with the trivial deformation as zero;

2. for each small extension $0 \to (t) \to A' \to A \to 0$ in $\mathbf{Art}_k$ and each object $x_A \in \mathcal{X}(A)$ lifting x, an *obstruction space* $T^2_{\mathcal{X}}(x)$ (a finite-dimensional k-vector space, the same for all A) and a canonical *obstruction class*

$$\mathrm{ob}(x_A, A') \in T^2_{\mathcal{X}}(x) \otimes_k (t)$$

that vanishes if and only if x_A lifts to an object over A', and, when it vanishes, the set of such lifts is a torsor under $T^1_{\mathcal{X}}(x) \otimes_k (t)$.

The theory is said to satisfy *openness of versality* if, whenever a finite-type S-scheme U carries an object $\xi \in \mathcal{X}(U)$ whose induced formal deformation at a point $u \in U$ is versal (a hull for the deformation functor of $x = \xi(u)$ in the sense of definition 5.3.1), there is an open neighborhood $u \in V \subseteq U$ on which ξ is versal at every point.

The two vector spaces $T^1_{\mathcal{X}}(x)$ and $T^2_{\mathcal{X}}(x)$ are the abstraction of the tangent-obstruction pair of Chapter 5, lifted from a fixed object to a point of a stack. Their defining property is exactly the content of theorem 5.1.2 and theorem 5.2.3(2): a first-order deformation is a tangent vector, a lift across a small extension exists exactly when an obstruction class in the degree-two space vanishes, and the lifts, when they exist, form a torsor under the tangent space tensored with the kernel. What the definition adds is that these spaces are attached to *every* point of $\mathcal{X}$ and are required to be finite-dimensional, which is the input that makes Schlessinger's finiteness condition (H3) automatic.

Openness of versality is the subtler ingredient and deserves its own reading. A hull is a formal object: it controls the deformation functor at a single point, order by order. The condition demands that versality, once achieved formally at a point through an actual family over a finite-type base, is not an isolated accident but propagates to a Zariski neighborhood. Geometrically this is what makes an atlas possible. A single versal formal deformation gives a smooth map to $\mathcal{X}$ from a formal neighborhood; openness upgrades "formal neighborhood" to "Zariski neighborhood", so that a versal family over a scheme is a smooth map to $\mathcal{X}$ on an open set rather than only at one point. Without openness one could algebraize each hull separately and get a collection of pointwise-smooth maps that never assemble into a single smooth atlas; with it, each algebraized hull is smooth on a whole open, and these opens cover $\mathcal{X}$.

Before the theorem it is worth grounding the definition in the running touchstones, where T^1 and T^2 are already computed. For the Hilbert scheme, a point $[Z]$ has tangent space $T^1 = H^0(Z, N_{Z/Y})$ (theorem 2.4.4), and the obstruction lives in $T^2 = H^1(Z, N_{Z/Y})$; these are the deformation and obstruction spaces of the closed subscheme Z. For the moduli of curves, a point $[C]$ has tangent space $T^1 = H^1(C, T_C)$ and obstruction space $T^2 = H^2(C, T_C)$ in the sense of theorem 5.1.2; and because a curve is one-dimensional, $H^2(C, T_C) = 0$, so $\mathcal{M}_g$ is unobstructed and its deformation theory has vanishing obstruction space at every point. These are exactly the inputs the criteria will consume.

Example 8.4.2: The Deformation Theory of the Hilbert Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective and let $[Z] \in \underline{\mathrm{Hilb}}^P_Y(k)$ correspond to a closed subscheme $Z \subseteq Y$ with Hilbert polynomial P. The deformation and obstruction theory of $\underline{\mathrm{Hilb}}^P_Y$ at $[Z]$ is

$$T^1_{\underline{\mathrm{Hilb}}}([Z]) = H^0(Z, N_{Z/Y}), \qquad T^2_{\underline{\mathrm{Hilb}}}([Z]) = H^1(Z, N_{Z/Y})$$

the global sections and first cohomology of the normal sheaf. The tangent identification is theorem 2.4.4. For the obstruction, a first-order deformation $\widetilde{Z} \subseteq Y_\varepsilon$ is a section of $N_{Z/Y}$; to lift it across a small extension $A' \to A$ one perturbs the local equations of Z, and the mismatch on overlaps is a Čech 1-cocycle with values in $N_{Z/Y}$, whose class in $H^1(Z, N_{Z/Y})$ vanishes exactly when the local perturbations glue. Both groups are finite-dimensional because $N_{Z/Y}$ is coherent on the projective scheme Z. When Z is a smooth complete intersection in $Y = \mathbb{P}^n$, the normal sheaf $N_{Z/Y} = \bigoplus_i \mathcal{O}_Z(d_i)$ splits as a sum of positive line bundles (from example 2.4.2 applied to each equation), and for $\dim Z \geq 2$ the arithmetic Cohen-Macaulayness of a complete intersection gives $H^1(Z, \mathcal{O}_Z(d_i)) = 0$ (the intermediate cohomology of $\mathcal{O}_Z(m)$ vanishes for every m), whence $H^1(Z, N_{Z/Y}) = 0$: the complete intersection is unobstructed, and Hilb^P_Y is smooth at $[Z]$ of dimension $\sum_i h^0(\mathcal{O}_Z(d_i))$. This is the local structure of the Hilbert scheme, read entirely off the deformation theory.

The example shows the deformation theory of a moduli functor is not new work but a repackaging of the cohomology already computed in Chapters 2 and 5. This is the point of phrasing the criteria in terms of T^1 and T^2: whoever can compute the tangent and obstruction cohomology of the objects has, without further effort, the deformation and obstruction theory the criteria require.

With the deformation-theoretic input named, everything assembles into Artin's theorem. The statement is a list of five conditions, numbered (0) through (4) to match their logical roles: (0) and (1) are the "the functor is reasonable" conditions (a stack, and one of finite presentation), (2) is the representability of the diagonal, (3) is the deformation-theoretic input with effectivity, and (4) is openness of versality. The theorem asserts their conjunction is equivalent to algebraicity.

Theorem 8.4.3: Artin's Representability Criteria

Let S be a locally Noetherian scheme and let

$$\mathcal{X} \longrightarrow (\mathbf{Sch}/S)$$

be a category fibered in groupoids, with structure morphism $\mathcal{X} \to S$ locally of finite presentation. Then $\mathcal{X}$ is an algebraic stack in the sense of definition 8.3.1 if and only if the following conditions hold.

(0) *Stack.* $\mathcal{X}$ is a stack for the fppf topology (equivalently, since the objects are S-schemes, for the étale topology): isomorphisms form a sheaf and descent data are effective.

(1) *Limit-preserving.* $\mathcal{X}$ is locally of finite presentation as a fibered category: for every filtered colimit $A = \varinjlim A_i$ of $\mathcal{O}_S$-algebras the natural functor

$$\varinjlim_i \mathcal{X}(\operatorname{Spec} A_i) \longrightarrow \mathcal{X}(\operatorname{Spec} A)$$

is an equivalence of groupoids.

(2) *Representable diagonal.* The diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is representable by algebraic spaces; equivalently the $\underline{\mathrm{Isom}}$ sheaves of any two objects over a base are algebraic spaces. On formal deformations this is the assertion that the automorphism-and-deformation functor of each object satisfies Schlessinger's conditions (H1)-(H4) of theorem 5.3.5, so that each has a hull.

(3) *Effective, coherent obstruction theory.* $\mathcal{X}$ admits a deformation and obstruction theory in the sense of definition 8.4.1, with finite-dimensional $T^1_{\mathcal{X}}(x)$ and $T^2_{\mathcal{X}}(x)$ depending in a coherent (constructible, base-change-compatible) way on x, and every formal object is *effective*: for each complete local Noetherian $\mathcal{O}_S$-algebra R with residue field k, the natural map

$$\mathcal{X}(\operatorname{Spec} R) \longrightarrow \varprojlim_n \mathcal{X}(\operatorname{Spec} R/\mathfrak{m}_R^{n+1})$$

is an equivalence, so a compatible system of objects over the Artinian quotients comes from an object over R itself.

(4) *Openness of versality.* The deformation and obstruction theory satisfies openness of versality: a formally versal deformation carried by a finite-type S-scheme is versal on an open neighborhood.

The theorem is due to Artin [2], building on the algebraization theorem of [1]; the modern formulation, with the coherence condition on the obstruction theory made precise, is that of Hall and Rydh [16], and a fully detailed treatment is in the Stacks Project [29]. Because the criteria characterize an intricate geometric property through purely functorial and infinitesimal data, they are the working tool by which every moduli stack in practice is shown to be algebraic: one never constructs an atlas by hand, one checks the list.

The full proof of the forward direction, that the criteria imply algebraicity, requires Artin approximation and Néron-Popescu desingularization, which are beyond the present scope. The deferral is warranted by the depth of the input: approximation is a theorem in commutative algebra about the density of algebraic solutions among formal ones over an excellent base, and its proof through the desingularization of a regular map of Noetherian rings is a long development in its own right (see [1], [16], [29], and for the commutative-algebra input [6]). What can and should be made precise is the mechanism by which the criteria assemble an atlas, since that mechanism is the entire conceptual content and is exactly what the verifications in the sequel exploit.

***Proof Key ideas*:**
The proof that Artin's criteria imply algebraicity constructs a smooth atlas point by point and then spreads it out; the analytic input, algebraization of the hull, is cited. The converse, that an algebraic stack satisfies the criteria, is comparatively direct.

Step 1: The criteria produce a hull at each point. Fix a point $x \in \mathcal{X}(k)$ over a residue field k of S. The deformation functor

$$D_x : \mathbf{Art}_k \longrightarrow \mathbf{Set}, \qquad D_x(A) = \{\text{objects of } \mathcal{X}(A) \text{ lifting } x\}/\cong$$

has $D_x(k) = \{*\}$. Condition (3) gives it a tangent space $T^1_{\mathcal{X}}(x)$, finite-dimensional, so Schlessinger's finiteness (H3) holds; condition (2) gives the fiber-product conditions (H1), (H2), (H4) through the representability of the diagonal (the Isom sheaves being algebraic spaces is what makes the automorphism-quotients form algebraic fiber products, which is the abstract form of Schlessinger's gluing axioms). By theorem 5.3.5, D_x has a hull: a complete local $\mathcal{O}_S$-algebra R_x with a formal versal deformation $\widehat{\xi}_x$, whose tangent space is $T^1_{\mathcal{X}}(x)$ and whose obstructions are captured by $T^2_{\mathcal{X}}(x)$. Concretely R_x is a quotient of the power series ring $\mathcal{O}_{S,s}[\![t_1, \ldots, t_d]\!]$, $d = \dim T^1_{\mathcal{X}}(x)$, by an ideal generated by $\dim T^2_{\mathcal{X}}(x)$ obstruction equations.

Step 2: The hull is effective and algebraizes. The hull $\widehat{\xi}_x$ is a compatible system of objects over the Artinian quotients $R_x/\mathfrak{m}^{n+1}$. Effectivity (the second half of condition (3)) promotes this system to a genuine object $\widehat{\xi}_x \in \mathcal{X}(\operatorname{Spec} R_x)$ over the complete local ring itself. This is where the formal becomes almost-algebraic: an object over a complete local ring, not yet over a finite-type scheme. *Artin approximation* now closes the gap. Because $\mathcal{X}$ is limit-preserving (condition (1)) and $\mathcal{X} \to S$ is locally of finite presentation, an object over the completion R_x can be approximated to any finite order by an object over a finite-type $\mathcal{O}_S$-algebra: there exist a finite-type $\mathcal{O}_S$-algebra B, a point b with $\mathcal{O}_{U,b}$ Henselian and $\widehat{\mathcal{O}_{U,b}} \cong R_x$, and an object $\xi_x \in \mathcal{X}(U)$, $U = \operatorname{Spec} B$, whose induced formal deformation at b agrees with $\widehat{\xi}_x$. This is the step whose proof rests on Néron-Popescu desingularization and is cited to [1], [16], [29]. The map $\xi_x : U \to \mathcal{X}$ is, at b, formally smooth of relative dimension controlled by $T^1_{\mathcal{X}}(x)$: it is a formally versal family.

Step 3: Openness of versality spreads the atlas. The map $\xi_x : U \to \mathcal{X}$ is versal at the single point b. Condition (4), openness of versality, provides an open neighborhood $b \in V_x \subseteq U$ on which ξ_x is versal at every point. Versality at every point of V_x is precisely the statement that the induced morphism $V_x \to \mathcal{X}$ is smooth (formal smoothness plus finite presentation is smoothness): a formally versal family over a scheme, versal at all points, is a smooth map to the stack. This is the key transition, and it is what condition (4) exists to guarantee. Without it ξ_x would be smooth only at b, giving no open piece of an atlas.

Step 4: Assembling the global atlas. Carrying out Steps 1-3 at every point x produces smooth maps $V_x \to \mathcal{X}$ from finite-type schemes, one through each point of $\mathcal{X}$. Their disjoint union

$$U = \coprod_x V_x \longrightarrow \mathcal{X}$$

is smooth (smoothness is checked componentwise) and surjective (every point of $\mathcal{X}$ lies in the image of some V_x, by construction). This is a smooth atlas. The representability of the diagonal, condition (2), is the other half of definition 8.3.1; combined with the smooth surjective atlas, it certifies $\mathcal{X}$ as an algebraic stack. If moreover the automorphism groups are unramified, the atlas can be taken étale and $\mathcal{X}$ is Deligne-Mumford (theorem 8.2.3).

Step 5: The converse. Suppose $\mathcal{X}$ is already algebraic with a smooth atlas $p\colon U \to \mathcal{X}$. Condition (0) is part of the definition of a stack. Condition (2) is the representability of the diagonal, again part of the definition. For condition (1), limit-preservation, one uses that U and the groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$ are of finite presentation over S (an atlas of an algebraic stack of finite presentation can be so chosen), and finite-presentation schemes commute with filtered colimits of rings by the standard limit formalism, which passes to the stack quotient. For condition (3), the deformation and obstruction theory is read off the atlas: the tangent space of $\mathcal{X}$ at $x = p(u)$ is the quotient of $T_u U$ by the image of the tangent space to the automorphism group (the fiber of $R \to U \times_S U$), and the obstruction theory descends from the smooth U, where it is the classical tangent-obstruction theory of theorem 5.1.2; effectivity holds because objects over a complete local ring are pulled back from the atlas through Grothendieck's existence theorem (section 2.3 and [12]). Condition (4), openness of versality, holds because smoothness of p is an open condition on U, so a versal family is versal on an open. Thus every algebraic stack satisfies the criteria, completing the proof.

The proof is the conceptual center of the chapter, and the division of labor in it is worth restating. Conditions (0) and (1) ensure the functor is manageable, a stack of finite presentation, so that the local-to-global and finite-order arguments have something to bite on. Condition (2) is one of the two defining data of an algebraic stack, the representable diagonal, and it feeds Schlessinger's theorem to produce a hull. Condition (3) supplies the deformation theory that gives the hull its shape and the effectivity that turns a formal system into an object over a complete ring. Artin approximation, the cited black box, algebraizes that object into a family over a finite-type scheme. Condition (4) then spreads the versality of that family from one point to an open, and the opens cover the stack. The single hardest step, and the only one deferred, is the algebraization; the rest is the deformation theory of Part II applied uniformly.

A reader who has followed Chapter 5 will recognize that conditions (2) and (3) are almost exactly Schlessinger's criteria promoted from a single deformation functor to a fibered category. Schlessinger asked when a functor of Artin rings has a hull; Artin asks the same of the deformation functor at every point, adds effectivity so the hull is an actual object, and adds openness so the pointwise hulls glue. The criteria are "Schlessinger with algebraization and spreading-out", and that is the precise sense in which they close the loop from Part II's formal deformation theory to Part III's algebraic stacks.

The payoff is the theorem the book has been building toward since Chapter 1: the moduli of curves is an algebraic stack, and so, recovered as a special case, are the Hilbert and Quot functors. The verification is a matter of checking the five conditions, and every one of them is discharged by material already developed, modulo the single cited approximation step.

Theorem 8.4.4: $\mathcal{M}_g$ and the Hilbert and Quot Functors Are Algebraic

Fix $g \geq 2$ and work over $S = \operatorname{Spec} \mathbb{Z}$ (or over a field). Then:

1. the moduli stack $\mathcal{M}_g$ of smooth projective curves of genus g, and its Deligne-Mumford compactification $\overline{\mathcal{M}}_g$ of stable curves, are algebraic stacks, indeed Deligne-Mumford stacks, smooth of dimension $3g-3$ over S;
2. the Hilbert functor $\underline{\mathrm{Hilb}}_Y^P$ (definition 2.1.6) and the Quot functor $\underline{\mathrm{Quot}}_{\mathcal{F}/Y}^P$ of Chapter 2 are algebraic; being functors of sets (their objects have no nontrivial automorphisms), they are algebraic spaces, and in fact schemes.

***Proof*:**
The proof checks the five criteria of theorem 8.4.3 for $\mathcal{M}_g$; the Hilbert and Quot cases are the same argument with the automorphisms absent, and are treated in example 8.4.5. The one cited input is Artin approximation, invoked inside theorem 8.4.3 and not re-opened here.

Step 1: $\mathcal{M}_g$ is a stack of finite presentation (conditions (0) and (1)). That $\mathcal{M}_g$ is a stack for the fppf topology is descent for families of smooth curves: a curve over a base is a scheme proper and smooth with one-dimensional connected fibers of genus g, and such families descend along fppf covers because proper smooth morphisms and the numerical invariants defining them descend (theorem 6.4.3). This is condition (0). For condition (1), a smooth projective curve of genus g over $\operatorname{Spec} A$, with $A = \varinjlim A_i$ a filtered colimit, is a finitely presented object: it is cut out in some $\mathbb{P}^N_A$ by finitely many equations (a genus-g curve with $g \geq 2$ is tricanonically embedded by $\omega_C^{\otimes 3}$ in $\mathbb{P}^{5g-6}$, so pluricanonical embedding gives a uniform projective model), and a finitely presented scheme over a colimit ring descends to a finite stage A_i, uniquely up to a further stage. Hence $\varinjlim \mathcal{M}_g(\operatorname{Spec} A_i) \to \mathcal{M}_g(\operatorname{Spec} A)$ is an equivalence: $\mathcal{M}_g$ is limit-preserving. The uniform projective model is exactly the boundedness input, and it is theorem 2.2.5 applied to the tricanonical Hilbert polynomial $P(m) = (6m-1)(g-1)$ that guarantees all genus-g curves embed in one bounded family inside a single Hilbert scheme.

Step 2: The diagonal is representable (condition (2)). The diagonal of $\mathcal{M}_g$ encodes the $\underline{\mathrm{Isom}}$ sheaves: for two families of curves C, C' over a base T, the functor $\underline{\mathrm{Isom}}_T(C, C')$ assigns to $T' \to T$ the set of isomorphisms $C_{T'} \cong C'_{T'}$. For smooth projective curves this functor is representable by a scheme, quasi-projective and unramified over T: an isomorphism of curves is a point of the Hilbert scheme of the product $C \times_T C'$ (its graph), and the graphs of isomorphisms form a locally closed, unramified subscheme because a curve of genus $g \geq 2$ has only finitely many automorphisms (Hurwitz's bound $|\mathrm{Aut}(C)| \leq 84(g-1)$). Representability of $\underline{\mathrm{Isom}}$ is representability of the diagonal, and its unramifiedness is what will make $\mathcal{M}_g$ Deligne-Mumford rather than only Artin. On the formal side, the deformation-and-automorphism functor of a curve satisfies Schlessinger's (H1)-(H4) because the tangent space $H^1(C, T_C)$ is finite-dimensional and the automorphism scheme is finite and reduced, so theorem 5.3.5 applies and each curve has a hull.

Step 3: Deformation and obstruction theory, with effectivity (condition (3)). The deformation and obstruction theory of $\mathcal{M}_g$ at a curve $[C]$ is

$$T^1_{\mathcal{M}_g}([C]) = H^1(C, T_C), \qquad T^2_{\mathcal{M}_g}([C]) = H^2(C, T_C)$$

by theorem 5.1.2, with $T_C = \mathcal{O}_C(-K_C)$ the tangent sheaf. Both are finite-dimensional (coherent cohomology on a projective curve), so the coherence and finiteness parts of condition (3) hold; and because C is one-dimensional, $H^2(C, T_C) = 0$, so the obstruction space vanishes and $\mathcal{M}_g$ is unobstructed. By Riemann-Roch on the curve, $\dim H^1(C, T_C) = h^0(C, \Omega_C^{\otimes 2}) = 3g - 3$ for $g \geq 2$ (the tangent sheaf has degree $2 - 2g$, and $h^0(T_C) = 0$ since $\deg T_C < 0$, while $h^1(T_C) = -\deg T_C + g - 1 = 3g - 3$). Thus the tangent space to $\mathcal{M}_g$ has dimension $3g - 3$ everywhere and the obstruction vanishes, which already forces smoothness of dimension $3g - 3$ once algebraicity is known. Effectivity of formal deformations is Grothendieck's existence theorem (section 2.3, and [12]): a compatible system of curves over the Artinian quotients $R/\mathfrak{m}^{n+1}$ of a complete local ring R algebraizes to a formal family, and because the family is polarized (each curve carries the ample tricanonical bundle, whose sections give the projective embedding), Grothendieck's existence theorem promotes the formal family to an actual projective curve over $\operatorname{Spec} R$. Polarizability is what makes formal deformations of curves effective, and it is a feature of curves not shared by every deformation problem.

Step 4: Openness of versality (condition (4)*).* Openness of versality for $\mathcal{M}_g$ follows from the unobstructedness established in Step 3 together with the uppersemicontinuity of $H^1(C, T_C)$. Concretely, take a family of curves $\mathcal{C} \to U$ over a finite-type base with a point u at which the Kodaira-Spencer map $T_u U \to H^1(C_u, T_{C_u})$ is surjective (the condition for formal versality at u). Surjectivity of the Kodaira-Spencer map is an open condition on U: it is the non-vanishing of a minor of a map of coherent sheaves whose ranks are uppersemicontinuous, so the locus where the map is surjective is open. On that open the family is versal at every point, which is openness of versality. The vanishing of $H^2(C, T_C)$ is what makes formal versality equivalent to Kodaira-Spencer surjectivity in the first place, so the unobstructedness of Step 3 is again the load-bearing fact.

Step 5: Conclusion for $\mathcal{M}_g$ and its compactification. With all five criteria verified, theorem 8.4.3 certifies $\mathcal{M}_g$ as an algebraic stack. The diagonal is unramified (Step 2, finite reduced automorphism schemes), so by theorem 8.2.3 the atlas can be taken étale and $\mathcal{M}_g$ is Deligne-Mumford. Its smoothness and dimension $3g-3$ come from Step 3: the tangent space has constant dimension $3g - 3$ and the obstruction vanishes, so the atlas is smooth of that dimension and $\mathcal{M}_g$ inherits smoothness. The same argument applies to $\overline{\mathcal{M}}_g$ with the single modification that stable curves have nodes, so the deformation theory at a nodal curve is governed by Ext groups rather than $H^i(T_C)$ (the tangent sheaf is replaced by the sheaf $\mathcal{H}om(\Omega_C, \mathcal{O}_C)$ and a local smoothing parameter is added at each node, computed in Chapter 5 as the T^1 of a node); the obstruction still vanishes because a nodal curve remains one-dimensional and its deformation-obstruction theory has no degree-two term. The finite automorphisms of a stable curve keep the diagonal unramified, so $\overline{\mathcal{M}}_g$ is Deligne-Mumford, smooth of dimension $3g - 3$, and, as Chapter 10 will show, proper. This establishes part (1).

Step 6: The Hilbert and Quot functors. For $\underline{\mathrm{Hilb}}_Y^P$ and $\underline{\mathrm{Quot}}_{\mathcal{F}/Y}^P$ the same five checks succeed, and are simpler because the objects (closed subschemes, coherent quotients) have no nontrivial automorphisms, so the diagonal is trivial (an isomorphism of subschemes fixing the ambient Y is the identity), the $\underline{\mathrm{Isom}}$ sheaves are trivial, and the stack is a functor of sets. Conditions (0) and (1) are the sheaf property and finite presentation from Chapter 2; condition (2) holds trivially because the diagonal of a sheaf of sets is a monomorphism; condition (3) is the deformation theory $T^1 = H^0(N)$, $T^2 = H^1(N)$ of example 8.4.2, effective by Grothendieck's existence theorem; condition (4) is again openness of Kodaira-Spencer surjectivity. A stack that is a sheaf of sets with a trivial diagonal and a smooth atlas is an algebraic space, and a quasi-projective algebraic space with a trivial diagonal is a scheme, recovering the Grothendieck existence theorem of section 2.3 that Hilb_Y^P and $\mathrm{Quot}_{\mathcal{F}/Y}^P$ are projective schemes. This is part (2), completing the proof.

This theorem is the destination of the whole book, and it closes the loop that Part II opened. In Chapter 4 and Chapter 5 the deformation theory of curves was developed as a self-contained study of infinitesimal structure: $H^1(C, T_C)$ was the space of first-order deformations, $H^2(C, T_C)$ was where obstructions would live, and the vanishing $H^2 = 0$ on a curve made the moduli of curves unobstructed. At the time this was a statement about a formal deformation functor, with no geometric object it was the deformation theory *of*. Artin's criteria supply the missing object: they take exactly that deformation data, H^1 as tangent space, H^2 as obstruction space, $3g-3$ as the dimension, add the boundedness of theorem 2.2.5 to make the family finite-type and the effectivity of Grothendieck existence to make formal deformations algebraize, and conclude that a smooth Deligne-Mumford stack $\mathcal{M}_g$ of dimension $3g-3$ exists. The deformation theory was never idle: it was the local model of a global object that only the criteria could produce.

The single dependence on machinery beyond this book is worth locating precisely. Every condition except the algebraization inside Artin approximation was verified from Chapters 1 through 7: descent (Chapter 6) for the stack property, boundedness (Chapter 2) for finite presentation, Hilbert-scheme representability (Chapter 2) for the diagonal, the tangent-obstruction theory (Chapter 5) for the deformation theory, and Grothendieck existence (Chapter 2) for effectivity. Only the passage from an object over a complete local ring to an object over a finite-type scheme, the algebraization step, uses Néron-Popescu desingularization from outside. This is the boundary of a first course in the subject: the geometry and the deformation theory are internal, the approximation theorem is imported.

The abstract verification acquires its full meaning when carried out on a single functor where every group can be named. The Hilbert functor is the ideal test case: its objects have no automorphisms, so the diagonal degenerates and the criteria reduce to their sheaf-of-sets form, and one sees exactly how Artin's list recovers the concrete representability theorem of Chapter 2.

Example 8.4.5: Verifying the Criteria for the Hilbert Functor
Fix a projective scheme $Y \subseteq \mathbb{P}_k^n$ and a Hilbert polynomial P, and check the five conditions of theorem 8.4.3 for $\underline{\mathrm{Hilb}}_Y^P$ (definition 2.1.6), a functor of sets.

1. *Condition* (0)*, stack.* A functor of sets is a stack for the fppf topology exactly when it is an fppf sheaf. For $\underline{\mathrm{Hilb}}_Y^P$ this is fppf descent for flat families of closed subschemes: a subscheme $Z \subseteq Y_T$ flat over T with fiberwise Hilbert polynomial P is determined by its ideal sheaf, and quasi-coherent sheaves with descent data glue (theorem 6.3.4), the flatness and Hilbert polynomial descending because they are checked on fibers. So $\underline{\mathrm{Hilb}}_Y^P$ is an fppf sheaf.
2. *Condition* (1)*, limit-preserving.* A flat family of subschemes over $\operatorname{Spec} A$, $A = \varinjlim A_i$, is cut out by finitely many equations of bounded degree (the ideal is generated in degrees $\leq m_0$, where m_0 is the uniform regularity bound of theorem 2.2.5), and finitely many equations over a colimit ring descend to a finite stage. Hence $\varinjlim \underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} A_i) \to \underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} A)$ is a bijection: limit-preservation is boundedness plus finite-presentation descent. This is where theorem 2.2.5 does its work.
3. *Condition* (2)*, diagonal.* The objects are closed subschemes of a fixed Y; an isomorphism of two such subschemes over T compatible with the inclusion into Y_T is forced to be the identity, since both are the same closed subscheme. So the $\underline{\mathrm{Isom}}$ sheaf is trivial, the diagonal is a monomorphism, and it is representable trivially. This is the sense in which the Hilbert

functor has "no automorphisms": its diagonal is trivial, and this is precisely why it is representable by a scheme rather than only by a stack.

4. *Condition* (3)*, deformation theory.* At a point $[Z]$ the tangent space is $T^1 = H^0(Z, N_{Z/Y})$ (theorem 2.4.4) and the obstruction space is $T^2 = H^1(Z, N_{Z/Y})$ (example 8.4.2), both finite-dimensional. The Schlessinger conditions (H1)-(H4) of theorem 5.3.5 hold: the deformation functor of Z inside Y is $A \mapsto \{Z_A \subseteq Y_A \text{ flat}\}$, and it satisfies the fiber-product axioms because gluing flat subschemes over a fiber product of Artinian rings is unobstructed by the sheaf property. Effectivity is Grothendieck's existence theorem (section 2.3): a compatible system of subschemes $Z_n \subseteq Y_{R/\mathfrak{m}^{n+1}}$ algebraizes to a subscheme $Z \subseteq Y_R$ because the ambient Y is projective and the ideal sheaves form a compatible system with the required coherence.

5. *Condition* (4)*, openness of versality.* A family $Z \subseteq Y_U$ over a finite-type base is versal at a point u when the map $T_uU \to H^0(Z_u, N_{Z_u/Y})$ is surjective, an open condition by uppersemicontinuity of the ranks. On the versal open the family is a smooth (indeed, since the diagonal is trivial, étale-onto-image) map to the functor.

All five conditions hold, so $\underline{\mathrm{Hilb}}_Y^P$ is algebraic. Because its diagonal is trivial, it is an algebraic space; because it is bounded and carries an ample sheaf (it sits inside a Grassmannian, hence a projective space), it is a scheme. This recovers the Grothendieck existence theorem of section 2.3, that Hilb_Y^P is a projective scheme, now derived as a special case of Artin's criteria. The lesson is that the Hilbert scheme is the degenerate case of the theory, where trivial automorphisms collapse the stack to a scheme, and the criteria specialize to the concrete representability already proved by hand in Chapter 2.

The example makes visible what the criteria buy and where. For the Hilbert functor, conditions (0) and (1) are descent and boundedness, condition (2) is vacuous because there are no automorphisms, condition (3) is the normal-sheaf cohomology $H^0(N)$ and $H^1(N)$, and condition (4) is the openness of a surjectivity. The single feature that separates the Hilbert functor from $\mathcal{M}_g$ is the diagonal: trivial for subschemes, unramified-but-nontrivial for curves. That difference is the whole difference between a scheme and a Deligne-Mumford stack, and Artin's criteria absorb it into a single condition on $\underline{\mathrm{Isom}}$, treating the representable and the stacky moduli problems by one uniform mechanism.

Exercise 8.4.1

1. Let $\mathcal{X}$ be an algebraic stack over a locally Noetherian S, of finite presentation, with a smooth atlas $U \to \mathcal{X}$. Show directly that $\mathcal{X}$ is limit-preserving (condition (1) of theorem 8.4.3), by reducing the claim for $\mathcal{X}$ to the corresponding claim for the finite-presentation schemes U and $R = U \times_{\mathcal{X}} U$ and using that a finitely presented scheme over $\varinjlim A_i$ descends to a finite stage. Explain in one sentence why finite presentation of the atlas is the hypothesis that makes this work.

2. For a smooth projective curve C of genus $g \geq 2$, use Riemann-Roch and Serre duality to compute $h^0(C, T_C)$ and $h^1(C, T_C)$, where $T_C = \mathcal{O}_C(-K_C)$ is the tangent sheaf. Conclude that $T^1_{\mathcal{M}_g}([C]) = H^1(C, T_C)$ has dimension $3g - 3$ and $T^2_{\mathcal{M}_g}([C]) = H^2(C, T_C) = 0$, and interpret the vanishing $h^0(T_C) = 0$ in terms of the automorphisms of C.

3. Explain why openness of versality (condition (4)) is not automatic from the other four conditions, by describing what could go wrong: give the geometric picture of a family that is formally versal at one point u but not at nearby points, and explain why, without condition

(4), the algebraized hulls of the proof of theorem 8.4.3 would fail to assemble into a smooth atlas. (You may argue heuristically; a rigorous counterexample is beyond the scope.)

4. Let $Z \subseteq \mathbb{P}^n$ be a smooth complete intersection of hypersurfaces of degrees $d_1, \dots, d_c$ with $\dim Z \geq 2$. Using $N_{Z/\mathbb{P}^n} = \bigoplus_{i=1}^c \mathcal{O}_Z(d_i)$ (from example 2.4.2), compute the tangent and obstruction spaces $H^0(Z, N_{Z/\mathbb{P}^n})$ and $H^1(Z, N_{Z/\mathbb{P}^n})$ in the sense of example 8.4.2, and use that a complete intersection is arithmetically Cohen-Macaulay to deduce that the Hilbert scheme is smooth at $[Z]$. Then treat the plane curve $Z \subseteq \mathbb{P}^2$ of degree d (a hypersurface, $c = 1$, $\dim Z = 1$) directly: compute the dimension of Hilb at $[Z]$ explicitly and compare it with the dimension $\binom{d+2}{2} - 1$ of the space of degree-d plane curves.

5. Condition (3) of theorem 8.4.3 demands that formal deformations be *effective*. Explain, using Grothendieck's existence theorem (section 2.3), why effectivity holds for $\mathcal{M}_g$ (families of curves) but requires the presence of an ample line bundle (the tricanonical polarization). Contrast with the stack $\mathfrak{Coh}_X$ of coherent sheaves, where effectivity of formal deformations of a non-projective object is a hypothesis to be checked rather than free.

6. Reconcile the two ways this book has produced the Hilbert scheme: the direct construction of Chapter 2 (as a locally closed subscheme of a Grassmannian, via boundedness and the flattening stratification) and the derivation of example 8.4.5 (as a special case of Artin's criteria). Which inputs are shared between the two, and which is logically stronger? Explain in particular why the Chapter 2 construction is needed for the criteria to apply to $\mathcal{M}_g$ at all, so that the two approaches are not two proofs of one fact but stand in a dependency relation.

8.5 Geometry on a Stack

The previous four sections built the objects: an algebraic space is a sheaf étale-locally a scheme (definition 8.1.3), a Deligne-Mumford stack carries an étale atlas (definition 8.2.1), an Artin stack a smooth one (definition 8.3.1), and Artin's criteria (theorem 8.4.3) certify when a moduli problem lands among them. What remains is to do geometry on these objects. A scheme comes with a structure sheaf, quasi-coherent modules, line bundles and a Picard group, sheaf cohomology, and adjectives like smooth, normal, separated, and proper. This section transports each of these to an algebraic stack. The mechanism is uniform and was prefigured throughout the chapter: a property or a structure on a stack $\mathcal{X}$ is a property or structure on an atlas $U \to \mathcal{X}$ that is compatible with the groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$ presenting $\mathcal{X}$. Descent (theorem 6.3.4, theorem 6.4.3) is exactly the guarantee that such atlas-level data glue to genuine data on the stack.

Two registers run in parallel. Quasi-coherent sheaves, the Picard group, and cohomology are *structures*: they are computed on the atlas and assembled by descent along $R \rightrightarrows U$. Adjectives split into two kinds. The atlas-local ones (smooth, normal, regular, locally of finite type) are read off any atlas because they are smooth-local on the source (definition 7.4.3); the separatedness-type ones (separated, proper) are conditions on the *diagonal* $\Delta_{\mathcal{X}}$, because separatedness of a space is a property of its diagonal, and a stack's diagonal encodes its $\underline{\mathrm{Isom}}$ sheaves. The section closes with the capstone the whole book has aimed at: $\overline{\mathcal{M}}_g$ is smooth and proper as a stack even though its coarse space $\overline{M}_g$ is singular. Smoothness is read on the atlas, properness from the valuative criterion, and neither survives the passage to the coarse space, which is why the stack is the right object.

Throughout, $\mathcal{X}$ is an algebraic stack over a base S with a presentation $R \rightrightarrows U$ by a groupoid of schemes, $s, t \colon R \to U$ the source and target maps (étale for a DM stack, smooth for an Artin stack), and $U \to \mathcal{X}$ the atlas. The two projections $\mathrm{pr}_1, \mathrm{pr}_2 \colon R \to U$ of a groupoid presentation are s and t; the composition law is a map $m \colon R \times_{s,U,t} R \to R$.

Quasi-coherent sheaves are the first structure to transport, and the guiding principle is that a sheaf on $\mathcal{X}$ should be a sheaf on the atlas that does not depend on which point of the atlas one uses to see a given point of $\mathcal{X}$. Two points $u_1, u_2 \in U$ mapping to the same point of $\mathcal{X}$ are joined by a point of R, and asking a sheaf to be insensitive to the choice is asking for descent data along $R \rightrightarrows U$. This is the same descent that Chapter 6 proved for the fppf topology, now organized by a groupoid rather than a cover.

Definition 8.5.1: Quasi-Coherent Sheaves on a Stack

Let $\mathcal{X}$ be an algebraic stack with presentation $R \rightrightarrows U$, source and target $s, t \colon R \to U$. A *quasi-coherent sheaf* $\mathcal{F}$ on $\mathcal{X}$ is a quasi-coherent sheaf $\mathcal{F}_U$ on the atlas U together with an isomorphism of quasi-coherent sheaves on R

$$\varphi \colon s^*\mathcal{F}_U \xrightarrow{\sim} t^*\mathcal{F}_U$$

the *descent datum*, satisfying the cocycle condition $\mathrm{pr}_{13}^*\varphi = \mathrm{pr}_{23}^*\varphi \circ \mathrm{pr}_{12}^*\varphi$ on $R \times_{s,U,t} R$ and the normalization $e^*\varphi = \mathrm{id}$ along the identity section $e \colon U \to R$. A *morphism* $\mathcal{F} \to \mathcal{G}$ of quasi-coherent sheaves is a morphism $\mathcal{F}_U \to \mathcal{G}_U$ on U compatible with the descent data. The category of such is $\mathrm{QCoh}(\mathcal{X})$. The *structure sheaf* $\mathcal{O}_{\mathcal{X}}$ is $(\mathcal{O}_U, \mathrm{can})$, with the canonical identification $s^*\mathcal{O}_U \cong \mathcal{O}_R \cong t^*\mathcal{O}_U$.
Equivalently, $\mathrm{QCoh}(\mathcal{X})$ is the category of quasi-coherent sheaves on the *lisse-étale site* $\mathcal{X}_{\text{lis-ét}}$, whose objects are smooth morphisms $V \to \mathcal{X}$ from schemes and whose coverings are smooth surjections; a quasi-coherent sheaf assigns to each such $V \to \mathcal{X}$ a quasi-coherent sheaf on V, compatibly with smooth pullback.

The definition is descent made into a category, and its independence of the presentation is what makes it well posed. A different atlas $U' \to \mathcal{X}$ presents the same stack by a groupoid R', and the two presentations are related by a common refinement; the descent categories are equivalent because descent data glue and the equivalence is exactly theorem 6.3.4 applied to the two atlases against a common cover. The lisse-étale reformulation makes this coordinate-free: a quasi-coherent sheaf is a rule on *all* smooth schemes over $\mathcal{X}$ at once, and any single atlas recovers it by restriction. The structure sheaf $\mathcal{O}_{\mathcal{X}}$ is the descent of $\mathcal{O}_U$, and a quasi-coherent sheaf is a module over it in the same atlas-plus-descent sense.

The first example is the one that makes the definition concrete and that will recur: on a classifying stack the descent datum is a group action, so a quasi-coherent sheaf is a representation.

Example 8.5.2: $\mathrm{QCoh}(BG)$ as Representations

Take G an affine group scheme over k and $BG = [\mathrm{pt}/G]$ (definition 7.3.2). The atlas is $\mathrm{pt} = \operatorname{Spec} k \to BG$, the universal torsor, and its groupoid is $R = \mathrm{pt} \times_{BG} \mathrm{pt} = G$, with $s, t \colon G \to \mathrm{pt}$ both the structure map and composition $m \colon G \times G \to G$ the multiplication. A quasi-coherent sheaf on BG is thus a k-vector space V (a quasi-coherent sheaf on pt) with a descent isomorphism $\varphi \colon s^*V \to t^*V$ on G, that is, a k-linear map

$$\varphi \colon V \otimes_k \mathcal{O}_G \xrightarrow{\sim} V \otimes_k \mathcal{O}_G$$

of $\mathcal{O}_G$-modules, subject to the cocycle condition on $G \times G$ and the normalization at the identity. Unwinding, φ is a comodule structure $V \to V \otimes_k \mathcal{O}_G$ over the Hopf algebra $\mathcal{O}_G$: the cocycle condition is coassociativity and the normalization is counitality. A comodule over $\mathcal{O}_G$ is precisely a *representation* of G (a linear action $G \times V \to V$), so

$$\mathrm{QCoh}(BG) \simeq \mathrm{Rep}(G)$$

the category of (algebraic) G-representations. For $G = \mathbb{G}_m$, with $\mathcal{O}_G = k[u, u^{-1}]$, a comodule is a $\mathbb{Z}$-grading $V = \bigoplus_{n\in\mathbb{Z}} V_n$ (the n-th weight space is where u acts by u^n), so $\mathrm{QCoh}(B\mathbb{G}_m)$ is the category of $\mathbb{Z}$-graded vector spaces. The structure sheaf $\mathcal{O}_{BG}$ is the trivial representation k.

The identification $\mathrm{QCoh}(BG) = \mathrm{Rep}(G)$ is the paradigm for all stacky geometry: a geometric construction on the stack is a representation-theoretic construction on the group, and conversely. A vector bundle on BG is a finite-dimensional representation; the structure sheaf is the trivial representation; tensor product of sheaves is tensor product of representations. For a quotient stack $[X/G]$ the same holds with X present: $\mathrm{QCoh}([X/G])$ is the category of G-equivariant quasi-coherent sheaves on X, a quasi-coherent sheaf $\mathcal{F}$ on X with an isomorphism $\sigma^*\mathcal{F} \cong \mathrm{pr}_X^*\mathcal{F}$ satisfying the cocycle condition, which is a G-linearization in the sense of the theory of Chapter 3. Equivariant geometry and stacky geometry are the same subject seen from two sides.

With sheaves in hand, line bundles and cohomology follow at once, since a line bundle is a quasi-coherent sheaf that is locally free of rank one and cohomology is the derived functor of global sections. The subtlety is that on a stack "global sections" means sections invariant under the groupoid, so $H^0(\mathcal{X}, \mathcal{F})$ is not $H^0(U, \mathcal{F}_U)$ but the R-invariant part, and the higher cohomology is the derived functor of this invariants functor. The two computational models, Čech on the groupoid and the lisse-étale site, agree, and each is useful.

Definition 8.5.3: Picard Group and Cohomology of a Stack

Let $\mathcal{X}$ be an algebraic stack with presentation $R \rightrightarrows U$.

1. The *Picard group* $\mathrm{Pic}\,(\mathcal{X})$ is the group of isomorphism classes of invertible sheaves (line bundles) on $\mathcal{X}$ under tensor product. By definition 8.5.1 a line bundle is a line bundle L on U with a descent datum $\varphi\colon s^*L \cong t^*L$ satisfying the cocycle condition, so

$$\mathrm{Pic}\,(\mathcal{X}) = \{(L, \varphi) : L \in \mathrm{Pic}\,(U),\ \varphi \text{ a groupoid descent datum}\}/\cong$$

2. For $\mathcal{F} \in \mathrm{QCoh}(\mathcal{X})$, the *cohomology* $H^i(\mathcal{X}, \mathcal{F})$ is the i-th right derived functor of the global sections functor $\Gamma(\mathcal{X}, -) = \mathrm{Hom}_{\mathrm{QCoh}(\mathcal{X})}(\mathcal{O}_{\mathcal{X}}, -)$, the functor of R-invariant sections. Equivalently it is the cohomology of $\mathcal{F}$ on the lisse-étale site $\mathcal{X}_{\text{lis-ét}}$.

When the nerve pieces U, R, $R \times_U R$, ... are all affine (as they are for BG, $B\mu_n$, $B\mathbb{G}_m$, or any quotient by an affine group, so that the fiber products are affine), $H^i(\mathcal{X}, \mathcal{F})$ is computed by the *Čech complex of the groupoid*, the cochain complex

$$\Gamma(U, \mathcal{F}_U) \longrightarrow \Gamma(R, s^*\mathcal{F}_U) \longrightarrow \Gamma(R \times_{s,U,t} R, \dots) \longrightarrow \cdots$$

whose p-th term is the global sections over the p-fold fiber product $R \times_U \cdots \times_U R$ (the degree-p piece of the nerve of the groupoid) and whose differential is the alternating sum of the simplicial face maps; its i-th cohomology is $H^i(\mathcal{X}, \mathcal{F})$.

The definition says cohomology on a stack is the cohomology of the *simplicial scheme* (the nerve of the groupoid), and this is what makes stacky cohomology computable and what connects it to classical constructions. For an affine scheme $\mathcal{X} = X$ presented by the trivial groupoid $X \rightrightarrows X$, the Čech complex collapses to $\Gamma(X, \mathcal{F})$ in degree zero and the definition returns ordinary sheaf cohomology ($H^{>0}$ vanishing on an affine); for a general scheme the derived-functor definition still returns ordinary sheaf cohomology, but the trivial-groupoid Čech complex sees only $\Gamma(X, \mathcal{F}) = H^0$ and one must resolve $\mathcal{F}$ (or pass to an affine atlas) to recover the higher cohomology. For a quotient stack the nerve is the bar construction of the action, and the cohomology is *equivariant* cohomology. For BG the nerve is the bar construction of G, and $H^i(BG, \mathcal{F})$ is the group cohomology (or, in the algebraic category, the derived-invariants $\mathrm{Ext}^i_{\mathrm{Rep}(G)}(k, V)$ of the representation $V = \mathcal{F}$) of G. The Picard group similarly becomes an equivariant Picard group, the group of G-linearized line bundles.

The computation that anchors this definition, and that will drive the examples, is the Picard group of $B\mathbb{G}_m$. It is the first place where a stack's Picard group differs from that of its coarse space (a point, with trivial Picard group), and the answer is dictated entirely by the representation theory of $\mathbb{G}_m$.

Proposition 8.5.4: The Picard Group of $B\mathbb{G}_m$

Over a field or local ring k, $\mathrm{Pic}\,(B\mathbb{G}_m) \cong \mathbb{Z}$ (over a general base ring the underlying rank-one modules contribute an extra factor $\mathrm{Pic}\,(k)$, giving $\mathrm{Pic}\,(B\mathbb{G}_m) \cong \mathrm{Pic}\,(k) \times \mathbb{Z}$). The isomorphism sends an integer n to the line bundle $\mathcal{O}(n)$ corresponding, under example 8.5.2, to the one-dimensional representation of $\mathbb{G}_m$ of weight n, and the weight is a complete invariant of a line bundle on $B\mathbb{G}_m$.

***Proof*:**
By example 8.5.2, $\mathrm{QCoh}(B\mathbb{G}_m)$ is the category of $\mathbb{Z}$-graded k-modules, with $\mathcal{F}$ corresponding to a $\mathbb{Z}$-graded module $V = \bigoplus_n V_n$. A line bundle is an invertible object of this category under the tensor product $(V \otimes W)_n = \bigoplus_{p+q=n} V_p \otimes W_q$. The invertible graded modules are exactly the modules that are free of rank one and concentrated in a single degree: if $V \otimes W \cong k$ (the unit, concentrated in degree 0), then comparing graded pieces forces V to be an invertible k-module concentrated in one degree n and W its inverse concentrated in degree $-n$. Over a general base k the underlying rank-one module contributes $\mathrm{Pic}\,(k)$; over a field or local ring it is free, and the only invariant is the degree $n \in \mathbb{Z}$. The tensor product of the degree-n line bundle with the degree-m one is concentrated in degree $n + m$, so $n \mapsto \mathcal{O}(n)$ is a group isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathrm{Pic}\,(B\mathbb{G}_m)$, and the weight n determines the bundle, as we sought to show.

The proposition is the smallest instance of a stack seeing more than its coarse space. The coarse space of $B\mathbb{G}_m$ is a single point, whose Picard group is trivial, yet $\mathrm{Pic}\,(B\mathbb{G}_m) = \mathbb{Z}$: the stack carries a whole $\mathbb{Z}$ of line bundles, indexed by the weight of the $\mathbb{G}_m$-action, none of which descend to the point. This is the line-bundle shadow of the automorphism group $\mathbb{G}_m$ that every point of $B\mathbb{G}_m$ carries. The general pattern is $\mathrm{Pic}\,(BG) = \widehat{G} = \mathrm{Hom}(G, \mathbb{G}_m)$, the character group of G, because a line bundle on BG is a one-dimensional representation and a one-dimensional representation is a character. The same mechanism gives $\mathrm{Pic}\,(B\mu_n) = \mathbb{Z}/n$ and $\mathrm{Pic}\,(BGL_n) = \mathbb{Z}$ (via the determinant character), so the Picard group of a classifying stack is a direct read-out of the character theory of the group.

The last structure to transport is the vocabulary of adjectives. A scheme is smooth, normal, separated, or proper; each of these should have a meaning for a stack, and the meaning is governed by a single dichotomy. Some properties are *smooth-local on the source*: they hold for a scheme if and only if they hold after a smooth surjective base change, and such a property is read off any atlas. Others are properties of the *diagonal*, and separatedness is the prototype, since a scheme is separated exactly when its diagonal is a closed immersion. On a stack the diagonal carries the Isom sheaves, so diagonal conditions are conditions on how objects are glued and how their automorphisms behave.

Definition 8.5.5: Geometric Properties of an Algebraic Stack

Let $\mathcal{X}$ be an algebraic stack with atlas $U \to \mathcal{X}$, presentation $R \rightrightarrows U$, and diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$.

1. *Atlas-local properties.* $\mathcal{X}$ is *smooth*, *regular*, *normal*, *reduced*, *locally of finite type*, or of a fixed *dimension* over S if the atlas U has the corresponding property. Each of these properties P is smooth-local on the source, so by definition 7.4.3 and theorem 6.4.3 the truth of P for U is independent of the atlas and defines P for $\mathcal{X}$.

2. *Diagonal conditions.* $\mathcal{X}$ is *quasi-separated* (resp. *separated*) if $\Delta_{\mathcal{X}}$ is quasi-compact and quasi-separated (resp. proper); equivalently $\mathcal{X}$ is separated when the Isom sheaves are proper, so automorphisms and isomorphisms are parametrized by proper algebraic spaces. $\mathcal{X}$ is *Deligne-Mumford* (resp. *Artin*) according to whether $\Delta_{\mathcal{X}}$ is unramified (resp. representable), by theorem 8.2.3.

3. *Properness.* $\mathcal{X}$ is *proper* over S if it is separated, of finite type, and *universally closed*, the last verified by the *valuative criterion for stacks*: for every discrete valuation ring A with fraction field K, every 2-commutative square

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ \operatorname{Spec} A & \longrightarrow & S \end{array}$$

admits, after a finite extension of A, a lift $\operatorname{Spec} A \to \mathcal{X}$, and any two lifts agreeing over $\operatorname{Spec} K$ are related by a unique isomorphism (no extension needed for the uniqueness clause); existence of the lift after a finite extension gives universal closedness (hence properness together with separatedness and finite type), and uniqueness up to unique isomorphism gives separatedness.

The dichotomy in the definition is the organizing fact of stacky geometry, and the reason properties split is instructive. Smoothness, normality, and regularity are conditions on the *local rings* of a scheme, and smooth morphisms induce nice (regular, flat with regular fibers) maps on local rings, so these conditions transfer up and down a smooth atlas without loss. That is why they can be checked on any atlas. Separatedness and properness are conditions on *global* behavior, on how the object sits in a product with itself (the diagonal) and on limits of families (the valuative criterion), and these are exactly the places where a stack differs from a scheme because its points have automorphisms. The valuative criterion for a stack differs from the scheme version in one word: the lift is unique *up to unique isomorphism*, not literally unique, because the objects being lifted have automorphisms. This one modification is what will make $\overline{\mathcal{M}}_g$ proper while its coarse space is only proper after forgetting automorphisms.

The valuative criterion is abstract until one sees it as a completeness statement about families, so the grounding example is a family of curves degenerating over a punctured disk. The lift over $\operatorname{Spec} A$ is a limit of the family, and the criterion says the limit exists in the stack after a finite base change: this is stable reduction, the input to the capstone of the book.

Example 8.5.6: The Valuative Criterion as Stable Reduction

Let A be a discrete valuation ring with fraction field K and residue field κ, geometrically the local ring of a smooth curve at a point, $\operatorname{Spec} K$ the punctured disk and $\operatorname{Spec} \kappa$ the central point. A morphism $\operatorname{Spec} K \to \overline{\mathcal{M}}_g$ is a family of stable curves over the punctured disk, that is, a smooth (or stable) curve C_K of genus g over K. The valuative criterion for $\overline{\mathcal{M}}_g$ (definition 8.5.5(3)) asks whether this family extends to a stable curve over the whole disk $\operatorname{Spec} A$, filling in the central fiber. The *stable reduction theorem*, which Chapter 10 develops, asserts exactly this: after a finite extension $A \subseteq A'$ (a ramified base change, replacing the disk by a cover) the family C_K acquires a unique stable limit $C_{A'}$ over $\operatorname{Spec} A'$, a stable curve whose central fiber is a nodal degeneration of the generic one. Existence of the limit is universal closedness, so $\overline{\mathcal{M}}_g$ is universally closed; the finite extension is the "up to a finite extension of A" clause of the criterion, forced because the limiting curve may acquire automorphisms not present over K; and the uniqueness of the stable limit (up to unique isomorphism) is separatedness. Thus properness of $\overline{\mathcal{M}}_g$ is stable reduction, translated into the valuative criterion.

The example is the mechanism by which the geometry of the moduli stack is governed by the geometry of degenerating curves, and it explains why the boundary $\overline{\mathcal{M}}_g \setminus \mathcal{M}_g$ of nodal curves is exactly what properness requires: a family of smooth curves can degenerate, and for the moduli space to be proper the degenerate limits must be points of it, which forces the boundary to consist of nodal (stable) curves. The reason a finite extension of A is unavoidable is worth isolating: over K a curve may have a monodromy that only becomes trivial after a base change, and the stable limit lives over the cover; this is precisely the failure of the coarse space to be proper in the naive sense and the reason a stack, whose valuative criterion tolerates a finite extension, is the correct object.

All the pieces are now in place to compute, and the examples below run from the simplest classifying stacks to the capstone. They are chosen to display each of the three registers: the Picard group and cohomology of $B\mathbb{G}_m$ and $B\mu_n$ (structures via descent), the tangent and canonical bundles of a quotient stack (equivariant geometry), and the smoothness and properness of $\overline{\mathcal{M}}_g$ (adjectives via atlas and diagonal). The last is the statement the entire book has aimed at.

Example 8.5.7: Picard Groups, Cohomology, Tangent Bundles, and $\overline{\mathcal{M}}_g$

1. *The weight of a line bundle on $B\mathbb{G}_m$.* By proposition 8.5.4, $\operatorname{Pic}(B\mathbb{G}_m) = \mathbb{Z}$, the isomorphism sending a line bundle to its *weight*, the integer n such that the corresponding one-dimensional representation of $\mathbb{G}_m$ is $t \mapsto t^n$. Concretely, the tautological line bundle $\mathcal{O}(1)$ on $B\mathbb{G}_m$ (the universal one-dimensional representation, weight 1) generates, and $\mathcal{O}(n) = \mathcal{O}(1)^{\otimes n}$. A section of $\mathcal{O}(n)$ over $B\mathbb{G}_m$ is a $\mathbb{G}_m$-invariant element of the weight-n representation $k_{(n)}$, which is 0 for $n \neq 0$ and k for $n = 0$; hence $H^0(B\mathbb{G}_m, \mathcal{O}(n)) = 0$ for $n \neq 0$ and $= k$ for $n = 0$, so $\mathcal{O}(n)$ has no nonzero global sections unless it is trivial. This is the stacky version of the fact that a nontrivial character has no invariant vectors.

2. *Cohomology of $B\mu_n$.* Work over any field k. The group scheme μ_n is diagonalizable (the Cartier dual of $\mathbb{Z}/n$, of multiplicative type), so by the $\operatorname{QCoh}(BG) = \operatorname{Rep}(G)$ dictionary (example 8.5.2) its representations are exactly the $\mathbb{Z}/n$-graded k-vector spaces $V = \bigoplus_{m \in \mathbb{Z}/n} V_m$, and $H^i(B\mu_n, \mathcal{F}) = \operatorname{Ext}^i_{\operatorname{Rep}(\mu_n)}(k, V)$ for $V = \mathcal{F}$. The category of $\mathbb{Z}/n$-graded vector spaces is semisimple in every characteristic (a graded submodule is a direct summand, namely the sum of the graded pieces it meets), so μ_n is linearly reductive over any base, char $k | n$

included; all higher Ext vanish, and

$$H^i(B\mu_n, \mathcal{F}) = \begin{cases} V^{\mu_n} = V_0 & i = 0 \\ 0 & i > 0 \end{cases}$$

the invariants (the weight-0 part of the $\mathbb{Z}/n$-graded module V) in degree 0 and nothing above. In particular $H^i(B\mu_n, \mathcal{O}) = k$ for $i = 0$ and 0 for $i > 0$: the structure sheaf has the cohomology of a point, and this holds in *every* characteristic. The contrast is with the *constant* group $\underline{\mathbb{Z}/n}$, whose representation category obeys Maschke's theorem and is semisimple only when the order n is invertible in k; there the higher group cohomology $H^i(\underline{\mathbb{Z}/n}, k)$ is nonzero for $i > 0$ once $\operatorname{char} k \big| n$. When $\operatorname{char} k \nmid n$ the two group schemes are isomorphic and agree; when $\operatorname{char} k \big| n$ they differ, μ_n becoming infinitesimal (nonreduced) but remaining linearly reductive, while $\underline{\mathbb{Z}/n}$ stays étale but ceases to be linearly reductive. The higher cohomology that appears in char $\big|n$ is thus a feature of $B\underline{\mathbb{Z}/n}$, not of $B\mu_n$.

3. *The tangent and canonical bundle of* $[X/G]$. Let G be a smooth affine group scheme acting on a smooth variety X over k with finite stabilizers (so that the infinitesimal action is injective), and $q\colon X \to [X/G]$ the atlas. The tangent sheaf $T_{[X/G]}$ is defined by descent: it is the G-equivariant sheaf on X whose fiber is the quotient $T_X/\mathfrak{g}_X$, where $\mathfrak{g}_X \subseteq T_X$ is the image of the infinitesimal action $\mathfrak{g} \otimes_k \mathcal{O}_X \to T_X$ of the Lie algebra $\mathfrak{g} = \operatorname{Lie}(G)$ (the distribution tangent to the orbits, a subbundle since the stabilizers are finite). There is a short exact sequence of equivariant sheaves on X

$$0 \to \mathfrak{g} \otimes_k \mathcal{O}_X \to T_X \to q^* T_{[X/G]} \to 0$$

whose descent to $[X/G]$ computes the tangent bundle of the stack. Taking top exterior powers, the canonical bundle is

$$\omega_{[X/G]} = \det(T_{[X/G]})^{-1}, \qquad q^*\omega_{[X/G]} \cong \omega_X \otimes (\det \mathfrak{g})_X$$

as G-equivariant line bundles on X, where $\det \mathfrak{g}$ is the one-dimensional representation $\Lambda^{\mathrm{top}}\mathfrak{g}$ (the modular character). For $X = \mathrm{pt}$ this gives $\dim BG = -\dim G$ and $\omega_{BG} = \det \mathfrak{g}$, consistent with the general formula ($\omega_{\mathrm{pt}} = \mathcal{O}$, so $q^*\omega_{BG} = \det \mathfrak{g}$) and with the tangent-complex count: the tangent complex of BG is $\mathfrak{g}[1]$, concentrated in degree -1, so $\det(T_{BG}) = (\det \mathfrak{g})^{-1}$ and $\omega_{BG} = \det(T_{BG})^{-1} = \det \mathfrak{g}$. This recovers the negative dimension of BG (definition 8.3.4): the stack BG has dimension $-\dim G$ because that tangent complex sits in degree -1, the automorphisms subtracting dimension.

4. $\overline{\mathcal{M}}_g$ *is smooth and proper as a stack.* The capstone. Fix $g \geq 2$ over a base where the relevant characteristics are excluded as in example 8.2.5. The stack $\overline{\mathcal{M}}_g$ is *smooth* of dimension $3g-3$ over the base: smoothness is an atlas-local property (definition 8.5.5(1)), and an atlas is the smooth locus of a Hilbert scheme of pluricanonically embedded stable curves (theorem 8.4.4), which is smooth because the deformation theory of a stable curve is unobstructed. A stable curve C is generically nodal, so Ω_C is not locally free and deformations are governed by $\operatorname{Ext}^i(\Omega_C, \mathcal{O}_C)$ rather than $H^i(C, T_C)$; the obstruction lives in $\operatorname{Ext}^2(\Omega_C, \mathcal{O}_C)$ and the tangent space is $\operatorname{Ext}^1(\Omega_C, \mathcal{O}_C)$ (these coincide with $H^i(C, T_C)$ only in the smooth locus, where Ω_C is locally free). The obstruction vanishes: a node is a hypersurface (local complete intersection) singularity, so Ω_C has projective dimension ≤ 1 locally and the local $\mathcal{E}xt$ sheaves $\mathcal{E}xt^j(\Omega_C, \mathcal{O}_C)$ vanish for $j \geq 2$, while $\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)$ is a skyscraper supported at

the nodes (the first-order smoothing of each node, Chapter 5 deformation theory of a node). The local-to-global spectral sequence $H^p(C, \mathcal{E}xt^q(\Omega_C, \mathcal{O}_C)) \Rightarrow \mathrm{Ext}^{p+q}(\Omega_C, \mathcal{O}_C)$ then has only the terms $H^p(C, \mathcal{H}om(\Omega_C, \mathcal{O}_C))$ $(q = 0)$ and $H^p(C, \mathcal{E}xt^1)$ $(q = 1)$; the $q = 0$ row vanishes above $p = 1$ because a curve has cohomological dimension one, and the $q = 1$ terms vanish for $p \geq 1$ because a skyscraper has no higher cohomology, so every contribution to $\mathrm{Ext}^2(\Omega_C, \mathcal{O}_C)$ is zero and the deformation theory is unobstructed. The tangent space $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ has constant dimension $3g - 3$. The stack $\overline{\mathcal{M}}_g$ is *proper* over the base: it is separated (the diagonal is finite, since stable curves have finite automorphism groups, so the Isom sheaves are finite hence proper, definition 8.5.5(2)), of finite type (boundedness of stable curves, theorem 2.2.5), and universally closed by the valuative criterion, which for $\overline{\mathcal{M}}_g$ is the stable reduction theorem (example 8.5.6). Thus $\overline{\mathcal{M}}_g$ is smooth and proper as a stack. Its coarse space $\overline{M}_g$ (theorem 8.2.4) is proper as a scheme but is *singular*: it has quotient singularities at the points corresponding to curves with automorphisms (the coarse space is locally $\overline{\mathcal{M}}_g$ modulo the automorphism group, and $\mathbb{A}^{3g-3}/\mathrm{Aut}(C)$ is generically a genuine quotient singularity when $\mathrm{Aut}(C) \neq 1$, the exception being the special case where the group acts by pseudo-reflections and the quotient stays smooth). Smoothness lives on the stack and is destroyed by passing to the coarse space, which is the precise sense in which $\overline{\mathcal{M}}_g$ is the right object and its coarse space is a lossy shadow.

The four computations exhibit the whole apparatus of the section at work and, taken together, justify the stack-theoretic language the book has built. The Picard group and cohomology of $B\mathbb{G}_m$ and $B\mu_n$ are pure descent: they are the representation theory of the automorphism group, invisible on the coarse space. The tangent and canonical bundles of $[X/G]$ are equivariant geometry, the orbit distribution subtracting the dimension of the group and the modular character twisting the canonical bundle. The smoothness and properness of $\overline{\mathcal{M}}_g$ are the payoff: smoothness read on a Hilbert atlas and driven by the unobstructedness of stable curves, properness read from the diagonal (finite automorphisms) and the valuative criterion (stable reduction), and both features destroyed by the coarse space, whose quotient singularities are the fingerprint of the automorphisms the stack retains. The moduli of curves, worked in full in Chapter 10, is the object where all three registers of stacky geometry meet, and the geometry on a stack developed here is what makes it a smooth proper object rather than a singular scheme.

Exercise 8.5.1

1. Let G be an affine group scheme over k and $BG = [\mathrm{pt}/G]$. Using the identification $\mathrm{QCoh}(BG) \simeq \mathrm{Rep}(G)$ (example 8.5.2), show that $\mathrm{Pic}\,(BG) \cong \widehat{G} = \mathrm{Hom}(G, \mathbb{G}_m)$, the character group of G. Deduce $\mathrm{Pic}\,(B\mu_n) = \mathbb{Z}/n$ and $\mathrm{Pic}\,(BGL_n) = \mathbb{Z}$, and identify a generator of each.

2. Let G act on a scheme X over k. Prove directly from definition 8.5.1 that $\mathrm{QCoh}([X/G])$ is equivalent to the category of G-equivariant quasi-coherent sheaves on X (a sheaf $\mathcal{F}$ on X with an isomorphism $\sigma^*\mathcal{F} \cong \mathrm{pr}_X^* \mathcal{F}$ satisfying the cocycle condition), where $\sigma \colon G \times X \to X$ is the action. Identify the structure sheaf $\mathcal{O}_{[X/G]}$ and describe what a G-invariant global section of $\mathcal{O}_{[X/G]}$ is in terms of X and G.

3. Show that a $\mathbb{Z}$-graded k-vector space $V = \bigoplus_n V_n$, regarded as an object of $\mathrm{QCoh}(B\mathbb{G}_m)$ via example 8.5.2, has $H^i(B\mathbb{G}_m, V) = 0$ for all $i > 0$ and $H^0(B\mathbb{G}_m, V) = V_0$. (Use that $\mathbb{G}_m$ is linearly reductive over any base: its representation category is semisimple, so the invariants functor is exact.) Conclude that $H^i(B\mathbb{G}_m, \mathcal{O}(n)) = 0$ for all $i > 0$ and every n, and $= 0$ in degree 0 unless $n = 0$.

4. For a discrete valuation ring A with fraction field K, state the valuative criterion (definition 8.5.5(3)) for a morphism of algebraic stacks $\mathcal{X} \to \mathcal{Y}$ to be *universally closed* and for it to be *separated*, being careful about the "up to unique isomorphism" and "after a finite extension of A" clauses. Explain, in one paragraph, why the scheme-theoretic valuative criterion (unique lift, no extension) is the special case where objects have no automorphisms, and why $\overline{\mathcal{M}}_g$ requires the stacky version.

5. Let G be a smooth affine group scheme of dimension d acting on a smooth n-dimensional variety X with finite stabilizers; assume in addition that the stabilizers are reduced (equivalently, that the diagonal is unramified, theorem 8.2.3), automatic in characteristic 0 but not in characteristic p (where e.g. μ_p-stabilizers are finite yet nonreduced and make $[X/G]$ Artin, not Deligne-Mumford), so that $[X/G]$ is Deligne-Mumford. Using the presentation with atlas $U = X$ and groupoid $R = G \times X$, compute $\dim[X/G] = \dim U - \dim R$ (definition 8.3.4), and confirm the answer $n - d$ agrees with the tangent-space count from the exact sequence $0 \to \mathfrak{g} \otimes \mathcal{O}_X \to T_X \to q^*T_{[X/G]} \to 0$ of example 8.5.7(3). Then specialize to $X = \mathrm{pt}$ to recover $\dim BG = -d$.

6. Explain why $\operatorname{Pic}(B\mathbb{G}_m) = \mathbb{Z}$ is not a contradiction with the fact that the coarse space of $B\mathbb{G}_m$ is a single point with trivial Picard group. Identify precisely which line bundles on the stack fail to descend to the coarse space, and relate this to the automorphism group $\mathbb{G}_m$ acting on the fibers. (Hint: a line bundle descends to the coarse space when $\mathbb{G}_m$ acts trivially on its fibers, that is, when its weight is 0.)

9

Gerbes and Twisted Sheaves

Torsors were the one-categorical trace of the automorphisms that stacks geometrize: a G-torsor is a sheaf locally isomorphic to G, and its class in H^1 measures the failure to be trivial. Raising this construction one categorical level produces a gerbe, a stack locally isomorphic to the classifying stack BG, whose obstruction to triviality lives in H^2. The gerbe was named at the end of the descent chapter and has since appeared unbidden at every turn: the residual gerbe $B\underline{\mathrm{Aut}}(x)$ is the local model of an algebraic stack at a point, and the extra automorphisms of a special elliptic curve make $\mathcal{M}_{1,1}$ carry a gerbe over its special points. This chapter studies gerbes in their own right, and in doing so closes Part III by connecting the moduli theory of the book to the Brauer group and to the twisted sheaves that the Brauer group governs.

The development runs in four movements. The first defines a gerbe and its band and proves that gerbes banded by an abelian sheaf A are classified by $H^2(X, A)$, exactly one degree above the torsor classification of Chapter 6. The second specializes the band to $\mathbb{G}_m$ and finds the Brauer group: the class of a $\mathbb{G}_m$-gerbe lives in $H^2(X, \mathbb{G}_m)$, an Azumaya algebra produces such a gerbe through the boundary map of $1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$, and the field case recovers the Brauer group of Galois cohomology. The third section defines twisted sheaves, the modules of a $\mathbb{G}_m$-gerbe, and shows that they form an abelian category in which the Brauer class is precisely the obstruction to the existence of a twisted line bundle. The fourth constructs their moduli, the twisted analogue of the moduli of sheaves, which is again an algebraic stack and which, because it can be a fine moduli space where the untwisted problem is only coarse, resolves questions such as the period-index problem that the untwisted theory cannot reach. A closing note records the topological shadow of the whole picture, the Dixmier-Douady class and twisted K-theory, which belongs to other books.

9.1 Torsors, Banded Gerbes, and H^2

The torsor is a sheaf that is locally isomorphic to a group G and whose isomorphism classes are counted by H^1. This chapter climbs one categorical level. A torsor is a sheaf of sets locally modeled on G; its two-dimensional refinement is a stack of groupoids locally modeled on the classifying stack BG, and its isomorphism classes are counted not by H^1 but by H^2. That object is a gerbe. It was named at the end of Chapter 6 (definition 6.5.4), where the passage from H^1 to H^2 was flagged as the two-dimensional analogue of the passage from a sheaf to a torsor, and its full development promised here. This section delivers that development for the case that matters most: a gerbe banded by an abelian sheaf A is classified by $H^2_\tau(X, A)$, exactly one degree above the torsor theorem (theorem 6.5.3), and the proof is the two-cocycle mechanism run in place of the one-cocycle mechanism.

Two threads from earlier parts of the book meet here. The first is descent: a gerbe is a stack, and the objects of a stack glue along a cover with the same cocycle bookkeeping that governed descent in Chapter 6, one dimension up. The second is the automorphism obstruction of Part I. When a moduli problem fails to be representable because its objects carry automorphisms, as for the elliptic curves of example 1.3.4, the correct object is not a scheme but a stack, and the local structure of that stack at a point with automorphism group G is precisely a gerbe banded by G. The residual gerbe of a stack point, built at the end of Chapter 8, is the H^2-shadow of the j-line's coarse-space collapse, and it is the concrete geometric object this section is written to name. Throughout, τ is the étale or fppf topology, X a scheme over the base S, and the site is $(\mathbf{Sch}/X)_\tau$; for the abelian bands of interest the two topologies agree.

Definition 9.1.1: Gerbe

A *gerbe* over the site $(\mathbf{Sch}/X)_\tau$ is a stack in groupoids $\mathcal{G} \to (\mathbf{Sch}/X)_\tau$ that is

1. *locally nonempty*: every object U of the site admits a covering family $\{U' \to U\}$ with $\mathcal{G}(U') \neq \varnothing$; and
2. *locally connected*: for every U and any two objects $x, y \in \mathcal{G}(U)$, there is a covering family $\{U' \to U\}$ such that $x|_{U'} \cong y|_{U'}$ in $\mathcal{G}(U')$ for each U'.

A gerbe $\mathcal{G}$ is *neutral* (or *trivial*) if it has a global object, that is, if $\mathcal{G}(X) \neq \varnothing$. In that case, for any global object $x \in \mathcal{G}(X)$ there is an equivalence of stacks

$$\mathcal{G} \simeq B\underline{\mathrm{Aut}}(x)$$

with the classifying stack (definition 7.3.2) of the automorphism sheaf $\underline{\mathrm{Aut}}(x)$ of x.

The two axioms are the exact two-dimensional analogues of the two conditions defining a torsor (definition 6.5.1). Local nonemptiness is the direct lift of τ-local nonemptiness of a torsor: a torsor is a sheaf of sets that is locally inhabited, a gerbe is a stack of groupoids that is locally inhabited. Local connectedness is the lift of simple transitivity. A torsor is *simply* transitive, so its fibers, once inhabited, are a single G-orbit with trivial stabilizer, a set with one isomorphism class and no symmetry; a gerbe relaxes this to *transitivity only*, so its fibers, once inhabited, are a single isomorphism class of objects but each object carries an automorphism group. That surviving automorphism group is the new one-dimensional datum a gerbe carries over a torsor, and it is what the band will record. Neutrality plays the role that triviality played for torsors: a global object trivializes the gerbe just as a global section trivialized a torsor, and the trivialized gerbe is $B\underline{\mathrm{Aut}}(x)$, the stacky point with automorphisms $\underline{\mathrm{Aut}}(x)$, exactly as a trivialized torsor was G itself.

The equivalence $\mathcal{G} \simeq B\underline{\mathrm{Aut}}(x)$ deserves a word, since it is the anchor for every example below. The classifying stack $B\underline{\mathrm{Aut}}(x)$ of definition 7.3.2 is the stackification of the prestack whose only object over each U is the trivial $\underline{\mathrm{Aut}}(x)$-torsor, with automorphisms $\underline{\mathrm{Aut}}(x)(U)$; its objects over a general U are the $\underline{\mathrm{Aut}}(x)$-torsors on U. The equivalence sends a $\underline{\mathrm{Aut}}(x)$-torsor P over U to the object of $\mathcal{G}(U)$ obtained by twisting $x|_U$ by P, that is, the object locally isomorphic to x and reglued by the transition cocycle of P; conversely an object $y \in \mathcal{G}(U)$ gives the torsor $\underline{\mathrm{Isom}}(x|_U, y)$ of local isomorphisms $x \to y$, which is a torsor under $\underline{\mathrm{Aut}}(x)$ precisely because $\mathcal{G}$ is locally connected (so the Isom sheaf is locally nonempty) and its objects have automorphism group $\underline{\mathrm{Aut}}(x)$. A neutral gerbe is therefore no more and no less than the moduli of torsors under a fixed automorphism sheaf, packaged as a stack.

Before the band, the definition should be grounded in the object it was written to describe. The classifying stack of a group over a point is the first gerbe, and it is neutral by construction.

Example 9.1.2: BG as the Neutral G-Gerbe

Let G be a sheaf of groups on $(\mathbf{Sch}/X)_\tau$, and let BG be its classifying stack (definition 7.3.2): the stack whose objects over U are the G-torsors on U, with morphisms the torsor isomorphisms. Then BG is a gerbe over X, and it is neutral.

Local nonemptiness holds because the trivial torsor $G|_U$ is an object of $BG(U)$ for every U, so $BG(U)$ is never empty; the site already has a global object, namely the trivial torsor over X itself. Local connectedness holds because any two G-torsors P, P' on U become isomorphic on a common trivializing cover: each is locally isomorphic to G, so on a cover refining the trivializations of both, $P|_{U'} \cong G \cong P'|_{U'}$. Thus BG satisfies both gerbe axioms. It is neutral because it has the global object $G|_X$, the trivial torsor; and taking $x = G|_X$, the automorphism sheaf $\underline{\mathrm{Aut}}(G|_X)$ is G acting on itself by right translation (proposition 6.5.2), so the equivalence of definition 9.1.1 reads

$$BG \simeq B\underline{\mathrm{Aut}}(G|_X) = BG$$

tautologically. The gerbe BG is the model neutral gerbe: every neutral gerbe is of this form for $G = \underline{\mathrm{Aut}}(x)$ its automorphism band, and the content of the next definition is to attach that band to a gerbe that need not be neutral.

A gerbe that is not neutral has no global object to read an automorphism sheaf off of. Yet every object of $\mathcal{G}(U)$, over every U and every cover, has an automorphism sheaf, and because the gerbe is locally connected these local automorphism sheaves are all locally isomorphic. The band is the invariant that records this common local automorphism sheaf, glued across the gerbe. For a nonabelian automorphism sheaf the gluing is only well-defined up to inner automorphism, and the band is the corresponding object; for an abelian sheaf inner automorphisms are trivial and the band is a genuine identification. The abelian case is the one this book develops in full.

Definition 9.1.3: Band of a Gerbe, and A-Gerbes

Let $\mathcal{G}$ be a gerbe over $(\mathbf{Sch}/X)_\tau$. For each object U of the site and each $x \in \mathcal{G}(U)$, the automorphism sheaf $\underline{\mathrm{Aut}}(x)$ is a sheaf of groups on U. Local connectedness provides, for any two objects x, y over a common cover, an isomorphism $x|_{U'} \cong y|_{U'}$, which induces an isomorphism $\underline{\mathrm{Aut}}(x)|_{U'} \cong \underline{\mathrm{Aut}}(y)|_{U'}$ of automorphism sheaves. This isomorphism is canonical up to conjugation by $\underline{\mathrm{Aut}}(y)$, because the isomorphism $x \cong y$ is unique only up to composition with an automorphism. The *band* (or *lien*) of $\mathcal{G}$ is the resulting sheaf of groups on X, well-defined up to inner automorphism, obtained by gluing these local automorphism sheaves. Let A be a sheaf of *abelian* groups on X. A *banding of $\mathcal{G}$ by A* is a choice, for every object $x \in \mathcal{G}(U)$, of an isomorphism $A|_U \cong \underline{\mathrm{Aut}}(x)$, compatible with restriction and with the isomorphisms induced by morphisms of $\mathcal{G}$. Because A is abelian, inner automorphisms of $\underline{\mathrm{Aut}}(x)$ are trivial, so a banding is a *canonical* identification with no conjugation ambiguity. A gerbe equipped with a banding by an abelian sheaf A is an A-*gerbe*; when the band A is understood, "gerbe" means "A-gerbe".

The conjugation ambiguity is the whole reason the band is a subtle invariant for nonabelian coefficients and a trivial one for abelian coefficients. When $\underline{\mathrm{Aut}}(x)$ is nonabelian, the isomorphism $x \cong y$ that transports one automorphism sheaf to another is unique only up to an automorphism of the target, and composing with that automorphism conjugates the transported identification. The band is therefore not a sheaf of groups but a sheaf of groups modulo inner automorphisms, an object Giraud calls a *lien*. The full nonabelian theory, in which a gerbe is banded by a lien and the classification lives in a nonabelian H^2 that is a set with a torsor action rather than a group, is developed in Giraud's foundational treatise [14]; it is stated here and not developed. For an abelian A the ambiguity evaporates: conjugation by any element of an abelian group is the identity, so the transported identification is unique, the band is a genuine abelian sheaf, and a banding is a plain isomorphism $A \cong \underline{\mathrm{Aut}}$. The classification then lands in the ordinary abelian H^2, and that is the theorem this section proves in full.

The definition is grounded by the object that motivated Part III: a point of an algebraic stack, whose automorphisms organize into a gerbe over its residue field.

Example 9.1.4: The Residual Gerbe of a Stack Point

Let $\mathcal{X}$ be an algebraic stack (definition 8.3.1) and let $x \in \mathcal{X}(k)$ be a point over a field k, with automorphism group scheme $G = \underline{\mathrm{Aut}}(x)$, a group scheme of finite type over k. The *residual gerbe* at x is the reduced locally closed substack $\mathcal{G}_x \hookrightarrow \mathcal{X}$ supported at the image point, and it is a gerbe over $\operatorname{Spec} k$ banded by G; when G is abelian it is a G-gerbe. Since $\operatorname{Spec} k$ has a single point and x is a global object of $\mathcal{G}_x$ over k, the residual gerbe is *neutral*, and by definition 9.1.1

$$\mathcal{G}_x \simeq BG = B\underline{\mathrm{Aut}}(x)$$

the classifying stack of the automorphism group over the residue field.

The running touchstone makes this concrete. Consider the moduli stack $\mathcal{M}_{1,1}$ of elliptic curves and its point $[E]$ at an elliptic curve E over $k = \bar{k}$ of characteristic $\neq 2, 3$. A generic elliptic curve has automorphism group $\mu_2 = \{\pm 1\}$, so the residual gerbe at a generic point is $B\mu_2$. At the two special j-invariants the automorphism group jumps: at $j = 0$ the curve $y^2 = x^3 + 1$ has $\underline{\mathrm{Aut}} = \mu_6$, and at $j = 1728$ the curve $y^2 = x^3 + x$ has $\underline{\mathrm{Aut}} = \mu_4$ (the values recorded in example 1.3.4). The residual gerbes at these points are therefore

$$\mathcal{G}_{[E], j=0} \simeq B\mu_6, \qquad \mathcal{G}_{[E], j=1728} \simeq B\mu_4$$

each a neutral abelian gerbe over $\operatorname{Spec} k$. This is the H^2-shadow that the j-line of theorem 1.3.6 discards: the coarse space $\mathbb{A}^1_j$ records only the point j, forgetting the automorphism group entirely, whereas the stack $\mathcal{M}_{1,1}$ remembers it as a residual gerbe. The automorphism obstruction to representability, first seen in Part I as the failure of a fine moduli space to exist, is now visible as a nonzero band on a residual gerbe.

The residual gerbes above are all neutral, because a point of a stack always carries the tautological global object x itself. The interesting gerbes, the ones that will realize the Brauer group in the next section, are the nonneutral ones, and the classification theorem is exactly the count of how nonneutral an A-gerbe can be. The count is a class in H^2, and the neutral gerbe is its base point, precisely as the trivial torsor was the base point of H^1.

The motivation for the classification is the same as for torsors, lifted one dimension. An A-gerbe is locally neutral by axiom: on a cover it acquires objects, and any two are locally isomorphic, so on a fine enough cover the gerbe looks like BA. What obstructs neutrality globally is the failure of these local objects to be chosen coherently, and that failure is measured by a Čech class. For a torsor the local objects were sections and their discrepancies were one-cochains; for a gerbe the local objects are objects of a groupoid, their pairwise discrepancies are isomorphisms, and the failure of those isomorphisms to compose coherently on triple overlaps is a two-cochain valued in the band. Its coherence on quadruple overlaps is the two-cocycle condition. This is the mechanism, and the theorem is its formalization.

Theorem 9.1.5: Classification of A-Gerbes by H^2

Let A be a sheaf of abelian groups on the site $(\mathbf{Sch}/X)_\tau$. There is a natural bijection

$$\{\text{equivalence classes of } A\text{-gerbes on } X \text{ in } \tau\} \xrightarrow{\sim} H^2_\tau(X, A)$$

between equivalence classes of gerbes banded by A and the second cohomology of the site with coefficients in A, under which the neutral gerbe BA corresponds to the zero class. The set of A-gerbes is an abelian group under the contracted product of gerbes, and the bijection is an isomorphism of abelian groups. For a nonabelian band the classification lives in Giraud's nonabelian H^2 and is a set with a distinguished class rather than a group; that generality is treated in [14].

***Proof*:**
The proof constructs the class of a gerbe as a Čech two-cocycle from local objects and their discrepancies, checks it is well-defined, and inverts the construction. It is the one-degree-up version of the proof of theorem 6.5.3, and the parallel is worth keeping in view: objects replace sections, isomorphisms of objects replace transition elements, and the coherence of those isomorphisms is one dimension higher. Recall the Čech description of H^2. For a covering family $\mathcal{U} = \{U_i \to X\}$, write $U_{ij} = U_i \times_X U_j$, and likewise U_{ijk}, U_{ijkl} for triple and quadruple overlaps. A 2-*cocycle* on $\mathcal{U}$ with values in A is a family $(\alpha_{ijk}) \in \prod_{i,j,k} A(U_{ijk})$ satisfying the cocycle condition

$$\alpha_{jkl} - \alpha_{ikl} + \alpha_{ijl} - \alpha_{ijk} = 0 \quad \text{on } U_{ijkl} \tag{9.1}$$

(the alternating sum over the four faces of the tetrahedron U_{ijkl}), and two cocycles are cohomologous if they differ by a coboundary $(\delta\beta)_{ijk} = \beta_{jk} - \beta_{ik} + \beta_{ij}$ of a 1-cochain $(\beta_{ij}) \in \prod_{i,j} A(U_{ij})$. The group of cocycles modulo coboundaries is $\check{H}^2(\mathcal{U}, A)$, and $H^2_\tau(X, A)$ is its colimit over refinements, which agrees with the derived-functor cohomology of A on the site by [10].

Step 1: From a gerbe to a two-cocycle. Let $\mathcal{G}$ be an A-gerbe. By local nonemptiness choose a covering family $\mathcal{U} = \{U_i \to X\}$ and objects $x_i \in \mathcal{G}(U_i)$, one over each cover piece. On a double overlap U_{ij} the restricted objects $x_i|_{U_{ij}}$ and $x_j|_{U_{ij}}$ are two objects of $\mathcal{G}(U_{ij})$; by local connectedness, after refining the cover so that each pair is already isomorphic over U_{ij}, choose an isomorphism

$$\varphi_{ij} : x_j|_{U_{ij}} \xrightarrow{\sim} x_i|_{U_{ij}}$$

the two-dimensional analogue of the transition element g_{ij} of a torsor. These isomorphisms need not compose coherently on triple overlaps. On U_{ijk} compare the two isomorphisms $x_k \to x_i$ obtained as $\varphi_{ij} \circ \varphi_{jk}$ and directly as φ_{ik}: both are isomorphisms $x_k|_{U_{ijk}} \to x_i|_{U_{ijk}}$, so they differ by a unique automorphism of x_i,

$$\varphi_{ij} \circ \varphi_{jk} = \alpha_{ijk} \cdot \varphi_{ik} \quad \text{on } U_{ijk} \tag{9.2}$$

where $\alpha_{ijk} \in \underline{\mathrm{Aut}}(x_i)(U_{ijk}) = A(U_{ijk})$, the last identification being the banding by A. The element α_{ijk} is the *discrepancy* of the transition isomorphisms on the triple overlap, and it is the exact two-dimensional analogue of the fact that for a torsor the transition elements composed on the nose (the one-cocycle condition $g_{ik} = g_{ij}g_{jk}$); here they compose only up to the automorphism α_{ijk}.

That (α_{ijk}) is a two-cocycle is the associativity of composition of the φ on the quadruple overlap U_{ijkl}. Compute the composite $x_l \to x_i$ in two ways. Bracketing as $(\varphi_{ij} \circ \varphi_{jk}) \circ \varphi_{kl}$ and applying (9.2) to the inner pair, then again, expresses it through φ_{il} and the automorphisms $\alpha_{ijk}, \alpha_{ikl}$; bracketing as $\varphi_{ij} \circ (\varphi_{jk} \circ \varphi_{kl})$ expresses it through φ_{il} and $\alpha_{jkl}, \alpha_{ijl}$. Since composition of isomorphisms is associative and A is abelian (so the automorphisms transported along φ_{ij} from $\underline{\mathrm{Aut}}(x_j)$ to $\underline{\mathrm{Aut}}(x_i)$ are unchanged, both being identified with A), equating the two bracketings gives

$$\alpha_{ijk} + \alpha_{ikl} = \alpha_{jkl} + \alpha_{ijl} \quad \text{on } U_{ijkl}$$

which is the cocycle condition (9.5). Thus (α_{ijk}) is a 2-cocycle on $\mathcal{U}$ with values in A.

Step 2: The class is independent of choices. Two kinds of choice were made: the objects x_i and the isomorphisms φ_{ij}. Changing the isomorphisms first, replace each φ_{ij} by $\varphi'_{ij} = \beta_{ij} \cdot \varphi_{ij}$ for some $\beta_{ij} \in \underline{\mathrm{Aut}}(x_i)(U_{ij}) = A(U_{ij})$. Substituting into (9.2) and using that A is abelian and central in each automorphism sheaf,

$$\varphi'_{ij} \circ \varphi'_{jk} = \beta_{ij}\, \varphi_{ij}\, \beta_{jk}\, \varphi_{jk} = (\beta_{ij} + \beta_{jk} + \alpha_{ijk})\, \varphi_{ik} = (\beta_{ij} + \beta_{jk} - \beta_{ik})\, \varphi'_{ik}$$

where the second equality moves β_{jk} past φ_{ij} using the banding (its image in $\underline{\mathrm{Aut}}(x_i)$ is the same $\beta_{jk} \in A$ because the band identifies all automorphism sheaves with A), and the third rewrites $\varphi_{ik} = \varphi'_{ik} - \beta_{ik}$ additively. Hence the new discrepancy is $\alpha'_{ijk} = \alpha_{ijk} + (\beta_{jk} - \beta_{ik} + \beta_{ij})$, differing from α_{ijk} by the coboundary $(\delta\beta)_{ijk}$: the class in $\check{H}^2(\mathcal{U}, A)$ is unchanged. Changing the objects x_i to isomorphic objects $x'_i \cong x_i$ over U_i transports the isomorphisms φ_{ij} along these isomorphisms and leaves the discrepancies invariant, again by centrality of A; and refining the cover restricts the cocycle to the refinement, changing nothing in the colimit. Thus $[(\alpha_{ijk})] \in H^2_\tau(X, A)$ depends only on the equivalence class of $\mathcal{G}$.

Step 3: From a two-cocycle to a gerbe. Conversely, let (α_{ijk}) be a two-cocycle on $\mathcal{U}$. Build a gerbe $\mathcal{G}_\alpha$ by descent for stacks (theorem 7.2.6) as follows. On each U_i take the neutral gerbe $BA|_{U_i}$, whose objects are A-torsors; the tautological object $\xi_i \in BA(U_i)$ is the trivial A-torsor.

Glue $BA|_{U_j}$ to $BA|_{U_i}$ over U_{ij} by the equivalence $\Phi_{ij}\colon BA|_{U_{ij}} \to BA|_{U_{ij}}$ that is the identity functor on objects and morphisms (both restrict to the same BA); the gluing carries ξ_j to ξ_i by the tautological isomorphism φ_{ij}. On a triple overlap the composite $\Phi_{ij} \circ \Phi_{jk}$ and Φ_{ik} agree as functors but their action on the tautological objects differs by the 2-cochain α_{ijk}: the natural transformation

$$\Phi_{ij} \circ \Phi_{jk} \Rightarrow \Phi_{ik}$$

is multiplication by $\alpha_{ijk} \in A(U_{ijk})$, acting through the band. The descent datum for a stack requires that these natural transformations satisfy the coherence pentagon on quadruple overlaps, and that coherence is exactly the two-cocycle condition (9.5): the alternating sum vanishing is the statement that the two ways of coherently associating four gluings agree. By theorem 7.2.6 the descent datum is effective, and the pieces glue to a stack $\mathcal{G}_\alpha$ over X.

The glued stack $\mathcal{G}_\alpha$ is a gerbe: it is locally nonempty because each $BA|_{U_i}$ has the object ξ_i, and locally connected because the objects of BA over any base, being A-torsors, are locally isomorphic. Its band is A, because the automorphism sheaf of the tautological object ξ_i is $\underline{\mathrm{Aut}}_{BA}(\text{trivial torsor}) = A$, glued by the banding. Thus $\mathcal{G}_\alpha$ is an A-gerbe. Applying Step 1 to $\mathcal{G}_\alpha$ with the objects ξ_i and the tautological isomorphisms φ_{ij} returns, by (9.2), the discrepancy α_{ijk}: the natural transformation $\Phi_{ij} \circ \Phi_{jk} \Rightarrow \Phi_{ik}$ was defined to be α_{ijk}. Hence the class of $\mathcal{G}_\alpha$ is $[(\alpha_{ijk})]$.

Step 4: The two constructions are mutually inverse. Step 1 sends a gerbe to a class; Step 3 sends a cocycle to a gerbe whose class is the given one. It remains to see that a gerbe $\mathcal{G}$ with class $[(\alpha_{ijk})]$ is equivalent to the gerbe $\mathcal{G}_\alpha$ built from that cocycle. The objects $x_i \in \mathcal{G}(U_i)$ and isomorphisms φ_{ij} chosen in Step 1 exhibit $\mathcal{G}$ as glued from the neutral gerbes $\mathcal{G}|_{U_i} \simeq B\underline{\mathrm{Aut}}(x_i) \simeq BA|_{U_i}$ along equivalences whose triple-overlap discrepancies are α_{ijk}, which is precisely the descent datum defining $\mathcal{G}_\alpha$. The equivalence sending x_i to the tautological object ξ_i is compatible with the gluing, hence extends to an equivalence $\mathcal{G} \simeq \mathcal{G}_\alpha$. Cohomologous cocycles give equivalent gerbes by Step 2 (a coboundary is a change of the isomorphisms φ_{ij}, which induces an equivalence of the glued gerbes), so the two assignments descend to mutually inverse bijections between equivalence classes of A-gerbes and $H^2_\tau(X, A)$. The neutral gerbe BA, having the global object $\xi =$ the trivial torsor, admits objects x_i all restricting from a single global object, so all transition isomorphisms can be chosen compatibly and every discrepancy α_{ijk} is trivial: the neutral gerbe maps to the zero class.

Step 5: The group structure. The abelian group structure on $H^2_\tau(X, A)$, addition of two-cocycles, transports to gerbes as the *contracted product*. Given A-gerbes $\mathcal{G}, \mathcal{G}'$ with cocycles $(\alpha_{ijk}), (\alpha'_{ijk})$, form the gerbe $\mathcal{G} \wedge^A \mathcal{G}'$ whose objects over U are pairs (x, x') of a local object of each, with automorphisms identified along the band by the codiagonal $A \times A \to A$ (adding the two automorphisms); its triple-overlap discrepancy is $\alpha_{ijk} + \alpha'_{ijk}$, so its class is the sum. The neutral gerbe BA is the identity, the gerbe with cocycle $(-\alpha_{ijk})$ is the inverse, and the bijection of Step 4 is a group isomorphism, completing the proof.

The theorem is the exact one-story ascent from the torsor classification, and the dictionary between the two is worth setting down, because every later computation reads it off. A torsor is trivialized by a global section; a gerbe is neutralized by a global object. The discrepancy of local sections on a double overlap is a transition element g_{ij}, a one-cochain; the discrepancy of local objects on a double overlap is an isomorphism φ_{ij}, and its own failure to compose is the two-cochain α_{ijk}. The one-cocycle condition $g_{ik} = g_{ij} g_{jk}$ was the associativity of transition on triple overlaps; the two-cocycle condition (9.5) is the associativity of transition one level up, the coherence of the isomorphisms on quadruple overlaps. The freedom to change local sections was a coboundary in degree one; the freedom to change local isomorphisms is a coboundary in degree two. Every structural feature of the torsor theorem reappears with its degree raised by one, which is the sense in which a gerbe is the two-dimensional torsor promised in Chapter 6.

The role of abelianness is the same as it was for torsors, and locating it explains what the nonabelian theory must repair. The proof used commutativity of A in exactly one place that mattered: transporting an automorphism along a transition isomorphism φ_{ij} changes it by conjugation, and for abelian A conjugation is trivial, so the band identifies all automorphism sheaves with one fixed A and the cochains live in a single abelian group. When the band is nonabelian this identification is ambiguous up to inner automorphism, the transported cochains no longer live in a fixed group, and the cocycle condition must be phrased for a lien rather than a sheaf of groups. The resulting nonabelian H^2 is not a group but a set on which a group of two-cocycles acts, with the neutral gerbe a distinguished point that need not be a zero; the full account, including the obstruction in H^3 to a band being realized by any gerbe at all, is Giraud's [14]. This book takes the abelian band as its object because the coefficient group that realizes the Brauer group in the next section, the multiplicative group $\mathbb{G}_m$, is abelian, and the entire $\mathbb{G}_m$-gerbe theory lives inside the theorem just proved.

The classification also completes the automorphism thread that has run since Part I, and the completion is best seen on a nonneutral gerbe, since the residual gerbes of example 9.1.4 were all neutral. A nonzero class in H^2 produces a gerbe with no global object, and the next example builds one by hand from a nonzero two-cocycle, so that the base-point structure of the theorem is not left abstract.

Example 9.1.6: A Nonneutral Gerbe from a Two-Cocycle

Let $A = \mu_n$ and suppose X has a nonzero class $\alpha \in H^2_\tau(X, \mu_n)$, represented by a two-cocycle (α_{ijk}) on a cover $\mathcal{U} = \{U_i \to X\}$. By theorem 9.1.5 the gerbe $\mathcal{G}_\alpha$ built from (α_{ijk}) in Step 3 of the proof is a μ_n-gerbe with class $\alpha \neq 0$, hence *not* neutral: it has no global object, because a global object would trivialize the class. Concretely $\mathcal{G}_\alpha$ has, on each U_i, the object ξ_i, and these fail to glue to a global object precisely because the isomorphisms φ_{ij} gluing ξ_j to ξ_i do not compose coherently: their triple-overlap discrepancy is α_{ijk}, and if a global object existed one could choose the φ_{ij} to compose on the nose, forcing α_{ijk} to be a coboundary and $\alpha = 0$.

A source of such classes is at hand from the Kummer sequence of Chapter 6. The connecting map of the long exact cohomology sequence of $1 \to \mu_n \to \mathbb{G}_m \to \mathbb{G}_m \to 1$ (the sequence (6.5) of Chapter 6) in degree two,

$$\operatorname{Pic}(X) = H^1(X, \mathbb{G}_m) \xrightarrow{\delta} H^2(X, \mu_n)$$

sends a line bundle L to the class of the μ_n-gerbe of n-th roots of L: the gerbe $\mathcal{G}_L^{1/n}$ whose objects over U are pairs (M, ι) of a line bundle M on U and an isomorphism $\iota \colon M^{\otimes n} \cong L|_U$. This gerbe is locally nonempty because an n-th root of L exists locally (any line bundle is locally trivial, hence locally an n-th power), and locally connected because any two local n-th roots differ by a μ_n-torsor; its band is μ_n, acting on a root M by scaling. It is neutral exactly when L has a global n-th root, so its class $\delta[L] \in H^2(X, \mu_n)$ vanishes iff $L \in n \cdot \operatorname{Pic}(X)$. On a variety with a line bundle that is not an n-th power, for instance $L = \mathcal{O}(1)$ on $\mathbb{P}^m$ with $n \geq 2$ (so $L \notin n \operatorname{Pic}(\mathbb{P}^m) = n\mathbb{Z}$), the gerbe $\mathcal{G}_L^{1/n}$ is a nonneutral μ_n-gerbe, an explicit representative of a nonzero class in $H^2(\mathbb{P}^m, \mu_n)$. This is the first nonneutral gerbe of the book, and the mechanism, an obstruction to extracting a global root, is the template for the Brauer classes of the next section.

Exercise 9.1.1

1. Let $\mathcal{G}$ be a gerbe over $(\mathbf{Sch}/X)_\tau$. Prove directly from definition 9.1.1 that $\mathcal{G}$ is neutral if and only if it has a global object, and that a global object $x \in \mathcal{G}(X)$ determines an equivalence $\mathcal{G} \simeq B\underline{\mathrm{Aut}}(x)$. Identify precisely which axiom of a gerbe supplies local nonemptiness of the Isom sheaf $\underline{\mathrm{Isom}}(x|_U, y)$ and which supplies its simple transitivity, so that it is a torsor under $\underline{\mathrm{Aut}}(x)$.
2. Show that the residual gerbe $\mathcal{G}_x$ of a point $x \in \mathcal{X}(k)$ of an algebraic stack (example 9.1.4) is always neutral, and hence always of the form $B\underline{\mathrm{Aut}}(x)$ over $\operatorname{Spec} k$. Explain why this does not contradict the existence of nonneutral gerbes: what feature of $\operatorname{Spec} k$, as opposed to a higher-dimensional base, forces neutrality?
3. Carry out in full the verification that the discrepancy family (α_{ijk}) of (9.2) satisfies the two-cocycle condition (9.5), by expanding the composite $\varphi_{ij} \circ \varphi_{jk} \circ \varphi_{kl}$ on the quadruple overlap U_{ijkl} in the two bracketings and equating them. State explicitly where the abelianness of the band A is used.
4. Using the connecting map $\delta\colon \mathrm{Pic}\,(X) \to H^2(X, \mu_n)$ of example 9.1.6, prove that the μ_n-gerbe $\mathcal{G}_L^{1/n}$ of n-th roots of a line bundle L is neutral if and only if L admits a global n-th root, and hence that $\delta[L] = 0$ if and only if $L \in n \cdot \mathrm{Pic}\,(X)$. Deduce that $\mathcal{G}^{1/2}_{\mathcal{O}(1)}$ on $\mathbb{P}^2_k$ is a nonneutral μ_2-gerbe.
5. Prove that the contracted product $\mathcal{G} \wedge^A \mathcal{G}'$ of two A-gerbes (theorem 9.1.5, Step 5) has class $[\mathcal{G}] + [\mathcal{G}']$ by computing its triple-overlap discrepancy from those of $\mathcal{G}$ and $\mathcal{G}'$. Identify the neutral gerbe BA as the identity element and describe the inverse of an A-gerbe both as a gerbe and at the level of its class.
6. Explain, by locating the exact step in the proof of theorem 9.1.5 where commutativity of A is used, why the classification is a group isomorphism for an abelian band but only a bijection of pointed sets (with a distinguished class that need not be a zero) for a nonabelian band. Contrast this with the analogous phenomenon for torsors in theorem 6.5.3, and state in one sentence what new invariant (a lien, and an H^3-obstruction) the nonabelian theory of [14] must introduce that the abelian theory does not need.

9.2 $\mathbb{G}_m$-Gerbes and the Brauer Group

The previous section classified A-gerbes on X by $H^2_\tau(X, A)$ for an abelian band A, one cohomological degree above the torsors of Chapter 6. This section specializes the band to the multiplicative group $\mathbb{G}_m$, where the arithmetic content appears. The reason $\mathbb{G}_m$ is the band to single out is that $H^2_{\text{ét}}(X, \mathbb{G}_m)$ is not an abstract cohomology group but a classical invariant in disguise: over a field it is the Brauer group, the group of central simple algebras up to Morita equivalence, and over a general scheme it records the failure of Azumaya algebras and PGL_n-bundles to lift to GL_n. A $\mathbb{G}_m$-gerbe is thus a geometric avatar of a Brauer class, and the two-way dictionary between algebras and gerbes is what this section makes precise. The central construction is the boundary map of the short exact sequence $1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$, which sends a PGL_n-torsor (equivalently a rank-n^2 Azumaya algebra) to the class in $H^2(X, \mathbb{G}_m)$ obstructing its lift to a vector bundle. This class is n-torsion, and the map it defines embeds the Azumaya Brauer group into its cohomological cousin. Gabber's theorem, that the embedding is a bijection onto the torsion subgroup, is stated and cited; the construction of the map and its torsion bound are carried out in full.

The band $\mathbb{G}_m$ deserves its own name at the level of gerbes because the entire theory of twisted sheaves in the next section is written for gerbes banded by it. Applying theorem 9.1.5 with $A = \mathbb{G}_m$ gives the classification for free; the definition records the terminology.

Definition 9.2.1: $\mathbb{G}_m$-Gerbe

Let X be a scheme and τ the étale (equivalently, for $\mathbb{G}_m$, the fppf) topology on $(\mathbf{Sch}/X)_\tau$. A $\mathbb{G}_m$-*gerbe* on X is a gerbe $\mathcal{G} \to (\mathbf{Sch}/X)_\tau$ banded by the multiplicative group $\mathbb{G}_m$ in the sense of definition 9.1.3: a gerbe together with a fixed identification $\mathbb{G}_m \cong \underline{\mathrm{Aut}}(x)$ of the automorphism sheaf of every local object x with $\mathbb{G}_m$, compatible with restriction. By theorem 9.1.5 the equivalence classes of $\mathbb{G}_m$-gerbes on X are in natural bijection with

$$H^2_{\text{ét}}(X, \mathbb{G}_m)$$

the neutral gerbe $B\mathbb{G}_{m,X}$ corresponding to the zero class. Given a class $\alpha \in H^2_{\text{ét}}(X, \mathbb{G}_m)$, write $\mathcal{G}_\alpha$ for a gerbe representing it.

The identification étale = fppf for $\mathbb{G}_m$-cohomology is Grothendieck's theorem that $H^i_{\text{ét}}(X, \mathbb{G}_m) = H^i_{\text{fppf}}(X, \mathbb{G}_m)$ for a smooth group scheme, so the choice of site does not matter and either may be used; this is why the definition brackets the two topologies together. The content of theorem 9.1.5 in this case is that a $\mathbb{G}_m$-gerbe is the same data as a class in a single cohomology group, and the neutral gerbe $B\mathbb{G}_{m,X}$, the classifying stack of the trivial $\mathbb{G}_m$-bundle, is the base point. A nonzero class is a gerbe with no global object, an obstruction to trivializing the automorphism-$\mathbb{G}_m$'s coherently across X.

The first example to have in hand is the automorphism gerbe of a stack point, which threads back to the $\mathcal{M}_{1,1}$ story of Chapter 1 and shows that $\mathbb{G}_m$-gerbes arise unbidden from the geometry already developed.

Example 9.2.2: The $\mathbb{G}_m$-Gerbe of a Stacky Point

Let k be a field and consider the classifying stack $B\mathbb{G}_{m,k} = [\operatorname{Spec} k/\mathbb{G}_m]$, the neutral $\mathbb{G}_m$-gerbe over $\operatorname{Spec} k$. Its single object over $\operatorname{Spec} k$ has automorphism group $\mathbb{G}_m(k) = k^\times$, the band, and it represents the zero class in $H^2_{\text{ét}}(\operatorname{Spec} k, \mathbb{G}_m)$. This is the trivial case: a point with a $\mathbb{G}_m$ of automorphisms, but with a global object, hence neutral.

A nonneutral example comes from an elliptic curve with extra automorphisms. Over the special points $j = 0$ and $j = 1728$ of the coarse space of $\mathcal{M}_{1,1}$ (theorem 1.3.6), the curve E has automorphism group larger than the generic $\{\pm 1\}$: cyclic of order 6 at $j = 0$ and of order 4 at $j = 1728$ (in characteristic 0, or away from $2, 3$; example 1.3.4). Over an algebraically closed field the residual gerbe at such a point is the classifying stack $B\mu_n$ (with $n = 6$ at $j = 0$, $n = 4$ at $j = 1728$, the order of $\underline{\mathrm{Aut}}(E)$), a μ_n-gerbe rather than a $\mathbb{G}_m$-gerbe: the automorphism group of the object bands the residual gerbe, and pushing forward along $\mu_n \hookrightarrow \mathbb{G}_m$ realizes it as a $\mathbb{G}_m$-gerbe whose class is the image of the μ_n-class. Over a field the class is torsion of order dividing n, and it is the algebraic shadow of the extra symmetry of the special elliptic curve.

Over an algebraically closed k this residual gerbe is neutral: the curve E is a k-rational object, so it is the classifying stack $B\mu_n$ of a global object and its class is zero. A *nonneutral* example lives over a subfield k_0 over which the special curve E does *not* descend to a k_0-rational elliptic curve.

There the residual gerbe is a nontrivial *form* of $B\mu_n$, a μ_n-gerbe with no global object rather than the classifying stack of any k_0-rational curve, and its class in $H^2(\operatorname{Spec} k_0, \mu_n)$ can be nonzero. The obstruction is whether the object E itself descends to k_0, not whether its automorphisms are rational: a classifying stack $B\mu_n$ always has the trivial torsor as a global object and so is neutral regardless of $\mu_n(k_0)$. Only when E fails to descend does the gerbe acquire a nonzero class, the algebraic shadow of the field of definition of the symmetric curve.

The example makes concrete the slogan that a $\mathbb{G}_m$-gerbe measures a failure of descent for objects with $\mathbb{G}_m$ of automorphisms. When the object exists, the gerbe is $B\mathbb{G}_m$ and the class is zero; when it exists only locally, the gluing discrepancies assemble into a 2-cocycle whose class is the obstruction. The same phenomenon over a field is the Brauer group, to which the discussion now turns.

The cohomological Brauer group is defined so that the field case reproduces the classical group-cohomological Brauer group of [9], and so that the torsion condition, invisible over a field where the whole group is torsion, becomes the operative restriction over a higher-dimensional base. Alongside it sits the Azumaya Brauer group, built from actual algebras rather than cohomology classes; the comparison of the two is the theorem of the section.

Definition 9.2.3: Brauer Group and Cohomological Brauer Group

Let X be a scheme. The *cohomological Brauer group* of X is the torsion subgroup

$$\operatorname{Br}'(X) = H^2_{\text{ét}}(X, \mathbb{G}_m)_{\text{tors}}$$

of the second étale cohomology group of $\mathbb{G}_m$. An *Azumaya algebra* on X is a sheaf $\mathcal{A}$ of $\mathcal{O}_X$-algebras that is, étale-locally on X, isomorphic to a matrix algebra $\operatorname{Mat}_n(\mathcal{O}_X)$ (equivalently, $\mathcal{A}$ is a locally free $\mathcal{O}_X$-algebra of finite rank whose fibers $\mathcal{A} \otimes \kappa(x)$ are central simple algebras over the residue fields $\kappa(x)$). Two Azumaya algebras $\mathcal{A}, \mathcal{B}$ are *Morita equivalent*, or *Brauer equivalent*, if there exist locally free sheaves $\mathcal{E}, \mathcal{F}$ of positive rank with

$$\mathcal{A} \otimes_{\mathcal{O}_X} \underline{\operatorname{End}}(\mathcal{E}) \cong \mathcal{B} \otimes_{\mathcal{O}_X} \underline{\operatorname{End}}(\mathcal{F})$$

The *(Azumaya) Brauer group* $\operatorname{Br}(X)$ is the set of Morita equivalence classes, a group under $\otimes_{\mathcal{O}_X}$ with identity the class of $\mathcal{O}_X$ and inverse the class of the opposite algebra $\mathcal{A}^{\text{op}}$. When $X = \operatorname{Spec} k$ is the spectrum of a field, $\operatorname{Br}(k)$ is the classical Brauer group of central simple k-algebras, and by [9]

$$\operatorname{Br}(\operatorname{Spec} k) = H^2\big(\operatorname{Gal}(k^{\text{sep}}/k), (k^{\text{sep}})^\times\big)$$

the Galois cohomology of the multiplicative group of a separable closure.

Two distinct groups now carry the name Brauer, and the distinction is the crux of the section. The Azumaya group $\operatorname{Br}(X)$ is built from algebras: étale-locally matrix algebras, glued by a PGL_n-cocycle since the automorphisms of Mat_n are PGL_n by the Skolem-Noether theorem. The cohomological group $\operatorname{Br}'(X)$ is a subgroup of a cohomology group, defined without reference to any algebra. Over a field the two coincide and both equal the Galois-cohomological group of [9], because over a field every class of $H^2(\operatorname{Gal}, (k^{\text{sep}})^\times)$ is torsion and is represented by a central simple algebra. Over a general scheme the coincidence is a theorem, Gabber's, and the torsion restriction in the definition of Br' is exactly what an Azumaya algebra can reach: a matrix-algebra bundle of rank n^2 produces an n-torsion class, so only torsion classes are candidates for algebras.

The Galois-cohomological description over a field is worth pausing on, since it grounds every field computation below. For k a field with absolute Galois group $G_k = \mathrm{Gal}(k^{\mathrm{sep}}/k)$, the group $\mathrm{Br}(k) = H^2(G_k, (k^{\mathrm{sep}})^\times)$ is built from the crossed-product construction: a 2-cocycle $c\colon G_k \times G_k \to (k^{\mathrm{sep}})^\times$ yields a central simple algebra $\bigoplus_{\sigma \in G} k^{\mathrm{sep}} u_\sigma$ with multiplication twisted by c, and every central simple algebra arises this way. This is the field model the gerbe picture generalizes: a $\mathbb{G}_m$-gerbe on X is the geometric object whose Čech 2-cocycle over an étale cover plays the role of the crossed-product cocycle over the field.

The bridge between the two Brauer groups is the boundary map of the defining sequence of PGL_n. This is the point where the gerbe interpretation does its work: the obstruction to lifting a PGL_n-torsor to a GL_n-torsor is a $\mathbb{G}_m$-gerbe, and its class in $H^2(X, \mathbb{G}_m)$ is the Brauer class of the corresponding Azumaya algebra. The following theorem constructs this map, bounds its image by n-torsion, and records that it embeds the Azumaya group into the cohomological one.

Theorem 9.2.4: Azumaya Algebras, PGL_n-Torsors, and $\mathbb{G}_m$-Gerbes

Let X be a scheme. Fix $n \geq 1$ and consider the short exact sequence of étale sheaves of groups

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{GL}_n \longrightarrow \mathrm{PGL}_n \longrightarrow 1 \tag{9.3}$$

where $\mathbb{G}_m$ is the center of GL_n (scalar matrices). Then:

1. The boundary map of nonabelian cohomology

$$\delta\colon H^1_{\text{ét}}(X, \mathrm{PGL}_n) \longrightarrow H^2_{\text{ét}}(X, \mathbb{G}_m)$$

assigns to a PGL_n-torsor P the class of the $\mathbb{G}_m$-gerbe $\mathcal{G}_P$ of local lifts of P to a GL_n-torsor. A rank-n^2 Azumaya algebra $\mathcal{A}$ corresponds to a PGL_n-torsor $P_{\mathcal{A}}$ (theorem 6.5.3 applied to the automorphism sheaf $\underline{\mathrm{Aut}}(\mathcal{A}) = \mathrm{PGL}_n$), and $\delta([P_{\mathcal{A}}]) = [\mathcal{A}] \in \mathrm{Br}(X)$ is its Brauer class.

2. The class $\delta([P]) \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ is n-torsion.

3. The assignment $[\mathcal{A}] \mapsto \delta([P_{\mathcal{A}}])$ is a well-defined injective group homomorphism

$$\mathrm{Br}(X) \hookrightarrow \mathrm{Br}'(X) = H^2_{\text{ét}}(X, \mathbb{G}_m)_{\mathrm{tors}}$$

from the Azumaya Brauer group to the cohomological Brauer group.

***Proof*:**
The three assertions are the construction of the boundary map, the torsion bound, and the injectivity of the induced homomorphism. They are proved in that order.

Step 1: The gerbe of local lifts and the boundary class. Let P be a PGL_n-torsor on X, classified by an étale-Čech 1-cocycle $\bar{g}_{ij} \in \mathrm{PGL}_n(U_{ij})$ on a cover $\{U_i \to X\}$ (theorem 6.5.3). Define a stack $\mathcal{G}_P$ over $(\mathbf{Sch}/X)_{\text{ét}}$ whose objects over $U \to X$ are the GL_n-torsors Q on U equipped with an isomorphism $Q/\mathbb{G}_m \cong P|_U$ of PGL_n-torsors, that is, the lifts of P across (9.3). Morphisms are isomorphisms of GL_n-torsors commuting with the identification of the associated PGL_n-torsor. Two facts make $\mathcal{G}_P$ a $\mathbb{G}_m$-gerbe.

First, $\mathcal{G}_P$ is locally nonempty: on each U_i the restriction $P|_{U_i}$ is trivial, so it lifts to the trivial GL_n-torsor, giving an object of $\mathcal{G}_P(U_i)$. Second, $\mathcal{G}_P$ is locally connected with band $\mathbb{G}_m$: if Q, Q'

are two lifts of $P|_U$, then étale-locally both are trivial, and any two lifts differ by an automorphism of P that lifts to GL_n, hence by an element of the kernel $\mathbb{G}_m$; concretely $Q' = Q \otimes L$ for a $\mathbb{G}_m$-torsor L, so any two objects are locally isomorphic and the automorphism sheaf of a lift is $\mathbb{G}_m$ (the scalars acting on Q that fix the induced PGL_n-torsor). Thus $\mathcal{G}_P$ is a $\mathbb{G}_m$-gerbe, and $\delta([P]) := [\mathcal{G}_P] \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ is its class under theorem 9.1.5.

Explicitly, choose set-theoretic lifts $g_{ij} \in \mathrm{GL}_n(U_{ij})$ of the cocycle $\bar{g}_{ij}$. The failure of the g_{ij} to satisfy the cocycle condition is measured by

$$\alpha_{ijk} = g_{ij}\, g_{jk}\, g_{ki} \in \mathrm{GL}_n(U_{ijk}) \tag{9.4}$$

which, because $\bar{g}_{ij}\bar{g}_{jk}\bar{g}_{ki} = 1$ in PGL_n, lies in the kernel $\mathbb{G}_m(U_{ijk})$ (it is a scalar matrix). A direct expansion on U_{ijkl}, using associativity of matrix multiplication, gives the 2-cocycle identity $\alpha_{jkl}\,\alpha_{ijl}\,\alpha_{ijk}^{-1}\,\alpha_{ikl}^{-1} = 1$, so (α_{ijk}) is a Čech 2-cocycle with values in $\mathbb{G}_m$, and its class is $\delta([P])$. Changing the lifts g_{ij} by scalars $\lambda_{ij} \in \mathbb{G}_m(U_{ij})$ alters α_{ijk} by the coboundary of (λ_{ij}), so the class is independent of the choices. For the Azumaya algebra $\mathcal{A}$ of rank n^2, the sheaf $\underline{\mathrm{Aut}}_{\mathcal{O}_X\text{-alg}}(\mathcal{A})$ is PGL_n by the Skolem-Noether theorem, so $\mathcal{A}$ is classified by a PGL_n-torsor $P_{\mathcal{A}}$; the lifts of $P_{\mathcal{A}}$ to GL_n are precisely the locally free sheaves $\mathcal{E}$ with $\underline{\mathrm{End}}(\mathcal{E}) \cong \mathcal{A}$, and $\delta([P_{\mathcal{A}}]) = [\mathcal{A}]$ is the obstruction to the existence of such an $\mathcal{E}$ globally, which is the Brauer class.

Step 2: The torsion bound. The determinant gives a homomorphism $\det\colon \mathrm{GL}_n \to \mathbb{G}_m$ whose restriction to the central $\mathbb{G}_m$ of scalars is the n-th power map $\lambda \mapsto \lambda^n$. Apply det to the scalar-valued cochain α_{ijk} of (9.4): since α_{ijk} is the scalar matrix $\alpha_{ijk} \cdot \mathrm{id}$,

$$\det(\alpha_{ijk}) = \alpha_{ijk}^n$$

On the other hand $\det(g_{ij}) \in \mathbb{G}_m(U_{ij})$ are units, and applying the multiplicative det to (9.4) gives $\det(\alpha_{ijk}) = \det(g_{ij})\det(g_{jk})\det(g_{ki})$, which is the Čech coboundary $\partial(\det g)_{ijk}$ of the 1-cochain $(\det g_{ij})$. Therefore $\alpha_{ijk}^n = \partial(\det g)_{ijk}$ is a coboundary, so $n \cdot \delta([P]) = n \cdot [\alpha] = [\alpha^n] = 0$ in $H^2_{\text{ét}}(X, \mathbb{G}_m)$. Hence $\delta([P])$ is n-torsion.

Step 3: Well-definedness, homomorphism, and injectivity. First, $[\mathcal{A}] \mapsto \delta([P_{\mathcal{A}}])$ is well-defined on Morita classes: replacing $\mathcal{A}$ by $\mathcal{A} \otimes \underline{\mathrm{End}}(\mathcal{E})$ replaces the rank n by nm (with $m = \mathrm{rk}\,\mathcal{E}$) and the torsor $P_{\mathcal{A}}$ by the torsor of $\mathcal{A} \otimes \underline{\mathrm{End}}(\mathcal{E})$, but the associated gerbe of local lifts is unchanged up to equivalence, because $\underline{\mathrm{End}}(\mathcal{E})$ is a matrix algebra with a global lift $\mathcal{E}$, so it contributes the zero class and $\delta([P_{\mathcal{A} \otimes \underline{\mathrm{End}}(\mathcal{E})}]) = \delta([P_{\mathcal{A}}])$. Thus the map descends to $\mathrm{Br}(X)$.

It is a homomorphism because tensor product of Azumaya algebras corresponds to the Baer sum of the associated gerbes: if $\mathcal{A}$ has rank n^2 with cocycle α_{ijk} and $\mathcal{B}$ has rank m^2 with cocycle β_{ijk}, then $\mathcal{A} \otimes \mathcal{B}$ has rank $(nm)^2$, its structure sheaf glues by the Kronecker product $g_{ij} \otimes h_{ij}$, and the scalar discrepancy of $g_{ij} \otimes h_{ij}$ is $\alpha_{ijk}\,\beta_{ijk}$, so $\delta([\mathcal{A} \otimes \mathcal{B}]) = \delta([\mathcal{A}]) + \delta([\mathcal{B}])$ in additive notation. The identity $\mathcal{O}_X$ maps to the class of the neutral gerbe, namely 0.

For injectivity, suppose $\delta([\mathcal{A}]) = 0$, so the gerbe $\mathcal{G}_{P_{\mathcal{A}}}$ of local lifts is neutral and has a global object: a globally defined GL_n-torsor Q with $Q/\mathbb{G}_m \cong P_{\mathcal{A}}$. Equivalently there is a globally defined locally free sheaf $\mathcal{E}$ of rank n with $\underline{\mathrm{End}}(\mathcal{E}) \cong \mathcal{A}$. Then $\mathcal{A} = \underline{\mathrm{End}}(\mathcal{E}) = \mathcal{O}_X \otimes \underline{\mathrm{End}}(\mathcal{E})$ is Morita equivalent to $\mathcal{O}_X$, so $[\mathcal{A}] = 0$ in $\mathrm{Br}(X)$. Hence δ has trivial kernel and the induced map $\mathrm{Br}(X) \hookrightarrow \mathrm{Br}'(X)$ is injective, completing the proof.

The theorem constructs an injection $\mathrm{Br}(X) \hookrightarrow \mathrm{Br}'(X)$ and identifies the Azumaya Brauer group with a subgroup of the torsion of $H^2_{\text{ét}}(X, \mathbb{G}_m)$. The gerbe $\mathcal{G}_P$ is the geometric carrier of the obstruction: it is neutral exactly when the PGL_n-torsor lifts to a vector bundle, that is, when the Azumaya algebra is a matrix algebra, and its class is the precise cohomological measure of the failure to lift. The torsion bound is not an accident of the proof but the structural fact that an Azumaya algebra of rank n^2 can only see n-torsion: the determinant collapses the n-th power of the obstruction to a coboundary. This is why the cohomological Brauer group is defined as the *torsion* subgroup and not all of $H^2(X, \mathbb{G}_m)$; the nontorsion classes, which occur on non-separated or non-quasicompact schemes, can never come from an algebra.

Whether the injection is surjective onto the torsion is a substantially harder question, answered by Gabber.

> **9.2.5 Note: Gabber's Theorem** ($\mathrm{Br} = \mathrm{Br}'$) Gabber's theorem states that for a quasi-compact scheme X admitting an ample line bundle (for instance any quasi-projective scheme over an affine base), the injection of theorem 9.2.4 is a bijection
>
> $$\mathrm{Br}(X) = \mathrm{Br}'(X) = H^2_{\text{ét}}(X, \mathbb{G}_m)_{\text{tors}}$$
>
> so every torsion class in $H^2_{\text{ét}}(X, \mathbb{G}_m)$ is the class of an Azumaya algebra. The proof is beyond the present scope: it constructs, from a torsion cohomology class, an Azumaya algebra representing it, using the theory of twisted sheaves developed in the next section together with a delicate argument producing a twisted vector bundle of the correct rank. The result and its proof are treated in [23] and the co-requisite [8]; the special case where X is the spectrum of a field is elementary and recovers the classical fact that every element of $\mathrm{Br}(k) = H^2(G_k, (k^{\text{sep}})^\times)$ is represented by a central simple algebra ([9]).

The surjectivity of $\mathrm{Br} \to \mathrm{Br}'$ is a theorem with content, not a formality: producing an algebra from a cohomology class reverses the natural direction. Gabber proved it for schemes carrying an ample line bundle, in particular quasi-projective ones (de Jong later gave a written proof); it can fail without such a hypothesis, where non-separated or otherwise pathological schemes carry torsion classes in Br' with no Azumaya representative, so $\mathrm{Br} \subsetneq \mathrm{Br}'$ in general. For the purposes of this book the injection is the working tool, and the distinction between Br and Br' can be elided precisely on the schemes where computations are done. The next section constructs the twisted sheaves that Gabber's proof requires, closing the loop from the algebra side back to the gerbe side.

With the correspondence established, the concrete content of the Brauer group is a series of computations over fields, where the group is fully torsion and coincides with the Galois cohomology of [9]. The quaternion algebra is the smallest nontrivial example and the model for everything: a μ_2-gerbe of order exactly 2, realized by an actual 4-dimensional division algebra.

Example 9.2.6: The Quaternion Algebra and Brauer Groups of Fields

The computations below are all over a field, where $\mathrm{Br}(k) = \mathrm{Br}'(k) = H^2(G_k, (k^{\text{sep}})^\times)$ is entirely torsion.

1. *The quaternion algebra as a gerbe of order* 2. Let k be a field of characteristic $\neq 2$ and $a, b \in k^\times$. The quaternion algebra

   $$(a, b)_k = k\langle i, j\rangle/(i^2 = a,\ j^2 = b,\ ij = -ji)$$

   is a 4-dimensional central simple k-algebra, split by the quadratic extension $k(\sqrt{a})$. Its class $[(a, b)_k] \in \mathrm{Br}(k)$ has order dividing 2: the opposite algebra $(a, b)_k^{\text{op}} \cong (a, b)_k$ (the map $i \mapsto -i$, $j \mapsto -j$ is an anti-isomorphism to itself), so $[(a, b)_k] + [(a, b)_k] = [(a, b)_k \otimes (a, b)_k] = 0$

because $(a,b)_k \otimes (a,b)_k^{\mathrm{op}} \cong \underline{\mathrm{End}}((a,b)_k)$ is a matrix algebra ($16 = 4^2$-dimensional). Under theorem 9.2.4 it is a $\mathbb{G}_m$-gerbe realized as a μ_2-gerbe: the boundary of the PGL_2-torsor of $(a,b)_k$, whose Čech 2-cocycle is the sign cocycle valued in $\mu_2 \subset \mathbb{G}_m$. The class is nonzero (order exactly 2) precisely when $(a,b)_k$ is a division algebra, that is, when b is not a norm from $k(\sqrt{a})$; for example $(-1,-1)_{\mathbb{R}} = \mathbb{H}$, Hamilton's quaternions, is the nonzero element of $\mathrm{Br}(\mathbb{R})$.

2. $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$. The absolute Galois group of $\mathbb{R}$ is $G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}/2$, generated by complex conjugation acting on $\mathbb{C}^\times$. Then

$$\mathrm{Br}(\mathbb{R}) = H^2(\mathbb{Z}/2, \mathbb{C}^\times) = \mathbb{R}^\times / N_{\mathbb{C}/\mathbb{R}}(\mathbb{C}^\times) = \mathbb{R}^\times/\mathbb{R}_{>0} = \{\pm 1\} \cong \mathbb{Z}/2$$

using the standard computation of the cohomology of a cyclic group as a norm quotient (the Herbrand quotient / Tate cohomology of $\mathbb{Z}/2$; see [9]): H^2 of a cyclic group $\langle\sigma\rangle$ acting on M is $M^\sigma/N(M)$, here $\mathbb{R}^\times$ modulo the norms $z\bar{z} = |z|^2 > 0$. The nontrivial class is $\mathbb{H}$, the quaternion algebra $(-1,-1)_{\mathbb{R}}$ of item (1).

3. $\mathrm{Br}(\mathbb{F}_q) = 0$. A finite field $\mathbb{F}_q$ has absolute Galois group $\widehat{\mathbb{Z}}$, procyclic, generated topologically by the Frobenius. Every central simple algebra over a finite field is a matrix algebra, since a finite division ring is a field by Wedderburn's little theorem, so

$$\mathrm{Br}(\mathbb{F}_q) = H^2(\widehat{\mathbb{Z}}, (\overline{\mathbb{F}}_q)^\times) = 0$$

the vanishing coming from the surjectivity of the norm map for the procyclic Galois group ([9]). Every Azumaya algebra over $\mathbb{F}_q$ splits, and every $\mathbb{G}_m$-gerbe over $\operatorname{Spec} \mathbb{F}_q$ is neutral.

4. *Local and global fields.* For a nonarchimedean local field K (a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$), local class field theory gives $\mathrm{Br}(K) = \mathbb{Q}/\mathbb{Z}$ via the invariant map, so every element has a representative division algebra of any prescribed order; for a global field K (a number field or a function field of a curve over $\mathbb{F}_q$), the fundamental exact sequence of global class field theory reads

$$0 \longrightarrow \mathrm{Br}(K) \longrightarrow \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

summing over all places v, so a global Brauer class is a finite tuple of local invariants summing to zero. These are the central results of [9] and are cited, not reproved here.

5. $\mathrm{Br}(\mathbb{P}^n_k) = \mathrm{Br}(k)$. For projective space over any field k, pullback along $\mathbb{P}^n_k \to \operatorname{Spec} k$ induces an isomorphism

$$\mathrm{Br}(k) \xrightarrow{\sim} \mathrm{Br}(\mathbb{P}^n_k)$$

so projective space carries no Brauer classes beyond those of the ground field. This is a computation in étale cohomology (using the projective bundle formula for $H^2(\mathbb{G}_m)$ and the vanishing of the relevant higher direct images), stated here and cited to [23] and the co-requisite [8]; it says that a Severi-Brauer variety over $\mathbb{P}^n_k$ contributes nothing new, and that the Brauer group is a birational invariant insensitive to adding projective-space factors.

The computations make the abstract correspondence tangible. Over $\mathbb{R}$ the Brauer group is a single $\mathbb{Z}/2$ carried by the quaternions, the unique division algebra; over a finite field it vanishes and every gerbe is neutral; over local and global fields it is the arithmetic heart of class field theory, the invariant map and the reciprocity sum being the organizing principles. That $\mathrm{Br}(\mathbb{P}^n_k) = \mathrm{Br}(k)$ shows the geometric Brauer group of a rational variety reduces to the arithmetic of the ground field, and it is the first instance of a computation where the base is a scheme rather than a point. The field cases are self-contained through the group cohomology of [9]; the higher-dimensional computations, where étale cohomology is unavoidable, are the province of [8], the co-requisite whose deep Brauer theory this book cites rather than develops. The next section builds the twisted sheaves that live on a $\mathbb{G}_m$-gerbe and turn the Brauer class into a category of geometric objects.

Exercise 9.2.1

1. Let P be a PGL_n-torsor on a scheme X with Čech 1-cocycle $\bar{g}_{ij}$, and let $g_{ij} \in \mathrm{GL}_n(U_{ij})$ be lifts. Verify in full the 2-cocycle identity for $\alpha_{ijk} = g_{ij}g_{jk}g_{ki}$ (equation (9.4)) on the quadruple overlaps U_{ijkl}, and confirm that changing the lifts g_{ij} by scalars $\lambda_{ij} \in \mathbb{G}_m(U_{ij})$ changes α_{ijk} by the coboundary of (λ_{ij}). Conclude that the class $\delta([P]) \in H^2(X, \mathbb{G}_m)$ is independent of the choice of lifts.

2. Give a second proof, distinct from the determinant argument of theorem 9.2.4, that $\delta([P])$ is n-torsion for a PGL_n-torsor P, by using instead the tensor-power structure: for a local rank-n lift $\mathcal{E}$ of P (so $\underline{\mathrm{End}}(\mathcal{E})$ is the Azumaya algebra of P), relate $n \cdot \delta([P])$ to the obstruction to gluing the determinant line bundles $\det \mathcal{E} = \wedge^n \mathcal{E}$, and explain why the argument via det is the cleaner one. (Hint: the class $n \cdot \delta([P])$ is the obstruction to gluing a rank-1 sheaf, whose torsor of lifts is a PGL_1-torsor and so visibly lifts.)

3. Compute $\mathrm{Br}(k)$ for k algebraically closed, and deduce that every Azumaya algebra over an algebraically closed field is a matrix algebra and every $\mathbb{G}_m$-gerbe over $\operatorname{Spec} k$ is neutral. Then explain, using theorem 9.2.4, why the residual μ_n-gerbe of the special point $j = 0$ of $\mathcal{M}_{1,1}$ (example 9.2.2) is neutral over $\overline{k}$ but may be nonneutral over a subfield.

4. Let $a, b \in k^\times$ with $\operatorname{char} k \neq 2$. Show that the quaternion algebra $(a, b)_k$ is split (isomorphic to $\mathrm{Mat}_2(k)$) if and only if the conic $ax^2 + by^2 = z^2$ has a k-rational point, if and only if b is a norm from $k(\sqrt{a})$. Deduce that $[(a, b)_k] = 0$ in $\mathrm{Br}(k)$ exactly in these cases, and compute $[(a, b)_k]$ for $k = \mathbb{R}$ in the four sign cases of $(a, b) \in \{\pm 1\}^2$.

5. Using the local-global exact sequence of [9] recalled in example 9.2.6(4), show that there is no division algebra over $\mathbb{Q}$ ramified at exactly one place, and construct explicitly a quaternion division algebra over $\mathbb{Q}$ ramified at exactly two places (say $\{2, \infty\}$ or $\{p, q\}$ for two odd primes). State the order of its class in $\mathrm{Br}(\mathbb{Q})$.

6. Explain why the cohomological Brauer group is defined as the *torsion* subgroup of $H^2(X, \mathbb{G}_m)$ rather than the whole group, by two routes: (i) from theorem 9.2.4, why no Azumaya algebra can produce a nontorsion class; and (ii) name the hypothesis in Gabber's theorem (box 9.2.5) under which $\mathrm{Br} = \mathrm{Br}'$ holds, and state what can go wrong without it. (You may cite the existence of nontorsion classes on non-quasi-projective schemes without constructing one.)

9.3 Twisted Sheaves

The previous two sections built the $\mathbb{G}_m$-gerbe $\mathcal{G}_\alpha$ attached to a class $\alpha \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ (definition 9.2.1, theorem 9.1.5) and identified the torsion part of that cohomology group with the Brauer group (definition 9.2.3, theorem 9.2.4). So far the Brauer class has been an obstruction seen from the outside: a cohomology class, a gerbe that fails to be neutral, an Azumaya algebra that fails to be a matrix algebra. This section supplies the objects that the gerbe was built to carry, and through them the Brauer class acquires a concrete meaning. On an ordinary scheme one studies quasi-coherent sheaves; on a $\mathbb{G}_m$-gerbe the natural objects are the quasi-coherent sheaves on which the band $\mathbb{G}_m$ acts by scaling. These are the *twisted sheaves*. They form an abelian category $\mathrm{Tw}_\alpha(X)$ that reduces to $\mathrm{QCoh}(X)$ when $\alpha = 0$, and whose failure to contain a twisted line bundle is exactly the nonvanishing of α. The Brauer class, which measured the failure of the gerbe to be neutral in section 9.2, now measures the failure of a twisted line bundle to exist, and the equivalence with modules over an Azumaya algebra turns the whole theory into linear algebra over a division algebra.

A quasi-coherent sheaf on any algebraic stack was defined in Chapter 8 (definition 8.5.1): a compatible assignment of a quasi-coherent sheaf $\mathcal{F}_\xi$ on U to every object $\xi \in \mathcal{G}_\alpha(U)$, together with isomorphisms along pullbacks. On a gerbe there is extra structure. Every object ξ has automorphism group $\mathbb{G}_m$ acting on it, and this automorphism acts on the fiber $\mathcal{F}_\xi$ through some integer power of the scaling character. The integer is locally constant, so on a connected gerbe it is a single integer, the *weight*. Splitting $\mathrm{QCoh}(\mathcal{G}_\alpha)$ into weight components is the first structural fact, and the weight-one part is the category this section studies.

Definition 9.3.1: α-Twisted Sheaf

Let $\mathcal{G}_\alpha \to X$ be a $\mathbb{G}_m$-gerbe of class $\alpha \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ (definition 9.2.1). A quasi-coherent sheaf $\mathcal{F}$ on $\mathcal{G}_\alpha$ (definition 8.5.1) has *weight* $w \in \mathbb{Z}$ if, for every object $\xi \in \mathcal{G}_\alpha(U)$, the tautological automorphism $t \in \mathbb{G}_m(U) = \underline{\mathrm{Aut}}(\xi)$ acts on the fiber $\mathcal{F}_\xi$ by the scalar t^w. An *α-twisted sheaf* is a quasi-coherent sheaf on $\mathcal{G}_\alpha$ of weight 1. The category of α-twisted sheaves is denoted $\mathrm{Tw}_\alpha(X)$ (equivalently $\mathrm{QCoh}(\mathcal{G}_\alpha, 1)$, the weight-one component of $\mathrm{QCoh}(\mathcal{G}_\alpha)$).
Concretely, choose an étale cover $\{U_i \to X\}$ trivializing α, with objects $x_i \in \mathcal{G}_\alpha(U_i)$ and gluing isomorphisms $\varphi_{ij} \colon x_j|_{U_{ij}} \xrightarrow{\sim} x_i|_{U_{ij}}$ whose failure of the cocycle condition is the chosen 2-cocycle representative $\alpha_{ijk} \in \mathbb{G}_m(U_{ijk})$ (theorem 9.1.5), so that

$$\varphi_{ij}\,\varphi_{jk} = \alpha_{ijk}\,\varphi_{ik} \tag{9.5}$$

Then an α-twisted sheaf is the same datum as a system $(\mathcal{E}_i, \psi_{ij})$ of quasi-coherent sheaves $\mathcal{E}_i$ on U_i and isomorphisms $\psi_{ij} \colon \mathcal{E}_j|_{U_{ij}} \xrightarrow{\sim} \mathcal{E}_i|_{U_{ij}}$ satisfying the *twisted cocycle condition*

$$\psi_{ij}\,\psi_{jk}\,\psi_{ki} = \alpha_{ijk} \cdot \mathrm{id}_{\mathcal{E}_i} \qquad \text{on } U_{ijk} \tag{9.6}$$

A morphism $(\mathcal{E}_i, \psi_{ij}) \to (\mathcal{E}'_i, \psi'_{ij})$ is a system of $\mathcal{O}_{U_i}$-linear maps $f_i \colon \mathcal{E}_i \to \mathcal{E}'_i$ commuting with the gluing, $f_i \psi_{ij} = \psi'_{ij} f_j$. When each $\mathcal{E}_i$ is locally free of finite rank r, the system is an *α-twisted vector bundle of rank* r, and $r = 1$ gives an *α-twisted line bundle*.

The two descriptions are the same object read two ways. The first is intrinsic: a sheaf on the gerbe, singled out by how the band acts. The second is the Čech translation, and it is the one used in computations. Its content is the single formula (9.6): the gluing isomorphisms ψ_{ij} do not satisfy the ordinary cocycle condition that would let them glue to a sheaf on X; they satisfy it only up to the scalar α_{ijk}. When $\alpha = 0$ one may choose the trivial cocycle $\alpha_{ijk} = 1$, and then (9.6) becomes the genuine cocycle condition and the $\mathcal{E}_i$ descend to an ordinary sheaf on X. For nonzero α they cannot descend, and the twisted sheaf lives only on the gerbe. This is the precise sense in which a twisted sheaf is "a sheaf that would exist on X if α vanished".

The rank of an α-twisted vector bundle is well-defined, since the ψ_{ij} are isomorphisms and the twist by α_{ijk} is a scalar; here and throughout, “rank” of a twisted sheaf means the rank r of the underlying locally free $\mathcal{E}_i$ as in definition 9.3.1, which over a splitting cover is its dimension as a vector space, not the smaller rank it may carry as a module over the division algebra or Azumaya algebra representing α. The two senses differ by the degree of that algebra, and where the module rank is meant it is named as such. The higher numerical invariants require more care. A twisted Chern character can be defined once one fixes an auxiliary datum: choosing a $\mathbb{G}_m$-linearized line bundle $\mathcal{L}$ on the gerbe of weight 1 (equivalently a rank-one twisted sheaf, when one exists) untwists everything by tensoring, but the point of the theory is that such an $\mathcal{L}$ need not exist. The correct statement is that the rational Chern character of an α-twisted sheaf lives in a B-twisted Hodge-type group determined by a lift B of α; the precise formalism is developed in [3] and is the twisted analogue of the ordinary Chern character. For the arithmetic applications the rank alone already carries the decisive information, as the period-index discussion of the next section shows.

Example 9.3.2: Twisted Modules over a Point

Take $X = \operatorname{Spec} k$ for a field k and a $\mathbb{G}_m$-gerbe $\mathcal{G}_\alpha$ of class $\alpha \in H^2(\mathrm{Gal}(k^{\mathrm{sep}}/k), (k^{\mathrm{sep}})^\times) = \mathrm{Br}(k)$ (definition 9.2.3). Trivialize α over $\operatorname{Spec} k^{\mathrm{sep}} \to \operatorname{Spec} k$, the limit of the finite étale covers $\operatorname{Spec} k' \to \operatorname{Spec} k$ over finite Galois k'/k, on which the 2-cocycle becomes a Galois 2-cocycle $\alpha_{\sigma,\tau} \in (k^{\mathrm{sep}})^\times$. A finite-dimensional α-twisted vector space is then a k^{sep}-vector space V together with semilinear maps $\psi_\sigma \colon {}^\sigma V \to V$ for each $\sigma \in \mathrm{Gal}$, satisfying

$$\psi_\sigma \circ {}^\sigma\psi_\tau = \alpha_{\sigma,\tau} \cdot \psi_{\sigma\tau}$$

the finite-cover form of (9.6). This is exactly the data of a module over the crossed-product algebra $(k^{\mathrm{sep}}/k, \alpha)$ built from the cocycle $\alpha_{\sigma,\tau}$, whose class in the Brauer group is α ([9, the crossed-product construction]). When $k = \mathbb{R}$ and α is the nontrivial class of order two, the crossed-product algebra is Hamilton’s quaternions $\mathbb{H}$, and an α-twisted vector space over $\mathbb{R}$ is precisely a right $\mathbb{H}$-module, that is, a quaternionic vector space. Its rank as an $\mathbb{H}$-module measures how many copies of $\mathbb{H}$ it is built from, and the smallest nonzero one, $\mathbb{H}$ itself, is $\mathbb{H}$-module rank one but twisted rank two: its rank as a twisted sheaf (definition 9.3.1) is the dimension of the descent vector space over $\mathbb{C} = \mathbb{R}^{\mathrm{sep}}$, which is two, the degree of $\mathbb{H}$. The Brauer obstruction has become the statement that the smallest twisted object is $\mathbb{H}$, not $\mathbb{R}$.

The example is the point-level model that the whole section generalizes. Over a point, an α-twisted sheaf is a module over the division algebra with Brauer class α, and the twisted rank of the smallest such module is the index of that division algebra. The passage from a field to a scheme replaces the division algebra by an Azumaya algebra and the module by a coherent $\mathcal{O}_X$-module structure, but the mechanism is identical: the Brauer class forces the twisted objects to be modules over an algebra that is not $\mathcal{O}_X$.

The first structural theorem organizes twisted sheaves into a category and records how the twist behaves under the sheaf operations. The category $\mathrm{Tw}_\alpha(X)$ ought to be abelian, since twisting is a global condition that kernels and cokernels respect; the tensor product of an α-twisted sheaf with a β-twisted sheaf ought to be $(\alpha+\beta)$-twisted, since the twisting scalars multiply; and the $\alpha = 0$ case ought to recover ordinary sheaves. The one substantive consequence is the last clause: a twisted *line* bundle exists if and only if the class vanishes. That statement is the geometric content of the Brauer group, and it is what elevates the twist from bookkeeping to obstruction.

Theorem 9.3.3: The Category of Twisted Sheaves

Let X be a scheme and $\alpha, \beta \in H^2_{\text{ét}}(X, \mathbb{G}_m)$.

1. The category $\mathrm{Tw}_\alpha(X)$ of α-twisted sheaves is abelian, and the full subcategory of coherent (or of locally free) α-twisted sheaves is closed under the operations that preserve those conditions.

2. Tensor product over $\mathcal{O}_X$ (computed componentwise on a common trivializing cover) sends an α-twisted sheaf and a β-twisted sheaf to an $(\alpha+\beta)$-twisted sheaf. In particular $\mathrm{Tw}_0(X) = \mathrm{QCoh}(X)$, and the internal hom $\mathcal{H}om_{\mathcal{O}_X}(-,-)$ sends an α-twisted sheaf and a β-twisted sheaf to a $(\beta-\alpha)$-twisted sheaf; the dual of an α-twisted sheaf is $(-\alpha)$-twisted.

3. There exists an α-twisted invertible sheaf if and only if $\alpha = 0$ in $H^2_{\text{ét}}(X, \mathbb{G}_m)$.

***Proof*:**

Fix an étale cover $\{U_i \to X\}$ on which α and β are represented by cocycles α_{ijk} and β_{ijk}, and describe twisted sheaves by the Čech data of definition 9.3.1.

Step 1: Abelianness. Let $f = (f_i)\colon (\mathcal{E}_i, \psi_{ij}) \to (\mathcal{E}'_i, \psi'_{ij})$ be a morphism of α-twisted sheaves. Form the kernel component by component, $\mathcal{K}_i = \ker(f_i\colon \mathcal{E}_i \to \mathcal{E}'_i)$. The gluing ψ_{ij} carries $\mathcal{E}_j$ isomorphically to $\mathcal{E}_i$ and, by the compatibility $f_i\psi_{ij} = \psi'_{ij} f_j$, carries $\ker f_j$ into $\ker f_i$; restricting ψ_{ij} therefore gives isomorphisms $\psi^K_{ij}\colon \mathcal{K}_j \to \mathcal{K}_i$. These inherit (9.6) from the ψ_{ij} because the identity is twisted by the same scalar α_{ijk} on the subsheaf as on the whole. Thus $(\mathcal{K}_i, \psi^K_{ij})$ is an α-twisted sheaf, and it is the kernel of f in $\mathrm{Tw}_\alpha(X)$: a morphism into $(\mathcal{E}_i, \psi_{ij})$ killed by f factors uniquely through it because it does so component by component and the factorizations are automatically compatible with the gluing. The cokernel is constructed identically with $\mathcal{C}_i = \mathrm{coker}(f_i)$, the restricted gluing again inheriting (9.6). The zero object is $(0, \mathrm{id})$ and biproducts are componentwise. Because kernels and cokernels are computed componentwise and $\mathrm{QCoh}(U_i)$ is abelian, every monomorphism is the kernel of its cokernel and every epimorphism the cokernel of its kernel, checked on each U_i; hence $\mathrm{Tw}_\alpha(X)$ is abelian. Coherence and local freeness are conditions on the $\mathcal{E}_i$ alone, preserved by the operations that preserve them on each U_i, which proves the closure statement.

Step 2: The tensor product adds classes. Let $(\mathcal{E}_i, \psi_{ij})$ be α-twisted and $(\mathcal{F}_i, \chi_{ij})$ be β-twisted on the common cover. Set $\mathcal{G}_i = \mathcal{E}_i \otimes_{\mathcal{O}_{U_i}} \mathcal{F}_i$ with gluing $\theta_{ij} = \psi_{ij} \otimes \chi_{ij}$. The twisted cocycle scalar of the tensor is the product of the two,

$$\theta_{ij}\theta_{jk}\theta_{ki} = (\psi_{ij}\psi_{jk}\psi_{ki}) \otimes (\chi_{ij}\chi_{jk}\chi_{ki}) = (\alpha_{ijk}\,\mathrm{id}) \otimes (\beta_{ijk}\,\mathrm{id}) = (\alpha_{ijk}\beta_{ijk})\,\mathrm{id}_{\mathcal{G}_i}$$

and since the cocycle of $\alpha+\beta$ is the product $\alpha_{ijk}\beta_{ijk}$ (the group law in H^2 is induced by multiplication in $\mathbb{G}_m$), the tensor product is $(\alpha+\beta)$-twisted. For $\alpha = \beta = 0$ the trivial cocycle makes (9.6) the ordinary cocycle condition, so a 0-twisted sheaf is descent data for an ordinary quasi-coherent sheaf on X, giving $\mathrm{Tw}_0(X) = \mathrm{QCoh}(X)$. For the internal hom, the gluing on $\mathcal{H}om(\mathcal{E}_i, \mathcal{F}_i)$ is $g \mapsto \chi_{ij} \circ g \circ \psi_{ij}^{-1}$, whose triple product carries the scalar $\beta_{ijk}\alpha_{ijk}^{-1}$ because the inverse gluing ψ^{-1} contributes α_{ijk}^{-1}; hence $\mathcal{H}om$ is $(\beta-\alpha)$-twisted, and the dual, the case $\mathcal{F} = \mathcal{O}_X$ (which is 0-twisted), is $(-\alpha)$-twisted.

Step 3: A twisted line bundle trivializes the class. Suppose $(\mathcal{L}_i, \lambda_{ij})$ is an α-twisted invertible sheaf:

each $\mathcal{L}_i$ is an invertible $\mathcal{O}_{U_i}$-module and $\lambda_{ij}\colon \mathcal{L}_j \to \mathcal{L}_i$ are isomorphisms with $\lambda_{ij}\lambda_{jk}\lambda_{ki} = \alpha_{ijk}\,\mathrm{id}$. Refining the cover if necessary, trivialize each $\mathcal{L}_i \cong \mathcal{O}_{U_i}$; then λ_{ij} becomes multiplication by a unit $\ell_{ij} \in \mathbb{G}_m(U_{ij})$, and the twisted cocycle condition reads

$$\ell_{ij}\,\ell_{jk}\,\ell_{ki} = \alpha_{ijk} \tag{9.7}$$

Equation (9.7) states exactly that α_{ijk} is the Čech coboundary $\delta(\ell)_{ijk}$ of the 1-cochain (ℓ_{ij}), so α is a coboundary and $[\alpha] = 0$ in $H^2_{\text{ét}}(X, \mathbb{G}_m)$.

Conversely, if $\alpha = 0$, choose a 1-cochain (ℓ_{ij}) with $\delta(\ell)_{ijk} = \alpha_{ijk}$, that is, with (9.7) holding. Set $\mathcal{L}_i = \mathcal{O}_{U_i}$ and $\lambda_{ij} = \ell_{ij}$; then $\lambda_{ij}\lambda_{jk}\lambda_{ki} = \alpha_{ijk}\,\mathrm{id}$, so $(\mathcal{L}_i, \lambda_{ij})$ is an α-twisted line bundle. Thus an α-twisted invertible sheaf exists if and only if α is a coboundary, that is, if and only if $\alpha = 0$, as required.

The three parts read as a dictionary between the cohomology of $\mathbb{G}_m$ and the linear algebra of twisted sheaves. Part (1) says the twist is a well-behaved decoration: it does not disturb the abelian structure, so all the homological constructions of ordinary sheaf theory transfer verbatim, and one may speak of exact sequences, extensions, and cohomology of twisted sheaves. Part (2) turns the group $H^2(X, \mathbb{G}_m)$ into a grading: the categories $\mathrm{Tw}_\alpha(X)$ are the graded pieces, the tensor product is the graded multiplication, and the degree-zero piece is ordinary sheaves. This is why the Brauer class is added, not multiplied, under tensor: the $\mathbb{G}_m$-weights compose additively in the exponent. Part (3) carries the geometric content. On an ordinary scheme, an invertible sheaf always exists, $\mathcal{O}_X$ itself. On the gerbe $\mathcal{G}_\alpha$, an α-twisted line bundle exists exactly when the gerbe is neutral, and its existence would furnish, by tensoring, an equivalence $\mathrm{Tw}_\alpha(X) \simeq \mathrm{QCoh}(X)$ that trivializes the whole theory. The Brauer class is the obstruction to that trivialization, seen now inside the category of sheaves rather than in the classification of gerbes.

The obstruction of part (3) is worth isolating in its cleanest form before building the Azumaya equivalence on top of it, because the two together explain why twisted sheaves of higher rank always exist even when line bundles do not.

Example 9.3.4: The Twisted Line Bundle Obstruction

Let $\mathcal{G}_\alpha \to X$ be a $\mathbb{G}_m$-gerbe. By theorem 9.3.3(3), an α-twisted line bundle exists exactly when $\alpha = 0$, and the same argument identifies twisted line bundles with a neutralization of the gerbe: a weight-one invertible sheaf on $\mathcal{G}_\alpha$ is a section of $\mathcal{G}_\alpha \to X$ up to the $\mathbb{G}_m$-band, so its existence is the statement that $\mathcal{G}_\alpha$ is neutral (theorem 9.1.5). The two obstructions, "the gerbe has a global object" and "there is a twisted line bundle", are the same vanishing $\alpha = 0$.

Two concrete instances make the obstruction visible.

1. Over $X = \operatorname{Spec} \mathbb{R}$ with $\alpha \in \mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$ nontrivial, there is no α-twisted line bundle: a one-dimensional α-twisted vector space over $\mathbb{R}$ would be a one-dimensional right $\mathbb{H}$-module (example 9.3.2), and $\mathbb{H}$ has no such module, since $\mathbb{H}$ is a division algebra of dimension four and its smallest nonzero module is $\mathbb{H}$ itself, of rank one over $\mathbb{H}$ but real dimension four (hence sheaf-rank two, the $\mathbb{C}$-dimension of the descent vector space V in the twisted-sheaf datum of example 9.3.2, which is the degree of $\mathbb{H}$ and half its real dimension, not the real dimension four itself). There is thus an α-twisted sheaf of $\mathbb{H}$-module rank one (namely $\mathbb{H}$ as a module over itself), but no α-twisted *invertible* sheaf, because "invertible" means trivializing to $\mathcal{O}$ locally on X, which the twist forbids.

2. Over a smooth projective surface X with a nonzero Brauer class α arising from a conic bundle

or an elliptic fibration, the same phenomenon holds globally: there are α-twisted vector bundles of every rank divisible by the index of α, but none of rank one, and the smallest rank of an α-twisted vector bundle equals the index of the Brauer class. The nonexistence of the twisted line bundle is a curved-space refinement of the $\mathbb{R}$-computation, and it is what the moduli of twisted sheaves of the next section measures.

In both cases the twist obstructs rank one but not higher rank: the Brauer class kills the invertible object while permitting the vector bundle of rank equal to its index.

The example already contains the shape of the general theory. Whenever a twisted line bundle fails to exist, a twisted vector bundle of the index rank exists in its place, and its sheaf of endomorphisms is an Azumaya algebra representing α. This is the bridge to the second main result of the section, the equivalence between twisted sheaves and modules over an Azumaya algebra. It recasts the twisted category, which is defined on the auxiliary gerbe, as the category of modules over an algebra living on X itself, and it is the tool that makes twisted sheaves computable in classical terms.

The construction runs in both directions, and its input is a locally free α-twisted sheaf $\mathcal{V}$ of rank n. Such a $\mathcal{V}$ exists whenever α lies in $\mathrm{Br}(X)$: the tautological weight-one sheaf on the gerbe $\mathcal{G}_\alpha$ is a locally free α-twisted sheaf of rank n, living on the gerbe rather than on X, and it is this global rank-n twisted bundle that the construction takes as input. Given a locally free α-twisted sheaf $\mathcal{V}$ of rank n, its endomorphism sheaf $\mathcal{A} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{V})$ is 0-twisted by theorem 9.3.3(2), so it descends to an sheaf of $\mathcal{O}_X$-algebras on X; étale-locally $\mathcal{V}$ is free of rank n and $\mathcal{A}$ is the matrix algebra $M_n(\mathcal{O}_X)$, which is the defining property of an Azumaya algebra of degree n. In the other direction, tensoring with $\mathcal{V}$ turns a module over $\mathcal{A}$ into a twisted sheaf. These two functors are mutually inverse equivalences.

Definition 9.3.5: The Azumaya Module Equivalence

Let $\alpha \in \mathrm{Br}(X) \subseteq H^2_{\text{ét}}(X, \mathbb{G}_m)$ be represented by an Azumaya algebra $\mathcal{A}$ of degree n (definition 9.2.3, theorem 9.2.4), and let $\mathcal{V}$ be a locally free α-twisted sheaf of rank n with $\mathcal{A} \cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{V})$. The *Azumaya module equivalence* is the pair of functors

$$\mathrm{Tw}_\alpha(X) \longrightarrow \mathrm{Mod}(\mathcal{A}), \qquad \mathcal{E} \longmapsto \mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{E})$$

$$\mathrm{Mod}(\mathcal{A}) \longrightarrow \mathrm{Tw}_\alpha(X), \qquad \mathcal{M} \longmapsto \mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}$$

where $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{E})$ is a right $\mathcal{A} = \mathcal{H}om(\mathcal{V},\mathcal{V})$-module by precomposition ($g \cdot a = g \circ a$, so $(g \cdot a) \cdot b = g \cdot (ab)$), so that $\mathrm{Mod}(\mathcal{A})$ is the category of right $\mathcal{A}$-modules, and $\mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}$ tensors such an $\mathcal{M}$ against the left $\mathcal{A}$-module $\mathcal{V}$, on which $\mathcal{A} = \mathcal{H}om(\mathcal{V},\mathcal{V})$ acts by evaluation.

Proposition 9.3.6: Twisted Sheaves are Azumaya Modules

The two functors of definition 9.3.5 are mutually inverse equivalences of categories $\mathrm{Tw}_\alpha(X) \simeq \mathrm{Mod}(\mathcal{A})$.

***Proof*:**
The argument establishes the equivalence first in the split case in full and then globally by descent.

Step 1: The split case. Suppose $\alpha = 0$, so $\mathcal{A} = M_n(\mathcal{O}_X)$ and $\mathcal{V} = \mathcal{O}_X^{\oplus n}$ is untwisted, with $\mathcal{A} = \mathcal{H}om(\mathcal{V}, \mathcal{V})$ the matrix algebra. The claim is that $\mathcal{E} \mapsto \mathcal{H}om(\mathcal{V}, \mathcal{E}) = \mathcal{E}^{\oplus n}$ and $\mathcal{M} \mapsto \mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}$ are inverse equivalences between $\mathrm{QCoh}(X)$ and $\mathrm{Mod}(M_n(\mathcal{O}_X))$. This is Morita equivalence for the matrix algebra: the bimodule $\mathcal{V} = \mathcal{O}_X^{\oplus n}$ is a left $M_n(\mathcal{O}_X)$-module (by evaluation) and a right $\mathcal{O}_X$-module, and its dual $\mathcal{V}^\vee$ is the reverse bimodule, a right $M_n(\mathcal{O}_X)$-module and a left $\mathcal{O}_X$-module, with $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{V}, \mathcal{E}) = \mathcal{V}^\vee \otimes_{\mathcal{O}_X} \mathcal{E}$ carrying the right M_n-action. The canonical maps

$$\mathcal{V} \otimes_{\mathcal{O}_X} \mathcal{V}^\vee \xrightarrow{\sim} M_n(\mathcal{O}_X), \qquad \mathcal{V}^\vee \otimes_{M_n(\mathcal{O}_X)} \mathcal{V} \xrightarrow{\sim} \mathcal{O}_X$$

are isomorphisms of bimodules, the first because an $n \times n$ matrix is a sum of rank-one matrices $v \otimes w^\vee$, the second because the trace pairing $\mathcal{V}^\vee \otimes \mathcal{V} \to \mathcal{O}_X$ is M_n-balanced and surjective. From these two isomorphisms the composite $\mathcal{H}om(\mathcal{V}, \mathcal{E}) \otimes_{\mathcal{A}} \mathcal{V} = (\mathcal{V}^\vee \otimes_{\mathcal{O}_X} \mathcal{E}) \otimes_{M_n} \mathcal{V} = \mathcal{E} \otimes_{\mathcal{O}_X} (\mathcal{V}^\vee \otimes_{M_n} \mathcal{V}) = \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{O}_X = \mathcal{E}$ recovers $\mathcal{E}$ naturally, and the reverse composite $\mathcal{H}om(\mathcal{V}, \mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}) = \mathcal{V}^\vee \otimes_{\mathcal{O}_X} (\mathcal{M} \otimes_{M_n} \mathcal{V}) = \mathcal{M} \otimes_{M_n} (\mathcal{V} \otimes_{\mathcal{O}_X} \mathcal{V}^\vee) = \mathcal{M} \otimes_{M_n} M_n = \mathcal{M}$ recovers $\mathcal{M}$; both natural in the argument. The functors are therefore inverse equivalences in the split case, which is the classical Morita theorem for $M_n(\mathcal{O}_X)$ applied over the structure sheaf.

Step 2: The general case by descent. For arbitrary $\alpha \in \mathrm{Br}(X)$, choose an étale cover $p \colon U \to X$ on which $\alpha|_U = 0$, so that over U the algebra $\mathcal{A}|_U \cong M_n(\mathcal{O}_U)$ splits and $\mathcal{V}|_U$ is the untwisted $\mathcal{O}_U^{\oplus n}$; such a cover exists because α is represented by the Azumaya algebra $\mathcal{A}$, which étale-locally is a matrix algebra (theorem 9.2.4). Over U, Step 1 gives an equivalence $\mathrm{Tw}_0(U) = \mathrm{QCoh}(U) \simeq \mathrm{Mod}(\mathcal{A}|_U)$, and it is natural in U: the two functors $\mathcal{H}om(\mathcal{V}, -)$ and $- \otimes_{\mathcal{A}} \mathcal{V}$ commute with the pullback along $U \times_X U \rightrightarrows U$ because $\mathcal{H}om$ and $\otimes$ commute with flat base change. Twisted sheaves on X are, by definition 9.3.1, quasi-coherent sheaves on the gerbe, and a quasi-coherent sheaf on a stack is a sheaf on any étale atlas together with descent data (definition 8.5.1); likewise $\mathcal{A}$-modules on X satisfy étale descent because $\mathcal{A}$ is a quasi-coherent sheaf of algebras. The two categories $\mathrm{Tw}_\alpha(X)$ and $\mathrm{Mod}(\mathcal{A})$ are thus the descent categories of $\mathrm{QCoh}(U)$ and $\mathrm{Mod}(\mathcal{A}|_U)$ along p, and an equivalence over U compatible with the descent data descends to an equivalence over X. The compatibility is exactly the naturality just noted. Therefore $\mathcal{H}om(\mathcal{V}, -)$ and $\mathcal{V} \otimes_{\mathcal{A}} -$ are inverse equivalences $\mathrm{Tw}_\alpha(X) \simeq \mathrm{Mod}(\mathcal{A})$, completing the proof.

The equivalence is the computational payoff of the section. It says that studying α-twisted sheaves is the same as studying modules over a fixed Azumaya algebra $\mathcal{A}$ of class α, an object living on X with no reference to the gerbe. The gerbe was the conceptual home of the twist, the site where the band $\mathbb{G}_m$ acted by scaling; the Azumaya algebra is its shadow on X, and the equivalence trades the extra dimension of the gerbe for the noncommutativity of $\mathcal{A}$. Over a point this recovers the statement of example 9.3.2 exactly: $\mathcal{A}$ is the division algebra D with $[D] = \alpha$, and $\mathrm{Mod}(\mathcal{A})$ is the category of D-modules, so a twisted vector space is a D-module. The twist has been converted into module theory over a division algebra, which is the form in which the arithmetic of the Brauer group is most transparent.

The equivalence also refines the twisted-line-bundle obstruction of theorem 9.3.3(3). An α-twisted line bundle corresponds under the equivalence to a right $\mathcal{A}$-module that is locally free of rank one over $\mathcal{O}_X$, which forces $\mathcal{A}$ to act on a rank-one module, hence $\mathcal{A} \cong \mathcal{O}_X$ and $\alpha = 0$. The obstruction and the equivalence are two views of the same fact: the Brauer class is nonzero precisely when the representing algebra is not the structure sheaf, and precisely when the twisted objects have no invertible member.

A twisted sheaf on a Severi-Brauer variety completes the picture and connects it back to the geometry that produced the Brauer class in the first place.

Example 9.3.7: Twisted Sheaves on a Severi-Brauer Variety

Let D be a central division algebra over a field k of degree n, with class $\alpha = [D] \in \mathrm{Br}(k)$, and let $P = \mathrm{SB}(D)$ be its Severi-Brauer variety, the form of $\mathbb{P}^{n-1}_k$ that becomes $\mathbb{P}^{n-1}$ over k^{sep} and whose functor of points sends a k-algebra R to the right ideals of $D \otimes_k R$ that are local direct summands of reduced dimension 1 (over $\bar{k}$ these are the right ideals of dimension n; for D a division algebra $\mathrm{SB}(D)(k) = \varnothing$, which is why the class is nonsplit). Over k^{sep} the twisted line bundle $\mathcal{O}(1)$ on $\mathbb{P}^{n-1}$ exists, but it does not descend to P, because its descent obstruction is exactly α: the Galois cocycle needed to glue $\mathcal{O}(1)$ over the twisting differs from the trivial one by $\alpha_{\sigma,\tau}$, so $\mathcal{O}(1)$ is a $p^*\alpha$-twisted line bundle on P rather than an ordinary one, where $p\colon P \to \operatorname{Spec} k$ is the structure map. What does descend is $\mathcal{O}(n) = \mathcal{O}(1)^{\otimes n}$, since $n\alpha = 0$ in $\mathrm{Br}(k)$ (the class is n-torsion, theorem 9.2.4), and more to the point the twisted sheaf $\mathcal{O}(1)$ on P is the geometric incarnation of the twisted objects over k: pushing it forward, $p_*\mathcal{O}(1)$ is an α-twisted vector space over k of twisted rank n, since $H^0(\mathbb{P}^{n-1}, \mathcal{O}(1))$ is n-dimensional over k^{sep}. Its endomorphism algebra recovers the division algebra, $\mathcal{H}om_k(p_*\mathcal{O}(1), p_*\mathcal{O}(1)) \cong D$ (definition 9.3.5 over the point), so $p_*\mathcal{O}(1)$ is the tautological twisted vector space whose associated Azumaya algebra is D itself, the point-level model of example 9.3.2 realized geometrically on P. The Severi-Brauer variety thus carries a canonical $p^*\alpha$-twisted line bundle whose existence on P, contrasted with its nonexistence on $\operatorname{Spec} k$, is the geometric statement that P splits α: pulling α back to P kills it, because on P the twisted line bundle $\mathcal{O}(1)$ finally exists.

The Severi-Brauer example closes the circle opened in section 9.2. There the Brauer class was produced as the obstruction in a PGL_n-torsor, which is the same data as the Severi-Brauer variety. Here that variety reappears as the space on which the twisted line bundle exists after pullback, and the splitting of the Brauer class by P is read off the sudden existence of $\mathcal{O}(1)$ as an ordinary object upstairs. Twisted sheaves make the passage between the two descriptions of a Brauer class, the cohomological and the geometric, into a statement about where a line bundle does and does not exist.

Exercise 9.3.1

1. Let $\mathcal{G}_\alpha \to X$ be a $\mathbb{G}_m$-gerbe and let $\mathcal{F}$ be a quasi-coherent sheaf on $\mathcal{G}_\alpha$ of weight w (definition 9.3.1). Show that tensoring $\mathcal{F}$ with a weight-w' sheaf gives a weight-$(w + w')$ sheaf, and deduce that the weight-zero sheaves are exactly the pullbacks of quasi-coherent sheaves from X. Conclude that $\mathrm{QCoh}(\mathcal{G}_\alpha)$ decomposes as a direct sum $\bigoplus_{w\in\mathbb{Z}} \mathrm{QCoh}(\mathcal{G}_\alpha, w)$ over the weight, and identify $\mathrm{QCoh}(\mathcal{G}_\alpha, 1)$ with $\mathrm{Tw}_\alpha(X)$.

2. Verify directly from the Čech description that the tensor product of two α-twisted sheaves is 2α-twisted, and that if $2\alpha \neq 0$ there is still no 2α-twisted line bundle. Using this, show that the assignment $\alpha \mapsto \mathrm{Tw}_\alpha(X)$ upgrades $H^2_{\text{ét}}(X, \mathbb{G}_m)$ to a grading on the category of all twisted sheaves, and explain why the tensor unit lives in degree 0.

3. Over $X = \operatorname{Spec} \mathbb{R}$ with $\alpha \in \mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$ the nontrivial class, describe the abelian category $\mathrm{Tw}_\alpha(\mathbb{R})$ of coherent α-twisted sheaves explicitly as the category of finite-dimensional right $\mathbb{H}$-modules. What are the simple objects? What is the smallest twisted rank, and why is there no object of twisted rank one that is invertible in the sense of theorem 9.3.3(3)?

4. Let $\mathcal{A}$ be an Azumaya algebra of degree n on X with class α, and let $\mathcal{V}$ be a locally free α-twisted sheaf of rank n with $\mathcal{A} = \mathcal{H}om(\mathcal{V}, \mathcal{V})$. Using the Azumaya module equivalence (definition 9.3.5), show that the α-twisted sheaves that are locally free of rank n over $\mathcal{O}_X$ correspond to the $\mathcal{A}$-modules that are locally free of rank one over $\mathcal{A}$, and hence that a global

such twisted bundle exists if and only if $\mathcal{A}$ has a rank-one free module, that is, if and only if $\mathcal{A} \cong \mathcal{H}om(\mathcal{W}, \mathcal{W})$ splits off a twisted bundle $\mathcal{W}$. Relate this to the index of α.

5. Let $P = \mathrm{SB}(D) \to \operatorname{Spec} k$ be the Severi-Brauer variety of a degree-n division algebra D (example 9.3.7). Show that the pullback $p^* : \mathrm{Br}(k) \to \mathrm{Br}(P)$ kills $\alpha = [D]$, by exhibiting on P the $p^*\alpha$-twisted line bundle $\mathcal{O}(1)$ and invoking theorem 9.3.3(3). Explain why this does not contradict the nonexistence of an α-twisted line bundle over $\operatorname{Spec} k$, and interpret the statement "P splits α" in terms of twisted line bundles.

6. The internal hom $\mathcal{H}om(\mathcal{E}, \mathcal{F})$ of an α-twisted sheaf $\mathcal{E}$ and a β-twisted sheaf $\mathcal{F}$ is $(\beta - \alpha)$-twisted (theorem 9.3.3(2)). Deduce that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{E})$ is 0-twisted for any α-twisted sheaf $\mathcal{E}$, hence descends to a genuine sheaf of $\mathcal{O}_X$-algebras on X, and that when $\mathcal{E}$ is locally free of rank n this algebra is Azumaya of degree n. Explain how this recovers the construction of the Azumaya algebra in definition 9.3.5 and why the class of that algebra is α rather than 0, despite $\mathcal{H}om(\mathcal{E}, \mathcal{E})$ itself being untwisted.

9.4 Moduli of Twisted Sheaves

This closing section reaches the payoff toward which the entire chapter has aimed. The previous three sections built the apparatus: a gerbe is the two-categorical object one level above a torsor, a $\mathbb{G}_m$-gerbe realizes a Brauer class geometrically, and an α-twisted sheaf is a quasi-coherent sheaf on the gerbe $\mathcal{G}_\alpha$ of class α on which the band acts by scalars. What has been missing is a moduli space. Chapter 2 constructed the Hilbert and Quot schemes as the fine moduli spaces of untwisted subschemes and quotients, and Chapter 8 assembled the whole framework into the statement that the stack $\mathfrak{Coh}$ of coherent sheaves on a projective X is algebraic. This section carries out the same program for twisted sheaves: it defines the stack $\mathcal{M}_\alpha$ of α-twisted sheaves, states that it is algebraic (the twisted analogue of Chapter 8, due to Lieblich), and then exhibits the reason the twisted theory earns its own development. The twisted moduli space is often a *fine* moduli space where the untwisted one is only coarse, and it converts the period-index problem for a Brauer class into a statement about the ranks of twisted sheaves.

The section is also where Part III closes. The gerbes and twisted sheaves of this chapter are the last of the non-representable phenomena the book set out to organize, and their moduli are the last of the algebraic stacks it constructs. Once the twisted moduli stack is in hand, only the capstone remains, in which the moduli of curves assembles every strand of the book into a single object.

The definition of the twisted moduli stack copies the untwisted one word for word, with "coherent sheaf" replaced by "α-twisted sheaf". This is the right way to think about it: the twist is a decoration on the objects, not a change to the moduli formalism. A family of α-twisted sheaves over a base T is an α-twisted sheaf on the gerbe pulled back to X_T, flat over T, and the stack records these families and their isomorphisms exactly as $\mathfrak{Coh}$ records families of ordinary sheaves.

Definition 9.4.1: Moduli Stack of Twisted Sheaves

Let X be a projective scheme over a field k with a fixed ample line bundle $\mathcal{O}_X(1)$, and let $\alpha \in \mathrm{Br}(X)$ be a Brauer class with a chosen $\mathbb{G}_m$-gerbe $\mathcal{G}_\alpha \to X$ representing it (definition 9.2.1). The *moduli stack of α-twisted sheaves*, denoted $\mathcal{M}_\alpha$, is the stack over $(\mathbf{Sch}/k)$ whose fiber over a k-scheme T is the groupoid

$$\mathcal{M}_\alpha(T) = \left\{ \begin{array}{l} \mathcal{E} \text{ an } \alpha\text{-twisted sheaf on } X_T = X \times_k T, \\ \text{quasi-coherent of finite presentation and flat over } T \end{array} \right\}$$

with morphisms the isomorphisms of twisted sheaves. Here an α-twisted sheaf on X_T means a quasi-coherent sheaf on the pulled back gerbe $\mathcal{G}_\alpha \times_X X_T$ of weight 1 under the band (definition 9.3.1, definition 8.5.1); flatness over T is checked on the underlying sheaf of any local trivialization. For a polynomial $P \in \mathbb{Q}[m]$ the open and closed substack $\mathcal{M}_\alpha^P \subseteq \mathcal{M}_\alpha$ is the locus of families whose fibers have *twisted Hilbert polynomial* P, where the twisted Hilbert polynomial of an α-twisted sheaf $\mathcal{E}$ on X is $P_{\mathcal{E}}(m) = \chi(X, \mathcal{E}(m))$ computed from the twisted Chern character and the untwisted Todd class through the twisted Riemann-Roch formula.

The definition is the twisted mirror of the stack $\mathfrak{Coh}$ of Chapter 8, and every ingredient is one already in hand. The objects are the twisted sheaves of definition 9.3.1; the family condition is flatness over the base, unchanged from the untwisted case because flatness of a twisted sheaf is flatness of its local trivializations; and the numerical invariant that cuts $\mathcal{M}_\alpha$ into finite-type pieces is the twisted Hilbert polynomial, the twisted analogue of the invariant that stratified $\mathfrak{Coh}$. The dependence on α is the reason for the construction: $\mathcal{M}_0$ is the untwisted stack $\mathfrak{Coh}$, while for $\alpha \neq 0$ the objects of $\mathcal{M}_\alpha$ are never sheaves on X, because theorem 9.3.3 showed that an α-twisted invertible sheaf, hence any α-twisted sheaf that trivializes the twist, exists only when $\alpha = 0$. Different Brauer classes give different moduli problems, and this is what makes the twisted theory more than a relabeling.

One subtlety deserves a word. The twisted Hilbert polynomial requires a notion of $\chi(X, \mathcal{E}(m))$ for a twisted sheaf $\mathcal{E}$, and $\mathcal{E}$ is not a sheaf on X, so its cohomology must be taken on the gerbe. The gerbe $\mathcal{G}_\alpha \to X$ has X proper as its base, and coherent cohomology of a weight-1 sheaf on it is finite-dimensional because the weight-1 part descends, on a cover U trivializing α, to a coherent sheaf on the trivializing cover, whose cohomology is finite-dimensional by the finiteness of coherent cohomology on the proper scheme X (definition 8.5.1, and the cohomology of a stack developed in Chapter 8). The $\mathbb{G}_m$-gerbe itself has non-representable structure map with $\mathbb{G}_m$ inertia, so this finiteness is a property of the weight-1 descent, not of a properness of the gerbe. The Euler characteristic $\chi(X, \mathcal{E}(m)) = \sum_i (-1)^i \dim_k H^i(\mathcal{G}_\alpha, \mathcal{E}(m))$ is therefore a well-defined integer, and it is a numerical polynomial in m by the twisted Riemann-Roch theorem, exactly as in the untwisted case. The leading coefficient encodes the twisted rank and the lower coefficients the twisted Chern character.

A first concrete instance grounds the definition before the general theory. The cleanest case is $X = \operatorname{Spec} k$ a point, where a twist is a class in the field Brauer group and a twisted sheaf is a module over a division algebra.

Example 9.4.2: Twisted Sheaves over a Point

Let $X = \operatorname{Spec} k$ and let $\alpha \in \mathrm{Br}(k)$ be represented by a central division algebra D over k of degree d (so $\dim_k D = d^2$ and the period of α divides d). By definition 9.3.5 and the computation of example 9.3.4, the category of α-twisted sheaves on the point is the category of finitely generated right D-modules. Every such module is a direct sum of copies of D, so an α-twisted sheaf of rank rd

(as a k-vector space it has dimension rd^2) is $D^{\oplus r}$, unique up to isomorphism, with automorphism group $\mathrm{GL}_r(D)$. The moduli stack $\mathcal{M}_\alpha$ over $\operatorname{Spec} k$ therefore decomposes as a disjoint union

$$\mathcal{M}_\alpha = \coprod_{r\geq 0} B\mathrm{GL}_r(D)$$

of classifying stacks of the groups $\mathrm{GL}_r(D)$, one for each isomorphism class of D-module. Each component $B\mathrm{GL}_r(D)$ is a smooth algebraic stack over k of dimension $-\dim_k \mathrm{GL}_r(D) = -r^2d^2$, negative because the automorphisms outnumber the moduli: there is a single object and a positive-dimensional group fixing it. When $\alpha = 0$, $D = k$ and this recovers $\coprod_r B\mathrm{GL}_r$, the stack of finite-dimensional vector spaces, which is the point $X = \operatorname{Spec} k$ case of the untwisted $\mathfrak{Coh}$. The effect of the twist is to replace the structure group GL_r by the inner form $\mathrm{GL}_r(D)$ cut out by the division algebra, which is exactly the twisting the Brauer class encodes.

The point case already displays the two features that make the twisted moduli theory work. First, the twist acts by replacing the automorphism groups of the objects with inner forms: over the point, GL_r becomes $\mathrm{GL}_r(D)$, and in general the twisted sheaves are the modules over an Azumaya algebra (definition 9.3.5) whose fiberwise structure group is an inner form of GL_n. Second, the stack is still built from classifying stacks and quotient stacks, so the machinery of Chapters 7 and 8 applies without modification. The general theorem, that $\mathcal{M}_\alpha$ over a projective X is an algebraic stack, is the twisted counterpart of the algebraicity of $\mathfrak{Coh}$, and its proof runs along the same lines: verify Artin's criteria, using a twisted deformation theory and a twisted boundedness in place of the untwisted ones.

The algebraicity of $\mathcal{M}_\alpha$ is Lieblich's theorem. Its proof is the twisted analogue of the verification of Artin's criteria for $\mathfrak{Coh}$ carried out in Chapter 8, and rather than reproduce that long argument in twisted dress, the theorem is stated with its key ideas and the full proof is cited. The deferral is warranted because the substance is not new mathematics but the transfer of an argument already given in full: the twisted deformation theory, the twisted boundedness, and the verification of the criteria all mirror the untwisted case, and Lieblich's paper carries out the transfer with the care the general Brauer class requires.

Theorem 9.4.3: Algebraicity of the Twisted Moduli Stack

Let X be a projective scheme over a field k with a fixed ample sheaf, and let $\alpha \in \mathrm{Br}(X)$. Then the moduli stack $\mathcal{M}_\alpha$ of definition 9.4.1 is an algebraic stack in the sense of definition 8.3.1, locally of finite type over k. For a fixed twisted Hilbert polynomial P the substack $\mathcal{M}_\alpha^P$ is of finite type. If moreover X is a smooth projective surface and a stability condition is fixed, the open substack $\mathcal{M}_\alpha^{P,s}$ of stable α-twisted sheaves with invariants P admits a coarse moduli space that is a quasi-projective scheme, and the substack of semistable sheaves has a proper good moduli space.

***Proof Key ideas*:**
The full proof is Lieblich's [22]; it verifies Artin's criteria (theorem 8.4.3) for $\mathcal{M}_\alpha$ exactly as Chapter 8 verified them for $\mathfrak{Coh}$ and $\mathcal{M}_g$, with each untwisted input replaced by its twisted counterpart. The transfer of the algebraization step, which for stacks over a general base requires the theory of coherent cohomology on the gerbe over its proper base, is the technical content and is carried out there; the conceptual mechanism is the following.

Step 1: Twisted sheaves form a stack. That $\mathcal{M}_\alpha$ is a stack for the fppf topology (condition (0) of theorem 8.4.3) is descent for twisted sheaves. A twisted sheaf on X_T is a quasi-coherent sheaf on the gerbe $\mathcal{G}_\alpha \times_X X_T$, and quasi-coherent sheaves on a stack satisfy descent because they are defined by descent on the atlas (definition 8.5.1); the weight-1 condition under the band and the flatness over T are local and descend. So a compatible family of twisted sheaves over an fppf cover of T glues to a twisted sheaf over T, which is condition (0).

Step 2: Twisted deformation theory. The deformation and obstruction theory of $\mathcal{M}_\alpha$ at a twisted sheaf $\mathcal{E}$ is the twisted Ext theory: the tangent space is $\mathrm{Ext}^1_{\mathcal{G}_\alpha}(\mathcal{E}, \mathcal{E})$ and the obstruction space is $\mathrm{Ext}^2_{\mathcal{G}_\alpha}(\mathcal{E}, \mathcal{E})$, the self-extension groups computed in the abelian category $\mathrm{Tw}_\alpha(X)$ of theorem 9.3.3. These are finite-dimensional because $\mathcal{E}$ is coherent and its weight-1 cohomology descends to the proper base X, and they satisfy the tangent-torsor and obstruction-vanishing properties by the same homological argument that governs ordinary coherent sheaves, since $\mathrm{Tw}_\alpha(X)$ is an abelian category with enough of the structure of $\mathrm{QCoh}(X)$. This is the deformation-theoretic input to condition (3), and it is where the twist enters: the extensions are taken among twisted sheaves, so the infinitesimal structure is that of the twisted category, not the untwisted one.

Step 3: Twisted boundedness. The finite-type claim for $\mathcal{M}^P_\alpha$ (part of conditions (1) and (3)) rests on a twisted boundedness theorem: the α-twisted sheaves with fixed twisted Hilbert polynomial P, or fixed twisted Chern character, that are semistable form a bounded family. The proof mirrors the untwisted Castelnuovo-Mumford regularity bound of Chapter 2, applied on the gerbe. On an fppf cover trivializing α, a twisted sheaf becomes an ordinary sheaf on the cover with a GL_n-descent datum, and the untwisted boundedness on the cover, made equivariant, bounds the twisted sheaves upstairs. Boundedness is what makes $\mathcal{M}^P_\alpha$ finite-type rather than only locally of finite type, and it is the twisted analogue of the boundedness that cut $\mathfrak{Coh}$ into finite-type pieces.

Step 4: Effectivity and openness. Effectivity of formal twisted sheaves (the rest of condition (3)) is Grothendieck's existence theorem applied to the weight-1 descent: a compatible system of twisted sheaves over the Artinian quotients of a complete local ring algebraizes, because the base X is proper over $\operatorname{Spec} R$ and a twisted sheaf, carrying the ample sheaf pulled back from X, is a compatible system of coherent sheaves on the proper X_R to which the existence theorem applies. Openness of versality (condition (4)) is the upper-semicontinuity of Ext^1 in twisted families, identical to the untwisted argument. With all five criteria verified in their twisted forms, theorem 8.4.3 certifies $\mathcal{M}_\alpha$ as an algebraic stack.

Step 5: The coarse moduli space. For a surface with a fixed stability condition, the stable locus $\mathcal{M}^{P,s}_\alpha$ has automorphism groups reduced to the twisted scalars $\mathbb{G}_m$ (a stable twisted sheaf is twisted-simple), so it is a $\mathbb{G}_m$-gerbe over an algebraic space, and that space is a quasi-projective scheme by the twisted version of the Gieseker-Maruyama construction, built as a GIT quotient of a twisted Quot scheme. The semistable locus has a proper good moduli space by the twisted analogue of the Simpson-Langton-good-moduli-space theory, which is the good moduli space construction of Chapter 8 applied to the twisted stack. The details are in [22], completing the account of the key ideas.

The theorem places the twisted moduli stack on exactly the footing the book has built for every moduli problem: it is algebraic, cut into finite-type pieces by a numerical invariant, and equipped with a coarse space when a stability condition rigidifies the automorphisms. The proof is a checklist rerun of Artin's criteria, and the only new ingredients, the twisted Ext deformation theory and the twisted boundedness, are the twisted shadows of the untwisted inputs from Chapters 2 and 5. This is the sense in which the twisted theory is not a separate subject but a decoration of the one already developed. What makes it worth developing separately is what the twisted moduli space can do that the untwisted one cannot, and that is the content of the next result.

The reason to construct $\mathcal{M}_\alpha$ rather than work with untwisted sheaves is that the twist can make a moduli problem *fine* where the untwisted version is only coarse, and that it converts an arithmetic question about the Brauer class, the period-index problem, into a geometric question about the twisted moduli space. Both phenomena are best seen on an example, and the canonical one is a Brauer class arising from an elliptic fibration, where the twisted moduli space of rank-one twisted sheaves is a new variety related to the original by the twist.

Example 9.4.4: Fine Moduli, Period, and Index

Three faces of the same construction show why the twisted moduli space earns its place.

1. *Fineness from the twist.* Let X be a smooth projective surface. The untwisted moduli space M of stable sheaves on X with fixed invariants is often only a coarse moduli space: a universal sheaf on $X \times M$ need not exist, because the stable sheaves have scalar automorphisms $\mathbb{G}_m$ that obstruct gluing a universal family, and the obstruction is a Brauer class $\alpha_M \in \mathrm{Br}(M)$ on the base M itself, the class of the $\mathbb{G}_m$-gerbe of Chapter 8's residual-gerbe type. Two distinct twists are in play and should be kept apart: a class $\alpha \in \mathrm{Br}(X)$ on X, which is what the stack $\mathcal{M}_\alpha$ of α-twisted sheaves is built from, and the class $\alpha_M \in \mathrm{Br}(M)$ on M, which is the obstruction to a universal family for the untwisted problem. The twist that repairs the coarse space is the second one, on M: there exists a $\mathrm{pr}_M^* \alpha_M$-twisted universal sheaf $\mathcal{U}$ on $X \times M$, and this $\mathcal{U}$ makes M a *fine* moduli space for α_M-twisted sheaves even though it was only coarse for untwisted ones. The obstruction α_M to a universal family, read as a Brauer class on M, is precisely the twist one absorbs to turn the coarse space fine.
2. *Period and index.* For $\alpha \in \mathrm{Br}(X)$ the *period* $\mathrm{per}(\alpha)$ is the order of α in $\mathrm{Br}(X)$, and the *index* $\mathrm{ind}(\alpha)$ is the greatest common divisor of the degrees of the Azumaya algebras representing α, equivalently the minimal rank of a locally free α-twisted sheaf. That $\mathrm{ind}(\alpha)$ is the minimal twisted rank is exactly definition 9.3.5: an Azumaya algebra of degree d representing α is $\underline{\mathrm{End}}$ of a locally free α-twisted sheaf of rank d, and conversely. So the period-index problem, the question of how large $\mathrm{ind}(\alpha)$ can be relative to $\mathrm{per}(\alpha)$, becomes the question of the minimal rank of a twisted sheaf, a question the moduli stack $\mathcal{M}_\alpha$ is designed to answer: the nonemptiness of $\mathcal{M}_\alpha^P$ for a P of small twisted rank is the existence of a twisted sheaf of that rank. The elementary bound $\mathrm{per}(\alpha) \mid \mathrm{ind}(\alpha)$ follows from definition 9.2.3 and the torsion bound of theorem 9.2.4: the class of a degree-d Azumaya algebra is d-torsion, so the period divides the index. The companion fact that period and index share the same prime factors is a further elementary consequence of the primary decomposition of the torsion Brauer group $\mathrm{Br}'(X)$ of definition 9.2.3 into its p-primary parts, obtained by applying the torsion bound of theorem 9.2.4 in each p-primary component rather than to α as a whole.
3. *A class from an elliptic fibration.* Let $\pi\colon S \to C$ be a relatively minimal elliptic surface over a smooth projective curve C, with a section absent so that π has no rational section but does have multisections. The relative Jacobian $J \to C$ is an elliptic surface with a section, and S is a torsor under J over C; the class of this torsor in the sheaf cohomology group $H^1(C, J)$, mapped into $\mathrm{Br}(J)$ through the Leray sequence of π, is a Brauer class $\alpha \in \mathrm{Br}(J)$ measuring the failure of S to have a section. A rank-one α-twisted sheaf on J is precisely the data that a genuine line bundle on S would provide if $S \to C$ had a section, and the twisted moduli space $\mathcal{M}_\alpha$ of rank-one α-twisted sheaves on J recovers S itself: $S \cong \mathcal{M}_\alpha^{P_1}$ for the twisted Hilbert polynomial P_1 of a rank-one twisted sheaf. The twisted moduli space of the Jacobian reconstructs the original fibration, and the Brauer class is the exact obstruction

that separates them. The order of α divides the multisection index of π, tying the period of the class to the geometry of the fibration.

The example is the justification for the chapter. In the first face, the twist converts a coarse moduli space into a fine one by absorbing the scalar automorphisms into a universal twisted sheaf; the obstruction to a universal family, which for untwisted sheaves is a genuine failure, becomes a Brauer class one can twist away. In the second, the period-index problem, a central question about how far a Brauer class is from being represented by a small algebra, is recast as a question about the minimal rank of a twisted sheaf, which the moduli stack is built to detect. In the third, the twisted moduli space of the Jacobian of an elliptic fibration reconstructs the original fibration, so the Brauer class is the precise geometric datum distinguishing a surface from its Jacobian. Each face is a problem the untwisted theory cannot even phrase cleanly, and each becomes a statement about $\mathcal{M}_\alpha$.

Two further points about the elliptic-fibration computation connect it to the book's running threads. The Brauer class α there is n-torsion where n is the multisection index, so it is a class of the kind theorem 9.2.4 produced from a PGL_n-torsor, and the twisted sheaves are the modules over the corresponding Azumaya algebra of definition 9.3.5. And the reconstruction $S \cong \mathcal{M}_\alpha^{P_1}$ is a twisted Fourier-Mukai phenomenon: the twisted derived category of J is equivalent to the untwisted derived category of S, an equivalence that lives one level below the moduli statement and is the twisted analogue of the classical Fourier-Mukai equivalence between an abelian variety and its dual. That derived-categorical shadow is the subject of the closing box.

The Brauer class and its twisted sheaves have a topological counterpart that the book will not develop but that is too close to pass over in silence. The same cohomological class that measures a Brauer obstruction reappears, over the complex numbers or in a purely topological setting, as a degree-three cohomology class, and the twisted sheaves reappear as twisted vector bundles and twisted K-theory. The exponential sequence is the bridge, and the analogy is precise enough to state as a dictionary and stop, since developing it belongs to the topological and operator-algebraic volumes of the series.

9.4.5 Note: The Topological Bridge and Twisted Derived Categories Two lineages run parallel to the algebraic-geometric one of this chapter, and the correspondence with each is worth naming precisely.

The Dixmier-Douady class and twisted K-theory. On a topological space or a complex manifold X, the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$ (or its continuous or smooth analogue) gives a boundary map $H^2(X, \mathcal{O}_X^\times) \to H^3(X, \mathbb{Z})$, and the image of a Brauer class $\alpha \in H^2(X, \mathbb{G}_m)$ under this map is its *topological* or *Dixmier-Douady class* in $H^3(X, \mathbb{Z})_{\mathrm{tors}}$. The dictionary is

$$\mathrm{Br}(X) = H^2(X, \mathbb{G}_m)_{\mathrm{tors}} \quad \longleftrightarrow \quad H^3(X, \mathbb{Z})_{\mathrm{tors}}$$

the algebraic Brauer group on the left, its topological shadow on the right, matched by the exponential boundary. Where an α-twisted sheaf lives on the algebraic side, an α-twisted vector bundle lives on the topological side, and the Grothendieck group of twisted bundles is *twisted K-theory*, the receptacle for D-brane charges in the physics that motivated its study. The operator-algebraic incarnation, where the class classifies continuous-trace C^*-algebras and the twist is a bundle of compact operators, belongs to the C^*-algebra volume of the series ('NateCStarAlgebras'), and the homotopy-theoretic twisted K-theory to the algebraic-topology volume ('NateAlgTop'). The three theories, algebraic, topological, and operator-algebraic, are the same class H^3_{tors} read in three categories.

Twisted derived categories and Fourier-Mukai. On the derived side, a Brauer class α twists the derived category: $D^b(X, \alpha)$, the bounded derived category of α-twisted coherent sheaves, is a triangulated category depending on α, reducing to $D^b(X)$ when $\alpha = 0$. Twisted moduli spaces produce equivalences of these: for the elliptic fibration $S \to J$ of example 9.4.4, the reconstruction $S \cong \mathcal{M}_\alpha$ upgrades to a *twisted Fourier-Mukai equivalence* $D^b(J, \alpha) \simeq D^b(S)$, the twisted analogue of the classical Fourier-Mukai transform between an abelian variety and its dual. This is the theory of [3], and its systematic development, together with the derived and spectral algebraic geometry it lives in, belongs to a future volume on derived algebraic geometry. The point to carry forward is only the analogy: the Brauer class that twists the moduli problem also twists the derived category, and the twisted moduli space is where a twisted derived equivalence is realized geometrically.

The box closes the chapter by placing the algebraic theory of twisted sheaves inside the wider landscape of which it is one facet. The Brauer group is simultaneously an object of arithmetic (central simple algebras, cited to 'NateGaloisCohom'), of algebraic geometry (the $\mathbb{G}_m$-gerbes and twisted sheaves of this chapter), of topology (the Dixmier-Douady class and twisted K-theory), and of derived categories (twisted Fourier-Mukai). This chapter owns the second of these, borrows the first, and points to the third and fourth. With the twisted moduli stack constructed and its neighbors named, Part III is complete: the framework of descent, stacks, gerbes, and their moduli is fully in place, and the capstone that follows will assemble it into the moduli of curves, the object toward which the entire book has been building.

Exercise 9.4.1

1. Let $X = \operatorname{Spec} k$ and let $\alpha \in \mathrm{Br}(k)$ be represented by a central division algebra D of degree d. Using example 9.4.2, describe the moduli stack $\mathcal{M}_\alpha$ of α-twisted sheaves completely: identify its connected components, the object and automorphism group of each, and the dimension of each component as an algebraic stack. Then explain why $\mathcal{M}_\alpha$ is not itself a scheme (its coarse space, a disjoint union of points $\operatorname{Spec} k$, discards all of the automorphism data), and

contrast this with the situation on a projective surface where a stability condition is imposed and the moduli space is a positive-dimensional scheme.

2. Show that $\mathcal{M}_0 = \mathfrak{Coh}(X)$: for the zero Brauer class, the moduli stack of 0-twisted sheaves is the untwisted stack of coherent sheaves of Chapter 8. Identify precisely which ingredient of definition 9.4.1 degenerates when $\alpha = 0$, and explain, using theorem 9.3.3, why for $\alpha \neq 0$ the objects of $\mathcal{M}_\alpha$ are never coherent sheaves on X.

3. The proof of theorem 9.4.3 identifies the tangent and obstruction spaces of $\mathcal{M}_\alpha$ at a twisted sheaf $\mathcal{E}$ as $\operatorname{Ext}^1_{\mathcal{G}_\alpha}(\mathcal{E},\mathcal{E})$ and $\operatorname{Ext}^2_{\mathcal{G}_\alpha}(\mathcal{E},\mathcal{E})$. Suppose X is a smooth projective surface and $\mathcal{E}$ is a stable α-twisted sheaf. Assuming the twisted analogue of Serre duality on the surface ($\operatorname{Ext}^2(\mathcal{E},\mathcal{E}) \cong \operatorname{Hom}(\mathcal{E},\mathcal{E}\otimes\omega_X)^\vee$) and that a stable twisted sheaf is twisted-simple ($\operatorname{Hom}(\mathcal{E},\mathcal{E}) = k$), compute the expected dimension of the moduli *stack* $\mathcal{M}_\alpha^{P,s}$ near $[\mathcal{E}]$ as $\dim \operatorname{Ext}^1 - \dim \operatorname{Ext}^0$, deduce that the coarse moduli *space* has expected dimension $\dim \operatorname{Ext}^1$ (rigidifying the $\mathbb{G}_m$-gerbe raises the dimension by one), and explain in one sentence why the $\mathbb{G}_m$ of twisted scalars is what makes the coarse space a scheme rather than a stack.

4. Let $\alpha \in \operatorname{Br}(X)$ have period n, and recall from example 9.4.4 that the index $\operatorname{ind}(\alpha)$ is the minimal rank of a locally free α-twisted sheaf. Using the torsion bound of theorem 9.2.4 (the class of a degree-d Azumaya algebra is d-torsion) and definition 9.3.5, prove that $\operatorname{per}(\alpha) \,\big|\, \operatorname{ind}(\alpha)$. Then, phrasing the period-index problem as a question about the moduli stack, state what geometric fact about $\mathcal{M}_\alpha^P$ would establish the period-index bound $\operatorname{ind}(\alpha) \,\big|\, \operatorname{per}(\alpha)^r$ for a specific exponent r (you need not prove any such bound; state the moduli-theoretic reformulation).

5. For the elliptic fibration $\pi\colon S \to C$ of example 9.4.4, with relative Jacobian $J \to C$ and Brauer class $\alpha \in \operatorname{Br}(J)$, explain why a rank-one α-twisted sheaf on J plays the role that a line bundle would play if $S \to C$ had a section, and why the twisted moduli space $\mathcal{M}_\alpha$ of rank-one twisted sheaves on J recovers S. In particular, identify what the order of α measures about the fibration, and connect the reconstruction $S \cong \mathcal{M}_\alpha$ to the twisted Fourier-Mukai equivalence $D^b(J,\alpha) \simeq D^b(S)$ named in box 9.4.5.

6. The box box 9.4.5 records the correspondence $\operatorname{Br}(X) = H^2(X,\mathbb{G}_m)_{\mathrm{tors}} \leftrightarrow H^3(X,\mathbb{Z})_{\mathrm{tors}}$ via the exponential boundary map. Write down the long exact cohomology sequence of the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$ around degree two, and use it to express the boundary map $H^2(X,\mathcal{O}_X^\times) \to H^3(X,\mathbb{Z})$ and to explain why its restriction to torsion is the map appearing in the dictionary. Then state, without proof, the topological analogue of an α-twisted sheaf and the topological invariant that plays the role of the twisted moduli space's Chern character.

10

Application: The Moduli of Curves

Part I built moduli spaces as functors and asked when a scheme represents them; Part II linearized the question, computing the deformations and obstructions of a single object; Part III replaced the scheme with the stack that finally represents the objects-with-automorphisms a moduli problem classifies. Each Part returned again and again to one example, the moduli of curves, and each left it in a more finished state: a functor whose non-representability was diagnosed by the automorphisms of an elliptic curve, then a tangent space of dimension $3g-3$ read off from the cohomology of the tangent sheaf, then an algebraic stack certified by Artin's criteria. This chapter assembles those threads into the theorem the whole book has been approaching.

The object is $\overline{\mathcal{M}}_g$, and its properties come from the three Parts in turn. Its smoothness and its dimension $3g-3$ are the unobstructed deformation theory of Part II. Its being a Deligne-Mumford stack, and the properness that makes it a good compactification, are the stack theory and the valuative criterion of Part III, with stable reduction supplying the limits and the deformation theory of a node supplying the local model of the boundary. Its coarse moduli space, a projective variety singular where the stack is smooth, is the geometric invariant theory quotient of Part I. The first section defines $\mathcal{M}_g$ and records that it is a smooth Deligne-Mumford stack; the second compactifies it by admitting stable curves; the third proves properness through stable reduction; and the fourth descends to the coarse space, revisits $\mathcal{M}_{1,1}$ and $\overline{\mathcal{M}}_{1,1}$ in full, and points toward the intersection theory of tautological classes that the next volume takes up. The moduli of curves is where deformation theory, geometric invariant theory, and stacks meet in a single object, and it is the natural place for this book to end.

10.1 The Moduli Stack $\mathcal{M}_g$

The book has built three machines, and this chapter runs them together on a single problem. Part I asked when a moduli problem is representable and found that objects with automorphisms defeat fine representability, the j-line of the elliptic curves being the first casualty. Part II linearized a moduli problem: the tangent space to a moduli functor is a cohomology group, and for a smooth curve C that group is $H^1(C, T_C)$, of dimension $3g-3$. Part III supplied the object that finally represents a moduli problem whose objects have automorphisms, the algebraic stack, together with its Deligne-Mumford refinement for objects with finite automorphism groups. This section assembles the three into the statement the whole book has pointed toward: for $g \geq 2$ the moduli of smooth genus-g curves is a smooth Deligne-Mumford stack of dimension $3g-3$. Every clause of that sentence is a theorem from an earlier chapter, and the work here is to see that the hypotheses of each apply to curves and to assemble the conclusions.

The starting point is the correct notion of a family. A single curve is a point of the moduli space; a family of curves over a base S is a map from S to it, and the moduli problem is the assignment $S \mapsto \{\text{families over } S\}$. For curves the right notion of family is forced by the demand that fibers be smooth curves and that the family vary algebraically, which is exactly a smooth proper morphism with the correct fibers. Throughout, k is an algebraically closed field, of characteristic 0 where a count of automorphisms requires it (this is flagged where it is used), and a curve means a smooth proper connected k-curve; $g \geq 2$ unless stated otherwise.

Definition 10.1.1: Family of Smooth Curves

A *family of smooth genus-g curves* over a scheme S is a morphism

$$\pi \colon \mathcal{C} \to S$$

that is smooth, proper, and whose geometric fibers are connected smooth curves of genus g. A morphism of families over a morphism $f \colon S' \to S$ of base schemes is a cartesian square

$$\begin{array}{ccc} \mathcal{C}' & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ S' & \xrightarrow{f} & S \end{array}$$

exhibiting $\mathcal{C}'$ as the pullback $f^*\mathcal{C} = \mathcal{C} \times_S S'$; in particular for each morphism f the pullback $f^*\mathcal{C} \to S'$ is again a family of smooth genus-g curves. A *family of n-pointed smooth genus-g curves* is such a π together with n disjoint sections $\sigma_1, \ldots, \sigma_n \colon S \to \mathcal{C}$ of π, with pairwise disjoint images, whose values on each geometric fiber are n distinct points.
The *moduli stack of smooth genus-g curves* $\mathcal{M}_g$ is the category fibered in groupoids over $(\mathbf{Sch}/k)$ whose objects over S are the families $\pi \colon \mathcal{C} \to S$ and whose morphisms are the cartesian squares above; the fiber $\mathcal{M}_g(S)$ is the groupoid of families over S with isomorphisms of families. The *moduli stack of n-pointed curves* $\mathcal{M}_{g,n}$ is defined identically with n-pointed families in place of families.

The definition makes three demands, and each earns its place. Smoothness of π guarantees the fibers are smooth curves and that they vary without acquiring singularities. Properness rules out families that lose points off to infinity, so that a fiber is a complete curve of a well-defined genus rather than an affine piece of one. Connectedness of the fibers keeps the genus a single invariant of each fiber rather than a sum over components. That π^* of a family is a family is the pullback stability recorded in definition 10.1.1: smoothness, properness, and the fiber conditions are all stable under base change, so $\mathcal{M}_g$ really is a fibered category, its objects glued by the cartesian squares. The passage from a functor to a groupoid is the correction Part III forced. A curve C has automorphisms, so two families can be isomorphic in more than one way; recording only isomorphism classes, as a set-valued functor does, discards the automorphisms and destroys the descent that makes the moduli problem geometric. The fibered category $\mathcal{M}_g$ keeps the isomorphisms, and the automorphisms of a curve reappear as the automorphisms of that curve viewed as an object of $\mathcal{M}_g(\operatorname{Spec} k)$.

The definition is abstract, so it should be grounded at once in the smallest family that is not a single curve: a one-parameter family whose fibers are elliptic curves, the case $g = 1$ that motivated the whole apparatus in Part I.

Example 10.1.2: The Legendre Family

Fix k algebraically closed of characteristic $\neq 2$, let $S = \mathbb{A}^1_k \setminus \{0, 1\}$ with coordinate λ, and let $\mathcal{C} \subset \mathbb{P}^2_k \times S$ be the family of plane cubics

$$y^2 z = x(x - z)(x - \lambda z)$$

with $\pi \colon \mathcal{C} \to S$ the projection to S. For each $\lambda \in S(k)$ the fiber is the smooth plane cubic $y^2 = x(x - 1)(x - \lambda)$, a genus-1 curve, smooth precisely because $\lambda \neq 0, 1$ keeps the three branch points $0, 1, \lambda$ (and ∞) distinct. The morphism π is proper, being a closed subscheme of $\mathbb{P}^2 \times S$ projected to S, and it is smooth of relative dimension 1 because the fiberwise discriminant $\lambda^2(\lambda-1)^2$ is invertible on S; the section $z = 0$, $x = 0$ (the point at infinity) marks each fiber, so (π, σ) is a family of 1-pointed genus-1 curves and defines a map $S \to \mathcal{M}_{1,1}$.

The family exhibits both features that force a stack. First, the j-invariant of the fiber over λ is

$$j(\lambda) = 256 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}$$

a degree-6 map $S \to \mathbb{A}^1_j$; two parameters λ, λ' give isomorphic fibers exactly when $j(\lambda) = j(\lambda')$, so the map to the coarse j-line (theorem 1.3.6) is six-to-one over a general j, recording the six values of λ permuted by the anharmonic group S_3 acting through $\lambda \mapsto 1 - \lambda, 1/\lambda, \ldots$. Second, every fiber carries the automorphism $y \mapsto -y$ fixing λ, the hyperelliptic involution, so the family admits a nontrivial automorphism over S itself. The coarse space forgets both the six-fold ambiguity and the involution; the stack $\mathcal{M}_{1,1}$ remembers them, and this is the running touchstone (example 1.3.4) that the general theory below organizes.

With the moduli stack defined, the central theorem can be assembled. Every adjective in "smooth Deligne-Mumford stack of dimension $3g-3$" is a property of stacks made precise in Part III, and each is verified by importing a result about curves. Algebraicity was proved in general in the algebraic-stacks chapter, by checking Artin's criteria; smoothness is the unobstructedness of curve deformations from Part II, together with the dimension count $\dim H^1(C, T_C) = 3g - 3$; the Deligne-Mumford property is the finiteness of $\operatorname{Aut}(C)$, which makes the diagonal unramified. The proof below is the assembly of these three imports, and its interest is entirely in seeing that each earlier theorem applies on the nose.

Theorem 10.1.3: $\mathcal{M}_g$ is a Smooth Deligne-Mumford Stack of Dimension $3g-3$

For $g \geq 2$, the moduli stack $\mathcal{M}_g$ is a smooth Deligne-Mumford stack over k, of dimension

$$\dim \mathcal{M}_g = 3g - 3$$

and $\mathcal{M}_{g,n}$ is a smooth Deligne-Mumford stack of dimension $3g-3+n$.

***Proof*:**
The three defining properties of the conclusion are established in turn: that $\mathcal{M}_g$ is an algebraic stack, that it is Deligne-Mumford, and that it is smooth of the stated dimension. Each step imports one theorem from an earlier chapter and checks its hypotheses for curves.

Step 1: $\mathcal{M}_g$ is an algebraic stack. That $\mathcal{M}_g$ is a stack in the fppf topology is descent for families of curves: a family $\pi\colon \mathcal{C} \to S$ is a scheme proper and smooth over S, and such schemes descend along fppf covers of S once a relative polarization is fixed, which the relative pluricanonical bundle $\omega_{\mathcal{C}/S}^{\otimes 3}$ supplies (it is relatively very ample for $g \geq 2$, since $\deg \omega_C^{\otimes 3} = 3(2g-2) = 6g-6 \geq 2g+1$ on each fiber). That $\mathcal{M}_g$ is moreover *algebraic* is theorem 8.4.4, where Artin's criteria were verified: $\mathcal{M}_g$ is a limit-preserving fppf stack with a deformation and obstruction theory ($H^1(C, T_C)$ and $H^2(C, T_C)$, from Part II) satisfying the Schlessinger-Artin conditions, and with effective formal versal deformations, so the criteria certify a smooth atlas $U \to \mathcal{M}_g$ from a scheme. Concretely the atlas is the locus in a Hilbert scheme of tricanonically embedded curves, on which the theorem of theorem 8.4.4 produces the smooth cover. Thus $\mathcal{M}_g$ is an algebraic stack.

Step 2: $\mathcal{M}_g$ is Deligne-Mumford. By theorem 8.2.3, an algebraic stack is Deligne-Mumford if and only if its diagonal is unramified, equivalently the automorphism group scheme $\underline{\mathrm{Aut}}(C)$ of every object is unramified over the base, equivalently finite and reduced for objects over a field. The object over $\operatorname{Spec} k$ is a curve C, and its automorphism group scheme is $\underline{\mathrm{Aut}}(C)$. For $g \geq 2$ this is a finite reduced group scheme: finiteness is definition 10.1.4 below (Hurwitz's bound $|\mathrm{Aut}(C)| \leq 84(g-1)$ makes the group finite), and reducedness holds because $H^0(C, T_C) = 0$ for $g \geq 2$, the tangent space to $\underline{\mathrm{Aut}}(C)$ at the identity being the space of infinitesimal automorphisms $H^0(C, T_C)$. That $H^0(C, T_C) = 0$ follows from $\deg T_C = \deg \omega_C^\vee = -(2g-2) < 0$: a global section of a line bundle of negative degree on a connected curve vanishes. Hence $\underline{\mathrm{Aut}}(C)$ is a finite reduced group scheme, i.e. a finite constant group scheme in characteristic 0, the diagonal of $\mathcal{M}_g$ is unramified, and $\mathcal{M}_g$ is Deligne-Mumford.

Step 3: $\mathcal{M}_g$ is smooth of dimension $3g-3$. Smoothness of a Deligne-Mumford stack is smoothness of a (equivalently, any) étale atlas, and by the deformation-theoretic criterion it is the unobstructedness of the deformation functor of every object together with a constant tangent-space dimension. The deformation functor of a curve C is unobstructed by proposition 5.4.1: the obstructions to lifting a first-order deformation lie in $H^2(C, T_C)$, which vanishes because a curve has cohomological dimension 1, so every first-order deformation extends and the functor is formally smooth. Its tangent space is the space of first-order deformations of C, computed in theorem 4.3.2 to be

$$T_{[C]}\mathcal{M}_g = H^1(C, T_C)$$

of dimension $\dim H^1(C, T_C) = 3g-3$ for $g \geq 2$. The dimension count is Riemann-Roch: $\chi(T_C) = \deg T_C + (1-g) = (2-2g) + (1-g) = 3-3g$, and since $h^0(T_C) = 0$ from Step 2, $h^1(T_C) = -\chi(T_C) = 3g-3$. A formally smooth Deligne-Mumford stack whose tangent space has constant dimension $3g-3$ is smooth of dimension $3g-3$, which is the claim.

> For $\mathcal{M}_{g,n}$ the same three steps apply verbatim, with two changes. The automorphism group of an n-pointed curve $(C, p_1, \ldots, p_n)$ is the subgroup of $\mathrm{Aut}(C)$ fixing each p_i, still finite for $g \geq 2$, so the stack is again Deligne-Mumford. The tangent space gains n dimensions, one for the position of each marked point moving in the curve: the deformations of an n-pointed curve are classified by $H^1(C, T_C(-p_1 - \cdots - p_n))$, whose dimension is $3g - 3 + n$ by Riemann-Roch for $g \geq 2$. Hence $\mathcal{M}_{g,n}$ is smooth Deligne-Mumford of dimension $3g - 3 + n$, as required.

The theorem is the payoff of the whole book, and the way its proof splits across the three Parts is the point. Nothing in it is new; the entire content is that each hypothesis of an imported theorem holds for curves, and the assembly of the conclusions is the statement. The number $3g - 3$ deserves emphasis: it is the dimension of the space in which a curve can be deformed, and it appears first as $\dim H^1(C, T_C)$, a cohomological count, and second as the dimension of the moduli stack, a geometric one, the two identified by the deformation theory of Part II. That the two agree is the content of "the tangent space to the moduli space is the deformation space", the principle that made the local structure of moduli computable. For $g = 2$ this gives $\dim \mathcal{M}_2 = 3$, for $g = 3$ it gives 6, and in general the moduli of curves grows linearly in the genus, a fact with no analogue for the finite lists of low genus but which governs the geometry of $\mathcal{M}_g$ for all large g.

The Deligne-Mumford property is worth isolating, because it is exactly what fails for $g \leq 1$ and exactly what the finiteness of automorphisms buys. A Deligne-Mumford stack has an étale atlas, so it looks étale-locally like a scheme and its points have finite automorphism groups; an Artin stack with positive- dimensional automorphism groups, like $B\mathbb{G}_m$, does not. The line $\deg T_C < 0 \Rightarrow H^0(T_C) = 0$ that killed the infinitesimal automorphisms is the same computation that, run at $g = 1$ where $\deg T_C = 0$, returns $H^0(T_C) = k$, the one-dimensional space of translations, and at $g = 0$ where $\deg T_C = 2 > 0$ returns $H^0(T_C) = H^0(\mathbb{P}^1, \mathcal{O}(2))$, the three-dimensional Lie algebra of PGL_2. The genus condition $g \geq 2$ is precisely the condition that this Lie algebra vanish, that curves be rigid enough to have finite automorphism groups, and hence that their moduli be Deligne-Mumford rather than only Artin.

That finiteness of automorphism groups is what separates the well-behaved $\mathcal{M}_g$ for $g \geq 2$ from the Artin cases below deserves its own definition and a sharp bound. The automorphism group of a curve of genus at least 2 is not just finite but uniformly bounded in terms of the genus, a classical theorem of Hurwitz, and the bound quantifies the rigidity that makes the moduli Deligne-Mumford.

Definition 10.1.4: Automorphism Group of a Curve; Hurwitz's Bound

Let C be a smooth proper connected curve of genus g over $k = \bar{k}$. Its *automorphism group* $\mathrm{Aut}(C)$ is the group of k-automorphisms $C \to C$, equivalently the k-points of the automorphism group scheme $\underline{\mathrm{Aut}}(C)$. For $g \geq 2$ the group $\mathrm{Aut}(C)$ is finite, and in characteristic 0 it satisfies *Hurwitz's bound*

$$|\mathrm{Aut}(C)| \leq 84(g-1)$$

A curve is *hyperelliptic* if it admits a degree-2 map $C \to \mathbb{P}^1$; the corresponding involution $\iota \colon C \to C$ exchanging the two sheets, the *hyperelliptic involution*, is a central element of $\mathrm{Aut}(C)$ intrinsic to C, being the unique involution whose quotient is $\mathbb{P}^1$, and its fixed points are the $2g + 2$ Weierstrass points lying over the branch points.

The finiteness alone is what the moduli theory needs, and it is worth seeing why $g \geq 2$ forces it, since the mechanism is exactly the tangent-space computation of theorem 10.1.3. An algebraic group acting faithfully on a variety is finite as soon as its Lie algebra vanishes, and the Lie algebra of

$\underline{\mathrm{Aut}}(C)$ is $H^0(C, T_C)$, which is 0 for $g \geq 2$ because $\deg T_C = 2 - 2g < 0$. Thus $\underline{\mathrm{Aut}}(C)$ is a group scheme of dimension 0, and being of finite type it is finite. The Hurwitz bound $84(g-1)$ is the sharp refinement, and it is proved by a covering-space argument worth recording.

***Proof of Hurwitz's bound*:**
Work in characteristic 0 and let $G = \mathrm{Aut}(C)$, which is finite by the preceding paragraph. The quotient $C \to C/G = \bar{C}$ is a degree-$|G|$ map to a smooth curve $\bar{C}$ of some genus $\bar{g}$, ramified over finitely many points $q_1, \ldots, q_r$ of $\bar{C}$ with ramification indices $e_1, \ldots, e_r$. The Riemann-Hurwitz formula for the quotient map reads

$$2g - 2 = |G| \left(2\bar{g} - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{e_i} \right) \right)$$

the term $1 - 1/e_i$ being the contribution of each branch point, since the fiber over q_i has $|G|/e_i$ points each of ramification index e_i. Because $g \geq 2$ the left side is positive, so the bracketed quantity

$$\chi = 2\bar{g} - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{e_i} \right)$$

is a strictly positive rational number, and $|G| = (2g-2)/\chi$; maximizing $|G|$ is minimizing the positive χ. A finite case analysis on $(\bar{g}, r, e_1, \ldots, e_r)$ bounds χ below. If $\bar{g} \geq 2$ then $\chi \geq 2$. If $\bar{g} = 1$ then $\chi = \sum(1 - 1/e_i) \geq 1 - 1/2 = 1/2$ once $r \geq 1$ (and $\chi > 0$ forces $r \geq 1$). If $\bar{g} = 0$ then $\chi = -2 + \sum(1 - 1/e_i)$, and positivity forces $r \geq 3$; the minimum positive value of $-2 + \sum_{i=1}^{r}(1 - 1/e_i)$ over integers $e_i \geq 2$ is attained at $r = 3$ with $(e_1, e_2, e_3) = (2, 3, 7)$, giving

$$\chi = -2 + \left(1 - \tfrac{1}{2}\right) + \left(1 - \tfrac{1}{3}\right) + \left(1 - \tfrac{1}{7}\right) = -2 + \tfrac{1}{2} + \tfrac{2}{3} + \tfrac{6}{7} = \frac{1}{42}$$

since $\frac{1}{2} + \frac{2}{3} + \frac{6}{7} = \frac{21+28+36}{42} = \frac{85}{42}$ and $-2 + \frac{85}{42} = \frac{1}{42}$. Any other triple (e_1, e_2, e_3) of integers ≥ 2 with positive χ gives $\chi \geq 1/42$, as a direct check of the nearby triples $(2,3,8), (2,4,5), (3,3,4), \ldots$ confirms, each exceeding $1/42$. Hence $\chi \geq 1/42$ in all cases, and

$$|G| = \frac{2g-2}{\chi} \leq 42(2g-2) = 84(g-1)$$

which is the bound, as claimed.

The bound is sharp: the Klein quartic $x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$ has genus 3 and automorphism group $\mathrm{PSL}_2(\mathbb{F}_7)$ of order $168 = 84 \cdot 2 = 84(3-1)$, realizing the $(2,3,7)$ triangle group that the extremal case picks out. The role of the bound in the moduli theory is qualitative, not quantitative: it certifies that $\mathrm{Aut}(C)$ is finite for every curve of genus ≥ 2, uniformly, which is precisely the input Step 2 of theorem 10.1.3 needed to conclude the diagonal is unramified and $\mathcal{M}_g$ is Deligne-Mumford. Where $\mathrm{Aut}(C)$ jumps, as at the Klein quartic or the hyperelliptic curves, the stack $\mathcal{M}_g$ acquires a point with larger automorphism group, a stacky point whose residual gerbe records the extra symmetry, exactly as the residual gerbe of $\mathcal{M}_{1,1}$ recorded the μ_4, μ_6 symmetry at the special j-invariants of example 1.3.4.

The hyperelliptic involution is the cleanest example of a curve automorphism, and it is worth seeing that hyperelliptic curves form a large family whose generic member has automorphism group exactly $\{1, \iota\}$, so that the involution is present on every hyperelliptic curve but is typically the whole automorphism group.

Example 10.1.5: The Hyperelliptic Involution and the Genus-2 Curve

Every curve of genus 2 is hyperelliptic. Riemann-Roch gives $h^0(C, \omega_C) = g = 2$, so the canonical map $C \to \mathbb{P}^1$ is a degree-2 map (the canonical bundle has degree $2g - 2 = 2$ and two sections), and this exhibits C as a double cover of $\mathbb{P}^1$. Explicitly, a genus-2 curve is

$$y^2 = f(x), \qquad f(x) = \prod_{i=1}^{6}(x - a_i)$$

a double cover of the x-line branched at the six roots $a_1, \ldots, a_6$ of a squarefree sextic (and at ∞ when $\deg f = 5$; six branch points on $\mathbb{P}^1$ in either normalization). The hyperelliptic involution is $\iota\colon (x, y) \mapsto (x, -y)$, exchanging the two sheets, with quotient the x-line $\mathbb{P}^1$ and six fixed points, the Weierstrass points lying over the branch points.

The moduli count matches theorem 10.1.3. A genus-2 curve is determined by its six branch points on $\mathbb{P}^1$ up to the action of PGL_2 on $\mathbb{P}^1$ (which is 3-dimensional) and the permutation S_6 of the points; so

$$\dim \mathcal{M}_2 = \dim\{6 \text{ points on } \mathbb{P}^1\} - \dim \mathrm{PGL}_2 = 6 - 3 = 3$$

exactly $3g - 3 = 3$ for $g = 2$. The generic such curve has automorphism group $\{1, \iota\} = \mathbb{Z}/2$, generated by the involution alone; special configurations of the six points (extra symmetry of the branch divisor under a subgroup of PGL_2) enlarge $\mathrm{Aut}(C)$, producing the stacky points where the automorphism group jumps. Since every genus-2 curve is hyperelliptic, the hyperelliptic locus is all of $\mathcal{M}_2$, of dimension 3; for $g \geq 3$ it is a proper substack, computed next.

The last low-genus phenomenon to record is the contrast between the moduli stacks for $g = 0, 1$ and the Deligne-Mumford stacks for $g \geq 2$, together with the dimensions and the hyperelliptic locus for small g. Assembling these in one place shows both why the theory bifurcates at $g = 2$ and how the number $3g - 3$ manifests concretely.

Example 10.1.6: The Moduli Stacks in Low Genus

The moduli stacks for small g exhibit the transition from Artin to Deligne-Mumford, and the dimension formula $3g - 3$ across the range.

1. *Genus* 0. The only smooth genus-0 curve over $k = \bar{k}$ is $\mathbb{P}^1$, with automorphism group $\mathrm{Aut}(\mathbb{P}^1) = \mathrm{PGL}_2$, a 3-dimensional algebraic group. A family of genus-0 curves is a $\mathbb{P}^1$-bundle, and the moduli stack is the classifying stack

$$\mathcal{M}_0 = B\mathrm{PGL}_2$$

an Artin stack of dimension -3, with a single point whose automorphism group is PGL_2. Its coarse space is a point: all $\mathbb{P}^1$'s are isomorphic. The dimension $3 \cdot 0 - 3 = -3$ is the negative of $\dim \mathrm{PGL}_2$, the signature of a point with a 3-dimensional automorphism group. Because PGL_2 is positive- dimensional, $\mathcal{M}_0$ is Artin but *not* Deligne-Mumford.

2. *Genus* 1. A smooth genus-1 curve has no distinguished point, and $\mathrm{Aut}(C)$ contains the 1-dimensional group of translations, so $\mathcal{M}_1$ (unpointed) is again not Deligne-Mumford. Rigidifying by a marked point kills the translations: $\mathcal{M}_{1,1}$, the moduli of elliptic curves, is a Deligne-Mumford stack of dimension $3 \cdot 1 - 3 + 1 = 1$, with generic automorphism

group $\mu_2 = \{\pm 1\}$ and special groups μ_4, μ_6 at $j = 1728, 0$. Its coarse space is the j-line $\mathbb{A}^1_j$ (theorem 1.3.6), one-dimensional, matching the stack dimension. This is the boundary case: $g = 1$ needs a marked point to become Deligne-Mumford, and the automorphism obstruction of example 1.3.4 is exactly the residual μ_2 that survives.

3. *Genus* 2. By example 10.1.5, $\dim \mathcal{M}_2 = 3 = 3 \cdot 2 - 3$, and $\mathcal{M}_2$ is a smooth Deligne-Mumford stack (theorem 10.1.3). Every genus-2 curve is hyperelliptic, so the hyperelliptic locus is all of $\mathcal{M}_2$.

4. *Genus* 3. Here $\dim \mathcal{M}_3 = 6 = 3 \cdot 3 - 3$. A generic genus-3 curve is a smooth plane quartic (canonical bundle very ample, embedding $C \hookrightarrow \mathbb{P}^2$ as a degree-4 curve), *not* hyperelliptic; the hyperelliptic curves form a proper substack.

The hyperelliptic locus $\mathcal{H}_g \subset \mathcal{M}_g$ of curves admitting a degree-2 map to $\mathbb{P}^1$ has dimension $2g - 1$ for all $g \geq 2$. A hyperelliptic curve is a double cover of $\mathbb{P}^1$ branched at $2g + 2$ points (Riemann-Hurwitz for a degree-2 cover, with one index-2 ramification point over each branch point: $2g - 2 = 2 \cdot (-2) + b \cdot (2 - 1) = -4 + b$ forces $b = 2g + 2$), and such a cover is determined by the branch divisor up to PGL_2 acting on $\mathbb{P}^1$:

$$\dim \mathcal{H}_g = (2g + 2) - \dim \mathrm{PGL}_2 = (2g + 2) - 3 = 2g - 1$$

For $g = 2$ this gives $2 \cdot 2 - 1 = 3 = \dim \mathcal{M}_2$, confirming that all genus-2 curves are hyperelliptic; for $g = 3$ it gives $2 \cdot 3 - 1 = 5 < 6 = \dim \mathcal{M}_3$, so the hyperelliptic locus is a divisor (codimension 1) in $\mathcal{M}_3$; and for large g the codimension $(3g - 3) - (2g - 1) = g - 2$ grows, so the generic curve is far from hyperelliptic.

The table across $g = 0, 1, 2, 3$ tells the whole story of why $g \geq 2$ is the regime of the theorem. At $g = 0$ and $g = 1$ the automorphism groups are positive-dimensional (all of PGL_2, or the translations), the moduli stacks are Artin and not Deligne-Mumford, and the coarse spaces collapse the moduli to a point or to the j-line. At $g \geq 2$ the automorphism groups are finite by Hurwitz, the stacks are Deligne-Mumford, and the dimension settles into the linear growth $3g - 3$ that governs all higher genus. The hyperelliptic locus interpolates: it is everything at $g = 2$, a divisor at $g = 3$, and a shrinking-codimension family thereafter, the last place where the double-cover description of Part I's curves survives before the general curve becomes too generic to be so described. The stacky structure that example 1.3.4 first exhibited for $g = 1$ persists at every genus as the residual gerbes over the curves with extra automorphisms, and it is these that force $\mathcal{M}_g$ to be a stack rather than a variety even for $g \geq 2$, notwithstanding the finiteness of every individual automorphism group.

Exercise 10.1.1

1. Verify the dimension formula $\dim \mathcal{M}_{g,n} = 3g - 3 + n$ for $g \geq 2$ directly from Riemann-Roch, by computing $h^1(C, T_C(-p_1 - \cdots - p_n))$ for n distinct points $p_1, \ldots, p_n$ on a curve C of genus g. Confirm that h^0 of this twisted tangent sheaf vanishes, so that the dimension is $-\chi$, and recover the unpointed count at $n = 0$.

2. Show that a smooth genus-0 curve over $k = \bar{k}$ is isomorphic to $\mathbb{P}^1$, and that $\mathrm{Aut}(\mathbb{P}^1) = \mathrm{PGL}_2$ has Lie algebra $H^0(\mathbb{P}^1, T_{\mathbb{P}^1})$ of dimension 3. Conclude that $\mathcal{M}_0$ is not Deligne-Mumford, and identify the single object of $\mathcal{M}_0(\operatorname{Spec} k)$ and its automorphism group.

3. Using the Klein quartic $C\colon x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$, verify that its genus is 3 (a smooth

plane curve of degree d has genus $\binom{d-1}{2}$) and that Hurwitz's bound $84(g-1)$ at $g=3$ equals 168. Identify which of the three cases $(\bar{g}, r, e_i)$ in the proof of Hurwitz's bound is realized by the quotient $C \to C/\mathrm{Aut}(C)$, given that $|\mathrm{Aut}(C)| = 168$.

4. Let $\mathcal{H}_g \subset \mathcal{M}_g$ be the hyperelliptic locus. Compute its dimension $2g-1$ from the double-cover description, and determine the codimension of $\mathcal{H}_g$ in $\mathcal{M}_g$ as a function of g. For which g is $\mathcal{H}_g$ a divisor, and for which is it all of $\mathcal{M}_g$?

5. Explain why the unpointed moduli stack $\mathcal{M}_1$ of genus-1 curves is not Deligne-Mumford, but $\mathcal{M}_{1,1}$ is. Compute $H^0(C, T_C)$ for a genus-1 curve C and interpret its dimension as the dimension of the automorphism group of C; explain how marking a point cuts this down to a finite group.

6. The Legendre family (example 10.1.2) gives a map $S = \mathbb{A}^1 \setminus \{0,1\} \to \mathcal{M}_{1,1}$. Explain why this map is neither injective nor a monomorphism of stacks: identify the finite fibers of the induced map to the coarse j-line and the automorphism over S that obstructs the family from being a monomorphism. What does this say about the difference between $\mathcal{M}_{1,1}$ and its coarse space?

10.2 Stable Curves and Deligne-Mumford Stability

The stack $\mathcal{M}_g$ of theorem 10.1.3 is smooth and Deligne-Mumford, but it is not proper: a family of smooth genus-g curves over a punctured disc need not extend to a family of smooth curves over the whole disc, because the limiting curve can degenerate. A curve moving in a one-parameter family can, in the limit, acquire a singularity, split into pieces, or send a subfamily off to infinity. To compactify $\mathcal{M}_g$ one must decide which singular curves to admit as limits. Admitting too few leaves $\mathcal{M}_g$ non-proper; admitting too many, such as arbitrary singular curves, produces a bloated space with no separation, since many different degenerations become indistinguishable. The correct class is the *stable curves*: curves with the mildest possible singularities, the nodes, subject to a rigidity condition that keeps their automorphism groups finite. This is the class Deligne and Mumford isolated, and the resulting compactification $\overline{\mathcal{M}}_g$ is the object toward which the chapter has been building.

This section develops the singular curves and the stability condition, and assembles the deformation-theoretic input to show that $\overline{\mathcal{M}}_g$ is a smooth Deligne-Mumford stack containing $\mathcal{M}_g$ as a dense open substack. The one property deferred is properness, which rests on the stable reduction theorem and is proved in section 10.3. Throughout, k is algebraically closed, of characteristic 0 where a count of automorphisms requires it, and $g \geq 2$ unless stated otherwise.

Definition 10.2.1: Nodal (Prestable) Curve

A *nodal curve* (or *prestable curve*) over k is a reduced, connected, proper k-scheme C of pure dimension 1 whose only singularities are *nodes*: points $p \in C$ at which the complete local ring is

$$\widehat{\mathcal{O}}_{C,p} \cong k[[x,y]]/(xy)$$

the local model of two smooth branches crossing transversally. Its *arithmetic genus* is

$$p_a(C) = \dim_k H^1(C, \mathcal{O}_C)$$

The *dual graph* Γ_C of C has one vertex for each irreducible component C_i, weighted by the geometric genus g_i of the normalization $\widetilde{C_i}$, and one edge for each node, joining the vertices of the two branches through that node (a self-edge, i.e. a loop, when both branches lie on the same component). The *dualizing sheaf* ω_C is the unique line bundle whose restriction to the smooth locus is the sheaf of Kähler differentials and which, near a node with branches $x = 0$ and $y = 0$, consists of the rational differentials η with at worst simple poles along each branch whose residues satisfy

$$\mathrm{Res}_{x=0}\eta + \mathrm{Res}_{y=0}\eta = 0$$

It is a line bundle of degree $\deg \omega_C = 2p_a(C) - 2$.

Three points in the definition carry the weight. The first is that a node is the mildest singularity a reduced curve can have: it is the singularity of two lines meeting at a point, and it has a single branch on each side rather than a cusp or worse. The second is that the arithmetic genus p_a, unlike the geometric genus, does not change when a smooth curve degenerates to a nodal one; it is the invariant that stays constant in a flat family, which is exactly why it, and not the geometric genus of the pieces, indexes the compactified moduli. The third is the dualizing sheaf ω_C. On a smooth curve ω_C is the canonical bundle Ω^1_C, but a nodal curve is singular, so Ω^1_C is no longer a line bundle; the residue condition at each node repairs this, producing a genuine line bundle ω_C that agrees with Ω^1_C away from the nodes and satisfies Serre duality and Riemann-Roch on the nodal curve. The degree formula $\deg \omega_C = 2p_a - 2$ is the nodal extension of $\deg \omega_C = 2g - 2$ for smooth curves, and it is what makes ω_C the right object to test for ampleness. The dualizing sheaf is developed in general in the megabook Part I; here $\deg \omega_C = 2p_a - 2$ and the residue description are the facts used, and the reader may take them from [12, the dualizing sheaf] or verify them in the worked cases below.

To ground the definition, consider the simplest nodal curve, the one-nodal degeneration of an elliptic curve, and read off its dual graph, arithmetic genus, and dualizing sheaf directly.

Example 10.2.2: The Nodal Cubic and Its Dual Graph

Let $C \subset \mathbb{P}^2$ be the nodal cubic $y^2 z = x^3 + x^2 z$, the projective closure of the affine curve $y^2 = x^3 + x^2 = x^2(x+1)$. Near the origin $(x, y) = (0, 0)$ the equation factors as $y^2 - x^2 = (y-x)(y+x)$ to leading order, so the two branches $y = x$ and $y = -x$ cross transversally: the origin is a node, and one checks that C is smooth elsewhere. The curve C is irreducible, so its dual graph Γ_C has a single vertex; the normalization $\widetilde{C} \to C$ separates the two branches at the node into two distinct points, and $\widetilde{C} \cong \mathbb{P}^1$ has geometric genus 0, so the vertex is weighted 0. The single node is a self-node on the one component, so Γ_C has one vertex of weight 0 and one loop.

The arithmetic genus is computed from the normalization sequence

$$0 \to \mathcal{O}_C \to \nu_*\mathcal{O}_{\widetilde{C}} \to k_p \to 0$$

where $\nu\colon \widetilde{C} \cong \mathbb{P}^1 \to C$ is the normalization and k_p is the skyscraper at the node measuring the two preimages glued to one point. Taking cohomology and using $H^1(\mathbb{P}^1, \mathcal{O}) = 0$ gives $H^1(C, \mathcal{O}_C) \cong k$, so $p_a(C) = 1$. The geometric genus of the normalization is 0; the node contributes the extra unit of arithmetic genus, so a node identifying two points of a genus-0 curve raises the arithmetic genus by one. The dualizing sheaf has degree $\deg \omega_C = 2p_a - 2 = 0$, and concretely a generating section is the differential $\omega = dx/y$ on the smooth locus, which on the normalization $\mathbb{P}^1$ has simple poles at the two points lying over the node with opposite residues ± 1, exactly the residue condition of definition 10.2.1. Thus $\omega_C \cong \mathcal{O}_C$ is trivial of degree 0, matching $2 \cdot 1 - 2 = 0$: the nodal cubic is the $j = \infty$ degeneration of the elliptic curves, and its dualizing sheaf is the limit of the trivial canonical bundle of a smooth genus-1 curve.

The nodal cubic has $\deg \omega_C = 0$, so ω_C is not ample: a genus-1 nodal curve is not stable on its own. Stability enters precisely to select, among nodal curves of arithmetic genus $g \geq 2$, those whose dualizing sheaf is ample, and the content of the definition is that this analytic condition on ω_C has two equivalent reformulations, one group-theoretic and one purely combinatorial in the dual graph. The equivalence is what makes stability both geometrically meaningful (ampleness of ω_C is what allows the pluricanonical embedding and coarse-space construction of section 10.4) and checkable by hand (the combinatorial condition is read off the dual graph).

Definition 10.2.3: Stable Curve

A *stable curve* of genus $g \geq 2$ is a nodal curve C of arithmetic genus $p_a(C) = g$ satisfying any one of the following equivalent conditions (their equivalence is theorem 10.2.4):

1. the dualizing sheaf ω_C is ample;
2. the automorphism group $\mathrm{Aut}(C)$ is finite;
3. *(combinatorial stability)* writing g_i for the geometric genus of the normalization $\widetilde{C_i}$ of each component, every component with $g_i = 0$ (normalization a smooth rational curve $\mathbb{P}^1$) meets the rest of C in at least 3 points (nodes on that component, counting a self-node twice), and every component with $g_i = 1$ meets the rest in at least 1 point; equivalently, every component satisfies $2g_i - 2 + k_i > 0$, where k_i is the number of those points.

A *semistable curve* weakens "≥ 3" to "≥ 2" for smooth rational components and imposes no condition on the higher-genus ones, equivalently requires ω_C nef rather than ample.

The combinatorial condition is the working criterion, so it deserves a translation; throughout it, "component" is read through its normalization, so that "genus" means the geometric genus g_i and a self-node of a component shows up as two of that component's attaching points. A smooth rational component is a copy of $\mathbb{P}^1$; on its own $\mathbb{P}^1$ has the 3-dimensional automorphism group PGL_2, so to rigidify it one must pin down at least three of its points, and the only points pinned down by the rest of the curve are the nodes where it attaches. A rational component meeting the rest in only one or two points therefore carries a positive-dimensional group of automorphisms fixing those attaching points, and this group acts on the whole curve, so $\mathrm{Aut}(C)$ is infinite. A genus-1 component (geometric genus $g_i = 1$) has the 1-dimensional translation group; one attaching point suffices to kill the translations, so genus-1 components need only meet the rest once. Every component of geometric genus ≥ 2 is already rigid with finite automorphisms and imposes no attaching condition. The unstable configurations are exactly the rational components with ≤ 2 special points and the genus-1 components with 0 special points, and stability forbids them.

The equivalence of the three conditions is the technical heart of the section, and it is proved by relating all three to the degree of ω_C on each component.

Theorem 10.2.4: Ampleness, Finite Automorphisms, and Combinatorial Stability

Let C be a nodal curve of arithmetic genus $g \geq 2$. The three conditions of definition 10.2.3 are equivalent:

$$\omega_C \text{ ample} \iff \mathrm{Aut}(C) \text{ finite} \iff C \text{ combinatorially stable}$$

***Proof*:**
The argument runs through the restriction of ω_C to each component. Fix an irreducible component C_i of C with normalization $\nu_i \colon \widetilde{C_i} \to C_i$ of geometric genus g_i, and let k_i be the number of points at which C_i meets the rest of C, that is the number of nodes lying on C_i, a self-node counted twice. The starting computation is the degree of ω_C on C_i.

Step 1: the degree of ω_C on each component. The adjunction description of the dualizing sheaf gives, on the normalization $\widetilde{C_i}$,

$$\nu_i^*(\omega_C|_{C_i}) \cong \omega_{\widetilde{C_i}}(D_i)$$

where D_i is the divisor of the k_i points of $\widetilde{C_i}$ lying over the nodes on C_i. Indeed a section of ω_C near a node is a differential with simple poles of opposite residue on the two branches, so pulling back to a single branch $\widetilde{C_i}$ allows a simple pole at each of the k_i marked points, which is exactly $\omega_{\widetilde{C_i}}(D_i)$. Taking degrees,

$$\deg\left(\omega_C|_{C_i}\right) = \deg \omega_{\widetilde{C_i}} + k_i = (2g_i - 2) + k_i$$

This is the local stability number of the component C_i. A line bundle on a proper curve is ample if and only if it has positive degree on every irreducible component, so

$$\omega_C \text{ ample} \iff (2g_i - 2) + k_i > 0 \text{ for every component } C_i \qquad (*)$$

which is the bridge to the other two conditions.

Step 2: ampleness $\iff$ combinatorial stability. Read the inequality $(2g_i-2)+k_i > 0$ component by component. If $g_i \geq 2$ then $2g_i - 2 \geq 2 > 0$ and the inequality holds with no condition on k_i. If $g_i = 1$ then $2g_i - 2 = 0$, and the inequality reads $k_i > 0$, i.e. $k_i \geq 1$: the genus-1 component must meet the rest in at least one point. If $g_i = 0$ then $2g_i - 2 = -2$, and the inequality reads $k_i > 2$, i.e. $k_i \geq 3$: the rational component must meet the rest in at least three points. These are precisely the clauses of combinatorial stability in definition 10.2.3, so $(*)$ is the statement that ω_C is ample if and only if C is combinatorially stable.

Step 3: finite automorphisms $\Rightarrow$ combinatorial stability. Argue by contrapositive: an unstable component produces infinitely many automorphisms. Suppose some component C_i violates combinatorial stability. If C_i is rational $(g_i = 0)$ with $k_i \leq 2$ special points, its normalization is $\mathbb{P}^1$ with at most two marked points; the subgroup of $\mathrm{PGL}_2 = \mathrm{Aut}(\mathbb{P}^1)$ fixing those ≤ 2 points is positive-dimensional (it is the full PGL_2 for $k_i \leq 1$ and the 1-dimensional torus $\mathbb{G}_m$ of automorphisms $z \mapsto \lambda z$ fixing $0, \infty$ for $k_i = 2$). Each such automorphism fixes the attaching nodes, so it extends by the identity on the rest of C to an automorphism of C; the resulting subgroup of $\mathrm{Aut}(C)$ is positive-dimensional, hence infinite. If C_i has arithmetic genus 1 with $k_i = 0$, it is a connected component of arithmetic genus 1 meeting nothing, so $C = C_i$ has arithmetic genus $1 < 2$, contradicting $g \geq 2$; thus the genus-1 clause is the statement that a genus-1 piece must attach, and its failure again yields the translations of the elliptic component as automorphisms fixing no attaching point. In every case $\mathrm{Aut}(C)$ is infinite, so finiteness of $\mathrm{Aut}(C)$ forces combinatorial stability.

Step 4: combinatorial stability $\Rightarrow$ finite automorphisms. Suppose C is combinatorially stable, so ω_C is ample by Steps 1 and 2. An automorphism $\sigma \in \mathrm{Aut}(C)$ preserves the canonical polarization ω_C, since ω_C is intrinsic to C, so $\mathrm{Aut}(C)$ is a group of automorphisms of the polarized variety (C, ω_C). For a projective variety with an ample line bundle the automorphism group preserving the polarization is a linear algebraic group of finite type, and its identity component acts trivially on the finite set of components and nodes, hence restricts to a subgroup of $\prod_i \mathrm{Aut}(\widetilde{C}_i, D_i)$, the automorphisms of each normalized component fixing its marked points. By ampleness each factor $(\widetilde{C}_i, D_i)$ has $\deg \omega_{\widetilde{C}_i}(D_i) = (2g_i - 2) + k_i > 0$, so $\deg T_{\widetilde{C}_i}(-D_i) = -((2g_i - 2) + k_i) < 0$ and hence $H^0(\widetilde{C}_i, T_{\widetilde{C}_i}(-D_i)) = 0$: a component with an ample marked canonical bundle has no infinitesimal automorphisms fixing its marked points. The Lie algebra of the identity component of $\mathrm{Aut}(C)$ therefore vanishes, so that identity component is trivial and $\mathrm{Aut}(C)$ is a finite group.

Steps 2, 3, and 4 close the cycle: ampleness $\iff$ combinatorial stability (Step 2), combinatorial stability $\Rightarrow$ finite automorphisms (Step 4), and finite automorphisms $\Rightarrow$ combinatorial stability (Step 3), so all three conditions are equivalent, as required.

The three faces of stability serve three purposes, and it is worth seeing which does what. Ampleness of ω_C is the geometric condition: it is what lets $\omega_C^{\otimes 3}$ embed a stable curve in projective space, so that stable curves become points of a Hilbert scheme and the coarse moduli space can be cut out by geometric invariant theory (section 10.4). Finiteness of $\mathrm{Aut}(C)$ is the stack condition: it is exactly what makes $\overline{\mathcal{M}}_g$ Deligne-Mumford rather than only Artin, the direct analogue of the finiteness of $\mathrm{Aut}(C)$ for smooth curves that made $\mathcal{M}_g$ Deligne-Mumford in theorem 10.1.3. Combinatorial stability is the computational condition: it is checked by drawing the dual graph and counting attaching points, and it is what one verifies in practice. The single inequality $(2g_i - 2) + k_i > 0$ per component ties all three together, and it is the degree of ω_C on that component, so the whole theory of stability is the demand that ω_C have positive degree everywhere.

With the definition in hand, the range of stable curves is best seen through examples, from the boundary case of genus 1 with a marked point up to the maximally degenerate genus-2 curves and the unstable curves that stability rejects.

Example 10.2.5: Stable and Unstable Curves in Low Genus

The following curves exhibit the stability condition across its range.

1. *The nodal cubic as a point of $\overline{\mathcal{M}}_{1,1}$.* The nodal cubic C of example 10.2.2 has arithmetic genus 1 and $\deg \omega_C = 0$, so ω_C is not ample and C is not stable as an unpointed curve. Adjoining one marked point p at a smooth point of C changes the count: the stability of a *pointed* curve tests $\omega_C(p) = \omega_C \otimes \mathcal{O}_C(p)$, of degree $0 + 1 = 1 > 0$, which is ample. Combinatorially, the single component is rational after normalization (geometric genus 0)

and carries the self-node (2 special points) plus the marked point p, for 3 special points, meeting the ≥ 3 threshold. So the pointed nodal cubic (C, p) is stable, and it is the point of $\overline{\mathcal{M}}_{1,1}$ lying over $j = \infty$, the degeneration adjoined to $\mathcal{M}_{1,1}$ to compactify it (section 10.4).

2. *A reducible stable genus-*2 *curve.* Take two smooth elliptic curves E_1, E_2 (genus 1 each) and glue them by identifying a point $q_1 \in E_1$ with a point $q_2 \in E_2$, forming a single node. The result $C = E_1 \cup_q E_2$ is nodal and connected; its dual graph is two vertices of weight 1 joined by one edge. Its arithmetic genus is $g_1 + g_2 + (\#\text{nodes} - \#\text{components} + 1) = 1 + 1 + (1 - 2 + 1) = 2$. Each component is genus 1 and meets the other in $k_i = 1$ point, so $(2 \cdot 1 - 2) + 1 = 1 > 0$ on each: combinatorial stability holds, and C is a stable genus-2 curve. Its automorphism group is finite: the automorphisms of each (E_i, q_i) fixing the node (generically the involution $\mathbb{Z}/2$, enlarged to $\mathbb{Z}/4$ or $\mathbb{Z}/6$ when E_i has $j = 1728$ or $j = 0$), together with the swap of the two components when $E_1 \cong E_2$ via an isomorphism carrying q_1 to q_2.

3. *An irreducible stable genus-*2 *curve with the maximal number of nodes.* A stable genus-g curve has at most $3g - 3$ nodes, attained on the maximally degenerate curves that are the deepest boundary points of $\overline{\mathcal{M}}_g$; for $g = 2$ the maximum is $3g - 3 = 3$. Realize it by an irreducible rational curve $\widetilde{C} \cong \mathbb{P}^1$ with three pairs of points glued to three nodes, so the dual graph is one vertex of weight 0 with three loops. The arithmetic genus is $g_i + \#\text{nodes} - \#\text{components} + 1 = 0 + 3 - 1 + 1 = 3$; to land at genus 2 take instead $\mathbb{P}^1$ with *two* pairs of points glued, giving one vertex of weight 0 with two loops and arithmetic genus $0 + 2 - 1 + 1 = 2$. The single rational component carries $k_i = 4$ special points (two nodes, each counted twice as a self-node), so $(2 \cdot 0 - 2) + 4 = 2 > 0$: the curve is stable. This two-noded irreducible curve and the one-node reducible curve of part (2) are two of the boundary strata of $\overline{\mathcal{M}}_2$; the three-noded stable genus-2 curves (two rational components meeting in three points, the theta graph, or a rational curve with a node attached to a nodal cubic, the dumbbell) are the maximally degenerate points, of codimension $3 = 3g - 3$, the dimension of $\overline{\mathcal{M}}_2$ itself.

4. *An unstable curve and its stabilization.* Let D be a smooth genus-2 curve and attach a smooth rational curve $R \cong \mathbb{P}^1$ to D at a single point r, forming a node. The curve $C = D \cup_r R$ is nodal of arithmetic genus $2 + 0 + (1 - 2 + 1) = 2$, but the rational *tail* R meets the rest in only $k_R = 1$ point, so $(2 \cdot 0 - 2) + 1 = -1 < 0$: on R the sheaf ω_C has negative degree, so ω_C is not ample and C is *unstable*. Concretely $R \cong \mathbb{P}^1$ carries the $\mathbb{G}_m$ of automorphisms fixing the node r and the point at infinity of R, an infinite group of automorphisms of C, so $\mathrm{Aut}(C)$ is infinite. The *stabilization* of C contracts the offending tail: collapse R to the point r, leaving the smooth genus-2 curve D, which is stable. Contracting a rational tail with one node, or a rational *bridge* with two nodes (a $\mathbb{P}^1$ meeting the rest in exactly two points), is the operation that turns any semistable curve into a stable one without changing the arithmetic genus, and it is the second half of the stable reduction procedure of section 10.3.

The examples exhibit the full range of stability: the boundary between stable and unstable is drawn component by component by the sign of $(2g_i - 2) + k_i$, the degree of ω_C there. The reducible genus-2 curve, the irreducible multinodal one, and the pointed nodal cubic are stable because every component clears the threshold; the tail curve is unstable because one component falls below it, and stabilization deletes exactly that component. The bound of $3g - 3$ nodes on a stable genus-g curve is not accidental: each node is a codimension-1 degeneration, and $3g - 3$ is the dimension of $\overline{\mathcal{M}}_g$, so a curve with $3g - 3$ nodes is a point of the deepest, 0-dimensional, boundary stratum. This is the combinatorial shadow of the dimension count $\dim \mathcal{M}_g = 3g - 3$ from theorem 10.1.3, and it foreshadows the boundary stratification of section 10.3.

The stable curves are now the objects of a moduli problem, and the deformation theory of Part II makes the resulting stack smooth. The key new input is the deformation theory of a node, computed in example 5.4.2: the node $k[x, y]/(xy)$ has a one-dimensional space of first-order deformations, spanned by the versal smoothing $xy = t$, which resolves the node into a smooth curve as t moves off 0. Each node thus contributes one deformation parameter that smooths it, and this parameter is transverse to the locus where the node persists, which is why the boundary is a divisor and why passing to stable curves does not destroy smoothness.

Theorem 10.2.6: $\overline{\mathcal{M}}_g$ is a Smooth Deligne-Mumford Stack

For $g \geq 2$, the moduli stack $\overline{\mathcal{M}}_g$ of stable genus-g curves is a smooth Deligne-Mumford stack over k of dimension

$$\dim \overline{\mathcal{M}}_g = 3g - 3$$

containing $\mathcal{M}_g$ as a dense open substack, with complement the boundary $\partial\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_g \setminus \mathcal{M}_g$ of curves with at least one node. Its properness is established in section 10.3.

***Proof*:**
A *family of stable curves* over S is a proper flat morphism $\pi\colon \mathcal{C} \to S$ whose geometric fibers are stable genus-g curves, and $\overline{\mathcal{M}}_g$ is the category fibered in groupoids over $(\mathbf{Sch}/k)$ whose objects over S are such families and whose morphisms are the cartesian squares, exactly as for $\mathcal{M}_g$ in definition 10.1.1 with "smooth" relaxed to "at worst nodal." The proof establishes in turn that it is algebraic and Deligne-Mumford, that it is smooth of dimension $3g - 3$, and that $\mathcal{M}_g$ is a dense open substack.

Step 1: algebraic and Deligne-Mumford. That $\overline{\mathcal{M}}_g$ is an algebraic stack follows as in theorem 10.1.3: families of stable curves descend along fppf covers once a relative polarization is fixed, and the relative pluricanonical bundle $\omega_{\mathcal{C}/S}^{\otimes 3}$ supplies one, since ω_C is ample of degree $2g-2$ on each stable fiber (Step 1 of theorem 10.2.4), so $\omega_C^{\otimes 3}$ has degree $6g-6 \geq 2g+1$ for $g \geq 2$; the degree bound is the smooth-curve heuristic, and on a nodal stable curve the very ampleness of $\omega_C^{\otimes 3}$ is the classical tricanonical embedding of stable curves, which embeds each stable curve in a fixed projective space. Artin's criteria are verified as for $\mathcal{M}_g$, with the deformation theory of stable curves in place of that of smooth curves. It is Deligne-Mumford by theorem 8.2.3: the diagonal is unramified because every stable curve has finite reduced automorphism group. Finiteness is definition 10.2.3(2), part of the definition of stability; reducedness holds because the tangent space to $\underline{\mathrm{Aut}}(C)$ at the identity is the space of infinitesimal automorphisms, which vanishes by the computation $H^0(\widetilde{C}_i, T_{\widetilde{C}_i}(-D_i)) = 0$ on each component from Step 4 of theorem 10.2.4. So $\underline{\mathrm{Aut}}(C)$ is finite and reduced, the diagonal is unramified, and $\overline{\mathcal{M}}_g$ is Deligne-Mumford.

Step 2: smooth of dimension $3g - 3$. By the deformation-theoretic criterion, smoothness of the Deligne-Mumford stack is the unobstructedness of the deformation functor of every stable curve together with a constant tangent-space dimension. A stable curve C is a nodal curve, and its deformation theory splits into deformations that preserve the nodes and deformations that smooth them. The deformations preserving the singularities are governed, as for smooth curves, by the cohomology of the tangent complex, and the local contribution of each node is the versal smoothing (example 5.4.2): the node $k[x, y]/(xy)$ has $T^1 = k$, spanned by the deformation

$xy = t$. The full first-order deformation space of C fits in the local-to-global sequence

$$0 \to H^1(C, T_C) \to \mathrm{Def}^1_C \to H^0(C, \mathcal{T}^1_C) \to 0$$

where $\mathcal{T}^1_C$ is the sheaf of local first-order deformations, supported at the nodes with stalk k at each node (its global sections count the smoothing parameters, one per node), and $H^1(C, T_C)$ classifies the locally trivial deformations that move the components and the node positions while keeping the nodes. Curves are unobstructed by proposition 5.4.1: the obstructions lie in the degree-2 term Def^2_C, which vanishes because C is a curve, of cohomological dimension 1, so both the global term $H^2(C, T_C)$ and the higher local terms vanish. Hence every first-order deformation of C extends and the deformation functor is formally smooth. Its tangent space then has dimension

$$\dim \mathrm{Def}^1_C = \dim H^1(C, T_C) + \#\{\text{nodes}\}$$

from the local-to-global sequence, and the two terms recombine to a constant. Smoothing a node trades one node-preserving parameter for one smoothing parameter: near a boundary point with δ nodes, C is the limit of the smooth curves obtained by the δ versal smoothings, so $\dim H^1(C, T_C) = (3g-3) - \delta$ node-preserving directions and the δ smoothing directions add to $3g-3$, each smoothing parameter being transverse to the boundary divisor along which its node persists. The total is thus $3g-3$, the same as in the smooth case $\delta = 0$, and constant across the boundary. So $\overline{\mathcal{M}}_g$ is a formally smooth Deligne-Mumford stack with constant tangent dimension $3g-3$, hence smooth of dimension $3g-3$.

Step 3: $\mathcal{M}_g$ is a dense open substack. A smooth curve is a stable curve with no nodes, so $\mathcal{M}_g$ is the substack of $\overline{\mathcal{M}}_g$ where the fibers are smooth, which is the locus where the discriminant of π is invertible: an open condition, so $\mathcal{M}_g \hookrightarrow \overline{\mathcal{M}}_g$ is an open immersion. It is dense because the smoothing parameter of Step 2 exhibits every boundary point as a limit of smooth curves: given a stable curve C_0 with a node, the versal smoothing $xy = t$ deforms it to a smooth curve for $t \neq 0$, so a punctured neighborhood of $[C_0]$ in $\overline{\mathcal{M}}_g$ lies in $\mathcal{M}_g$. Thus $\mathcal{M}_g$ meets every open set and is dense, with complement the boundary $\partial\overline{\mathcal{M}}_g$ of curves carrying at least one node. This is the open immersion and the density, as required.

The theorem is the structural payoff of the compactification, and the way its proof reuses the earlier chapters is the point. Smoothness comes entirely from Part II: the unobstructedness of curve deformations (proposition 5.4.1) and the node's one-dimensional smoothing space (example 5.4.2) are the only inputs, and they combine so that $\dim = 3g-3$ holds uniformly across the boundary, with each node's smoothing parameter transverse to the boundary divisor it crosses. The Deligne-Mumford property comes from stability itself, the finiteness of $\mathrm{Aut}(C)$ being condition (2) of the definition, so that the compactified stack is Deligne-Mumford for the same reason $\mathcal{M}_g$ was. The one property not yet in hand is properness: nothing above shows that a family of stable curves over a punctured disc extends to the whole disc, and that is exactly the content of stable reduction, deferred to section 10.3.

The smoothness of $\overline{\mathcal{M}}_g$ as a stack should be held against the singularity of its coarse space. The coarse moduli space $\overline{M}_g$ (section 10.4) is singular in general, its singularities coming from the curves with extra automorphisms whose stabilizers act nontrivially on the deformation space; the stack $\overline{\mathcal{M}}_g$ resolves them by remembering the automorphism groups, and it is smooth precisely because the deformation space $\mathrm{Def}^1_C = k^{3g-3}$ is a smooth chart at every point, before the automorphism group is quotiented out. This is the sharpest form of the principle that a stack is the right object: the stack is smooth where its coarse space is singular, and the singularity of the coarse space is the shadow of the automorphism groups the stack keeps and the coarse space forgets. It is the same phenomenon that made $\mathcal{M}_{1,1}$ a smooth stack with a stacky point at $j = 0, 1728$ while its coarse space, the smooth j-line, hid the μ_4, μ_6 symmetry (example 1.3.4); the boundary of $\overline{\mathcal{M}}_g$ extends the same picture to the nodal curves.

Exercise 10.2.1

1. Verify the degree formula $\deg \omega_C = 2p_a(C) - 2$ for the reducible stable genus-2 curve $C = E_1 \cup_q E_2$ of example 10.2.5(2), by computing $\deg(\omega_C|_{E_i})$ on each elliptic component from the adjunction formula $\deg(\omega_C|_{C_i}) = (2g_i - 2) + k_i$ and summing. Confirm that ω_C has positive degree on each component, so ω_C is ample.

2. Show that a stable curve of genus g has at most $3g - 3$ nodes. (Use the degree bound: each node contributes to the special-point count k_i on its components, and combinatorial stability forces $(2g_i - 2) + k_i \geq 1$; sum $\deg \omega_C = 2g - 2$ over components and bound the number of nodes.) For $g = 2$, exhibit the dual graph of a stable curve attaining the maximum $3g - 3 = 3$.

3. Let C be a nodal curve consisting of a smooth rational curve $R \cong \mathbb{P}^1$ meeting a smooth genus-g curve D (with $g \geq 2$) at exactly two points, forming two nodes (a rational *bridge*). Compute the arithmetic genus of C, decide whether C is stable, and describe its stabilization. Contrast with the rational *tail* of example 10.2.5(4).

4. Using the local-to-global sequence $0 \to H^1(C, T_C) \to \mathrm{Def}^1_C \to H^0(C, \mathcal{T}^1_C) \to 0$ from the proof of theorem 10.2.6, compute $\dim \mathrm{Def}^1_C$ for the one-node reducible curve $E_1 \cup_q E_2$ of example 10.2.5(2). Confirm the total is $3g - 3 = 3$, and identify how many of the three parameters are node-preserving and how many smooth the node.

5. Explain why the nodal cubic (example 10.2.2) is not a stable curve of genus 1 on its own, but becomes stable once a smooth marked point is added, by comparing $\deg \omega_C$ with $\deg \omega_C(p)$ and checking the combinatorial condition for the pointed curve. Why does the genus-1 case require a marked point where the genus-g ($g \geq 2$) case does not?

6. Show directly that the automorphism group of the rational-tail curve $C = D \cup_r R$ of example 10.2.5(4) is infinite, by exhibiting a one-parameter family of automorphisms of C supported on the tail R. Explain how this failure of finiteness matches the failure of ampleness of ω_C on R, illustrating the equivalence of theorem 10.2.4.

10.3 Stable Reduction and Properness

The two previous sections built $\overline{\mathcal{M}}_g$ and proved it a smooth Deligne–Mumford stack of dimension $3g - 3$ containing $\mathcal{M}_g$ as a dense open (theorem 10.2.6), with the properness clause of that theorem deferred to this section. What has not yet been shown is the property that makes the compactification worth the name: that $\overline{\mathcal{M}}_g$ is *proper*. Properness is the algebraic incarnation of compactness, and for a moduli space it has a concrete meaning. A one-parameter degeneration of smooth curves, a family over a punctured disc whose fibers are smooth genus-g curves, ought to have a well-defined limit inside the compactified moduli space, and that limit ought to be unique. This section proves exactly that. The existence of the limit is the *stable reduction theorem*, and its uniqueness is the separatedness of the stack; together they are the two halves of the valuative criterion for stacks (definition 8.5.5), and together they say $\overline{\mathcal{M}}_g$ is proper.

The section then reads off the geometry of the compactification. The complement $\partial\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_g \setminus \mathcal{M}_g$, the locus of singular stable curves, is a divisor whose components are indexed by the ways a stable curve of genus g can degenerate along a single node. Its deeper strata are indexed by dual graphs, with codimension equal to the number of nodes. The local model of the whole picture is the node smoothing $xy = t$ of Part II (example 5.4.2): each node contributes one boundary-transverse coordinate, so the boundary is normal crossings in the stack. This is the payoff promised at the start of the chapter. As a stack $\overline{\mathcal{M}}_g$ is smooth and proper, its boundary a normal-crossings divisor, even though, as the next section shows, its coarse space $\overline{M}_g$ acquires singularities.

The central input is a theorem whose statement is elementary and whose proof draws on the semistable reduction of curves, a result at the boundary of what a first course develops. The setup is a family of smooth curves parametrized by a punctured disc, or algebraically over the punctured spectrum of a discrete valuation ring, together with the question of what happens as the parameter approaches the puncture. Without allowing singular curves the limit need not exist: a family of smooth plane cubics whose j-invariant runs off to infinity has no smooth limit, because the limiting curve is nodal. Deligne and Mumford's insight was that admitting stable curves, and permitting a finite base change to absorb monodromy, produces a limit that always exists and is unique.

Theorem 10.3.1: Stable Reduction

Let R be a discrete valuation ring with fraction field K and residue field k, and write $\eta = \operatorname{Spec} K$ for the generic point and $0 = \operatorname{Spec} k$ for the closed point of $S = \operatorname{Spec} R$. Let $C_\eta \to \eta$ be a smooth proper geometrically connected curve of genus $g \geq 2$ over K, or more generally a stable curve of genus g over K. Then there exists a finite extension $R \subseteq R'$ of discrete valuation rings, with fraction field K' a finite extension of K, such that the base-changed curve $C_{\eta'} = C_\eta \times_K K'$ extends to a family of *stable* curves

$$\mathcal{C} \longrightarrow S' = \operatorname{Spec} R'$$

whose generic fiber is $C_{\eta'}$ and whose special fiber $\mathcal{C}_0$ is a stable curve of genus g over the residue field k' of R'. The stable extension is unique: any two stable families over S' (after possibly enlarging R' further) with the same generic fiber are canonically isomorphic.

Geometrically the theorem says that a family of smooth curves over a punctured disc, however wildly it degenerates at the origin, can be completed to a family over the whole disc whose central fiber is a stable curve, provided one is willing to pass to a finite cover of the base disc first. The finite base change is not a defect of the method but a necessity: it is the algebraic reflection of the monodromy of the family around the puncture, and a multivalued limit becomes single-valued only after passing to a cover on which the monodromy is unipotent. The uniqueness clause is what pins down the limit and makes $\overline{\mathcal{M}}_g$ separated; it is the statement that a smooth family has at most one stable degeneration.

***Proof Key ideas*:**
The full proof rests on the semistable reduction theorem for curves, whose complete argument uses resolution of singularities of arithmetic surfaces and the structure theory of the minimal regular model, tools beyond the scope of this book. The complete proofs are in [13] (the original, via the reduction to the case of a curve with a level structure and the Néron model of its Jacobian) and in [17] (a modern exposition organized around the two steps below). The strategy is a two-step construction, and each step is instructive in itself.

Step 1: Semistable reduction. After a finite base change $R \subseteq R'$, the curve $C_{\eta'}$ extends to a flat proper family $\mathcal{X} \to S'$ whose total space $\mathcal{X}$ is a regular scheme and whose special fiber $\mathcal{X}_0$ is a *semistable* curve: a reduced nodal curve, but not necessarily one with ample dualizing sheaf. The base change is where monodromy is absorbed. The family C_η over the punctured disc carries a monodromy representation on the first cohomology of its fibers, and by the local monodromy theorem this monodromy is quasi-unipotent; passing to the finite cover $S' \to S$ on which it becomes unipotent is precisely what allows the total space to be resolved to a family with reduced nodal special fiber. Concretely one takes any flat proper extension of C_η over S and resolves the singularities of its total space, producing a regular arithmetic surface whose special fiber is a divisor with normal crossings. One defect can remain: the special fiber may be *nonreduced*, its components carrying multiplicities, with local model

$$x^a y^b = t$$

in $R[x, y]$ (a regular total space, since t is expressed by x, y) at a crossing of components of multiplicities a and b, whose special fiber $x^a y^b = 0$ is the node taken with those multiplicities. The finite base change is what clears the multiplicities. Passing to the cover $t = (t')^n$, where n is chosen so that the monodromy, quasi-unipotent by the local monodromy theorem, becomes unipotent, turns this local model into $x^a y^b = (t')^n$, a surface singularity of type A; taking its minimal resolution replaces the point by a chain of rational curves and yields a regular total space whose special fiber is now reduced and nodal, with each surviving node in the versal-smoothing model

$$xy = t'$$

the node from Part II (example 5.4.2). This is why the node is the local model of the entire compactification.

Step 2: Contraction to the stable model. The semistable special fiber $\mathcal{X}_0$ may fail to be stable in exactly one way: it may contain *unstable components*, smooth rational curves meeting the rest of $\mathcal{X}_0$ in only one or two points, on which the dualizing sheaf has non-positive degree (definition 10.2.3). These are the chains and tails of $\mathbb{P}^1$'s introduced by the resolution and base change. Each such component is contracted: a (-1)-curve or a chain of rational components is blown down within the family, replacing it by a smooth point or a node. The contraction is carried out on the whole family $\mathcal{X} \to S'$, not just the special fiber, and it changes nothing over the generic point η' (where the fiber is already smooth and has no such components). After finitely many contractions no unstable component remains, and the resulting special fiber has ample dualizing sheaf: it is a stable curve. On the contracted family $\mathcal{C} \to S'$ the relative dualizing sheaf $\omega_{\mathcal{C}/S'}$ is relatively ample, so $\mathcal{C} \to S'$ is a family of stable curves.

Uniqueness. Two stable families over S' with isomorphic generic fibers are isomorphic. The stable model is the relative canonical model of the family: it is Proj of the sheaf of algebras $\bigoplus_{n \geq 0} \pi_* \omega_{\mathcal{C}/S'}^{\otimes n}$, and this construction depends only on the family away from a codimension-two locus, hence only on the generic fiber together with the requirement of stability. Since the canonical model is unique, so is the stable limit. This is the clause that makes the limiting stable curve well-defined, completing the key ideas of the proof.

The theorem is the geometric engine of the compactification, and its structure deserves a closer look. The finite base change and the uniqueness clause are the two features that make the limit both exist and be unique, and they correspond precisely to the two halves of the valuative criterion below. That the intermediate semistable model is not unique, while the final stable model is, is the reason

$\overline{\mathcal{M}}_g$ rather than a moduli space of semistable curves is the correct compactification: only the stable condition rigidifies the limit. The reader should hold onto the node model $xy = t'$ of Step 1, since it is the shape every boundary point takes locally, and the transversality of its smoothing parameter t' is what makes the boundary a divisor rather than a higher-codimension locus.

Before the concrete degenerations, the passage from stable reduction to properness deserves to be spelled out, because it is the whole reason the theorem was proved. Properness for a Deligne–Mumford stack is checked, as for a scheme, by a valuative criterion: a map from the punctured spectrum of a valuation ring into the stack must extend across the puncture, uniquely up to unique isomorphism, after possibly a finite base change on the source. For a moduli stack a map from $\eta = \operatorname{Spec} K$ is exactly a family of objects over K, and the extension across $0 = \operatorname{Spec} k$ is exactly a limiting object. Stable reduction is this valuative criterion, read for the moduli of stable curves.

Theorem 10.3.2: $\overline{\mathcal{M}}_g$ is Proper and Separated

For $g \geq 2$ the Deligne–Mumford stack $\overline{\mathcal{M}}_g$ is proper and separated over the base field k. Consequently $\overline{\mathcal{M}}_g$ is a smooth proper Deligne–Mumford stack of dimension $3g - 3$, and $\mathcal{M}_g$ is an open (but not proper) substack.

***Proof*:**
Properness of a Deligne–Mumford stack of finite type over k is the conjunction of three properties: finite type, separatedness, and the existence half of the valuative criterion (universal closedness). These are the conditions packaged in the definition of a proper morphism of stacks (definition 8.5.5). Each is established from what the chapter has built.

Step 1: Finite type. The stack $\overline{\mathcal{M}}_g$ is of finite type over k. Every stable curve of genus g has ω_C ample (definition 10.2.3), and its tricanonical embedding by $\omega_C^{\otimes 3}$ realizes it as a curve of fixed degree $3(2g-2)$ and fixed Hilbert polynomial in a fixed projective space $\mathbb{P}^{5g-6}$. The stable curves therefore form a bounded family: they are parametrized by a locally closed subscheme of a single Hilbert scheme (definition 2.1.6), whose quotient by the projective linear group presents $\overline{\mathcal{M}}_g$. Boundedness of this Hilbert-scheme locus gives finite type. This is the same boundedness that will produce the coarse projective space in the next section.

Step 2: Separatedness (uniqueness of limits). Separatedness of a Deligne–Mumford stack is checked by the valuative criterion in its uniqueness form: for a discrete valuation ring R with fraction field K, any two extensions to $\operatorname{Spec} R$ of a given object over $\operatorname{Spec} K$ are canonically isomorphic. For $\overline{\mathcal{M}}_g$ an object over $\operatorname{Spec} R$ is a family of stable curves $\mathcal{C} \to \operatorname{Spec} R$, and an object over $\operatorname{Spec} K$ is its generic fiber. The uniqueness clause of theorem 10.3.1 is exactly this: two families of stable curves over $\operatorname{Spec} R$ with isomorphic generic fibers are isomorphic. Since the diagonal of a Deligne–Mumford stack is unramified and the automorphism groups of stable curves are finite (theorem 10.2.6), the isomorphism is unique, and $\overline{\mathcal{M}}_g$ is separated.

Step 3: Universal closedness (existence of limits). The existence half of the valuative criterion requires that *every* object over $\operatorname{Spec} K$ extend, after a finite base change $K \subseteq K'$, to an object over $\operatorname{Spec} R'$ for the corresponding valuation ring R'. An object over $\operatorname{Spec} K$ is a stable curve C_η over K; since a $\operatorname{Spec} K \to \overline{\mathcal{M}}_g$ can land in the boundary as well as in the open $\mathcal{M}_g$, the curve C_η may be smooth or already singular. Both are covered by theorem 10.3.1 in the generality stated there: a stable curve over K, smooth or nodal, acquires after a finite base change $R \subseteq R'$ a family of stable curves $\mathcal{C} \to \operatorname{Spec} R'$ with generic fiber $C_{\eta'}$.

When C_η is smooth this is the principal case of the theorem, the finite base change absorbing the monodromy of a smooth degeneration. When C_η is already a nodal stable curve there is no smooth-locus monodromy to absorb, but a base change may still be forced, now by monodromy permuting the components and nodes of C_η or acting on the Jacobians of the components; in either event the theorem supplies a stable limit. So every object over $\operatorname{Spec} K$ extends after the permitted finite base change. That base change is precisely the latitude the valuative criterion for stacks allows, and which distinguishes properness of a stack from properness of a scheme. Thus $\overline{\mathcal{M}}_g$ satisfies the existence half of the criterion and is universally closed.

Combining the three steps, $\overline{\mathcal{M}}_g$ is of finite type, separated, and universally closed over k, hence proper. With the smoothness and the open embedding of $\mathcal{M}_g$ from theorem 10.2.6, it is a smooth proper Deligne–Mumford stack of dimension $3g-3$, and $\mathcal{M}_g$ is open but omits the boundary, hence is not itself proper, as required.

This is the payoff the chapter was aimed at. The moduli of smooth curves $\mathcal{M}_g$ is not compact, for the reason any space of nondegenerate objects fails to be compact: degenerations escape it. Enlarging to $\overline{\mathcal{M}}_g$ by admitting stable curves plugs every such escape, and the plug is exactly a stable limit supplied by theorem 10.3.1. The finite base change in the existence step is where the stack framework proves its worth. A map into a scheme from a punctured disc extends without base change or not at all; a map into a Deligne–Mumford stack is allowed a finite cover of the source, and that latitude is exactly what a family with nontrivial monodromy requires. A properness that is invisible at the level of the coarse space, which sees only the underlying point set, is manifest at the level of the stack, which sees the automorphisms and the base changes they permit.

With properness in hand the geometry of the added locus can be described. The boundary $\partial\overline{\mathcal{M}}_g$ is the complement of the smooth locus, the closed substack parametrizing singular stable curves. Its structure is dictated by the possible degenerations: a generic point of the boundary parametrizes a stable curve with a single node, and there are exactly two ways a single node can arise. Either the node is non-disconnecting, so that the normalization of the curve is connected of genus $g-1$, or it is disconnecting, splitting the curve into two components of genera i and $g-i$. This dichotomy is the origin of the boundary divisors.

Definition 10.3.3: The Boundary and its Stratification

The *boundary* of $\overline{\mathcal{M}}_g$ is the reduced closed substack

$$\partial\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_g \setminus \mathcal{M}_g$$

parametrizing stable curves that are singular. It is a divisor (pure codimension one), with irreducible components the *boundary divisors*:

1. Δ_0, the closure of the locus of irreducible stable curves with a single non-disconnecting node, whose normalization is a smooth connected curve of genus $g-1$; and
2. Δ_i for $1 \leq i \leq \lfloor g/2 \rfloor$, the closure of the locus of stable curves with a single disconnecting node splitting the curve into a component of genus i and a component of genus $g-i$ (with $\Delta_i = \Delta_{g-i}$, whence the range).

The whole boundary is stratified by *topological type*: to each stable curve C associate its *dual graph* Γ_C, the graph with one vertex of weight p_g for each irreducible component of geometric genus p_g and one edge for each node. The locally closed stratum $\mathcal{M}_\Gamma \subseteq \overline{\mathcal{M}}_g$ of curves with a fixed dual graph Γ has

$$\operatorname{codim} \mathcal{M}_\Gamma = \#\{\text{edges of } \Gamma\} = \#\{\text{nodes}\}$$

and its closure is the union of the strata whose graphs are obtained from Γ by smoothing edges. The boundary $\partial\overline{\mathcal{M}}_g$ is a *normal-crossings divisor* in the stack: at a curve with δ nodes, the δ smoothing parameters are independent local coordinates cutting out the δ branches of the boundary transversally.

The stratification is the combinatorial skeleton of the compactification, and the local deformation theory of Part II is what makes it precise. The tangent space to $\overline{\mathcal{M}}_g$ at a stable curve C with δ nodes decomposes: the deformations of C that preserve all δ nodes form a subspace of codimension δ, and the remaining δ directions each smooth one node independently, along the versal parameter t of the node model $xy = t$ (example 5.4.2). Each node thus contributes exactly one transverse coordinate to the boundary, which is why the number of nodes equals the codimension and why the δ boundary branches cross normally. The deepest strata, the curves with the maximal number $3g-3$ of nodes, are the 0-dimensional strata: totally degenerate stable curves whose components are all rational and whose dual graph is trivalent with $3g-3$ edges.

The dichotomy Δ_0 versus Δ_i is the first thing to compute, and it is entirely combinatorial. Smoothing the single node of a curve in Δ_0 raises the geometric genus from $g-1$ to g while keeping the curve connected, so Δ_0 has codimension one and its generic curve is irreducible. Smoothing the node of a curve in Δ_i merges the two components into one connected curve of genus $i + (g-i) = g$, again codimension one; here the generic curve is reducible with two components. The number of boundary divisors is therefore $1 + \lfloor g/2 \rfloor$: one Δ_0 and one Δ_i for each i from 1 to $\lfloor g/2 \rfloor$.

Example 10.3.4: The Boundary Divisors of $\overline{\mathcal{M}}_2$

Take $g = 2$, so $\dim \overline{\mathcal{M}}_2 = 3g - 3 = 3$. The boundary divisors are indexed by Δ_0 and Δ_i for $1 \leq i \leq \lfloor 2/2 \rfloor = 1$, namely Δ_0 and Δ_1, two divisors of dimension 2.

1. Δ_0: an irreducible curve of arithmetic genus 2 with one non-disconnecting node, whose

normalization is a smooth genus-1 curve with two marked points (the two preimages of the node) glued. Its generic point is stable because the nodal curve has finite automorphism group: an automorphism restricts to one of the genus-1 normalization preserving the two-point preimage set, and only finitely many translations fix that pair, together with the involution swapping the two branches at the node. Smoothing the node yields a smooth genus-2 curve, so Δ_0 is a divisor.

2. Δ_1: two smooth genus-1 curves E_1, E_2 joined at a single disconnecting node (one point of each glued). This is stable exactly because each genus-1 component carries one node, meeting the rest in one point, which is the combinatorial threshold for a genus-1 component (definition 10.2.3). Its moduli within Δ_1 are the j-invariants of E_1 and E_2; the choice of gluing point on each component contributes nothing, since the translation automorphisms of an elliptic curve act transitively on its points, so the two j-invariants alone give $\dim \Delta_1 = 2$.

The dual graph of a generic Δ_0-curve is a single vertex of weight 1 with one loop; that of a generic Δ_1-curve is two vertices of weight 1 joined by one edge. The deepest stratum of $\overline{\mathcal{M}}_2$ is 0-dimensional, reached by curves with $3g-3=3$ nodes: for instance two rational components ($\mathbb{P}^1$'s) joined at three nodes (the dual graph is the "theta graph" with two trivalent vertices and three edges), which is stable since each rational component meets the rest in three points. The codimensions line up: 3 nodes, codimension 3, a point in the 3-dimensional $\overline{\mathcal{M}}_2$.

The example makes the transversality concrete: at the two-node and three-node strata one literally counts nodes and reads off codimension. The remaining task is to see stable reduction in action on a family, so that the abstract limit of theorem 10.3.1 becomes a computation one can watch happen. The cleanest case is the one that motivated the whole story, a family of elliptic curves whose j-invariant degenerates. The genus is 1 rather than ≥ 2, so the relevant stack is $\overline{\mathcal{M}}_{1,1}$ with its marked point, but the mechanism, semistable reduction followed by contraction, is identical and lays bare the base change.

Example 10.3.5: Stable Reduction of a Degenerating Family

The Weierstrass family with $j \to \infty$. Over the punctured disc $\operatorname{Spec} R$ with $R = k[[t]]$, $K = k((t))$, take the Weierstrass family

$$E_\eta : \quad y^2 = x^3 + x^2 + t$$

over K. For $t \neq 0$ this is a smooth elliptic curve, with discriminant $-64t - 432t^2 = -64t(1 + \frac{27}{4}t)$, a unit times t on the nose, so $\operatorname{ord}_t \Delta = 1$ and the j-invariant satisfies $j(E_\eta) \sim c/t \to \infty$ as $t \to 0$. Setting $t = 0$ naively gives $y^2 = x^3 + x^2 = x^2(x+1)$, a nodal cubic with a node at the origin: the Weierstrass model already extends to a family whose special fiber is nodal, and the total space $y^2 = x^3 + x^2 + t$ is regular at that node (its equation has nonvanishing differential). The special fiber is the nodal cubic, an irreducible stable curve of arithmetic genus 1 (stable as a point of $\overline{\mathcal{M}}_{1,1}$, carrying the marked point at infinity). No base change is needed here because the monodromy of this degeneration is already unipotent, a single Dehn twist about the vanishing cycle; the limit is the nodal cubic, the point $j = \infty$ of the coarse space $\mathbb{P}^1 = \overline{M}_{1,1}$. This is the algebraic form of the Tate curve, whose analytic uniformization $E_\eta \cong \mathbb{G}_m / q^{\mathbb{Z}}$ with $q = q(t) \to 0$ degenerates to $\mathbb{G}_m$ with 0 and ∞ glued, again the nodal cubic; see [7] for the Tate parametrization and the q-expansion of j.

When a base change *is* forced, the mechanism appears in full. Take the family

$$E_\eta : \quad y^2 = x^3 + t$$

over $K = k((t))$ with $\operatorname{char} k \neq 2, 3$, whose discriminant is $-432t^2$ and whose j-invariant is identically 0: this is a family of curves each with extra automorphisms, degenerating with additive reduction. The naive limit $y^2 = x^3$ is a cuspidal cubic, which is *not* nodal, hence not semistable. Semistable reduction requires the base change $t = (t')^6$: over $R' = k[[t']]$ the substitution $x = (t')^2 u$, $y = (t')^3 v$ transforms the equation into $v^2 = u^3 + 1$, a smooth elliptic curve over k with good reduction. The finite base change of degree 6 is exactly the order of the automorphism group μ_6 that obstructed the naive limit; it absorbs the monodromy and produces a smooth (in particular stable) special fiber. This is the base change of theorem 10.3.1 made explicit.

*A pencil of genus-*2 *curves.* For $g = 2$, consider a pencil of smooth genus-2 curves specializing to a curve of the shape in example 10.3.4(2): two elliptic curves joined at a node. A genus-2 curve is hyperelliptic, $y^2 = f_6(x)$ with $\deg f_6 = 6$, so it is the double cover of $\mathbb{P}^1$ branched at the six roots of f_6. The disconnecting node of Δ_1 arises when the six branch points do not collide in a single pair but split into *two triples clustered over two distinct values* of x: the base $\mathbb{P}^1$ pinches into two $\mathbb{P}^1$'s joined at a node, and each carries three of the branch points *together with the node itself as a fourth.* A double cover of $\mathbb{P}^1$ must be branched over an even number of points (Riemann–Hurwitz), so three clustered branch points cannot bound a component on their own; the node is forced to be the fourth branch point, and the cover over each $\mathbb{P}^1$, branched at four points, is a genus-1 curve. Over $R = k[[t]]$ with $\operatorname{char} k \neq 2, 3$, take

$$y^2 = \big((x-1)^3 - t\big)\big((x+1)^3 - t\big)$$

For $t \neq 0$ the equation $(x-1)^3 = t$ has three distinct roots clustered near $x = 1$ and $(x+1)^3 = t$ three distinct roots near $x = -1$, six simple branch points in all, so the generic fiber is a smooth genus-2 curve. As $t \to 0$ the two triples collapse onto the distinct values $x = 1$ and $x = -1$, and the naive fiber $y^2 = (x-1)^3(x+1)^3$ is non-nodal at both points. This is where the base change of theorem 10.3.1 is forced, exactly as in the additive case above: after $t = (t')^6$ the substitution $x = 1 + (t')^2 u$ turns the local factor $(x-1)^3 - t$ into $(t')^6(u^3 - 1)$, resolving the collision over $x = 1$ into a smooth elliptic tail $v^2 = (u^3 - 1)h(u)$, and symmetrically over $x = -1$. Each tail is elliptic precisely because the local cubic $v^2 = u^3 - 1$ has odd degree in u, hence is ramified at $u = \infty$, the point toward the neck, which supplies the fourth branch point counted above. The cover is therefore *ramified* over the neck of the base, not unramified. Because a single ramification point of a double cover has one preimage, the semistable model that separates the two tails carries a rational bridge over the neck of the base $\mathbb{P}^1$, a component meeting each tail in one point; this neck $\mathbb{P}^1$ is an unstable rational piece (two special points, non-positive dualizing degree), and contracting it, the contraction step of theorem 10.3.1, fuses the two tails at a single disconnecting node. The stable limit is therefore two genus-1 curves meeting at a node, a point of $\Delta_1 \subseteq \overline{\mathcal{M}}_2$. It is unique by theorem 10.3.1, and it lands in the boundary divisor Δ_1 described in example 10.3.4: the degeneration exhibits a boundary point of $\overline{\mathcal{M}}_2$ as the limit of smooth genus-2 curves.

The two elliptic examples show the two regimes cleanly: multiplicative degeneration ($j \to \infty$) needs no base change and lands on Δ_0-type boundary, the nodal curve; additive degeneration ($j = 0$ with a cusp) needs a base change, whose degree is the order of the extra automorphisms, and after it lands back in the interior with good reduction. The genus-2 pencil shows the disconnecting degeneration landing on Δ_1. Every stable degeneration one meets is a combination of these: a base change to kill monodromy, a semistable model, a contraction of unstable rational pieces, and a stable limit sitting in a boundary stratum indexed by the dual graph of the limit. This is the concrete content of the properness proved in theorem 10.3.2.

Exercise 10.3.1

1. State the valuative criterion for properness of a Deligne–Mumford stack (definition 8.5.5), separating its existence and uniqueness halves. Explain, in one sentence each, which half is supplied by the existence clause of theorem 10.3.1 and which by its uniqueness clause, and why the finite base change permitted in the criterion is essential for a stack but has no analogue in the criterion for a scheme.
2. For $g = 3$, list all boundary divisors of $\overline{\mathcal{M}}_3$ and give the dual graph and generic geometric type of each. What is $\dim \overline{\mathcal{M}}_3$, and what is the dimension of each boundary divisor?
3. Consider the family $E_\eta \colon y^2 = x^3 + t^2 x$ over $K = k((t))$ with $\operatorname{char} k \neq 2, 3$. Compute its j-invariant, determine whether the naive limit at $t = 0$ is semistable, and find the smallest base change $t = (t')^m$ after which the family acquires good (or at least semistable) reduction. Interpret m in terms of the automorphism group of the generic fiber.
4. Let C be a stable curve of genus g whose dual graph Γ has v vertices of geometric genera $p_1, \ldots, p_v$ and e edges. Prove the genus formula
$$g = \sum_{i=1}^{v} p_i + e - v + 1$$
and use it to show that a totally degenerate stable curve (all components rational) lies in a 0-dimensional stratum of $\overline{\mathcal{M}}_g$, with dual graph a trivalent graph having $3g - 3$ edges and $2g - 2$ vertices.
5. Explain why $\mathcal{M}_g$ is *not* proper, by exhibiting a one-parameter family of smooth genus-g curves with no limit inside $\mathcal{M}_g$, and identify where its stable limit sits in $\overline{\mathcal{M}}_g$. Contrast this with the statement of theorem 10.3.2.
6. The boundary $\partial\overline{\mathcal{M}}_g$ is a normal-crossings divisor in the stack, yet its image in the coarse space $\overline{M}_g$ need not be. Using the node model $xy = t$ (example 5.4.2) and the fact that a stable curve with δ nodes has δ independent smoothing parameters, explain why the δ branches of the boundary cross transversally in the stack. Why does this transversality account for $\operatorname{codim} \mathcal{M}_\Gamma = \#\{\text{nodes}\}$?

10.4 The Coarse Space and Low Genus

The previous three sections built the stack: $\mathcal{M}_g$ is a smooth Deligne-Mumford stack of dimension $3g-3$ (theorem 10.1.3), its compactification $\overline{\mathcal{M}}_g$ by stable curves is smooth and proper (theorem 10.2.6, theorem 10.3.2), and the boundary $\partial\overline{\mathcal{M}}_g$ stratifies by the combinatorics of nodes (definition 10.3.3). The stack is the object that remembers everything, including the automorphism group of each curve, and it is the object on which the deformation theory of Part II and the descent theory of Part III act cleanly. This closing section returns to the older question the whole book began with: is there a variety, a scheme of finite type over k, whose points are the isomorphism classes of stable curves? The answer is yes, but it is the coarse space, not the stack, and the passage from one to the other is where the automorphisms are paid for.

Two constructions of that coarse space run side by side, and this section presents both because each carries information the other does not. The Keel-Mori theorem (theorem 8.2.4) produces $\overline{M}_g$

abstractly, as an algebraic space receiving the stack, from the single hypothesis of finite inertia; it is fast and general but says nothing about projectivity. Geometric invariant theory produces $\overline{M}_g$ concretely, as a quotient of a Hilbert scheme of pluricanonically embedded curves by a projective linear group, and this construction sees the projective embedding directly. The section then spends the rest of its length making the abstract picture completely explicit in the one case where every number can be written down: $g = 1$, the moduli of elliptic curves, where the stack is $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$, the coarse space of $\overline{\mathcal{M}}_{1,1}$ is $\mathbb{P}^1$, and the generic automorphism μ_2 is visible in a fractional degree. A first look at the tautological classes λ, ψ, κ closes the book's technical development and points at the intersection theory that lies just beyond it.

The coarse moduli space is the best scheme-theoretic approximation to a stack: the universal target for maps that forget automorphisms. Its existence for $\mathcal{M}_g$ was promised in Part III as an application of Keel-Mori, and its projectivity is the last classical fact this book owes the reader about the moduli of curves. The two are separate theorems with separate proofs, and it is worth stating them together so the division of labor is visible: existence and the algebraic-space structure are Keel-Mori in full; projectivity is geometric invariant theory, with the hard stability computation cited.

Theorem 10.4.1: The Coarse Space $\overline{M}_g$ is a Projective Variety

Let $g \geq 2$ and k algebraically closed. The Deligne-Mumford stack $\overline{\mathcal{M}}_g$ admits a coarse moduli space

$$\pi\colon \overline{\mathcal{M}}_g \longrightarrow \overline{M}_g$$

with the following properties.

1. *Existence and the space structure.* $\overline{M}_g$ exists as a proper algebraic space of finite type over k, and π is universal among morphisms from $\overline{\mathcal{M}}_g$ to algebraic spaces: any morphism $\overline{\mathcal{M}}_g \to Z$ to an algebraic space factors uniquely through π. The map π induces a bijection between the k-points of $\overline{M}_g$ and the isomorphism classes of stable genus-g curves over k.

2. *Projectivity.* $\overline{M}_g$ is a projective variety over k; the open subvariety $M_g \subset \overline{M}_g$, the coarse space of $\mathcal{M}_g$, is quasi-projective.

3. *Singularities.* $\overline{M}_g$ is normal, but singular in general: at a point $[C]$ whose curve has a nontrivial automorphism group $\mathrm{Aut}(C)$, the coarse space looks locally like the quotient of the smooth deformation space $H^1(C, T_C)$ by the finite group $\mathrm{Aut}(C)$, so $\overline{M}_g$ carries at worst finite-quotient singularities, while the stack $\overline{\mathcal{M}}_g$ is smooth everywhere, including at the very points where the coarse space fails to be.

***Proof*:**
The three parts are proved in order, the first in full from the Part III theorem and the third by a direct local computation; the second cites the stability computation whose full form is beyond the scope here, as noted before the statement.

1. By theorem 10.2.6 the stack $\overline{\mathcal{M}}_g$ is a Deligne-Mumford stack, and by theorem 10.3.2 it is proper and separated over k; in particular it is of finite type with finite diagonal. A separated Deligne-Mumford stack of finite type over a field has finite inertia: the automorphism group scheme of a stable curve is finite (definition 10.2.3), which is exactly

the finite inertia hypothesis. The Keel-Mori theorem (theorem 8.2.4) applies verbatim to any such stack and produces a coarse moduli space $\pi\colon \overline{\mathcal{M}}_g \to \overline{M}_g$ that is an algebraic space, together with the universal property recorded in the statement and the bijection on geometric points. Properness of $\overline{M}_g$ follows from properness of $\overline{\mathcal{M}}_g$ because π is proper and surjective, and properness descends along such maps (definition 8.5.5); finite type is preserved for the same reason. This establishes (1) in full.

2. The coarse space $\overline{M}_g$ is constructed as a geometric invariant theory quotient, following the plan laid out in Part I. Fix an integer $\nu \geq 3$. For a stable curve C the dualizing sheaf ω_C is ample (definition 10.2.3), and $\omega_C^{\otimes\nu}$ is very ample with

$$N_\nu + 1 := h^0(C, \omega_C^{\otimes\nu}) = (2\nu - 1)(g - 1)$$

for $\nu \geq 2$, by Riemann-Roch on the nodal curve ([12]) applied to the line bundle $\omega_C^{\otimes\nu}$ of degree $\nu(2g-2)$. The complete linear system $|\omega_C^{\otimes\nu}|$ embeds C as a curve of degree $\nu(2g-2)$ in $\mathbb{P}^{N_\nu}$, its *ν-canonical model*, with a fixed Hilbert polynomial $P(t) = \nu(2g-2)t - (g-1)$ independent of C. The locus of such embedded curves is a locally closed subscheme $H \subset \mathrm{Hilb}^P(\mathbb{P}^{N_\nu})$ of the Hilbert scheme of Ch. 2 (definition 2.1.6, section 2.3), and two points of H give isomorphic abstract curves precisely when they lie in the same orbit of $\mathrm{PGL}_{N_\nu+1}$ acting by change of projective coordinates. Thus

$$\overline{M}_g \;=\; H^{\mathrm{ss}} /\!\!/ \mathrm{PGL}_{N_\nu+1}$$

is a candidate description as a projective GIT quotient (theorem 3.3.4), once one knows the Hilbert points of ν-canonical stable curves are semistable for the natural linearization. That last input is the stability theorem for curves: for ν large the Hilbert point (equivalently, by the Hilbert-Mumford numerical criterion theorem 3.4.5, the asymptotic Chow point) of a ν-canonically embedded stable curve is Hilbert-stable, and conversely a semistable Hilbert point in H is the point of a stable curve, so on H the semistable and stable loci coincide and parametrize exactly the stable curves (theorem 3.5.5). The verification of asymptotic stability is a one-parameter-subgroup computation on the Hilbert point, carried out in full in [25] and [17]; the key idea is that a curve fails to be Hilbert-semistable exactly when it has a subcurve on which $\omega_C^{\otimes\nu}$ is too positive relative to the rest, which for $\nu \gg 0$ recovers the combinatorial stability condition of definition 10.2.3. Granting theorem 3.5.5, theorem 3.3.4 presents $\overline{M}_g$ as the projective quotient of the semistable locus, hence a projective variety; the stable locus, which contains M_g, has a geometric quotient that is the quasi-projective M_g. The two coarse spaces produced, by Keel-Mori and by GIT, agree by the uniqueness half of the coarse-space universal property in (1).

3. Normality of $\overline{M}_g$ follows from smoothness of the stack: $\overline{\mathcal{M}}_g$ is smooth (theorem 10.2.6), and the coarse space of a smooth Deligne-Mumford stack with finite inertia is normal, being étale-locally a quotient of a smooth scheme by a finite group. For the local structure at a point $[C]$, work in a versal deformation. The deformation theory of Part II gives a smooth versal base $\operatorname{Spec} k[[t_1, \ldots, t_{3g-3}]]$ for C, with tangent space $H^1(C, T_C)$ of dimension $3g-3$ and no obstructions (proposition 5.4.1), and the automorphism group $\mathrm{Aut}(C)$ acts on this base fixing the origin. Because $\overline{\mathcal{M}}_g$ is Deligne-Mumford, $\mathrm{Aut}(C)$ is finite (definition 10.2.3), and an étale neighborhood of $[C]$ in the stack is the quotient stack $[H^1(C, T_C)/\mathrm{Aut}(C)]$ (definition 7.3.1). Passing to coarse spaces, an étale neighborhood of $[C]$ in $\overline{M}_g$ is the quotient variety $H^1(C, T_C)/\mathrm{Aut}(C)$, a finite-quotient singularity. When $\mathrm{Aut}(C)$ acts freely away from the origin, as for a generic curve where $\mathrm{Aut}(C)$ is trivial, this quotient is smooth;

when the action has nontrivial stabilizers, the quotient acquires a genuine singularity, and the stack is exactly the object that resolves it by remembering the group. This is what is meant by saying the stack is smooth where the coarse space is singular, completing the proof.

The theorem is the reconciliation the book was built toward, and reading it against the narrative arc is worth the pause. Part I posed the question of representability and diagnosed automorphisms as the obstruction to a fine moduli space; the coarse space is the concession one makes, a variety that represents the moduli problem up to the loss of the automorphism data. Part III showed that the stack loses nothing, and the price of the stack is that it is not a scheme. The Keel-Mori theorem is the bridge between the two worlds: it says that whenever the automorphisms are finite, one can always descend from the stack to a scheme-like object, paying only the passage from smooth to singular. The projectivity, by contrast, is not a formal consequence of anything; it is the geometric input that ω_C is ample on a stable curve, run through the machine of geometric invariant theory. The reason stable curves were defined by ampleness of ω_C in definition 10.2.3, rather than by the combinatorial condition alone, is exactly so that this GIT construction would have a polarization to work with; the combinatorial condition and the ampleness condition were proved equivalent there precisely so that the intrinsic definition and the projective one would coincide.

The description of $\overline{M}_g$ as a GIT quotient is also the origin of the first invariants of the moduli space. The linearization used to form the quotient descends to an ample line bundle on $\overline{M}_g$, and its restriction to the boundary and its intersection numbers are the beginning of the intersection theory of the moduli space. Before turning to that, the entire construction is made completely explicit in genus one, where the coarse space is a familiar curve and every automorphism can be named.

The moduli of elliptic curves is the one case where the stack, the coarse space, the compactification, and every automorphism group can be written down by hand, and it has served as the running touchstone since Part I. It is worth assembling the whole picture in one place, now that the language of stacks is available, because $\overline{\mathcal{M}}_{1,1}$ is the smallest nontrivial instance of every phenomenon in this chapter: a smooth proper Deligne-Mumford stack whose coarse space is a smooth projective curve, whose generic point carries a nontrivial automorphism, and whose boundary is a single point representing a nodal degeneration. The genus-one theory sits slightly outside the $g \geq 2$ conventions of this chapter, since an elliptic curve has $\dim H^1(T) = 1$ rather than $3g-3=0$ and infinitely many automorphisms as a group scheme when unpointed; the pointed stack $\mathcal{M}_{1,1}$ is the object that behaves like the higher-genus $\mathcal{M}_g$.

Example 10.4.2: $\mathcal{M}_{1,1}$ and $\overline{\mathcal{M}}_{1,1}$ in Full

Work over $k = \bar{k}$ of characteristic $\neq 2, 3$. An elliptic curve over k has a Weierstrass model $y^2 = x^3 + ax + b$ with discriminant $\Delta = -16(4a^3 + 27b^2) \neq 0$, and two such models give isomorphic pointed curves precisely when they are related by $(a, b) \mapsto (u^4 a, u^6 b)$ for some $u \in \mathbb{G}_m$, the scaling $x \mapsto u^2 x$, $y \mapsto u^3 y$. The moduli stack of elliptic curves is therefore the quotient stack

$$\mathcal{M}_{1,1} \;\simeq\; \left[\, (\mathbb{A}^2_{a,b} \setminus \{\Delta = 0\}) \,/\, \mathbb{G}_m \,\right]$$

with $\mathbb{G}_m$ acting by $u \cdot (a, b) = (u^4 a, u^6 b)$ (definition 7.3.1). This is the concrete realization of the elliptic curve as a family over its Weierstrass parameters, with the scaling ambiguity turned into a group action rather than divided out on the nose.

The coarse space is the j-line. The $\mathbb{G}_m$-invariant functions on $\mathbb{A}^2_{a,b} \setminus \{\Delta = 0\}$ are generated by the

j-invariant

$$j \;=\; 1728\,\frac{4a^3}{4a^3+27b^2}$$

a weight-zero rational function of (a,b), and the ring of invariants is exactly $k[j]$. The coarse space is Spec of this ring of invariants, the affine j-line

$$M_{1,1} \;=\; (\mathbb{A}^2 \setminus \{\Delta = 0\})/\!\!/\mathbb{G}_m \;\cong\; \mathbb{A}^1_j$$

recovering theorem 1.3.6: two elliptic curves are isomorphic if and only if they share the same j-invariant, and every value $j \in k$ is attained. The coarse map $\mathcal{M}_{1,1} \to \mathbb{A}^1_j$ is the universal map to a scheme, and it forgets exactly the automorphisms.

The compactification adds the nodal cubic at $j = \infty$. The stack $\overline{\mathcal{M}}_{1,1}$ of stable pointed genus-one curves adjoins to $\mathcal{M}_{1,1}$ the single isomorphism class of the nodal cubic, the curve $y^2 = x^3 + x^2$ (a rational curve with one node, marked at the smooth point at infinity), which is stable because its only pointed automorphisms form a finite group and ω_C is ample with the marking. As $j \to \infty$ a family of smooth elliptic curves degenerates to this nodal cubic, the boundary point of the moduli problem; the Tate-curve uniformization makes the degeneration explicit, presenting the nearby smooth curves as $\mathbb{G}_m/q^{\mathbb{Z}}$ with $q \to 0$ (see [7]). The coarse space of the compactified stack is therefore the projective j-line

$$\overline{M}_{1,1} \;=\; \mathbb{A}^1_j \cup \{j = \infty\} \;\cong\; \mathbb{P}^1$$

a smooth projective curve, in agreement with theorem 10.4.1(2): the coarse space is projective. In this genus the coarse space happens to be smooth, because the singularities of a coarse space are finite-quotient singularities of the deformation space, and here that space $H^1(E, T_E)$ is one-dimensional, on which any finite cyclic automorphism group acts by a character, and a one-dimensional linear quotient by a finite cyclic group is again a smooth line.

The stacky structure is the automorphism gerbe. The coarse space $\mathbb{P}^1$ forgets that every point carries automorphisms. A generic elliptic curve E has automorphism group $\mathrm{Aut}(E) = \mu_2 = \{\pm 1\}$, the inversion $P \mapsto -P$ fixing the marked origin, so the generic point of $\mathcal{M}_{1,1}$ is a μ_2-gerbe over its image in $\mathbb{A}^1_j$: the whole stack is generically a $B\mu_2$ living over the j-line, as recorded in example 1.3.4. At two special values the automorphism group jumps: at $j = 0$ the curve $y^2 = x^3 + 1$ has $\mathrm{Aut} = \mu_6$ (extra automorphisms from the sixth roots of unity acting by $x \mapsto \zeta^2 x$, $y \mapsto \zeta^3 y$), and at $j = 1728$ the curve $y^2 = x^3 + x$ has $\mathrm{Aut} = \mu_4$. These are the residual gerbes $B\mu_6$ and $B\mu_4$ of the two orbifold points. The coarse space $\mathbb{P}^1$ sees three ordinary points $j = 0, 1728, \infty$ where the stack sees $B\mu_6$, $B\mu_4$, and the nodal boundary; the passage to the coarse space discards exactly this gerbe structure, which is the automorphism obstruction of example 1.3.4 made geometric. The stack $\mathcal{M}_{1,1}$ remembers the μ_2, μ_4, μ_6; the scheme $\mathbb{A}^1_j$ does not.

The genus-one picture is the whole book in miniature: a moduli problem whose objects have automorphisms, whose stack is a smooth quotient $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$, whose coarse space is the classical j-line, and whose compactification is projective and glues in a nodal degeneration exactly as $\overline{\mathcal{M}}_g$ glues in stable curves. The one feature it does not exhibit, because its deformation space $H^1(E, T_E)$ is one-dimensional, is a singular coarse space; that first appears in genus three, where an automorphism can act on the base of dimension $3g - 3$ without pseudoreflections and so produce a genuine quotient singularity. Even in genus one the automorphisms leave a numerical trace: they force a fractional degree.

The intersection theory of $\overline{\mathcal{M}}_g$ is the subject that begins where this book ends, and it is organized by a distinguished collection of cohomology classes built from the universal curve. These are the tautological classes. They are defined here, and one of them is computed on $\overline{\mathcal{M}}_{1,1}$ to give the reader a concrete number, but their theory, the structure of the ring they generate and the theorems that compute their intersection numbers, is the content of a different volume and is only gestured at. The classes are built from three pieces of universal data: the Hodge bundle, the cotangent lines at the marked points, and their pushforwards.

Definition 10.4.3: Tautological Classes on $\overline{\mathcal{M}}_g$

Let $\pi\colon \overline{\mathcal{C}}_g \to \overline{\mathcal{M}}_g$ be the universal stable curve, the family whose fiber over a point $[C]$ is the curve C itself, with relative dualizing sheaf $\omega_\pi = \omega_{\overline{\mathcal{C}}_g/\overline{\mathcal{M}}_g}$.

1. The *Hodge bundle* is the rank-g vector bundle
$$\mathbb{E} \;=\; \pi_*\omega_\pi$$
on $\overline{\mathcal{M}}_g$, whose fiber over $[C]$ is the space $H^0(C, \omega_C)$ of holomorphic differentials, of dimension g. Its first Chern class is the *lambda class*
$$\lambda \;=\; \lambda_1 \;=\; c_1(\mathbb{E}) \;\in\; H^2(\overline{\mathcal{M}}_g)$$
and more generally $\lambda_i = c_i(\mathbb{E})$.

2. On the moduli of pointed curves $\overline{\mathcal{M}}_{g,n}$, with universal sections $\sigma_1, \ldots, \sigma_n\colon \overline{\mathcal{M}}_{g,n} \to \overline{\mathcal{C}}_{g,n}$ marking the n points, the *psi-classes* are
$$\psi_i \;=\; c_1\big(\sigma_i^*\omega_\pi\big) \;\in\; H^2(\overline{\mathcal{M}}_{g,n})$$
the first Chern class of the cotangent line to the curve at the i-th marked point.

3. The *kappa classes* are the pushforwards
$$\kappa_a \;=\; \pi_*\big(\psi^{a+1}\big) \;\in\; H^{2a}(\overline{\mathcal{M}}_g)$$
where $\psi = c_1(\omega_\pi(\sum_i \sigma_i))$ is the relative cotangent class on the universal curve and π_* is integration over the fiber; on $\overline{\mathcal{M}}_g$ with no marked points the twist $\sum_i \sigma_i$ is empty, so $\psi = c_1(\omega_\pi)$ and κ_a reduces to Mumford's $\pi_*(c_1(\omega_\pi)^{a+1})$, while on $\overline{\mathcal{M}}_{g,n}$ the sections σ_i contribute the twist.

The subring of the cohomology (or Chow) ring generated by the λ_i, ψ_i, and κ_a, together with the classes of the boundary strata Δ_i, is the *tautological ring* of $\overline{\mathcal{M}}_g$.

The definition packages the natural bundles a family of curves carries. The Hodge bundle is the most basic: a family of curves has a family of spaces of differentials, and $\mathbb{E}$ is that family, its determinant $\det\mathbb{E}$ the natural line bundle whose Chern class λ measures how the differentials twist over the moduli space. The psi-classes record the same twisting for the cotangent line at a marked point, and the kappa classes are what one gets by integrating powers of the cotangent class along the fibers of the universal curve. The tautological ring is the subring these generate, and the central discovery of the subject, that intersection numbers of these classes satisfy recursions governed by an integrable hierarchy, is the Witten-Kontsevich theorem; it and the full structure of the tautological ring belong to intersection theory on moduli spaces, treated in [24] and pursued in the intersection-theory volume of this series. This book states the classes and stops, so that the reader who continues arrives with the definitions already in hand.

The one computation worth doing now is on $\overline{\mathcal{M}}_{1,1}$, where the Hodge bundle has rank one and the answer is a fraction, exhibiting the stacky structure of example 10.4.2 numerically. It is grounded in the following low-genus data, and stated with the caveat that the number is an orbifold degree.

Example 10.4.4: Low Genus: Dimensions, Boundary, and λ on $\overline{\mathcal{M}}_{1,1}$

The dimension $\dim \overline{\mathcal{M}}_g = 3g - 3$ and the number of boundary divisors $\Delta_0, \Delta_1, \ldots, \Delta_{\lfloor g/2 \rfloor}$ of definition 10.3.3 for small g are as follows.

g	$\dim \overline{\mathcal{M}}_g = 3g-3$	$\#\{\Delta_i\}$	boundary divisors
2	3	2	$\Delta_0,\ \Delta_1$
3	6	2	$\Delta_0,\ \Delta_1$
4	9	3	$\Delta_0,\ \Delta_1,\ \Delta_2$
5	12	3	$\Delta_0,\ \Delta_1,\ \Delta_2$
6	15	4	$\Delta_0,\ \Delta_1,\ \Delta_2,\ \Delta_3$

In each row Δ_0 is the closure of the locus of irreducible curves with a single node (normalizing to a smooth curve of genus $g-1$ with two points glued), and Δ_i for $1 \le i \le \lfloor g/2 \rfloor$ is the closure of the locus of curves splitting into a genus-i and a genus-$(g-i)$ component meeting at one node. The count is $1 + \lfloor g/2 \rfloor$, so $g = 2$ has the two divisors Δ_0 and Δ_1: the irreducible one-nodal genus-two curves, and the pairs of elliptic curves joined at a point.

The boundary of $\overline{\mathcal{M}}_2$. A stable genus-two curve that is not smooth is one of two types. A point of Δ_1 is two elliptic curves E_1, E_2 joined transversally at a single node; its arithmetic genus is $1 + 1 = 2$ and it is stable because each component is a genus-one curve meeting the node (a genus-one component needs only one special point). A point of Δ_0 is an irreducible curve of geometric genus one with one node, the image of an elliptic curve with two points identified. These are the two codimension-one boundary strata; deeper strata, of higher codimension, glue in more nodes, and a point where the curve is a maximally degenerate stable curve with three nodes, all of whose components are rational, sits in codimension three, the deepest stratum in $\dim = 3$. There are two such configurations: the theta graph, two $\mathbb{P}^1$'s meeting at three points, and the dumbbell, two $\mathbb{P}^1$'s each carrying a self-node, joined by a bridge; both have first Betti number $b_1 = 2$ of the dual graph, which is what makes the arithmetic genus equal to 2. A genus-one component cannot appear in this stratum: it carries a one-dimensional modulus, its j-invariant, so any stratum containing one has dimension at least 1, hence codimension at most 2.

The degree of λ on $\overline{\mathcal{M}}_{1,1}$. On $\overline{\mathcal{M}}_{1,1}$ the Hodge bundle $\mathbb{E} = \pi_* \omega_\pi$ has rank one, its fiber over $[E]$ being the one-dimensional $H^0(E, \omega_E)$, and for a marked genus-one curve the cotangent line at the marked point coincides with the space of differentials, so $\lambda = \psi$ as classes on $\overline{\mathcal{M}}_{1,1}$. The degree of λ is computed by working on the coarse space $\mathbb{P}^1$ of example 10.4.2 and correcting for the generic automorphism. Over the Weierstrass presentation, a differential on $y^2 = x^3 + ax + b$ is a multiple of dx/y, which scales by u^{-1} under $(a, b) \mapsto (u^4 a, u^6 b)$; the Hodge line bundle is therefore the weight-one character of $\mathbb{G}_m$, and its degree over the compactified j-line is a single power of the boundary point. As a divisor class on the stack,

$$\deg_{\overline{\mathcal{M}}_{1,1}} \lambda \;=\; \frac{1}{24}$$

a fractional number, and the fraction records the automorphism orders. It is not a defect of the computation but the correct orbifold degree: the coarse space $\mathbb{P}^1$ has three special points $j = 0, 1728, \infty$ where the stack-automorphism groups are μ_6, μ_4, and, at the boundary node, μ_2 of order 2, and integrating a class over the stack divides the naive count by the order of the automorphism group at each point. The denominator 24 is exactly the number appearing in the theory of modular forms: the modular discriminant Δ, a modular form of weight 12, is a global section of $\mathbb{E}^{\otimes 12}$ over $\mathcal{M}_{1,1}$ with a simple zero at the boundary point $j = \infty$ (see [7]), so 12λ equals the class of the single boundary point, and dividing by the order 2 of that point's own stack-automorphism group μ_2 gives $\deg \lambda = \frac{1}{12} \cdot \frac{1}{2} = \frac{1}{24}$. Read against the coarse space, the moral is exact: a class that would have integer degree on a scheme has fractional degree on a stack, and the denominator is built from the automorphism orders that the coarse space theorem 10.4.1 threw away.

The fractional degree is the last word of the book's technical development, and it closes the circle opened in Part I. The automorphism obstruction was first seen as the failure of a fine moduli space to exist; it was then pinned to the nontrivial finite automorphism group $\mathrm{Aut}(C)$ that each stable curve carries, the source of the stackiness even when $H^0(C, T_C) = 0$ leaves no infinitesimal automorphisms; it was repaired by passing to a stack in Part III; and it is now visible as a number, the denominator 24 that a scheme could never produce. The stack is the object on which every one of these statements is clean, and the coarse space is the shadow it casts on the world of varieties. What lies beyond, the intersection theory of these tautological classes, the recursions among their integrals, and the moduli of higher-dimensional varieties for which the same three-act structure of representability, deformation, and stacks recurs, is the subject the closing chapter points the reader toward.

Exercise 10.4.1

1. Using theorem 10.4.1, explain precisely why $\overline{\mathcal{M}}_g$ admits a coarse moduli space that is an algebraic space but the stack itself is in general not an algebraic space. Which hypothesis of the Keel-Mori theorem (theorem 8.2.4) fails for a stack with a nonfinite inertia, and give an example from earlier in the book of a stack to which Keel-Mori does not apply.

2. Let C be a stable curve with automorphism group $\mathrm{Aut}(C)$ of order $m > 1$ acting on $H^1(C, T_C) \cong k^{3g-3}$. Describe the local structure of $\overline{M}_g$ near $[C]$ as a quotient singularity, and give a genus and a curve for which the quotient is smooth and one for which it is singular. Explain in one sentence why $\overline{\mathcal{M}}_g$ is smooth at $[C]$ regardless.

3. For $\mathcal{M}_{1,1} \simeq [(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ of example 10.4.2, verify that $j = 1728 \cdot 4a^3/(4a^3 + 27b^2)$ is $\mathbb{G}_m$-invariant under $(a, b) \mapsto (u^4 a, u^6 b)$, and confirm that it takes every value in k and that $j = 0$, $j = 1728$ correspond to the curves with $a = 0$, $b = 0$ respectively. Identify the automorphism group at each of these two points.

4. Compute $\dim \overline{\mathcal{M}}_g$ and the number of boundary divisors for $g = 2, 3, 6, 10$ using example 10.4.4. For $g = 2$, describe the two boundary divisors Δ_0 and Δ_1 as loci of stable curves, and state the codimension in $\overline{\mathcal{M}}_2$ of the deepest boundary stratum.

5. On $\overline{\mathcal{M}}_{g,n}$, let $\psi_i = c_1(\sigma_i^* \omega_\pi)$ be the psi-class of definition 10.4.3. Explain why on $\overline{\mathcal{M}}_{1,1}$ one has $\lambda = \psi$, where $\psi = \psi_1$ is the psi-class of the single marked point, and use this to state the value of $\int_{\overline{\mathcal{M}}_{1,1}} \psi$ as an orbifold degree. Why is the answer not an integer?

6. The tautological ring of definition 10.4.3 is generated by the λ_i, ψ_i, κ_a, and the boundary classes

Δ_i. Explain why the Hodge bundle $\mathbb{E} = \pi_* \omega_\pi$ has rank exactly g, and why $\lambda_i = c_i(\mathbb{E}) = 0$ for $i > g$. State, without proof, what further structure the classes λ_i satisfy that makes the tautological ring finitely described (naming the theorem you are pointing at).

11

What Next?

A book on moduli theory could continue almost without end, because the three tools it has assembled, the deformation theory that linearizes a classification problem, the geometric invariant theory that builds quotients, and the stacks that remember automorphisms, are the standard equipment of a large part of modern algebraic geometry. This closing chapter points to where they lead. It proves nothing; it is a map of the surrounding country, with the roads that leave from $\overline{\mathcal{M}}_g$ marked and the neighboring volumes of this series named where they carry the story on.

11.1 Intersection Theory on the Moduli of Curves

The tautological classes introduced at the end of the last chapter, the Hodge class λ and the ψ- and κ-classes, are the beginning of a subject in their own right. The Chow and cohomology rings of $\overline{\mathcal{M}}_{g,n}$ carry a distinguished subring, the tautological ring, generated by these classes and by the boundary strata, and the central problems ask for its structure and for the numbers obtained by integrating monomials in ψ and κ against the fundamental class. Those intersection numbers are governed by deep structures: the Witten conjecture, proved by Kontsevich, identifies their generating function with a solution of the Korteweg-de Vries hierarchy, and the ELSV formula ties them to Hurwitz numbers counting branched covers. The enumerative geometry of $\overline{\mathcal{M}}_g$ begun in Mumford's foundational essay [24] is developed systematically in the intersection-theory volume of this series ([11]), where the Grothendieck-Riemann-Roch computation of the classes λ_i and the structure of the tautological ring are carried out with the machinery of Chern classes and virtual fundamental classes.

11.2 Other Moduli Spaces

The moduli of curves is one of a family of spaces built by the same method. Principally polarized abelian varieties of dimension g form a Deligne-Mumford stack $\mathcal{A}_g$, smooth of dimension $\binom{g+1}{2}$, whose deformation theory was already glimpsed in the treatment of unobstructed objects; the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ sending a curve to its Jacobian ties the two together, and the study of its image is a subject with a long history. Level structures on abelian varieties lead to the Siegel modular varieties and, more generally, to the Shimura varieties at the confluence of Hodge theory, automorphic forms, and arithmetic; these are the subject of projected volumes on abelian and on Shimura varieties. In a different direction, the moduli of vector bundles and of principal G-bundles on a fixed curve, the stack $\mathfrak{Bun}_G$ met in the discussion of Artin stacks, is the geometric arena of the geometric Langlands program, and its cousins, moduli of Higgs bundles and of local systems, carry the non-abelian Hodge theory of Simpson. The twisted sheaves of the gerbes chapter reappear here: moduli of bundles are frequently only coarsely represented until one twists by a Brauer class, exactly as the period-index discussion foreshadowed.

11.3 Foundations Pushed Further

This book stopped its deformation theory at the naive cotangent complex and Schlessinger's criteria, enough for smooth objects and for the node, but the full theory reaches further. Illusie's cotangent complex $L_{X/S}$, a genuine object of the derived category rather than its first two terms, computes the deformations and obstructions of an arbitrary morphism and repairs the failures of the naive version at singular and non-l.c.i. points. Its natural home is derived algebraic geometry, where a scheme is replaced by a functor on simplicial or spectral rings and the tangent complex acquires higher homotopy; in that setting Artin's criteria become the representability theorem of Lurie, and the moduli functors of this book acquire derived enhancements whose classical truncations are the stacks constructed here. The categorical prerequisites, ∞-categories and the homotopy theory that supports them, belong to the category-theory volume ([5]) and to a projected volume on derived and spectral algebraic geometry, and they are the right language for the geometric Langlands correspondence and for the moduli of objects in a differential graded category.

11.4 Deformation Theory in Neighboring Subjects

The deformation theory of Part II is a local instrument that recurs wherever one asks how an object moves. The deformations of a subvariety, governed by the normal sheaf as in the Hilbert-scheme computation, control the geometry of curves on a surface and of complete intersections, the classical subject of Mumford's lectures and of Vistoli's work on the deformation theory of the Quot scheme ([30]). The deformations of a map, rather than of an object, are the engine of the theory of rational curves: the dimension estimate for a space of maps $\mathbb{P}^1 \to X$, together with the bend-and-break technique that produces rational curves by degenerating them, underlies Mori's proof of the cone theorem and the whole minimal model program, developed in Kollár's book on rational curves and in a projected volume on birational geometry. In each of these the pattern is the one established here: a tangent space and an obstruction space, each a cohomology group, that together linearize a moduli problem and predict the dimension of its solution.

11.5 Period Maps and Hodge Theory

A smooth projective family carries a variation of Hodge structure on its cohomology, and the period map records how the Hodge filtration moves as the fibers vary. The target of the period map is itself a moduli space, a quotient of a period domain by an arithmetic group, and the infinitesimal Torelli problem asks whether the period map is an immersion, a question answered by the same cohomological deformation theory used throughout this book. Hodge theory, the Hodge conjecture, and the algebraicity results of the theory of variations of Hodge structure are the subject of a projected volume on Hodge theory; the present book supplies the moduli-theoretic foundation on which the period map is defined.

11.6 A Short Reading List

The reader who wishes to go deeper has excellent guides. For the moduli of curves specifically, the book of Harris and Morrison [17] is the standard companion, and the original paper of Deligne and Mumford [13] remains the source of the stable-curve compactification and of stable reduction. For geometric invariant theory the reference is Mumford's own [25]. For descent and the foundations of stacks, the Grothendieck seminar as explained by Vistoli [30], the book of Laumon and Moret-Bailly [21], and the modern account of Olsson [26] are the standard treatments, with the Stacks Project [29] an encyclopedic resource. For deformation theory the books of Hartshorne [18] and Sernesi [28] carry the subject well past where this book leaves it. Each of these was cited in the chapters where its material was used, and together they chart the paths out of the country this book has surveyed.

The three ideas are worth restating one last time, because they are what the reader takes away. A moduli problem is a functor, and asking whether it is representable is asking for a space whose points are the objects classified. When automorphisms obstruct representability, the remedy is to remember them, and the object that does so is a stack. And the local structure of a moduli space, its smoothness and its dimension, is deformation theory, a computation in the cohomology of a single object. Every moduli space in the subjects above is built by holding these three ideas together, and the reader who has followed them here is equipped to meet the rest.

Index

Bibliography

[1] Michael Artin. “Algebraization of Formal Moduli, I”. In: *Global Analysis (Papers in Honor of K. Kodaira), University of Tokyo Press* (1969), pp. 21–71.

[2] Michael Artin. “Versal Deformations and Algebraic Stacks”. In: *Inventiones Mathematicae* 27 (1974), pp. 165–189.

[3] Andrei Căldăraru. “Derived Categories of Twisted Sheaves on Calabi–Yau Manifolds”. PhD thesis. Cornell University, 2000.

[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Groups and Representations*. 2026.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Category Theory*. 2026.

[6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Commutative Algebra*. 2026.

[7] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elliptic Curves*. 2026.

[8] Nathanael Chwojko-Srawley. *Everything You Need To Know About Étale Cohomology*. Corequisite; in preparation. 2026.

[9] Nathanael Chwojko-Srawley. *Everything You Need To Know About Group and Galois Cohomology*. 2026.

[10] Nathanael Chwojko-Srawley. *Everything You Need To Know About Homological Algebra*. 2026.

[11] Nathanael Chwojko-Srawley. *Everything You Need To Know About Intersection Theory*. Part II of the Algebraic Geometry megabook (NateAlgGeo). 2026.

[12] Nathanael Chwojko-Srawley. *Everything You Need To Know About Modern Algebraic Geometry*. Part I of the Algebraic Geometry megabook (NateAlgGeo). 2026.

[13] Pierre Deligne and David Mumford. “The Irreducibility of the Space of Curves of Given Genus”. In: *Publications Mathématiques de l’IHÉS* 36 (1969), pp. 75–109.

[14] Jean Giraud. *Cohomologie Non Abélienne*. Vol. 179. Grundlehren der mathematischen Wissenschaften. Springer-Verlag, 1971.

[15] Alexander Grothendieck. “Technique de descente et théorèmes d’existence en géométrie algébrique, I–VI”. In: *Fondements de la Géométrie Algébrique (Séminaire Bourbaki)*. Secrétariat mathématique, Paris, 1962.

[16] Jack Hall and David Rydh. “Artin’s Criteria for Algebraicity, Revisited”. In: *Algebra & Number Theory* 13.4 (2019), pp. 749–796.

[17] Joe Harris and Ian Morrison. *Moduli of Curves*. Vol. 187. Graduate Texts in Mathematics. Springer-Verlag, 1998.

[18] Robin Hartshorne. *Deformation Theory*. Vol. 257. Graduate Texts in Mathematics. Springer, 2010.

[19] Seán Keel and Shigefumi Mori. “Quotients by Groupoids”. In: *Annals of Mathematics* 145.1 (1997), pp. 193–213.

[20] Donald Knutson. *Algebraic Spaces*. Vol. 203. Lecture Notes in Mathematics. Springer-Verlag, 1971.

[21] Gérard Laumon and Laurent Moret-Bailly. *Champs Algébriques*. Vol. 39. Ergebnisse der Mathematik und ihrer Grenzgebiete. Springer-Verlag, 2000.

[22] Max Lieblich. “Moduli of Twisted Sheaves”. In: *Duke Mathematical Journal* 138.1 (2007), pp. 23–118.

[23] James S. Milne. *Étale Cohomology*. Vol. 33. Princeton Mathematical Series. Princeton University Press, 1980.

[24] David Mumford. “Towards an Enumerative Geometry of the Moduli Space of Curves”. In: *Arithmetic and Geometry (M. Artin and J. Tate, eds.)* Vol. 36. Progress in Mathematics. Birkhäuser, 1983, pp. 271–328.

[25] David Mumford, John Fogarty, and Frances Kirwan. *Geometric Invariant Theory*. 3rd ed. Ergebnisse der Mathematik und ihrer Grenzgebiete. Springer-Verlag, 1994.

[26] Martin Olsson. *Algebraic Spaces and Stacks*. Vol. 62. American Mathematical Society Colloquium Publications. American Mathematical Society, 2016.

[27] Michael Schlessinger. “Functors of Artin Rings”. In: *Transactions of the American Mathematical Society* 130 (1968), pp. 208–222.

[28] Edoardo Sernesi. *Deformations of Algebraic Schemes*. Vol. 334. Grundlehren der mathematischen Wissenschaften. Springer, 2006.

[29] The Stacks project authors. *The Stacks Project*. 2026. URL: https://stacks.math.columbia.edu.

[30] Angelo Vistoli. “Grothendieck Topologies, Fibered Categories and Descent Theory”. In: *Fundamental Algebraic Geometry: Grothendieck’s FGA Explained*. Vol. 123. Mathematical Surveys and Monographs. American Mathematical Society, 2005, pp. 1–104.